1.1 Vectors in Two- and Three-Dimensional Space

Note Title 11/16/2015

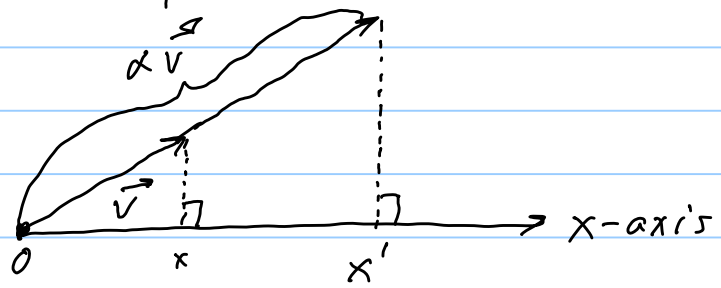
4. **4.** $(2, 3, 5) - 4\mathbf{i} + 3\mathbf{j} = (?, ?, ?)$

$(2,3,5) - 4\hat{i} + 3\hat{j} = (2,3,5) - (4,0,0) + (0,3,0)$

$= \underline{(-2, 6, 5)}$

12.6. (b) Using an argument based on similar triangles, prove that $\alpha\mathbf{v} = (\alpha x, \alpha y, \alpha z)$ when $\mathbf{v} = (x, y, z)$.

Consider just the vectors $\vec{v}$, $\alpha\vec{v}$ and the x-axis. Drop perpendiculars from $\vec{v}$ to the x-axis, and from $\alpha\vec{v}$ to the x-axis:



Assume $\alpha > 0$

x = 1st component of $\vec{v}$

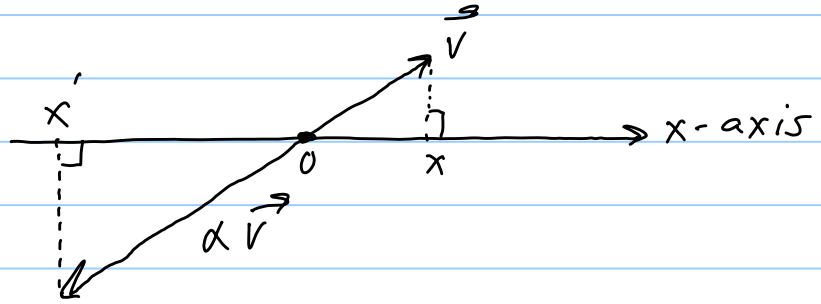
x' = 1st component of $\alpha\vec{v}$

By similar triangles $\dfrac{\text{length } \alpha\vec{v}}{\text{length } \vec{v}} = \dfrac{x'}{x}$

But length $\alpha\vec{v} = \alpha$ (length $\vec{v}$)

$\therefore \frac{\text{length } \alpha\vec{v}}{\text{length } \vec{v}} = \alpha = \frac{x'}{x}$, $\therefore x' = \alpha x$

$\alpha < 0$:



By similar triangles $\frac{\text{length } \alpha\vec{v}}{\text{length } \vec{v}} = |\alpha| = \frac{|x'|}{x}$

$\therefore |x'| = |\alpha| x = -\alpha x$

Since $x' < 0$, $-x' = |x'|$. $\therefore -x' = -\alpha x$,

$\therefore x' = \alpha x$

The analysis is similar for the y and z components.

$\therefore$ For $\alpha \neq 0$, $\alpha\vec{v} = (\alpha x, \alpha y, \alpha z)$

For $\alpha = 0$, $\alpha\vec{v}$ is the point vector $(0,0,0)$

and $(\alpha x, \alpha y, \alpha z) = (0,0,0)$.

19. **19.** The parallelogram whose adjacent sides are the vectors $\mathbf{i} + 3\mathbf{k}$ and $-2\mathbf{j}$

One side: $r(\hat{i} + 3\hat{k})$, $0 \le r \le 1$

Other side: $s(-2\hat{j})$, $0 \le s \le 1$

$\therefore r(\hat{i} + 3\hat{k}) + s(-2\hat{j})$

$= r\hat{i} - 2s\hat{j} + 3r\hat{k}$

$= (r, -2s, 3r)$, $0 \le r \le 1$, $0 \le s \le 1$

20. **20.** Show that $\mathbf{l}_1(t) = (1, 2, 3) + t(1, 0, -2)$ and $\mathbf{l}_2(t) = (2, 2, 1) + t(-2, 0, 4)$ parametrize the same line.

It is sufficient to show that a point $\vec{p}$ is a point on $l_1 \Leftrightarrow$ it is a point on l_2.

i.e., for every t_1 for l_1, there is a t_2 for l_2, and for every t_2 for l_2, there is a t_1 for l_1

$\therefore (1,2,3) + t_1(1,0,-2) = (2,2,1) + t_2(-2,0,4)$

$\therefore t_1(1,0,-2) - t_2(-2,0,4) = (1,0,-2)$

or $t_1 + 2t_2 = 1$ [1]

$t_1(0) - t_2(0) = 0$ (true for all t_1, t_2) [2]

$-2t_1 - 4t_2 = -2$ [3]

$\therefore \quad t_1 + 2t_2 = 1 \qquad [1]$

$2t_1 + 4t_2 = 2 \iff t_1 + 2t_2 = 1 \qquad [3]$

$\therefore$ [1] and [3] are equivalent.

$\therefore$ given any t_1, $t_2 = \frac{1 - t_1}{2}$

$\therefore$ given any point on l_1, there is a unique t_2 s.t. the point is on l_2

given any t_2, $t_1 = 1 - 2t_2$

so given any point on l_2, there is a unique t_1 s.t. the point is on l_1.

$\therefore$ the lines are the same.

21.

21. Do the points $(2, 3, -4)$, $(2, 1, -1)$, and $(2, 7, -10)$ lie on the same line?

Line connecting $(2,3,-4)$ and $(2,1,-1)$ is:

$$(2,3,-4) + t[(2,3,-4) - (2,1,-1)] =$$

$$(2,3,-4) + t(0,2,-3)$$

Is there a value of t s.t. $(2,7,-10) = (2,3,-4) + t(0,2,-3)$?

$(2,7,-10) - (2,3,-4) = (0,4,-6) = 2(0,2,-3)$.

$\therefore$ If $t = 2$, then $(2,7,-10)$ is on the same line.

$\therefore$ All 3 points lie on the same line.

25.

25. Show that there are no points (x, y, z) satisfying $2x - 3y + z - 2 = 0$ and lying on the line $\mathbf{v} = (2, -2, -1) + t(1, 1, 1)$.

Normal to the plane $2x - 3y + z - 2 = 0$ is $(2,-3,1)$.

The normal is perpendicular to the vector parallel to the line as $(2,-3,1) \cdot (1,1,1) = 0$.

$\therefore$ The line is parallel to the plane

Also, a point of the line, $(2,-2,1)$ is not in the plane since it doesn't satisfy

$2x - 3y + z - 2 = 0 : 2(2) - 3(-2) + (1) - 2 = 9 \neq 0$

$\therefore$ line is not in plane and is parallel to it.

Alternative method:

$\vec{v} = (2,-2,-1) + t(1,1,1) = (2+t, -2+t, -1+t)$

$\therefore x = 2 + t, \quad y = -2 + t, \quad z = -1 + t$

$\therefore$ Substituting into $2x - 3y + z - 2 = 0$,

$2(2+t) - 3(-2+t) + (-1+t) - 2 = 0,$

or $4 + 2t + 6 - 3t + (-1) + t - 2 = 0,$

or $7 = 0$, i.e., there is no value of t to make a point on the line lie in the plane.

26.

26. Show that every point on the line $\mathbf{v} = (1, -1, 2) + t(2, 3, 1)$ satisfies the equation $5x - 3y - z - 6 = 0$.

$\vec{v} = (1 + 2t, -1 + 3t, 2 + t)$, or

$x = 1 + 2t, \quad y = -1 + 3t, \quad z = 2 + t$

$\therefore$ Substituting into $5x - 3y - z - 6 = 0$,

$5(1+2t) - 3(-1+3t) - (2+t) - 6 = 0,$ or

$5 + 10t + 3 - 9t - 2 - t - 6 = 0,$ or

$8 - 8 + 10t - 10t = 0,$ or $0 = 0$ for all t

∴ For all t, the points $\vec{v}$ satisfy $5x-3y-z-6=0$,

28.

28. Do the lines $(x, y, z) = (t + 4, 4t + 5, t - 2)$ and $(x, y, z) = (2s + 3, s + 1, 2s - 3)$ intersect?

Solve: $t + 4 = 2s + 3$ [x]

$4t + 5 = s + 1$ [y]

$t - 2 = 2s - 3$ [z]

$\therefore t = 2s - 1$ [x]

$t = \frac{1}{4}s - 1$ [y]

$t = 2s - 1$ [z]

From [x], [y], $2s - 1 = \frac{1}{4}s - 1$, $s = 0$

$\therefore t = -1$

∴ For $s = 0$ and $t = -1$, the lines intersect at the point: $(3, 1, -3)$

31.

31. The plane determined by the three points (x_0, y_0, z_0), (x_1, y_1, z_1), and (x_2, y_2, z_2)

Let $\vec{v} = (x_1, y_1, z_1) - (x_0, y_0, z_0)$

$\vec{w} = (x_2, y_2, z_2) - (x_0, y_0, z_0)$

Assuming points are not colinear (i.e.,

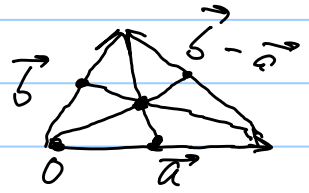
There is no value c s.t. $\vec{v} = c\vec{w}$),

The plane is described by:

$(x_0, y_0, z_0) + r\vec{v} + s\vec{w}$, for all real values r, s.

34. **34.** The medians of a triangle intersect at a point, and this point divides each median in a ratio of 2 : 1.

Let the sides of a triangle be: $\vec{a}$, $\vec{b}$, $\vec{b}-\vec{a}$



0 is the origin.

Median from $\vec{b}$: $\vec{a} - \frac{1}{2}\vec{b}$ [1]

Median from $\vec{a}$: $\vec{b} - \frac{1}{2}\vec{a}$ [2]

Median from $\vec{b}-\vec{a}$: $\vec{a} + \frac{1}{2}(\vec{b}-\vec{a}) - (0,0) = \frac{1}{2}\vec{a} + \frac{1}{2}\vec{b}$ [3]

Consider median from $\vec{a}$:

$$\frac{1}{2}\vec{a} + \frac{1}{3}\left(\vec{b} - \frac{1}{2}\vec{a}\right) = \frac{1}{3}\vec{a} + \frac{1}{3}\vec{b}$$

Consider median from $\vec{b}$:

$$\frac{1}{2}\vec{b} + \frac{1}{3}\left(\vec{a} - \frac{1}{2}\vec{b}\right) = \frac{1}{3}\vec{b} + \frac{1}{3}\vec{a}$$

∴ The above two medians meet at the same point, $\frac{1}{3}\vec{a} + \frac{1}{3}\vec{b}$, which was obtained by going $\frac{1}{3}$ of the way from the origin of the medians (i.e., 2:1 ratio).

Consider median from $\vec{b} - \vec{a}$:

This vector goes from the origin to the point $\frac{1}{2}$ along $(\vec{b} - \vec{a})$, or: $\frac{1}{2}\vec{a} + \frac{1}{2}\vec{b}$ from [3].

2/3 along this vector is $\frac{2}{3}\left(\frac{1}{2}\vec{a} + \frac{1}{2}\vec{b}\right) = \frac{1}{3}\vec{a} + \frac{1}{3}\vec{b}$.

∴ The point of intersection of the first two medians is on the third median, and the point is $\frac{1}{3}$ from the median point (or, $\frac{2}{3}$ from the origin).

∴ All 3 median intersect at one point, and that point is $\frac{1}{3}$ along the segment from the base (2:1 ratio).

37.

37. Find a line that lies entirely in the set defined by the equation $x^2 + y^2 - z^2 = 1$.

For different values of z, $x^2+y^2 = 1+z^2$, or circles of radii $\sqrt{1+z^2}$, parallel to xy-plane.

For $x = 0$, $y^2 - z^2 = 1$, an hyperbola.

$\therefore$ Line can't be parallel to a coordinate plane, so it must be oblique.

Consider the point $(1, 0, 0)$. This is on the surface, since $1^2 + 0^2 - 0^2 = 1$

Consider the vector $t(0, 1, 1) = (0, t, t)$

For all values t, $(1, 0, 0) + t(0, 1, 1) = (1, t, t)$

and $(1, t, t)$ is on the surface, since

$1^2 + t^2 - t^2 = 1$.

$\therefore$ All points on the line $\underline{(1,0,0) + t(0,1,1)}$ lie on the surface $x^2 + y^2 - z^2 = 1$

1.2 The Inner Product, Length, and Distance

Note Title 11/18/2015

5.

5. Is $\|8\mathbf{i} - 12\mathbf{k}\| \cdot \|6\mathbf{j} + \mathbf{k}\| - |(8\mathbf{i} - 12\mathbf{k}) \cdot (6\mathbf{j} + \mathbf{k})|$ equal to zero? Explain.

Since $\vec{a} \cdot \vec{b} = \|a\| \|b\| \cos\theta$,

Let $\vec{a} = (8\hat{i} - 12\hat{k})$, $\vec{b} = (6\hat{j} + \hat{k})$

$\therefore \|\vec{a}\| \cdot \|\vec{b}\| - |\vec{a} \cdot \vec{b}| = \|\vec{a}\| \cdot \|\vec{b}\| - |\|\vec{a}\| \cdot \|\vec{b}\| \cos\theta|$

$= \|\vec{a}\| \|\vec{b}\| (1 - |\cos\theta|)$

This will only be 0 when $\cos\theta = \pm 1$,

or $\theta = 0, \pi$, since $\vec{a} \neq 0$, $\vec{b} \neq 0$.

$\therefore$ for $\theta = 0, \pi$, $\vec{a} = r\vec{b}$, r some real $\neq 0$.

Since $8\hat{i} - 12\hat{k} \neq r(6\hat{j} + \hat{k})$, Then

The value is not zero.

12.

12. Let $\mathbf{v} = (2, 3)$. Suppose $\mathbf{w} \in \mathbb{R}^2$ is perpendicular to $\mathbf{v}$, and that $\|\mathbf{w}\| = 5$. This determines $\mathbf{w}$ up to sign. Find one such $\mathbf{w}$.

$(-3, 2) \perp (2, 3)$ since $(-3, 2) \cdot (2, 3) = -6 + 6 = 0$

$\|(-3, 2)\| = \sqrt{13}$. $\therefore \pm \frac{5}{\sqrt{13}} (-3, 2)$

15. **15.** What is the geometric relation between the vectors $\mathbf{v}$ and $\mathbf{w}$ if $\mathbf{v} \cdot \mathbf{w} = -\|\mathbf{v}\|\,\|\mathbf{w}\|$?

$\vec{v} \cdot \vec{w} = \|\vec{v}\|\,\|\vec{w}\| \cos\theta = -\|\vec{v}\|\,\|\vec{w}\| \Rightarrow \cos\theta = -1.$

$\therefore \theta = \pi$, so $\vec{v}, \vec{w}$ are parallel, and point in opposite directions.

21. **21.** Find the projection of $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$ onto $\mathbf{u} = -\mathbf{i} + \mathbf{j} + \mathbf{k}$.

Let θ = angle $(0 \le \theta \le \pi)$ between $\vec{v}, \vec{u}$.

$\therefore \|\vec{v}\| \cos\theta$ = length and direction of projection of $\vec{v}$ onto $\vec{u}$.

$\vec{v} \cdot \vec{u} = |\vec{v}|\,|\vec{u}| \cos\theta, \quad \therefore \|\vec{v}\| \cos\theta = \dfrac{\vec{v} \cdot \vec{u}}{\|\vec{u}\|}$

$\therefore$ Projection is $\dfrac{\vec{v} \cdot \vec{u}}{\|\vec{u}\|} \cdot \dfrac{\vec{u}}{\|\vec{u}\|} = \dfrac{\vec{v} \cdot \vec{u}}{\|\vec{u}\|^2}\,\vec{u}$

$\therefore \vec{v} \cdot \vec{u} = (2, 1, -3) \cdot (-1, 1, 1) = -2 + 1 - 3 = -4$

$\|\vec{u}\|^2 = 3. \quad \therefore$ Projection $\vec{v} = -\frac{4}{3}(-1, 1, 1)$

$= -\frac{4}{3}\hat{i} + \frac{4}{3}\hat{j} + \frac{4}{3}\hat{k}$

24.

24. Let $\mathbf{b} = (3, 1, 1)$ and P be the plane through the origin given by $x + y + 2z = 0$.

(a) Find an orthogonal basis for P. That is, find two nonzero orthogonal vectors $\mathbf{v}_1, \mathbf{v}_2 \in P$.

(b) Find the orthogonal projection of $\mathbf{b}$ onto P. That is, find $\text{Proj}_{\mathbf{v}_1}\mathbf{b} + \text{Proj}_{\mathbf{v}_2}\mathbf{b}$.

(a) $(1,-1,0)$ is on the plane

$(1,1,-1)$ is on the plane.

$(1,-1,0)\cdot(1,1,-1) = 0$, so vectors are $\perp$.

$\therefore$ Let $\vec{v_1} = (1,-1,0)$, $\vec{v_2} = (1,1,-1)$

(b) $\text{Proj}_{v_1} b = \dfrac{\vec{b}\cdot\vec{v_1}}{\|\vec{v_1}\|^2}\vec{v_1} = \dfrac{(3,1,1)\cdot(1,-1,0)}{2}(1,-1,0)$

$= (1,-1,0)$

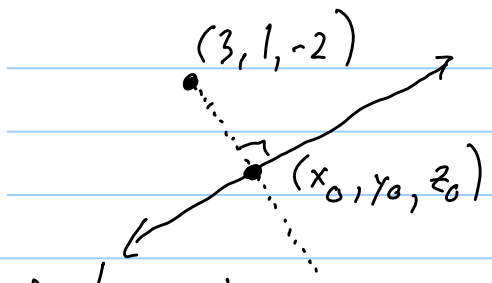
$\text{Proj}_{v_2} b = \dfrac{\vec{b}\cdot\vec{v_2}}{\|\vec{v_2}\|^2}\vec{v_2} = \dfrac{(3,1,1)\cdot(1,1,-1)}{3}(1,1,-1)$

$= (1,1,-1)$

$\therefore$ Projection $\vec{b}$ onto plane $= (1,-1,0)+(1,1,-1)$

$= (2,0,-1)$

26. **26.** Find the line through $(3, 1, -2)$ that intersects and is perpendicular to the line $x = -1 + t$, $y = -2 + t$, $z = -1 + t$. [HINT: If (x_0, y_0, z_0) is the point of intersection, find its coordinates.]



Let (x_0, y_0, z_0) be point of intersection.

$\therefore (x_0, y_0, z_0)$ is on line $l_1 = (-1, -2, -1) + t(1, 1, 1)$

The projection of $(3, 1, -2) - (x_0, y_0, z_0) = \vec{w}$ onto $\vec{v} = (1, 1, 1)$ must be zero.

$\therefore \vec{w} \cdot \vec{v} = 0$.

$\therefore (3 - x_0, 1 - y_0, -2 - z_0) \cdot (1, 1, 1) = 0$, or

$3 - x_0 + 1 - y_0 - 2 - z_0 = 0$, or

$x_0 + y_0 + z_0 = 2$ [1]

But $(x_0, y_0, z_0) = (-1, -2, -1) + t(1, 1, 1)$, some t

$\therefore x_0 = -1 + t$, $y_0 = -2 + t$, $z_0 = -1 + t$ [2]

Substituting [2] into [1],

$(-1 + t) + (-2 + t) + (-1 + t) = 2$, or

$-4 + 3t = 2$, $t = 2$

$$\therefore (x_0, y_0, z_0) = (-1,-2,-1) + 2(1,1,1) = (1,0,1)$$

$$\therefore \ell_2 = (3,1,-2) + s\left[(3,1,-2) - (x_0, y_0, z_0)\right]$$

$$= (3,1,-2) + s\left[(3,1,-2) - (1,0,1)\right]$$

$$\therefore \underline{\ell_2 = (3,1,-2) + s(2,1,-3)}$$

28.

28. For $\mathbf{v} = (v_1, v_2, v_3)$ let α, β, γ denote the angles between $\mathbf{v}$ and the x, y, and z axes, respectively. Show that $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

$$\vec{v} \cdot \hat{i} = \|\vec{v}\| \cdot \|\hat{i}\| \cos\alpha \quad \therefore v_1 = \sqrt{v_1^2 + v_2^2 + v_3^2}\cos\alpha$$

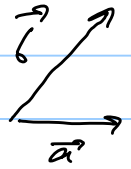
$$\text{or} \quad \cos\alpha = \frac{v_1}{\sqrt{v_1^2 + v_2^2 + v_3^2}}$$

$$\vec{v} \cdot \hat{j} = v_2 \quad \therefore \cos\beta = \frac{v_2}{\sqrt{v_1^2 + v_2^2 + v_3^2}}$$

$$\vec{v} \cdot \hat{k} = v_3 \quad \therefore \cos\gamma = \frac{v_3}{\sqrt{v_1^2 + v_2^2 + v_3^2}}$$

$$\therefore \cos^2\alpha + \cos^2\beta + \cos^2\gamma = \frac{v_1^2 + v_2^2 + v_3^2}{\left(\sqrt{v_1^2 + v_2^2 + v_3^2}\right)^2} = 1$$

38. **38.** Show that in any parallelogram the sum of the squares of the lengths of the four sides equals the sum of the squares of the lengths of the two diagonals.



Let $\vec{a}$, $\vec{b}$ be two non-parallel vectors

The diagonals are $\vec{a}+\vec{b}$ and $\vec{b}-\vec{a}$

$\therefore \|\vec{a}+\vec{b}\|^2 + \|\vec{b}-\vec{a}\|^2 =$

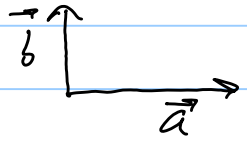
$(\vec{a}+\vec{b})\cdot(\vec{a}+\vec{b}) + (\vec{b}-\vec{a})\cdot(\vec{b}-\vec{a}) =$

$[\vec{a}\cdot\vec{a} + 2\vec{a}\cdot\vec{b} + \vec{b}\cdot\vec{b}] + [\vec{b}\cdot\vec{b} - 2\vec{a}\cdot\vec{b} + \vec{a}\cdot\vec{a}]$

$= 2\vec{a}\cdot\vec{a} + 2\vec{b}\cdot\vec{b} = 2\|\vec{a}\|^2 + 2\|\vec{b}\|^2$

$= \|\vec{a}\|^2 + \|\vec{a}\|^2 + \|\vec{b}\|^2 + \|\vec{b}\|^2$

39. **39.** Using vectors, show that the diagonals of a rectangle are perpendicular if and only if the rectangle is a square.



Let $\vec{a}$, $\vec{b}$ be non-parallel vectors representing

the sides of a rectangle. $\therefore \vec{a}\cdot\vec{b} = 0$

Diagonals are: $\vec{a}+\vec{b}$, $\vec{b}-\vec{a}$

$\therefore (\vec{a}+\vec{b})\cdot(\vec{b}-\vec{a}) = 0 \iff$

$\vec{a}\cdot\vec{b} - \vec{a}\cdot\vec{a} + \vec{b}\cdot\vec{b} - \vec{b}\cdot\vec{a} = \vec{b}\cdot\vec{b} - \vec{a}\cdot\vec{a} = 0$

since $\vec{a}\cdot\vec{b} = 0$ (a rectangle)

$$\therefore (\vec{a}+\vec{b})\cdot(\vec{b}-\vec{a}) = 0 \iff \vec{b}\cdot\vec{b} = \vec{a}\cdot\vec{a}$$

$$\iff \|\vec{b}\|^2 = \|\vec{a}\|^2$$

i.e., diagonals perpendicular $\iff$ sides of rectangle are equal

i.e., diagonals perpendicular $\iff$ rectangle is a square

1.3 Matrices, Determinants, and the Cross Product

Note Title 11/23/2015

3.

3. Compute $\mathbf{a} \times \mathbf{b}$, where $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\mathbf{b} = 2\mathbf{i} + \mathbf{j} + \mathbf{k}$.

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ 2 & 1 & 1 \end{vmatrix} = (-2-1, -(1-2), 1-(-4))$$

$$= (-3, 1, 5)$$

6.

6. A triangle has vertices (0, 0, 0), (1, 1, 1), and (0, −2, 3). Find its area.

$$\vec{a} = (1,1,1) - (0,0,0) = (1,1,1)$$

$$\vec{b} = (0,-2,3) - (0,0,0) = (0,-2,3)$$

$$\text{Area} = \frac{1}{2}\|\vec{a} \times \vec{b}\|, \quad \vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 1 \\ 0 & -2 & 3 \end{vmatrix} = (5,-3,-2)$$

$$\therefore \|(5,-3,-2)\| = \sqrt{5^2+3^2+2^2} = \sqrt{38}$$

$$\therefore \text{Area} = \frac{1}{2}\sqrt{38}$$

8.

8. What is the volume of the parallelepiped with sides $\mathbf{i}$, $3\mathbf{j} - \mathbf{k}$, and $4\mathbf{i} + 2\mathbf{j} - \mathbf{k}$?

$$\text{Volume} = \left| \hat{i} \cdot \left[(3\hat{j} - \hat{k}) \times (4\hat{i} + 2\hat{j} - \hat{k}) \right] \right|$$

$$= \text{abs}\left(\begin{vmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 4 & 2 & -1 \end{vmatrix}\right) = |-3-(-2)| = 1$$

$\therefore$ Volume = 1

15.

15. Find an equation for the plane that

(a) is perpendicular to $\mathbf{v} = (1, 1, 1)$ and passes through $(1, 0, 0)$.

(b) is perpendicular to $\mathbf{v} = (1, 2, 3)$ and passes through $(1, 1, 1)$.

(c) is perpendicular to the line $\mathbf{l}(t) = (5, 0, 2)t + (3, -1, 1)$ and passes through $(5, -1, 0)$.

(d) is perpendicular to the line $\mathbf{l}(t) = (-1, -2, 3)t + (0, 7, 1)$ and passes through $(2, 4, -1)$.

(a) $(1,1,1)\cdot(x,y,z) - (1,1,1)\cdot(1,0,0) =$

$$x + y + z - 1 = 0$$

(b) $(1,2,3)\cdot(x,y,z) - (1,2,3)\cdot(1,1,1) =$

$$x + 2y + 3z - 6 = 0$$

(c) $l(t) = \vec{a} + t\vec{b}$. Perpendicular to $l(t)$ means $\perp \vec{b}$. $\therefore (5,0,2)\cdot(x,y,z) - (5,0,2)\cdot(5,-1,0)$

$$= 5x + 2z - 25 = 0$$

(d) $\ell(t) = \vec{a} + t\vec{b}$, $\vec{b} = (-1, -2, 3)$.

$\therefore (-1, -2, 3) \cdot (x, y, z) - (-1, -2, 3) \cdot (2, 4, -1) =$

$-x - 2y + 3z + 13 = 0$

17. **17.** Show that the points $(0, -2, -1)$, $(1, 4, 0)$, $(2, 10, 1)$ do not determine a unique plane.

The points must be colinear

$(0, -2, -1) - (1, 4, 0) = (-1, -6, -1)$

$(0, -2, -1) - (2, 10, 1) = (-2, -12, -2)$

$(-2, -12, -2) = 2(-1, -6, -1)$

$\therefore$ All 3 points are colinear

19. **19.** (a) Show that two parallel planes are either identical or they never intersect.

(b) How do two nonparallel planes intersect?

(a) 2 parallel planes have equations:

$P_1: Ax + By + Cz + D_1 = 0$ [1]

$P_2: Ax + By + Cz + D_2 = 0$ [2]

where (A, B, C) is the normal to each plane

If $D_1 = D_2$, then the planes are identical,

for $(x,y,z) \in P_1 \Leftrightarrow (A,B,C)\cdot(x,y,z) = -D_1 = -D_2$
$\Leftrightarrow (x,y,z) \in P_2$

If $D_1 \neq D_2$, then

if $(x,y,z) \in P_1$, then $(A,B,C)\cdot(x,y,z) = -D_1 \neq D_2$

$\therefore (x,y,z) \notin P_2$

and if $(x,y,z) \in P_2$, then $(A,B,C)\cdot(x,y,z) = -D_2 \neq -D_1$

$\therefore (x,y,z) \notin P_2$

$\therefore$ Planes never intersect.

(3) Two nonparallel planes intersect in a line.

If $\vec{n}_1$ is a normal for P_1, $\vec{p}_1$ any point of P_1
$\vec{n}_2$ is a normal for P_2, $\vec{p}_2$ any point of P_2

Let $\vec{x}$ be any point of the intersection.

Consider $\vec{x} + t(\vec{n}_1 \times \vec{n}_2)$, the equation of a line containing $\vec{x}$ and parallel to $\vec{n}_1 \times \vec{n}_2$, t any real number.

$\therefore$ If $\vec{x} \in P_1$ and $\vec{x} \in P_2$, then so is

$\vec{x} + t(\vec{n}_1 \times \vec{n}_2)$, for any t, for

$$\left[\vec{x} + t(\vec{n}_1 \times \vec{n}_2) - \vec{\rho}_1\right] \cdot \vec{n}_1 =$$

$$(\vec{x} - \vec{\rho}_1) \cdot \vec{n}_1 + t\left[(\vec{n}_1 \times \vec{n}_2)\right] \cdot \vec{n}_1 =$$

$$0 + 0 = 0$$

since $(\vec{x} - \vec{\rho}_1) \cdot \vec{n}_1 = 0$ as $\vec{x}, \vec{\rho}_1 \in P_1$

and $(\vec{n}_1 \times \vec{n}_2) \cdot \vec{n}_1 = 0$ since $\vec{n}_1 \times \vec{n}_2$ is $\perp$ to both $\vec{n}_1, \vec{n}_2$

$\therefore$ Whenever $\vec{x} \in P_1 \cap P_2$, then

$$\vec{x} + t(\vec{n}_1 \times \vec{n}_2) \in P_1$$

Similarly, $\vec{x} + t(\vec{n}_1 \times \vec{n}_2) \in P_2$

$\therefore \vec{x} + t(\vec{n}_1 \times \vec{n}_2) \in P_1 \cap P_2$

But this line must me all of $P_1 \cap P_2$,

for if $\vec{y}$ is any other intersection point,

then since $\vec{x}, \vec{y} \in P_1 \cap P_2$, then

$(\vec{y} - \vec{x}) \cdot \vec{n}_1 = 0$ and $(\vec{y} - \vec{x}) \cdot \vec{n}_2 = 0$

$\therefore \vec{y} - \vec{x}$ is parallel to $\vec{n}_1 \times \vec{n}_2 \Rightarrow$

$\vec{y} - \vec{x} = t(\vec{n}_1 \times \vec{n}_2)$ some real t.

$\therefore \vec{y} = \vec{x} + t(\vec{n}_1 \times \vec{n}_2)$ and so $\vec{y}$ is on the same line of $\vec{x} + t(\vec{n}_1 \times \vec{n}_2)$

20. **20.** Find the intersection of the planes $x + 2y + z = 0$ and $x - 3y - z = 0$.

For $x + 2y + z = 0$, $\vec{n}_1 = (1, 2, 1)$

$x - 3y - z = 0$, $\vec{n}_2 = (1, -3, -1)$

$$\therefore \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 1 & -3 & -1 \end{vmatrix} = (1, 2, -5)$$

A common point on each plane is $(0, 0, 0)$

$\therefore \ell(t) = (0, 0, 0) + t(1, 2, -5)$

$\therefore \ell(t) = t(1, 2, -5)$

23. **23.** (a) Prove the two triple-vector-product identities

$$(\mathbf{a} \times \mathbf{b}) \times \mathbf{c} = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{b} \cdot \mathbf{c})\mathbf{a}$$

and

$$\mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \cdot \mathbf{c})\mathbf{b} - (\mathbf{a} \cdot \mathbf{b})\mathbf{c}.$$

(a) Let $a = (a_1, a_2, a_3)$

$b = (b_1, b_2, b_3)$

$c = (c_1, c_2, c_3)$

$$a \times b = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2, \\ a_3 b_1 - a_1 b_3, \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

$$\therefore (a \times b) \times c = \begin{vmatrix} i & j & k \\ a_2 b_3 - a_3 b_2 & a_3 b_1 - a_1 b_3 & a_1 b_2 - a_2 b_1 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= \Big[(a_3 b_1 - a_1 b_3) c_3 - (a_1 b_2 - a_2 b_1) c_2,$$

$$(a_1 b_2 - a_2 b_1) c_1 - (a_2 b_3 - a_3 b_2) c_3,$$

$$(a_2 b_3 - a_3 b_2) c_2 - (a_3 b_1 - a_1 b_3) c_1 \Big]$$

$$= \Big[a_3 c_3 b_1 - a_1 c_3 b_3 - a_1 c_2 b_2 + a_2 c_2 b_1,$$

$$a_1 c_1 b_2 - a_2 c_1 b_1 - a_2 c_3 b_3 + a_3 c_3 b_2,$$

$$a_2 c_2 b_3 - a_3 c_2 b_2 - a_3 c_1 b_1 + a_1 c_1 b_3 \Big]$$

$$= \Big[(a_2 c_2 + a_3 c_3) b_1 - (c_2 b_2 + c_3 b_3) a_1,$$
$$(a_1 c_1 + a_3 c_3) b_2 - (c_1 b_1 + c_3 b_3) a_2,$$
$$(a_1 c_1 + a_2 c_2) b_3 - (c_1 b_1 + c_2 b_2) a_3 \Big]$$

$$= \Big[a_1c_1b_1 + (a_2c_2 + a_3c_3)b_1 - c_1b_1a_1 - (c_2b_2 + c_3b_3)a_1 ,$$

$$a_2c_2b_2 + (a_1c_1 + a_3c_3)b_2 - c_2b_2a_2 - (c_1b_1 + c_3b_3)a_2 ,$$

$$a_3c_3b_3 + (a_1c_1 + a_2c_2)b_3 - c_3b_3a_3 - (c_1b_1 + c_2b_2)a_3 \Big]$$

$$= \Big[(a\cdot c)b_1 - (c\cdot b)a_1 ,$$

$$(a\cdot c)b_2 - (c\cdot b)a_2 ,$$

$$(a\cdot c)b_3 - (c\cdot b)a_3 \Big]$$

$$= (a\cdot c)(b_1, b_2, b_3) - (c\cdot b)(a_1, a_2, a_3)$$

$$= (a\cdot c)b - (b\cdot c)a$$

$$\therefore (a \times b) \times c = (a\cdot c)b - (b\cdot c)a$$

Similarly, $a \times (b \times c) = -(b \times c) \times a$

$$= -\big[(a\cdot b)c - (a\cdot c)b\big]$$

$$= (a\cdot c)b - (a\cdot b)c ,$$

since $u \times v = -v \times u$

(b) Prove $(\mathbf{u} \times \mathbf{v}) \times \mathbf{w} = \mathbf{u} \times (\mathbf{v} \times \mathbf{w})$ if and only if $(\mathbf{u} \times \mathbf{w}) \times \mathbf{v} = \mathbf{0}$.

(6) Assume $u, v, w \neq 0$, since if any one is 0, then statement clearly true.

(1) By (a), $(u \times v) \times w = u \times (v \times w) \Rightarrow$

$$(u \cdot w) v - (v \cdot w) u = (u \cdot w) v - (u \cdot v) w \Rightarrow$$

$$(v \cdot w) u = (u \cdot v) w \qquad [a]$$

If $(v \cdot w) \neq 0$, then $u = \frac{u \cdot v}{v \cdot w} w$

$\therefore (u \times w) \times v = \left(\frac{u \cdot v}{v \cdot w} w \times w \right) \times v = 0,$

since $(\alpha w \times w) = \alpha (w \times w) = \alpha \cdot 0 = 0.$

If $v \cdot w = 0$, then $u \cdot v = 0$ from [a].

$\therefore v \perp w, v \perp u, \therefore v = \alpha (u \times w)$

$\therefore (u \times w) \times v = (u \times w) \times [\alpha (u \times w)]$
$= \alpha [(u \times w) \times (u \times w)] = 0$

$\therefore (u \times v) w = u \times (v \times w) \Rightarrow (u \times w) \times v = 0$

(2) Assume $(u \times w) \times v = 0$. $\therefore$ By (a),

$$(u \cdot v) w - (w \cdot v) u = 0 \Rightarrow$$

$$-(v \cdot w) u = -(u \cdot v) w \Rightarrow$$

$$(u\cdot w)v-(v\cdot w)u=(u\cdot w)v-(u\cdot v)w \Rightarrow$$

$$(u\times v)\times w = u\times(v\times w)$$

$$\therefore (u\times w)\times v = 0 \Rightarrow (u\times v)\times w = u\times(v\times w)$$

(c) Also prove that
$(\mathbf{u}\times\mathbf{v})\times\mathbf{w}+(\mathbf{v}\times\mathbf{w})\times\mathbf{u}+(\mathbf{w}\times\mathbf{u})\times\mathbf{v}=\mathbf{0}$
(called the ***Jacobi identity***).

(c) $(u\times v)\times w = (u\cdot w)v-(v\cdot w)u$ [1]

$(v\times w)\times u = (u\cdot v)w-(u\cdot w)v$ [2]

$(w\times u)\times v = (v\cdot w)u-(u\cdot v)w$ [3]

Adding the right sides of [1], [2], [3],

$[(u\cdot w)v-(v\cdot w)u]+[(u\cdot v)w-(u\cdot w)v]+[(v\cdot w)u-(u\cdot v)w]$

$=[(u\cdot w)v-(u\cdot w)v]+[(u\cdot v)w-(u\cdot v)w]+[(v\cdot w)u-(v\cdot w)u]$

$=0+0+0=0$

$\therefore (u\times v)\times w+(v\times w)\times u+(w\times u)\times v=0$

24.

24. (a) Prove, without recourse to geometry, that

$$\mathbf{u}\cdot(\mathbf{v}\times\mathbf{w}) = \mathbf{v}\cdot(\mathbf{w}\times\mathbf{u}) = \mathbf{w}\cdot(\mathbf{u}\times\mathbf{v}) = -\mathbf{u}\cdot(\mathbf{w}\times\mathbf{v}) = -\mathbf{w}\cdot(\mathbf{v}\times\mathbf{u}) = -\mathbf{v}\cdot(\mathbf{u}\times\mathbf{w}).$$

(b) Use part (a) and Exercise 23(a) to prove that

$$(\mathbf{u}\times\mathbf{v})\cdot(\mathbf{u}'\times\mathbf{v}') = (\mathbf{u}\cdot\mathbf{u}')(\mathbf{v}\cdot\mathbf{v}') - (\mathbf{u}\cdot\mathbf{v}')(\mathbf{u}'\cdot\mathbf{v}) = \begin{vmatrix} \mathbf{u}\cdot\mathbf{u}' & \mathbf{u}\cdot\mathbf{v}' \\ \mathbf{u}'\cdot\mathbf{v} & \mathbf{v}\cdot\mathbf{v}' \end{vmatrix}.$$

(a) $u\cdot(v\times w) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$, if $u=(u_1,u_2,u_3)$, $v=(v_1,v_2,v_3)$, $w=(w_1,w_2,w_3)$

$$= -\begin{vmatrix} v_1 & v_2 & v_3 \\ u_1 & u_2 & u_3 \\ w_1 & w_2 & w_3 \end{vmatrix} = \begin{vmatrix} v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \\ u_1 & u_2 & u_3 \end{vmatrix} = -\begin{vmatrix} w_1 & w_2 & w_3 \\ v_1 & v_2 & v_3 \\ u_1 & u_2 & u_3 \end{vmatrix}$$

using single row exchanges.

$\therefore u\cdot(v\times w) = -v\cdot(u\times w) = v\cdot(w\times u) = -w\cdot(v\times u)$

Using $a\times b = -b\times a$, we also get,

$-w\cdot(v\times u) = w\cdot(u\times v)$, $u\cdot(v\times w) = -u\cdot(w\times v)$

$\therefore u\cdot(v\times w) = v\cdot(w\times u) = w\cdot(u\times v) =$
$-u\cdot(w\times v) = -v\cdot(u\times w) = -w\cdot(v\times u)$

(b) From (a) above, $(u\times v)\cdot(u'\times v') =$

$u'\cdot[v'\times(u\times v)] =$

$$u' \cdot [(v' \cdot v)u - (v' \cdot u)v] \quad \text{, by 23(a)}$$

$$= (v' \cdot v)(u' \cdot u) - (v' \cdot u)(u' \cdot v)$$

$$= \begin{vmatrix} u \cdot u' & u \cdot v' \\ u' \cdot v & v \cdot v' \end{vmatrix}$$

28.

28. Find an equation for the plane that passes through the point $(2, -1, 3)$ and is perpendicular to the line $\mathbf{v} = (1, -2, 2) + t(3, -2, 4)$.

The vector $(3, -2, 4)$, parallel to the line, is perpendicular to the plane.

$\therefore 3x - 2y + 4z + D = 0$

$3(2) - 2(-1) + 4(3) = 20 = -D$

$\therefore 3x - 2y + 4z - 20 = 0$

30.

30. Find the equation of the line that passes through the point $(1, -2, -3)$ and is perpendicular to the plane $3x - y - 2z + 4 = 0$.

$(3, -1, -2)$ is normal to the plane, and so is parallel to line in question.

$\therefore$ line : $(1, -2, -3) + t(3, -1, -2)$

31.

31. Find an equation for the plane containing the two (parallel) lines

$$\mathbf{v}_1 = (0, 1, -2) + t(2, 3, -1)$$

and

$$\mathbf{v}_2 = (2, -1, 0) + t(2, 3, -1).$$

The lines are parallel as they have the same generating vector, $(2,3,-1)$, but go through different points.

The vector, $(0,1,-2) - (2,-1,0) = (-2,2,-2)$

is parallel to the plane, as is the line generating vector, $(2,3,-1)$.

$\therefore (-2,2,-2) \times (2,3,-1)$ is perpendicular to the plane, & is $\therefore$ a normal for the plane.

$$\therefore \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 2 & -2 \\ 2 & 3 & -1 \end{vmatrix} = (4, -6, -10) = 2(2, -3, -5)$$

$\therefore (2, -3, -5)$ serves as a normal to the plane.

$\therefore 2x - 3y - 5z + D = 0.$

Since $(0,1,-2)$, from the first line, is in the plane,

$2(0) - 3(1) - 5(-2) + D = 0$, $D = -7$

$\therefore$ $2x - 3y - 5z - 7 = 0$

34. **34.** Find the distance from the point $(2, 1, -1)$ to the plane $x - 2y + 2z + 5 = 0$.

A point in the plane is $(-5, 0, 0)$

$\therefore$ Find projection of $(-5, 0, 0) - (2, 1, -1) = (-7, -1, 1)$ onto the unit normal to the plane, $\frac{(1, -2, 2)}{\sqrt{1+2^2+2^2}} =$

$\frac{1}{3}(1, -2, 2) = \vec{n}$.

The projection is: $\|(-7, -1, 1)\| \cos\theta$, θ = angle between $(-7, -1, 1)$ and $\vec{n}$.

Since $\|(-7, -1, 1)\| \, \|\vec{n}\| \cos\theta = (-7, -1, 1) \cdot \vec{n}$

$\therefore \|(-7, -1, 1)\| \cos\theta = \frac{(-7, -1, 1) \cdot \vec{n}}{\|n\|} = \frac{(-7, -1, 1) \cdot \vec{n}}{1}$

$= (-7, -1, 1) \cdot \left(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}\right) = -\frac{7}{3} + \frac{2}{3} + \frac{2}{3} = -1$

$\therefore$ Distance equals length of projection of $(-7, -1, 1)$ onto $\vec{n}$ $= |\|(-7, -1, 1)\| \cos\theta| = |-1| = \underline{1}$

35. **35.** Find an equation for the plane that contains the line $\mathbf{v} = (-1, 1, 2) + t(3, 2, 4)$ and is perpendicular to the plane $2x + y - 3z + 4 = 0$.

Note the normal to plane $2x + y - 3z + 4 = 0$ is

parallel to the plane in question, as is the vector $(3,2,4)$ of the line.

$\therefore (2,1,-3) \times (3,2,4)$ is perpendicular to these two vectors, and $\therefore$ is a normal to the plane in question.

$$\therefore \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -3 \\ 3 & 2 & 4 \end{vmatrix} = (10,-17,1) = \text{normal.}$$

$$\therefore 10x - 17y + z + D = 0$$

The plane contains the point $(-1,1,2)$ of the line.

$$\therefore 10(-1) - 17(1) + 1(2) + D = 0, \quad D = 25$$

$$\therefore \underline{10x - 17y + z + 25 = 0}$$

38.

38. Given vectors **a** and **b**, do the equations $\mathbf{x} \times \mathbf{a} = \mathbf{b}$ and $\mathbf{x} \cdot \mathbf{a} = \|\mathbf{a}\|$ determine a unique vector **x**? Argue both geometrically and analytically.

Let $x = (x_1, x_2, x_3)$, $a = (a_1, a_2, a_3)$, $b = (b_1, b_2, b_3)$

Analytically:

$$\vec{x} \times \vec{a} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 & x_2 & x_3 \\ a_1 & a_2 & a_3 \end{vmatrix} = \begin{pmatrix} a_3 x_2 - a_2 x_3, \\ a_1 x_3 - a_3 x_1, \\ a_2 x_1 - a_1 x_2 \end{pmatrix}$$

$$\therefore \quad a_3 x_2 - a_2 x_3 = b_1 \qquad [1]$$

$$-a_3 x_1 + a_1 x_3 = b_2 \qquad [2]$$

$$a_2 x_1 - a_1 x_2 = b_3 \qquad [3]$$

Multiplying [1] by a_1, [2] by a_2 and adding,

$$-a_2 a_3 x_1 + a_1 a_3 x_2 = a_1 b_1 + a_2 b_2 \qquad [4]$$

$$a_2 x_1 - a_1 x_2 = b_3 \qquad [3]$$

If multiply [3] by a_3 and add to [4],

$$0 = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$\therefore \vec{x} \times \vec{a} = \vec{b}$ is indeterminate for $\vec{x}$, and [1], [2], and [3] alone cannot solve for x_1, x_2, x_3

$$\det \begin{bmatrix} 0 & a_3 & -a_2 \\ -a_3 & 0 & a_1 \\ a_2 & -a_1 & 0 \end{bmatrix} = 0, \text{ so can't solve}$$

$$\begin{bmatrix} 0 & a_3 & -a_2 \\ -a_3 & 0 & a_1 \\ a_2 & -a_1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Using $\vec{x} \cdot \vec{a} = \|\vec{a}\|$, $a_1 x_1 + a_2 x_2 + a_3 x_3 = \sqrt{a_1^2 + a_2^2 + a_3^2}$,

and for simplicity, let $K = \|\vec{a}\|$, a constant.

Look at $\begin{bmatrix} 0 & a_3 & -a_2 \\ -a_3 & 0 & a_1 \\ a_1 & a_2 & a_3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ K \end{bmatrix}$

$$\det \begin{bmatrix} 0 & a_3 & -a_2 \\ -a_3 & 0 & a_1 \\ a_1 & a_2 & a_3 \end{bmatrix} = -a_3(-a_3^2 - a_1^2) - a_2(-a_2 a_3)$$

$$= a_3^3 + a_1^2 a_3 + a_2^2 a_3$$

$$= a_3(a_1^2 + a_2^2 + a_3^2)$$

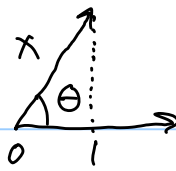
Assuming $a_3 \neq 0$ and $\|a\| \neq 0$, then the solution is determined, and is unique (using Cramer's rule).

Geometrically,

Assume $\vec{a}, \vec{b} \neq 0$.

Since $x \cdot a = \|a\| = \|x\| \|a\| \cos\theta$, then

$\|x\| \cos\theta = 1$, so the projection of $\vec{x}$ onto $\vec{a}$ has length 1



$\therefore$ Assume $\vec{a}$ is along the x-axis.

If P is a plane perpendicular to $\vec{a}$, intersecting the x-axis at $(1,0,0)$, then $\vec{x}$ could be any vector from the origin to any point on P.

Using $\vec{x} \times \vec{a} = \vec{b}$, mean $\|x\|\|a\| \sin\theta = \|b\|$,

so $\|x\| \sin\theta = \frac{\|b\|}{\|a\|}$. Assuming $\|a\| \neq 0$,

then $\|x\| \sin\theta$ is fixed, meaning

$\vec{x}$ describes a fixed circle on plane P, of radius $\|x\| \sin\theta$.

Since $\vec{x} \times \vec{a} = \vec{b}$, there is only one possibility of $\vec{x} \times \vec{a}$ such that $\vec{x} \times \vec{a}$ is in the direction of $\vec{b}$.

$\therefore$ $\vec{x}$ is uniquely determined.

39.

39. Determine the distance from the plane $12x + 13y + 5z + 2 = 0$ to the point $(1, 1, -5)$.

$(12, 13, 5)$ is a normal to the plane, and

$\vec{n} = \frac{(12,13,5)}{\sqrt{12^2+13^2+5^2}} = \frac{1}{13\sqrt{2}}(12,13,5)$ is a unit normal.

The point $(0,0,-\frac{2}{5})$ is in the plane.

$\therefore$ Project $(1,1,-5) - (0,0,-\frac{2}{5}) = (1,1,-\frac{23}{5}) = \vec{p}$ onto $\vec{n}$

$\therefore \vec{n}\cdot\vec{p} = \|\vec{n}\|\,\|\vec{p}\|\cos\theta$, $\therefore \|\vec{p}\|\cos\theta = \frac{\vec{n}\cdot\vec{p}}{\|\vec{n}\|} = \vec{n}\cdot\vec{p}$

$= (1,1,-\frac{23}{5})\cdot(\frac{12}{13\sqrt{2}}, \frac{13}{13\sqrt{2}}, \frac{5}{13\sqrt{2}})$

$= \frac{12}{13\sqrt{2}} + \frac{13}{13\sqrt{2}} - \frac{23}{5}\cdot\frac{5}{13\sqrt{2}} = \frac{12+13-23}{13\sqrt{2}}$

$= \frac{2}{13\sqrt{2}} = \frac{\sqrt{2}}{13}$

40.

40. Find the distance to the point (6, 1, 0) from the plane through the origin that is perpendicular to $\mathbf{i} - 2\mathbf{j} + \mathbf{k}$.

$(1,-2,1)$ is a normal to the plane.

$\therefore x - 2y + z + D = 0$. Since $(0,0,0)$ is in the plane, Then $D = 0$.

$\therefore x - 2y + z = 0$

$\therefore$ Project $(6,1,0) - (0,0,0) = (6,1,0)$ onto $(1,-2,1)$

$$\therefore \vec{n}\cdot\vec{\rho} = |\vec{n}||\vec{\rho}|\cos\theta, \quad |\vec{\rho}|\cos\theta = \frac{\vec{n}\cdot\vec{\rho}}{|\vec{n}|}$$

$$= \frac{(1,-2,1)\cdot(6,1,0)}{\sqrt{1^2+(-2)^2+1^2}} = \frac{6-2+0}{\sqrt{6}} = \frac{4}{\sqrt{6}}$$

42. **42.** Show that the plane that passes through the three points $A = (a_1, a_2, a_3)$, $B = (b_1, b_2, b_3)$, and $C = (c_1, c_2, c_3)$ consists of the points $P = (x, y, z)$ given by

$$\begin{vmatrix} a_1 - x & a_2 - y & a_3 - z \\ b_1 - x & b_2 - y & b_3 - z \\ c_1 - x & c_2 - y & c_3 - z \end{vmatrix} = 0.$$

(HINT: Write the determinant as a triple product.)

All 4 points, A, B, C, P, are in the plane.

∴ The parallelpiped of sides A-P, B-P, C-P must have a zero volume.

$$\therefore (\vec{A}-\vec{P})\cdot[(\vec{B}-\vec{P})\times(\vec{C}-\vec{P})] = 0$$

$$\therefore \begin{vmatrix} a_1-x & a_2-y & a_3-z \\ b_1-x & b_2-y & b_3-z \\ c_1-x & c_2-y & c_3-z \end{vmatrix} = 0$$

46. **46.** Suppose $\mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ are orthogonal unit vectors. Let $\mathbf{u} = \mathbf{v} \times \mathbf{w}$. Show that $\mathbf{w} = \mathbf{u} \times \mathbf{v}$ and $\mathbf{v} = \mathbf{w} \times \mathbf{u}$.

Since $V \perp W$, $V \cdot W = 0$ Now use #23 above

(1) $\therefore u \times v = (v \times w) \times v = (v \cdot v) w - (w \cdot v) v$

$= \|v\|^2 w - (0) v$

$= 1 \cdot w - 0 = w$

$\therefore w = u \times v$

(2) $w \times u = w \times (v \times w) = (w \cdot w) v - (w \cdot v) w$

$= 1 \cdot v - 0$

$= v$

$\therefore v = w \times u$

1.4 Cylindrical and Spherical Coordinates

Note Title 12/1/2015

3.

3. (a) The following points are given in cylindrical coordinates; express each in rectangular coordinates and spherical coordinates: $(1, 45°, 1)$, $(2, \pi/2, -4)$, $(0, 45°, 10)$, $(3, \pi/6, 4)$, $(1, \pi/6, 0)$, and $(2, 3\pi/4, -2)$. (Only the first point is solved in the Study Guide.)

(a) $(1, 45°, 1)$: $r\cos\theta = x = \sqrt{2}/2$

$r\sin\theta = y = \sqrt{2}/2$

$z = 1$

$\therefore$ rectangular: $(\sqrt{2}/2, \sqrt{2}/2, 1)$

spherical: $\rho = \sqrt{r^2+z^2} = \sqrt{2}$

$\theta = 45°$

$\phi = \arccos\left(\frac{z}{\rho}\right) = \arccos\left(\frac{1}{\sqrt{2}}\right) = 45°$

$\therefore$ spherical: $(\sqrt{2}, 45°, 45°)$

$\left(2, \frac{\pi}{2}, -4\right)$: $2\cos\frac{\pi}{2} = x = 0$

$2\sin\pi/2 = y = 2$

$z = -4$

$\therefore$ rectangular: $(0, 2, -4)$

spherical: $\rho = \sqrt{r^2+z^2} = \sqrt{4+16} = 2\sqrt{5}$

$\theta = \frac{\pi}{2} = 90°$

$\phi = \arccos\left(\frac{-4}{2\sqrt{5}}\right) = 153.4° = 2.68 \text{ rad}$

$\therefore (2\sqrt{5}, 90°, 153.4°)$

$(0, 45^\circ, 10)$:

Rectangular: $0 \cdot \cos 45^\circ = x = 0$

$0 \cdot \sin 45^\circ = y = 0$

$z = 10$

$\therefore (0, 0, 10)$

Spherical: $\rho = \sqrt{0^2 + 10^2} = 10$

$\theta = 45^\circ$

$\phi = \arccos\left(\frac{10}{10}\right) = 0^\circ$

$\therefore (10, 45^\circ, 0^\circ)$

$\left(3, \frac{\pi}{6}, 4\right)$: Rectangular: $x = 3\cos\frac{\pi}{6} = \frac{3\sqrt{3}}{2}$

$y = 3\sin\frac{\pi}{6} = \frac{3}{2}$

$z = 4$

$\therefore \left(\frac{3\sqrt{3}}{2}, \frac{3}{2}, 4\right)$

Spherical: $\rho = \sqrt{r^2 + z^2} = \sqrt{9 + 16} = 5$

$\theta = \frac{\pi}{6} = 30^\circ$

$\phi = \arccos\left(\frac{4}{5}\right) = 36.9^\circ$

$\therefore (5, 30^\circ, 36.9^\circ)$

$(1, \frac{\pi}{6}, 0)$: Rectangular: $x = 1 \cdot \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

$y = 1 \cdot \sin\frac{\pi}{6} = \frac{1}{2}$

$z = 0$

$\therefore \left(\sqrt{3}/2, \frac{1}{2}, 0\right)$

Spherical: $\rho = \sqrt{r^2 + z^2} = 1$

$\theta = \frac{\pi}{6} = 30^\circ$

$\phi = \arccos\left(\frac{0}{1}\right) = 90^\circ$

$\therefore (1, 30^\circ, 90^\circ)$

$(2, \frac{3}{4}\pi, -2)$: Rectangular: $x = 2\cos\frac{3}{4}\pi = -\sqrt{2}$

$y = 2\sin\frac{3}{4}\pi = \sqrt{2}$

$z = -2$

$\therefore (-\sqrt{2}, \sqrt{2}, -2)$

Spherical: $\rho = \sqrt{r^2 + z^2} = 2\sqrt{2}$

$\theta = \frac{3}{4}\pi = 135^\circ$

$\phi = \arccos\left(\frac{-2}{2\sqrt{2}}\right) = 135^\circ$

$\therefore (2\sqrt{2}, 135^\circ, 135^\circ)$

Summary:

Cylindrical	Rectangular	Spherical
$(1, 45^\circ, 1)$	$(\sqrt{2}/2, \sqrt{2}/2, 1)$	$(\sqrt{2}, 45^\circ, 45^\circ)$
$(2, \frac{\pi}{2}, -4)$	$(0, 2, -4)$	$(2\sqrt{5}, 90^\circ, 153.4^\circ)$
$(0, 45^\circ, 10)$	$(0, 0, 10)$	$(10, 45^\circ, 0^\circ)$
$(3, \frac{\pi}{6}, 4)$	$(\frac{3\sqrt{3}}{2}, \frac{3}{2}, 4)$	$(5, 30^\circ, 36.9^\circ)$
$(1, \frac{\pi}{6}, 0)$	$(\sqrt{3}/2, \frac{1}{2}, 0)$	$(1, 30^\circ, 90^\circ)$
$(2, \frac{3}{4}\pi, -2)$	$(-\sqrt{2}, \sqrt{2}, -2)$	$(2\sqrt{2}, 135^\circ, 135^\circ)$

(6) (b) Change each of the following points from rectangular coordinates to spherical coordinates and to cylindrical coordinates: $(2, 1, -2)$, $(0, 3, 4)$, $(\sqrt{2}, 1, 1)$, $(-2\sqrt{3}, -2, 3)$. (Only the first point is solved in the Study Guide.)

$(2, 1, -2)$: Spherical: $\rho = \sqrt{4+1+4} = 3$

$\theta = \arctan(\frac{1}{2}) = 26.6^\circ$

$\phi = \arccos(-\frac{2}{3}) = 131.8^\circ$

$\therefore (3, 26.6^\circ, 131.8^\circ)$

Cylindrical: $r = \sqrt{4+1} = \sqrt{5}$

$\theta = \arctan\left(\frac{1}{2}\right) = 26.6^\circ$

$z = -2$

$\therefore (\sqrt{5}, 26.6^\circ, -2)$

$(0,3,4)$: Spherical: $\rho = \sqrt{0+9+16} = 5$

$\theta = \frac{\pi}{2} = 90^\circ$

$\phi = \arccos\left(\frac{4}{5}\right) = 36.9^\circ$

$\therefore (5, 90^\circ, 36.9^\circ)$

Cylindrical: $r = \sqrt{0+9} = 3$

$\theta = 90^\circ$

$z = 4$

$\therefore (3, 90^\circ, 4)$

$(\sqrt{2}, 1, 1)$: Spherical: $\rho = \sqrt{2+1+1} = 2$

$\theta = \arctan\left(\frac{1}{\sqrt{2}}\right) = 35.3^\circ$

$\phi = \arccos\left(\frac{1}{2}\right) = 60^\circ$

$\therefore (2, 35.3^\circ, 60^\circ)$

Cylindrical: $r = \sqrt{2+1} = \sqrt{3}$

$\theta = 35.3°$

$z = 1$

$\therefore (\sqrt{3}, 35.3°, 1)$

$(-2\sqrt{3}, -2, 3)$: Spherical: $\rho = \sqrt{12+4+5} = 5$

$\theta = \arctan\left(\frac{-2}{-2\sqrt{3}}\right) = 210°$

$\phi = \arccos\left(\frac{3}{5}\right) = 53.1°$

$\therefore (5, 210°, 53.1°)$

Cylindrical: $r = \sqrt{12+4} = 4$

$\theta = 210°$

$z = 3$

$\therefore (4, 210°, 3)$

Summary:

Rectangular	Spherical	Cylindrical
$(2, 1, -2)$	$(3, 26.6°, 131.8°)$	$(\sqrt{5}, 26.6°, -2)$
$(0, 3, 4)$	$(5, 90°, 36.9°)$	$(3, 90°, 4)$
$(\sqrt{2}, 1, 1)$	$(2, 35.3°, 60°)$	$(\sqrt{3}, 35.3°, 1)$
$(-2\sqrt{3}, -2, 3)$	$(5, 210°, 53.1°)$	$(4, 210°, 3)$

4.

4. Describe the geometric meaning of the following mappings in cylindrical coordinates:

(a) $(r, \theta, z) \mapsto (r, \theta, -z)$

(b) $(r, \theta, z) \mapsto (r, \theta + \pi, -z)$

(c) $(r, \theta, z) \mapsto (-r, \theta - \pi/4, z)$

(a) Reflection with respect to xy-plane

(b) Rotation counterclockwise by 180° about the z-axis, with reflection about xy-plane

(c) Reflection about the z-axis, with clockwise rotation by 45° about z-axis.

5.

5. Describe the geometric meaning of the following mappings in spherical coordinates:

(a) $(\rho, \theta, \phi) \mapsto (\rho, \theta + \pi, \phi)$

(b) $(\rho, \theta, \phi) \mapsto (\rho, \theta, \pi - \phi)$

(c) $(\rho, \theta, \phi) \mapsto (2\rho, \theta + \pi/2, \phi)$

(a) Counterclockwise rotation by 180° about z-axis.

(b) Reflection about the xy-plane

(c) Counterclockwise rotation about z-axis by 90°, with radial expansion (zooming) by factor of 2.

12.

12. Using cylindrical coordinates and the orthonormal (orthogonal normalized) vectors $\mathbf{e}_r$, $\mathbf{e}_\theta$, and $\mathbf{e}_z$ (see Figure 1.4.8),

(a) express each of $\mathbf{e}_r$, $\mathbf{e}_\theta$, and $\mathbf{e}_z$ in terms of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and (x, y, z); and

(b) calculate $\mathbf{e}_\theta \times \mathbf{j}$ both analytically, using part (a), and geometrically.

(a) $\hat{e}_r = \cos\theta\,\hat{i} + \sin\theta\,\hat{j}$

$$= \frac{x}{\sqrt{x^2+y^2}}\,\hat{i} + \frac{y}{\sqrt{x^2+y^2}}\,\hat{j}$$

$\hat{e}_\theta = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$

$$= -\frac{y}{\sqrt{x^2+y^2}}\,\hat{i} + \frac{x}{\sqrt{x^2+y^2}}\,\hat{j}$$

$\hat{e}_z = \hat{k}$

(b) $\hat{e}_\theta \times \hat{j} = (-\sin\theta\,\hat{i} + \cos\theta\,\hat{j}) \times \hat{j}$

$= -\sin\theta\,(\hat{i}\times\hat{j}) + \cos\theta\,(\hat{j}\times\hat{j})$

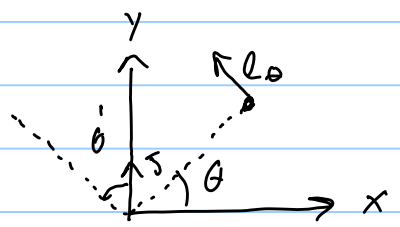
$= -\sin\theta\,\hat{k} = -\frac{y}{\sqrt{x^2+y^2}}\,\hat{k}$

Geometrically: $\|\hat{e}_\theta \times \hat{j}\| = \|\hat{e}_\theta\|\,\|\hat{j}\|\,|\sin\theta'|$

$= |\sin\theta'|,$

where θ' = angle between $\hat{e}_\theta$ and $\hat{j}$.

But $\theta' = \theta$



and $\hat{e}_\theta \times \hat{j}$ points toward $-\hat{k}$.

$\therefore \hat{e}_\theta \times \hat{j}$ points toward the negative z-axis and has magnitude $|\sin\theta|$.

13. **13.** Using spherical coordinates and the orthonormal (orthogonal normalized) vectors $\mathbf{e}_\rho$, $\mathbf{e}_\theta$, and $\mathbf{e}_\phi$ [see Figure 1.4.8(b)],

(a) express each of $\mathbf{e}_\rho$, $\mathbf{e}_\theta$, and $\mathbf{e}_\phi$ in terms of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ and (x, y, z); and

(b) calculate $\mathbf{e}_\theta \times \mathbf{j}$ and $\mathbf{e}_\phi \times \mathbf{j}$ both analytically and geometrically.

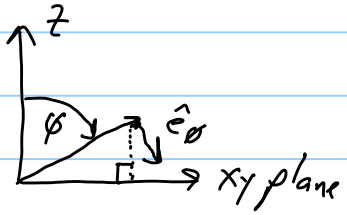
(a) $\hat{e}_\rho = \frac{x}{\sqrt{x^2+y^2+z^2}}\hat{i} + \frac{y}{\sqrt{x^2+y^2+z^2}}\hat{j} + \frac{z}{\sqrt{x^2+y^2+z^2}}\hat{k}$

$\hat{e}_\theta = -\sin\theta\,\hat{i} + \cos\theta\,\hat{j}$

$= \frac{-y}{\sqrt{x^2+y^2}}\hat{i} + \frac{x}{\sqrt{x^2+y^2}}\hat{j}$

$\hat{e}_\phi$: k-component is

$-|\hat{e}_\phi|\cos\left(\frac{\pi}{2}-\phi\right) = -\sin\phi$



xy-plane component is $|\hat{e}_\phi|\sin\left(\frac{\pi}{2}-\phi\right) = \cos\phi$

$\therefore \hat{e}_\phi = \cos\phi\cos\theta\,\hat{i} + \cos\phi\sin\theta\,\hat{j} - \sin\phi\,\hat{k}$

$\therefore \hat{e}_\phi = \frac{z}{\sqrt{x^2+y^2+z^2}} \cdot \frac{x}{\sqrt{x^2+y^2}}\hat{i} +$

$$\frac{z}{\sqrt{x^2+y^2+z^2}} \cdot \frac{y}{\sqrt{x^2+y^2}}\,\hat{j} -$$

$$\frac{\sqrt{x^2+y^2}}{\sqrt{x^2+y^2+z^2}}\,\hat{k}$$

(b) $\hat{e}_\theta \times \hat{j} = (-\sin\theta\,\hat{i} + \cos\theta\,\hat{j}) \times \hat{j}$

$= -\sin\theta\,(\hat{i} \times \hat{j}) + \cos\theta\,(\hat{j} \times \hat{j})$

$= -\sin\theta\,\hat{k} = -\frac{y}{\sqrt{x^2+y^2}}\,\hat{k}$

$\hat{e}_\phi \times \hat{j} = \left(\frac{zx}{\rho r}\,\hat{i} + \frac{zy}{\rho r}\,\hat{j} - \frac{r}{\rho}\,\hat{k}\right) \times \hat{j}$

$= \frac{zx}{\rho r}\,\hat{k} + \frac{r}{\rho}\,\hat{i}$

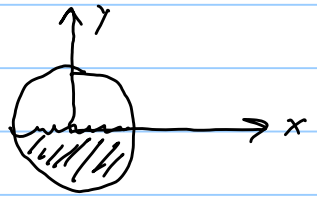
Geometrically, $\hat{e}_\theta \times \hat{j}$, as in #12 above, points towards $-\hat{k}$ and has magnitude $|\sin\theta|$

$\hat{e}_\phi \times \hat{j}$ is in the xz plane

18. **18.** A tank in the shape of a right-circular cylinder of radius 10 ft and height 16 ft is half filled and lying on its side. Describe the air space inside the tank by suitably chosen cylindrical coordinates.

Choose z-axis along height, going through center of cylinder.

$\therefore \quad 0 \le z \le 16$ ft



Look at cylinder from base

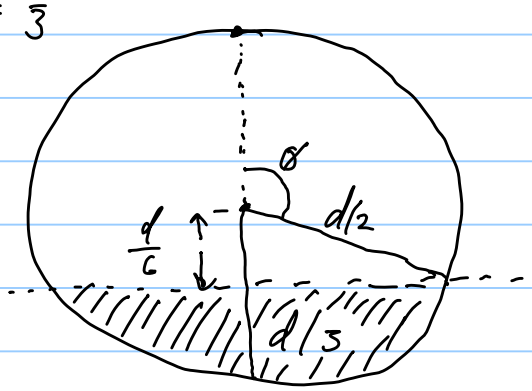
$\therefore \quad 0 \le r \le 10$ ft, $-\pi \le \theta \le 0$

19. **19.** A vibrometer is to be designed that withstands the heating effects of its spherical enclosure of diameter d, which is buried to a depth $d/3$ in the earth, the upper portion being heated by the sun (assume the surface is flat). Heat conduction analysis requires a description of the buried portion of the enclosure in spherical coordinates. Find it.

Look at cross-section of sphere:

Let surface of earth be parallel to xy-plane. Describe "volume" under surface.

$\frac{d}{2} - \frac{d}{3} = \frac{d}{6} \qquad \cos(\pi - \phi) = \frac{d/6}{d/2} = \frac{1}{3}$



Choose coordinate system w/ center of sphere as origin.

$\therefore \quad \pi - \arccos\left(\frac{1}{3}\right) \le \phi \le \pi$

$0 \le \theta \le 2\pi$ for θ.

For ρ, since $\phi > \frac{\pi}{2}$, then $\cos\phi \le 0$.

$\therefore$ Minimum of $-\rho\cos\phi$ is $\frac{d}{6}$

and the maximum of ρ is $\frac{d}{2}$

$\therefore -\frac{d}{6\cos\phi} \le \rho \le \frac{d}{2}$

Altogether, $-\frac{d}{6\cos\phi} \le \rho \le \frac{d}{2}$, $0 \le \theta \le 2\pi$,

$\pi - \arccos\left(\frac{1}{3}\right) \le \phi \le \pi$

20.

20. An oil filter cartridge is a porous right-circular cylinder inside which oil diffuses from the axis to the outer curved surface. Describe the cartridge in cylindrical coordinates, if the diameter of the filter is 4.5 inches, the height is 5.6 inches, and the center of the cartridge is drilled (all the way through) from the top to admit a $\frac{5}{8}$-inch-diameter bolt.

Let z-axis be along the drilled hole.

$\therefore \frac{5}{8}'' \le r \le 4.5''$, $0 \le \theta \le 2\pi$, $0 \le z \le 5.6''$

21.

21. Describe the surface given in spherical coordinates by $\rho = \cos 2\theta$.

$0 \le \cos 2\theta \le 1$, and $\rho = \sqrt{x^2+y^2+z^2}$.

Consider $z = 0$. $r = \cos 2\theta$ is a 4-leaf rose

contained in a unit circle ($x^2+y^2=1$)

ρ doesn't depend on ϕ. $\therefore$ for any θ,

$\rho = \cos 2\theta$ describes a semicircle from $\phi = 0$ to $\phi = \pi$, a semicircle of radius $\cos 2\theta$.

These semicircles "fan" out from tiny (radius near 0 when $\cos 2\theta$ is near 0) to a radius near 1 when $\cos 2\theta$ is near 1.

Essentially 3-D dumbbells that shrink to zero in size near origin.

22.

22. (a) Find all points $\mathbf{p} \in \mathbb{R}^3$ that have the same representation in both Cartesian and spherical coordinates.

(b) Find all points $\mathbf{p} \in \mathbb{R}^3$ that have the same representation in both Cartesian and cylindrical coordinates.

(a) Since $(x,y,z) = (\rho, \theta, \phi)$, then

$x = \rho = \sqrt{x^2+y^2+z^2} \rightarrow y^2+z^2 = 0 \Rightarrow y=0, z=0$

$\therefore (x,0,0) = (\rho, \theta, \phi) \rightarrow \theta = 0, \phi = 0$

Since $x = \rho \sin\phi\cos\theta$, then $x = \rho \sin(0)\cos(0)$

$\Rightarrow x = 0$.

$\therefore \underline{(0,0,0)}$ is the only point

(6) $(x, y, z) = (r, \theta, z)$

$\therefore x = \sqrt{x^2 + y^2} \Rightarrow y = 0 \Rightarrow \theta = 0$

$\therefore$ All points of form $(x, 0, z)$, or all points in xz-plane

1.5. n-Dimensional Euclidean Space

Note Title 12/2/2015

2.

2. In $\mathbb{R}^n$ show that

(a) $2\|\mathbf{x}\|^2 + 2\|\mathbf{y}\|^2 = \|\mathbf{x}+\mathbf{y}\|^2 + \|\mathbf{x}-\mathbf{y}\|^2$ (This is known as the ***parallelogram law***.)

(b) $\|\mathbf{x}-\mathbf{y}\|\|\mathbf{x}+\mathbf{y}\| \le \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$

(c) $4\langle \mathbf{x}, \mathbf{y}\rangle = \|\mathbf{x}+\mathbf{y}\|^2 - \|\mathbf{x}-\mathbf{y}\|^2$ (This is called the ***polarization identity***.)

(a) $\|x+y\|^2 + \|x-y\|^2 = (x+y)\cdot(x+y) + (x-y)\cdot(x-y)$

$= x\cdot x + 2x\cdot y + y\cdot y + x\cdot x - 2x\cdot y + y\cdot y$

$= 2\|x\|^2 + 2\|y\|^2$

(b) $\|x-y\|^2\|x+y\|^2 =$

$[\|x\|^2 - 2x\cdot y + \|y\|^2][\|x\|^2 + 2x\cdot y + \|y\|^2]$

$= \|x\|^4 - 2x\cdot y\|x\|^2 + \|x\|^2\|y\|^2 + 2x\cdot y\|x\|^2 - 4(x\cdot y)^2 + 2x\cdot y\|y\|^2 + \|x\|^2\|y\|^2 - 2x\cdot y\|y\|^2 + \|y\|^4$

$= \|x\|^4 + 2\|x\|^2\|y\|^2 + \|y\|^4 - 4(x\cdot y)^2$

$= [\|x\|^2 + \|y\|^2]^2 - 4(x\cdot y)^2$

$\le [\|x\|^2 + \|y\|^2]^2$, since $-4(x\cdot y)^2 \le 0$

$\therefore \|x-y\|^2\|x+y\|^2 \le [\|x\|^2 + \|y\|^2]^2$

$\therefore \|x-y\|\|x+y\| \leq \|x\|^2 + \|y\|^2$, as both sides are positive.

(c) $\|x+y\|^2 - \|x-y\|^2 =$

$(x+y)\cdot(x+y) - (x-y)\cdot(x-y)$

$= \|x\|^2 + 2x\cdot y + \|y\|^2 - [\|x\|^2 - 2x\cdot y + \|y\|^2]$

$= 4x\cdot y = 4\langle x, y\rangle$

$\therefore 4\langle x, y\rangle = \|x+y\|^2 - \|x-y\|^2$

Interpret these results geometrically in terms of the parallelogram formed by **x** and **y**.

(a) The sum of the squares of the diagonals of a parallelogram = The sum of the squares of all sides of the parallelogram

(b) The product of the diagonals of a parallelogram is $\leq$ The sum of the squares of two adjacent sides.

(c) The difference between the squares of the diagonals of a parallelogram equals four times the inner product of the sides.

Verify the Cauchy–Schwarz inequality and the triangle inequality for the vectors in Exercises 3 to 6.

3. **3.** $\mathbf{x} = (2, 0, -1), \mathbf{y} = (4, 0, -2)$

Note: $y = 2x$

(a) $|x \cdot y| = |2 \cdot 4 + 0 \cdot 0 + (-1)(-2)| = 10$

$\|x\| = \sqrt{2^2 + 0^2 + (-1)^2} = \sqrt{5}$ $\quad \|y\| = \sqrt{4^2 + 0^2 + (-2)^2} = \sqrt{20}$

$\|x\| \, \|y\| = \sqrt{5} \cdot \sqrt{20} = 10$

$\therefore |x \cdot y| \leq \|x\| \, \|y\|$

(b) $\|x + y\| = \|(6, 0, -3)\| = \sqrt{6^2 + 0^2 + (-3)^2} = \sqrt{45} = 3\sqrt{5}$

$\|x\| + \|y\| = \sqrt{5} + \sqrt{20} = 3\sqrt{5}$

$\therefore \|x + y\| \leq \|x\| + \|y\|$

7. **7.** Let $\mathbf{v}, \mathbf{w} \in \mathbb{R}^n$. If $\|\mathbf{v}\| = \|\mathbf{w}\|$, show that $\mathbf{v} + \mathbf{w}$ and $\mathbf{v} - \mathbf{w}$ are orthogonal.

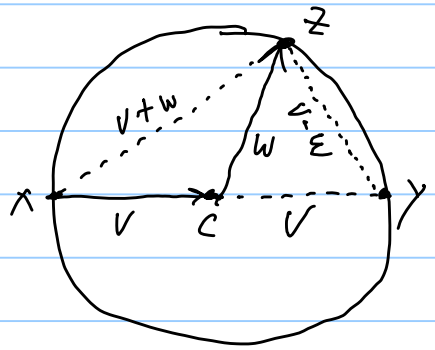
$(v + w) \cdot (v - w) = v \cdot v - v \cdot w + w \cdot v - w \cdot w$

$= -v \cdot w + w \cdot v$, since $v \cdot v = w \cdot w$

$= 0$, since $v \cdot w = w \cdot v$

$\therefore$ Dot product $= 0 \Rightarrow (v + w) \perp (v - w)$

8. **8.** Suppose T is a triangle formed by placing three points on a circle, two of which lie on the circle's diameter. Use the previous problem to show T is a right triangle.



Let the points be x, y, z, where x, y line on the circles diameter. Let C be the center of the circle.

Let $\vec{v} = \vec{xc}$, $\vec{w} = \vec{cz}$. Then $\vec{cy}$ also $= \vec{v}$

and $\|\vec{v}\| = \|\vec{w}\|$ since these are all radii.

$\therefore \vec{xz} = \vec{v} + \vec{w}$, and $\vec{zy} = \vec{v} - \vec{w}$

By #7, $\vec{xz} \perp \vec{zy}$. $\therefore \triangle xyz$ is a right triangle.

12. **12.** For matrix A in the previous problem, find a nonzero $\mathbf{x} \in \mathbb{R}^3$ such that $A\mathbf{x} = \mathbf{0}$.

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 1 \\ 0 & 3 & 3 \end{bmatrix} \quad \text{Let } x = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

$$\therefore Ax = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

13. **13.** Use induction on k to prove that if $\mathbf{x}_1, \ldots, \mathbf{x}_k \in \mathbb{R}^n$, then

$$\|\mathbf{x}_1 + \cdots + \mathbf{x}_k\| \leq \|\mathbf{x}_1\| + \cdots + \|\mathbf{x}_k\|.$$

(1) $k = 1$: Clearly, $\|x_1\| \leq \|x_1\|$

(2) $K=2$: $\|x_1 + x_2\| \leq \|x_1\| + \|x_2\|$

This is true by the triangle inequality.

(3) True for $K>2 \Rightarrow$ true for $K+1$

Suppose $\|x_1 + \ldots + x_K\| \leq \|x_1\| + \ldots + \|x_K\|$

$\therefore \|x_1 + \ldots + x_{K-1} + x_K + x_{K+1}\| =$

$\|x_1 + \ldots + x_{K-1} + (x_K + x_{K+1})\|$

$\leq \|x_1\| + \ldots + \|x_{K-1}\| + \|x_K + x_{K+1}\|$, by assumption

$\leq \|x_1\| + \ldots + \|x_{K-1}\| + \|x_K\| + \|x_{K+1}\|$, since

$\|x_K + x_{K+1}\| \leq \|x_K\| + \|x_{K+1}\|$, by (2)

$\therefore$ When true for $K>2$, true for $K+1$.

$\therefore$ By (1), (2), & (3), true for all K

14. **14.** Using algebra, prove the ***identity of Lagrange***. For real numbers $x_1, \ldots, x_n$ and $y_1, \ldots, y_n$.

$$\left(\sum_{i=1}^{n} x_i y_i\right)^2 = \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) - \sum_{i<j} (x_i y_j - x_j y_i)^2.$$

First, clarify term $\sum_{i<j} (x_i y_j - x_j y_i)^2$

If $n=2$, $\sum_{i<j} (x_i y_j - x_j y_i)^2 = (x_1 y_2 - x_2 y_1)^2$

If $n=3$, it's $\underset{i=1,j=2}{(x_1 y_2 - x_2 y_1)^2} + \underset{i=1,j=3}{(x_1 y_3 - x_3 y_1)^2} + \underset{i=2,j=3}{(x_2 y_3 - x_3 y_2)^2}$

If instead of using $\sum_{i<j}$ we used $\sum_{i,j=1}$,

Then for $n=2$, $\sum_{i,j=1} (x_i y_j - x_j y_i)^2 =$

$$\sum_{i=1}^{2} \left[\sum_{j=1}^{2} (x_i y_j - x_j y_i)^2 \right] =$$

$$\underset{i=1,j=1}{(x_1 y_1 - x_1 y_1)^2} + \underset{i=1,j=2}{(x_1 y_2 - x_2 y_1)^2} + \underset{i=2,j=1}{(x_2 y_1 - x_1 y_2)^2} + \underset{i=2,j=2}{(x_2 y_2 - x_2 y_2)^2}$$

$$= 0 + 2(x_1 y_2 - x_2 y_1)^2 + 0$$

$$= 2 \sum_{i<j}^{2} (x_i y_j - x_j y_i)^2$$

Thus, in general, $\sum_{i,j=1}^{n} (x_i y_j - x_j y_i)^2 = 2 \sum_{i<j}^{n} (x_i y_j - x_j y_i)^2$

because when $i=j$, $(x_i y_j - x_j y_i)^2 = 0$, and when

$i \neq j$, $(x_i y_j - x_j y_i)^2 = (x_j y_i - x_i y_j)^2$

and so this latter quantity is counted twice
(e.g., when $i=2, j=5$ and $i=5, j=2$)

$\therefore$ Need to prove $\frac{1}{2}\sum_{i,j=1}^{n}(x_i y_j - x_j y_i)^2 = \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) - \left(\sum_{i=1}^{n} x_i y_i\right)^2$

or, $\sum_{i,j=1}^{n}(x_i y_j - x_j y_i)^2 = 2\left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) - 2\left(\sum_{i=1}^{n} x_i y_i\right)^2$

The $\sum_{i,j=1}^{n}$ seems conceptually easier to understand.

Now, $(x_i y_j - x_j y_i)^2 = x_i^2 y_j^2 + x_j^2 y_i^2 - 2 x_i x_j y_i y_j$

$\therefore \sum_{i,j=1}^{n}(x_i y_j - x_j y_i)^2 = \sum_{i,j=1}^{n}(x_i^2 y_j^2 + x_j^2 y_i^2 - 2 x_i x_j y_i y_j)$

$= \sum_{i,j=1}^{n} x_i^2 y_j^2 + \sum_{i,j=1}^{n} x_j^2 y_i^2 - 2\sum_{i,j=1}^{n} x_i x_j y_i y_j$

But $\sum_{i,j=1}^{n} x_i^2 y_j^2 = \sum_{i,j=1}^{n} x_j^2 y_i^2$ since both are

$(x_1^2 + \ldots + x_n^2)(y_1^2 + \ldots + y_n^2) = \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right)$

$\therefore \sum_{i,j=1}^{n}(x_i y_j - x_j y_i)^2 = 2\left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) - 2\sum_{i,j=1}^{n} x_i x_j y_i y_j$

Now examine $\sum_{i,j=1}^{n} x_i x_j y_i y_j =$

$$x_1 y_1 \underbrace{(x_1 y_1 + \dots + x_n y_n)}_{i=1,\ j=1\dots n} + x_2 y_2 \underbrace{(x_1 y_1 + \dots + x_n y_n)}_{i=2,\ j=1\dots n} + \dots +$$

$$x_n y_n \underbrace{(x_1 y_1 + \dots + x_n y_n)}_{i=n,\ j=1\dots n}$$

$$= (x_1 y_1 + \dots + x_n y_n)(x_1 y_1 + \dots + x_n y_n)$$

$$= \left(\sum_{i=1}^{n} x_i y_i\right)^2$$

$$\therefore \sum_{i,j=1}^{n} (x_i y_j - x_j y_i)^2 = 2\left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) - 2\left(\sum_{i=1}^{n} x_i y_i\right)^2,$$

$$\text{or } \left(\sum_{i=1}^{n} x_i y_i\right)^2 = \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) - \frac{1}{2}\sum_{i,j=1}^{n} (x_i y_j - x_j y_i)^2$$

which was to be proved.

Use this to give another proof of the Cauchy–Schwarz inequality in $\mathbb{R}^n$,

The above term, $\frac{1}{2}\sum_{i,j=1}^{n} (x_i y_j - x_j y_i)^2$ contains all non-negative values.

$$\therefore \left(\sum_{i=1}^{n} x_i y_i\right)^2 \le \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right)$$

All terms are non-negative. $\therefore$ Taking square roots, $|\vec{x} \cdot \vec{y}| \le \|\vec{x}\| \|\vec{y}\|$

15. **15.** Prove that if A is an $n \times n$ matrix, then

(a) $\det(\lambda A) = \lambda^n \det A$; and

(b) if B is a matrix obtained from A by multiplying any row or column by a scalar λ, then $\det B = \lambda \det A$.

(a)(1) For a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, $\lambda A = \begin{bmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{bmatrix}$

$\det A = ad - bc$, $\det(\lambda A) = \lambda^2 ad - \lambda^2 bc$

$= \lambda^2(ad - bc) = \lambda^2 \det A$

(2) Assume true for $n = k \geq 2$

i.e., for $k \times k$ matrix, $\det(\lambda A) = \lambda^k \det(A)$

Consider a $n \times n$ matrix A, where $n = k+1$

By definition $\det(\lambda A) =$

$$\lambda a_{11} \det(\lambda A_1) - \lambda a_{12} \det(\lambda A_2) + \ldots + (-1)^{i+1} \lambda a_{1i} \det(\lambda A_i) + \ldots + (-1)^{n+1} \lambda a_{1n} \det(\lambda A_n)$$

But all the A_i are $k \times k$ matrices and by assumption $\det(\lambda A_i) = \lambda^k \det(A_i)$

$$\therefore \det(\lambda A) = \lambda a_{11} \lambda^k \det(A_1) + \ldots + (-1)^{i+1} \lambda a_{1i} \lambda^k \det(A_i) + \ldots + (-1)^{n+1} \lambda a_{1n} \lambda^k \det(A_n)$$

$$= a_{11} \lambda^{k+1} \det(A_1) + \ldots + (-1)^{i+1} a_{1i} \lambda^{k+1} \det(A_i) + \ldots + (-1)^{n+1} a_{1n} \lambda^{k+1} \det(A_n)$$

$$= \lambda^{k+1} \det(A)$$

$\therefore$ By (1) + (2), $\det(\lambda A) = \lambda^n \det(A)$

for all $n > 2$

(obviously true for $n = 1$)

(5) Let B be obtained from A by multiplying the "k"th row of A by λ.

$$\therefore \det B = (-1)^{k+1} b_{k1} \det(B_1) + \ldots + (-1)^{k+n} b_{kn} \det(B_n)$$

$$= (-1)^{k+1} \lambda a_{k1} \det(B_1) + \ldots + (-1)^{k+n} \lambda a_{kn} \det(B_n)$$

since $b_{ki} = \lambda a_{ki}$

$$= (-1)^{k+1} \lambda a_{k1} \det(A_1) + \ldots + (-1)^{k+n} \lambda a_{kn} \det(A_n)$$

since $A_i = B_i$

$$= \lambda \left[(-1)^{k+1} a_{k1} \det(A_1) + \ldots + (-1)^{k+n} a_{kn} \det(A_n) \right]$$

$$= \lambda \det(A)$$

The $A_i = B_i$ since the matrices are otherwise identical except for row k.

The same holds if it column k that is changed, using:

$$\det A = (-1)^{1+k} a_{1k} \det(A_1) + \ldots + (-1)^{n+k} a_{nk} \det(A_n)$$

16. **16.** Is $\det(A+B) = \det A + \det B$? Give a proof or counterexample.

In general, $\det(A+B) \neq \det A + \det B$

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\therefore \det(A+B) = \det \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} = 4$

$\det A = 1$, $\det B = 1$. $4 \neq 1+1$

17. **17.** Does $(A+B)(A-B) = A^2 - B^2$?

In general, $(A+B)(A-B) \neq A^2 - B^2$

$(A+B)(A-B) = A^2 + BA - AB - B^2$,

and $BA \neq AB$ in general.

18. **18.** Assuming the law $\det(AB) = (\det A)(\det B)$, prove that $\det(ABC) = (\det A)(\det B)(\det C)$.

$\det(ABC) = \det(A)\det(BC)$

$= \det(A)[\det(B)\det(C)]$

$= \det(A)\det(B)\det(C)$

19.

19. (This exercise assumes a knowledge of integration of continuous functions of one variable.) Note that the proof of the Cauchy–Schwarz inequality (Theorem 4)

depends only on the properties of the inner product listed in Theorem 1. Use this observation to establish the following inequality for continuous functions $f, g: [0, 1] \to \mathbb{R}$:

$$\left| \int_0^1 f(x)g(x)\ dx \right| \le \sqrt{\int_0^1 [f(x)]^2\ dx} \sqrt{\int_0^1 [g(x)]^2\ dx}.$$

Do this by

(a) verifying that the space of continuous functions from $[0, 1]$ to $\mathbb{R}$ forms a vector space; that is, we may think of functions f, g abstractly as "vectors" that can be added to each other and multiplied by scalars.

(a) Let f be any continuous function $f: [0,1] \to \mathbb{R}$, and let V = set of all numbers

$$v = \int_0^1 f(x)\,dx$$

Define addition for V: $v + w = \int_0^1 f(x)dx + \int_0^1 g(x)dx$,

Define scalar multiplication on V:

$$\alpha v = \alpha \int_0^1 f(x)dx$$

where f, g are continuous functions $: [0,1] \to \mathbb{R}$ and $\alpha \in \mathbb{R}$

(1) $\therefore$ whenever $v \in V$, so is αv, $\alpha \in \mathbb{R}$, since

$$\alpha v = \alpha \int_0^1 f(x)dx = \int_0^1 \alpha f(x)dx,$$ and since

$\alpha f(x)$ is continuous on $[0,1]$, $\int_0^1 \alpha f(x)dx \in V$

(2) Whenever $v, w \in V$, Then $v+w \in V$

Let $v = \int_0^1 f(x)\,dx$, $w = \int_0^1 g(x)\,dx$,

f, g some continous functions on $[0,1]$

$$\therefore v+w = \int_0^1 f(x)\,dx + \int_0^1 g(x)\,dx$$

$$= \int_0^1 [f(x)+g(x)]\,dx \in V,$$

since $f+g$ is continuous on $[0,1]$.

(3) Commutivity

Let $v, w \in V$. $\therefore v+w = w+v$

since $\int f + \int g = \int g + \int f$

(4) Associativity (a) Let $u, v, w \in V$, $u = \int_0^1 f$, $v = \int_0^1 g$, $w = \int_0^1 h$,
for some continuous $f, g, h : [0,1] \to \mathbb{R}$

$$\therefore (u+v)+w = \left(\int f + \int g\right) + \int h = \int (f+g) + \int h =$$

$$\int [(f+g)+h] = \int [f+(g+h)] = \int f + \int (g+h)$$

$$= \int f + \left(\int g + \int h\right) = u + (v+w)$$

(b) Now let $a, b \in \mathbb{R}$.

$$\therefore (ab)v = (ab)\int_0^1 f = a\left(b\int_0^1 f\right) = a(bv)$$

(5) Additive Identity

Let $g(x): [0,1] \to \mathbb{R}$ be s.t. $g(x) = 0$. $\therefore \int_0^1 g(x)\,dx = 0$

$\therefore$ Define $0 \in V$ as $0 = \int_0^1 0\,dx$

$$v + \int_0^1 0\,dx = \int_0^1 (f(x) + 0)\,dx = \int_0^1 f(x)\,dx = v,$$

f any continuous function on $[0,1]$

(6) Additive Inverse

Let $v \in V$. $\therefore$ There is a continuous function $f: [0,1] \to \mathbb{R}$ s.t. $v = \int_0^1 f(x)\,dx$

$\therefore$ Let $w = -\int_0^1 f(x)\,dx$

$\therefore v + w = \int_0^1 f(x) + \left(-\int_0^1 f(x)\,dx\right) = 0$

Since $-\int_0^1 f(x)\,dx = \int_0^1 -f(x)\,dx$, and $-f$ is continuous on $[0,1]$, then $\int_0^1 -f(x)\,dx \in V$,

and $\therefore w \in V$, so an addiv inverse exists for every $v \in V$.

(7) Multiplicative Identity

By definition, for any $v \in V$, $1v = 1\int_0^1 f(x)\,dx$

$= \int_0^1 1 \cdot f(x)\,dx = \int_0^1 f(x)\,dx = V$, f a continuous function on $[0,1]$.

(8) Distributive properties

Let $v, w \in V$, $\alpha, \beta \in R$.

$\therefore \exists\ f, g : [0,1] \to R$ s.t. $v = \int_0^1 f(x)\,dx$, $w = \int_0^1 g(x)\,dx$

$\therefore \alpha(v+w) = \alpha\left(\int_0^1 f(x)\,dx + \int_0^1 g(x)\,dx\right)$

$= \alpha\int_0^1 f(x)\,dx + \alpha\int_0^1 g(x)\,dx$

$= \alpha v + \alpha w$

Also, $(\alpha+\beta)v = (\alpha+\beta)\int_0^1 f(x)\,dx$

$= \alpha\int_0^1 f(x)\,dx + \beta\int_0^1 f(x)\,dx$

$= \alpha v + \beta v$

$\therefore V$ is a vector space

(6)

(b) introducing the inner product of functions

$$f \cdot g = \int_0^1 f(x)g(x)\,dx$$

and verifying that it satisfies conditions (i) to (iv) of Theorem 3.

Let $u, v, w \in V$. $\therefore \exists$ continuous $f, g, h : [0,1] \to \mathbb{R}$

s.t. $u = \int_0^1 f(x)\,dx$, $v = \int_0^1 g(x)\,dx$, $w = \int_0^1 h(x)\,dx$

Define $u \cdot v = \int_0^1 f(x)\,g(x)\,dx$

Since $f(x)g(x)$ is continuous on $[0,1]$, so

$u \cdot v \in V$.

Let $\alpha, \beta \in \mathbb{R}$

(i) Consider $(\alpha u + \beta v) \cdot w$

$$\alpha u + \beta v = \alpha \int_0^1 f + \beta \int_0^1 g = \int_0^1 (\alpha f + \beta g)$$

$$\therefore (\alpha u + \beta v) \cdot w = \int_0^1 (\alpha f + \beta g) \cdot h$$

$$= \int_0^1 \alpha f \cdot h + \beta g \cdot h = \alpha \int_0^1 f \cdot h + \beta \int_0^1 g \cdot h$$

$$= \alpha(u \cdot w) + \beta(v \cdot w)$$

(ii) $u \cdot v = \int_0^1 f \cdot g = \int_0^1 g \cdot f = v \cdot u$

(iii) $u \cdot u = \int_0^1 f(x) \cdot f(x)\,dx \geq 0$ since $f(x)^2 \geq 0$ on $[0,1]$ and $f(x)^2$ is continuous on $[0,1]$

(iv) (a) If $u \cdot u = 0$, Then $\int_0^1 f(x)^2 dx = 0$ by def of $u \cdot u$. Since $f(x)^2 \geq 0$ on $[0,1]$, and since $f(x)^2$ is continuous, Then $\int_0^1 f(x)^2 dx = 0 \Rightarrow f(x) = 0$ on $[0,1]$. $\therefore u = \int_0^1 f(x)\,dx = \int_0^1 0\,dx = 0$

$\therefore u \cdot u = 0 \Rightarrow u = 0$

(b) If $u = 0$, Then $u = \int_0^1 0\,dx$, from above definition of additive identity vector.

$\therefore u \cdot u = \int_0^1 0 \cdot 0\,dx = 0$

Note: proof of $|u \cdot v| \leq \|u\| \|v\|$ depends on (iva) above, not (ivb).

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The proof on pp. 61-62 of the text shows that

given any vector space V, with an inner product defined in such a way that properties (i)-(iv a) are satisfied, then if $x, y \in V$, then

$$(x \cdot y)^2 \leq (x \cdot x) \cdot (y \cdot y)$$

Proof: Let $x = \int_0^1 f(x)\,dx$, $y = \int_0^1 g(x)\,dx$

(a) If $y \cdot y = 0$, then by (iv a), $y = 0$

$\therefore x \cdot y = x \cdot 0 = \int f \cdot 0 = \int 0 = 0$

and $(x \cdot x) \cdot (y \cdot y) = (x \cdot x) \cdot \int 0^2 = (x \cdot x) \cdot 0$

$= 0 \left(\int f^2 dx \right)$ by (ii)

$= \int_0^1 0(f^2)\,dx = 0$

$\therefore (x \cdot y)^2 = 0^2 \leq 0 = (x \cdot x) \cdot (y \cdot y)$

(b) Let $a = y \cdot y$, $b = -x \cdot y$, and assume $a \neq 0$

$\therefore 0 \leq (ax + by) \cdot (ax + by)$ by (iii)

$= a^2 x \cdot x + 2ab\, x \cdot y + b^2 y \cdot y$ by (i), (ii), and 4(6), (8) above

$= (y \cdot y)^2 (x \cdot x) - 2(y \cdot y)(x \cdot y)^2 + (x \cdot y)^2 (y \cdot y)$

$$= (y\cdot y)^2(x\cdot x) - (y\cdot y)(x\cdot y)^2$$

since, for $r, s \in \mathbb{R}$, $-2rs^2 + s^2 r = -rs^2$

$\therefore (y\cdot y)(x\cdot y)^2 \le (y\cdot y)^2(x\cdot x)$

Since $y\cdot y > 0$, divide both sides by $y\cdot y$.

$\therefore (x\cdot y)^2 \le (y\cdot y)(x\cdot x)$

Taking square roots on real numbers,

$$|(x\cdot y)| \le \sqrt{y\cdot y}\,\sqrt{x\cdot x}$$

$$\therefore \left| \int_0^1 f(x)g(x)\,dx \right| \le \sqrt{\int_0^1 f(x)^2\,dx}\,\sqrt{\int_0^1 g(x)^2\,dx}$$

Note the proof above does not depend on $[0,1]$, so that it is true for any $[a,b]$.

20.

20. Define the transpose A^T of an $n \times n$ matrix A as follows: the ijth element of A^T is a_{ji} where a_{ij} is the ijth entry of A. Show that A^T is characterized by the following property: For all $\mathbf{x}$, $\mathbf{y}$ in $\mathbb{R}^n$,

$$(A^T\mathbf{x})\cdot\mathbf{y} = \mathbf{x}\cdot(A\mathbf{y}).$$

$$(A^T x)\cdot y = \sum_{i=1}^{n} (A^T x)_i \cdot y_i$$

$$\text{But } (A^T x)_i = \sum_{k=1}^{n} (A^T)_{ik} x_k = \sum_{k=1}^{n} A_{ki} x_k$$

$$\therefore (A^T x) \cdot y = \sum_{i=1}^{n} \left(\sum_{k=1}^{n} A_{ki} x_k \right) \cdot y_i$$

$$= \sum_{i=1}^{n} \left(\sum_{k=1}^{n} x_k A_{ki} \right) \cdot y_i$$

$$= \sum_{i=1}^{n} \left(x_1 A_{1i} y_i + x_2 A_{2i} y_i + \dots + x_n A_{ni} y_i \right)$$

$$= x_1 A_{11} y_1 + x_2 A_{21} y_1 + \dots + x_n A_{n1} y_1$$

$$+ x_1 A_{12} y_2 + x_2 A_{22} y_2 + \dots + x_n A_{n2} y_2$$

$$+ \quad \vdots \qquad\qquad \vdots \qquad\qquad \vdots$$

$$+ x_1 A_{1n} y_n + x_2 A_{2n} y_n + \dots + x_n A_{nn} y_n$$

$$= x_1 \left(\sum_{k=1}^{n} A_{1k} y_k \right) + x_2 \left(\sum_{k=1}^{n} A_{2k} y_k \right) + \dots + y_n \left(\sum_{k=1}^{n} A_{nk} y_k \right)$$

$$= \sum_{i=1}^{n} x_i \left(\sum_{k=1}^{n} A_{ik} y_k \right)$$

$$= \sum_{i=1}^{n} x_i \cdot (Ay)_i = x \cdot (Ay)$$

21.

21. Verify that the inverse of

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{is} \quad \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

(a) $$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ad-bc & bd-bd \\ -ca+ac & ad-bc \end{bmatrix}$$

$$= \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}$$

$$\therefore \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{ad-bc}\begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(b) $$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\left(\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right) =$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} \frac{d}{ad-bc} & \frac{-b}{ad-bc} \\ \frac{-c}{ad-bc} & \frac{a}{ad-bc} \end{bmatrix} = \begin{bmatrix} \frac{ad-bc}{ad-bc} & \frac{-ab+ab}{ad-bc} \\ \frac{cd-cd}{ad-bc} & \frac{-bc+ad}{ad-bc} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

22.

22. Use your answer in Exercise 21 to show that the solution of the system

$$ax + by = e$$
$$cx + dy = f$$

is

$$\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} e \\ f \end{bmatrix}.$$

$ax + by = e$ $cx + dy = f$ can be written as $\begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e \\ f \end{bmatrix}$

$\therefore$ If $\begin{bmatrix} x \\ y \end{bmatrix} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} e \\ f \end{bmatrix}$, Then

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}\left(\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} e \\ f \end{bmatrix}\right) =$$

$$\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right)\begin{bmatrix} e \\ f \end{bmatrix},$$ since $A(BC) = (AB)C$

Exercise #21 showed $\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$$\therefore \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\right)\begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} e \\ f \end{bmatrix} = \begin{bmatrix} e \\ f \end{bmatrix}$$

$\therefore$ This $\begin{bmatrix} x \\ y \end{bmatrix}$ is a solution.

Note, if $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, from #21,

$A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

$\therefore A^{-1}A\begin{bmatrix} x \\ y \end{bmatrix} = A^{-1}\begin{bmatrix} e \\ f \end{bmatrix}$, or $I\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} = A^{-1}\begin{bmatrix} e \\ f \end{bmatrix}$

Since inverses are unique, $A^{-1}\begin{bmatrix} e \\ f \end{bmatrix}$ is the only solution.

23.

23. Assuming the law $\det(AB) = (\det A)(\det B)$, verify that $(\det A)(\det A^{-1}) = 1$ and conclude that if A has an inverse, then $\det A \neq 0$.

If A has an inverse, then $AA^{-1} = I$

$\therefore \det(AA^{-1}) = \det(A)\det(A^{-1}) = \det(I) = 1$,

which means both $\det(A)$ and $\det(A^{-1})$ are nonzero.

24.

24. Find two 2×2 matrices A and B such that $AB = 0$ but $BA \neq 0$.

Let $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$

$\therefore AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ $BA = \begin{bmatrix} 2 & 2 \\ -2 & -2 \end{bmatrix}$

Review Exercises for Chapter 1

Note Title 12/10/2015

4. **4.** (a) Find the equation of the line through (0, 1, 0) in the direction of $3\mathbf{i} + \mathbf{k}$.

(b) Find the equation of the line passing through (0, 1, 1) and (0, 1, 0).

(c) Find an equation for the plane perpendicular to the vector (−1, 1, −1) and passing through the point (1, 1, 1).

(a) $(0,1,0) + t(3,0,1)$, t any real value

(b) $(0,1,1) - (0,1,0) = (0,0,1)$

$\therefore (0,1,1) + t(0,0,1)$, t any real value

(c) $(-1,1,-1)\cdot[(x,y,z) - (1,1,1)] = 0$

$\therefore -x + y - z + (1 + (-1) + 1) = 0$, or

$x - y + z - 1 = 0$

5. **5.** Find an equation for the plane containing the points (2, 1, −1), (3, 0, 2), and (4, −3, 1).

$(3,0,2) - (2,1,-1) = (1,-1,3)$

$(4,-3,1) - (2,1,-1) = (2,-4,2)$

$\therefore \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -1 & 3 \\ 2 & -4 & 2 \end{vmatrix} = (10,4,-2) =$ normal to plane

$\therefore (10,4,-2)\cdot[(x,y,z) - (2,1,-1)] = 0$, or

$10x + 4y - 2z - (20 + 4 + 2) = 0$, or

$10x + 4y - 2z - 26 = 0$, or $\underline{5x + 2y - z - 13 = 0}$

6. **6.** Find an equation for a line that is parallel to the plane $2x - 3y + 5z - 10 = 0$ and passes through the point $(-1, 7, 4)$. (There are lots of them.)

Normal to plane: $(2, -3, 5)$

$\therefore$ Vector perpendicular to $(2, -3, 5)$: $(1, -1, -1)$

since $(1, -1, -1) \cdot (2, -3, 5) = 0$

$\therefore$ One such line: $(-1, 7, 4) + t(1, -1, -1)$,
t any real value.

11. **11.** Use vector notation to describe the triangle in space whose vertices are the origin and the endpoints of vectors **a** and **b**.

One side of the triangle is described by $s\vec{a}$, $0 \le s \le 1$.

Another side by $t\vec{b}$, $0 \le t \le 1$

The third side is described by $\vec{a} + r(\vec{b} - \vec{a})$, $0 \le r \le 1$,
or $(1-r)\vec{a} + r\vec{b}$.

Using similar triangles, the triangular region

can be described as the base sliding continuously from the vertex to the base.

Use the $(1-r)\vec{a} + r\vec{b}$ vector as the side that slides down in a parallel fashion from apex (the origin) to the base.

$\therefore s[(1-r)\vec{a} + r\vec{b}]$, $0 \le s \le 1$, $0 \le r \le 1$

If s is fixed, as r varies between 0 and 1, the segment is described by $(1-r)\vec{a} + r\vec{b}$, and is parallel to $\vec{b} - \vec{a}$

$\therefore s(1-r)\vec{a} + sr\vec{b}$, $0 \le s \le 1$, $0 \le r \le 1$

12. **12.** Show that three vectors **a, b, c** lie in the same plane through the origin if and only if there are three scalars α, β, γ, not all zero, such that $\alpha\mathbf{a} + \beta\mathbf{b} + \gamma\mathbf{c} = \mathbf{0}$.

12.

(1) If all 3 line in a line, then after choosing one vector, say $\vec{a}$, the other two can be described as a scalar multiple of $\vec{a}$.

Let $\vec{b} = \frac{1}{\beta}\vec{a}$, $\vec{c} = \frac{1}{\gamma}\vec{a}$.

$\therefore \beta\vec{b} + \gamma\vec{c} - 2\vec{a} = 0$.

$\therefore$ Assume one, $\vec{a}$, is not collinear to $\vec{b}$ and $\vec{c}$.

$\therefore$ All points in plane can be described by

$\vec{a}$ and $\vec{b}$ ($\alpha\vec{a} + \beta\vec{b}$). Since $\vec{c}$ is in the plane, then there must be values for α, β, s.t.

$$\alpha\vec{a} + \beta\vec{b} = \vec{c}, \text{ or}$$

$$\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = 0, \quad \gamma = -1.$$

(2) Suppose there exists scalars α, β, γ, not all zero, s.t.

$$\alpha\vec{a} + \beta\vec{b} + \gamma\vec{c} = 0.$$

Choose a scalar that is not zero, say α.

$$\therefore \frac{\beta}{\alpha}\vec{b} + \frac{\gamma}{\alpha}\vec{c} = \vec{a}.$$

$\therefore$ The plane through the origin perpendicular to $\vec{b} \times \vec{c}$ contains $\vec{b}$, $\vec{c}$ and $\vec{a}$,

since $\vec{a} \cdot (\vec{b} \times \vec{c}) = \frac{\beta}{\alpha}\vec{b} \cdot (\vec{b} \times \vec{c}) + \frac{\gamma}{\alpha}\vec{c} \cdot (\vec{b} \times \vec{c}) = 0$

since $\vec{b} \cdot (\vec{b} \times \vec{c}) = (\vec{b} \times \vec{b}) \cdot \vec{c} = 0$

$$\vec{c} \cdot (\vec{b} \times \vec{c}) = -\vec{c} \cdot (\vec{c} \times \vec{b}) = -(\vec{c} \times \vec{c}) \cdot \vec{b} = 0$$

14.

14. Let $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ be unit vectors that are orthogonal to each other. If $\mathbf{a} = \alpha\mathbf{u} + \beta\mathbf{v} + \gamma\mathbf{w}$, show that

$$\alpha = \mathbf{a} \cdot \mathbf{u}, \qquad \beta = \mathbf{a} \cdot \mathbf{v}, \qquad \gamma = \mathbf{a} \cdot \mathbf{w}.$$

Interpret the results geometrically.

$$\vec{a}\cdot\vec{u} = (\alpha\vec{u} + \beta\vec{v} + \gamma\vec{w})\cdot\vec{u}$$

$$= \alpha\vec{u}\cdot\vec{u} + \beta\vec{v}\cdot\vec{u} + \gamma\vec{w}\cdot\vec{u}$$

$$= \alpha\,\vec{u}\cdot\vec{u}, \text{ since } \vec{v}\cdot\vec{u} = 0,\ \vec{w}\cdot\vec{u} = 0$$

$$= \alpha, \text{ since } \vec{u}\cdot\vec{u} = \|\vec{u}\|\|\vec{u}\|\cos\theta = 1\cdot 1\cdot 1\ (\theta = 0)$$

similarly $\vec{a}\cdot\vec{v} = \beta$, $\vec{a}\cdot\vec{w} = \gamma$

=

$$\alpha = \vec{a}\cdot\vec{u} = \|\vec{a}\|\|\vec{u}\|\cos\theta = \|\vec{a}\|\cos\theta$$

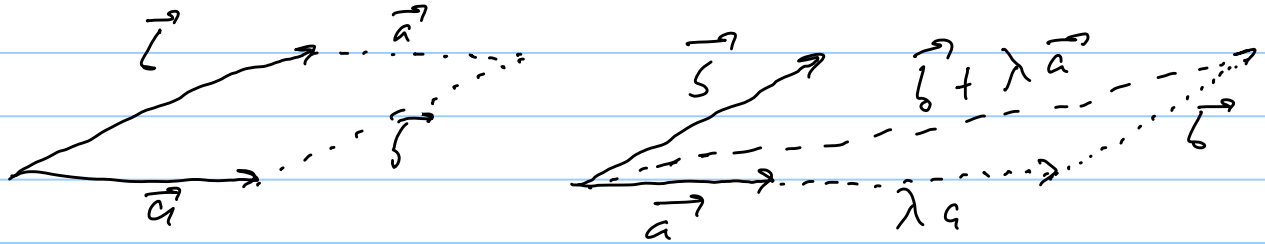
$\therefore \alpha$ = scalar value of projection of $\vec{a}$ onto $\vec{u}$

similarly for β, γ.

(similar to direction numbers).

17.

17. Let **a**, **b** be two vectors in the plane, $\mathbf{a} = (a_1, a_2)$, $\mathbf{b} = (b_1, b_2)$, and let λ be a real number. Show that the area of the parallelogram determined by **a** and $\mathbf{b} + \lambda\mathbf{a}$ is the same as that determined by **a** and **b**. Sketch. Relate this result to a known property of determinants.



Same height, same base ($\vec{a}$)

$$\text{Area}(1) = \|\vec{a} \times \vec{b}\| = (\|b\| \sin\theta)\|a\|$$

$$\text{Area}(2) = \|(\vec{b} + \lambda\vec{a}) \times \vec{a}\|$$

$$= \|(\vec{b} \times \vec{a}) + \lambda(\vec{a} \times \vec{a})\|$$

$$= \|(\vec{b} \times \vec{a}) + 0\| = (\|\vec{b}\| \sin\theta)\|\vec{a}\|$$

$$\therefore \text{Area}(1) = \text{Area}(2)$$

Property: when add scalar multiple of one row to another row, determinant stays unchanged.

19. **19.** Given nonzero vectors **a** and **b** in $\mathbb{R}^3$, show that the vector $\mathbf{v} = \|\mathbf{a}\|\mathbf{b} + \|\mathbf{b}\|\mathbf{a}$ bisects the angle between **a** and **b**.

show 2 angles are equal.

Angle between $\vec{v}, \vec{a}$: $\vec{v} \cdot \vec{a} = \|\vec{a}\|\vec{b} \cdot \vec{a} + \|\vec{b}\|\vec{a} \cdot \vec{a}$

$$= \|\vec{v}\| \|\vec{a}\| \cos\theta_{va}$$

$$\therefore \cos\theta_{va} = \frac{\vec{b} \cdot \vec{a} + \|\vec{b}\| \|\vec{a}\|}{\|\vec{v}\|}$$

Angle between $\vec{v}, \vec{b}$: $\vec{v} \cdot \vec{b} = \|\vec{a}\|\vec{b} \cdot \vec{b} + \|\vec{b}\|\vec{a} \cdot \vec{b}$

$$= \|\vec{v}\| \|\vec{b}\| \cos\theta_{vb}$$

$$\therefore \cos\theta_{vb} = \frac{\|\vec{a}\| \|\vec{b}\| + \vec{a} \cdot \vec{b}}{\|\vec{v}\|}$$

$\therefore \cos\theta_{vA} = \cos\theta_{vb}$, $\therefore \theta_{vA} = \pm\theta_{vb}$, or $|\theta_{vA}| = |\theta_{vb}|$

$\therefore \vec{v}$ bisects angle between $\vec{a}, \vec{b}$

21.

21. Use the triangle inequality to show that $\|\mathbf{v} - \mathbf{w}\| \geq \big|\|\mathbf{v}\| - \|\mathbf{w}\|\big|$.

Since $\|a + b\| \leq \|a\| + \|b\|$

(1) Let $\vec{a} = \vec{w}$, $\vec{b} = \vec{v} - \vec{w}$

$\therefore \|\vec{v}\| = \|\vec{w} + (\vec{v} - \vec{w})\| \leq \|w\| + \|\vec{v} - \vec{w}\|$

$\therefore \|\vec{v}\| - \|\vec{w}\| \leq \|\vec{v} - \vec{w}\|$

(2) Now let $\vec{a} = \vec{v}$, $\vec{b} = \vec{w} - \vec{v}$

$\therefore \|\vec{w}\| = \|\vec{v} + (\vec{w} - \vec{v})\| \leq \|\vec{v}\| + \|\vec{w} - \vec{v}\|$

$\therefore \|\vec{w}\| - \|\vec{v}\| \leq \|\vec{w} - \vec{v}\| = \|\vec{v} - \vec{w}\|$

$\therefore (1) \& (2) \Rightarrow \big|\|\vec{w}\| - \|\vec{v}\|\big| \leq \|\vec{v} - \vec{w}\|$

20.

20. Show that the vectors $\|\mathbf{b}\|\mathbf{a} + \|\mathbf{a}\|\mathbf{b}$ and $\|\mathbf{b}\|\mathbf{a} - \|\mathbf{a}\|\mathbf{b}$ are orthogonal.

$$(\|\vec{b}\|\vec{a} + \|\vec{a}\|\vec{b}) \cdot (\|\vec{b}\|\vec{a} - \|\vec{a}\|\vec{b}) =$$

$$\|\vec{b}\|^2 \vec{a}\cdot\vec{a} - \|\vec{b}\|\|\vec{a}\|\vec{a}\cdot\vec{b} + \|\vec{a}\|\|\vec{b}\|\vec{b}\cdot\vec{a} - \|\vec{a}\|^2\vec{b}\cdot\vec{b}$$

$$= \|\vec{b}\|^2\vec{a}\cdot\vec{a} - \|\vec{a}\|^2\vec{b}\cdot\vec{b} = \|\vec{b}\|^2\|\vec{a}\|^2 - \|\vec{a}\|^2\|\vec{b}\|^2 = 0$$

22. **22.** Use vector methods to prove that the distance from the point (x_1, y_1) to the line $ax + by = c$ is

$$\frac{|ax_1 + by_1 - c|}{\sqrt{a^2 + b^2}}.$$

Let (x_0, y_0) be a point on the line. $\therefore ax_0 + by_0 = c$

$\therefore a(x - x_0) + b(y - y_0) = 0$, or

$(a, b) \cdot (x - x_0, y - y_0) = 0$

$\therefore$ The line consists of all points (x, y) s.t. $(x - x_0, y - y_0)$ is $\perp$ to (a, b).

Let $\vec{n} = (a, b)$, the normal to the line.

$\therefore$ Distance from (x_1, y_1) to line is the magnitude of the projection of $(x_1, y_1) - (x_0, y_0)$ onto $\vec{n}$, or

$\|(x_1, y_1) - (x_0, y_0)\| \cos\theta$, θ = angle between $\vec{n}$ and $(x_1, y_1) - (x_0, y_0)$

But $\vec{n} \cdot [(x_1, y_1) - (x_0, y_0)] = \vec{n} \cdot (x_1, y_1) - \vec{n} \cdot (x_0, y_0)$

$= (a, b) \cdot (x_1, y_1) - (a, b) \cdot (x_0, y_0) =$

$ax_1 + by_1 - c$

And $\vec{n} \cdot [(x_1, y_1) - (x_0, y_0)] =$

$$\|\vec{n}\| \cdot \|(x_1, y_1) - (x_0, y_0)\| \cos\theta$$

$$\therefore \quad ax_1 + by_1 - c = \|\vec{n}\| \, \|(x_1, y_1) - (x_0, y_0)\| \cos\theta$$

$$\therefore \quad \frac{ax_1 + by_1 - c}{\|\vec{n}\|} = \|(x_1, y_1) - (x_0, y_0)\| \cos\theta$$

and $\|\vec{n}\| = \sqrt{a^2 + b^2}$

$$\therefore \quad \frac{|ax_1 + by_1 - c|}{\sqrt{a^2 + b^2}} = \left| \|(x_1, y_1) - (x_0, y_0)\| \cos\theta \right|$$

= magnitude of projection of $(x_1, y_1) - (x_0, y_0)$ onto $\vec{n}$

= distance of (x_1, y_1) to line.

23. **23.** Verify that the direction of $\mathbf{b} \times \mathbf{c}$ is given by the right-hand rule, by choosing $\mathbf{b}$, $\mathbf{c}$ to be two of the vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.

Choose $\vec{b}$ & $\vec{c}$ to be in xy-plane, with $\vec{b}$ along x-axis. $\therefore \vec{b} = b\hat{i}$ and $\vec{c} = c_1\hat{i} + c_2\hat{j}$, b, c_1, c_2 some real numbers.

$$\therefore \vec{b} \times \vec{c} = (b\hat{i}) \times (c_1\hat{i} + c_2\hat{j})$$

$$= (bc_1)\,\hat{i} \times \hat{i} + (bc_2)\,\hat{i} \times \hat{j}$$

$$= 0 + bc_2\hat{k} = bc_2\hat{k}$$

$\therefore$ Only component of $\vec{b} \times \vec{c}$ is along $\hat{k}$, same as right-hand rule.

24. **24.** (a) Suppose $\mathbf{a} \cdot \mathbf{b} = \mathbf{a}' \cdot \mathbf{b}$ for all $\mathbf{b}$. Show that $\mathbf{a} = \mathbf{a}'$.
(b) Suppose $\mathbf{a} \times \mathbf{b} = \mathbf{a}' \times \mathbf{b}$ for all $\mathbf{b}$. Is it true that $\mathbf{a} = \mathbf{a}'$?

(a) Let $\vec{a} = (a_1, a_2, a_3)$, $\vec{a}' = (a_1', a_2', a_3')$

$\vec{a} \cdot (1,0,0) = a_1$ $\quad \vec{a}' \cdot (1,0,0) = a_1'$ $\quad \therefore a_1 = a_1'$

$\vec{a} \cdot (0,1,0) = a_2$ $\quad \vec{a}' \cdot (0,1,0) = a_2'$ $\quad \therefore a_2 = a_2'$

$\vec{a} \cdot (0,0,1) = a_3$ $\quad \vec{a}' \cdot (0,0,1) = a_3'$ $\quad \therefore a_3 = a_3'$

$\therefore \vec{a} = \vec{a}'$

(b) Since $\vec{a} \times \vec{a}' = \vec{a}' \times \vec{a}' = 0$, and assuming $\vec{a}, \vec{a}' \neq 0$,

Then $\|\vec{a} \times \vec{a}'\| = \|\vec{a}\|\|\vec{a}'\| \sin\theta \Rightarrow \sin\theta = 0$.

$\therefore \vec{a}' = \lambda \vec{a}$, some scalar $\lambda \neq 0$.

But $\vec{a} \times \hat{i} = \vec{a}' \times \hat{i}$

$$\therefore \|\vec{a} \times \hat{i}\| = \|\vec{a}\| \sin\theta = \|\vec{a}' \times \hat{i}\|$$

$$= \|\lambda \vec{a} \times \hat{i}\| = |\lambda| \, \|\vec{a}\| \sin\theta$$

$$\therefore |\lambda| = 1$$

If $\lambda = -1$, then $\vec{a} \times \vec{b} = -\vec{a} \times \vec{b}$, for all $\vec{b}$,

$\therefore 2\,\vec{a} \times \vec{b} = 0$, or $\vec{a} \times \vec{b} = 0$ for all $\vec{b}$.

Since $\vec{a} \neq 0$, then $\lambda \neq -1$.

$\therefore \lambda = 1$, and $\therefore \vec{a} = \vec{a}'$

$\therefore$ True, $\vec{a} = \vec{a}'$

Perhaps an easier proof:

$\vec{a} \times \hat{i} = \vec{a}' \times \hat{i}$ means

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ 1 & 0 & 0 \end{vmatrix} = (0, a_3, -a_2) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1' & a_2' & a_3' \\ 1 & 0 & 0 \end{vmatrix} = (0, a_3', -a_2')$$

$$\therefore a_3 = a_3', \; a_2 = a_2'$$

Similarly, $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ 0 & 1 & 0 \end{vmatrix} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1' & a_2' & a_3' \\ 0 & 1 & 0 \end{vmatrix}$

from $\vec{a} \times \hat{j} = \vec{a}' \times \hat{j}$, and $\therefore$

$(-a_3, 0, a_1) = (-a_3', 0, a_1')$, $\therefore a_1 = a_1'$

$\therefore (a_1, a_2, a_3) = (a_1', a_2', a_3')$, or $\underline{\underline{\vec{a} = \vec{a}'}}$

25.

25. (a) Using vector methods, show that the distance between two nonparallel lines l_1 and l_2 is given by

$$d = \frac{|(\mathbf{v}_2 - \mathbf{v}_1) \cdot (\mathbf{a}_1 \times \mathbf{a}_2)|}{\|\mathbf{a}_1 \times \mathbf{a}_2\|},$$

where $\mathbf{v}_1$, $\mathbf{v}_2$ are any two points on l_1 and l_2, respectively, and $\mathbf{a}_1$ and $\mathbf{a}_2$ are the directions of l_1 and l_2. [HINT: Consider the plane through l_2 that is parallel to l_1. Show that the vector $(\mathbf{a}_1 \times \mathbf{a}_2)/\|\mathbf{a}_1 \times \mathbf{a}_2\|$ is a unit normal for this plane; now project $\mathbf{v}_2 - \mathbf{v}_1$ onto this normal direction.]

(b) Find the distance between the line l_1 determined by the points $(-1, -1, 1)$ and $(0, 0, 0)$ and the line l_2 determined by the points $(0, -2, 0)$ and $(2, 0, 5)$.

$l_1 : \vec{v}_1 + r\vec{a}_1$

$l_2 : \vec{v}_2 + s\vec{a}_2$

$r, s \in \mathbb{R}$

(a) A vector perpendicular to l_1 and l_2 is $\vec{a}_1 \times \vec{a}_2$

There is a plane, P_1, containing $l_1 \perp \vec{a}_1 \times \vec{a}_2$,

and there is a plane, P_2, containing $l_2 \perp \vec{a}_1 \times \vec{a}_2$

$\therefore$ Consider the distance between l_1 and l_2 as the distance between the planes P_1, P_2.

$\vec{v}_1 \in P_1$ since l_1 is in P_1, $\vec{v}_2 \in P_2$ since l_2 is in P_2.

$\therefore$ The magnitude of the projection of $\vec{v}_2 - \vec{v}_1$ onto $\vec{a}_1 \times \vec{a}_2$, a vector perpendicular to

So P_1 and P_2, is the distance between P_1 & P_2, and $\therefore$ the distance between l_1 & l_2.

$|\|\vec{v}_2 - \vec{v}_1\| \cos\theta|$ is the magnitude of the projection of $\vec{v}_2 - \vec{v}_1$ onto $\vec{a}_1 \times \vec{a}_2$, θ = angle between $(\vec{v}_2 - \vec{v}_1)$ and $\vec{a}_1 \times \vec{a}_2$

But $(\vec{v}_2 - \vec{v}_1) \cdot (\vec{a}_1 \times \vec{a}_2) = \|\vec{v}_2 - \vec{v}_1\| \, \|\vec{a}_1 \times \vec{a}_2\| \cos\theta$

$$\therefore \|\vec{v}_2 - \vec{v}_1\| \cos\theta = \frac{(\vec{v}_2 - \vec{v}_1) \cdot (\vec{a}_1 \times \vec{a}_2)}{\|\vec{a}_1 \times \vec{a}_2\|}$$

$$\therefore |\|v_2 - v_1\| \cos\theta| = \frac{|(\vec{v}_2 - \vec{v}_1) \cdot (\vec{a}_1 \times \vec{a}_2)|}{\|\vec{a}_1 \times \vec{a}_2\|}$$

(6) $l_1 : (0,0,0) + r(-1,-1,-1)$

$l_2 : (2,0,5) + s\{(2,0,5) - (0,-2,0)\} = (2,0,5) + s(2,2,5)$

$\therefore \vec{v}_2 - \vec{v}_1 = (2,0,5) - (0,0,0) = (2,0,5)$

$$\vec{a}_1 \times \vec{a}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -1 & -1 \\ 2 & 2 & 5 \end{vmatrix} = (-3, 3, 0)$$

$$\|\vec{a}_1 \times \vec{a}_2\| = \sqrt{9+9+0} = 3\sqrt{2}$$

$$(\vec{v}_2 - \vec{v}_1) \cdot (\vec{a}_1 \times \vec{a}_2) = (2,0,5) \cdot (-3,3,0) = -6$$

$$\therefore d = \frac{|-6|}{3\sqrt{2}} = \underline{\underline{\sqrt{2}}}$$

26.

26. Show that two planes given by the equations $Ax + By + Cz + D_1 = 0$ and $Ax + By + Cz + D_2 = 0$ are parallel, and that the distance between them is

$$\frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}.$$

Normal for each plane is: (A, B, C)

$\therefore$ Same normal $\Rightarrow$ planes parallel.

Let (x_0, y_0, z_0) be a point on $Ax + By + Cz + D_1 = 0$

$\therefore Ax_0 + By_0 + Cz_0 + D_1 = 0,$

$\therefore -D_1 = Ax_0 + By_0 + Cz_0.$

Distance between (x_0, y_0, z_0) and second plane is:

$$d = \frac{|Ax_0 + By_0 + Cz_0 + D_2|}{\sqrt{A^2 + B^2 + C^2}} = \frac{|-D_1 + D_2|}{\sqrt{A^2 + B^2 + C^2}}$$

But $|-D_1 + D_2| = |D_2 - D_1| = |D_1 - D_2|$

$$\therefore d = \frac{|D_1 - D_2|}{\sqrt{A^2 + B^2 + C^2}}$$

27.

27. (a) Prove that the area of the triangle in the plane with vertices (x_1, y_1), (x_2, y_2), (x_3, y_3) is the absolute value of

$$\frac{1}{2}\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}.$$

(b) Find the area of the triangle with vertices $(1, 2)$, $(0, 1)$, $(-1, 1)$.

Let $P_1 = (x_1, y_1, 0)$
$P_2 = (x_2, y_2, 0)$
$P_3 = (x_3, y_3, 0)$

(a) $\therefore$ Area $= \frac{1}{2}\left|(\vec{P_2} - \vec{P_1}) \times (\vec{P_3} - \vec{P_1})\right|$

$$= \frac{1}{2}\left\| \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_2 - x_1 & y_2 - y_1 & 0 \\ x_3 - x_1 & y_3 - y_1 & 0 \end{vmatrix} \right\|$$

$$= \frac{1}{2}\left\| \left(0, 0, (x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)\right) \right\|$$

$$= \frac{1}{2}\sqrt{0^2 + 0^2 + \left[(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)\right]^2}$$

$$= \frac{1}{2}\left|(x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_2 - y_1)\right|$$

$$= \frac{1}{2}\left| \begin{vmatrix} x_2 - x_1 & x_3 - x_1 \\ y_2 - y_1 & y_3 - y_1 \end{vmatrix} \right|$$

$$= \frac{1}{2}\left| \begin{vmatrix} 0 & 0 & 1 \\ x_2 - x_1 & x_3 - x_1 & x_1 \\ y_2 - y_1 & y_3 - y_1 & y_1 \end{vmatrix} \right|$$

= after adding Column 3 to Column 1

$$= \frac{1}{2}\left|\begin{vmatrix} 1 & 0 & 1 \\ x_2 & x_3 - x_1 & x_1 \\ y_2 & y_3 - y_1 & y_1 \end{vmatrix}\right|$$

= after adding Column 3 to Column 2

$$= \frac{1}{2}\left|\begin{vmatrix} 1 & 1 & 1 \\ x_2 & x_3 & x_1 \\ y_2 & y_3 & y_1 \end{vmatrix}\right|$$

= after swapping Column 3 for Column 2, Then swapping Column 2 for Column 1

$$= \frac{1}{2}\left|\begin{vmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{vmatrix}\right|$$

(6) Area $= \frac{1}{2}\left|\begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & -1 \\ 2 & 1 & 1 \end{vmatrix}\right| = \frac{1}{2}|1 - 3 + 1| = \frac{1}{2}$

28.

28. Convert the following points from Cartesian to cylindrical and spherical coordinates and plot:

(a) $(0, 3, 4)$

(b) $(-\sqrt{2}, 1, 0)$

(c) $(0, 0, 0)$

(d) $(-1, 0, 1)$

(e) $(-2\sqrt{3}, -2, 3)$

(a) Cylindrical: $r=\sqrt{0^2+3^2}=3$, $\theta=\frac{\pi}{2}$. $\therefore (3, \frac{\pi}{2}, 4)$

Spherical: $\rho=\sqrt{0^2+3^2+4^2}=5$, $\theta=\frac{\pi}{2}$, $\phi=\arcsin\frac{r}{\rho}$

$\therefore (5, \frac{\pi}{2}, \arcsin\frac{3}{5})$

(b) Cylindrical: $r=\sqrt{\sqrt{2}^2+1^2}=\sqrt{3}$, $\theta=\arctan\frac{1}{\sqrt{2}}=\pi-\arctan\frac{\sqrt{2}}{2}$

$\therefore (\sqrt{3}, \pi-\arctan\frac{\sqrt{2}}{2}, 0)$

spherical: $\rho=\sqrt{3}$, $\theta=\pi-\arctan\frac{\sqrt{2}}{2}$, $\phi=\frac{\pi}{2}$

$\therefore (\sqrt{3}, \pi-\arctan\frac{\sqrt{2}}{2}, \frac{\pi}{2})$

(c) Cylindrical: $r=0$, $\theta=$ indeterminate, choose $\theta=0$, $z=0$

$\therefore (0,0,0)$

Spherical: $\rho=0$, θ, ϕ indeterminate, choose 0.

$\therefore (0,0,0)$

(d) Cylindrical: $r=1$, $\theta=\pi$, $z=1$. $\therefore (1, \pi, 1)$

Spherical: $\rho=\sqrt{2}$, $\theta=\pi$, $\phi=\arcsin\frac{1}{\sqrt{2}}=\frac{\pi}{4}$

$\therefore (\sqrt{2}, \pi, \frac{\pi}{4})$

(e) Cylindrical: $r=\sqrt{12+4}=4$, $\theta=\pi+\arctan\frac{2}{2\sqrt{3}}$

$\therefore (4, \pi+\arctan\sqrt{3}/3, 3)$

Spherical: $\rho=5$, $\theta=\pi+\arctan\sqrt{3}/3$, $\phi=\arcsin\frac{4}{5}$

$\therefore \left(5, \arctan\frac{\sqrt{3}}{3}, \arcsin\frac{4}{5}\right)$

Summary:

Cartesian	Cylindrical	Spherical
$(0, 3, 4)$	$(3, \frac{\pi}{2}, 4)$	$(5, \frac{\pi}{2}, \arcsin\frac{3}{5})$
$(-\sqrt{2}, 1, 0)$	$(\sqrt{3}, \pi - \arctan\frac{\sqrt{2}}{2}, 0)$	$(\sqrt{3}, \pi - \arctan\frac{\sqrt{2}}{2}, \frac{\pi}{2})$
$(0, 0, 0)$	$(0, 0, 0)$	$(0, 0, 0)$
$(-1, 0, 1)$	$(1, \pi, 1)$	$(\sqrt{2}, \pi, \frac{\pi}{4})$
$(-2\sqrt{3}, -2, 3)$	$(4, \pi + \arctan\sqrt{3}/3, 3)$	$(5, \pi + \arctan\frac{\sqrt{3}}{3}, \arcsin\frac{4}{5})$

29.

29. Convert the following points from cylindrical to Cartesian and spherical coordinates and plot:

(a) $(1, \pi/4, 1)$
(b) $(3, \pi/6, -4)$
(c) $(0, \pi/4, 1)$
(d) $(2, -\pi/2, 1)$
(e) $(-2, -\pi/2, 1)$

(a) Cartesian: $x = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$, $y = \sin\frac{\pi}{4} = \frac{\sqrt{2}}{2}$, $z = 1$

$\therefore \left(\sqrt{2}/2, \sqrt{2}/2, 1\right)$

Spherical: $\rho^2 = r^2 + z^2 = 2$, $\theta = \frac{\pi}{4}$, $\phi = \arcsin\frac{r}{\rho} = \arcsin\frac{1}{\sqrt{2}}$

$\therefore \left(\sqrt{2}, \frac{\pi}{4}, \frac{\pi}{4}\right)$

(b) Cartesian: $x = 3\cos\frac{\pi}{6} = \frac{3\sqrt{3}}{2}$, $y = 3\sin\frac{\pi}{6} = \frac{3}{2}$, $z = -4$

$\therefore \left(\frac{3\sqrt{3}}{2}, \frac{3}{2}, -4\right)$

Spherical: $\rho^2 = 3^2 + (-4)^2 = 5^2$, $\theta = \frac{\pi}{6}$, $\phi = \arccos\frac{-4}{5}$

$$\therefore \left(5, \frac{\pi}{6}, \pi - \arccos\frac{4}{5}\right)$$

(c) Cartesian: $x = 0\cos\frac{\pi}{4} = 0$, $y = 0\sin\frac{\pi}{4} = 0$, $z = 1$

$$\therefore (0, 0, 1)$$

Spherical: $\rho^2 = 0 + 1^2 = 1$, $\theta = \frac{\pi}{4}$, $\phi = \arcsin\frac{0}{1} = 0$

$$\therefore \left(1, \frac{\pi}{4}, 0\right)$$

(d) Cartesian: $x = 2\cos\frac{-\pi}{2} = 0$, $y = 2\sin\frac{-\pi}{2} = -2$, $z = 1$

$$\therefore (0, -2, 1)$$

Spherical: $\rho^2 = 2^2 + 1^2 = 5$, $\theta = \frac{3}{2}\pi$, $\phi = \arccos\frac{1}{\sqrt{5}}$

$$\therefore \left(\sqrt{5}, \frac{3}{2}\pi, \arccos\frac{1}{\sqrt{5}}\right)$$

(e) Cartesian: $x = -2\cos\frac{-\pi}{2} = 0$, $y = -2\sin\frac{-\pi}{2} = 2$, $z = 1$

$$\therefore (0, 2, 1)$$

Spherical: $\rho^2 = (-2)^2 + 1^2 = 5$, $\theta = \frac{\pi}{2}$, $\phi = \arccos\frac{1}{\sqrt{5}}$

$$\therefore \left(\sqrt{5}, \frac{\pi}{2}, \arccos\frac{1}{\sqrt{5}}\right)$$

Summary:

Cylindrical	Cartesian	Spherical
$(1, \pi/4, 1)$	$(\sqrt{2}/2, \sqrt{2}/2, 1)$	$(\sqrt{2}, \pi/4, \pi/4)$
$(3, \pi/6, -4)$	$(\frac{3\sqrt{3}}{2}, \frac{3}{2}, -4)$	$(5, \pi/6, \pi - \arccos\frac{4}{5})$
$(0, \pi/4, 1)$	$(0, 0, 1)$	$(1, \pi/4, 0)$
$(2, -\pi/2, 1)$	$(0, -2, 1)$	$(\sqrt{5}, 3\pi/2, \arccos\frac{1}{\sqrt{5}})$
$(-2, -\pi/2, 1)$	$(0, 2, 1)$	$(\sqrt{5}, \pi/2, \arccos 1/\sqrt{5})$

30.

30. Convert the following points from spherical to Cartesian and cylindrical coordinates and plot:

(a) $(1, \pi/2, \pi)$
(b) $(2, -\pi/2, \pi/6)$
(c) $(0, \pi/8, \pi/35)$
(d) $(2, -\pi/2, -\pi)$
(e) $(-1, \pi, \pi/6)$

(a) Cartesian: $r = 1\sin\pi = 0 \therefore x = y = 0,\ z = 1\cos\pi = -1$

$\therefore (0, 0, -1)$

Cylindrical: $r = 0,\ \theta = \frac{\pi}{2},\ z = 1\cos\pi = -1$

$\therefore (0, \pi/2, -1)$

(b) Cartesian: $r = 2\sin\frac{\pi}{6} = 1$ $x = r\cos(-\frac{\pi}{2}) = 0,$

$y = r\sin(-\frac{\pi}{2}) = -1$

$z = 2\cos\frac{\pi}{6} = \sqrt{3}$

$\therefore (0, -1, \sqrt{3})$

Cylindrical: $r = 2\sin\frac{\pi}{6} = 1,\ \theta = -\frac{\pi}{2},\ z = 2\cos\frac{\pi}{6} = \sqrt{3}$

$\therefore (1, -\frac{\pi}{2}, \sqrt{3})$

(c) Cartesian: $r = 0\sin\phi = 0 \Rightarrow x = y = 0,\ z = 0\cos\frac{\pi}{35} = 0$

$\therefore (0, 0, 0)$

Cylindrical: $r = 0\sin\phi = 0,\ \theta = \frac{\pi}{8},\ z = 0\cos\phi = 0$

$\therefore \left(0, \frac{\pi}{8}, 0\right)$

(d) Cartesian: $r = 2\sin(-\pi) = 0 \Rightarrow x = y = 0,\ z = 2\cos(-\pi) = -2$

$\therefore (0, 0, -2)$

Cylindrical: $r = 2\sin(-\pi) = 0,\ \theta = -\frac{\pi}{2},\ z = 2\cos(-\pi) - 2$

$\therefore \left(0, -\frac{\pi}{2}, -2\right)$

(e) Cartesian: $r = (-1)\sin\frac{\pi}{6} = -\frac{1}{2},\ x = -\frac{1}{2}\cos\pi = \frac{1}{2}$

$y = -\frac{1}{2}\sin\pi = 0$

$z = (-1)\cos\frac{\pi}{6} = -\sqrt{3}/2$

$\therefore \left(\frac{1}{2}, 0, -\frac{\sqrt{3}}{2}\right)$

Cylindrical: $r = (-1)\sin\frac{\pi}{6} = \frac{-1}{2},\ \theta = \pi,\ z = (-1)\cos\frac{\pi}{6} = -\frac{\sqrt{3}}{2}$

$\therefore \left(-\frac{1}{2}, \pi, -\sqrt{3}/2\right)$

Summary:

Spherical	Cartesian	Cylindrical
$(1, \pi/2, \pi)$	$(0, 0, -1)$	$(0, \pi/2, -1)$
$(2, -\pi/2, \pi/6)$	$(0, -1, \sqrt{3})$	$(1, -\pi/2, \sqrt{3})$
$(0, \pi/8, \pi/35)$	$(0, 0, 0)$	$(0, \pi/8, 0)$
$(2, -\pi/2, -\pi)$	$(0, 0, -2)$	$(0, -\pi/2, -2)$
$(-1, \pi, \pi/6)$	$\left(\frac{1}{2}, 0, -\sqrt{3}/2\right)$	$\left(-\frac{1}{2}, \pi, -\frac{\sqrt{3}}{2}\right)$

31. **31.** Rewrite the equation $z = x^2 - y^2$ using cylindrical and spherical coordinates.

Cylindrical: $x = r\cos\theta$, so $x^2 = r^2\cos^2\theta$

$y = r\sin\theta$, $\therefore y^2 = r^2\sin^2\theta$

$\therefore z = r^2(\cos^2\theta - \sin^2\theta)$

But $\cos 2\theta = \cos\theta\cos\theta - \sin\theta\sin\theta$

$\therefore \underline{z = r^2\cos 2\theta}$

Spherical: $z = \rho\cos\phi$

$x = \rho\sin\phi\cos\theta$, $\therefore x^2 = \rho^2\sin^2\phi\cos^2\theta$

$y = \rho\sin\phi\sin\theta$, $\therefore y^2 = \rho^2\sin^2\phi\sin^2\theta$

$\therefore \rho\cos\phi = \rho^2\sin^2\phi\cos^2\theta - \rho^2\sin^2\phi\sin^2\theta$

or, $\cos\phi = \rho\sin^2\phi(\cos^2\theta - \sin^2\theta)$

$\therefore \underline{\cos\phi = \rho\sin^2\phi\cos 2\theta}$

32. **32.** Using spherical coordinates, show that

$$\phi = \cos^{-1}\left(\frac{\mathbf{u}\cdot\mathbf{k}}{\|\mathbf{u}\|}\right),$$

where $\mathbf{u} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Interpret geometrically.

$\vec{u}\cdot\hat{k} = \|\vec{u}\|\|\hat{k}\|\cos\phi$, ϕ = angle between $\vec{u}$, $\hat{k}$

and $\|\hat{k}\| = 1$. $\therefore \vec{u} \cdot \hat{k} = \|\vec{u}\| \cos\phi$

$\therefore \cos\phi = \dfrac{\vec{u} \cdot \hat{k}}{\|\vec{u}\|}$, or $\phi = \cos^{-1}\left(\dfrac{\vec{u} \cdot \hat{k}}{\|\vec{u}\|}\right)$

ϕ is angle between $\vec{u}$ and z-axis.

35.

35. (a) Show that for two $n \times n$ matrices A and B, and $\mathbf{x} \in \mathbb{R}^n$,

$$(AB)\mathbf{x} = A(B\mathbf{x}).$$

(b) What does the equality in part (a) imply about the relationship between the composition of the mappings $\mathbf{x} \mapsto B\mathbf{x}$, $\mathbf{y} \mapsto A\mathbf{y}$, and matrix multiplication?

(a) x is an $n \times 1$ column vector

$$\therefore [(AB)x]_{il} = \sum_{k=1}^{n} (AB)_{ik} x_{kl} \quad \text{by def. of } (AB)x$$

$$= \sum_{k=1}^{n} \left(\sum_{r=1}^{n} A_{ir} B_{rk} \right) x_{kl} \quad \text{by def. of } AB$$

$$= \sum_{k=1}^{n} \left[\sum_{r=1}^{n} (A_{ir} B_{rk} x_{kl}) \right] \quad \text{by distrib. law}$$

$$= \sum_{r=1}^{n} \left[\sum_{k=1}^{n} A_{ir} (B_{rk} x_{kl}) \right] \quad \text{by assoc. law}$$

$$= \sum_{r=1}^{n} A_{ir} \left[\sum_{k=1}^{n} B_{rk} x_{kl} \right] \quad \text{by distrib. law}$$

$$= \sum_{r=1}^{n} A_{ir}(Bx)_{r1} \qquad \text{by def. of } Bx$$

$$= A(Bx)_{i1} \qquad \text{by def of } A(Bx)$$

i.e., for each i, $1 \le i \le n$, $[(AB)x]_{i1} = A(Bx)_{i1}$

$\therefore (AB)x = A(Bx)$

(6) Matrix of a composition of mappings equals matrix multiplication of the matrices of each mapping.

37. **37.** (For students with some knowledge of linear algebra.) Verify that a linear mapping T of $\mathbb{R}^n$ to $\mathbb{R}^n$ is determined by an $n \times n$ matrix.

Let $\vec{e}_i$, $i = 1 \ldots n$, be a basis vector for R^n.

Consider $T(\vec{e}_i) \in R^n$. Let $T(\vec{e}_i) = \vec{c}_i$, an $n \times 1$ vector

$\therefore$ Consider an $n \times n$ matrix M in which each column of M is $T(\vec{e}_i)$.

$\therefore M\vec{e}_i = \vec{c}_i$. $\therefore M_{ji} = T(\vec{e}_i)_j$

Now consider $\vec{v} \in \mathbb{R}^n$, an $n \times 1$ vector.

$\therefore \vec{v} = v_1 \vec{e}_1 + \dots v_n \vec{e}_n$

$\therefore T\vec{v} = T(v_1 \vec{e}_1 + \dots + v_n \vec{e}_n)$

$= v_1 T(\vec{e}_1) + \dots + v_n T(\vec{e}_n)$, because T is linear

$= v_1 \vec{c}_1 + \dots + v_n \vec{c}_n = v_1 (M\vec{e}_1) + \dots v_n (M\vec{e}_n)$

$= M(v_1 \vec{e}_1) + \dots + M(v_n \vec{e}_n) = M(v_1 \vec{e}_1 + \dots v_n \vec{e}_n)$

$= M\vec{v}$

$\therefore T\vec{v} = M\vec{v}$, for all $\vec{v} \in \mathbb{R}^n$, where $M_{ji} = T(\vec{e}_i)_j$

38. **38.** Find an equation for the plane that contains $(3, -1, 2)$ and the line with equation $\mathbf{v} = (2, -1, 0) + t(2, 3, 0)$.

Just need two points of line in plane: choose $t = 0, 1$.

$\therefore$ Points of plane: $(3,-1,2)$, $(2,-1,0)$, $(4,2,0)$

$\therefore$ Let $\vec{a} = (3,-1,2) - (2,-1,0) = (1,0,2)$

$\vec{b} = (3,-1,2) - (4,2,0) = (-1,-3,2)$

$\therefore \vec{a} \times \vec{b}$ = normal to plane

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 0 & 2 \\ -1 & -3 & 2 \end{vmatrix} = (6, -4, -3) = \vec{n}$$

Let (x,y,z) be any point in plane.

$\therefore \vec{n} \cdot [(x,y,z) - (3,-1,2)] = 0$, could pick any point in plane. Use $(3,-1,2)$

$(6,-4,-3) \cdot [(x,y,z) - (3,-1,2)] = 0$, or

$6x - 4y - 3z = (18 + 4 + (-6)) = 16$

$\therefore 6x - 4y - 3z - 16 = 0$

39.

39. The work W done in moving an object from (0, 0) to (7, 2) subject to a constant force $\mathbf{F}$ is $W = \mathbf{F} \cdot \mathbf{r}$, where $\mathbf{r}$ is the vector with its head at (7, 2) and tail at (0, 0). The units are feet and pounds.

(a) Suppose the force $\mathbf{F} = 10 \cos\theta\,\mathbf{i} + 10 \sin\theta\,\mathbf{j}$. Find W in terms of θ.

(b) Suppose the force $\mathbf{F}$ has magnitude of 6 lb and makes an angle of $\pi/6$ rad with the horizontal, pointing right. Find W in foot-pounds.

(a) $\vec{F} \cdot \vec{r} = (10\cos\theta, 10\sin\theta) \cdot (7,2)$

$= 70\cos\theta + 20\sin\theta$

(b) $\vec{F} = 6\cos\frac{\pi}{6}\,\hat{i} + 6\sin\frac{\pi}{6}\,\hat{j} = (3\sqrt{3}, 3)$

$\therefore \vec{F} \cdot \vec{r} = (3\sqrt{3}, 3) \cdot (7,2) = 21\sqrt{3} + 6$ ft-lbs.

40.

40. If a particle with mass m moves with velocity $\mathbf{v}$, its *momentum* is $\mathbf{p} = m\mathbf{v}$. In a game of marbles, a marble with mass 2 grams (g) is shot with velocity 2 meters per second (m/s), hits two marbles with mass 1 g each, and comes to a dead halt. One of the marbles flies off with a velocity of 3 m/s at an angle of 45° to the incident direction of the larger marble, as in Figure 1.R.1. Assuming that the total momentum before and after the collision is the same (according to the law of conservation of momentum), at what angle and speed does the second marble move?

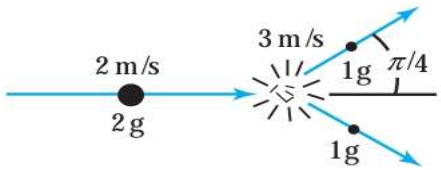


figure **1.R.1** Momentum and marbles.

Momentum of original particle: $(2g)(2\,m/sec)\,\hat{i}$

$= 4\ g\text{-}m/sec\ \hat{i}$

It transfers all its momentum to the two marbles (since it comes to a halt).

One marble has momentum:

$(1g)(3\,m/sec)\cos 45^\circ\,\hat{i} + (1g)(3\,m/sec)\sin 45^\circ\,\hat{j}$

$= (3\sqrt{2}/2\ g\text{-}m/sec,\ 3\sqrt{2}/2\ g\text{-}m/sec)$

$\therefore$ Other marble has momentum:

$(4\ g\text{-}m/sec, 0) - (3\sqrt{2}/2\ g\text{-}m/sec, 3\sqrt{2}/2\ g\text{-}m/sec) =$

$(4 - 3\sqrt{2}/2, -3\sqrt{2}/2)\ g\text{-}m/sec = \vec{p}$

$\therefore$ Speed $\dfrac{\|\vec{p}\|}{m} = \sqrt{(4-3\sqrt{2}/2)^2 + (3\sqrt{2}/2)^2}\ m/sec$

$= 2.83 \text{ m/sec}$

Angle: $\arctan\left(\frac{-3\sqrt{2}/2}{4-\frac{3\sqrt{2}}{2}}\right) = -48.5^\circ$

∴ other marble flies off at 2.83 m/sec

at an angle of -48.5° to the incident direction.

41. **41.** Show that for all x, y, z,

$$\begin{vmatrix} x+2 & y & z \\ z & y+1 & 10 \\ 5 & 5 & 2 \end{vmatrix} = -\begin{vmatrix} y & x+2 & z \\ 1 & z-x-2 & 10-z \\ 5 & 5 & 2 \end{vmatrix}.$$

$$\begin{vmatrix} x+2 & y & z \\ z & y+1 & 10 \\ 5 & 5 & 2 \end{vmatrix} = -\begin{vmatrix} y & x+2 & z \\ y+1 & z & 10 \\ 5 & 5 & 2 \end{vmatrix}$$ swap columns 1 & 2 (∴ change sign)

$$= -\begin{vmatrix} y & x+2 & z \\ 1 & z-x-2 & 10-z \\ 5 & 5 & 2 \end{vmatrix}$$ subtract row 1 from row 2 (no sign change)

42. **42.** Show that

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} \neq 0$$

if x, y, and z are all different.

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \\ 1 & z & z^2 \end{vmatrix} = \begin{vmatrix} 1 & x & x^2 \\ 0 & y-x & y^2-x^2 \\ 0 & z-x & z-x^2 \end{vmatrix}$$ subtracting row 1 from row 2 and row 3

$= (y-x)(z^2-x^2) - (z-x)(y^2-x^2)$

If determinant $=0$, then

$(z-x)(y^2-x^2) = (y-x)(z^2-x^2)$, or, since $x \neq y$, $x \neq z$

$\frac{y^2-x^2}{y-x} = \frac{z^2-x^2}{z-x}$, or

$\frac{(y+x)(y-x)}{y-x} = \frac{(z+x)(z-x)}{z-x}$, or $y+x = z+x$

or $y = z$, contrary to assumption.

$\therefore$ Determinant $\neq 0$.

44. **44.** Show that

$$\begin{vmatrix} n & n+1 & n+2 \\ n+3 & n+4 & n+5 \\ n+6 & n+7 & n+8 \end{vmatrix}$$

has the same value no matter what n is. What is this value?

$\begin{vmatrix} n & n+1 & n+2 \\ n+3 & n+4 & n+5 \\ n+6 & n+7 & n+8 \end{vmatrix}$ = subtract row 1 from row 2 and subtract row 1 from row 3

$\begin{vmatrix} n & n+1 & n+2 \\ 3 & 3 & 3 \\ 6 & 6 & 6 \end{vmatrix}$ = subtracting 2x row 2 from row 3

$$\begin{vmatrix} n & n+1 & n+2 \\ 3 & 3 & 3 \\ 0 & 0 & 0 \end{vmatrix} = 0$$ as all of row 3 is zero.

46. **46.** Find a 4×4 matrix C such that for every 4×4 matrix A we have $CA = 3A$.

Since $IA = A$, and $3IA = 3A$,

Then choose $C = 3I$.

48. **48.** Suppose $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is invertible and has integer entries.

What conditions must be satisfied for $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1}$ to have integer entries?

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The condition is that the determinant, $ad-bc$ must divide evenly each entry

Use the following definition for Exercises 50 and 51: Let $\mathbf{r}_1, \ldots, \mathbf{r}_n$ be vectors in $\mathbb{R}^3$ from $\mathbf{0}$ to the masses $m_1, \ldots, m_n$. The ***center of mass*** *is the vector*

$$\mathbf{c} = \frac{\sum_{i=1}^{n} m_i \mathbf{r}_i}{\sum_{i=1}^{n} m_i}.$$

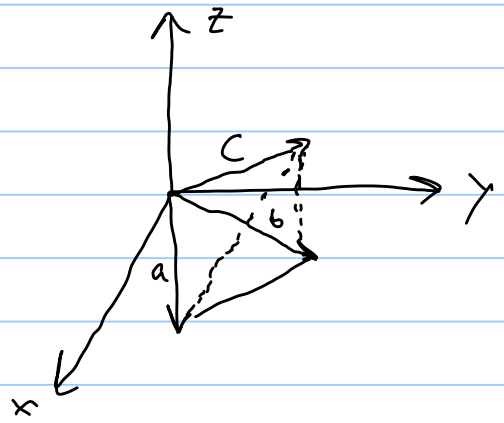
50.

50. A tetrahedron sits in xyz coordinates with one vertex at $(0, 0, 0)$, and the three edges concurrent at $(0, 0, 0)$ are coincident with the vectors **a, b, c**.

(a) Draw a figure and label the heads of the vectors **a**, **b**, **c**.

(b) Find the center of mass of each of the four triangular faces of the tetrahedron if a unit mass is placed at each vertex.

(a)



(b) For the base $\vec{a}+\vec{b}$: $\dfrac{\vec{0}+\vec{a}+\vec{b}}{3} = \dfrac{\vec{a}+\vec{b}}{3}$

Similarly, $\dfrac{\vec{a}+\vec{c}}{3}$, $\dfrac{\vec{b}+\vec{c}}{3}$

For the last face, $\dfrac{\vec{a}+\vec{b}+\vec{c}}{3}$

51.

51. Show that for any vector $\mathbf{r}$, the center of mass of a system satisfies

$$\sum_{i=1}^{n} m_i \|\mathbf{r}-\mathbf{r}_i\|^2 = \sum_{i=1}^{n} m_i \|\mathbf{r}_i-\mathbf{c}\|^2 + m\|\mathbf{r}-\mathbf{c}\|^2,$$

where $m = \sum_{i=1}^{n} m_i$ is the total mass of the system.

Note:

$$\vec{c} = \frac{\sum_{i=1}^{n} m_i \vec{r}_i}{\sum_{i=1}^{n} m_i} = \frac{\sum_{i=1}^{n} m_i \vec{r}_i}{m}$$

$$\therefore m\vec{c} = \sum_{i=1}^{n} m_i \vec{r}_i$$

$$\therefore \sum_{i=1}^{n} m_i \|\vec{r}_i - \vec{c}\|^2 + m\|\vec{r} - \vec{c}\|^2 = \sum_{i=1}^{n} m_i (\vec{r}_i - \vec{c}) \cdot (\vec{r}_i - \vec{c}) + m(\vec{r} - \vec{c}) \cdot (\vec{r} - \vec{c})$$

$$= \sum_{i=1}^{n} m_i (\vec{r}_i \cdot \vec{r}_i - 2\vec{r}_i \cdot \vec{c} + \vec{c} \cdot \vec{c}) + m(\vec{r} - \vec{c}) \cdot (\vec{r} - \vec{c})$$

$$= \sum_{i=1}^{n} m_i \vec{r}_i \cdot \vec{r}_i - 2\sum_{i=1}^{n} m_i \vec{r}_i \cdot \vec{c} + \sum_{i=1}^{n} m_i \vec{c} \cdot \vec{c} + m(\vec{r} \cdot \vec{r} - 2\vec{r} \cdot \vec{c} + \vec{c} \cdot \vec{c})$$

$$= \sum_{i=1}^{n} m_i \vec{r}_i \cdot \vec{r}_i - 2c \cdot \sum_{i=1}^{n} m_i \vec{r}_i + \vec{c} \cdot \vec{c} \sum_{i=1}^{n} m_i + m\vec{r} \cdot \vec{r} - 2m\vec{r} \cdot \vec{c} + m\vec{c} \cdot \vec{c}$$

$$= \sum_{i=1}^{n} m_i \vec{r}_i \cdot \vec{r}_i - 2\vec{c} \cdot (m\vec{c}) + m\vec{c} \cdot \vec{c} + m\vec{r} \cdot \vec{r} - 2m\vec{r} \cdot \vec{c} + m\vec{c} \cdot \vec{c}$$

$$= \sum_{i=1}^{n} m_i \vec{r}_i \cdot \vec{r}_i + m\vec{r} \cdot \vec{r} - 2m\vec{r} \cdot \vec{c}$$

$$= \sum_{i=1}^{n} m_i \vec{r}_i \cdot \vec{r}_i + m\vec{r} \cdot \vec{r} - 2\vec{r} \cdot \left(\sum_{i=1}^{n} m_i \vec{r}_i\right)$$

$$= \sum_{i=1}^{n} m_i \vec{r}_i \cdot \vec{r}_i + \vec{r} \cdot \vec{r} \sum_{i=1}^{n} m_i - 2\vec{r} \cdot \left(\sum_{i=1}^{n} m_i \vec{r}_i\right)$$

$$= \sum_{i=1}^{n} m_i \vec{r} \cdot \vec{r} - \sum_{i=1}^{n} 2 m_i \vec{r} \cdot \vec{r}_i + \sum_{i=1}^{n} m_i \vec{r}_i \cdot \vec{r}_i$$

$$= \sum_{i=1}^{n} \left(m_i \vec{r} \cdot \vec{r} - 2m_i \vec{r} \cdot \vec{r}_i + m_i \vec{r}_i \cdot \vec{r}_i\right)$$

$$= \sum_{i=1}^{n} m_i \left(\vec{r} \cdot \vec{r} - 2\vec{r} \cdot \vec{r}_i + \vec{r}_i \cdot \vec{r}_i\right)$$

$$= \sum_{i=1}^{n} m_i (\vec{r} - \vec{r}_i) \cdot (\vec{r} - \vec{r}_i) = \sum_{i=1}^{n} m_i \|\vec{r} - \vec{r}_i\|^2$$

In Exercises 52 to 57, find a unit vector that has the given property.

52. **52.** Parallel to the line $x = 3t + 1$, $y = 16t - 2$, $z = -(t + 2)$

$(x, y, z) = (1, -2, -2) + t(3, 16, -1)$

$\therefore$ parallel to $(3, 16, -1)$. $\sqrt{3^2 + 16^2 + (-1)^2} = \sqrt{266}$

$266 = 2 \times 7 \times 19$

$\therefore \frac{1}{\sqrt{266}}(3, 16, -1)$

53. **53.** Orthogonal to the plane $x - 6y + z = 12$ $(1, -6, 1) \cdot (x, y, z) = 12$

Normal to plane: $(1, -6, 1)$ $\sqrt{1^2 + (-6)^2 + 1^2} = \sqrt{38}$

$\therefore \frac{1}{\sqrt{38}}(1, -6, 1)$

54. **54.** Parallel to both the planes $8x + y + z = 1$ and $x - y - z = 0$

The normals to each plane: $(8, 1, 1)$ and $(1, -1, -1)$.

$\therefore$ a vector perpendicular to each normal will be parallel to each plane.

$\therefore (8, 1, 1) \times (1, -1, -1) = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 8 & 1 & 1 \\ 1 & -1 & -1 \end{vmatrix} = (0, 9, -9)$

or, $(0,1,-1)$. $\therefore$ Unit vector: $\frac{1}{\sqrt{2}}(0,1,-1)$

55. **55.** Orthogonal to $\mathbf{i}+2\mathbf{j}-\mathbf{k}$ and to $\mathbf{k}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 1 \\ 1 & 2 & -1 \end{vmatrix} = (-2,1,0) \quad \sqrt{(-2)^2+1^2+0^2} = \sqrt{5}$$

$\therefore \frac{1}{\sqrt{5}}(-2,1,0)$

56. **56.** Orthogonal to the line $x = 2t-1,\ y = -t-1,\ z = t+2$, and the vector $\mathbf{i}-\mathbf{j}$

line: $(x,y,z) = (-1,-1,2) + t(2,-1,1)$

$\therefore$ orthogonal to $(2,-1,1)$ and $(1,-1,0)$

$$\therefore \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -1 & 1 \\ 1 & -1 & 0 \end{vmatrix} = (1,1,-1) \cdot \sqrt{1^2+1^2+(-1)^2} = \sqrt{3}$$

$\therefore \frac{1}{\sqrt{3}}(1,1,-1)$

57. **57.** At an angle of 30° to $\mathbf{i}$ and making equal angles with $\mathbf{j}$ and $\mathbf{k}$

Let $\vec{v}$ be the vector $= (v_1, v_2, v_3)$

$\therefore \vec{v} \cdot \hat{i} = \|\vec{v}\| \|\hat{i}\| \cos 30^\circ$, or $v_1 = \sqrt{v_1^2 + v_2^2 + v_3^2}\left(\frac{\sqrt{3}}{2}\right)$

Or, $\frac{4}{3} v_1^2 = v_1^2 + v_2^2 + v_3^2$,

$\therefore v_1^2 = 3v_2^2 + 3v_3^2$

Equal angles w/ $\hat{j}$ and $\hat{k}$: $\frac{\vec{v} \cdot \hat{j}}{\|v\| \|\hat{j}\|} = \frac{\vec{v} \cdot \hat{k}}{\|\vec{v}\| \|\hat{k}\|} = \cos\theta$

$\therefore \frac{v_2}{\|v\|} = \frac{v_3}{\|v\|}$, or $v_2 = v_3$

$\therefore v_1^2 = 3v_2^2 + 3v_3^2 = 6v_2^2$

$\therefore v_1 = \pm\sqrt{6}\, v_2$. Choose $+\sqrt{6}\, v_2$ since

$\vec{v_1} \cdot \hat{i} = \|v\| \frac{\sqrt{3}}{2}$

Let $v_2 = v_3 = 1$. $\therefore (\sqrt{6}, 1, 1)$

Unit vector: $\frac{1}{\sqrt{8}}(\sqrt{6}, 1, 1) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2\sqrt{2}}, \frac{1}{2\sqrt{2}}\right)$

2.1 The Geometry of Real-Valued Functions

Note Title 1/4/2016

2.

2. Are the following functions vector-valued or scalar-valued?

(a) $f(u, v, w) = (u^2 v,\ we^u,\ 5v)$

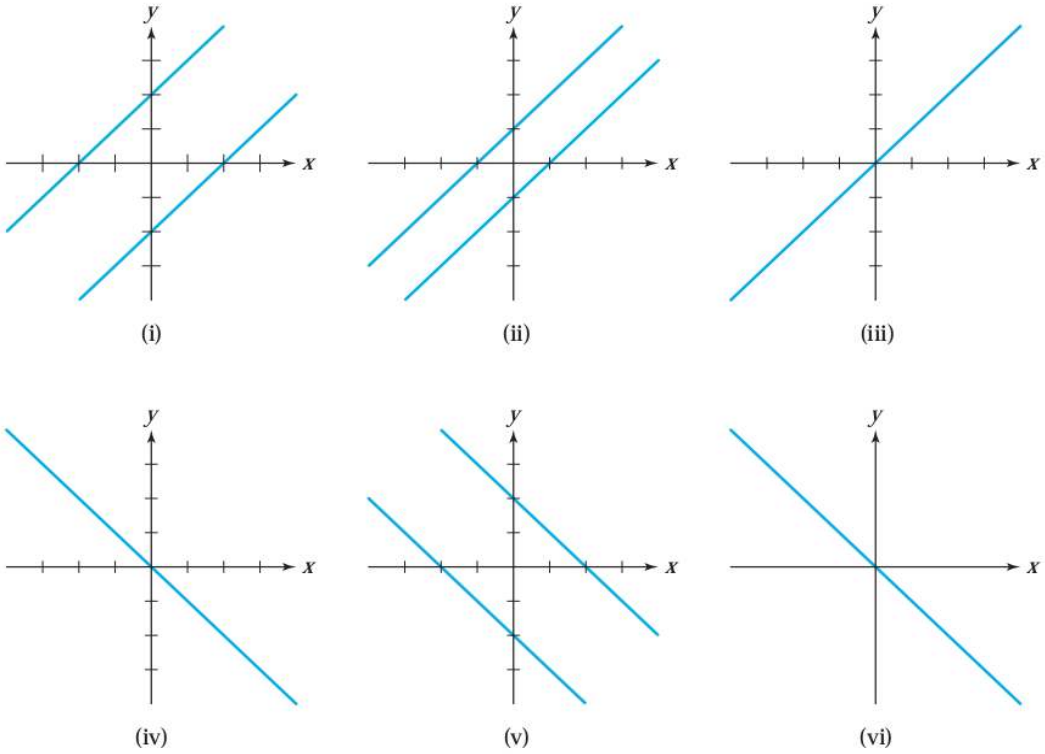
(b) $g(x) = \log \sqrt{x}$

(c) $h(x, y) = x^5 y^{-3}$

(a) vector-valued
(b) scalar-valued
(c) scalar-valued

4.

4. (a) $f(x, y) = (x - y)^2 = c, \quad c = 0, 1, 4$ (b) $f(x, y) = (x + y)^2 = c, \quad c = 0, 1, 4$



(a) $(x-y)^2 = c \iff x - y = \pm\sqrt{c}$, or $y = x \pm \sqrt{c}$

$\therefore$ $c = 0$: (iii) $y = 0$

$c = 1$: (ii) $y = x + 1,\ y = x - 1$

$c = 4$: (i) $y = x + 2,\ y = x - 2$

(6) $(x+y)^2 = C \iff x+y = \pm\sqrt{C}$, or $y = -x \pm \sqrt{C}$

$\therefore$ $C=0$: $y=-x$ (vi) or (iv)

$C=1$: $y=-x+1$, $y=-x-1$ (v) assuming each tick is $\frac{1}{2}$

$C=4$: $y=-x+2$, $y=-x-2$ (v)

6. **6.** Let $f(x, y) = 9x^2 + y^2$. Sketch the following.

(a) The level curves for f of values $c = 0, 1, 9$

(b) The sections of the graph of f in the planes $x = -1, x = 0, x = 1$

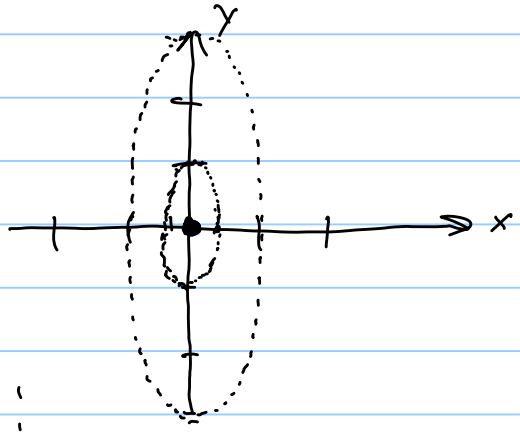
(c) The sections of the graph of f in the planes $y = -1, y = 0, y = 1$

(d) The graph of f

(a) $0 = 9x^2 + y^2 \Rightarrow x, y = 0$

$9x^2 + y^2 = 1 \iff \frac{x^2}{(\frac{1}{3})^2} + y^2 = 1$

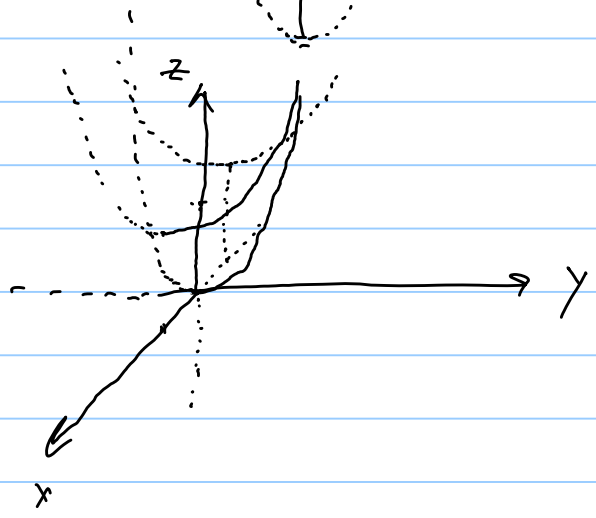
$9x^2 + y^2 = 9 \iff x^2 + \frac{y^2}{3^2} = 1$



(b) $x=-1$: $z = 9 + y^2$

$x=0$: $z = y^2$

$x=1$: $z = 9 + y^2$

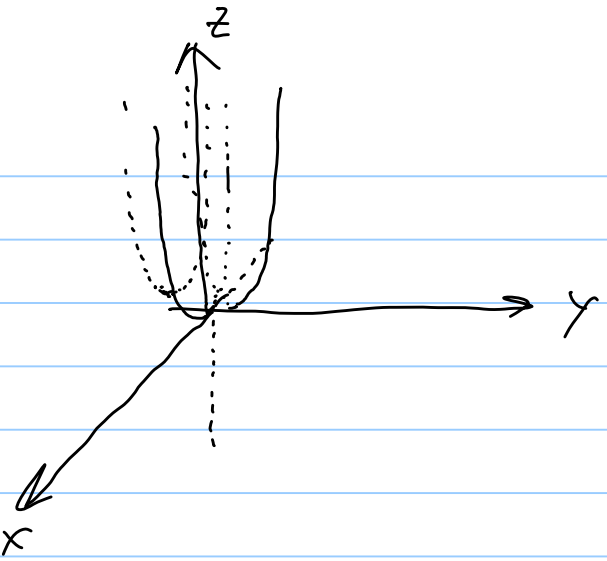


parabolas, which dip down to origin when $x=0$

(c) $y=-1$: $z=9x^2+1$

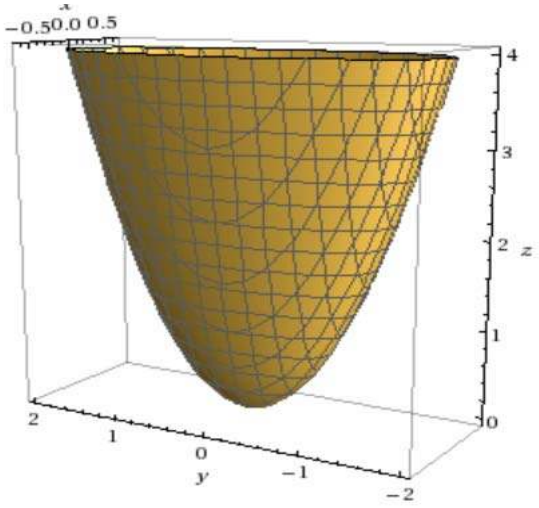
$y=0$: $z=9x^2$

$y=1$: $z=9x^2+1$

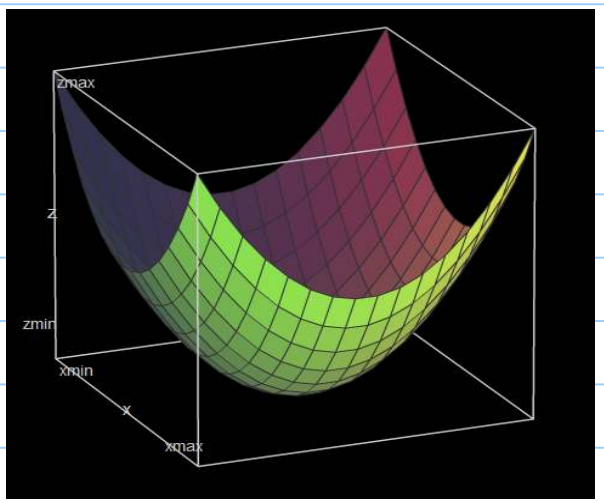


parabolas, which dip to origin when $y=0$.

(d) An elliptic paraboloid



or



7.

7. Sketch the level curves and graphs of the following functions:

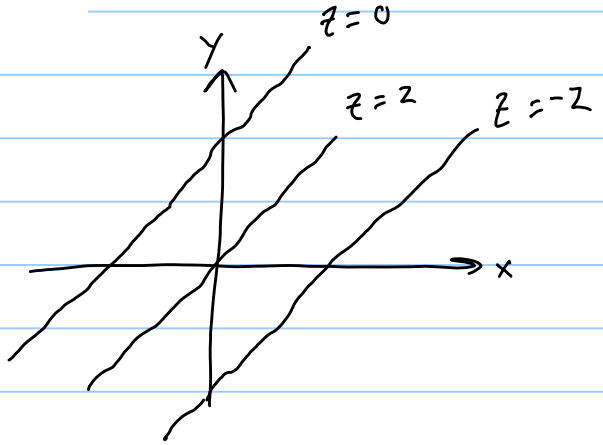
(a) $f: \mathbb{R}^2 \to \mathbb{R},\ (x, y) \mapsto x - y + 2$

(b) $f: \mathbb{R}^2 \to \mathbb{R},\ (x, y) \mapsto x^2 + 4y^2$

(c) $f: \mathbb{R}^2 \to \mathbb{R},\ (x, y) \mapsto -xy$

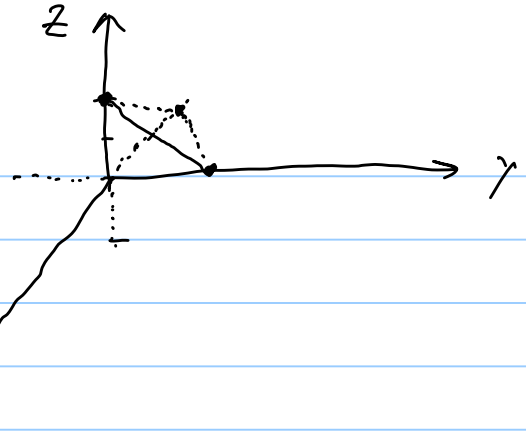
(a) Level curves:

parallel lines

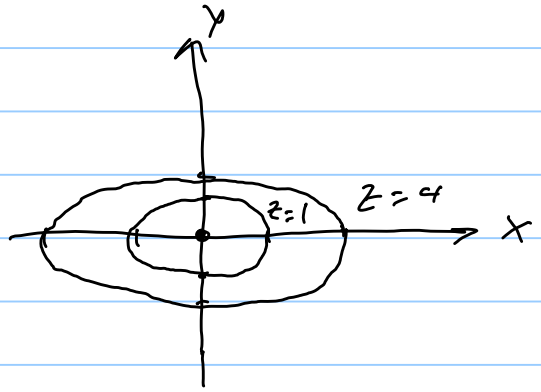


Graph:

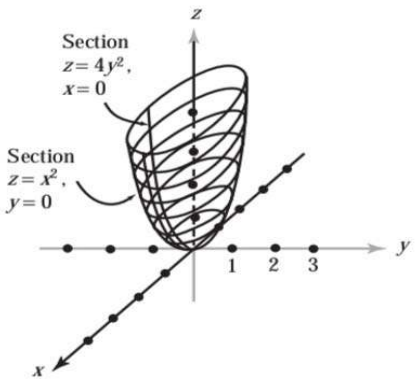
a plane seen behind yz plane.



(b) Level curves:

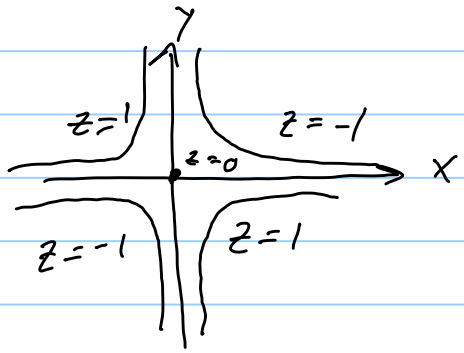


Graph:

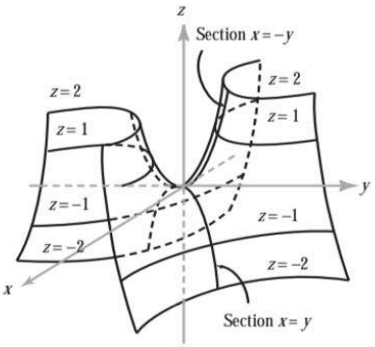


(c) Level curves:

hyperbolas dipping down for $z < 0$ in quadrants I, III
dipping up for quadrants II, IV

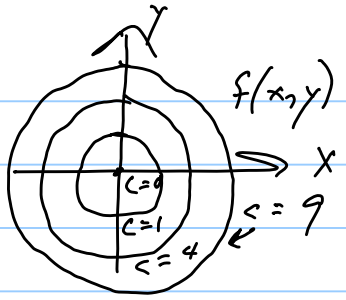


Graph: saddle or hyperbolic paraboloid

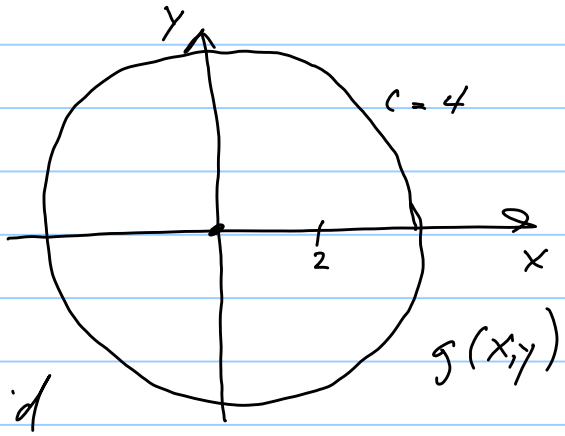


8.

8. Sketch level sets of values $c = 0, 1, 4, 9$ for both $f(x, y) = x^2 + y^2$ and $g(x, y) = \sqrt{x^2 + y^2}$. How are the graphs of f and g different? How are their sections different?

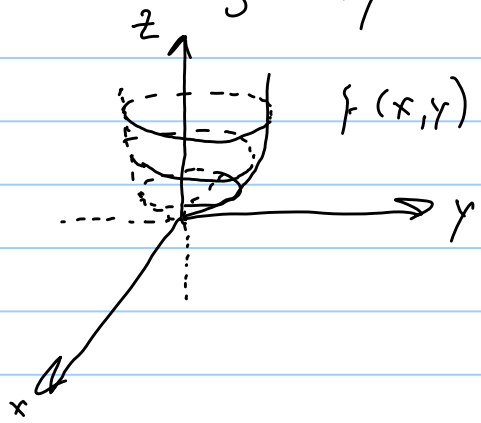


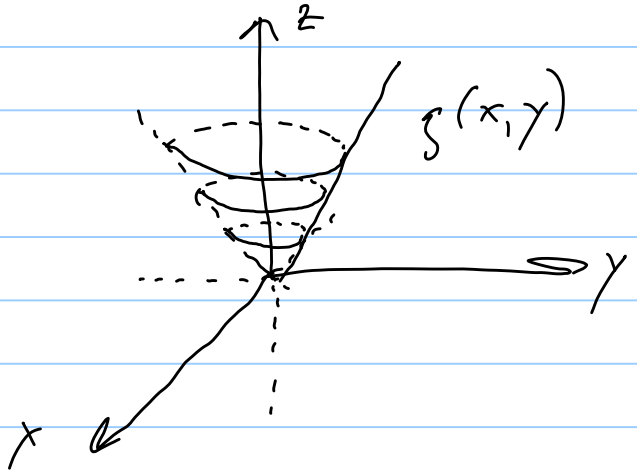
Level sets look similar (circles), but $g(x,y)$ curves are "expanded", or larger for same level values.



Graphs: $f(x,y)$: paraboloid

$g(x,y)$: cone





9.

9. Let S be the surface in $\mathbb{R}^3$ defined by the equation $x^2 y^6 - 2z = 3$.

(a) Find a real-valued function $f(x, y, z)$ of three variables and a constant c such that S is the level set of f of value c.

(b) Find a real-valued function $g(x, y)$ of two variables such that S is the graph of g.

(a) Let $f(x,y,z) = x^2y^6 - 2z$, let $c = 3$

$\therefore S = \{(x,y,z) \mid f(x,y,z) = 3\} \subset \mathbb{R}^3$

(5) $x^2y^6 - 2z = 3 \iff x^2y^6 - 3 = 2z \iff z = \frac{x^2y^6 - 3}{2}$

$\therefore$ let $g(x,y) = \frac{x^2y^6 - 3}{2}$

$\therefore$ graph $g = \{(x, y, g(x,y)) \subset \mathbb{R}^3\} = S$

i.e., graph of g = level set of f at $c = 3$.

Sketch or describe the surfaces in $\mathbb{R}^3$ of the equations presented in Exercises 25 to 37. (27 to 39)

28. **28.** $x + 2z = 4$

y doesn't matter. For $y = 0$, this is a line with intercepts at $(4,0,0)$ and $(0,0,2)$.

$\therefore$ A plane parallel to y-axis intersecting x-axis at $(4,0,0)$ and z-axis at $(0,0,2)$

30. **30.** $x^2 + y^2 - 2x = 0$

Completing the square, $x^2 + y^2 - 2x = 0 \iff (x-1)^2 + y^2 = 1$

When $z = 0$, this is a circle with center at $(1,0)$.

z doesn't matter. $\therefore$ A circular cylinder

perpendicular to xy-plane, centered at $(1,0,z)$.

32. **32.** $\frac{y^2}{9}+\frac{z^2}{4}=1+\frac{x^2}{16}$ $\qquad \frac{y^2}{3^2}+\frac{z^2}{2^2}-\frac{x^2}{4^2}=1$

When $z=0$, $\frac{y^2}{3^2}-\frac{x^2}{4^2}=1$ is an hyperbola intersecting y-axis at ± 3

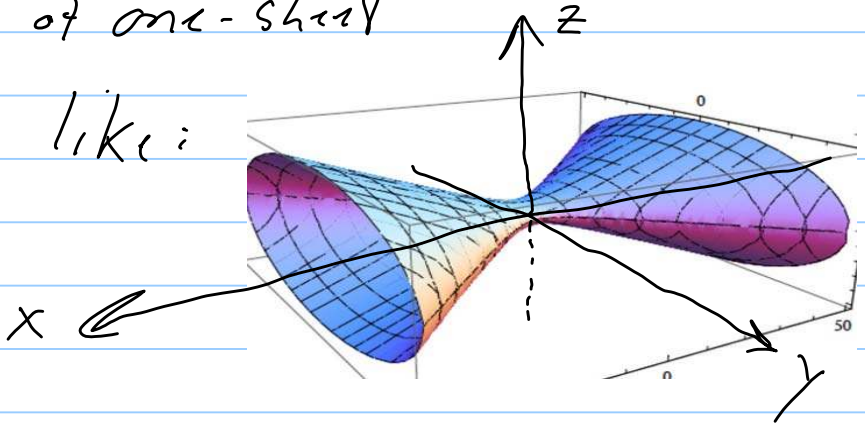
When $y=0$, $\frac{z^2}{2^2}-\frac{x^2}{4^2}=1$ is an hyperbola intersecting z-axis at ± 2

When $x=0$, $\frac{y^2}{3^2}+\frac{z^2}{2^2}=1$, an ellipse in yz-plane intercepts at $(0,\pm 3,0)$ and $(0,0,\pm 2)$

As $\pm x$ gets bigger, the ellipse proportionately expands, like a cylinder parallel to x-axis.

∴ Hyperboloid of one-sheet

Something like:



34. **34.** $y^2 + z^2 = 4$

x doesn't matter. For any plane $x = c$, $y^2 + z^2 = 4$ is a circle of radius 2.

∴ A circular cylinder parallel to x-axis.

36. **36.** $y^2 = x^2 + z^2$

For any $y = \pm c$, $x^2 + z^2 = c^2$ is a circle of radius c.

The origin $(0,0,0)$ is on the graph.

∴ A circular cone, parallel to y axis, centered at origin

38. **38.** $\frac{x^2}{9} + \frac{y^2}{12} + \frac{z^2}{9} = 1$

for $x = 0$, or $y = 0$, or $z = 0$, the section is an ellipse.

∴ An ellipsoid with intercepts at

$(\pm 3, 0, 0)$, $(0, \pm\sqrt{12}, 0)$, $(0, 0, \pm 3)$

40. **40.** Using polar coordinates, describe the level curves of the function defined by

$f(x, y) = 2xy/(x^2 + y^2)$ if $(x, y) \neq (0, 0)$ and $f(0, 0) = 0$.

$x = r\cos\theta$, $y = r\cos\theta$. ∴ $x^2 + y^2 = r^2$

$2xy = 2r^2\cos\theta\sin\theta$

$\therefore \dfrac{2xy}{x^2+y^2} \Leftrightarrow \dfrac{2r^2\cos\theta\sin\theta}{r^2} = 2\cos\theta\sin\theta$

$= \sin 2\theta$

$\therefore z = f(x,y) = g(r,\theta) = \sin 2\theta$

r does not matter.

$\therefore$ Visualize a ray ($r \geq 0$) sweeping out a surface as θ increases from 0 to 2π.

z increases from $\theta = 0$ to $\frac{\pi}{4}$, then decreases to 0 from $\frac{\pi}{4}$ to $\frac{\pi}{2}$, then becomes negative from $\frac{\pi}{2}$ to $\frac{3}{4}\pi$, etc.

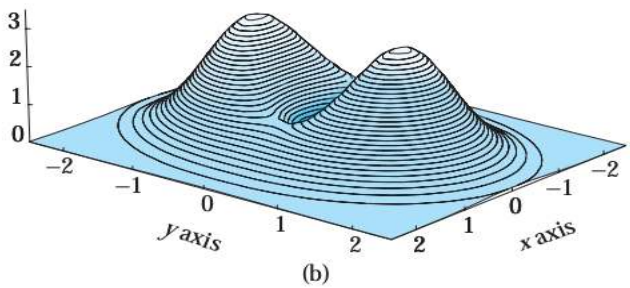
$\therefore$ An undulating surface centered at (0,0,0)

42

42. Show that in Figure 2.1.15, the level "curve" $z = 3$ consists of two points.

$z = (x^2 + 3y^2)\exp(1 - x^2 - y^2)$

Note that for any z, the graph is symmetric to the x-axis as $\pm y$ values leave z unchanged, and



(b)

also symmetric to y-axis as $\pm x$ values leave z unchanged.

Consider $x=0$. $\therefore z = 3y^2 e^{1-y^2}$

If $y=1$, $z=3$.

$$\frac{dz}{dy} = 6y e^{1-y^2} + 3y^2 e^{1-y^2}(-2y) = 6ye^{1-y^2} - 6y^3 e^{1-y^2}$$

and $\therefore \left.\frac{dz}{dy}\right|_{y=1} = 6-6 = 0$

$$\frac{d^2z}{dy^2} = 6e^{1-y^2} + 6ye^{1-y^2}(-2y) - 18y^2 e^{1-y^2} - 6y^3 e^{1-y^2}(-2y)$$

$$\therefore \left.\frac{d^2z}{dy^2}\right|_{y=1} = 6 - 12 - 18 + 12 = -12 < 0.$$

$\therefore y=1$ a local max in $x=0$ plane

Consider $y=1$ plane. $\therefore z = (x^2+3)e^{-x^2} = \frac{x^2+3}{e^{x^2}}$

At $x=0$, $z=3$. Since e^{x^2} increases faster than x^2 as x moves away from 0, $x=0$ is a local max in $y=1$ plane.

Now consider circles of constant value in xy-plane, $x^2+y^2 = k$.

$$\therefore z = \frac{k+2y^2}{e^{k-1}}, \text{ or } z = k' + k''y^2$$

∴ z depends on y^2, like a parabola, and increases the same for $\pm y$.

Since $e^{x^2+y^2}$ increases faster than x^2+3y^2, z must get smaller on these circles.

As shown above, for $x=0$, $z=3$ was a local max, and so on the circle with $(0, \pm 1)$, z is a local max at $z=3$.

∴ Just two points at $z=3$

2.2 Limits and Continuity

Note Title 1/14/2016

1. **1.** Let $f: \mathbb{R}^2 \to \mathbb{R}$ and suppose that $\lim_{(x,y)\to(1,3)} f(x, y) = 5$. What can you say about the value $f(1, 3)$?

Since $f(x,y)$ is not defined, you don't know if $(1,3)$ is even in domain of f. $\therefore$ Can't say anything about $f(1,3)$.

2. **2.** Let $f: \mathbb{R}^2 \to \mathbb{R}$ is continuous and suppose that $\lim_{(x,y)\to(1,3)} f(x, y) = 5$. What can you say about the value $f(1, 3)$?

Since f is continuous, then the limit at (x,y) = the value at (x,y). $\therefore$ $f(1,3)$ is defined, and $\therefore$ $f(1,3) = 5$.

3. **3.** Compute the limits:

(a) $\lim_{(x,y)\to(0,1)} x^3 y$

(b) $\lim_{x\to 0} \dfrac{\cos x - 1}{x^2}$

(c) $\lim_{h\to 0} \dfrac{e^h - 1}{h}$

(a) x^3y is continuous, so $\lim_{(x,y)\to(0,1)} x^3y = 0^3 \cdot 1 = \underline{0}$

(b) Using L'Hopital's Rule, $\lim_{x\to 0} \dfrac{\cos x - 1}{x^2} = \lim_{x\to 0} \dfrac{-\sin x}{2x} =$

$\lim_{x\to 0} \dfrac{-\cos x}{2} = \underline{-\dfrac{1}{2}}$

(c) Using L'Hopital's Rule, $\lim_{h\to 0} \dfrac{e^h - 1}{h} = \lim_{h\to 0} \dfrac{e^h}{1} = e^0 = \underline{1}$

4.

4. Compute the following limits:

(a) $\lim_{(x,y)\to(0,1)} e^x y$

(b) $\lim_{x\to 0} \frac{\sin^2 x}{x}$

(c) $\lim_{x\to 0} \frac{\sin^2 x}{x^2}$

(a) $\left[\lim_{(x,y)\to(0,1)} e^x\right]\left[\lim_{(x,y)\to(0,1)} y\right] = [1][1] = \underline{1}$

(b) $\left[\lim_{x\to 0} \frac{\sin x}{x}\right]\left[\lim_{x\to 0} \sin x\right] = [1][0] = \underline{0}$

(c) $\left[\lim_{x\to 0} \frac{\sin x}{x}\right]^2 = 1^2 = \underline{1}$

5.

5. Compute the following limits:

(a) $\lim_{x\to 3}(x^2 - 3x + 5)$

(b) $\lim_{x\to 0} \sin x$

(c) $\lim_{h\to 0} \frac{(x+h)^2 - x^2}{h}$

(a) $\lim_{x\to 3}(x^2 - 3x + 5) = 3^2 - 3(3) + 5 = \underline{5}$

(b) $\lim_{x\to 0} \sin x = \sin 0 = \underline{0}$

(c) $\lim_{h\to 0} (x+h)^2 - x^2 = 0$, $\lim_{h\to 0} h = 0$, $\therefore$ use L'Hopital's

$D_h\left[(x+h)^2 - x^2\right] = 2(x+h) \qquad D_h h = 1.$

$\therefore \lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h\to 0} \frac{2(x+h)}{1} = \underline{2x}$

6.

6. Let

$$f(x, y) = \begin{cases} \frac{xy^3}{x^2+y^6} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

(a) Compute the limit as $(x, y) \to (0, 0)$ of f along the path $x = 0$.

(b) Compute the limit as $(x, y) \to (0, 0)$ of f along the path $x = y^3$.

(c) Show that f is not continuous at $(0, 0)$.

(a) When $x=0$, $f(x,y) = \frac{0}{y^6} = 0$

$\therefore \lim_{(x,y)\to(0,0)} 0 = \underline{0}$

(b) When $x=y^3$, $f(x,y) = \frac{y^3 y^3}{y^6+y^6} = \frac{1}{2}$

$\therefore \lim_{(x,y)\to(0,0)} f(x,y) = \underline{\frac{1}{2}}$

(c) if $\lim_{(x,y)\to(0,0)} f(x,y) = f(0,0) = 0$, then limit along any path toward $(0,0)$ should be same.

(a) & (b) show this not to be so, so

$f(x,y)$ not continuous at $(0,0)$.

i.e., there is some $\epsilon > 0$ (take $\epsilon = \frac{1}{4}$) s.t. for every $\delta > 0$ there is some x with $0 < |x-0| < \delta$ but not $|f(0,0) - 0| < \epsilon$

So, from (6), for every $\delta > 0$, with $x = y^3$, $(x,y) \neq (0,0)$, $f(x,y) - 0 = \frac{1}{2}$, so

that $|f(x,y) - 0| = \frac{1}{2} < \frac{1}{4} = \epsilon$ is not true.

7.

7. Let $f(x, y, z) = \dfrac{e^{x+y}}{1+z^2}$. Compute $\lim_{h\to 0} \dfrac{f(1,2+h,3) - f(1,2,3)}{h}$.

$$f(1,2,3) = \frac{e^3}{10}$$

$$f(1,2+h,3) = \frac{e^{3+h}}{10}$$

$$\therefore \lim_{h\to 0} \frac{\frac{e^{3+h}}{10} - \frac{e^3}{10}}{h} = \frac{1}{10} \lim_{h\to 0} \frac{e^{3+h} - e^3}{h}$$

$$= \frac{1}{10} \lim_{h\to 0} \frac{D_h(e^{3+h} - e^3)}{D_h(h)} = \frac{1}{10} \lim_{h\to 0} e^{3+h} = \frac{e^3}{10}$$

8.

8. Compute the following limits if they exist:

(a) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{(x+y)^2 - (x-y)^2}{xy}$

(b) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{\sin xy}{y}$

(c) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^3 - y^3}{x^2 + y^2}$

(a) $$\lim_{(x,y)\to(0,0)} \frac{(x+y)^2 - (x-y)^2}{xy} = \lim_{(x,y)\to(0,0)} \frac{(x^2+2xy+y^2) - (x^2-2xy+y^2)}{xy}$$

$$= \lim_{(x,y)\to(0,0)} \frac{4xy}{xy} = 4$$

(b) For any x, $\displaystyle\lim_{y\to 0} \frac{\sin xy}{y} = \lim_{y\to 0} \frac{x\cos xy}{1} = x$

$\therefore$ For small $|x|$, it seems $\displaystyle\lim_{(x,y)\to(0,0)} \frac{\sin xy}{y} = 0$

Note also, $\sin(xy) = xy - \frac{(xy)^3}{3!} + \frac{(xy)^5}{5!} - \frac{(xy)^7}{7!} + \dots$

$\therefore \frac{\sin(xy)}{y} = x - \frac{x^3y^2}{3!} + \dots$, so $\lim_{(x,y)\to(0,0)} = 0$

$\therefore$ Let $\epsilon > 0$

Since $\lim_{x\to 0} x = 0$, choose $\delta_1 = \epsilon$

$\therefore$ if $|x| < \delta_1$ then $|x| < \epsilon$

Note $x - \sin x \geq 0$ for $0 \leq x < \frac{\pi}{2}$

since $\frac{d}{dx}(x - \sin x) = 1 - \cos x > 0$

since $\cos x < 1$ for $x < \frac{\pi}{2}$

$\therefore |\sin xy| \leq |xy|$ for $|xy| < \frac{\pi}{2}$

$\therefore$ For the above ϵ, choose $\delta = \min\{\delta_1, \frac{1}{2}\}$

$\therefore$ if $0 < \sqrt{x^2+y^2} < \delta$, and $y \neq 0$, then

$|x| < \sqrt{x^2+y^2} < \frac{1}{2}$, and $|y| < \frac{1}{2}$

$\therefore |xy| < \frac{\pi}{2}$

$\therefore |\sin xy| < |xy|$

$\therefore \left|\frac{\sin xy}{y}\right| < \frac{|xy|}{|y|} = |x| < \delta_1 = \epsilon$

$\therefore \left|\frac{\sin xy}{y}\right| < \epsilon$

$$\therefore \lim_{(x,y)\to(0,0)} \frac{\sin xy}{y} = 0$$

(c) First prove $\lim_{(x,y)\to(0,0)} \frac{x^3}{x^2+y^2} = 0$.

Let $\epsilon > 0$. Choose $\delta = \epsilon$

$\therefore$ if $0 < |(x,y)-(0,0)| = \sqrt{x^2+y^2} < \delta = \epsilon$,

then $|x| < \sqrt{x^2+y^2}$, so $|x| < \epsilon$

$\therefore |x^3| < \epsilon x^2 < \epsilon(x^2+y^2)$

$\therefore \left|\frac{x^3}{x^2+y^2}\right| < \epsilon$, or $\left|\frac{x^3}{x^2+y^2} - 0\right| < \epsilon$

$\therefore \lim_{(x,y)\to(0,0)} \frac{x^3}{x^2+y^2} = 0$ Similarly, $\lim_{(x,y)\to(0,0)} \frac{y^3}{x^2+y^2} = 0$

$\therefore \lim_{(x,y)\to(0,0)} \frac{x^3-y^3}{x^2+y^2} = 0$

9.

9. Compute the following limits if they exist:

(a) $\lim_{(x,y)\to(0,0)} \frac{e^{xy}-1}{y}$

(b) $\lim_{(x,y)\to(0,0)} \frac{\cos(xy)-1}{x^2y^2}$

(c) $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2+2}$

(a) Since $\lim_{y\to 0} \frac{e^{xy}-1}{y} = \lim_{y\to 0} \frac{xe^{xy}}{1} = x$, for all x

using L'Hopital's Rule.

$\therefore$ Let $\epsilon > 0$. $\exists \delta'$ s.t. if $0 < |y| < \delta'$, then

$$\left| \frac{e^{xy}-1}{y} - x \right| < \frac{\epsilon}{2}, \text{ or } \left| \frac{e^{xy}-1}{y} \right| < \frac{\epsilon}{2} + |x|, \text{ all } x$$

since $|a|-|b| \le |a-b|$, by triangle inequality

$\therefore$ Choose $\delta = \min\left\{ \frac{\epsilon}{2}, \delta' \right\}$

$\therefore$ if $0 < |(x,y)-(0,0)| = \sqrt{x^2+y^2} < \delta$, and $y \neq 0$,

then $0 < |y| < \delta < \delta'$ and $0 \le |x| < \delta \le \frac{\epsilon}{2}$

$$\therefore \left| \frac{e^{xy}-1}{y} - x \right| < \frac{\epsilon}{2}, \text{ and}$$

$$\therefore \left| \frac{e^{xy}-1}{y} - 0 \right| < \frac{\epsilon}{2} + |x| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$$

$$\therefore \lim_{(x,y)\to(0,0)} \frac{e^{xy}-1}{y} = 0$$

(b) Since $\lim_{x\to 0} \frac{\cos x - 1}{x^2} = \lim_{x\to 0} \frac{-\sin x}{2x} = \lim_{x\to 0} \frac{-\cos x}{2} = -\frac{1}{2}$

by L'Hopital's Rule, then

Given any $\epsilon > 0$, $\exists \delta'$ s.t. if $0 < |x| < \delta'$

Then $\left|\frac{\cos x - 1}{x^2} + \frac{1}{2}\right| < \epsilon$

$\therefore$ Choose $\delta = \min\{\delta', 0.9\}$. $\therefore \delta < 1$.

$\therefore$ if $0 < |(x,y)-(0,0)| < \delta$, and $(x,y) \neq (0,0)$, then

$0 < \sqrt{x^2+y^2} < \delta \Rightarrow 0 < |x| < \delta, 0 < |y| < \delta$

and $\therefore 0 < |xy| < \delta^2 < \delta \leq \delta'$

$\therefore \left|\frac{\cos(xy) - 1}{(xy)^2} - \left(-\frac{1}{2}\right)\right| < \epsilon$

$\therefore \lim_{(x,y)\to(0,0)} \frac{\cos(xy)-1}{x^2y^2} = -\frac{1}{2}$

(c) $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2+2} = \frac{\lim_{(x,y)\to(0,0)} xy}{\lim_{(x,y)\to(0,0)} (x^2+y^2+2)} = \frac{0}{2} = 0$

10.

10. Compute the following limits if they exist:

(a) $\lim_{(x,y)\to(0,0)} \frac{e^{xy}}{x+1}$

(b) $\lim_{(x,y)\to(0,0)} \frac{\cos x - 1 - (x^2/2)}{x^4+y^4}$

(c) $\lim_{(x,y)\to(0,0)} \frac{(x-y)^2}{x^2+y^2}$

(a) $\lim_{(x,y)\to(0,0)} \frac{e^{xy}}{x+1} = \frac{\lim_{(x,y)\to(0,0)} e^{xy}}{\lim_{(x,y)\to(0,0)} x+1} = \frac{e^0}{0+1} = 1$

(5) With $y=0$, $\lim_{x\to 0} \frac{\cos x - 1 - \frac{x^2}{2}}{x^4} = \lim_{x\to 0} \frac{-\sin x - x}{4x^3} =$

$\lim_{x\to 0} \frac{-\cos x - 1}{12x^2} = -\infty$.

$\therefore$ Limit doesn't exist

(c) With $y=0$, $\lim_{(x,y)\to(0,0)} \frac{(x-y)^2}{x^2+y^2} = \lim_{x\to 0} \frac{x^2}{x^2} = 1$

With $y=x$, $\lim_{(x,y)\to(0,0)} \frac{(x-y)^2}{x^2+y^2} = \lim_{x\to 0} \frac{(x-x)^2}{x^2+x^2} = 0$

$\therefore$ Limit doesn't exist

11.

11. Compute the following limits if they exist:

(a) $\lim_{(x,y)\to(0,0)} \frac{\sin xy}{xy}$

(b) $\lim_{(x,y,z)\to(0,0,0)} \frac{\sin(xyz)}{xyz}$

(c) $\lim_{(x,y,z)\to(0,0,0)} f(x,y,z)$, where $f(x,y,z) = (x^2+3y^2)/(x+1)$

(a) $\lim_{(x,y)\to(0,0)} \frac{\sin(xy)}{xy} = 1$

pf: Since $\lim_{z\to 0} \frac{\sin z}{z} = 1$, and $\lim_{(x,y)\to(0,0)} xy = 0$

then use limit of composite function.

If need $\frac{\sin xy}{xy}$ to be continuous at $(0,0)$,

define $\frac{\sin xy}{xy} = 1$ at $(x,y) = (0,0)$.

$\therefore$ $f: \mathbb{R}^2 \to \mathbb{R}$, where $f(x,y) = (x,y)$

$g: \mathbb{R} \to \mathbb{R}$, $g(z) = \frac{\sin z}{z}$.

$\therefore$ $g \circ f : \mathbb{R}^2 \to \mathbb{R}$, $(g \circ f)(x,y) = \frac{\sin(xy)}{xy}$

(5) $\lim_{(x,y,z)\to(0,0,0)} \frac{\sin(xyz)}{xyz} = 1$

Again, use limit of composite functions, and $\lim_{z\to 0} \frac{\sin z}{z} = 1$.

Let $f(x,y,z) = xyz$. $\therefore \lim_{(x,y,z)} xyz = 0$.

Let $g(z) = \frac{\sin z}{z}$. $\therefore (g \circ f)(x,y,z) = \frac{\sin(xyz)}{xyz}$

Using ϵ-δ, let $\epsilon > 0$. $\exists\, \epsilon' > 0$ s.t. if $0 < |r| < \epsilon'$, Then

$\left|\frac{\sin r}{r} - 1\right| < \epsilon$. Given $\epsilon' > 0$, $\exists\, \delta > 0$ s.t. if

$0 < |xyz| < \delta$ Then $|xyz| < \epsilon'$.

$\therefore$ Letting $r = xyz$, if $0 < |xyz| < \delta$, Then

$\left|\frac{\sin(xyz)}{xyz} - 1\right| < \epsilon$. $\therefore \lim_{(x,y,z)\to(0,0,0)} \frac{\sin(xyz)}{xyz} = 1$

(c) $\lim_{(x,y,z)\to(0,0,0)} \frac{x^2+3y^2}{x+1} = \frac{\lim_{(x,y,z)\to(0,0,0)} x^2+3y^2}{\lim_{(x,y,z)\to(0,0,0)} x+1} = \frac{0}{1}$

$= 0$

12. **12.** Compute the following limits if they exist:

(a) $\lim_{x\to 0} \frac{\sin 2x - 2x}{x^3}$

(b) $\lim_{(x,y)\to(0,0)} \frac{\sin 2x - 2x + y}{x^3 + y}$

(c) $\lim_{(x,y,z)\to(0,0,0)} \frac{2x^2 y \cos z}{x^2 + y^2}$

(a) Using L'Hopital's Rule, $\lim_{x\to 0} \frac{\sin 2x - 2x}{x^3} =$

$\lim_{x\to 0} \frac{2\cos 2x - 2}{3x^2} = \lim_{x\to 0} \frac{-4\sin 2x}{6x} = \lim_{x\to 0} \frac{-8\cos 2x}{6} = -\frac{4}{3}$

(b) When $y=0$, $\lim_{x\to 0} \frac{\sin 2x - 2x}{x^3} = -\frac{4}{3}$ from (a)

when $x=0$, $\lim_{y\to 0} \frac{y}{y} = 1$.

∴ Limit does not exist

(c) Note: $\lim_{(y,z)\to(0,0)} 2y\cos z = 0$.

Note also if $y=z=0$, then $\lim_{(x,y,z)\to(0,0,0)} \frac{2x^2 y\cos z}{x^2+y^2} = 0$

$\therefore$ Assume $(y,z) \neq (0,0)$

$\therefore$ Given any $\epsilon > 0$, $\exists \delta > 0$ s.t. if

$0 < |(y,z) - (0,0,)| = \sqrt{y^2+z^2} < \delta$, then

$|2y\cos z| < \epsilon$

$\therefore$ Choose above δ.

$\therefore$ If $(x,y) \neq (0,0)$, and $\sqrt{x^2+y^2+z^2} < \delta$

then $\sqrt{y^2+z^2} \leq \sqrt{x^2+y^2+z^2}$

so $\sqrt{y^2+z^2} < \delta$, $(\therefore |2y\cos z| < \epsilon)$

and $x^2+y^2 \geq x^2$ and $x^2+y^2 \neq 0$

$\therefore \left| \frac{2x^2 y\cos z}{x^2+y^2} \right| \leq \left| \frac{2x^2 y\cos z}{x^2} \right| = |2y\cos z| < \epsilon$

$\therefore \lim_{(x,y,z)\to(0,0,0)} \frac{2x^2 y\cos z}{x^2+y^2} = 0$

13.

13. Compute $\lim_{\mathbf{x}\to\mathbf{x}_0} f(\mathbf{x})$, if it exists, for the following cases:

(a) $f\colon \mathbb{R} \to \mathbb{R}$, $x \mapsto |x|$, $x_0 = 1$

(b) $f\colon \mathbb{R}^n \to \mathbb{R}$, $\mathbf{x} \mapsto \|\mathbf{x}\|$, arbitrary $\mathbf{x}_0$

(c) $f\colon \mathbb{R} \to \mathbb{R}^2$, $x \mapsto (x^2, e^x)$, $x_0 = 1$

(d) $f\colon \mathbb{R}^2\backslash\{(0,0)\} \to \mathbb{R}^2$, $(x,y) \mapsto (\sin(x-y), e^{x(y+1)} - x - 1)/\|(x,y)\|$, $\mathbf{x}_0 = (0,0)$

(a) $\lim_{x \to x_0} |x| = |x_0|$ ∴ For $x_0 = 1$, $\lim_{x \to 1} |x| = \lim_{x \to 1} x = 1$

(b) $f(\vec{x}) = \sqrt{x_1^2 + x_2^2 + \ldots + x_n^2}$

Since $\lim_{\vec{x} \to \vec{c}} \vec{a} \cdot \vec{x} = \vec{a} \cdot \vec{c}$, $\vec{a}, \vec{x}, \vec{c} \in R^n$

then $\lim_{\vec{x} \to \vec{c}} x_i = c_i$, where $x = (x_1, x_2, \ldots, x_n)$
$c = (c_1, c_2, \ldots, c_n)$

and ∴ $\lim_{\vec{x} \to \vec{c}} x_i^2 = c_i^2$, so

$\lim_{\vec{x} \to \vec{c}} x_1^2 + x_2^2 + \ldots + x_n^2 = c_1^2 + c_2^2 + \ldots + c_n^2$

∴ $\lim_{\vec{x} \to \vec{c}} \sqrt{x_1^2 + \ldots + x_n^2} = \sqrt{c_1^2 + \ldots + c_n^2}$

∴ $\lim_{\vec{x} \to \vec{x}_0} \|\vec{x}\| = \|\vec{x}_0\|$

(c) $f(x) = (g(x), h(x))$, where $g(x) = x^2$, $h(x) = e^x$

Both $g(x)$ and $h(x)$ are continuous for all $x \in R$.

∴ $\lim_{x \to x_0} f(x) = (g(x_0), h(x_0))$ since $\lim_{x \to x_0} g(x) = g(x_0)$
and $\lim_{x \to x_0} h(x) = h(x_0)$

$$\therefore \lim_{x \to 1} f(x) = f(1) = (1^2, e^1) = \underline{(1, e)}$$

(d) $f(x,y) = \dfrac{(\sin(x-y),\ e^{x(y+1)} - x - 1)}{\sqrt{x^2+y^2}}$

Consider $\lim_{(x,y)\to(0,0)} \dfrac{\sin(x-y)}{\sqrt{x^2+y^2}}$

When $y=0$, $\lim_{x\to 0^+} \dfrac{\sin x}{x} = 1$, $\lim_{x\to 0^-} \dfrac{\sin x}{|x|} = -1$

$\therefore \lim_{x\to 0} \dfrac{\sin x}{|x|}$ doesn't exist.

$\therefore$ for $f(x,y) = (g(x,y), h(x,y))$,

$\lim_{(x,y)\to(0,0)} g(x,y)$ doesn't exist,

$\therefore \lim_{(x,y)\to(0,0)} f(x,y)$ doesn't exist

14.

14. Let $f(x, y, z) = \frac{1}{x^2+y^2+z^2-1}$. Describe geometrically the set in $\mathbb{R}^3$ where f fails to be continuous.

$f(x,y,z)$ is not defined for $x^2+y^2+z^2 = 1$, so

$f(x,y,z)$ is not continuous for the sphere centered at the origin with radius of 1.

15. **15.** Where is the function $f(x, y) = \frac{1}{x^2+y^2}$ continuous?

$f(x,y)$ is not defined, and $\therefore$ not continuous, at $x^2+y^2=0$, or $(x,y)=(0,0)$.

16. **16.** Let $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

(a) Considering $A: \mathbb{R}^2 \to \mathbb{R}^2$ as a linear map, explicitly write the component functions of A.

(b) Show that A is continuous on all of $\mathbb{R}^2$.

(a) $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+2y \\ 3x+4y \end{bmatrix}$ $\therefore f(x,y) = (x+2y,\ 3x+4y)$

(b) Both $g(x,y) = x+2y$ and $h(x,y) = 3x+4y$ are continuous. $\therefore f(x,y) = (g(x,y), h(x,y))$ is continuous, for all (x,y)

17. **17.** Find $\lim_{(x,y)\to(0,0)} (3x^2 + 3y^2)\log(x^2 + y^2)$. (HINT: Use polar coordinates.)

Note: $\lim_{r\to 0} r^2 \log r = \lim_{r\to 0} \frac{\log r}{\frac{1}{r^2}} = \lim_{r\to 0} \frac{\frac{1}{r}}{\frac{-2}{r^3}}$

using L'Hopital's Rule, and $\lim_{r\to 0} \frac{\frac{1}{r}}{\frac{-2}{r^3}} = \lim_{r\to 0} -\frac{1}{2}r^2 = 0$.

$\therefore \lim_{r\to 0} 3r^2 \log r^2 = 6 \lim_{r\to 0} r^2 \log r = 6 \cdot 0 = 0.$

$\therefore$ Given any $\epsilon > 0$, $\exists\, \delta > 0$ s.t. if $0 < |r| < \delta$,

Then $|3r^2 \log r^2| < \epsilon$

$\therefore$ Let $r = \sqrt{x^2+y^2}$. $\therefore r^2 = x^2+y^2$

$\therefore$ if $0 < \sqrt{x^2+y^2} < \delta$, then

$|3(x^2+y^2)\log(x^2+y^2)| < \epsilon$

$\therefore \lim_{(x,y)\to(0,0)} (3x^2+3y^2)\log(x^2+y^2) = \underline{0}$

Show that the subsets of the plane in Exercises 18–21 are open:

18.

18. $A = \{(x, y) \mid -1 < x < 1, -1 < y < 1\}$

Let $\vec{P_0} = (x_0, y_0) \in A$. $\therefore -1 < x_0 < 1$ and $-1 < y < 1$.

Let $r_x = \min\{1-x_0, x_0+1\}$

$r_y = \min\{1-y_0, y_0+1\}$

Since $x_0 < 1$, then $0 < 1-x_0$
$-1 < x_0$, then $0 < x_0+1$ $\Big\}$ $\therefore r_x > 0$

Since $y_0 < 1$, then $0 < 1-y_0$
$-1 < y_0$, then $0 < y_0+1$ $\Big\}$ $\therefore r_y > 0$

Let $r = \min\{r_x, r_y\}$

Consider $D_r(\vec{P_0})$. Let $\vec{X} = (x,y) \in D_r(\vec{P_0})$

$\therefore |\vec{x} - \vec{p}_0| < r$, or $\sqrt{(x-x_0)^2 + (y-y_0)^2} < r$,

$\therefore (x-x_0)^2 + (y-y_0)^2 < r^2$

$\therefore |x-x_0| < r \leq r_x$ and $|y-y_0| < r \leq r_y$

$\therefore -r_x < x - x_0 < r_x$

But $r_x \leq 1 - x_0$, and $r_x \leq x_0 + 1$

$\therefore -1 - x_0 \leq -r_x$

$\therefore -1 - x_0 \leq -r_x < x - x_0 < r_x \leq 1 - x_0$

$\therefore -1 - x_0 < x - x_0$ and $x - x_0 < 1 - x_0$

$\therefore -1 < x$ and $x < 1$ [1]

For y: $-r_y < y - y_0 < r_y$

But $r_y \leq 1 - y_0$ and $r_y \leq y_0 + 1$

$\therefore -1 - y_0 \leq -r_y$

$\therefore -1 - y_0 \leq -r_y < y - y_0 < r_y \leq 1 - y_0$

$\therefore -1 - y_0 < y - y_0$ and $y - y_0 < 1 - y_0$

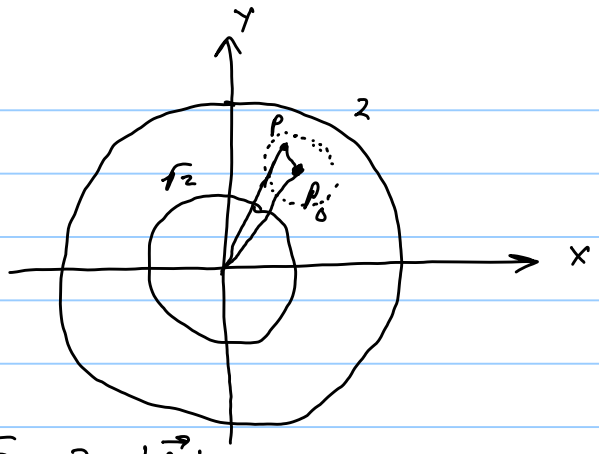
$\therefore -1 < y$ and $y < 1$ [2]

[1] & [2]: $-1 < x < 1$, $-1 < y < 1 \Rightarrow \vec{x} \in A$

$\therefore$ For any $\vec{p}_0 \in A$, $\exists\, r > 0$ s.t. $D_r(\vec{p}_0) \subset A$

$\therefore$ A is open

20. **20.** $C = \{(x, y) \mid 2 < x^2 + y^2 < 4\}$



Let $\vec{\rho}_0 = (x_0, y_0) \in C$

Look at distances of $\vec{\rho}_0$ to edge of circles:

$|\vec{\rho}_0| - \sqrt{2}$, $2 - |\vec{\rho}_0|$

Note: $\vec{\rho}_0 \in C \Rightarrow 2 < |\vec{\rho}_0|^2 < 4$, so $\sqrt{2} < |\vec{\rho}_0|$ and $|\vec{\rho}_0| < 2$. $\therefore 0 < |\vec{\rho}_0| - \sqrt{2}$, and $0 < 2 - |\vec{\rho}_0|$

$\therefore$ Let $0 < r < \min\{|\vec{\rho}_0| - \sqrt{2}, 2 - |\vec{\rho}_0|\}$

Consider $D_r(\vec{\rho}_0)$, let $\vec{\rho} = (x, y) \in D_r(\vec{\rho}_0)$

Must show $\sqrt{2} < |\vec{\rho}| < 2$

Note from triangle inequality,

$$|\vec{\rho}| - |\vec{\rho}_0| \le |\vec{\rho} - \vec{\rho}_0|$$

and $$|\vec{\rho}_0| - |\vec{\rho}| \le |\vec{\rho}_0 - \vec{\rho}| = |\vec{\rho} - \vec{\rho}_0|$$

If $\vec{\rho} \in D_r(\vec{\rho}_0)$, then $|\vec{\rho} - \vec{\rho}_0| < r$

But $|\vec{\rho}| - |\vec{\rho}_0| \le |\vec{\rho} - \vec{\rho}_0| < r < 2 - |\vec{\rho}_0|$

$\therefore |\vec{\rho}| - |\vec{\rho}_0| < 2 - |\vec{\rho}_0| \Rightarrow |\vec{\rho}| < 2$

Also, $|\vec{\rho}_0| - |\vec{\rho}| \le |\vec{\rho} - \vec{\rho}_0| < r < |\vec{\rho}_0| - \sqrt{2}$

$\therefore |\vec{p}_0| - |\vec{p}| < |\vec{p}_0| - \sqrt{2} \Rightarrow \sqrt{2} < |\vec{p}|$

$\therefore \sqrt{2} < |\vec{p}| < 2 \Rightarrow 2 < |\vec{p}|^2 < 4 \Rightarrow \vec{p} \in C$

$\therefore$ Given any $\vec{p}_0 \in C$, $\exists\, r > 0$ s.t. $D_r(\vec{p}_0) \subset C$

$\therefore$ C is open

22.

22. Let $A \subset \mathbb{R}^2$ be the open unit disk $D_1(0, 0)$ with the point $\mathbf{x}_0 = (1, 0)$ added, and let $f\colon A \to \mathbb{R}$, $\mathbf{x} \mapsto f(\mathbf{x})$ be the constant function $f(\mathbf{x}) = 1$. Show that $\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = 1$.

Given any $\epsilon > 0$, need to find an $\delta > 0$ s.t. if $0 < \|\vec{x} - \vec{x}_0\| < \delta$ and $\vec{x} \in A$, then $|f(\vec{x}) - 1| < \epsilon$.

But for all $\vec{x} \in A$, $f(\vec{x}) = 1$ $\therefore |f(\vec{x}) - 1| = |0|$, so $|f(\vec{x}) - 1| < \epsilon$ will always be true for $\vec{x} \in A$.

Here, we can choose $\delta = \epsilon$, and $\therefore$ must show that for this δ, there are values of $\vec{x} \in A$ in which $0 < \|\vec{x} - (1,0)\| < \delta$. Then, every $\vec{x} \in A \cap D_\delta(1,0)$ will be within δ of $(1,0)$, and will also be in A, so $|f(\vec{x}) - 1| < \epsilon$ since $f(\vec{x}) = 1$. That is, $A \cap D_\epsilon(1,0) \neq \emptyset$ for all $\epsilon > 0$.

$\therefore$ Let $\epsilon > 0$ and let $\vec{x} = (x_1, x_2)$.

Choose $\delta = \min\left\{1, \frac{\epsilon}{2}\right\}$

$\therefore$ Let $x_2 = 0$ and $x_1 = 1-\delta$. $\therefore x_2^2 = 0$.

Since $\delta < 1$, then $1-\delta > 0$, so $x_1 > 0$.

Also, since $x_1 + \delta = 1$, $x_1 < 1$.

$\therefore 0 < x_1^2 < 1$, and so $0 < x_1^2 + x_2^2 < 1$

$\therefore (x_1, x_2) \in A$ [1]

Since $x_1 - 1 = -\delta$, $(x_1-1)^2 = \delta^2$

$\therefore (x_1-1)^2 + x_2^2 = \delta^2$

$\therefore \| \vec{x} - (1,0) \| = \delta < \epsilon$

$\therefore \vec{x} \in D_\epsilon(\vec{x}_0)$ [2]

$\therefore \vec{x} \in A \cap D_\epsilon(\vec{x}_0)$, so $A \cap D_\epsilon(\vec{x}_0) \neq \emptyset$ for any $\epsilon > 0$.

$\therefore$ Given any $\epsilon > 0$, $\exists\, \delta > 0$ s.t. if $\vec{x} \in A$ and $0 < \| \vec{x} - \vec{x}_0 \| < \delta$, then $|f(\vec{x}) - 1| < \epsilon$

$\therefore \lim_{x \to x_0} f(\vec{x}) = 1$

25.

25. (a) Can $[\sin(x+y)]/(x+y)$ be made continuous by suitably defining it at $(0, 0)$?

(b) Can $xy/(x^2+y^2)$ be made continuous by suitably defining it at $(0, 0)$?

(c) Prove that $f\colon \mathbb{R}^2 \to \mathbb{R}$, $(x, y) \mapsto ye^x + \sin x + (xy)^4$ is continuous.

(a) $\lim_{(x,y)\to(0,0)} \frac{\sin(x+y)}{x+y} = 1$ since $\lim_{r\to 0} \frac{\sin r}{r} = 1$.

$\therefore$ Define $\frac{\sin(x+y)}{x+y} = 1$ at $(x,y) = (0,0)$

(b) $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2}$ doesn't exist.

For $x = 0$, $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = 0$

For $y = x$, $\lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{(x,y)\to(0,0)} \frac{x^2}{2x^2} = \frac{1}{2}$.

$\therefore$ Can't give $\frac{xy}{x^2+y^2}$ a unique value at $(0,0)$ so that limit $\neq$ unique value.

(c) $\lim_{(x,y)\to(a,b)} y = b$, $\lim_{(x,y)\to(a,b)} e^x = e^a$, $\lim_{(x,y)\to(a,b)} \sin x = \sin a$,

$\lim_{(x,y)\to(a,b)} (xy)^4 = (ab)^4$.

$\therefore$ Components are continuous, $\therefore$ sum of components is continuous.

26.

26. Using either ε's and δ's or spherical coordinates, show that

$$\lim_{(x,y,z)\to(0,0,0)} \frac{xyz}{x^2+y^2+z^2} = 0.$$

Let $\epsilon > 0$. Choose $\delta = \epsilon$.

If $0 < \|(x,y,z)-(0,0,0)\| = \sqrt{x^2+y^2+z^2} < \delta$, then

$|x| \le \sqrt{x^2+y^2+z^2} < \delta$

$\therefore \quad |x| < \delta \Rightarrow |x^3| < \delta x^2$

Similarly $|y^3| < \delta y^2$, $|z^3| < \delta z^2$

Letting $s = \max\{|x|, |y|, |z|\}$,

then $|x| \le s$, $|y| \le s$, $|z| \le s$, so $|xyz| \le s^3$

and s^3 is one of $|x^3|$, $|y^3|$, or $|z^3|$

$\therefore s^3 \le |x^3| + |y^3| + |z^3|$

$< \delta x^2 + \delta y^2 + \delta z^2 = \delta(x^2+y^2+z^2)$

$\therefore |xyz| < \delta(x^2+y^2+z^2)$

$\therefore \left| \frac{xyz}{x^2+y^2+z^2} - 0 \right| = \left| \frac{xyz}{x^2+y^2+z^2} \right| < \delta = \epsilon$

$\therefore \lim_{(x,y,z)\to(0,0,0)} \frac{xyz}{x^2+y^2+z^2} = 0$

Using spherical coordinates:

Let $\rho = \sqrt{x^2+y^2+z^2}$, $z = \rho\cos\phi$, $x = \rho\sin\phi\cos\theta$, $y = \rho\sin\phi\sin\theta$

$\therefore xyz = \rho^3\cos\phi\sin^2\phi\cos\theta\sin\theta$

$x^2+y^2+z^2 = \rho^2$

$\therefore \dfrac{xyz}{x^2+y^2+z^2} = \rho\cos\phi\sin^2\phi\cos\theta\sin\theta$

$\therefore \lim\limits_{(x,y,z)\to(0,0,0)} \dfrac{xyz}{x^2+y^2+z^2} = \lim\limits_{\rho\to 0} \rho\cos\phi\sin^2\phi\cos\theta\sin\theta$

$= (\cos\phi\sin^2\phi\cos\theta\sin\theta)\lim\limits_{\rho\to 0}\rho = 0$

Note: $\|(x,y,z)-(0,0,0)\| < \delta \iff \sqrt{x^2+y^2+z^2} < \delta$

$\iff \rho < \delta \iff |\rho - 0| < \delta$

27. **27.** Use the ε-δ formulation of limits to prove that $x^2 \to 4$ as $x \to 2$. Give another proof using Theorem 3.

(a) $\lim\limits_{x\to 2} x^2 = 4$

Proof: Let $\epsilon > 0$. Choose $\delta = \min\left\{\sqrt{\frac{\epsilon}{2}}, \frac{\epsilon}{8}, 1\right\}$

$\therefore$ if $0 < |x-2| < \delta$, then $-\delta < x-2 < \delta$

$\therefore 2-\delta < x < 2+\delta$

$4-\delta < x+2 < 4+\delta$

Since $\delta \leq 1$, then $4-\delta > 0$.

$\therefore 0 < x+2 < 4+\delta$

$\therefore |x+2| < 4+\delta$

$\therefore |x-2||x+2| < \delta(4+\delta)$

Or, $|x^2-4| < \delta^2 + 4\delta$

But $\delta \leq \sqrt{\frac{\epsilon}{2}}$, so $\delta^2 \leq \frac{\epsilon}{2}$

and $\delta \leq \frac{\epsilon}{8} \Rightarrow 4\delta \leq \frac{\epsilon}{2}$

$\therefore |x^2-4| < \delta^2 + 4\delta \leq \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$

$\therefore 0 < |x-2| < \delta \Rightarrow |x^2-4| < \epsilon$

(b) $\lim_{x\to 2} x = 2$, $\therefore \lim_{x\to 2} x^2 = \left(\lim_{x\to 2} x\right)\left(\lim_{x\to 2} x\right) = 2 \cdot 2 = 4$

28.

28. (a) Prove that for $\mathbf{x} \in \mathbb{R}^n$ and $s < t$, $D_s(\mathbf{x}) \subset D_t(\mathbf{x})$.

(b) Prove that if U and V are neighborhoods of $\mathbf{x} \in \mathbb{R}^n$, then so are $U \cap V$ and $U \cup V$.

(c) Prove that the boundary points of an open interval $(a, b) \subset \mathbb{R}$ are the points a and b.

(a) $D_s(\vec{x}) = \{\vec{y} \in R^n : \|\vec{y} - \vec{x}\| < s\}$

Since $s < t$, then for all $\vec{y} \in D_s(\vec{x})$,

$\|\vec{y} - \vec{x}\| < s < t$. $\therefore y \in D_t(\vec{x})$

$\therefore D_s(\vec{x}) \subset D_t(\vec{x})$

(b) $U = B_r(\vec{x})$, some r, $V = B_s(\vec{x})$, some s.

If $r < s$, then by (a), $B_r(\vec{x}) \subset B_s(\vec{x})$

$\therefore U \cap V = B_r(\vec{x}) \cap B_s(\vec{x}) = B_r(\vec{x})$,
a neighborhood of $\vec{x}$.

Similarly, if $s < r$, $U \cap V = B_s(\vec{x}) \cap B_r(\vec{x}) = B_s(\vec{x})$,
a neighborhood of $\vec{x}$.

If $r = s$, $U \cap V = B_r(\vec{x}) \cap B_s(\vec{x}) = B_r(\vec{x}) = B_s(\vec{x})$,
a neighborhood of $\vec{x}$.

(c) For each $x \in (a,b)$, there is an r s.t. $B_r(x) \subset (a,b)$.
$\therefore$ No point of (a,b) is a boundary point.

However, every $B_r(a)$ and $B_r(b)$ contains
elements of (a,b), and elements not in (a,b).
$\therefore$ a and b are boundary points.

e.g.: $B_r(a) = \{x : |x-a| < r\}$, any $r > 0$.

$\therefore -r < x-a < r$, or $a-r < x < a+r$

$\therefore x = a + \frac{1}{2}r < a+r$ and $a < a + \frac{1}{2}r = x$.
$\therefore x \in (a,b)$ and $x \in B_r(a)$

And $x = a - \frac{1}{2}r < a$, and $a - r < a - \frac{1}{2}r$.
$\therefore x \notin (a,b)$ and $x \in B_r(a)$

29. **29.** Suppose $\mathbf{x}$ and $\mathbf{y}$ are in $\mathbb{R}^n$ and $\mathbf{x} \neq \mathbf{y}$. Show that there is a continuous function $f\colon \mathbb{R}^n \to \mathbb{R}$ with $f(\mathbf{x}) = 1$, $f(\mathbf{y}) = 0$, and $0 \leq f(\mathbf{z}) \leq 1$ for every $\mathbf{z}$ in $\mathbb{R}^n$.

Since $\vec{x} \neq \vec{y}$, $\|\vec{x} - \vec{y}\| \neq 0$. Let $r = \|\vec{x} - \vec{y}\|$

$\therefore$ Let $f(\vec{z}) = \dfrac{\|\vec{z} - \vec{y}\|}{\|\vec{x} - \vec{y}\|}$ $\therefore f(\vec{y}) = 0$, $f(\vec{x}) = 1$.

But $f(\vec{z})$ will only be ≤ 1 when $\|\vec{z} - \vec{y}\| \leq \|\vec{x} - \vec{y}\|$

$\therefore$ Define $f(\vec{z}) = \begin{cases} \dfrac{\|\vec{z} - \vec{y}\|}{\|\vec{x} - \vec{y}\|}, & \text{for } \vec{z} \in D_r(\vec{y}) \\ 1, & \text{for } \vec{z} \notin D_r(\vec{y}) \end{cases}$

$f(\vec{z})$ is continuous, since

$f(\vec{z}) = \frac{1}{r}\sqrt{(z_1 - y_1)^2 + \ldots + (z_n - y_n)^2}$ is a continuous function of $\vec{z} \in R^n$

and $f(\vec{z}) = 1$ is continuous for $\vec{z} \in R^n$ and $f(\vec{z}) = 1$ at the boundary of $D_r(\vec{y})$

30. **30.** Let $f\colon A \subset \mathbb{R}^n \to \mathbb{R}$ be given and let $\mathbf{x}_0$ be a boundary point of A. We say that $\lim_{\mathbf{x} \to \mathbf{x}_0} f(\mathbf{x}) = \infty$ if for every $N > 0$ there is a $\delta > 0$ such that $0 < \|\mathbf{x} - \mathbf{x}_0\| < \delta$ and $\mathbf{x} \in A$ implies $f(\mathbf{x}) > N$.

(a) Prove that $\lim_{x \to 1} (x - 1)^{-2} = \infty$.

(b) Prove that $\lim_{x \to 0} 1/|x| = \infty$. Is it true that $\lim_{x \to 0} 1/x = \infty$?

(c) Prove that $\lim_{(x,y) \to (0,0)} 1/(x^2 + y^2) = \infty$.

(a) Let $N > 0$. Look at $\frac{1}{(x-1)^2} > N$

$\therefore \frac{1}{N} > (x-1)^2$, or $(x-1)^2 < \frac{1}{N}$, $|x-1| < \frac{1}{\sqrt{N}}$

$\therefore$ Choose $\delta = \frac{1}{\sqrt{N}}$.

$\therefore$ if $0 < |x-1| < \frac{1}{\sqrt{N}}$, Then $|x-1|^2 = (x-1)^2 < \frac{1}{N}$,

and $\therefore f(x) = \frac{1}{(x-1)^2} > N$ $\therefore \lim_{x \to 1} \frac{1}{(x-1)^2} = \infty$

(b) Let $N > 0$. Look at $\frac{1}{|x|} > N$, or $\frac{1}{N} > |x|$,

i.e. $|x| < \frac{1}{N}$. $\therefore$ Choose $\delta = \frac{1}{N}$.

$\therefore$ If $||x| - 0| = ||x|| = |x| < \delta = \frac{1}{N}$,

Then $\frac{1}{|x|} = f(x) > N$.

$\therefore \lim_{x \to 0} \frac{1}{|x|} = \infty$

$\lim_{x \to 0^+} \frac{1}{x} = +\infty$, $\lim_{x \to 0^-} \frac{1}{x} = -\infty$. $\therefore \lim_{x \to 0} \frac{1}{x} \neq \infty$

(c) Let $N > 0$ Choose $\delta = \frac{1}{\sqrt{N}}$

$\therefore$ if $0 < |(x,y) - (0,0)| < \delta$, Then

$0 < \sqrt{x^2+y^2} < \frac{1}{\sqrt{N}} \Rightarrow x^2 + y^2 < \frac{1}{N}$,

and $\therefore \frac{1}{x^2+y^2} > N$ $\therefore \lim_{(x,y) \to (0,0)} \frac{1}{x^2+y^2} = \infty$

31.

31. Let $b \in \mathbb{R}$ and $f: \mathbb{R}\backslash[b] \to \mathbb{R}$ be a function. We write $\lim_{x \to b-} f(x) = L$ and say that L is the ***left-hand limit*** of f at b if for every $\varepsilon > 0$ there is a $\delta > 0$ such that $x < b$ and $0 < |x - b| < \delta$ implies $|f(x) - L| < \varepsilon$.

(a) Formulate a definition of ***right-hand limit***, or $\lim_{x \to b+} f(x)$.

(b) Find $\lim_{x \to 0-} 1/(1 + e^{1/x})$ and $\lim_{x \to 0+} 1/(1 + e^{1/x})$.

(c) Sketch the graph of $1/(1 + e^{1/x})$.

(a) Let $b \in \mathbb{R}$ and $f: \mathbb{R}\backslash[b] \to \mathbb{R}$ be a function.

$\lim_{x \to b^+} f(x) = L \iff$ for every $\epsilon > 0$ there is a $\delta > 0$ s.t. $b < x$ and $0 < x - b < \delta$ imply $|f(x) - L| < \epsilon$.

(b) $\lim_{x \to 0^-} \frac{1}{1 + e^{1/x}}$

$\lim_{x \to 0^-} \frac{1}{x} = -\infty$, and $\lim_{x \to -\infty} e^x = 0$

$\therefore \lim_{x \to 0^-} e^{\frac{1}{x}} = 0$. $\therefore \lim_{x \to 0^-} \frac{1}{1 + e^{1/x}} = 1$

$\lim_{x \to 0^+} \frac{1}{1 + e^{1/x}} = \lim_{x \to 0^+} \frac{e^{-\frac{1}{x}}}{e^{-1/x} + 1}$

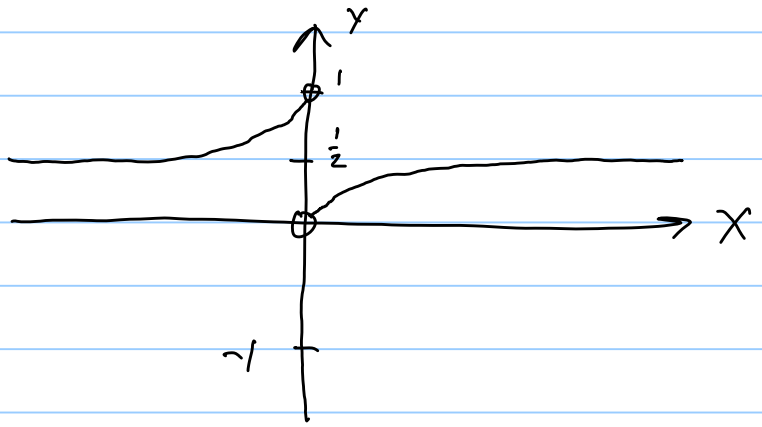
$\lim_{x \to 0^+} -\frac{1}{x} = -\infty$. $\lim_{x \to -\infty} e^x = 0$. $\therefore \lim_{x \to 0^+} e^{-\frac{1}{x}} = 0$

$\therefore \lim_{x \to 0^+} \frac{e^{-\frac{1}{x}}}{e^{-1/x} + 1} = \frac{0}{0 + 1} = 0 = \lim_{x \to 0^+} \frac{1}{1 + e^{1/x}}$

(c) As $x \to +\infty$, $\frac{1}{1+e^{1/x}} \to \frac{1}{2}$

As $x \to -\infty$, $\frac{1}{1+e^{1/x}} \to \frac{1}{2}$

$\therefore$



32. **32.** Show that f is continuous at $\mathbf{x}_0$ if and only if

$$\lim_{\mathbf{x} \to \mathbf{x}_0} \| f(\mathbf{x}) - f(\mathbf{x}_0) \| = 0.$$

Note that $\| f(\vec{x}) - f(\vec{x}_0) \|$ is equivalent to

$\| \| f(\vec{x}) - f(\vec{x}_0) \| - 0 \|$

$\therefore$ Given $\epsilon > 0$

(1) If $\lim_{\vec{x} \to \vec{x}_0} f(\vec{x}) = f(\vec{x}_0)$, then $\exists \delta > 0$ s.t. if

$0 < \| \vec{x} - \vec{x}_0 \| < \delta$ then $\| f(\vec{x}) - f(\vec{x}_0) \| < \epsilon$, and

$\therefore \| \| f(\vec{x}) - f(\vec{x}_0) \| - 0 \|$. $\therefore \lim_{\vec{x} \to \vec{x}_0} \| f(\vec{x}) - f(\vec{x}_0) \| = 0$

(2) If $\lim_{\vec{x} \to \vec{x}_0} \| f(\vec{x}) - f(\vec{x}_0) \| = 0$, then $\exists \delta > 0$ s.t.

if $0 < \|\vec{x} - \vec{x}_0\| < \delta$ then $\|\|f(\vec{x}) - f(\vec{x}_0)\| - 0\| < \epsilon$,

and $\therefore \|f(\vec{x}) - f(\vec{x}_0)\| < \epsilon$. $\therefore \lim_{\vec{x} \to \vec{x}_0} f(\vec{x}) = f(\vec{x}_0)$

33. **33.** Let $f: A \subset \mathbb{R}^n \to \mathbb{R}^m$ satisfy $\|f(\mathbf{x}) - f(\mathbf{y})\| \le K\|\mathbf{x} - \mathbf{y}\|^\alpha$ for all $\mathbf{x}$ and $\mathbf{y}$ in A for positive constants K and α. Show that f is continuous. (Such functions are called ***Hölder-continuous*** or, if $\alpha = 1$, ***Lipschitz-continuous***.)

For a given function f, K and α are known and the condition is true for all $\vec{x}, \vec{y} \in A$.

$\therefore$ Let $\epsilon > 0$ and let $\vec{x}_0$ be any element in A.

Choose $\delta = \sqrt[\alpha]{\frac{\epsilon}{K}}$. Note $K, \alpha > 0$.

$\therefore$ if $\|\vec{x} - \vec{x}_0\| < \delta$, then

$\|\vec{x} - \vec{x}_0\| < \sqrt[\alpha]{\frac{\epsilon}{K}}$, so $K\|\vec{x} - \vec{x}_0\|^\alpha < \epsilon$

$\therefore \|f(\vec{x}) - f(\vec{x}_0)\| \le K\|\vec{x} - \vec{x}_0\| < \epsilon$

$\therefore \lim_{\vec{x} \to \vec{x}_0} f(\vec{x}) = f(\vec{x}_0)$ and $\therefore f$ is continuous

at x_0, so f is continuous on A.

34. **34.** Show that $f: \mathbb{R}^n \to \mathbb{R}^m$ is continuous at all points if and only if the inverse image of every open set is open.

(a) Suppose f is continuous at all points of R^n.
Let B be any open set of R^m containing values of $f(\vec{x})$, where $\vec{x} \in R^n$. Since B is open, $\exists\, D_\epsilon(f(\vec{x})) \subset B$, $\epsilon > 0$. Since f is continuous, $\exists\, \delta > 0$ s.t. for $\vec{y} \in D_\delta(\vec{x})$, $f(\vec{y}) \in D_\epsilon(f(\vec{x}))$.
Note that $D_\delta(\vec{x})$ is open.
$\therefore$ For each point $f(\vec{x})$ in B, let $A =$ union of all such corresponding $D_\delta(\vec{x})$.
The union of open sets is open. $\therefore$ The union of all such $D_\delta(\vec{x})$ is open, and the union of all the $D_\delta(\vec{x})$ represents the inverse image of B.

(b) Assume inverse image of every open set of R^m is open.
Let $\vec{x}_0 \in R^n$ and let $\epsilon > 0$. $D_\epsilon(f(\vec{x}_0))$ is open. Let A be the inverse image of

$D_\epsilon(f(\vec{x}_0))$. Since $\vec{x}_0 \in A$ and A is open, $\exists\, \delta > 0$ s.t. $D_\delta(\vec{x}_0) \subset A$. But since A is the inverse image, then all of the image of $D_\delta(\vec{x}_0)$ is contained in $D_\epsilon(f(\vec{x}_0))$.

$\therefore$ Given any $\epsilon > 0$, $\exists\, \delta$ s.t. if $\vec{x} \in D_\delta(\vec{x}_0)$ then $\|\vec{x} - \vec{x}_0\| < \delta$ by definition of $D_\delta(\vec{x}_0)$, and $\|f(\vec{x}) - f(\vec{x}_0)\| < \epsilon$ since $f(\vec{x}) \in D_\epsilon(f(\vec{x}_0))$.

$\therefore \lim_{\vec{x} \to \vec{x}_0} f(\vec{x}) = f(\vec{x}_0)$, so f is continuous.

35.

35. (a) Find a specific number $\delta > 0$ such that if $|a| < \delta$, then $|a^3 + 3a^2 + a| < 1/100$.

(b) Find a specific number $\delta > 0$ such that if $x^2 + y^2 < \delta^2$, then

$$|x^2 + y^2 + 3xy + 180xy^5| < 1/10{,}000.$$

(a) $|a^3 + 3a^2 + a| \le |a^3| + 3|a^2| + |a|$

If $|a| < 1$, then $|a^2| < |a|$, $|a|^3 < a$.

$\therefore |a^3| + 3|a^2| + |a| < |a| + 3|a| + |a| = 5|a|$

$\therefore$ if $|a| < \frac{1}{500}$, then $5|a| < \frac{1}{100}$. $\therefore$ let $\delta = \frac{1}{500}$

(6) $|x^2+y^2+3xy+180xy^5| \le |x^2+y^2| + 3|xy| + 180|xy|y^4$

Note if $x^2+y^2<\delta^2$, then $x^2<\delta^2$, $y^2<\delta^2$, so

$|x|<\delta$, $|y|<\delta$, $\therefore |xy|<\delta^2$

If $\delta<1$, then $y^2<\delta^2 \Rightarrow y^4<\delta^4<\delta^2$

$\therefore |x^2+y^2| + 3|xy| + 180|xy|y^4 < \delta^2+3\delta^2+180\delta^2\delta^2$

$< 184\delta^2 < (14\delta)^2$

$\therefore$ if $(14\delta)^2 < \frac{1}{10^4}$, or $14\delta < \frac{1}{10^2}$, or

$\underline{\delta < \frac{1}{1400}} \approx 0.0007$, then

$|x^2+y^2+3xy+180xy^5| < \frac{1}{10,000}$

2.3 Differentiation

Note Title 2/2/2016

1.

1. Find $\partial f/\partial x$, $\partial f/\partial y$ if

(a) $f(x, y) = xy$

(b) $f(x, y) = e^{xy}$

(c) $f(x, y) = x \cos x \cos y$

(d) $f(x, y) = (x^2 + y^2) \log (x^2 + y^2)$

(a) $f_x = y$, $f_y = x$

(b) $f_x = ye^{xy}$, $f_y = xe^{xy}$

(c) $f_x = (\cos y)\dfrac{\partial}{\partial x}(x\cos x) = (\cos y)(\cos x - x\sin x)$

$= \cos x \cos y - x \sin x \cos y$

$f_y = -x \cos x \sin y$

(d) $f_x = \dfrac{\partial (x^2+y^2)}{\partial x}\left[\log (x^2+y^2)\right] + \left[x^2+y^2\right]\dfrac{\partial \log (x^2+y^2)}{\partial x}$

$= 2x\left[\log (x^2+y^2)\right] + \left[x^2+y^2\right]\dfrac{1}{x^2+y^2}(2x)$

$= 2x\left[1 + \log (x^2+y^2)\right]$

$f_y = 2y \log (x^2+y^2) + (x^2+y^2)\dfrac{1}{x^2+y^2}(2y)$

$= 2y\left[1 + \log (x^2+y^2)\right]$

2.

2. Evaluate the partial derivatives $\partial z/\partial x$, $\partial z/\partial y$ for the given function at the indicated points.

(a) $z = \sqrt{a^2 - x^2 - y^2}$; $(0, 0)$, $(a/2, a/2)$

(b) $z = \log \sqrt{1 + xy}$; $(1, 2)$, $(0, 0)$

(c) $z = e^{ax} \cos(bx + y)$; $(2\pi/b, 0)$

(a) $f_x = \frac{1}{2}(a^2 - x^2 - y^2)^{-\frac{1}{2}}(-2x) = -x(a^2 - x^2 - y^2)^{-\frac{1}{2}}$

$f_y = -y(a^2 - x^2 - y^2)^{-\frac{1}{2}}$

$\therefore f_x(0,0) = 0, \quad f_y(0,0) = 0$

$f_x\left(\frac{a}{2}, \frac{a}{2}\right) = -\frac{a}{2}\left(a^2 - \frac{a^2}{4} - \frac{a^2}{4}\right)^{-\frac{1}{2}} = -\frac{\frac{a}{2}}{\sqrt{\frac{a^2}{2}}}$

$= -\frac{a}{|a|}\frac{\sqrt{2}}{2}$

$f_y\left(\frac{a}{2}, \frac{a}{2}\right) = -\frac{a}{|a|}\frac{\sqrt{2}}{2}$

(b) $f_x = \frac{1}{\sqrt{1+xy}} \cdot \frac{1}{2}(1+xy)^{-\frac{1}{2}}(y) = \frac{y}{2(1+xy)}$

$f_y = \frac{x}{2(1+xy)}$

$\therefore f_x(1,2) = \frac{2}{2(1+2)} = \frac{1}{3} \qquad f_y(1,2) = \frac{1}{6}$

$f_x(0,0) = 0 \qquad f_y(0,0) = 0$

(c) $f_x = ae^{ax}\cos(6x+y) - e^{ax}\,6\sin(6x+y)$

$f_y = -e^{ax}\sin(6x+y)$

$\therefore f_x\left(\frac{2\pi}{6}, 0\right) = ae^{\frac{2a\pi}{6}}$

$f_y\left(\frac{2\pi}{6}, 0\right) = 0$

3.

3. In each case following, find the partial derivatives $\partial w/\partial x$, $\partial w/\partial y$.

(a) $w = xe^{x^2+y^2}$

(b) $w = \dfrac{x^2+y^2}{x^2-y^2}$

(c) $w = e^{xy}\log(x^2+y^2)$

(d) $w = x/y$

(e) $w = \cos(ye^{xy})\sin x$

(a) $w_x = e^{x^2+y^2} + xe^{x^2+y^2}(2x)$

$= (2x^2+1)e^{x^2+y^2}$

$w_y = xe^{x^2+y^2}(2y) = 2xye^{x^2+y^2}$

(b) $w_x = \dfrac{(x^2-y^2)(2x) - (x^2+y^2)(2x)}{(x^2-y^2)^2}$

$= \dfrac{-4xy^2}{(x^2+y^2)^2}$

$w_y = \dfrac{(x^2-y^2)(2y) - (x^2+y^2)(-2y)}{(x^2-y^2)^2} = \dfrac{4x^2y}{(x^2+y^2)^2}$

(c) $w_x = ye^{xy}\log(x^2+y^2) + e^{xy}\cdot\frac{1}{x^2+y^2}\cdot 2x$

$w_y = xe^{xy}\log(x^2+y^2) + e^{xy}\cdot\frac{1}{x^2+y^2}\cdot 2y$

(d) $w_x = \frac{1}{y} \quad w_y = -\frac{x}{y^2}$

(e) $w_x = -\sin(ye^{xy})\cdot(y^2e^{xy})\sin x + \cos(ye^{xy})\cos x$

$w_y = -\sin(ye^{xy})\cdot(e^{xy} + xye^{xy})\sin x$

4.

4. Decide which of the following functions are C^1, which are just differentiable.

(a) $f(x, y) = \dfrac{2xy}{(x^2+y^2)^2}$

(b) $f(x, y) = \dfrac{x}{y} + \dfrac{y}{x}$

(c) $f(r, \theta) = \frac{1}{2}r\sin 2\theta,\ r > 0$

(d) $f(x, y) = \dfrac{xy}{\sqrt{x^2+y^2}}$

(e) $f(x, y) = \dfrac{x^2y}{x^4+y^2}$

(a) $f_x = \dfrac{(x^2+y^2)^2 2y - 2xy[2(x^2+y^2)\cdot 2x]}{(x^2+y^2)^4}$

$f_y = \dfrac{(x^2+y^2)^2 2x - 2xy[2(x^2+y^2)2y]}{(x^2+y^2)^4}$

$\therefore f_x$ and f_y are continuous for $(x,y) \neq (0,0)$.

$\therefore f(x,y)$ is differentiable and C^1 in its domain: $\mathbb{R}^2 - (0,0)$.

(b) $f_x = \frac{1}{y} - \frac{y}{x^2}$, $f_y = -\frac{x}{y^2} + \frac{1}{x}$

$\therefore$ f_x and f_y are continuous for all $(x,y) \neq (0,0)$.

$\therefore$ $f(x,y)$ is differentiable and C^1 in its domain: $R^2 - (0,0)$

(c) $f_r = \frac{1}{2}\sin 2\theta$, $f_\theta = \frac{1}{2} r (2\cos 2\theta) = r\cos 2\theta$

f_r, f_θ are continuous for $r > 0$, all θ.

$\therefore$ $f(r,\theta)$ is C^1 for its domain.

(d) $f_x = \dfrac{(x^2+y^2)^{1/2} y - xy\left[\frac{1}{2}(x^2+y^2)^{-\frac{1}{2}} 2x\right]}{(x^2+y^2)}$

$f_y = \dfrac{(x^2+y^2)^{1/2} x - xy\left[\frac{1}{2}(x^2+y^2)^{-\frac{1}{2}} 2y\right]}{(x^2+y^2)}$

f_x, f_y are continuous for all $(x,y) \neq (0,0)$

$\therefore$ $f(x,y)$ is C^1 in its domain: $R^2 - (0,0)$

(e) $f_x = \dfrac{(x^4+y^2)2xy - x^2y(4x^3)}{(x^4+y^2)^2}$, continuous for $(x,y) \neq (0,0)$

$f_y = \dfrac{(x^4+y^2)x^2 - x^2y(2y)}{(x^4+y^2)^2}$, continuous for $(x,y) \neq (0,0)$

$\therefore$ $f(x,y)$ is C^1 if its domain; $R^2 - (0,0)$

5. **5.** Find the equation of the plane tangent to the surface $z = x^2 + y^3$ at (3, 1, 10).

$f_x = 2x$, $\therefore f_x(3,1) = 6$

$f_y = 3y^2$, $\therefore f_y(3,1) = 3$

$\therefore z = f(3,1) + f_x(3,1)(x-3) + f_y(3,1)(y-1)$

$\therefore z = 10 + 6(x-3) + 3(y-1)$

$= 10 + 6x - 18 + 3y - 3$

$\therefore \underline{z = 6x + 3y - 11}$

6. **6.** Let $f(x, y) = e^{x+y}$. Find the equation for the tangent plane to the graph of f at the point (0, 0).

$f_x = f_y = e^{x+y}$ $\quad f_x(0,0) = f_y(0,0) = e^0 = 1$

$f(0,0) = e^0 = 1$. $\quad \therefore z = 1 + 1(x-0) + 1(y-0)$

$\therefore \underline{z = x + y + 1}$

7. **7.** Let $f(x, y) = e^{x-y}$. Find the equation for the tangent plane to the graph of f at the point (1, 1).

$f(1,1) = e^0 = 1$, $f_x = e^{x-y}$, $f_y = -e^{x-y}$

$\therefore f_x(1,1) = e^0 = 1$, $f_y(1,1) = -e^0 = -1$

$$\therefore z = 1 + 1(x-1) - 1(y-1)$$
$$= 1 + x - 1 - y + 1$$

$$\therefore \underline{z = x - y + 1}$$

8. **8.** Using the respective functions in Exercise 1, compute the plane tangent to the graphs at the indicated points.

(a) (0, 0)

(b) (0, 1)

(c) $(0, \pi)$

(d) (0, 1)

(a) $f(x,y) = xy$. $f_x = y$, $f_y = x$ $\quad f(0,0) = 0$

$f_x(0,0) = f_y(0,0) = 0$

$\therefore z = 0 + 0(x-0) + 0(y-0)$

$\therefore \underline{z = 0}$

(b) $f(x,y) = e^{xy}$. $\therefore f_x = ye^{xy}$, $f_y = xe^{xy}$

$f(0,1) = e^0 = 1$ $\quad f_x(0,1) = 1 \cdot e^0 = 1$ $\quad f_y(0,1) = 0 \cdot e^0 = 0$

$\therefore z = 1 + 1(x-0) + 0(y-1)$

$\therefore \underline{z = x + 1}$

(c) $f(x,y) = x\cos x \cos y$ $\quad f(0,\pi) = 0$

$f_x = \cos x \cos y - x \sin x \cos y$ $\quad f_x(0,\pi) = -1$

$f_y = -x \cos x \sin y$ $\quad f_y(0,\pi) = 0$

$\therefore z = 0 - 1(x-0) + 0(y-\pi)$

$\therefore \underline{z = -x}$

(d) $f(x,y) = (x^2+y^2)\log(x^2+y^2) \quad f(0,1) = 0$

$f_x = 2x\left[1 + \log(x^2+y^2)\right] \quad f_x(0,1) = 0$

$f_y = 2y\left[1 + \log(x^2+y^2)\right] \quad f_y(0,1) = 2$

$\therefore z = 0 + 0(x-0) + 2(y-1)$

$\therefore \underline{z = 2y - 2}$

9.

9. Compute the matrix of partial derivatives of the following functions:

(a) $f: \mathbb{R}^2 \to \mathbb{R}^2,\ f(x, y) = (x, y)$

(b) $f: \mathbb{R}^2 \to \mathbb{R}^3,\ f(x, y) = (xe^y + \cos y,\ x,\ x + e^y)$

(c) $f: \mathbb{R}^3 \to \mathbb{R}^2,\ f(x, y, z) = (x + e^z + y,\ yx^2)$

(d) $f: \mathbb{R}^2 \to \mathbb{R}^3,\ f(x, y) = (xye^{xy},\ x\sin y,\ 5xy^2)$

(a) $f_1(x,y) = x \quad \therefore f_{1x} = 1 \quad f_{1y} = 0$

$f_2(x,y) = y \quad f_{2x} = 0 \quad f_{2y} = 1$

$\therefore \underline{\int f(\vec{x}) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}}$

(b) $f_1(x,y) = xe^y + \cos y \quad \therefore f_{1x} = e^y,\ f_{1y} = xe^y - \sin y$

$f_2(x,y) = x \quad f_{2x} = 1,\ f_{2y} = 0$

$f_3(x,y) = x + e^y \quad f_{3x} = 1,\ f_{3y} = e^y$

$$\therefore Df(\vec{x}) = \begin{bmatrix} e^y & xe^y - \sin y \\ 1 & 0 \\ 1 & e^y \end{bmatrix}$$

(c) $f_1(x,y,z) = x + e^z + y \quad \therefore f_{1x} = 1 \quad f_{1y} = 1 \quad f_{1z} = e^z$

$f_2(x,y,z) = yx^2 \quad f_{2x} = 2xy \quad f_{2y} = x^2 \quad f_{2z} = 0$

$$\therefore Df(\vec{x}) = \begin{bmatrix} 1 & 1 & e^z \\ 2xy & x^2 & 0 \end{bmatrix}$$

(d) $f_1(x,y) = xye^{xy} \quad f_{1x} = ye^{xy} + xy^2e^{xy} \quad f_{1y} = xe^{xy} + x^2ye^{xy}$

$f_2(x,y) = x\sin y \quad f_{2x} = \sin y \quad f_{2y} = x\cos y$

$f_3(x,y) = 5xy^2 \quad f_{3x} = 5y^2 \quad f_{3y} = 10xy$

$$\therefore Df(\vec{x}) = \begin{bmatrix} ye^{xy} + xy^2e^{xy} & xe^{xy} + x^2ye^{xy} \\ \sin y & x\cos y \\ 5y^2 & 10xy \end{bmatrix}$$

10.

10. Compute the matrix of partial derivatives of

(a) $f(x, y) = (e^x, \sin xy)$

(b) $f(x, y, z) = (x - y, y + z)$

(c) $f(x, y) = (x + y, x - y, xy)$

(d) $f(x, y, z) = (x + z, y - 5z, x - y)$

(a) $f_1(x,y) = e^x \quad f_{1x} = e^x \quad f_{1y} = 0$

$f_2(x,y) = \sin xy \quad f_{2x} = y\cos xy \quad f_{2y} = x\cos xy$

$$\therefore Df(\vec{x}) = \begin{bmatrix} e^x & 0 \\ y\cos xy & x\cos xy \end{bmatrix}$$

(b) $f_1(x,y,z) = x - y \quad f_{1x} = 1 \quad f_{1y} = -1 \quad f_{1z} = 0$

$f_2(x,y,z) = y + z \quad f_{2x} = 0 \quad f_{2y} = 1 \quad f_{2z} = 1$

$$\therefore Df(\vec{x}) = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

(c) $f_1(x,y) = x + y \quad f_{1x} = 1 \quad f_{1y} = 1$

$f_2(x,y) = x - y \quad f_{2x} = 1 \quad f_{2y} = -1$

$f_3(x,y) = xy \quad f_{3x} = y \quad f_{3y} = x$

$$\therefore Df(\vec{x}) = \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ y & x \end{bmatrix}$$

(d) $f_1(x,y,z) = x + z \quad f_{1x} = 1 \quad f_{1y} = 0 \quad f_{1z} = 1$

$f_2(x,y,z) = y - 5z \quad f_{2x} = 0 \quad f_{2y} = 1 \quad f_{2z} = -5$

$f_3(x,y,z) = x - y \quad f_{3x} = 1 \quad f_{3y} = -1 \quad f_{3z} = 0$

$$\therefore Df(\vec{x}) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -5 \\ 1 & -1 & 0 \end{bmatrix}$$

11. **11.** Find the equation of the tangent plane to $f(x, y) = x^2 - 2xy + 2y^2$ having slope 2 in the positive x direction and slope 4 in the positive y direction.

$f_x = 2x - 2y \qquad f_y = -2x + 4y$

Let (x,y) be the point of tangency

$\therefore f_x(x,y) = 2 = 2x - 2y$

$f_y(x,y) = 4 = -2x + 4y$

Solving, $6 = 2y \Rightarrow y = 3$

$\therefore 2x - 2(3) = 2 \Rightarrow x = 4$

$\therefore$ tangent point $= (4,3)$

$\therefore f_x(4,3) = 2, \quad f_y(4,3) = 4$

$f(4,3) = 4^2 - 2(4)(3) + 2(3)^2 = 16 - 24 + 18 = 10$

$\therefore z = 10 + 2(x-4) + 4(y-3)$

$= 10 + 2x - 8 + 4y - 12$

$\therefore \underline{z = 2x + 4y - 10}$

12. **12.** Let $f(x, y) = e^{(2x+3y)}$.

(a) Find the tangent plane to f at $(0, 0)$.

(b) Use this to approximate $f(.1, 0)$ and $f(0, .1)$.

(c) With a calculator, find the exact values of $f(.1, 0)$ and $f(0, .1)$.

(a) $f(0,0) = e^0 = 1$

$f_x = 2e^{(2x+3y)} \qquad f_y = 3e^{(2x+3y)}$

$f_x(0,0) = 2 \qquad f_y(0,0) = 3$

$\therefore z = 1 + 2(x-0) + 3(y-0)$

$= 2x + 3y + 1$

(6) $f(.1, 0)$: $\Delta x = 0.1, \Delta y = 0$

$\therefore f(\Delta x, \Delta y) \approx 1 + 2\Delta x + 3\Delta y$

$= 1 + 2(0.1) + 3(0)$

$\therefore f(.1, 0) \approx 1.2$

$f(0, .1)$: $\Delta x = 0, \Delta y = 0.1$

$\therefore f(\Delta x, \Delta y) \approx 1 + 2\Delta x + 3\Delta y$

$= 1 + 2(0) + 3(0.1)$

$\therefore f(0, .1) \approx 1.3$

(c) $f(.1, 0) = e^{(2(.1) + 3(0))} = e^{.2} = 1.2214$

$f(0, .1) = e^{(2(0) + 3(.1))} = e^{.3} = 1.34986$

13.

13. Where does the plane tangent to $z = e^{x-y}$ at (1, 1, 1) meet the z axis?

$f(x, y) = e^{x-y}$ $\quad f(1, 1) = 1$

$f_x = e^{x-y}$ $\quad f_x(1, 1) = e^0 = 1$

$f_y = -e^{x-y}$ $\quad f_y(1, 1) = -e^0 = -1$

$\therefore$ Plane: $z = 1 + 1(x-1) - 1(y-1)$

$= x - y + 1$

Meet z-axis at $(0,0,z)$

$\therefore z = (0) - (0) + 1 = 1$

$\therefore$ Meet z-axis at $(0,0,1)$

14. **14.** Why should the graphs of $f(x, y) = x^2 + y^2$ and $g(x, y) = -x^2 - y^2 + xy^3$ be called "tangent" at $(0, 0)$?

$f(0,0) = 0 \quad f_x = 2x \quad f_y = 2y \quad f_x(0,0) = 0, \; f_y(0,0) = 0$

Tangent plane: $z = 0 + 0(x-0) + 0(y-0) = 0$

$\therefore z = 0$

$g(0,0) = 0 \quad g_x = -2x + y^3 \quad g_y = -2y + 3xy^2$

$g_x(0,0) = 0 \quad g_y(0,0) = 0$

Tangent plane: $z = 0 + 0(x-0) + 0(y-0) = 0$

$\therefore z = 0.$

Both $f(x,y)$ and $g(x,y)$ have a tangent plane of $z = 0$ at $(0,0)$.

15. **15.** Let $f(x, y) = e^{xy}$. Show that $x(\partial f/\partial x) = y(\partial f/\partial y)$.

$f_x = y e^{xy} \quad f_y = x e^{xy}.$

$\therefore x \dfrac{\partial f}{\partial x} = xy e^{xy} = yx e^{xy} = y \dfrac{\partial f}{\partial y}$

16. **16.** Use the linear approximation to approximate a suitable function $f(x, y)$ and thereby estimate the following:

(a) $(0.99e^{0.02})^8$

(b) $(0.99)^3 + (2.01)^3 - 6(0.99)(2.01)$

(c) $\sqrt{(4.01)^2 + (3.98)^2 + (2.02)^2}$

(a) Let $f(x,y) = (xe^y)^8 = x^8 e^{8y}$

Let $x=1,\ y=0,\ \Delta x = -0.01,\ \Delta y = 0.02$

$\therefore f(1,0) = 1^8 e^0 = 1 \quad f_x = 8x^7 e^{8y} \quad f_y = 8x^8 e^{8y}$

$\therefore f_x(1,0) = 8 \quad f_y(1,0) = 8$

$\therefore \quad z = 1 + 8(x-1) + 8(y-0)$

or $f(x,y) \approx 1 + 8(x-1) + 8(y-0)$

$\therefore f(1+\Delta x, 0+\Delta y) \approx 1 + 8\Delta x + 8\Delta y$

$\therefore f(0.99, 0.02) = f(1+(-0.01), 0+0.02)$

$\approx 1 + 8(-0.01) + 8(0.02)$

$= 1 - 0.08 + 0.16$

$= 1.08$

$\therefore (0.99e^{0.02})^8 \approx 1.08$

(b) $f(x,y) = x^3 + y^3 - 6xy$

$x = 1.00,\ \Delta x = -0.01 \Rightarrow 0.99$

$y = 2.00,\ \Delta y = 0.01 \rightarrow 2.01$

$$f(1,2) = 1^3 + 2^3 - 6(1)(2) = -3$$

$$f_x = 3x^2 - 6y \qquad f_x(1,2) = -9$$
$$f_y = 3y^2 - 6x \qquad f_y(1,2) = 6$$

$$\therefore z = -3 - 9(x-1) + 6(y-2)$$

$$\therefore f(1+\Delta x, 2+\Delta y) \simeq -3 - 9\Delta x + 6\Delta y$$

$$\therefore f(0.99, 2.01) \simeq -3 - 9(-0.01) + 6(0.01)$$
$$= -3 + 0.09 + 0.06$$
$$= 2.85$$

$$\therefore (0.99)^3 + (2.01)^3 - 6(0.99)(2.01) \simeq \underline{2.85}$$

(c) $f(x,y,z) = \sqrt{x^2+y^2+z^2}$

$x = 4.00 \quad \Delta x = 0.01 \Rightarrow 4.01$
$y = 4.00 \quad \Delta y = -0.02 \Rightarrow 3.98$
$z = 2.00 \quad \Delta z = 0.02 \Rightarrow 2.02$

$$f(4,4,2) = \sqrt{4^2+4^2+2^2} = \sqrt{36} = 6$$

$$f_x = \tfrac{1}{2}(x^2+y^2+z^2)^{-\frac{1}{2}}(2x) = x/\sqrt{x^2+y^2+z^2}$$
$$\therefore f_x(4,4,2) = \tfrac{4}{6} = \tfrac{2}{3}$$

$$f_y = y/\sqrt{x^2+y^2+z^2} \qquad f_y(4,4,2) = 2/3$$

$$f_z = z/\sqrt{x^2+y^2+z^2} \qquad f_z(4,4,2) = \tfrac{2}{6} = \tfrac{1}{3}$$

$$\therefore z = 6 + \tfrac{2}{3}(x-4) + \tfrac{2}{3}(y-4) + \tfrac{1}{3}(z-2)$$

$\therefore f(4+\Delta x, 4+\Delta y, 2+\Delta z) = 6 + \frac{2}{3}\Delta x + \frac{2}{3}\Delta y + \frac{1}{3}\Delta z$

$\therefore f(4.01, 3.98, 2.02) \simeq 6 + \frac{2}{3}(0.01) + \frac{2}{3}(-0.02) + \frac{1}{3}(0.02)$

$= 6 + \frac{2 - 4 + 2}{3(100)} = 6 + 0 = 6$

$\therefore \sqrt{(4.01^2 + (3.98)^2 + (2.02)^2} \simeq 6.00$

17. **17.** Let P be the tangent plane to the graph of $g(x, y) = 8 - 2x^2 - 3y^2$ at the point $(1, 2, -6)$. Let $f(x, y) = 4 - x^2 - y^2$. Find the point on the graph of f which has tangent plane parallel to P.

$g(1,2) = -6 \quad g_x = -4x \quad g_x(1,2) = -4$

$g_y = -6y \quad g_y(1,2) = -12$

$\therefore z = -6 - 4(x-1) - 12(y-2)$

or $4(x-1) + 12(y-2) + z = -6$

or $(4, 12, 1) \cdot (x-1, y-2, z) = -6$ (dot product)

$\therefore$ normal to plane $= (4, 12, 1)$

$f_x = -2x \quad f_y = -2y$

$\therefore z = f(x_0, y_0) - 2x_0(x - x_0) - 2y_0(y - y_0)$

$\therefore (2x_0, 2y_0, 1) \cdot (x - x_0, y - y_0, z) = f(x_0, y_0)$ (Dot product)

Make normals parallel

$\therefore (2x_0, 2y_0, 1) = (4, 12, 1) \quad \therefore x_0 = 2, \; y_0 = 6$

$\therefore f(2,6) = -36 \quad \therefore$ At $(2, 6, -36)$

18.

18. Let $f(x, y) = xe^{y^2} - ye^{x^2}$.

(a) Find the equation for the tangent plane to the graph of f at $(1, 2)$.

(b) What point on the surface $z = x^2 - y^2$ has a tangent plane parallel to the plane found in part (a)?

(a) $f(1,2) = e^4 - 2e$

$f_x = e^{y^2} - 2xye^{x^2} \qquad f_y = 2xye^{y^2} - e^{x^2}$

$f_x(1,2) = e^4 - 4e \qquad f_y(1,2) = 4e^4 - e$

$\therefore z = (e^4 - 2e) + (e^4 - 4e)(x-1) + (4e^4 - e)(y-2)$

Or, $(4e - e^4, e - 4e^4, 1) \cdot (x-1, y-2, z) = e^4 - 2e$

$\therefore$ Normal is: $(4e - e^4, e - 4e^4, 1)$

(b) Let (x_0, y_0, z_0) be the tangent point, $f(x,y) = x^2 - y^2$

$\therefore f_x = 2x \quad f_y = -2y$

$\therefore$ Tangent plane: $z = (x_0^2 - y_0^2) + 2x_0(x - x_0) - 2y_0(y - y_0)$

or $(-2x_0, 2y_0, 1) \cdot (x - x_0, y - y_0, z) = x_0^2 - y_0^2$

$\therefore$ Normal at (x_0, y_0) is $(-2x_0, 2y_0, 1)$

Make normals parallel.

$\therefore (-2x_0, 2y_0, 1) = (4e - e^4, e - 4e^4, 1)$

$\therefore x_0 = \frac{4e - e^4}{-2} = \frac{1}{2}e^4 - 2e$

$y_0 = \frac{e - 4e^4}{2} = \frac{1}{2}e - 2e^4$

$\therefore$ at $\left(\frac{1}{2}e^4 - 2e, \frac{1}{2}e - 2e^4, z_0\right)$

$z_0 = \left(\frac{1}{2}e^4 - 2e\right)^2 - \left(\frac{1}{2}e - 2e^4\right)^2$ (from $z = f(x,y)$)

$= \frac{1}{4}e^8 - 2e^5 + 4e^2 - \left[\frac{1}{4}e^2 - 2e^5 + 4e^8\right]$

$= -\frac{15}{4}e^8 + \frac{15}{4}e^2$

$\therefore$ at $\left(\frac{1}{2}e^4 - 2e, \frac{1}{2}e - 2e^4, -\frac{15}{4}e^8 + \frac{15}{4}e^2\right)$

19.

19. Compute the gradients of the following functions:

(a) $f(x, y, z) = x \exp(-x^2 - y^2 - z^2)$ (Note that $\exp u = e^u$.)

(b) $f(x, y, z) = \dfrac{xyz}{x^2 + y^2 + z^2}$

(c) $f(x, y, z) = z^2 e^x \cos y$

(a) $f_x = e^{-x^2-y^2-z^2} - 2x^2 e^{-x^2-y^2-z^2}$

$f_y = -2xy\, e^{-x^2-y^2-z^2}$ $\quad f_z = -2xz\, e^{-x^2-y^2-z^2}$

$\therefore \nabla f = e^{-x^2-y^2-z^2}(1 - 2x^2, -2xy, -2xz)$

(b) $f_x = \dfrac{(x^2+y^2+z^2)(yz) - xyz(2x)}{(x^2+y^2+z^2)^2}$

$f_y = \dfrac{(x^2+y^2+z^2)(xz) - xyz(2y)}{(x^2+y^2+z^2)^2}$

$f_z = \dfrac{(x^2+y^2+z^2)(xy) - xyz(2z)}{(x^2+y^2+z^2)^2}$

$\therefore \nabla f = \dfrac{1}{x^2+y^2+z^2}\left[yz(-x^2+y^2+z^2),\ xz(x^2-y^2+z^2),\ xy(x^2+y^2-z^2)\right]$

(c) $f_x = (z^2\cos y)e^x \quad f_y = -z^2 e^x \sin y \quad f_z = 2ze^x\cos y$

$\therefore \nabla f = \left[z^2e^x\cos y,\ -z^2e^x\sin y,\ 2ze^x\cos y\right]$

20. **20.** Compute the tangent plane at (1, 0, 1) for each of the functions in Exercise 19. [The solution to part (c) only is in the Study Guide.]

(a) $\nabla f(1,0,1) = e^{-2}(-1,0,-2) = (-e^{-2}, 0, -2e^{-2})$

$\therefore (e^{-2}, 0, -2e^{-2}) \cdot (x-1, y-0, z-1) = 0 \quad [\vec{n}\cdot\vec{r} = 0]$

Or, $e^{-2}(x-1) - 2e^{-2}(z-1) = 0 \quad$ [now multiply by e^2]

Or, $(x-1) - 2(z-1) = 0$, or $x - 2z + 1 = 0$

(b) $\nabla f(1,0,1) = \frac{1}{2}[0, 2, 0] = (0, 1, 0)$

$\therefore (0,1,0)\cdot(x-1, y-0, z-1) = 0$

Or, $\underline{y = 0}$

(c) $\nabla f(1,0,1) = (e, 0, 2e)$

$\therefore (e,0,2e)\cdot(x-1, y-0, z-1) = 0$

Or, $e(x-1) + 2e(z-1) = 0, \quad \therefore x - 1 + 2z - 2 = 0$

$\therefore \underline{x + 2z - 3 = 0}$

21.

21. Find the equation of the tangent plane to $z = x^2 + 2y^3$ at (1, 1, 3).

Let $z = f(x,y) \quad \therefore f(1,1) = 3 \quad f_x = 2x \quad f_y = 6y^2$

$f_x(1,1) = 2 \quad f_y(1,1) = 6$

$\therefore z = 3 + 2(x-1) + 6(y-1)$

Or, $\underline{2x + 6y - z - 5 = 0}$

22.

22. Let

$$f(x, y) = \begin{cases} \frac{x^2 y^4}{x^4 + 6y^8} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

(a) Show that $\frac{\partial f}{\partial x}(0, 0)$ and $\frac{\partial f}{\partial y}(0, 0)$ exist.

(b) Show that f is not differentiable at (0,0) by showing that f is not continuous at (0,0).

(a) $\frac{\partial f}{\partial x}(0,0) = \lim_{h\to 0} \frac{f(x+h,0)-f(x,0)}{h}\Big|_{x=0}$

$$\lim_{h\to 0} \frac{\frac{(0+h)^2 0^4}{(0+h)^4+6^8} - \frac{0^2 0^4}{0^4+6^8}}{h} = \lim_{h\to 0} \frac{\frac{0}{h^4+0} - 0}{h}$$

$$= \lim_{h\to 0} 0 = 0 \quad \therefore \frac{\partial f}{\partial x}(0,0) = 0$$

Similarly, $\frac{\partial f}{\partial y}(0,0) = \lim_{h\to 0} \frac{f(0,y+h)-f(0,y)}{h}\Big|_{y=0}$

$$\lim_{h\to 0} \frac{\frac{0^2(0+h)^4}{0^4+6(0+h)^8} - \frac{0^2 0^4}{0^4+6^8}}{h} = \lim_{h\to 0} 0 = 0$$

$$\therefore \frac{\partial f}{\partial y}(0,0) = 0$$

(b) Along the path $y = \sqrt{x}$,

$$\lim_{(x,y)\to(0,0)} \frac{x^2 y^4}{x^4+6y^8} = \lim_{x\to 0} \frac{x^2(\sqrt{x})^4}{x^4+6(\sqrt{x})^8} = \lim_{x\to 0} \frac{x^4}{x^4+6x^4}$$

$$= \lim_{x\to 0} \frac{x^4}{7x^4} = \lim_{x\to 0} \frac{1}{7} = \frac{1}{7} \neq 0.$$

$$\therefore \lim_{(x,y)\to(0,0)} f(x,y) \neq f(0,0) \quad \therefore f(x,y) \text{ is not continuous}$$

23.

23. Let P be the tangent plane to $f(x, y) = x^2 y^3$ at $(1, 2, 8)$. Let l be the line contained in P which passes through the point $(1, 3, 20)$ and passes directly above $(2, 1)$. That is, l contains the point $(1, 3, 20)$ and a point of the form $(2, 1, z)$. Find a parametrization for l.

The tangent plane contains the vector from

$(1,2,8)$ to $(1,3,20) = (0,1,12)$

and the vector from $(1,2,8)$ to $(2,1,z) = (1,-1,z-8)$

These two vectors are perpendicular to normal vector for P.

$\therefore$ cross product is parallel to normal.

Cross product: $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 1 & 12 \\ 1 & -1 & z-8 \end{vmatrix} = (z+4, 12, -1)$

To find normal to tangent plane:

$f_x = 2xy^3 \quad f_y = 3y^2x^2$

$f_x(1,2) = 16 \quad f_y(1,2) = 12 \quad f(1,2) = 8$

$\therefore$ Tangent plane: $z = 8 + 16(x-1) + 12(y-2)$

or, $16x + 12y - z = 32$

$\therefore$ Normal to plane: $(16, 12, -1)$

$\therefore (16, 12, -1) = (z+4, 12, -1)$

$\therefore$ For line l, $z = 12$, and $(2,1,z) = (2,1,12)$

$\therefore$ l contains $(1,3,20)$ and $(2,1,12)$
and is parallel to vector $(-1,2,8)$

$\therefore$ $\underline{l(t) = (1,3,20) + t(-1,2,8)}$

24. **24.** Calculate $\nabla h(1, 1, 1)$ if $h(x, y, z) = (x + z)\, e^{x-y}$.

$h_x = e^{x-y} + (x+z)e^{x-y}$ $\quad h_x(1,1,1) = 1 + 2 = 3$

$h_y = -(x+z)e^{x-y}$ $\quad h_y(1,1,1) = -2$

$h_z = e^{x-y}$ $\quad h_z(1,1,1) = 1$

$\therefore$ $\nabla h(1,1,1) = \underline{(3,-2,1)}$

25. **25.** Let $f(x, y, z) = x^2 + y^2 - z^2$. Calculate $\nabla f(0, 0, 1)$.

$f_x = 2x$ $\quad f_x(0,0,1) = 0$

$f_y = 2y$ $\quad f_y(0,0,1) = 0$

$f_z = -2z$ $\quad f_z(0,0,1) = -2$

$\therefore$ $\nabla f(0,0,1) = \underline{(0,0,-2)}$

26. **26.** Evaluate the gradient of $f(x, y, z) = \log(x^2 + y^2 + z^2)$ at $(1, 0, 1)$.

$f_x = \frac{2x}{x^2+y^2+z^2}$ $\quad f_x(1,0,1) = 1$

$f_y = \frac{2y}{x^2+y^2+z^2}$ $\quad f_y(1,0,1) = 0$

$f_z = \frac{2z}{x^2+y^2+z^2}$ $\quad f_z(1,0,1) = 1$

$\therefore \nabla f(1,0,1) = \underline{(1,0,1)}$

27. **27.** Describe all Hölder-continuous functions with $\alpha > 1$ (see Exercise 33, Section 2.2). (HINT: What is the derivative of such a function?)

From Exercise 33, Section 2.2.,

> **33.** Let $f: A \subset \mathbb{R}^n \to \mathbb{R}^m$ satisfy $\| f(\mathbf{x}) - f(\mathbf{y}) \| \leq K \|\mathbf{x} - \mathbf{y}\|^\alpha$ for all $\mathbf{x}$ and $\mathbf{y}$ in A for positive constants K and α. Show that f is continuous. (Such functions are called ***Hölder-continuous*** or, if $\alpha = 1$, ***Lipschitz-continuous***.)

$\therefore$ Let $f: A \subset \mathbb{R}^n \to \mathbb{R}^m$. Since it's Hölder-continuous, $\exists\, K, \alpha$ s.t. $\| f(\vec{x}) - f(\vec{y}) \| \leq K \|\vec{x} - \vec{y}\|^\alpha$, $\forall \vec{x}, \vec{y} \in A$, and $\alpha > 1$.

Note: $\frac{\| f(\vec{x}) - f(\vec{y}) \|}{\|\vec{x} - \vec{y}\|} \leq K \|x - y\|^{\alpha-1}$, for $\vec{x} \neq \vec{y}$.

Let $\epsilon > 0$. Choose $\delta = \sqrt[\alpha-1]{\frac{\epsilon}{K}}$. Note $\alpha - 1 > 0$

Let any $\vec{y} \in A$, and let $\vec{y} = (y_1, \ldots, y_n)$.

$\therefore$ if $\|\vec{x}-\vec{y}\| < \delta$, and $x \in A$, then

$\|\vec{x}-\vec{y}\| < \sqrt[\alpha-1]{\frac{\epsilon}{k}}$, so $\|\vec{x}-\vec{y}\|^{\alpha-1} < \frac{\epsilon}{k}$

(since for $0<a<b$, $r>0$, then $a^r < b^r$)

$\therefore$ $k\|\vec{x}-\vec{y}\|^{\alpha-1} < \epsilon \Rightarrow k\|\vec{x}-\vec{y}\|^{\alpha} < \epsilon\|\vec{x}-\vec{y}\|$

But $\|f(\vec{x})-f(\vec{y})\| < k\|\vec{x}-\vec{y}\|^{\alpha}$

So $\|f(\vec{x})-f(\vec{y})\| < \epsilon\|\vec{x}-\vec{y}\|$

Note, for $i=1,\dots,m$, $|f_i(\vec{x})-f_i(\vec{y})| \le \|f(\vec{x})-f(\vec{y})\|$

and for $j=1,\dots,n$, $\|(y_1,\dots,y_j+h,\dots,y_n)-(y_1,\dots,y_j,\dots,y_n)\| = |h|$

$\therefore$ Consider $\vec{x_j} = (y_1,\dots,y_j+h,\dots,y_n)$

Given the $\epsilon>0$ above and $\delta = \sqrt[\alpha-1]{\frac{\epsilon}{k}}$ as above,

if $\|\vec{x_j}-\vec{y}\| = |h| < \delta$, then

$|f_i(\vec{x_j})-f_i(\vec{y})| \le \|f(\vec{x_j})-f(\vec{y})\| < \epsilon\|\vec{x_j}-\vec{y}\| = \epsilon|h|$

$\therefore$ $\left|\frac{f_i(\vec{x_j})-f(\vec{y})}{h} - 0\right| < \epsilon$

or $\left|\frac{f_i(y_1,\dots,y_j+h,\dots,y_n)-f_i(y_1,\dots,y_j,\dots y_n)}{h} - 0\right| < \epsilon$

$$\therefore \lim_{h \to 0} \frac{f_i(y_1, \ldots, y_j + h, \ldots, y_n) - f_i(y_1, \ldots, y_j, \ldots, y_n)}{h} = 0$$

$$\therefore \frac{\partial f_i}{\partial y_j}(\vec{y}) = 0 \text{ , for all } y \in A, \text{ and } j = 1, 2, \ldots, n$$

$\therefore f_i(\vec{y})$ is constant along the y_j component of $\vec{y}$. Call this vector $\vec{y_j}$. $f_i(\vec{y_j}) = c_{ij}$

Similarly, $\frac{\partial f_i}{\partial y_k}(\vec{y}) = 0 \Rightarrow f_i(\vec{y_k}) = c_{ik}$, $j \neq k$.

But $f(\vec{y})$ is continuous (Exercise 33, sec. 2.2)

$\therefore f_i(\vec{y})$ is continuous, and in particular is continuous at $(y_1, \ldots, y_j, \ldots, y_k, \ldots, y_n)$ so that $c_{ij} = c_{ik}$.

That is, the value of $f_i(\vec{y})$ along the j component must equal the value along the k component

$\therefore$ for all $y \in A$, $f_i(\vec{y}) = c_i$ $(c_i = c_{ij} = c_{ik})$.

$\therefore f(\vec{y}) = (c_1, c_2, \ldots, c_m)$, a constant for all $y \in A$.

$\therefore$ Hölder-continuous function with $\alpha > 1 \Rightarrow$ function is constant

Note: $\alpha > 1$ was needed for $0 < a < b \Rightarrow a^{\alpha-1} < b^{\alpha-1}$, since $\alpha - 1 > 0$.

28.

28. Suppose $f: \mathbb{R}^n \to \mathbb{R}^m$ is a linear map. What is the derivative of f?

For $\vec{x} \in \mathbb{R}^n$, $\vec{x} = (x_1, \ldots, x_n) = x_1 \hat{e}_1 + \ldots + x_n \hat{e}_n$,

where $\hat{e}_i = (0, \ldots, \underset{\uparrow}{1}, \ldots 0)$, the natural basis.
ith component

$\therefore f(\vec{x}) = f(x_1 \hat{e}_1 + \ldots + x_n \hat{e}_n) = x_1 f(\hat{e}_1) + \ldots + x_n f(\hat{e}_n)$.

Note $f(\vec{x}) = [f_1(\vec{x}), \ldots, f_m(\vec{x})]$

$\therefore f(\vec{x}) = x_1 [f_1(\hat{e}_1), \ldots, f_m(\hat{e}_1)] + \ldots + x_n [f_1(\hat{e}_n), \ldots, f_m(\hat{e}_n)]$

$$\therefore f(\vec{x}) = \begin{bmatrix} f_1(\hat{e}_1) & f_1(\hat{e}_2) & \ldots & f_1(\hat{e}_n) \\ f_2(\hat{e}_1) & f_2(\hat{e}_2) & \ldots & f_2(\hat{e}_n) \\ \vdots & \vdots & & \vdots \\ f_m(\hat{e}_1) & f_m(\hat{e}_2) & & f_m(\hat{e}_n) \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Let T be the matrix $\begin{bmatrix} f_1(\hat{e}_1) & \cdots & f_1(\hat{e}_n) \\ \vdots & & \vdots \\ f_m(\hat{e}_1) & \ldots & f_m(\hat{e}_n) \end{bmatrix}$. $\therefore f(\vec{x}) = T\vec{x}$

Must prove: (1) Partial derivatives of f exist.
(2) Above matrix T contains the partial derivatives of f
(3) $\lim_{\vec{x} \to \vec{x}_0} \dfrac{\| f(\vec{x}) - f(\vec{x}_0) - T(\vec{x} - \vec{x}_0) \|}{\| \vec{x} - \vec{x}_0 \|} = 0$

Then the derivative of f will be the matrix T.

(3) Assuming (1) & (2) are true, letting $\vec{h} = \vec{x} - \vec{x}_0$,

then $$\lim_{\vec{h}\to\vec{0}} \frac{\| f(\vec{x}_0+\vec{h}) - f(\vec{x}_0) - T\vec{h}\|}{\|\vec{h}\|} =$$

$$\lim_{\vec{h}\to\vec{0}} \frac{\| f(\vec{x}_0) + f(\vec{h}) - f(\vec{x}_0) - T\vec{h}\|}{\|\vec{h}\|} =$$

$$\lim_{\vec{h}\to\vec{0}} \frac{\| f(\vec{h}) - T\vec{h}\|}{\|\vec{h}\|} = \lim_{\vec{h}\to\vec{0}} \frac{\| T\vec{h} - T\vec{h}\|}{\|\vec{h}\|} = 0$$

since, as defined above, $f(\vec{x}) = T\vec{x}$ since f is linear.

(1) Note: if $g: \mathbb{R}^n \to \mathbb{R}^m$, where

$g(\vec{h}) = (g_1(\vec{h}), \dots, g_m(\vec{h}))$, $\vec{h} \in \mathbb{R}^n$, $g_i: \mathbb{R}^n \to \mathbb{R}$

Then $\lim_{\vec{h}\to\vec{0}} \frac{\|g(\vec{h})\|}{\|\vec{h}\|} = 0 \iff \lim_{\vec{h}\to\vec{0}} \frac{|g_i(\vec{h})|}{\|\vec{h}\|} = 0$,

$i = 1, 2, \dots, m$

$\therefore$ Letting $g(\vec{h}) = f(\vec{x}_0 + \vec{h}) - f(\vec{x}_0) - T\vec{h}$,

only need to prove: $\lim_{\vec{h}\to\vec{0}} \frac{|f_i(\vec{x}_0+\vec{h}) - f_i(\vec{x}_0) - (T\vec{h})_i|}{\|\vec{h}\|} = 0$

where $i = 1, \dots, m$

and $(T\vec{h})_i$ = ith entry in $m \times 1$ column vector.

But $f_i(\vec{x}_0 + \vec{h}) = f_i(\vec{x}_0) + f_i(\vec{h})$ since

f is linear, and $\therefore$ so is f_i

$$\therefore \left| f_i(\vec{x}_0 + \vec{h}) - f_i(\vec{x}_0) - (T\vec{h})_i \right| = \left| f_i(\vec{h}) - (T\vec{h})_i \right|$$

Let $\vec{h} = (h_1, \ldots, h_n) = h_1 \hat{e}_1 + \ldots h_n \hat{e}_n$

Then $f_i(\vec{h}) = h_1 f_i(\hat{e}_1) + \ldots + h_n f_i(\hat{e}_n)$

But $(T\vec{h})_i = h_1 f_i(\hat{e}_1) + \ldots + h_n f_i(\hat{e}_n)$

$$\therefore f_i(\vec{h}) - (T\vec{h})_i = 0$$

$$\therefore \lim_{\vec{h} \to \vec{0}} \frac{\left| f_i(\vec{x}_0 + \vec{h}) - f_i(\vec{x}_0) - (T\vec{h})_i \right|}{\|\vec{h}\|} =$$

$$\lim_{\vec{h} \to \vec{0}} \frac{0}{\|\vec{h}\|} = 0$$

Now consider $\vec{h} = a\hat{e}_j$, $j = 1, \ldots, n$

$$\therefore \|\vec{h}\| = |a|, \text{ and } \|\vec{h}\| \to \vec{0} \Leftrightarrow a \to 0.$$

$$\therefore \lim_{a \to 0} \frac{\left| f_i(\vec{x}_0 + a\hat{e}_j) - f_i(\vec{x}_0) - T(a\hat{e}_j)_i \right|}{|a|} = 0$$

$$\text{or } \lim_{a \to 0} \left| \frac{f_i(\vec{x}_0 + a\hat{e}_j) - f_i(\vec{x}_0) - a(T\hat{e}_j)_i}{a} \right| = 0$$

$$\text{or } \lim_{a \to 0} \left| \frac{f_i(\vec{x}_0 + a\hat{e}_j) - f_i(\vec{x}_0)}{a} - (T\hat{e}_j)_i \right| = 0$$

$$\therefore \lim_{a \to 0} \frac{f_i(\vec{x}_0 + a\hat{e}_j) - f_i(\vec{x}_0)}{a} = (T\hat{e}_j)_i$$

$$\text{But } \frac{\partial f_i}{\partial x_j} = \lim_{a \to 0} \frac{f_i(\vec{x}_0 + a\hat{e}_j) - f_i(\vec{x}_0)}{a}$$

$$\therefore \frac{\partial f_i}{\partial x_j} \text{ exists and is } (T\hat{e}_j)_i = f_i(\hat{e}_j)$$

i.e., The partials exist, proving (1), and T is the matrix of partial derivatives

$$(2) \quad \begin{bmatrix} f_1(\hat{e}_1) & \cdots & f_1(\hat{e}_n) \\ \vdots & & \\ f_m(\hat{e}_1) & \cdots & f_m(\hat{e}_n) \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

$\therefore$ For linear $f: \mathbb{R}^n \to \mathbb{R}^m$, the derivative is the linear map itself, represented by its matrix T.

Example: $f(\vec{x}) = (\vec{a}\cdot\vec{x}, \vec{b}\cdot\vec{x}, \ldots \vec{\rho}\cdot\vec{x})$

$$\therefore T = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \\ b_1 & b_2 & \cdots & b_n \\ \vdots & & & \\ \rho_1 & \rho_2 & \cdots & \rho_n \end{bmatrix}$$

and so $\dfrac{\partial f_2}{\partial x_7} = b_7$ (assuming $n \geq 7$)

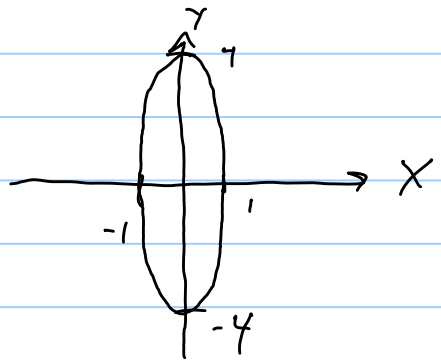
2.4 Introduction to Paths and Curves

Note Title 2/9/2016

Sketch the curves that are the images of the paths in Exercises 1 to 4.

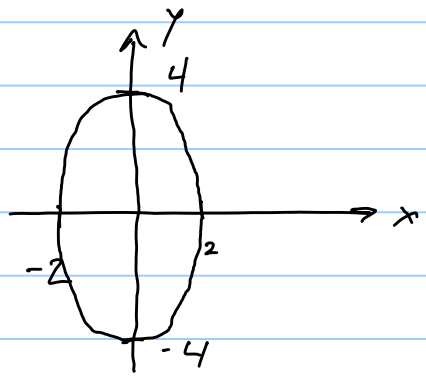
1. **1.** $x = \sin t,\ y = 4\cos t,$ where $0 \le t \le 2\pi$

At $t = 0 : (0, 4)$ ∴ Ellipse
$t = \pi/2 : (1, 0)$
$t = \pi : (0, -4)$ $\left(\frac{x}{1}\right)^2 + \left(\frac{y}{4}\right)^2 = 1$
$t = 3\pi/2 : (-1, 0)$



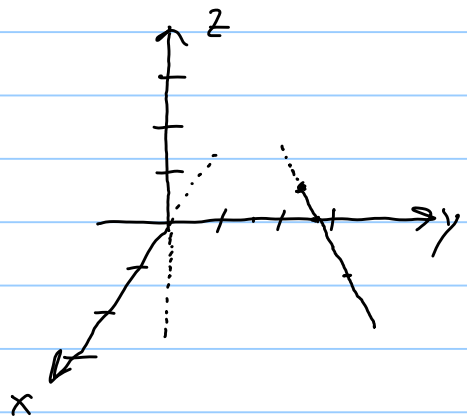
2. **2.** $x = 2\sin t,\ y = 4\cos t,$ where $0 \le t \le 2\pi$

$t = 0 : (0, 4)$ ∴ Ellipse
$t = \pi/2 : (2, 0)$
$t = \pi : (0, -4)$ $\left(\frac{x}{2}\right)^2 + \left(\frac{y}{4}\right)^2 = 1$
$t = 3\pi/2 : (-2, 0)$



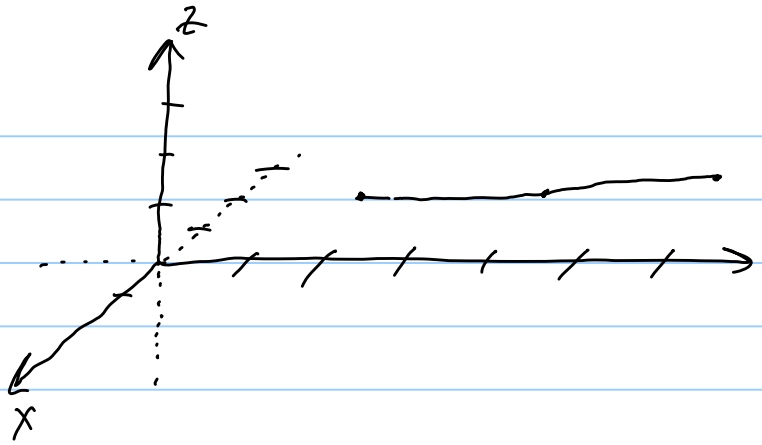
3. **3.** $\mathbf{c}(t) = (2t - 1,\ t + 2,\ t)$

$t = 0 : (-1, 2, 0)$ $(-1, 2, 0) + t(2, 1, 1)$
$t = 1 : (1, 3, 1)$ a line
$t = 2 : (3, 4, 2)$



4. **4.** $\mathbf{c}(t) = (-t,\ 2t,\ 1/t),$ where $1 \le t \le 3$

$t = 1 : (-1, 2, 1)$ $t = 2 : \left(-2, 4, \frac{1}{2}\right)$ $t = 3 : \left(-3, 6, \frac{1}{3}\right)$



Part of "$\frac{1}{x}$" curve that gets closer to xy-plane as projection onto xy-plane goes out toward $(-\infty, +\infty)$.

5.

5. Consider the circle C of radius 2, centered at the origin.

(a) Find a parametrization for C inducing a counterclockwise orientation and starting at (2, 0).

$\vec{c}(t) = 2(m, m)$, where m is some function of $\cos t$ or $\sin t$

at $t = 0$, $(2, 0)$, so $2(\cos m, \sin m)$

$\therefore \underline{\vec{c}(t) = 2(\cos t, \sin t)}$

(b) Find a parametrization for C inducing a clockwise orientation and starting at (0, 2).

(c) Find a parametrization for C if it is now centered at the point (4, 7).

(b) To shift starting point $\frac{\pi}{2}$ radians from (a),

use $2\left(\cos\left(\frac{\pi}{2} + t\right), \sin\left(\frac{\pi}{2} + t\right)\right)$

To make it clockwise, use a reflection with respect to y-axis: $(x, y) \to (-x, y)$.

$\therefore 2\left[-\cos\left(\frac{\pi}{2} + t\right), \sin\left(\frac{\pi}{2} + t\right)\right]$

To simplify: $\cos\left(\frac{\pi}{2}+x\right) = \cos\frac{\pi}{2}\cos x - \sin\frac{\pi}{2}\sin x$

$= -\sin x$

$\sin\left(\frac{\pi}{2}+x\right) = \sin\frac{\pi}{2}\cos x + \cos\frac{\pi}{2}\sin x$

$= \cos x$

$\therefore \vec{c}(t) = 2(\sin t, \cos t)$

(c) From (a), $\vec{c}(t) = 2(\cos t, \sin t) + (4, 7)$

$= (4 + 2\cos t, 7 + 2\sin t)$

6.

6. Give a parametrization for each of the following curves:

(a) The line passing through (1, 2, 3) and (−2, 0, 7)

(b) The graph of $f(x) = x^2$

(c) The square with vertices (0, 0), (0, 1), (1, 1), and (1, 0) (Break it up into line segments.)

(d) The ellipse given by $\frac{x^2}{9} + \frac{y^2}{25} = 1$

(a) $(1,2,3) - (-2,0,7) = (3,2,-4)$.

$\therefore \vec{c}(t) = (1,2,3) + t(3,2,-4)$

(b) $\vec{c}(t) = (t, t^2)$

(c) $(0,0)$ to $(0,1)$: $(0,1) - (0,0) = (0,1)$.

$\therefore (0,0) + s(0,1) = (0,s)$, $0 \le s \le 1$

$(0,1)$ to $(1,1)$: $(1,1) - (0,1) = (1,0)$

$\therefore (0,1) + s(1,0) = (s,1)$, $0 \le s \le 1$

or $(s-1, 1)$, $1 \le s \le 2$

$(1,1)$ to $(1,0)$: $(1,0)-(1,1)=(0,-1)$

$\therefore (1,1)+s(0,-1)=(1,1-s),\ 0\le s\le 1$

or, $(1,1-(s-2))=(1,3-s),\ 2\le s\le 3$

$(1,0)$ to $(0,0)$: $(0,0)-(1,0)=(-1,0)$

$\therefore (1,0)+s(-1,0)=(1-s,0),\ 0\le s\le 1$

or $(1-(s-3),0)=(4-s,0),\ 3\le s\le 4$

$\therefore$

$$\vec{c}(t)=\begin{cases}(0,s), & 0\le s\le 1\\ (s-1,1), & 1\le s\le 2\\ (1,3-s), & 2\le s\le 3\\ (4-s,0), & 3\le s\le 4\end{cases}$$

(d) If $x=3\cos\theta$, $y=5\sin\theta$, $0\le\theta\le 2\pi$, then

$$\left(\frac{x}{3}\right)^2+\left(\frac{y}{5}\right)^2=\frac{9\cos^2\theta}{9}+\frac{25\sin^2\theta}{25}=1.$$

$\therefore \vec{c}(\theta)=(3\cos\theta,5\sin\theta),\ 0\le\theta\le 2\pi$

In Exercises 7 to 10, determine the velocity vector of the given path.

7. **7.** $\mathbf{c}(t)=6t\mathbf{i}+3t^2\mathbf{j}+t^3\mathbf{k}$

$$\vec{c}'(t)=6\hat{i}+6t\hat{j}+3t^2\hat{k}$$

8. **8.** $\mathbf{c}(t)=(\sin 3t)\mathbf{i}+(\cos 3t)\mathbf{j}+2t^{3/2}\mathbf{k}$

$$\vec{c}'(t)=(3\cos t)\hat{i}-(3\sin t)\hat{j}+3t^{\frac{1}{2}}\hat{k}$$

9. **9.** $\mathbf{r}(t) = (\cos^2 t, 3t - t^3, t)$

$\vec{r}\,'(t) = (2\cos t(-\sin t), 3 - 3t^2, 1)$

since $\sin 2\theta = 2\sin\theta\cos\theta$

$= (-\sin 2t, 3 - 3t^2, 1)$

10. **10.** $\mathbf{r}(t) = (4e^t, 6t^4, \cos t)$ Assume typo, and e^t means e^t

$\vec{r}\,'(t)\ (4e^t, 24t^3, -\sin t)$

In Exercises 11 to 14, compute the tangent vectors to the given path.

11. **11.** $\mathbf{c}(t) = (e^t, \cos t)$

$\vec{c}\,'(t) = (e^t, -\sin t)$

12. **12.** $\mathbf{c}(t) = (3t^2, t^3)$

$\vec{c}\,(t) = (6t, 3t^2)$

13. **13.** $\mathbf{c}(t) = (t\sin t, 4t)$

$\vec{c}\,'(t) = (\sin t + t\cos t, 4)$

14. **14.** $\mathbf{c}(t) = (t^2, e^2)$

$\vec{c}\,(t) = (2t, 0)$

15. **15.** When is the velocity vector of a point on the rim of a rolling wheel *horizontal*? What is the speed at this point?

From Example 4, p. 119, $\mathbf{c}(t) = \left(vt - r\sin\frac{vt}{R},\ R - r\cos\frac{vt}{R}\right)$.

(a) For a point on a wheel, $r = R$

$\therefore \vec{c}(t) = \left(vt - R\sin\frac{vt}{R},\ R - R\cos\frac{vt}{R}\right)$

$\therefore \vec{c}'(t) = \left(v - v\cos\frac{vt}{R},\ vt\sin\frac{vt}{R}\right)$

$= v\left(1 - \cos\frac{vt}{R},\ t\sin\frac{vt}{R}\right)$

$\vec{c}'(t)$ is horizontal when $t\sin\frac{vt}{R} = 0$.

Neglecting $t = 0$, $\sin\frac{vt}{R} = 0$ when

$\frac{vt}{R} = n\pi$, $n = 0, 1, 2, 3, \ldots$

$\therefore t = \frac{nR\pi}{v}$, $n = 0, 1, 2, 3, \ldots$

(b) Velocity at these points is

$\vec{c}'(t) = v(1 - \cos n\pi,\ 0)$

$= 0$, for $n = 0, 2, 4, \ldots$ (even)

$2v$, for $n = 1, 3, 5, \ldots$ (odd)

$\therefore$ Speed: 0 when rim contacts road

$2v$ when point of rim at top.

16. **16.** If the position of a particle in space is $(6t, 3t^2, t^3)$ at time t, what is its velocity vector at $t = 0$?

$\vec{c}'(t) = (6, 6t, 3t^2)$. $\therefore \vec{c}'(0) = (6,0,0)$

In Exercises 17 and 18, determine the equation of the tangent line to the given path at the specified value of t.

17. **17.** $(\sin 3t, \cos 3t, 2t^{5/2})$; $t = 1$

$\vec{c}'(t) = (3\cos 3t, -3\sin 3t, 5t^{3/2})$

$\therefore \vec{c}'(1) = (3\cos 3, -3\sin 3, 5)$

$\vec{c}(1) = (\sin 3, \cos 3, 2)$

$\therefore$ tangent line at $t=1$:

$\vec{l}(s) = (\sin 3, \cos 3, 2) + (s-1)(3\cos 3, -3\sin 3, 5)$

18. **18.** $(\cos^2 t, 3t - t^3, t)$; $t = 0$

$\vec{c}(0) = (1,0,0)$

$\vec{c}'(t) = (2\cos t(-\sin t), 3-3t^2, 1)$

$\therefore \vec{c}'(0) = (0,3,1)$

$\therefore$ tangent line at point $t=0$:

$\vec{l}(s) = (1,0,0) + s(0,3,1) = (1, 3s, s)$

In Exercises 19 to 22, suppose that a particle following the given path $\mathbf{c}(t)$ *flies off on a tangent at* $t = t_0$. *Compute the position of the particle at the given time* t_1.

19.

19. $\mathbf{c}(t) = (t^2, t^3 - 4t, 0)$, where $t_0 = 2$, $t_1 = 3$

$\vec{c}(t_0) = (4, 0, 0) \quad \vec{c}'(t) = (2t, 3t^2 - 4, 0)$

$\therefore \vec{c}'(t_0) = (4, 8, 0)$

$\therefore$ tangent line at $t_0 = 2$:

$\vec{l}(s) = (4, 0, 0) + (s-2)(4, 8, 0)$

$\therefore \vec{l}(3) = (4, 0, 0) + (3-2)(4, 8, 0) = (8, 8, 0)$

$\therefore$ at $\underline{(8, 8, 0)}$

20.

20. $\mathbf{c}(t) = (e^t, e^{-t}, \cos t)$, where $t_0 = 1$, $t_1 = 2$

$\vec{c}(t_0) = (e, \frac{1}{e}, \cos 1) \quad \vec{c}'(t) = (e^t, -e^{-t}, -\sin t)$

$\therefore \vec{c}'(t_0) = (e, -\frac{1}{e}, -\sin 1)$

$\therefore$ Tangent line at $t_0 = 1$:

$\vec{l}(s) = (e, \frac{1}{e}, \cos 1) + (s-1)(e, -\frac{1}{e}, -\sin 1)$

$\therefore \vec{l}(2) = (e, \frac{1}{e}, \cos 1) + (2-1)(e, -\frac{1}{e}, -\sin 1)$

$= \underline{(2e, 0, \cos 1 - \sin 1)}$

21.

21. $\mathbf{c}(t) = (4e^t, 6t^4, \cos t)$, where $t_0 = 0$, $t_1 = 1$

$\vec{c}(t_0) = (4, 0, 1)$ $\quad \vec{c}'(t) = (4e^t, 24t^3, -\sin t)$

$\vec{c}'(t_0) = (4, 0, 0)$

$\therefore$ Tangent line at $t_0 = 0$:

$\vec{l}(s) = (4,0,1) + (s-0)(4,0,0)$

$\therefore \vec{l}(1) = (4,0,1) + (4,0,0) = \underline{(8,0,1)}$

22. **22.** $\mathbf{c}(t) = (\sin e^t, t, 4 - t^3)$, where $t_0 = 1$, $t_1 = 2$

$\vec{c}(t_0) = (\sin e, 1, 3)$ $\quad \vec{c}'(t) = (e^t \cos e^t, 1, -3t^2)$

$\vec{c}'(t_0) = (e\cos e, 1, -3e^2)$

$\therefore$ Tangent line at $t_0 = 1$:

$\vec{l}(s) = (\sin e, 1, 3) + (s-1)(e\cos e, 1, -3e^2)$

$\therefore \vec{l}(2) = (\sin e, 1, 3) + (e\cos e, 1, -3e^2)$

$= \underline{(\sin e + e\cos e, 2, 3 - 3e^2)}$

23. **23.** The position vector for a particle moving on a helix is $\mathbf{c}(t) = (\cos(t), \sin(t), t^2)$.

(a) Find the speed of the particle at time $t_0 = 4\pi$.

(b) Is $\mathbf{c}'(t)$ ever orthogonal to $\mathbf{c}(t)$?

(c) Find a parametrization for the tangent line to $\mathbf{c}(t)$ at $t_0 = 4\pi$.

(d) Where will this line intersect the xy plane?

(a) $\vec{c}'(t) = (-\sin t, \cos t, 2t)$ $\quad \therefore \vec{c}'(4\pi) = (0, 1, 8\pi)$

$\therefore$ Speed $= \|\vec{c}\,'(4\pi)\| = \sqrt{0^2 + 1^2 + (8\pi)^2} = \sqrt{1 + 64\pi^2}$

(b) $\vec{c}(t) \cdot \vec{c}\,'(t) = [-\cos t \sin t,\ \sin t \cos t,\ 2t^3]$

$2t^3 = 0$ only when $t = 0$.

$\therefore \vec{c}(t) \perp \vec{c}\,'(t)$ at $t = 0$

(c) $\vec{c}(t_0) = (1, 0, 16\pi^2)$, From (a), $\vec{c}\,'(t_0) = (0, 1, 8\pi)$

$\therefore \vec{l}(t) = (1, 0, 16\pi^2) + (t - 4\pi)(0, 1, 8\pi)$

(d) When z-component is 0, $\vec{l}(t)$ intersects xy-plane.

$\therefore 16\pi^2 + (t - 4\pi)\, 8\pi = 0$

$\therefore \dfrac{-16\pi^2}{8\pi} = t - 4\pi,\ t = -2\pi + 4\pi = 2\pi$

$\therefore \vec{l}(2\pi) = (1, 0, 16\pi^2) + (2\pi - 4\pi)(0, 1, 8\pi)$

$= (1, 0, 16\pi^2) + (0, -2\pi, -16\pi^2)$

$= (1, -2\pi, 0)$

24. **24.** Consider the spiral given by $\mathbf{c}(t) = (e^t \cos(t),\ e^t \sin(t))$. Show that the angle between $\mathbf{c}$ and $\mathbf{c}'$ is constant.

$$\vec{c}\,'(t) = \left(e^t\cos(t) - e^t\sin(t),\ e^t\sin(t) + e^t\cos(t)\right)$$

$$= e^t\left(\cos t - \sin t,\ \sin t + \cos t\right)$$

$$\vec{c}\,(t) = e^t(\cos t,\ \sin t)$$

$$\therefore\ \vec{c}\,(t)\cdot\vec{c}\,'(t) = e^{2t}\left[\cos^2 t - \sin t\cos t + \sin^2 t + \sin t\cos t\right]$$

$$= e^{2t}$$

$$\|\vec{c}\,'(t)\| = \sqrt{e^{2t}\left(\cos^2 t + \sin^2 t\right)} = e^t$$

$$\|c'(t)\| = e^t\sqrt{\cos^2 t - 2\cos t\sin t + \sin^2 t + \sin^2 t + 2\sin t\cos t + \cos^2 t}$$

$$= e^t\sqrt{2}$$

$$\therefore\ \cos\theta = \frac{\vec{c}\,(t)\cdot\vec{c}\,'(t)}{\|\vec{c}\,(t)\|\ \|\vec{c}\,'(t)\|} = \frac{e^{2t}}{(e^t)(e^t\sqrt{2})} = \frac{\sqrt{2}}{2}$$

$$\therefore\ \theta = \frac{\pi}{4}\ \text{radians}$$

$\therefore$ For all t, $\theta = \frac{\pi}{4}$ = angle between $\vec{c}\,(t)$ and $\vec{c}\,'(t)$ is constant

25.

25. Let $\mathbf{c}(t) = (t^3, t^2, 2t)$ and $f(x, y, z) = (x^2 - y^2, 2xy, z^2)$.

(a) Find $(f \circ \mathbf{c})(t)$.

(b) Find a parametrization for the tangent line to the curve $f \circ \mathbf{c}$ at $t = 1$.

(a) $(f \circ \vec{c}\,)(t) = f(t^3, t^2, 2t)$

$$= \left[(t^3)^2 - (t^2)^2,\ 2(t^3)(t^2),\ (2t)^2 \right]$$

$$= \underline{\left(t^6 - t^4,\ 2t^5,\ 4t^2 \right)}$$

(5) $(f \circ \vec{c})'(t) = (6t^5 - 4t^3,\ 10t^2,\ 8t)$

$(f \circ \vec{c})(1) = (0, 2, 4)$

$(f \circ \vec{c})'(1) = (2, 10, 8)$

$\therefore$ Tangent line of $f \circ \vec{c}$ at $t = 1$:

$$\underline{\vec{\ell}(s) = (0, 2, 4) + (s-1)(2, 10, 8)}$$

2.5 Properties of the Derivative

Note Title 2/16/2016

1. **1.** If $f: U \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable, prove that $\mathbf{x} \mapsto f^2(\mathbf{x}) + 2f(\mathbf{x})$ is differentiable as well, and compute its derivative in terms of $\mathbf{D}f(\mathbf{x})$.

By Theorem 10 of the text, using the product rule,

$$Df^2(\vec{x}) = [Df(\vec{x})]f(\vec{x}) + f(\vec{x})Df(\vec{x})$$

where $Df(\vec{x})$ is a $1 \times n$ matrix, $f(\vec{x}) \in \mathbb{R}$

Using the constant multiple rule,

$$D(2f(\vec{x})) = 2Df(\vec{x}),$$

and using the sum rule,

$$D[f^2(\vec{x}) + 2f(\vec{x})] = [Df(\vec{x})]f(\vec{x}) + f(\vec{x})Df(\vec{x}) + 2Df(\vec{x})$$

$$= 2f(\vec{x})Df(\vec{x}) + 2Df(\vec{x})$$

$$= \underline{(2f(\vec{x}) + 2)Df(\vec{x})}$$

2. **2.** Prove that the following functions are differentiable, and find their derivatives at an arbitrary point:

(a) $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto 2$

(b) $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x + y$

(c) $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto 2 + x + y$

(d) $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2$

(e) $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto e^{xy}$

(f) $f: U \to \mathbb{R}, (x, y) \mapsto \sqrt{1 - x^2 - y^2}$, where $U = \{(x, y) \mid x^2 + y^2 < 1\}$

(g) $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^4 - y^4$

(a) $\partial f/\partial x = 0$, $\partial f/\partial y = 0$, both are continuous, so f is differentiable. $Df(x,y) = \underline{[0 \ 0]}$

(b) $\partial f/\partial x = 1$, $\partial f/\partial y = 1$, both are continuous,

$\therefore Df(x,y) = \underline{[1 \ 1]}$. Could use sum rule as well.

(c) $\partial f/\partial x = 1$, $\partial f/\partial y = 1$. Both partial are continuous, f is differentiable. $Df(x,y) = [1 \ 1]$

(d) $\partial f/\partial x = 2x$, $\partial f/\partial y = 2y$. Both continuous $\Rightarrow f$ is differentiable. $Df(x,y) = \underline{[2x \ 2y]}$

(e) x is differentiable, y is differentiable, $\therefore$ by product rule, xy is differentiable.
By chain rule, composite function e^{xy} is differentiable.

$\partial f/\partial x = ye^{xy}$ $\partial f/\partial y = xe^{xy}$

$\therefore Df(x,y) = \underline{[ye^{xy} \ xe^{xy}]}$

By chain rule: Let $h(x,y) = xy$, $g(x) = e^x$

$\therefore f(x,y) = (g \circ h)(x,y)$.

$\therefore Df(x,y) = Dg(h(x,y))\,Dg(x,y)$

$= [e^{xy}][y \ x] = [ye^{xy} \ xe^{xy}]$

(f) $1-x^2-y^2$ is differentiable by product rule (for x^2, y^2), sum rule (for $1-x^2-y^2$), and $\therefore f$ is differentiable by chain rule (for $\sqrt{x}, x>0$).

Let $h(x,y) = 1-x^2-y^2$ $\therefore Dh(x,y) = \begin{bmatrix} -2x & -2y \end{bmatrix}$

$g(z) = \sqrt{z}$ $\therefore Dg(z) = \begin{bmatrix} \frac{1}{2\sqrt{z}} \end{bmatrix}$

$\therefore Df(x,y) = Dg(h(x,y))\, Dh(x,y)$

$$= \begin{bmatrix} \frac{1}{2\sqrt{1-x^2-y^2}} \end{bmatrix} \begin{bmatrix} -2x & -2y \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-x}{\sqrt{1-x^2-y^2}} & \frac{-y}{\sqrt{1-x^2-y^2}} \end{bmatrix}, \quad 1-x^2-y^2 > 0$$

(g) By product rule (for x^4, y^4) and sum rule (for x^4-y^4), f is differentiable. $\partial f/\partial x = 4x^3$ $\partial f/\partial y = -4y^3$

$\therefore Df(x,y) = \begin{bmatrix} 4x^3 & -4y^3 \end{bmatrix}$

3.

3. Verify the first special case of the chain rule for the composition $f \circ \mathbf{c}$ in each of the cases:

(a) $f(x, y) = xy$, $\mathbf{c}(t) = (e^t, \cos t)$

(b) $f(x, y) = e^{xy}$, $\mathbf{c}(t) = (3t^2, t^3)$

(c) $f(x, y) = (x^2 + y^2) \log \sqrt{x^2 + y^2}$, $\mathbf{c}(t) = (e^t, e^{-t})$

(d) $f(x, y) = x \exp(x^2 + y^2)$, $\mathbf{c}(t) = (t, -t)$

"The first special case" is according to text on page 127, which is $f: \mathbb{R}^3 \to \mathbb{R}$, $c: \mathbb{R} \to \mathbb{R}^3$

(a) (1) $f \circ c = e^t \cos t$.

$\therefore (f \circ c)'(t) = e^t \cos t - \sin t\, e^t$

(2) $Df(\vec{c}(t))\, D\vec{c}(t)$

$$Df(x,y) = \begin{bmatrix} y & x \end{bmatrix} \quad D\vec{c}(t) = \begin{bmatrix} e^t \\ -\sin t \end{bmatrix}$$

$$\therefore Df(\vec{c}(t))\, D\vec{c}(t) = \begin{bmatrix} \cos t & e^t \end{bmatrix} \begin{bmatrix} e^t \\ -\sin t \end{bmatrix}$$

$$= \cos t\, e^t - \sin t\, e^t$$

(1) = (2)

(b) (1) $(f \circ c)(t) = e^{(3t^2)(t^3)} = e^{3t^5}$

$\therefore (f \circ c)'(t) = 15t^4 e^{3t^5}$

(2) $Df(x,y) = \begin{bmatrix} y e^{xy} & x e^{xy} \end{bmatrix} \quad D\vec{c}(t) = \begin{bmatrix} 6t \\ 3t^2 \end{bmatrix}$

$$\therefore Df(\vec{c}(t))\, D\vec{c}(t) = \begin{bmatrix} t^3 e^{3t^5} & 3t^2 e^{3t^5} \end{bmatrix} \begin{bmatrix} 6t \\ 3t^2 \end{bmatrix}$$

$$= 6t^4 e^{3t^5} + 9t^4 e^{3t^5} = 15t^4 e^{3t^5}$$

$\therefore$ (1) = (2)

(c) (c) $f(x, y) = (x^2 + y^2) \log \sqrt{x^2 + y^2}$, $\mathbf{c}(t) = (e^t, e^{-t})$

(1) $(f \circ c)(t) = (e^{2t} + e^{-2t}) \log \sqrt{e^{2t} + e^{-2t}}$

$\therefore (f \circ c)'(t) = 2(e^{2t} - e^{-2t}) \log \sqrt{e^{2t} + e^{-2t}}$

$+ (e^{2t} + e^{-2t}) \frac{1}{\sqrt{e^{2t} + e^{-2t}}} \left(\frac{1}{2} \frac{2e^{2t} - 2e^{-2t}}{\sqrt{e^{2t} + e^{-2t}}} \right)$

$= 2(e^{2t} - e^{-2t}) \log \sqrt{e^{2t} + e^{-2t}} + (e^{2t} - e^{-2t})$

$= (e^{2t} - e^{-2t}) \left(2 \log \sqrt{e^{2t} + e^{-2t}} + 1 \right)$

(2) $\partial f / \partial x = 2x \log \sqrt{x^2 + y^2} + \frac{(\frac{1}{2}) 2x (x^2 + y^2)}{x^2 + y^2}$

$\partial f / \partial y = 2y \log \sqrt{x^2 + y^2} + \frac{(\frac{1}{2})(2y)(x^2 + y^2)}{x^2 + y^2}$

$D \vec{c}(t) = \begin{bmatrix} e^t \\ -e^{-t} \end{bmatrix}$

$\therefore D f(\vec{c}(t)) D \vec{c}(t) =$

$\begin{bmatrix} 2e^t \log \sqrt{e^{2t} + e^{-2t}} + e^t & 2e^{-t} \log \sqrt{e^{2t} + e^{-2t}} + e^{-t} \end{bmatrix} \begin{bmatrix} e^t \\ -e^{-t} \end{bmatrix}$

$= 2e^{2t} \log \sqrt{e^{2t} + e^{-2t}} + e^{2t} - 2e^{-2t} \log \sqrt{e^{2t} + e^{-2t}} - e^{-2t}$

$= 2(e^{2t} - e^{-2t}) \log \sqrt{e^{2t} + e^{-2t}} + (e^{2t} - e^{-2t})$

$= (e^{2t} - e^{-2t}) \left(2 \log \sqrt{e^{2t} + e^{-2t}} + 1 \right)$

$\therefore (1) = (2)$ Note: $2 \log \sqrt{e^{2t} + e^{-2t}} = \log (e^{2t} + e^{-2t})$

(d) (d) $f(x, y) = x\exp(x^2 + y^2)$, $\mathbf{c}(t) = (t, -t)$

(1) $(f \circ c)(t) = t e^{t^2 + t^2} = t e^{2t^2}$

$\therefore (f \circ c)'(t) = e^{2t^2} + t(4t)e^{2t^2} = (1 + 4t^2)e^{2t^2}$

(2) $\partial f/\partial x = e^{x^2+y^2} + 2x^2 e^{x^2+y^2}$

$\partial f/\partial y = 2xy e^{x^2+y^2}$

$D\vec{c}(t) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\therefore Df(\vec{c}(t)) = \begin{bmatrix} e^{2t^2} + 2t^2 e^{2t^2} & -2t^2 e^{2t^2} \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$= e^{2t^2} + 2t^2 e^{2t^2} + 2t^2 e^{2t^2}$

$= (1 + 4t^2)e^{2t^2}$

$\therefore (1) = (2)$

4. **4.** What is the velocity vector for each path $\mathbf{c}(t)$ in Exercise 3? [The solution to part (b) only is in the Study Guide to this text.]

(a) $\mathbf{c}(t) = (e^t, \cos t)$ $\quad \vec{c}'(t) = (e^t, -\sin t)$

(b) $\mathbf{c}(t) = (3t^2, t^3)$ $\quad \vec{c}'(t) = (6t, 3t^2)$

(c) $\vec{c}(t) = (e^t, e^{-t})$ $\quad \vec{c}'(t) = (e^t, -e^{-t})$

(d) $\mathbf{c}(t) = (t, -t)$ $\quad \vec{c}'(t) = (1, -1)$

5. **5.** Let $f: \mathbb{R}^3 \to \mathbb{R}$ and $g: \mathbb{R}^3 \to \mathbb{R}$ be differentiable. Prove that

$$\nabla(fg) = f\nabla g + g\nabla f.$$

$$\nabla(fg) = \left[\frac{\partial fg}{\partial x}, \frac{\partial fg}{\partial y}, \frac{\partial fg}{\partial z}\right], \text{ now use product rule}$$

$$= \left[g\frac{\partial f}{\partial x} + f\frac{\partial g}{\partial x}, g\frac{\partial f}{\partial y} + f\frac{\partial g}{\partial y}, g\frac{\partial f}{\partial z} + f\frac{\partial g}{\partial z}\right]$$

$$= \left[f\frac{\partial g}{\partial x}, f\frac{\partial g}{\partial y}, f\frac{\partial g}{\partial z}\right] + \left[g\frac{\partial f}{\partial x}, g\frac{\partial f}{\partial y}, g\frac{\partial f}{\partial z}\right]$$

$$= f\nabla g + g\nabla f$$

6. **6.** Let $f: \mathbb{R}^3 \to \mathbb{R}$ be differentiable. Making the substitution

$$x = \rho\cos\theta\sin\phi, \quad y = \rho\sin\theta\sin\phi, \quad z = \rho\cos\phi$$

(spherical coordinates) into $f(x, y, z)$, compute $\partial f/\partial\rho$, $\partial f/\partial\theta$, and $\partial f/\partial\phi$ in terms of $\partial f/\partial x$, $\partial f/\partial y$, and $\partial f/\partial z$.

$$\frac{\partial f}{\partial \rho} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \rho} + \frac{\partial f}{\partial y}\cdot\frac{\partial y}{\partial \rho} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial \rho}$$

$$= \cos\theta\sin\phi\frac{\partial f}{\partial x} + \sin\theta\sin\phi\frac{\partial f}{\partial y} + \cos\phi\frac{\partial f}{\partial z}$$

$$\frac{\partial f}{\partial \theta} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \theta} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \theta} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial \theta}$$

$$= -\rho \sin\theta \sin\phi \frac{\partial f}{\partial x} + \rho \cos\theta \sin\phi \frac{\partial f}{\partial y}$$

$$\frac{\partial f}{\partial \phi} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial \phi} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial \phi} + \frac{\partial f}{\partial z}\frac{\partial z}{\partial \phi}$$

$$= \rho \cos\theta \cos\phi \frac{\partial f}{\partial x} + \rho \sin\theta \cos\phi \frac{\partial f}{\partial y} - \rho \sin\phi \frac{\partial f}{\partial \phi}$$

7.

7. Let $f(u, v) = (\tan(u-1) - e^v,\ u^2 - v^2)$ and $g(x, y) = (e^{x-y},\ x - y)$. Calculate $f \circ g$ and $\mathbf{D}(f \circ g)(1, 1)$.

(a) $(f \circ g)(x,y) = \left[\tan(e^{x-y}-1) - e^{x-y},\ (e^{x-y})^2 - (x-y)^2\right]$

$$= \left[\tan(e^{x-y}-1) - e^{x-y},\ e^{2x-2y} - x^2 + 2xy + y^2\right]$$

(b) $Df(u,v) = \begin{bmatrix} \sec^2(u-1) & -e^v \\ 2u & -2v \end{bmatrix}$

$g(1,1) = (e^0, 0) = (1, 0)$

$\therefore Df(g(1,1)) = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}$

$Dg(x,y) = \begin{bmatrix} e^{x-y} & -e^{x-y} \\ 1 & -1 \end{bmatrix} \quad \therefore Dg(1,1) = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$

$$\therefore Df(g(1,1))\,Dg(1,1) = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix} = \underline{\begin{bmatrix} 0 & 0 \\ 2 & -2 \end{bmatrix}}$$

8.

8. Let $f(u, v, w) = (e^{u-w}, \cos(v + u) + \sin(u + v + w))$ and $g(x, y) = (e^x, \cos(y - x), e^{-y})$. Calculate $f \circ g$ and $\mathbf{D}(f \circ g)(0, 0)$.

(a) $(f \circ g)(x,y) =$

$$\left[e^{e^x - e^{-y}},\ \cos\left(e^x + \cos(y-x)\right) + \sin\left(e^x + \cos(y-x) + e^{-y}\right)\right]$$

(b) $$Df(u,v,w) = \begin{bmatrix} e^{u-w} & 0 & -e^{u-w} \\ -\sin(v+u) + \cos(u+v+w) & -\sin(v+u) + \cos(u+v+w) & \cos(u+v+w) \end{bmatrix}$$

$$g(0,0) = (e^0, \cos 0, e^0) = (1,1,1)$$

$$\therefore Df(g(0,0)) = Df(1,1,1) = \begin{bmatrix} 1 & 0 & -1 \\ \cos 3 - \sin 2 & \cos 3 - \sin 2 & \cos 3 \end{bmatrix}$$

$$Dg(x,y) = \begin{bmatrix} e^x & 0 \\ \sin(y-x) & -\sin(y-x) \\ 0 & -e^{-y} \end{bmatrix} \quad Dg(0,0) = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\therefore Df(g(0,0))\,Dg(0,0) = \begin{bmatrix} 1 & 0 & 1 \\ \cos 3 - \sin 2 & \cos 3 - \sin 2 & \cos 3 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & -1 \end{bmatrix}$$

$$= \underline{\begin{bmatrix} 1 & -1 \\ \cos 3 - \sin 2 & -\cos 3 \end{bmatrix}}$$

9. **9.** Find $(\partial/\partial s)(f \circ T)(1, 0)$, where $f(u, v) = \cos u \sin v$ and $T: \mathbb{R}^2 \to \mathbb{R}^2$ is defined by $T(s, t) = (\cos(t^2 s), \log\sqrt{1 + s^2})$.

$$f \circ T(s,t) = \cos(\cos(t^2 s)) \sin(\log\sqrt{1+s^2})$$

$$\therefore \frac{\partial\, f\circ T}{\partial s} = \sin(\log\sqrt{1+s^2})\left[-\sin(\cos(t^2 s))(-\sin(t^2 s))\, t^2\right] + \cos(\cos(t^2 s))\left[\cos(\log\sqrt{1+s^2})\left(\frac{1}{\sqrt{1+s^2}}\right)\left(\frac{2s}{2\sqrt{1+s^2}}\right)\right]$$

$$\therefore \frac{\partial\, f\circ T}{\partial s}(1,0) = \sin(\log 2)[0] + \cos 1\left[\cos(\log\sqrt{2})\left(\tfrac{1}{2}\right)\right] = \underline{\tfrac{1}{2}\cos 1\,(\cos(\log\sqrt{2}))}$$

Another way: $Df(s,t) = \begin{bmatrix} -\sin(t^2 s)(t^2) & -\sin(t^2 s)(2st) \\ \frac{s}{1+s^2} & 0 \end{bmatrix}$

$$Df(u,v) = \begin{bmatrix} -\sin u \sin v & \cos u \cos v \end{bmatrix}$$

$f \circ T: \mathbb{R}^2 \to \mathbb{R}$, $\therefore Df\circ T = \begin{bmatrix} \frac{\partial\, f\circ T}{\partial s} & \frac{\partial\, f\circ T}{\partial t} \end{bmatrix}$

$$Df\circ T = Df(T(s,t)) \cdot DT(s,t)$$

$$= Df(u,v)\cdot DT(s,t) \;\Big|\; \begin{matrix} u = \cos(t^2 s) \\ v = \log\sqrt{1+s^2} \end{matrix} \Big|\; \begin{matrix} s=1 \\ t=0 \end{matrix}$$

$$= Df(u,v)\cdot DT(1,0) \;\Big|\; \begin{matrix} u = 1 \\ v = \log\sqrt{2} \end{matrix}$$

$$= Df(1, \log\sqrt{2}) \cdot DT(1,0)$$

$$= \begin{bmatrix} -\sin 1 \sin(\log\sqrt{2}) & \cos 1 \cos(\log\sqrt{2}) \end{bmatrix} \begin{bmatrix} 0 & 0 \\ \frac{1}{2} & 0 \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{\cos 1 \cos(\log\sqrt{2})}{2} & 0 \end{bmatrix}$$

$$\therefore \frac{\partial\, foT}{\partial s}(1,0) = \frac{\cos(1)\cos(\log\sqrt{2})}{2}$$

10.

10. Suppose that the temperature at the point (x, y, z) in space is $T(x, y, z) = x^2 + y^2 + z^2$. Let a particle follow the right-circular helix $\sigma(t) = (\cos t, \sin t, t)$ and let $T(t)$ be its temperature at time t.

(a) What is $T'(t)$?

(b) Find an approximate value for the temperature at $t = (\pi/2) + 0.01$.

(a) $T(t) = \cos^2 t + \sin^2 t + t^2 = 1 + t^2$

$\therefore T'(t) = 2t$

(b) $T(t) \simeq T(t_0) + T'(t_0)(t - t_0)$

$\therefore$ If $t_0 = \pi/2$, $t = \frac{\pi}{2} + 0.01$

$$\therefore T\left(\frac{\pi}{2} + 0.01\right) \simeq T\left(\frac{\pi}{2}\right) + 2\left(\frac{\pi}{2}\right)(0.01)$$

$$= 1 + \frac{\pi^2}{4} + \pi(0.01) \simeq 3.5$$

11. Let $f(x, y, z) = (3y + 2, x^2 + y^2, x + z^2)$. Let $\mathbf{c}(t) = (\cos(t), \sin(t), t)$.

(a) Find the path $\mathbf{p} = f \circ \mathbf{c}$ and the velocity vector $\mathbf{p}'(\pi)$.

(b) Find $\mathbf{c}(\pi)$, $\mathbf{c}'(\pi)$ and $\mathbf{D}f(-1, 0, \pi)$.

(c) Thinking of $\mathbf{D}f(-1, 0, \pi)$ as a linear map, find $\mathbf{D}f(-1, 0, \pi)\,(\mathbf{c}'(\pi))$.

(a) $\vec{p}(t) = f \circ \vec{c}(t) = (3\sin t + 2, \cos^2 t + \sin^2 t, \cos t + t^2)$

$= (3\sin t + 2, 1, \cos t + t^2)$

$\vec{p}'(t) = (3\cos t, 0, -\sin t + 2t)$

$\therefore p'(\pi) = (-3, 0, 2\pi)$

(b) $\vec{c}(\pi) = (\cos\pi, \sin\pi, \pi) = (-1, 0, \pi)$

$\vec{c}'(t) = (-\sin t, \cos t, 1) \quad \therefore \vec{c}'(\pi) = (0, -1, 1)$

$Df(x, y, z) = \begin{bmatrix} 0 & 3 & 0 \\ 2x & 2y & 0 \\ 1 & 0 & 2z \end{bmatrix} \quad \therefore Df(-1, 0, \pi) = \begin{bmatrix} 0 & 3 & 0 \\ -2 & 0 & 0 \\ 1 & 0 & 2\pi \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 3 & 0 \\ -2 & 0 & 0 \\ 1 & 0 & 2\pi \end{bmatrix} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} -3 \\ 0 \\ 2\pi \end{bmatrix}$

12. **12.** Let $h: \mathbb{R}^3 \to \mathbb{R}^5$ and $g: \mathbb{R}^2 \to \mathbb{R}^3$ be given by $h(x, y, z) = (xyz, e^{xz}, x\sin(y), \frac{-9}{x}, 17)$ and $g(u, v) = (v^2 + 2u, \pi, 2\sqrt{u})$. Find $\mathbf{D}(h \circ g)(1, 1)$.

$$Dh(x,y,z) = \begin{bmatrix} yz & xz & xy \\ ze^{xz} & 0 & xe^{xz} \\ \sin(y) & x\cos(y) & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad Dg(u,v) = \begin{bmatrix} 2 & 2v \\ 0 & 0 \\ \frac{1}{\sqrt{u}} & 0 \end{bmatrix}$$

$$Dg(1,1) = \begin{bmatrix} 2 & 2 \\ 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$g(1,1) = (3, \pi, 2)$$

$$\therefore Dh(3,\pi,2) = \begin{bmatrix} 2\pi & 6 & 3\pi \\ 2e^6 & 0 & 3e^6 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\therefore D(h \circ g)(1,1) = Dh(3,\pi,2)\,Dg(1,1) =$$

$$\begin{bmatrix} 2\pi & 6 & 3\pi \\ 2e^6 & 0 & 3e^6 \\ 0 & -3 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 0 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 7\pi & 4\pi \\ 7e^6 & 4e^6 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

13. **13.** Suppose that a duck is swimming in the circle $x = \cos t$, $y = \sin t$ and that the water temperature is given by the formula $T = x^2 e^y - xy^3$. Find dT/dt, the rate of change in temperature the duck might feel: (a) by the chain rule; (b) by expressing T in terms of t and differentiating.

(a) $D\,T(x,y) = \begin{bmatrix} 2xe^y - y^3 & x^2e^y - 3xy^2 \end{bmatrix}$

Let $f(t) = (\cos t, \sin t)$. $\therefore Df(t) = \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix}$

$\therefore D(T \circ f)(t) = \begin{bmatrix} 2xe^y - y^3 & x^2e^y - 3xy^2 \end{bmatrix} \begin{bmatrix} -\sin t \\ \cos t \end{bmatrix}_{\substack{x=\cos t \\ y=\sin t}}$

$= -\sin t\,(2xe^y - y^3) + \cos t\,(x^2e^y - 3xy^2) \Big|_{\substack{x=\cos t \\ y=\sin t}}$

$= -\sin t\,(2(\cos t)e^{\sin t} - \sin^3 t) + \cos t\,((\cos^2 t)e^{\sin t} - 3\cos t \sin^2 t)$

$\therefore T'(t) = e^{\sin t}(\cos^3 t - 2\sin t \cos t) + \sin^4 t - 3\cos^2 t \sin^2 t$

(b) $T(t) = (\cos^2 t)e^{\sin t} - (\cos t)\sin^3 t$

$T'(t) = e^{\sin t}(2\cos t(-\sin t) + e^{\sin t}(\cos t)(\cos^2 t)$

$+ (\sin t)(\sin^3 t) - (\cos t)(3\sin^2 t)(\cos t)$

$= e^{\sin t}(\cos^3 t - 2\sin t \cos t) + \sin^4 t - 3\cos^2 t \sin^2 t$

14. **14.** Let $f: \mathbb{R}^n \to \mathbb{R}^m$ be a linear mapping so that (by Exercise 28, Section 2.3) $\mathbf{D}f(\mathbf{x})$ is the matrix of f. Check the validity of the chain rule directly for linear mappings.

Let $f: \mathbb{R}^n \to \mathbb{R}^m$ be $T\vec{x}$ It was shown in Ex. 28 of

Sec. 2.3 that $T = Df = \begin{bmatrix} f_1(\hat{e}_1) \cdots f_1(\hat{e}_n) \\ \vdots \\ f_m(\hat{e}_1) \cdots f_m(\hat{e}_n) \end{bmatrix} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} \cdots \frac{\partial f_1}{\partial x_n} \\ \vdots \\ \frac{\partial f_m}{\partial x_1} \cdots \frac{\partial f_m}{\partial x_n} \end{bmatrix}$

Let $g: \mathbb{R}^m \to \mathbb{R}^p$ be $M\vec{y}$ so that

$$M = Dg = \begin{bmatrix} g_1(\hat{e}_1) \cdots g_1(\hat{e}_m) \\ \vdots \\ g_p(\hat{e}_1) \cdots g_p(\hat{e}_m) \end{bmatrix} = \begin{bmatrix} \frac{\partial g_1}{\partial y_1} \cdots \frac{\partial g_1}{\partial y_m} \\ \vdots \quad\quad \vdots \\ \frac{\partial g_p}{\partial y_1} \cdots \frac{\partial g_p}{\partial y_m} \end{bmatrix}$$

The derivative is the linear map (the matrix)

Note: $f(\vec{x}) = T\vec{x}$, $Df(\vec{x}) = T$, not $T\vec{x}$

T is composed of real numbers, it has no variables to evaluate for.
So, "Df at $\vec{x}$ is T".

Let $\vec{y} = f(\vec{x}) = T\vec{x}$

$\therefore (g \circ f)(\vec{x}) = g(f(\vec{x})) = g(\vec{y})$

$= g(T\vec{x}) = M(T\vec{x}) = (MT)\vec{x}$

$\therefore g \circ f = MT : \mathbb{R}^n \to \mathbb{R}^p$

so $D(g \circ f)(\vec{x}) = MT$

By chain rule, $D(g \circ f)(\vec{x}) = Dg(f(\vec{x}))\, Df(\vec{x})$

= "Derivative of g at $f(\vec{x})$" times

"Derivative of f at $\vec{x}$"

$= MT$

Again, $Dg(f(\vec{x})) = Dg(\vec{y}) = M$, $Df(\vec{x}) = T$

since $Dg = M$ and M has only real number entries, no variables.

Just like if $f(x) = 3x$, $f' = 3$, so $f'(1) = f'(12) = 3$.

15. **15.** Let $f: \mathbb{R}^2 \to \mathbb{R}^2; (x, y) \mapsto (e^{x+y}, e^{x-y})$. Let $\mathbf{c}(t)$ be a path with $\mathbf{c}(0) = (0, 0)$ and $\mathbf{c}'(0) = (1, 1)$. What is the tangent vector to the image of $\mathbf{c}(t)$ under f at $t = 0$?

$D(f \circ \vec{c})(0) = Df(\vec{c}(0)) \cdot D\vec{c}(0)$

$$Df(x,y) = \begin{bmatrix} e^{x+y} & e^{x+y} \\ e^{x-y} & -e^{x-y} \end{bmatrix}$$

$$\therefore Df(\vec{c}(0)) = Df((0,0)) = \begin{bmatrix} e^0 & e^0 \\ e^0 & -e^0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$D\vec{c}(0) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \therefore Df(\vec{c}(0)) \cdot D\vec{c}(0) = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

or, $\underline{(2,0)}$

16. **16.** Let $f(x, y) = 1/\sqrt{x^2 + y^2}$. Compute $\nabla f(x, y)$.

$$\nabla f(x,y) = \begin{bmatrix} f_x & f_y \end{bmatrix}$$

$$= \begin{bmatrix} -\frac{1}{2}(x^2+y^2)^{-3/2}(2x) & -\frac{1}{2}(x^2+y^2)^{-3/2}(2y) \end{bmatrix}$$

$$= \begin{bmatrix} -x(x^2+y^2)^{-3/2} & -2y(x^2+y^2)^{-\frac{3}{2}} \end{bmatrix}$$

17. **17.** Write out the chain rule for each of the following functions and justify your answer in each case using Theorem 11.

(a) $\partial h/\partial x$, where $h(x, y) = f(x, u(x, y))$

(b) dh/dx, where $h(x) = f(x, u(x), v(x))$

(c) $\partial h/\partial x$, where $h(x, y, z) = f(u(x, y, z), v(x, y), w(x))$

(a) Let $v(x,y) = x$. $\therefore h(x,y) = f(v(x,y), u(x,y))$

Note $\frac{\partial v(x,y)}{\partial x} = 1$, $\frac{\partial v}{\partial y}(x,y) = 0$

Let $g(x,y) = (v(x,y), u(x,y))$

$\therefore$ have $f(u,v)$, and $h(x,y) = (f \circ g)(x,y)$

$\therefore Dh(x,y) = Df(g(x,y)) \cdot Dg(x,y)$

$Dh(x,y) = \begin{bmatrix} \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} \end{bmatrix}$, $Df(v,u) = \begin{bmatrix} \frac{\partial f}{\partial v} & \frac{\partial f}{\partial u} \end{bmatrix}$

$$D g(x,y) = \begin{bmatrix} \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{bmatrix}$$

$$\therefore \begin{bmatrix} \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial v} & \frac{\partial f}{\partial u} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\partial f}{\partial v} + \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} & \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} \end{bmatrix}$$

$$\therefore \underline{\frac{\partial h}{\partial x} = \frac{\partial f}{\partial v} + \frac{\partial f}{\partial u}\frac{\partial u}{\partial x}}, \text{ where } f(v,u) = f(x, u(x,y))$$

(5) Let $w(x) = x$, so $h(x) = f(w(x), u(x), v(x))$

where $f: R^3 \to R$, let $g(x) = (w(x), u(x), v(x))$

$g: R \to R^3$

$\therefore h(x) = (f \circ g)(x)$, $h: R \to R$

By chain rule, $Dh(x) = Df(g(x)) \cdot Dg(x)$

$$Dg(x) = \begin{bmatrix} w'(x) \\ u'(x) \\ v'(x) \end{bmatrix} \quad Df = \begin{bmatrix} \frac{\partial f}{\partial w} & \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \end{bmatrix}$$

and $Dh(x) = \begin{bmatrix} \frac{dh}{dx} \end{bmatrix}$

$$\therefore Dh(x) = \left[\frac{dh}{dx}\right] = \begin{bmatrix} \frac{\partial f}{\partial w} & \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} \end{bmatrix} \begin{bmatrix} w'(x) \\ u'(x) \\ v'(x) \end{bmatrix}$$

$$= \frac{\partial f}{\partial w}\frac{dw}{dx} + \frac{\partial f}{\partial u}\cdot\frac{du}{dx} + \frac{\partial f}{\partial v}\cdot\frac{dv}{dx}$$

But $\frac{dw}{dx} = 1$ since $w(x) = x$

$$\therefore \frac{dh}{dx} = \frac{\partial f}{\partial w} + \frac{\partial f}{\partial u}\cdot\frac{du}{dx} + \frac{\partial f}{\partial v}\cdot\frac{dv}{dx},$$

where $f(w,u,v) = f(x, u(x), v(x))$

So, could write $\underline{\frac{dh}{dx} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial u}\cdot\frac{du}{dx} + \frac{\partial f}{\partial v}\cdot\frac{dv}{dx}}$

(c) (c) $\partial h/\partial x$, where $h(x, y, z) = f(u(x, y, z), v(x, y), w(x))$

Let $g: \mathbb{R}^3 \to \mathbb{R}^3$ by $g(x,y,z) = (u(x,y,z), v(x,y), w(x))$

$f: \mathbb{R}^3 \to \mathbb{R}^1$, $f(u,v,w)$ $\therefore h(x,y,z) = (f \circ g)(x,y,z)$

By chain rule, $Dh(x,y,z) = Df(g(x,y,z)) \cdot Dg(x,y,z)$

$$Dh = \begin{bmatrix} \frac{\partial h}{\partial x} & \frac{\partial h}{\partial y} & \frac{\partial h}{\partial z} \end{bmatrix} \quad Df = \begin{bmatrix} \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} & \frac{\partial f}{\partial w} \end{bmatrix}$$

$$Dg = \begin{pmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{pmatrix} = \begin{bmatrix} u_x & u_y & u_z \\ v_x & v_y & 0 \\ w_x & 0 & 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} \frac{\partial h}{\partial x} \end{bmatrix} = \begin{bmatrix} \frac{\partial f}{\partial u} & \frac{\partial f}{\partial v} & \frac{\partial f}{\partial w} \end{bmatrix} \begin{bmatrix} u_x \\ v_x \\ w_x \end{bmatrix}$$

$$= \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} + \frac{\partial f}{\partial w} \cdot \frac{dw}{dx}$$

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18. Verify the chain rule for $\partial h/\partial x$, where $h(x, y) = f(u(x, y), v(x, y))$ and

$$f(u, v) = \frac{u^2 + v^2}{u^2 - v^2}, \qquad u(x, y) = e^{-x-y}, \qquad v(x, y) = e^{xy}.$$

(a) $$h(x,y) = \frac{\left(e^{-x-y}\right)^2 + \left(e^{xy}\right)^2}{\left(e^{-x-y}\right)^2 - \left(e^{xy}\right)^2} = \frac{e^{-2x-2y} + e^{2xy}}{e^{-2x-2y} - e^{2xy}}$$

$$\therefore \frac{\partial h}{\partial x} = \frac{\left(e^{-2x-2y} - e^{2xy}\right)\left(-2e^{-2x-2y} + 2ye^{2xy}\right) - \left(e^{-2x-2y} + e^{2xy}\right)\left(-2e^{-2x-2y} - 2ye^{2xy}\right)}{\left(e^{-2x-2y} - e^{2xy}\right)^2}$$

$$= \frac{-2e^{-4x-4y} + 2e^{-2x-2y+2xy} + 2ye^{-2x-2y+2xy} - 2ye^{4xy}}{e^{-4x-4y} - 2e^{-2x-2y+2xy} + e^{4xy}}$$

$$- \frac{\left(-2e^{-4x-4y} - 2e^{-2x-2y+2xy} - 2ye^{-2x-2y+2xy} - 2ye^{4xy}\right)}{e^{-4x-4y} - 2e^{-2x-2y+2xy} + e^{4xy}}$$

$$= \frac{4e^{-2x-2y+2xy} + 4ye^{-2x-2y+2xy}}{e^{-4x-4y} - 2e^{-2x-2y+2xy} + e^{4xy}}$$

(6) $\frac{\partial h}{\partial x} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}$

$\frac{\partial u}{\partial x} = -e^{-x-y} \qquad \frac{\partial v}{\partial x} = ye^{xy}$

$$\left.\begin{aligned}\frac{\partial f}{\partial u} &= \frac{(u^2-v^2)(2u) - (u^2+v^2)(2u)}{(u^2-v^2)^2}\\ \frac{\partial f}{\partial v} &= \frac{(u^2-v^2)(2v) - (u^2+v^2)(-2v)}{(u^2-v^2)^2}\end{aligned}\right\} \begin{aligned}u &= e^{-x-y}\\ v &= e^{xy}\end{aligned}$$

$$\therefore \frac{\partial f}{\partial u} = \frac{\left(e^{-2x-2y} - e^{2xy}\right)\left(2e^{-x-y}\right) - \left(e^{-2x-2y} + e^{2xy}\right)\left(2e^{-x-y}\right)}{\left(e^{-2x-2y} - e^{2xy}\right)^2}$$

$$= \frac{2e^{-3x-3y} - 2e^{-x-y+2xy} - 2e^{-3x-3y} - 2e^{-x-y+2xy}}{e^{-4x-4y} - 2e^{-2x-2y+2xy} + e^{4xy}}$$

$$= \frac{-4e^{-x-y+2xy}}{e^{-4x-4y} - 2e^{-2x-2y+2xy} + e^{4xy}}$$

$$\frac{\partial f}{\partial v} = \frac{\left(e^{-2x-2x} - e^{2xy}\right)\left(2e^{xy}\right) - \left(e^{-2x-2y} + e^{2xy}\right)\left(-2e^{xy}\right)}{e^{-4x-4y} - 2e^{-2x-2y+2xy} + e^{4xy}}$$

$$= 2e^{-2x-2y+xy} - 2e^{3xy} + 2e^{-2x-2y+xy} + 2e^{3xy}$$

$$= \frac{4e^{-2x-2y+xy}}{e^{-4x-4y} - 2e^{-2x-2y+2xy} + e^{4xy}}$$

$$\therefore \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x} =$$

$$\frac{-4e^{-x-y}+2xy}{e^{-4x-4y}-2e^{-2x-2y+2xy}+e^{4xy}}\left(-e^{-x-y}\right)$$

$$+\ \frac{4e^{-2x-2y+xy}}{e^{-4x-4y}-2e^{-2x-2y+2xy}+e^{4xy}}\left(ye^{xy}\right)$$

$$= \frac{4e^{-2x-2y+2xy} + 4ye^{-2x-2y+2xy}}{e^{-4x-4y}-2e^{-2x-2y+2xy}+e^{4xy}}$$

Amazingly, (a) = (b) = $\frac{\partial h}{\partial x}$

19.

19. (a) Let $y(x)$ be defined implicitly by $G(x, y(x)) = 0$, where G is a given function of two variables. Prove that if $y(x)$ and G are differentiable, then

$$\frac{dy}{dx} = -\frac{\partial G/\partial x}{\partial G/\partial y} \quad \text{if} \quad \frac{\partial G}{\partial y} \neq 0.$$

(a) Let $h(x) = G(x, y(x)) = 0$.

By the chain rule, $h'(x) = \frac{\partial G}{\partial x}\cdot\frac{dx}{dx} + \frac{\partial G}{\partial y}\cdot\frac{dy}{dx} = 0$

Note $\frac{dx}{dx} = 1$.

$$\therefore \frac{\partial G}{\partial x} \cdot 1 + \frac{\partial G}{\partial y} \cdot \frac{dy}{dx} = 0, \quad \frac{\partial G}{\partial y} \cdot \frac{dy}{dx} = -\frac{\partial G}{\partial x}$$

$$\therefore \frac{dy}{dx} = -\frac{\partial G/\partial x}{\partial G/\partial y}, \text{ assuming } \frac{\partial G}{\partial y}(x) \neq 0.$$

(6) (b) Obtain a formula analogous to that in part (a) if y_1, y_2 are defined implicitly by

$$G_1(x, y_1(x), y_2(x)) = 0,$$
$$G_2(x, y_1(x), y_2(x)) = 0.$$

$$\frac{\partial G_1}{\partial x} \cdot \frac{dx}{dx} + \frac{\partial G_1}{\partial y_1} \cdot \frac{dy_1}{dx} + \frac{\partial G_1}{\partial y_2} \frac{dy_2}{dx} = 0$$

$$\frac{\partial G_2}{\partial x} \cdot \frac{dx}{dx} + \frac{\partial G_2}{\partial y_1} \frac{dy_1}{dx} + \frac{\partial G_2}{\partial y_2} \frac{dy_2}{dx} = 0$$

$$\therefore \begin{bmatrix} \frac{\partial G_1}{\partial y_1} & \frac{\partial G_1}{\partial y_2} \\ \frac{\partial G_2}{\partial y_1} & \frac{\partial G_2}{\partial y_2} \end{bmatrix} \begin{bmatrix} y_1' \\ y_2' \end{bmatrix} = \begin{bmatrix} -\frac{\partial G_1}{\partial x} \\ -\frac{\partial G_2}{\partial x} \end{bmatrix}$$

$$\therefore \begin{bmatrix} \frac{dy_1}{dx} \\ \frac{dy_2}{dx} \end{bmatrix} = \frac{1}{\frac{\partial G_1}{\partial y_1}\frac{\partial G_2}{\partial y_2} - \frac{\partial G_1}{\partial y_2}\frac{\partial G_2}{\partial y_1}} \begin{bmatrix} \frac{\partial G_2}{\partial y_2} & -\frac{\partial G_1}{\partial y_2} \\ -\frac{\partial G_2}{\partial y_1} & \frac{\partial G_1}{\partial y_1} \end{bmatrix} \begin{bmatrix} -\frac{\partial G_1}{\partial x} \\ \frac{\partial G_2}{\partial x} \end{bmatrix}$$

assuming $\frac{\partial G_1}{\partial y_1}\frac{\partial G_2}{\partial y_2} - \frac{\partial G_1}{\partial y_2}\frac{\partial G_2}{\partial y_1} \neq 0$

(c) (c) Let y be defined implicitly by

$$x^2 + y^3 + e^y = 0.$$

Compute dy/dx in terms of x and y.

Let $G(x, y(x)) = x^2 + y^3 + e^y = 0$

By (a), $\frac{dy}{dx} = -\frac{\partial G/\partial x}{\partial G/\partial y} = -\frac{2x}{3y^2 + e^y}$

20. **20.** Thermodynamics texts[4] use the relationship

$$\left(\frac{\partial y}{\partial x}\right)\left(\frac{\partial z}{\partial y}\right)\left(\frac{\partial x}{\partial z}\right) = -1.$$

Explain the meaning of this equation and prove that it is true. [HINT: Start with a relationship $F(x, y, z) = 0$ that defines $x = f(y, z)$, $y = g(x, z)$, and $z = h(x, y)$ and differentiate implicitly.]

Let $G(y, z) = F(x(y, z), y, z) = 0$ [1]

$H(x, z) = F(x, y(x, z), z) = 0$ [2]

$J(x, y) = F(x, y, z(x, y)) = 0$ [3]

From [1], $G_y = F_x \cdot \frac{\partial x}{\partial y} + F_y \frac{dy}{dy} + F_z \frac{\partial z}{\partial y} = 0$

Since for $G(y, z)$, z is not a function of y, $\frac{\partial z}{\partial y} = 0$

Also, $\frac{dy}{dy} = 1$

$\therefore G_y = F_x \frac{\partial x}{\partial y} + F_y = 0$

Similarly, $G_z = F_x \frac{\partial x}{\partial z} + F_z = 0$ $\left(\frac{\partial y}{\partial z} = 0\right)$

[1'] $\therefore \frac{\partial x}{\partial y} = -\frac{F_y}{F_x}$, $\frac{\partial x}{\partial z} = -\frac{F_z}{F_x}$, assuming $F_x \neq 0$

From [2], $H_x = F_x \frac{dx}{dx} + F_y \frac{\partial y}{\partial x} + F_z \frac{\partial z}{\partial x}$

Or, $F_x + F_y \frac{\partial y}{\partial x} = 0 \quad \left(\frac{\partial z}{\partial x} = 0\right)$

Similarly, $H_z = F_y \frac{\partial y}{\partial z} + F_z = 0 \quad \left(\frac{\partial x}{\partial z} = 0\right)$

[2'] $\therefore \frac{\partial y}{\partial x} = -\frac{F_x}{F_y} \quad \frac{\partial y}{\partial z} = -\frac{F_z}{F_y}$, $F_y \neq 0$

From [3], $J_x = F_x \frac{dx}{dx} + F_z \frac{\partial z}{\partial x} = 0 \quad \left(\frac{\partial y}{\partial x} = 0\right)$

Or, $F_x + F_z \frac{\partial z}{\partial x} = 0$

Similarly, $J_y = F_y + F_z \frac{\partial z}{\partial y} = 0$

[3'] $\therefore \frac{\partial z}{\partial x} = -\frac{F_x}{F_z}$, $\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$, $F_z \neq 0$

From [2'] $\frac{\partial y}{\partial x} = -\frac{F_x}{F_y}$ From [3'] $\frac{\partial z}{\partial y} = -\frac{F_y}{F_z}$

From [1'] $\frac{\partial x}{\partial z} = -\frac{F_z}{F_x}$

$$\therefore \left(\frac{\partial y}{\partial x}\right)\left(\frac{\partial z}{\partial y}\right)\left(\frac{\partial x}{\partial z}\right) = \left(-\frac{F_x}{F_y}\right)\left(-\frac{F_y}{F_z}\right)\left(-\frac{F_z}{F_x}\right) = -1$$

assuming $F_x, F_y, F_z \neq 0$

21.

21. Dieterici's equation of state for a gas is

$$P(V-b)\,e^{a/RVT} = RT,$$

where a, b, and R are constants. Regard volume V as a function of temperature T and pressure P and prove that

$$\frac{\partial V}{\partial T} = \left(R + \frac{a}{TV}\right) \Big/ \left(\frac{RT}{V-b} - \frac{a}{V^2}\right).$$

Rewrite equation: $\log(V-b) + a/RVT = \log\left(\frac{RT}{P}\right)$

$$\therefore \frac{\partial \log(V-b)}{\partial T} + \frac{\partial}{\partial T}\left(\frac{a}{RVT}\right) = \frac{\partial}{\partial T}\left(\log \frac{RT}{P}\right)$$

$$\frac{1}{V-b}\frac{\partial V}{\partial T} + \frac{a}{R}\left(-\frac{1}{(VT)^2}\right)\left(V + T\frac{\partial V}{\partial T}\right) = \frac{1}{\frac{RT}{P}} \cdot \frac{R}{P}$$

$$\therefore \frac{1}{V-b}\frac{\partial V}{\partial T} - \frac{a}{RVT^2} - \frac{a}{RTV^2}\frac{\partial V}{\partial T} = \frac{1}{T}$$

$$\therefore \frac{\partial V}{\partial T}\left(\frac{1}{V-b} - \frac{a}{RTV^2}\right) = \frac{1}{T} + \frac{a}{RVT^2}$$

$$\therefore \frac{\partial V}{\partial T} = \frac{\frac{1}{T} + \frac{a}{RVT^2}}{\frac{1}{V-b} - \frac{a}{RTV^2}} \cdot \frac{RT}{RT} = \frac{R + \frac{a}{VT}}{\frac{RT}{V-b} - \frac{a}{V^2}}$$

22.

22. This exercise gives another example of the fact that the chain rule is not applicable if f is not differentiable. Consider the function

$$f(x, y) = \begin{cases} \dfrac{xy^2}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0). \end{cases}$$

Show that

(a) $\partial f/\partial x$ and $\partial f/\partial y$ exist at $(0, 0)$.

(b) If $\mathbf{g}(t) = (at, bt)$ for constants a and b, then $f \circ \mathbf{g}$ is differentiable and $(f \circ \mathbf{g})'(0) = ab^2/(a^2 + b^2)$, but $\nabla f(0, 0) \cdot \mathbf{g}'(0) = 0$.

(a) Computing f_x directly won't help since

$$f_x = \frac{(x^2+y^2)y^2 - xy^2(2x)}{(x^2+y^2)^2} = \frac{(-x^2+y^2)y^2}{(x^2+y^2)^2}, \quad x, y \neq 0$$

and nothing cancels to get rid of denominator.

So, look at original definition:

$$\frac{\partial f}{\partial x}(0,0) = \lim_{h \to 0} \frac{f(0+h, 0) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{\frac{(0+h)0^2}{(0+h)^2+0^2} - 0}{h} = \lim_{h \to 0} 0 = 0$$

$$\frac{\partial f}{\partial y}(0,0) = \lim_{h \to 0} \frac{f(0, 0+h) - f(0,0)}{h}$$

$$= \lim_{h \to 0} \frac{\frac{0(0+h)^2}{0+(0+h)^2} - 0}{h} = \lim_{h \to 0} 0 = 0$$

(b) $f \circ \vec{g}(t) = \dfrac{(at)(bt)^2}{(at)^2 + (bt)^2} = \dfrac{ab^2t^3}{a^2t^2 + b^2t^2} = \dfrac{ab^2t}{a^2+b^2}$

$$\therefore \frac{d}{dt}(f\circ g)(0) = \frac{ab^2}{a^2+b^2} \neq 0 \text{ if } a,b \neq 0$$

But, by chain rule, $D(f\circ\vec{g})(0) = Df(\vec{g}(0))\cdot g'(0)$

$$= Df(0,0)\cdot g'(0) = \begin{bmatrix} f_x(0,0) & f_y(0,0) \end{bmatrix}\begin{bmatrix} a \\ b \end{bmatrix}$$

$= 0$ since $f_x(0,0)=0, f_y(0,0)$ by (a).

$$\therefore D(f\circ\vec{g})(0) = \nabla f(0,0)\cdot\vec{g}'(0) = 0$$

So $D(f\circ\vec{g})(0) \neq \frac{d}{dt}(f\circ\vec{g})(0)$

23.

23. Prove that if $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{x}_0 \in U$, there is a neighborhood V of $\mathbf{0} \in \mathbb{R}^n$ and a function $R_1\colon V \to \mathbb{R}$ such that for all $\mathbf{h} \in V$, we have $\mathbf{x}_0 + \mathbf{h} \in U$,

$$f(\mathbf{x}_0 + \mathbf{h}) = f(\mathbf{x}_0) + [\mathbf{D}f(\mathbf{x}_0)]\mathbf{h} + R_1(\mathbf{h})$$

and

$$\frac{R_1(\mathbf{h})}{\|\mathbf{h}\|} \to 0 \quad \text{as} \quad \mathbf{h} \to \mathbf{0}.$$

Since f is differentiable at $\vec{x}_0 \in U$, then given any $\epsilon > 0$, $\exists\, \delta > 0$ s.t. if $0 < \|\vec{x}-\vec{x}_0\| < \delta$, then

[1] $\dfrac{\|f(\vec{x}) - f(\vec{x}_0) - [Df(\vec{x}_0)](\vec{x}-\vec{x}_0)\|}{\|\vec{x}-\vec{x}_0\|} < \epsilon$, by definition

$\therefore$ Define $V = B_\delta(\vec{0})$, and define $\vec{h} = \vec{x}-\vec{x}_0$, where

$\vec{x} \in D_\delta(\vec{x}_0)$ $\therefore$ $\vec{h} \in V \Leftrightarrow \vec{x}_0 + \vec{h} \in D_\delta(\vec{x}_0)$, as $\vec{h} \in V$ means $\|\vec{h}\| < \delta$

Define $R_1(\vec{h}) = f(\vec{x}_0 + \vec{h}) - f(\vec{x}_0) - [Df(\vec{x}_0)]\vec{h}$

$\therefore$ Given the above $\epsilon > 0$, choose the above δ, and if $0 < \|\vec{h}\| < \delta$, then $0 < \|\vec{x} - \vec{x}_0\| < \delta$, since $\vec{h} = \vec{x} - \vec{x}_0$

and $\therefore$ [1] above $\Rightarrow$

$$\frac{\| f(\vec{x}_0 + \vec{h}) - f(\vec{x}_0) - [Df(\vec{x}_0)]\vec{h} \|}{\|\vec{h}\|} < \epsilon \text{, or}$$

$$\frac{|R_1(\vec{h})|}{\|\vec{h}\|} < \epsilon \text{, since } \|R_1(\vec{h})\| = |R_1(\vec{h})| \text{ as } R_1(\vec{h}) : \mathbb{R}^n \to \mathbb{R}$$

$$\therefore \lim_{\vec{h} \to \vec{0}} \frac{|R_1(\vec{h})|}{\|\vec{h}\|} = 0$$

Since $\frac{||R_1(\vec{h})| - 0|}{\|\vec{h}\|} = \frac{|R_1(\vec{h}) - 0|}{\|\vec{h}\|}$, $\lim_{\vec{h} \to \vec{0}} \frac{R_1(\vec{h})}{\|\vec{h}\|} = 0$

$\therefore \forall \vec{h} \in V$, $\vec{x}_0 + \vec{h} \in U$, $f(\vec{x}_0 + \vec{h}) = f(\vec{x}_0) + [Df(\vec{x}_0)]\vec{h} + R_1(\vec{h})$

and $\lim_{\vec{h} \to \vec{0}} \frac{R_1(\vec{h})}{\|\vec{h}\|} = 0$

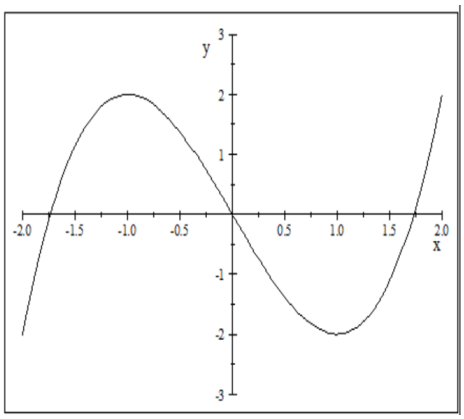
24.

24. Suppose $\mathbf{x}_0 \in \mathbb{R}^n$ and $0 \le r_1 < r_2$. Show that there is a C^1 function $f: \mathbb{R}^n \to \mathbb{R}$ such that $f(\mathbf{x}) = 0$ for $\|\mathbf{x} - \mathbf{x}_0\| \ge r_2$; $0 < f(\mathbf{x}) < 1$ for $r_1 < \|\mathbf{x} - \mathbf{x}_0\| < r_2$; and $f(\mathbf{x}) = 1$ for $\|\mathbf{x} - \mathbf{x}_0\| \le r_1$. [HINT: Apply a cubic polynomial with $g(r_1^2) = 1$ and $g(r_2^2) = g'(r_2^2) = g'(r_1^2) = 0$ to $\|\mathbf{x} - \mathbf{x}_0\|^2$ when $r_1 < \|\mathbf{x} - \mathbf{x}_0\| < r_2$.]

$f(x) = ax^3 + bx^2$ are C^1 functions, $a, b \neq 0$.

A plot of, e.g., $y = x^3 - 3x$

looks like:



The idea is to use the part of the function between the max & min, where slope = 0, and define the function to be 0 to the right, and 1 to the left, using r_1 and r_2.

We want r_1 to be the local max and r_2 to be the local min (since $r_1 < r_2$).

$\therefore$ set $y' = (x - r_1)(x - r_2) = 0$

or $y' = x^2 - (r_1 + r_2)x + r_1 r_2 = 0$

$y'' = 2x - (r_1 + r_2)$, $y''(r_2) = r_2 - r_1 > 0$, $y''(r_1) = r_1 - r_2 < 0$

$\therefore$ r_2 will be local min, r_1 a local max.

Integrate y': $y = \frac{1}{3}x^3 - \frac{(r_1 + r_2)}{2}x^2 + r_1 r_2 x + C$

or $f(x) = x^3 - 3(r_1 + r_2)x^2 + 6r_1 r_2 x + K$

We want $f(r_2) = 0$, so

$$f(r_2) = r_2^3 - 3(r_1+r_2)r_2^2 + 6r_1r_2^2 + k = 0$$

$$= r_2^3 - 3r_2^3 - 3r_1r_2^2 + 6r_1r_2^2 + k = 0$$

$$= -2r_2^3 + 3r_1r_2^2 + k = 0, \quad k = 2r_2^3 - 3r_1r_2^2$$

$$\therefore f(x) = x^3 - 3(r_1+r_2)x^2 + 6r_1r_2x + (2r_2^3 - 3r_1r_2^2)$$

We want $f(r_1) = 1$, so

$$f(r_1) = r_1^3 - 3r_1^3 - 3r_1^2r_2 + 6r_1^2r_2 + (2r_2^3 - 3r_1r_2^2)$$

$$= -2r_1^3 + 3r_1^2r_2 + 2r_2^3 - 3r_1r_2^2$$

$$= 2(r_2^3 - r_1^3) - 3r_1r_2(r_2 - r_1)$$

$$= 2(r_2-r_1)(r_2^2 + r_2r_1 + r_1^2) - (r_2-r_1)(3r_1r_2)$$

$$= (r_2-r_1)(2r_2^2 - r_1r_2 + 2r_1^2)$$

$$= (r_2-r_1)\left[r_2(2r_2-r_1) + 2r_1^2\right] > 0$$

since $r_2 > r_1 > 0$

The above $f(x)$ is centered around 0. Now define it by shifting according to x_0.

$\therefore$ Let $a = (r_2-r_1)\left[r_2(2r_2-r_1) + 2r_1^2\right]$

$$f(\vec{x}) = \begin{cases} 1, & \text{for } \|\vec{x} - \vec{x}_0\| \le r_1 \\ \frac{1}{a}\left[\|\vec{x} - \vec{x}_0\|^3 - 3(r_1 + r_2)\|\vec{x} - \vec{x}_0\|^2 + 6r_1 r_2 \|\vec{x} - \vec{x}_0\| + (2r_2^2 - 3r_1 r_2^2)\right] & \\ \qquad \text{for } r_1 < \|\vec{x} - \vec{x}_0\| < r_2 & \\ 0, & \text{for } r_2 \le \|\vec{x}\| \end{cases}$$

The above $f(\vec{x})$ meets the requirement of $0 < f(\vec{x}) < 1$ for $r_1 < \|\vec{x} - \vec{x}_0\| < r_2$, from the construction above.

$f(\vec{x})$ is also C^1 since, along any x_i, $f(\vec{x})$ is the cubic polynomial from local max to local min, and $\therefore$ the partial derivative at the endpoints (r_1, r_2) exists (and is 0).

25. **25.** Find a C^1 mapping $f: \mathbb{R}^3 \to \mathbb{R}^3$ that takes the vector $\mathbf{i} + \mathbf{j} + \mathbf{k}$ emanating from the origin to $\mathbf{i} - \mathbf{j}$ emanating from (1, 1, 0) and takes $\mathbf{k}$ emanating from (1, 1, 0) to $\mathbf{k} - \mathbf{i}$ emanating from the origin.

From the answer in the back of the book:

From #24 above, let $g_1(\vec{x}): \mathbb{R}^3 \to \mathbb{R}$ and $g_2(\vec{x}): \mathbb{R}^3 \to \mathbb{R}$ be C^1 functions such that:

$$g_1(\vec{x}) = \begin{cases} 1, & \|\vec{x}\| \le r_2/3 \\ \text{cubic polynomial described in Ex. \#24}, & \frac{r_2}{3} < \|x\| < \frac{2r_2}{3} \\ 0, & \frac{2r_2}{3} \le \|\vec{x}\| \end{cases}$$

and

$$g_2(\vec{x}) = \begin{cases} 1, & \|\vec{x}-(1,1,0)\| \le \frac{\sqrt{2}}{3} \\ \text{cubic polynomial described in Ex. \#24, for } & \frac{\sqrt{2}}{3} < \|\vec{x}-(1,1,0)\| < \frac{2\sqrt{2}}{3} \\ 0, & \frac{2\sqrt{2}}{3} \le \|\vec{x}-(1,1,0)\| \end{cases}$$

In the above, $r_1 = \frac{\sqrt{2}}{3}$, $r_2 = \frac{2\sqrt{2}}{3}$, using Ex. #24 above.

$g_1(\vec{x})$ is for vectors arising from the origin

$g_2(\vec{x})$ is for vectors arising from $(1,1,0)$

Let $h_1(\vec{x}) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$, for vectors arising from $(0,0,0)$

$$h_2(\vec{x}) = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \text{ for vectors arising from } (1,1,0)$$

Note $h_1(\vec{x})$ and $h_2(\vec{x})$ are C^1, since, if $j(\vec{x}) = T\vec{x}$, then $Dj(\vec{x}) = T$, and so the matrix T of partial derivatives (just real numbers in this case) are all continuous (since they are real constants).

$\therefore$ Let $f(\vec{x}) = g_1(\vec{x})h_1(\vec{x}) + g_2(\vec{x})h_2(\vec{x})$

This is the sum and product of C^1 functions, and so is a C^1 function itself.

26.

26. What is wrong with the following argument? Suppose $w = f(x, y, z)$ and $z = g(x, y)$. By the chain rule,

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial x} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial x} = \frac{\partial w}{\partial x} + \frac{\partial w}{\partial z}\frac{\partial z}{\partial x}.$$

Hence, $0 = (\partial w/\partial z)(\partial z/\partial x)$, and so $\partial w/\partial z = 0$ or $\partial z/\partial x = 0$, which is, in general, absurd.

Write $W(x,y) = f(x, y, g(x,y))$

The $\frac{\partial W}{\partial x}$ on the left side means the derivative from holding the 2nd independent variable constant.

The $\frac{\partial W}{\partial x}$ on the right is really $\frac{\partial f}{\partial x}$, and means the derivative of the 1st independent variable while holding the 2nd & 3rd independent variables constant.

Thus, it's an abuse of terminology, and the two are different derivatives.

27.

27. Prove rules (iii) and (iv) of Theorem 10. (HINT: Use the same addition and subtraction tricks as in the one-variable case and Theorem 8.)

(iii) **Product Rule**. Let f: $U \subset \mathbb{R}^n \to \mathbb{R}$ and g: $U \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable at $\mathbf{x}_0$ and let $h(\mathbf{x}) = g(\mathbf{x})\, f(\mathbf{x})$. Then h: $U \subset \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\mathbf{x}_0$ and

$$\mathbf{D}h(\mathbf{x}_0) = g(\mathbf{x}_0)\mathbf{D}f(\mathbf{x}_0) + f(\mathbf{x}_0)\mathbf{D}g(\mathbf{x}_0).$$

(Note that each side of this equation is a $1 \times n$ matrix; a more general product rule is presented in Exercise 31 at the end of this section.)

Note: $Dh(\vec{x}_0)$, $Df(\vec{x}_0)$, $Dg(\vec{x}_0)$ are $1 \times n$ matrices

$$h(\vec{x}) - h(\vec{x}_0) - \left[g(\vec{x}_0)Df(\vec{x}_0) + f(\vec{x}_0)Dg(\vec{x}_0)\right](\vec{x}-\vec{x}_0)$$

$$= g(\vec{x})f(\vec{x}) - g(\vec{x}_0)f(\vec{x}_0) - \left[g(\vec{x}_0)Df(\vec{x}_0) + f(\vec{x}_0)Dg(\vec{x}_0)\right](\vec{x}-\vec{x}_0) \quad [0]$$

$$= g(\vec{x})f(\vec{x}) - g(\vec{x}_0)f(\vec{x}) + g(\vec{x}_0)f(\vec{x}) \quad \{\text{adding/subtracting } g(\vec{x}_0)f(\vec{x})\}$$

$$- g(\vec{x}_0)f(\vec{x}_0) - \left[g(\vec{x}_0)Df(\vec{x}_0) + f(\vec{x}_0)Dg(\vec{x}_0)\right](\vec{x}-\vec{x}_0)$$

$$= \left[g(\vec{x}) - g(\vec{x}_0)\right]f(\vec{x}) + g(\vec{x}_0)\left[f(\vec{x}) - f(\vec{x}_0)\right]$$

$$- \left[g(\vec{x}_0)Df(\vec{x}_0) + f(\vec{x}_0)Dg(\vec{x}_0)\right](\vec{x}-\vec{x}_0)$$

$\{\text{now subtract/add } [Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)]f(\vec{x})\}$

$$= \left[g(\vec{x}) - g(\vec{x}_0) - Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\right]f(\vec{x}) + \left[Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\right]f(\vec{x})$$

$\{\text{and subtract/add } g(\vec{x}_0)[Df(\vec{x}_0)(\vec{x}-\vec{x}_0)]\}$

$$+ g(\vec{x}_0)\left[f(\vec{x}) - f(\vec{x}_0) - Df(\vec{x}_0)(\vec{x}-\vec{x}_0)\right] + g(\vec{x}_0)\left[Df(\vec{x}_0)(\vec{x}-\vec{x}_0)\right]$$

$$- \left[g(\vec{x}_0)Df(\vec{x}_0) + f(\vec{x}_0)Dg(\vec{x}_0)\right](\vec{x}-\vec{x}_0)$$

$$= \left[g(\vec{x}) - g(\vec{x}_0) - Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\right]f(\vec{x}) \quad [1]$$

$$+ g(\vec{x}_0)\left[f(\vec{x}) - f(\vec{x}_0) - Df(\vec{x}_0)(\vec{x}-\vec{x}_0)\right] \quad [2]$$

$$+ \left[f(\vec{x}) - f(\vec{x}_0)\right]\left[Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\right] \quad [3]$$

For [1], $\lim_{\vec{x}\to\vec{x}_0} \frac{[g(\vec{x}) - g(\vec{x}_0) - Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)]}{\|\vec{x}-\vec{x}_0\|} f(\vec{x}) =$

$$\lim_{\vec{x}\to\vec{x}_0} \frac{[g(\vec{x}) - g(\vec{x}_0) - Dg(x_0)(\vec{x}-\vec{x}_0)}{\|\vec{x}-\vec{x}_0\|} \cdot \lim_{\vec{x}\to\vec{x}_0} f(\vec{x}) =$$

$$0 \cdot f(\vec{x}_0) = 0$$

as $f(\vec{x})$ is continuous at $\vec{x}_0$ since it is differentiable at x_0. Also, the first limit is from the definition of the derivative of $g(\vec{x})$.

Note: as shown in file "Chapter 2 Notes",

$$\lim_{\vec{x}\to\vec{x}_0} \frac{\|f(\vec{x}) - f(\vec{x}_0) - T(\vec{x}-\vec{x}_0)\|}{\|\vec{x}-\vec{x}_0\|} = 0 \iff \lim_{\vec{x}\to\vec{x}_0} \frac{f(\vec{x}) - f(\vec{x}_0) - T(\vec{x}-\vec{x}_0)}{\|\vec{x}-\vec{x}_0\|} = \vec{0}$$

For [2] above, since f is differentiable at $\vec{x}_0$

$$\lim_{\vec{x}\to\vec{x}_0} g(\vec{x}_0) \frac{[f(\vec{x}) - f(\vec{x}_0) - Df(\vec{x}_0)(\vec{x}-\vec{x}_0)]}{\|\vec{x}-\vec{x}_0\|} =$$

$$g(\vec{x}_0) \cdot 0 = 0$$

For [3], consider $\lim_{\vec{x}\to\vec{x}_0} [f(\vec{x}) - f(\vec{x}_0)] \frac{[Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)]}{\|\vec{x}-\vec{x}_0\|}$

$\lim_{\vec{x}\to\vec{x}_0} [f(\vec{x}) - f(\vec{x}_0)] = 0$, as f is continuous at $\vec{x}_0$ since f is differentiable at $\vec{x}_0$.

In "Chapter 2 Notes", it is shown that for any matrix T, $\|T\vec{x}\| \le K\|\vec{x}\|$, some real value K specific to T. Actually, $K = \|T\hat{e}_1\| + \dots + \|T\hat{e}_n\|$, where $\hat{e}_i$ is the natural basis, and T is $m \times n$ and $T_{ij} \in R$.

$\therefore$ Let K be s.t. $\|Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\| \le K\|\vec{x}-\vec{x}_0\|$

$\therefore$ Let $\epsilon > 0$. $\exists \delta > 0$ s.t. if $0 < \|\vec{x}-\vec{x}_0\| < \delta$, then

$$|f(\vec{x}) - f(\vec{x}_0)| < \frac{\epsilon}{K} \qquad \{f(\vec{x}) \text{ continuous at } \vec{x}_0\}$$

$$\therefore |f(\vec{x}) - f(\vec{x}_0)|\,\|Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\| < \frac{\epsilon}{K} \cdot K\|\vec{x}-\vec{x}_0\| = \epsilon\|\vec{x}-\vec{x}_0\|$$

$$\therefore \left| \frac{|f(\vec{x}) - f(\vec{x}_0)|\,\|Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\| - 0}{\|\vec{x}-\vec{x}_0\|} \right| < \epsilon$$

$$\therefore \lim_{\vec{x}\to\vec{x}_0} [f(\vec{x}) - f(\vec{x}_0)] \frac{[Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)]}{\|\vec{x}-\vec{x}_0\|} = 0$$

$\therefore$ [1], [2], and [3] $\Rightarrow$ for [0],

$$\lim_{\vec{x}\to\vec{x}_0} \frac{f(\vec{x})g(\vec{x}) - f(\vec{x}_0)g(\vec{x}_0) - [g(\vec{x}_0)Df(\vec{x}_0) + f(\vec{x}_0)Dg(\vec{x}_0)](\vec{x}-\vec{x}_0)}{\|\vec{x}-\vec{x}_0\|} = 0$$

$$\therefore \boxed{Dh(\vec{x}_0) = g(\vec{x}_0)Df(\vec{x}_0) + f(\vec{x}_0)Dg(\vec{x}_0)}$$

(iv) **Quotient Rule.** With the same hypotheses as in rule (iii), let $h(\mathbf{x}) = f(\mathbf{x})/g(\mathbf{x})$ and suppose g is never zero on U. Then h is differentiable at $\mathbf{x}_0$ and

$$\mathbf{D}h(\mathbf{x}_0) = \frac{g(\mathbf{x}_0)\mathbf{D}f(\mathbf{x}_0) - f(\mathbf{x}_0)\mathbf{D}g(\mathbf{x}_0)}{[g(\mathbf{x}_0)]^2}.$$

It will probably be easier to prove just for $\frac{1}{g(\vec{x})}$, and then apply (iii) for $\frac{f(\vec{x})}{g(\vec{x})}$, so prove:

$$D\left(\frac{1}{g}\right)(\vec{x}_0) = -\frac{1}{g(\vec{x}_0)^2} Dg(\vec{x}_0)$$

$\therefore$ Look at $\frac{1}{g(\vec{x})} - \frac{1}{g(\vec{x}_0)} - \left(-\frac{1}{g(\vec{x}_0)^2} Dg(\vec{x}_0)\right)(\vec{x}-\vec{x}_0)$ [0]

$$= \frac{g(\vec{x}_0) - g(\vec{x})}{g(\vec{x})\,g(\vec{x}_0)} + \frac{Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)}{g(\vec{x}_0)^2}$$

{adding/subtracting $\frac{g(\vec{x}_0)-g(\vec{x})}{g(\vec{x}_0)^2}$}

$$= \frac{g(\vec{x}_0) - g(\vec{x}) + Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)}{g(\vec{x}_0)^2} - \frac{g(\vec{x}_0)-g(\vec{x})}{g(\vec{x}_0)^2} + \frac{g(\vec{x}_0)-g(\vec{x})}{g(\vec{x})\,g(\vec{x}_0)}$$

$$= -\frac{1}{g(\vec{x}_0)^2}\left[g(\vec{x}) - g(\vec{x}_0) - Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\right] \quad [1]$$

$$+ \left[\frac{g(\vec{x}) - g(\vec{x}_0)}{g(\vec{x}_0)}\right]\left[\frac{1}{g(\vec{x}_0)} - \frac{1}{g(\vec{x})}\right] \quad [2]$$

For [1], since $g(\vec{x})$ is differentiable at $\vec{x}_0$,

$$\lim_{\vec{x}\to\vec{x}_0}\left[-\frac{1}{g(\vec{x}_0)^2}\right]\left[\frac{g(\vec{x})-g(\vec{x}_0)-Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)}{\|\vec{x}-\vec{x}_0\|}\right]$$

$$=\left[-\frac{1}{g(\vec{x}_0)^2}\right]\lim_{\vec{x}\to\vec{x}_0}\frac{g(\vec{x})-g(\vec{x}_0)-Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)}{\|\vec{x}-\vec{x}_0\|}$$

$$=-\frac{1}{g(\vec{x}_0)^2}\cdot 0=0$$

For [2], $\left[\frac{g(\vec{x})-g(\vec{x}_0)}{g(\vec{x}_0)}\right]\left[\frac{1}{g(\vec{x}_0)}-\frac{1}{g(\vec{x})}\right]$

{adding/substracting $Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)$}

$$=\left[\frac{g(\vec{x})-g(\vec{x}_0)-Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)+Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)}{g(\vec{x}_0)}\right]\left[\frac{1}{g(\vec{x}_0)}-\frac{1}{g(\vec{x})}\right]$$

$$=\frac{1}{g(\vec{x}_0)}\left[g(\vec{x})-g(\vec{x}_0)-Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\right]\left[\frac{1}{g(\vec{x}_0)}-\frac{1}{g(\vec{x})}\right] \quad [3]$$

$$+\frac{1}{g(\vec{x}_0)}\left[Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\right]\left[\frac{1}{g(x_0)}-\frac{1}{g(\vec{x})}\right] \quad [4]$$

For [3], $\lim_{\vec{x}\to\vec{x}_0}\frac{1}{g(\vec{x}_0)}\left[\frac{g(\vec{x})-g(\vec{x}_0)-Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)}{\|\vec{x}-\vec{x}_0\|}\right]\left[\frac{1}{g(\vec{x}_0)}-\frac{1}{g(\vec{x})}\right]$

$$=\frac{1}{g(\vec{x}_0)}\cdot 0\cdot 0=0$$

$g(\vec{x})$ differentiable at $\vec{x}_0$; $g(\vec{x})$ continuous at $\vec{x}_0$

For [4], using $\|T\vec{x}\| \le k\|\vec{x}\|$, where T is a matrix and k is a real number specific to T,

Let k be s.t. $\|Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\| \le k\|x-x_0\|$

Let $\epsilon > 0$. Since $\lim_{\vec{x}\to\vec{x}_0}\left[\frac{1}{g(\vec{x})} - \frac{1}{g(\vec{x}_0)}\right] = 0$ as

g is continuous at x_0, $\exists\, \delta > 0$ s.t.

if $0 < \|\vec{x}-\vec{x}_0\| < \delta$, then $\left|\frac{1}{g(\vec{x})} - \frac{1}{g(\vec{x}_0)}\right| < \frac{\epsilon}{k}\cdot|g(\vec{x}_0)|$

$\therefore \left|\frac{1}{g(\vec{x}_0)}\right| \|Dg(x_0)(x-x_0)\| \left|\frac{1}{g(\vec{x})} - \frac{1}{g(\vec{x}_0)}\right|$

$$\le \left|\frac{1}{g(\vec{x}_0)}\right| k\|\vec{x}-\vec{x}_0\| \frac{\epsilon}{k}|g(\vec{x}_0)| = \epsilon\|\vec{x}-\vec{x}_0\|$$

$\therefore$ Given the above $\epsilon > 0$, $\exists\, \delta > 0$ s.t. if $0 < \|\vec{x}-\vec{x}_0\| < \delta$, then

$$\frac{\left|\frac{1}{g(\vec{x}_0)}\right| \|Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)\| \left|\frac{1}{g(\vec{x})} - \frac{1}{g(\vec{x}_0)}\right|}{\|\vec{x}-\vec{x}_0\|} < \epsilon$$

$$\therefore \lim_{\vec{x}\to\vec{x}_0} \frac{1}{g(\vec{x}_0)} \frac{Dg(\vec{x}_0)(\vec{x}-\vec{x}_0)}{\|\vec{x}-\vec{x}_0\|}\left[\frac{1}{g(\vec{x})} - \frac{1}{g(\vec{x}_0)}\right] = 0$$

$\therefore$ [3] & [4] $\Rightarrow$

$$\lim_{\vec{x}\to\vec{x}_0} \frac{\left[\frac{g(\vec{x})-g(\vec{x}_0)}{g(\vec{x}_0)}\right]\left[\frac{1}{g(\vec{x}_0)} - \frac{1}{g(\vec{x})}\right]}{\|\vec{x}-\vec{x}_0\|} = 0$$

which is [2].

$\therefore$ [1] & [2] $\Rightarrow$

$$\lim_{\vec{x}\to\vec{x}_0} \frac{\frac{1}{g(\vec{x})} - \frac{1}{g(\vec{x}_0)} - \left[\left(-\frac{1}{g(\vec{x}_0)^2}\right) Dg(\vec{x}_0)\right](\vec{x}-\vec{x}_0)}{\|\vec{x}-\vec{x}_0\|} = 0$$

which is [0]

$$\therefore \quad D\left(\frac{1}{g}\right)(x_0) = -\frac{1}{g(\vec{x}_0)^2} Dg(\vec{x}_0)$$

28.

28. Show that h: $\mathbb{R}^n \to \mathbb{R}^m$ is differentiable if and only if each of the m components h_i: $\mathbb{R}^n \to \mathbb{R}$ is differentiable. (HINT: Use the coordinate projection function and the chain rule for one implication and consider

$$\left[\frac{\|h(\mathbf{x}) - h(\mathbf{x}_0) - \mathbf{D}h(\mathbf{x}_0)(\mathbf{x}-\mathbf{x}_0)\|}{\|\mathbf{x}-\mathbf{x}_0\|}\right]^2 = \frac{\sum_{i=1}^{m}[h_i(\mathbf{x}) - h_i(\mathbf{x}_0)\mathbf{D}h_i(\mathbf{x}_0)(\mathbf{x}-\mathbf{x}_0)]^2}{\|\mathbf{x}-\mathbf{x}_0\|^2}$$

to obtain the other.)

should be,

$$\in \left[h_i(\vec{x}) - h_i(\vec{x}_0) - Dh_i(\vec{x}_0)(\vec{x}-\vec{x}_0)\right]^2$$

Note: $|a_i| \le \sqrt{a_1^2 + \dots + a_m^2} \le |a_1| + \dots + |a_m|$, $i = 1, 2, \dots, m$

This is easily proved by squaring all sides.

(1) Assume $h: R^n \to R^m$ is differentiable, and let $\vec{x}_0 \in R^n$

$\therefore$ Given $\epsilon > 0$, $\exists \delta$ s.t. if $0 < \|\vec{x}-\vec{x}_0\| < \delta$, then

$$\frac{\|h(\vec{x}) - h(\vec{x}_0) - Dh(\vec{x}_0)(\vec{x}-\vec{x}_0)\|}{\|\vec{x}-\vec{x}_0\|} < \epsilon$$

Note: $h(\vec{x}) - h(\vec{x}_0) - Dh(\vec{x}_0)(\vec{x} - \vec{x}_0) =$

$$\begin{bmatrix} h_1(\vec{x}) \\ \vdots \\ h_m(\vec{x}) \end{bmatrix} - \begin{bmatrix} h_1(\vec{x}_0) \\ \vdots \\ h_m(\vec{x}_0) \end{bmatrix} - \begin{bmatrix} \frac{\partial h_1}{\partial x_1}(\vec{x}_0) & \cdots & \frac{\partial h_1}{\partial x_n}(\vec{x}_0) \\ \vdots & & \vdots \\ \frac{\partial h_m}{\partial x_1}(\vec{x}_0) & & \frac{\partial h_m}{\partial x_n}(\vec{x}_0) \end{bmatrix} \begin{bmatrix} (\vec{x} - \vec{x}_0)_1 \\ \vdots \\ (\vec{x} - \vec{x}_0)_n \end{bmatrix}$$

Let a_i = value of the "i" row:

$$a_i = h_i(\vec{x}) - h_i(\vec{x}_0) - \left[\frac{\partial h_i}{\partial x_1}(\vec{x}_0), \ldots, \frac{\partial h_i}{\partial x_n}(\vec{x}_0)\right] \cdot \left[(\vec{x} - \vec{x}_0)_1, \ldots, (\vec{x} - \vec{x}_0)_n\right]$$

(dot product)

$$\therefore \sqrt{a_1^2 + \ldots + a_m^2} = \| h(\vec{x}) - h(\vec{x}_0) - Dh(\vec{x}_0)(\vec{x} - \vec{x}_0) \|$$

and $|a_i| = |h_i(\vec{x}) - h_i(\vec{x}_0) - Dh_i(x_0)(\vec{x} - \vec{x}_0|$

Since $|a_i| \le \sqrt{a_1^2 + \ldots + a_m^2}$, then, if $0 < \|\vec{x} - \vec{x}_0\| < \delta$,

$$\frac{|h_i(\vec{x}) - h_i(\vec{x}_0) - Dh_i(\vec{x}_0)(\vec{x} - \vec{x}_0)|}{\|\vec{x} - \vec{x}_0\|} \le \frac{\|h(\vec{x}) - h(\vec{x}_0) - Dh(\vec{x}_0)(\vec{x} - \vec{x}_0)\|}{\|\vec{x} - \vec{x}_0\|} < \epsilon$$

$$\therefore \lim_{\vec{x} \to \vec{x}_0} \frac{|h_i(\vec{x}) - h_i(\vec{x}_0) - Dh_i(x_0)(\vec{x} - \vec{x}_0)|}{\|\vec{x} - \vec{x}_0\|} = 0$$

So $h_i : \mathbb{R}^n \to \mathbb{R}^m$ is differentiable wherever h is.

(2) Assume for $i = 1, \ldots, m$, $h_i : \mathbb{R}^n \to \mathbb{R}^m$ are differentiable. Let $\vec{x}_0 \in \mathbb{R}^n$.

Let $\epsilon > 0$. Then $\exists\, \delta > 0$ s.t. if $0 < \|\vec{x} - \vec{x}_0\| < \delta_i$, then

$$\frac{|h_i(\vec{x}) - h_i(\vec{x}_0) - Dh_i(\vec{x}_0)(\vec{x}-\vec{x}_0)|}{\|\vec{x}-\vec{x}_0\|} < \frac{\epsilon}{m}$$

Let $a_i = h_i(\vec{x}) - h_i(\vec{x}_0) - Dh_i(\vec{x}_0)(\vec{x}-\vec{x}_0)$, $i = 1, \ldots, m$

The above says $\frac{|a_i|}{\|\vec{x}-\vec{x}_0\|} < \epsilon$

But $\sqrt{a_1^2 + \ldots + a_m^2} = \|h(\vec{x}) - h(\vec{x}_0) - Dh(\vec{x}_0)(\vec{x}-\vec{x}_0)\|$

and $\sqrt{a_1^2 + \ldots + a_m^2} \le |a_1| + \ldots + |a_m|$

$\therefore$ Let $\delta = \min\{\delta_1, \ldots, \delta_m\}$. $\therefore$ if $0 < \|\vec{x}-\vec{x}_0\| < \delta$, then

$$\therefore \frac{\|h(\vec{x}) - h(\vec{x}_0) - Dh(\vec{x}_0)(\vec{x}-\vec{x}_0)\|}{\|\vec{x}-\vec{x}_0\|} \le \frac{|a_1|}{\|\vec{x}-\vec{x}_0\|} + \ldots + \frac{|a_m|}{\|\vec{x}-\vec{x}_0\|} < \frac{\epsilon}{m} + \ldots + \frac{\epsilon}{m} = \epsilon$$

$$\therefore \lim_{\vec{x}\to\vec{x}_0} \frac{\|h(\vec{x}) - h(\vec{x}_0) - Dh(\vec{x}_0)(\vec{x}-\vec{x}_0)\|}{\|\vec{x}-\vec{x}_0\|} = 0$$

$\therefore h: \mathbb{R}^n \to \mathbb{R}^m$ is differentiable wherever $h_i: \mathbb{R}^n \to \mathbb{R}^m$ are.

29.

29. Use the chain rule and differentiation under the integral sign, namely,

$$\frac{d}{dx}\int_a^b f(x, y)\, dy = \int_a^b \frac{\partial f}{\partial x}(x, y)\, dy,$$

to show that

$$\frac{d}{dx}\int_0^x f(x, y)\, dy = f(x, x) + \int_0^x \frac{\partial f}{\partial x}(x, y)\, dy.$$

To understand $\frac{d}{dx}\int_a^b f(x,y)\,dy = \int_a^b \frac{\partial f}{\partial x}(x,y)\,dy$,

look at Riemann sum:

$$\frac{d}{dx}\sum_{i=1}^{n} f(x,\xi_i)\Delta x_i = \sum_{i=1}^{n} \frac{\partial}{\partial x}\left[f(x,\xi_i)\Delta x_i\right]$$

where Δx_i goes from a to b. Since Δx_i only depend on a and b, not x, then $\frac{\partial}{\partial x}\left[f(x,\xi_i)\Delta x_i\right]$
$= \left[\frac{\partial f(x,\xi_i)}{\partial x}\right]\Delta x_i$

The limit, as the norm of the partition goes to 0, exists if $f(x,y)$ is sufficiently "nice".

Apparently, in this problem, $f(x,y)$ is sufficiently "nice."

For this problem, consider splitting up the two references to x. i.e.,

Let
$$F(a,x) = \int_0^a f(x,y)\,dy \qquad [1]$$

$\therefore$ If $a = x$, $F(x,x) = \int_0^x f(x,y)\,dy$

By the Fundamental Theorem of Calculus,

if $F(x) = \int_0^x f(t)\,dt$, then $F'(x) = f(x)$.

$\therefore$ Holding x constant in [1], by the Fundamental Theorem,

$$\frac{\partial F}{\partial a}(a, x) = \frac{d}{da}\int_0^a f(x,y)\,dy = f(x,a) \quad [2]$$

Differentiating under the integral,

$$\frac{\partial F}{\partial x}(a,x) = \frac{d}{dx}\int_0^a f(x,y)\,dy = \int_0^a \frac{\partial}{\partial x} f(x,y)\,dy \quad [3]$$

Now let $a(x) = x$. By chain rule,

$$\therefore \frac{d}{dx} F(a(x), x) = F_a \cdot \frac{da}{dx} + F_x \cdot \text{But } \frac{da}{dx} = 1$$

$$\therefore \frac{d}{dx}\int_0^x f(x,y)\,dy = \frac{d}{dx} F(a(x), x) = F_a + F_x$$

using [2], [3]

$$= f(x, a(x)) + \int_0^{a(x)} \frac{\partial}{\partial x} f(x,y)\,dy$$

$$= f(x,x) + \int_0^x \frac{\partial}{\partial x} f(x,y)\,dy$$

30.

30. For what integers $p > 0$ is

$$f(x) = \begin{cases} x^p \sin(1/x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

differentiable? For what p is the derivative continuous?

(a) For $x \neq 0$, and all integers $p > 0$, $f(x)$ is differentiable.

For $x = 0$, look at $\lim_{x \to 0} \frac{x^p \sin(\frac{1}{x}) - 0}{x - 0} = \lim_{x \to 0} \frac{x^p \sin(\frac{1}{x})}{x}$

Let $\epsilon > 0$. Since $|\sin \frac{1}{x}| \leq 1$ for all $x \neq 0$,

Then $|x^p \sin \frac{1}{x}| \leq |x^p|$ for all $x \neq 0$.

$\therefore \left| \frac{x^p \sin \frac{1}{x}}{x} \right| \leq |x^{p-1}| = |x|^{p-1}$, all $x \neq 0$.

$|x|^{p-1} < \epsilon \iff |x| < \sqrt[p-1]{\epsilon}$, $x \neq 0$, $p - 1 > 0$.

$\therefore$ For all $p \geq 2$, choose $\delta = \sqrt[p-1]{\epsilon}$.

Then if $0 < |x| < \delta$, $\left| \frac{x^p \sin \frac{1}{x}}{x} \right| < \epsilon$

$\therefore$ $f(x)$ differentiable $\forall x$ for all integers $\underline{p \geq 2}$

(b) Look at $f'(x) = p x^{p-1} \sin(\frac{1}{x}) + x^p \cos(\frac{1}{x})(-\frac{1}{x^2})$

$= p x^{p-1} \sin(\frac{1}{x}) - x^{p-2} \cos(\frac{1}{x})$

From (a) $x^{p-1}\sin(\frac{1}{x})$ is differentiable if $p-1 \geq 2$, or $p \geq 3$

Similar reasoning to (a) shows $x^{p-2}\cos(\frac{1}{x})$ is differentiable if $p-2 \geq 2$, or $p \geq 4$

$\therefore$ Derivative continuous for $p \geq 4$.

31.

31. Suppose $f: \mathbb{R}^n \to \mathbb{R}$ and $g: \mathbb{R}^n \to \mathbb{R}^m$ are differentiable. Show that the product function $h(\mathbf{x}) = f(\mathbf{x})g(\mathbf{x})$ from $\mathbb{R}^n$ to $\mathbb{R}^m$ is differentiable and that if $\mathbf{x}_0$ and $\mathbf{y}$ are in $\mathbb{R}^n$, then $[\mathbf{D}h(\mathbf{x}_0)]\mathbf{y} = f(\mathbf{x}_0)\{[\mathbf{D}g(\mathbf{x}_0)]\mathbf{y}\} + \{[\mathbf{D}f(\mathbf{x}_0)]\mathbf{y}\}g(\mathbf{x}_0)$.

Consider $g: \mathbb{R}^n \to \mathbb{R}^m$. If $\vec{x} \in \mathbb{R}^n$, then

$g(\vec{x}) = (g_1(\vec{x}), \ldots, g_m(\vec{x}))$, where $g_i: \mathbb{R}^n \to \mathbb{R}$, $i = 1, \ldots, m$

$\therefore$ For $f: \mathbb{R}^n \to \mathbb{R}$,

$h(\vec{x}) = f(\vec{x})g(\vec{x}) = (f(\vec{x})g_1(\vec{x}), \ldots, f(\vec{x})g_m(\vec{x}))$

As shown in problem #28, $h(\vec{x})$ is differentiable $\leftrightarrow$ each $f(\vec{x})g_i(\vec{x})$ is differentiable.

But by Theorem 10 (iii), and shown in problem #27, each $f(\vec{x})g_i(\vec{x})$ is differentiable

since $f(\vec{x})$ is differentiable and $g(\vec{x})$ is differentiable (and $\therefore$ so is each $g_i(\vec{x})$).

$\therefore h(\vec{x})$ is differentiable

If $h_i(\vec{x}) = f(\vec{x}) g_i(\vec{x})$, by the product rule,

$$\underset{[1\times n]}{D h_i(\vec{x})} = \underset{[1\times 1]}{f(\vec{x})} \underset{[1\times n]}{D g_i(\vec{x})} + \underset{[1\times 1]}{g_i(\vec{x})} \underset{[1\times n]}{D f(\vec{x})}$$

where the matrix sizes are shown.

$\therefore Dh(\vec{x}_0)$

$$f(\vec{x}_0)\overset{m\times n}{\begin{bmatrix} g_{1x_1}(\vec{x}_0) & \cdots & g_{1x_n}(\vec{x}_0) \\ \vdots & & \\ g_{mx_1}(\vec{x}_0) & \cdots & g_{mx_n}(\vec{x}_0) \end{bmatrix}} + \overset{m\times n}{\begin{bmatrix} g_1(\vec{x}_0) f_{x_1}(\vec{x}_0) & \cdots & g_1(\vec{x}_0) f_{x_n}(\vec{x}_0) \\ \vdots & & \\ g_m(\vec{x}_0) f_{x_1}(\vec{x}_0) & \cdots & g_m(\vec{x}_0) f_{x_n}(\vec{x}_0) \end{bmatrix}}$$

$\therefore$ for $\vec{y} \in \mathbb{R}^n$, $Dh(\vec{x}_0)$, an $m\times n$ matrix, maps $\vec{y}$, an $n\times 1$ matrix, as follows:

$$f(\vec{x}_0)\begin{bmatrix} g_{1x_1}(\vec{x}_0) & \cdots & g_{1x_n}(\vec{x}_0) \\ \vdots & & \\ g_{mx_1}(\vec{x}_0) & \cdots & g_{mx_n}(\vec{x}_0) \end{bmatrix}\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} + \begin{bmatrix} g_1(\vec{x}_0) f_{x_1}(\vec{x}_0) & \cdots & g_1(\vec{x}_0) f_{x_n}(\vec{x}_0) \\ \vdots & & \\ g_m(\vec{x}_0) f_{x_1}(\vec{x}_0) & \cdots & g_m(\vec{x}_0) f_{x_n}(\vec{x}_0) \end{bmatrix}\begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$

$$= \underbrace{f(\vec{x}_0)}_{[1\times 1]} \underbrace{[Dg(\vec{x}_0)]\vec{y}}_{m\times 1} + \underbrace{\begin{bmatrix} g_1(\vec{x}_0) f_{x_1}(\vec{x}_0) y_1 + \dots + g_1(\vec{x}_0) f_{x_n}(\vec{x}_0) y_n \\ \vdots \\ g_m(\vec{x}_0) f_{x_1}(\vec{x}_0) y_1 + \dots + g_m(\vec{x}_0) f_{x_n}(\vec{x}_0) y_n \end{bmatrix}}_{m\times 1}$$

$$= f(\vec{x}_0)\left[Dg(\vec{x}_0)\vec{y}\right] + \underbrace{\left(f_{x_1}(\vec{x}_0) y_1 + \dots + f_{x_n}(\vec{x}_0) y_n\right)}_{[1\times 1],\text{a dot product}} \underbrace{\begin{bmatrix} g_1(\vec{x}_0) \\ \vdots \\ g_m(\vec{x}_0) \end{bmatrix}}_{m\times 1}$$

$$= \underbrace{f(\vec{x}_0)}_{[1\times 1]}\left[\underbrace{Dg(\vec{x}_0)}_{[m\times n]}\underbrace{\vec{y}}_{[n\times 1]}\right] + \left[\underbrace{Df(\vec{x}_0)}_{[1\times n]}\cdot\underbrace{\vec{y}}_{[n\times 1]}\right]\underbrace{\vec{g}(\vec{x}_0)}_{[m\times 1]}$$

32.

32. Let $g(u, v) = (e^u, u + \sin v)$ and $f(x, y, z) = (xy, yz)$. Compute $\mathbf{D}(g \circ f)$ at $(0, 1, 0)$ using the chain rule.

First, $f(0,1,0) = (0,0)$

$\therefore D(g\circ f)(0,1,0) = Dg(0,0)\cdot Df(0,1,0)$

$$Dg = \begin{bmatrix} g_{1u} & g_{1v} \\ g_{2u} & g_{2v} \end{bmatrix} = \begin{bmatrix} e^u & 0 \\ 1 & \cos v \end{bmatrix} \quad \therefore Dg(0,0) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$Df = \begin{bmatrix} f_{1x} & f_{1y} & f_{1z} \\ f_{2x} & f_{2y} & f_{2z} \end{bmatrix} = \begin{bmatrix} y & x & 0 \\ 0 & z & y \end{bmatrix} \quad \therefore Df(0,1,0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore D(g\circ f)(0,1,0) = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

33. **33.** Let $f: \mathbb{R}^4 \to \mathbb{R}$ and $\mathbf{c}(t): \mathbb{R} \to \mathbb{R}^4$. Suppose $\nabla f(1, 1, \pi, e^6) = (0, 1, 3, -7)$, $\mathbf{c}(\pi) = (1, 1, \pi, e^6)$, and $\mathbf{c}'(\pi) = (19, 11, 0, 1)$. Find $\dfrac{d(f \circ \mathbf{c})}{dt}$ when $t = \pi$.

$$\frac{d}{dt}(f \circ c)(\pi) = Df(c(\pi)) \cdot c'(\pi)$$

$$= Df(1, 1, \pi, e^6) \cdot (19, 11, 0, 1)$$

$$= (0, 1, 3, -7) \cdot (19, 11, 0, 1)$$

$$= 0 + 11 + 0 - 7 = \underline{4}$$

34. **34.** Suppose $f: \mathbb{R}^n \to \mathbb{R}^m$ and $g: \mathbb{R}^p \to \mathbb{R}^q$.

(a) What must be true about the numbers n, m, p, and q for $f \circ g$ to make sense?

(b) What must be true about the numbers n, m, p, and q for $g \circ f$ to make sense?

(c) When does $f \circ f$ make sense?

(a) $g(\vec{x}) \in R^q$ must be in domain of $f \subset R^n$

$\therefore \underline{q = n}$ p and m can be any positive integer.

(b) $f(\vec{x}) \in R^m$ must be in domain of $g \subset R^p$.

$\therefore \underline{m = p}$ n and q can be any positive integer.

(c) When range is a subset of domain.

$\therefore \underline{m = n}$

35. **35.** If $z = f(x - y)$, use the chain rule to show that $\dfrac{\partial z}{\partial x} + \dfrac{\partial z}{\partial y} = 0$.

Let $z(x,y) = f(x-y)$, so $z: \mathbb{R}^2 \to \mathbb{R}$, $f: \mathbb{R} \to \mathbb{R}$

Let $g: \mathbb{R}^2 \to \mathbb{R}$ by $g(x,y) = x - y$

$\therefore z = (f \circ g)(x,y)$

$\therefore Dz = [z_x \; z_y] = f'(g(x,y)) [g_x \; g_y]$

$= f'(g(x,y)) [1 \; -1]$

$= [f'(g(x,y)) \quad -f'(g(x,y))]$

$\therefore \dfrac{\partial z}{\partial x} + \dfrac{\partial z}{\partial y} = f'(g(x,y)) - f'(g(x,y)) = 0$

36. **36.** Let $w = x^2 + y^2 + z^2$, $x = uv$, $y = u\cos v$, $z = u\sin v$. Use the chain rule to find $\dfrac{\partial w}{\partial u}$ when $(u, v) = (1, 0)$.

$w(x,y,z) = x^2 + y^2 + z^2$ Let $f(u,v) = (uv, u\cos v, u\sin v)$

$\dfrac{\partial w}{\partial u} = \dfrac{\partial w}{\partial x}\dfrac{\partial x}{\partial u} + \dfrac{\partial w}{\partial y}\cdot\dfrac{\partial y}{\partial u} + \dfrac{\partial w}{\partial z}\cdot\dfrac{\partial z}{\partial u}$

$= 2x(v) + 2y(\cos v) + 2z(\sin v)$

At $(u,v) = (1,0)$, $x = 0$, $y = 1$, $z = 0$

$$\therefore \frac{\partial w}{\partial u} = 2x(0) + 2y(1) + 2z(0) = 2y = \underline{2}$$

Or, $w = (uv)^2 + (u\cos v)^2 + (u\sin v)^2$

$$= u^2v^2 + u^2\cos^2 v + u^2\sin^2 v$$

$$\therefore w_u = 2uv^2 + 2u\cos^2 v + 2u\sin^2 v$$

$$\therefore w_u(1,0) = 0 + 2(1)(1) + 0 = \underline{2}$$

2.6 Gradients and Directional Derivatives

Note Title 2/29/2016

1. **1.** Show that the directional derivative of $f(x, y, z) = z^2x + y^3$ at $(1, 1, 2)$ in the direction $(1/\sqrt{5})\mathbf{i} + (2/\sqrt{5})\mathbf{j}$ is $2\sqrt{5}$.

$\nabla f = (z^2, 3y^2, x)$. $\therefore \nabla f(1,1,2) = (4, 3, 1)$

$\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0\right)$ is a unit vector.

$\therefore \nabla f(1,1,2) \cdot \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0\right) = (4,3,1) \cdot \left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}, 0\right) =$

$\frac{4}{\sqrt{5}} + \frac{6}{\sqrt{5}} + 0 = \frac{10}{\sqrt{5}} = \underline{2\sqrt{5}}$

2. **2.** Compute the directional derivatives of the following functions at the indicated points in the given directions:

(a) $f(x, y) = x + 2xy - 3y^2$, $(x_0, y_0) = (1, 2)$, $\mathbf{v} = \frac{3}{5}\mathbf{i} + \frac{4}{5}\mathbf{j}$

(b) $f(x, y) = \log \sqrt{x^2 + y^2}$, $(x_0, y_0) = (1, 0)$, $\mathbf{v} = (1/\sqrt{5})(2\mathbf{i} + \mathbf{j})$

(c) $f(x, y) = e^x \cos(\pi y)$, $(x_0, y_0) = (0, -1)$, $\mathbf{v} = -(1/\sqrt{5})\mathbf{i} + (2/\sqrt{5})\mathbf{j}$

(d) $f(x, y) = xy^2 + x^3y$, $(x_0, y_0) = (4, -2)$, $\mathbf{v} = (1/\sqrt{10})\mathbf{i} + (3/\sqrt{10})\mathbf{j}$

(a) $\nabla f = (1+2y, 2x-6y)$. $\nabla f(1,2) = (5, -10)$

$\therefore \nabla f(1,2) \cdot \left(\frac{3}{5}, \frac{4}{5}\right) = \frac{15}{5} - \frac{40}{5} = \frac{-25}{5} = \underline{-5}$

(b) $f_x = \frac{1}{\sqrt{x^2+y^2}} \cdot \frac{1}{2}(x^2+y^2)^{-\frac{1}{2}} \cdot 2x = \frac{x}{x^2+y^2}$. $f_y = \frac{y}{x^2+y^2}$

$\therefore \nabla f(1,0) = (1, 0)$. $\therefore \nabla f(1,0) \cdot \vec{v} = \underline{\frac{2}{\sqrt{5}}}$

(c) $\nabla f = (e^x \cos(\pi y), -\pi e^x \sin(\pi y))$

$\therefore \nabla f(0,-1) = (-1, 0)$. $\therefore \nabla f(0,-1) \cdot \left(-\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right) =$

$\frac{1}{\sqrt{5}} + 0 = \underline{\frac{1}{\sqrt{5}}}$

(d) $\nabla f = (y^2 + 3x^2 y, 2xy + x^3)$ $\therefore \nabla f(4,-2) = (-92, 48)$

$\therefore \nabla f(4,-2) \cdot \left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right) = -\frac{92}{\sqrt{10}} + \frac{144}{\sqrt{10}} = \underline{\frac{52}{\sqrt{10}}}$

3.

3. Compute the directional derivatives of the following functions along unit vectors at the indicated points in directions *parallel* to the given vector:

(a) $f(x, y) = x^y$, $(x_0, y_0) = (e, e)$, $\mathbf{d} = 5\mathbf{i} + 12\mathbf{j}$

(b) $f(x, y, z) = e^x + yz$, $(x_0, y_0, z_0) = (1, 1, 1)$, $\mathbf{d} = (1, -1, 1)$

(c) $f(x, y, z) = xyz$, $(x_0, y_0, z_0) = (1, 0, 1)$, $\mathbf{d} = (1, 0, -1)$

(a) $f_x = y x^{y-1}$ $f_y = x^y \ln x$, assuming $x > 0$.

$\therefore \nabla f(e,e) = (e e^{e-1}, e^e \ln e) = (e^e, e^e)$

$\frac{\vec{d}}{\|\vec{d}\|} = \left(\frac{5}{13}, \frac{12}{13}\right)$. $\therefore \nabla f(e,e) \cdot \left(\frac{5}{13}, \frac{12}{13}\right) = \underline{\frac{17}{13} e^e}$

(b) $\nabla f = (e^x, z, y)$. $\therefore \nabla f(1,1,1) = (e, 1, 1)$

$\frac{\vec{d}}{\|\vec{d}\|} = \frac{1}{\sqrt{3}}(1, -1, 1)$. $\therefore \nabla f(1,1,1) \cdot \frac{1}{\sqrt{3}}(1,-1,1) = \underline{\frac{e}{\sqrt{3}}}$

(c) $\nabla f = (yz, xz, xy)$. $\therefore \nabla f(1,0,1) = (0, 1, 0)$

$\frac{\vec{d}}{\|\vec{d}\|} = \frac{1}{\sqrt{2}}(1, 0, -1)$ $\therefore \nabla f(1,0,1) \cdot \frac{1}{\sqrt{2}}(1,0,-1) = \underline{0}$

4. **4.** You are walking on the graph of $f(x, y) = y\cos(\pi x) - x\cos(\pi y) + 10$, standing at the point (2, 1, 13). Find an x, y-direction you should walk in to stay at the same level.

$f_x = -\pi y \sin(\pi x) - \cos(\pi y) \quad f_x(2,1) = 1$

$f_y = \cos(\pi x) + \pi x \sin(\pi y) \quad f_y(2,1) = 1$

$\therefore \nabla f(2,1) = (1, 1)$.

To stay at same level, go perpendicular to ∇f (i.e., in between steepest ascent & descent).

$\therefore \pm(1,-1)$, since $(1,-1)\cdot(1,1) = 0$

or $\hat{i} - \hat{j}$ or $-\hat{i} + \hat{j} = \pm(1,-1)$

5. **5.** (a) Let $f: \mathbb{R}^3 \to \mathbb{R}$, $\mathbf{x}_0 \in \mathbb{R}^3$. If $\mathbf{v}$ is a unit vector in $\mathbb{R}^3$, show that the maximum value of the directional derivative of f at $\mathbf{x}_0$ along $\mathbf{v}$ is $||\nabla f(\mathbf{x}_0)||$.

(b) Let $f(x, y, z) = x^3 - y^3 + z^3$. Find the maximum value for the directional derivative of f at the point (1, 2, 3).

(a) $\nabla f(\vec{x}_0) \cdot \vec{v} = \|\nabla f(\vec{x}_0)\| \, \|\vec{v}\| \cos\theta$

But $\|\vec{v}\| = 1$, and $\cos\theta$ is max at $\cos\theta = 1$.

$\therefore$ Max value is $\|\nabla f(\vec{x}_0)\|$

(b) $f_x = 3x^2$, $f_y = -3y^2$, $f_z = 3z^2$.

$\nabla f(1,2,3) = (3,-12,27)$

$\therefore \|\nabla f(1,2,3)\| = \sqrt{3^2+12^2+27^2} = \sqrt{882} = 21\sqrt{2}$

6. **6.** Find a vector which is normal to the curve $x^3 + xy + y^3 = 11$ at (1, 2).

$f_x = 3x^2 + y$, $f_y = x + 3y^2$. $\therefore \nabla f(1,2) = (5,13)$

7. **7.** Find the rate of change of $f(x, y, z) = xyz$ in the direction normal to the surface $yx^2 + xy^2 + yz^2 = 3$ at (1, 1, 1).

Let $g(x,y,z) = yx^2 + xy^2 + yz^2$

$g_x = 2yx + y^2$ $\quad g_y = x^2 + 2xy + z^2$ $\quad g_z = 2yz$

$\therefore \nabla g(1,1,1) = (3,4,2)$ = normal to surface at $(1,1,1)$.

$\therefore$ unit normal $= \dfrac{(3,4,2)}{\sqrt{3^2+4^2+2^2}} = \dfrac{1}{\sqrt{29}}(3,3,2) = \vec{n}$

for $f(x,y,z)$, $f_x = yz$, $f_y = xz$, $f_z = xy$.

$\therefore \nabla f(1,1,1) \cdot \vec{n} = (1,1,1) \cdot \dfrac{1}{\sqrt{29}}(3,4,2) = \dfrac{9}{\sqrt{29}}$

8. **8.** Find the planes tangent to the following surfaces at the indicated points:

(a) $x^2 + 2y^2 + 3xz = 10$, at the point $(1, 2, \frac{1}{3})$

(b) $y^2 - x^2 = 3$, at the point $(1, 2, 8)$

(c) $xyz = 1$, at the point $(1, 1, 1)$

(a) $\nabla f = (2x + 3z, 4y, 3x)$. $\therefore \nabla f(1, 2, \frac{1}{3}) = (2+1, 8, 3) = (3, 8, 3)$

$\therefore (3, 8, 3) \cdot (x-1, y-2, z-\frac{1}{3}) = 0$, or

$3(x-1) + 8(y-2) + 3z - 1$, or

$3x + 8y + 3z - 20 = 0$

(b) $\nabla f(x, y, z) = (-2x, 2y, 0)$. $\therefore \nabla f(1, 2, 8) = (-2, 4, 0)$

$\therefore (-2, 4, 0) \cdot (x-1, y-2, z-8) = 0$, or

$-2x + 2 + 4y - 8 = 0$, or $2x - 4y + 6 = 0$

(c) $\nabla f = (yz, xz, xy)$, $\therefore \nabla f(1, 1, 1) = (1, 1, 1)$

$\therefore (1, 1, 1) \cdot (x-1, y-1, z-1) = 0$, $\therefore x + y + z - 3 = 0$

9. **9.** Find the equation for the plane tangent to each surface $z = f(x, y)$ at the indicated point:

(a) $z = x^3 + y^3 - 6xy$, at the point $(1, 2, -3)$

(b) $z = (\cos x)(\cos y)$, at the point $(0, \pi/2, 0)$

(c) $z = (\cos x)(\sin y)$, at the point $(0, \pi/2, 1)$

(a) Let $g(x, y, z) = x^3 + y^3 - 6xy - z = 0$ $\nabla g = (3x^2 - 6y, 3y^2 - 6x, -1)$

$\therefore \nabla g(1,2,-3) = (3-12, 12-6, -1) = (-9,6,-1)$.

$\therefore (-9,6,-1)\cdot(x-1, y-2, z+3) = 0$, or

$-9x+9+6y-12-z-3=0$, or

$\underline{9x-6y+z+6=0}$

(b) $g(x,y,z) = (\cos x)(\cos y) - z = 0$, $\nabla g = (-\sin x\cos y, -\cos x\sin y, -1)$

$\therefore \nabla g(0, \frac{\pi}{2}, 0) = (0,-1,-1)$

$\therefore (0,-1,-1)\cdot(x-0, y-\frac{\pi}{2}, z-0) = 0$, or

$-y+\frac{\pi}{2}-z=0$, or $\underline{y+z-\frac{\pi}{2}=0}$

(c) $g(x,y,z) = \cos x \sin y - z$, $\nabla g = (-\sin x\sin y, \cos x\cos y, -1)$

$\therefore \nabla g(0, \frac{\pi}{2}, 1) = (0,0,-1)$.

$\therefore (0,0,-1)\cdot(x-0, y-\frac{\pi}{2}, z-1) = 0$, or $\underline{z-1=0}$

10.

10. Compute the gradient ∇f for each of the following functions:

(a) $f(x, y, z) = 1/\sqrt{x^2+y^2+z^2}$

(b) $f(x, y, z) = xy + yz + xz$

(c) $f(x, y, z) = \dfrac{1}{x^2+y^2+z^2}$

(a) $\nabla f(x,y,z) = -\dfrac{1}{x^2+y^2+z^2}\cdot\dfrac{1}{\sqrt{x^2+y^2+z^2}}(x,y,z)$

(b) $\nabla f(x,y,z) = (y+z,\ x+z,\ y+x)$

(c) $\nabla f(x,y,z) = -\dfrac{2}{(x^2+y^2+z^2)^2}(x,y,z)$

11.

11. For the functions in Exercise 10, what is the direction of fastest increase at (1, 1, 1)? [The solution to part (c) only is in the Study Guide to this text.]

(a) $\nabla f(1,1,1) = -\dfrac{1}{3\sqrt{3}}(1,1,1)$

(b) $\nabla f(1,1,1) = (2,2,2)$

(c) $\nabla f(1,1,1) = -\frac{2}{9}(1,1,1)$

∴ For (a) & (c), direction is same as $(-1,-1,-1)$

For (b), direction is $(1,1,1)$

12.

12. Show that a unit normal to the surface $x^3y^3 + y - z + 2 = 0$ at (0, 0, 2) is given by $\mathbf{n} = (1/\sqrt{2})(\mathbf{j} - \mathbf{k})$.

$f_x = 3x^2y^3 \quad f_y = 3y^2x^3 + 1 \quad f_z = -1$

$\therefore \nabla f(0,0,2) = (0,1,-1) \quad \sqrt{1^2+(-1)^2} = \sqrt{2}$

$\therefore$ normal: $\dfrac{1}{\sqrt{2}}(0,1,-1) = \dfrac{1}{\sqrt{2}}(\hat{j} - \hat{k})$

13. **13.** Find a unit normal to the surface $\cos(xy) = e^z - 2$ at $(1, \pi, 0)$.

$f(x,y,z) = \cos(xy) - e^z + 2 = 0$

$f_x = -y\sin(xy) \quad f_y = -x\sin(xy) \quad f_z = -e^z$

$\therefore \nabla f(1, \pi, 0) = (-\pi\sin(\pi), -\sin(\pi), -e^0) = (0, 0, -1)$

$\therefore$ $(0,0,-1)$ or $(0,0,1)$

14. **14.** Verify Theorems 13 and 14 for $f(x, y, z) = x^2 + y^2 + z^2$.

$f(x,y,z)$ is a sphere, so unit normal pointing at fastest rate of increase should be pointing directly away from origin (e.g., an expanding bubble). Also, normal to tangent plane should point away from origin.

$\nabla f(x,y,z) = (f_x, f_y, f_z) = (2x, 2y, 2z) = 2(x,y,z)$.

Unit normal: $\frac{1}{\sqrt{3}}(x,y,z)$, and

This points away from origin.

Alternatively, let $\vec{x_0} = (x_0, y_0, z_0)$ be a point on the sphere, let $\vec{n}$ be a unit vector in some

direction at $\vec{x}_0$. So the line through $\vec{x}_0$ in the $\vec{v}$ direction is: $\vec{x}_0 + t\vec{n}$, $\vec{v} = (n_1, n_2, n_3)$

$$\lim_{t \to 0} \frac{f(\vec{x}_0 + t\vec{n}) - f(\vec{x}_0)}{t} =$$

$$\lim_{t \to 0} \frac{(x_0 + tn_1)^2 + (y_0 + tn_2)^2 + (z_0 + tn_3)^2 - (x_0^2 + y_0^2 + z_0)^2}{t} =$$

$$\lim_{t \to 0} \frac{2x_0 n_1 t + 2y_0 n_2 t + 2z_0 n_3 t + t^2(n_1^2 + n_2^2 + n_3^2)}{t} =$$

$$\lim_{t \to 0} \left[2x_0 n_1 + 2y_0 n_2 + 2z_0 n_3 + t(n_1^2 + n_2^2 + n_3^2)\right] =$$

$$2x_0 n_1 + 2y_0 n_2 + 2z_0 n_3 = 2\vec{x}_0 \cdot \vec{n}$$

$$= \nabla f(\vec{x}_0) \cdot \vec{n}$$

$\vec{x}_0 = (x_0, y_0, z_0)$ which points away from origin.

$\nabla f(\vec{x}_0) \cdot \vec{n}$ is max when $\vec{n}$ is parallel to $\vec{x}_0$,

i.e., away from origin.

15. **15.** Show that the definition following Theorem 14 yields, as a special case, the formula for the plane tangent to the graph of $f(x, y)$ by regarding the graph as a level surface of $F(x, y, z) = f(x, y) - z$ (see Section 2.3).

If $F(x, y, z) = f(x, y) - z$, then $\nabla F = (f_x, f_y, -1)$

$\therefore$ Tangent plane at $F(x_0, y_0, z_0) = f(x_0, y_0) - z_0$

is $\nabla F(x_0, y_0, z_0) \cdot (x - x_0, y - y_0, z - z_0) = 0$

or $\left[f_x(x_0, y_0),\ f_y(x_0, y_0),\ -1 \right] \cdot (x - x_0, y - y_0, z - z_0) = 0$

or $f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0) - (z - z_0) = 0,$

or $\underline{z = z_0 + f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)}$

which is the tangent plane definition on p. 110.

16.

16. Let $f(x, y) = -(1 - x^2 - y^2)^{1/2}$ for (x, y) such that $x^2 + y^2 < 1$. Show that the plane tangent to the graph of f at $(x_0, y_0, f(x_0, y_0))$ is orthogonal to the vector with components $(x_0, y_0, f(x_0, y_0))$. Interpret this geometrically.

Let $F(x, y, z) = f(x, y) - z$.

$\therefore F_x = -\frac{1}{2}(1 - x^2 - y^2)^{-\frac{1}{2}} \cdot 2x = -\frac{x}{\sqrt{1 - x^2 - y^2}}$

$F_y = -\frac{y}{\sqrt{1 - x^2 - y^2}}$

$F_z = -1$

$\therefore$ Normal to tangent plane at $[x_0, y_0, f(x_0, y_0)]$ is:

$$\left(\frac{-x_0}{\sqrt{1 - x_0^2 - y_0^2}},\ \frac{-y_0}{\sqrt{1 - x_0^2 - y_0^2}},\ -1 \right) = \vec{N}.$$

Multiply $\vec{N}$ by the constant $-\sqrt{1 - x_0^2 - y_0^2} = k$

and get $k\vec{N} = (x_0, y_0, \sqrt{1-x_0^2-y_0^2}) = (x_0, y_0, f(x_0, y_0))$

$\therefore \vec{N}$ is parallel to $(x_0, y_0, f(x_0, y_0))$.

$\therefore$ The tangent plane, perpendicular to $\vec{N}$, is perpendicular to $(x_0, y_0, f(x_0, y_0))$.

Geometrically, $z = -(1-x^2-y^2)^{\frac{1}{2}}$ is the bottom of a sphere $(x^2+y^2+z^2=1)$. The tangent plane to a point on a sphere is perpendicular to the radius vector from origin to the point.

17.

17. For the following functions $f: \mathbb{R}^3 \to \mathbb{R}$ and $\mathbf{g}: \mathbb{R} \to \mathbb{R}^3$, find ∇f and $\mathbf{g}'$ and evaluate $(f \circ \mathbf{g})'(1)$.

(a) $f(x, y, z) = xz + yz + xy$, $\mathbf{g}(t) = (e^t, \cos t, \sin t)$

(b) $f(x, y, z) = e^{xyz}$, $\mathbf{g}(t) = (6t, 3t^2, t^3)$

(c) $f(x, y, z) = (x^2 + y^2 + z^2) \log \sqrt{x^2 + y^2 + z^2}$, $\mathbf{g}(t) = (e^t, e^{-t}, t)$

(a) $\nabla f = (z+y, z+x, x+y)$, $g'(t) = (e^t, -\sin t, \cos t)$

$(f \circ \vec{g})'(1) = \nabla f(\vec{g}(1)) \cdot \vec{g}'(1)$

$= \nabla f(e, \cos 1, \sin 1) \cdot (e, -\sin 1, \cos 1)$

$= (\cos 1 + \sin 1, e + \sin 1, e + \cos 1) \cdot (e, -\sin 1, \cos 1)$

$= (e\cos 1 + e\sin 1) - e\sin 1 - \sin^2 1 + (e\cos 1 + \cos^2 1)$

$= 2e\cos 1 + \cos^2 1 - \sin^2 1$

(6) $\nabla f = (yze^{xyz}, xze^{xyz}, xye^{xyz})$, $\vec{\gamma}'(t) = (6, 6t, 3t^2)$

$(f\circ\vec{\gamma})'(1) = \nabla f(\vec{\gamma}(1)) \cdot \vec{\gamma}'(1)$

$= \nabla f(6,3,1) \cdot (6,6,3)$

$= (3e^{18}, 6e^{18}, 18e^{18}) \cdot (6,6,3)$

$= e^{18}(18 + 36 + 54) = 108e^{18}$

(c) $f_x = 2x\log\sqrt{x^2+y^2+z^2} + (x^2+y^2+z^2)\cdot\frac{1}{\sqrt{x^2+y^2+z^2}}\cdot\frac{1}{2}(x^2+y^2+z^2)^{-\frac{1}{2}}\cdot 2x$

$= x\log(x^2+y^2+z^2) + x$

$f_y = y\log(x^2+y^2+z^2) + y$

$f_z = z\log(x^2+y^2+z^2) + z$

$\vec{\gamma}'(t) = (e^t, -e^{-t}, 1)$ $\quad\quad \vec{\gamma}(1) = (e, e^{-1}, 1)$

$\therefore (f\circ\vec{\gamma})'(1) = \nabla f(\vec{\gamma}(1)) \cdot \vec{\gamma}'(1)$

$= \left[e\log(e^2+e^{-2}+1)+e,\ e^{-1}\log(e^2+e^{-2}+1)+e^{-1},\ \log(e^2+e^{-2}+1)+1\right)\cdot(e, e^{-1}, 1)$

$= e^2\log(e^2+e^{-2}+1) + e^2 + e^{-2}\log(e^2+e^{-2}+1) + e^{-2} + \log(e^2+e^{-2}+1) + 1$

$= (e^2+e^{-2}+1)\log(e^2+e^{-2}+1) + (e^2+e^{-2}+1)$

$$= (e^2 + e^{-2} + 1)\left[\log(e^2 + e^{-2} + 1) + 1\right]$$

18. **18.** Compute the directional derivative of f in the given directions $\mathbf{v}$ at the given points P.

(a) $f(x, y, z) = xy^2 + y^2z^3 + z^3x$, P $= (4, -2, -1)$, $\mathbf{v} = 1/\sqrt{14}(\mathbf{i} + 3\mathbf{j} + 2\mathbf{k})$

(b) $f(x, y, z) = x^{yz}$, P $= (e, e, 0)$, $\mathbf{v} = \frac{12}{13}\mathbf{i} + \frac{3}{13}\mathbf{j} + \frac{4}{13}\mathbf{k}$

Note $\vec{v}$ is a unit vector for (a) & (b).

(a) $\nabla f = (y^2 + z^3,\ 2xy + 2yz^3,\ 3y^2z^2 + 3xz^2)$

$\therefore \nabla f(4, -2, -1) = (3, -16 + 4, 6 + 12) = (3, -12, 18)$

$\therefore \nabla f(4, -2, -1) \cdot \vec{v} = (3, -12, 18) \cdot (1, 3, 2)\frac{1}{\sqrt{14}}$

$= \frac{3 - 36 + 36}{\sqrt{14}} = \frac{3}{\sqrt{14}}$

(b) $\nabla f = (yzx^{yz-1},\ (z\log x)x^{yz},\ (y\log x)x^{yz})$

$\therefore \nabla f(e, e, 0) = (0, 0, e \cdot e^0) = (0, 0, e)$

$\therefore \nabla f(e, e, 0) \cdot \vec{v} = (0, 0, e) \cdot (12, 3, 4)\frac{1}{13}$

$= \frac{4}{13}e$

19. **19.** You are standing on the graph of $f(x, y) = 100 - 2x^2 - 3y^2$ at the point $(2, 3, 65)$.

(a) What are the xy coordinates of the highest point on the graph?

(b) Show that the gradient of f is the zero vector at the point found in (a).

(a) x and y can only subtract from 100. ∴ max value is when $x = y = 0$.
∴ $(0, 0, 100)$.

(b) $\nabla f(x,y) = (-4x, -6y)$. ∴ $\nabla f(0,0) = (0,0)$

20. **20.** Find the two points on the hyperboloid $x^2 + 4y^2 - z^2 = 4$, where the tangent plane is parallel to the plane $2x + 2y + z = 5$.

The plane can be written as $(2,2,1)\cdot(x-0, y-0, z-5) = 0$.
∴ Normal to plane is $(2,2,1)$.

Tangent planes parallel to above plane will have same normal!

∴ $\nabla f(x,y,z) = (2x, 8y, -2z) = \pm(2,2,1)$

∴ $(1, \frac{1}{4}, -\frac{1}{2})$ gives $\nabla f = (2,2,1)$

$(-1, -\frac{1}{4}, \frac{1}{2})$ gives $\nabla f = -(2,2,1)$

∴ $(1, \frac{1}{4}, -\frac{1}{2})$ and $(-1, -\frac{1}{4}, \frac{1}{2})$

21. **21.** Let $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ and $r = \|\mathbf{r}\|$. Prove that

$$\nabla\left(\frac{1}{r}\right) = -\frac{\mathbf{r}}{r^3}.$$

$r = \|\vec{r}\| = \sqrt{x^2+y^2+z^2}$ $\therefore \frac{1}{r(x,y,z)} = (x^2+y^2+z^2)^{-\frac{1}{2}}$

$\frac{\partial\left(\frac{1}{r}\right)}{\partial x} = -\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}}(2x) = -\frac{x}{(\sqrt{x^2+y^2+z^2})^3} = -\frac{x}{r^3}$

$\therefore \frac{\partial\left(\frac{1}{r}\right)}{\partial y} = -\frac{y}{r^3}$, $\frac{\partial\left(\frac{1}{r}\right)}{\partial z} = -\frac{z}{r^3}$

$\therefore \nabla\left(\frac{1}{r}\right) = -\left(\frac{x}{r^3}, \frac{y}{r^3}, \frac{z}{r^3}\right) = -\frac{\vec{r}}{r^3}$

22. **22.** Captain Ralph is in trouble near the sunny side of Mercury. The temperature of the ship's hull when he is at location (x, y, z) will be given by $T(x, y, z) = e^{-x^2-2y^2-3z^2}$, where x, y, and z are measured in meters. He is currently at $(1, 1, 1)$.

(a) In what direction should he proceed in order to decrease the temperature most rapidly?

(b) If the ship travels at e^8 meters per second, how fast will be the temperature decrease if he proceeds in that direction?

(c) Unfortunately, the metal of the hull will crack if cooled at a rate greater than $\sqrt{14}e^2$ degrees per second. Describe the set of possible directions in which he may proceed to bring the temperature down at no more than that rate.

(a) $T_x = -2xe^{-x^2-2y^2-3z^2}$ $T_y = -4ye^{-x^2-2y^2-3z^2}$

$T_z = -6ze^{-x^2-2y^2-3z^2}$ $T(1,1,1) = e^{-1-2-3} = e^{-6}$

$\therefore \nabla T(1,1,1) = e^{-6}(-2,-4,-6)$.

$\therefore -\nabla T(1,1,1) = e^{-6}(2,4,6) \equiv K(1,2,3) \equiv \frac{(1,2,3)}{\sqrt{14}}$

$\therefore$ proceed toward $\frac{1}{\sqrt{14}}(1,2,3)$

(b) The rate of change in the direction of (a) is

$$-\|\nabla T(1,1,1)\| = -\|e^{-6}(2,4,6)\| = -e^{-6}\sqrt{2^2+4^2+6^2}$$

$$= -2e^{-6}\sqrt{14} \text{ degrees/m.}$$

$\therefore$ If travelling at e^8 m/sec, then rate of change is $(e^8 \text{ m/sec})(-2e^{-6}\sqrt{14} \text{ degrees/m})$

$$= -2\sqrt{14}e^2 \text{ degrees/sec}$$

(c) Look at $\nabla T(1,1,1)\cdot\vec{v}$, where $\|\vec{v}\| = e^8$

$$\therefore \nabla T(1,1,1)\cdot\vec{v} = \|\nabla T(1,1,1)\|\,\|\vec{v}\|\cos\theta$$

$$= \left[e^{-6}\sqrt{(-2)^2+(-4)^2+(-6)^2}\right]e^8\cos\theta$$

$$= e^2\sqrt{56}\cos\theta = 2e^2\sqrt{14}\cos\theta$$

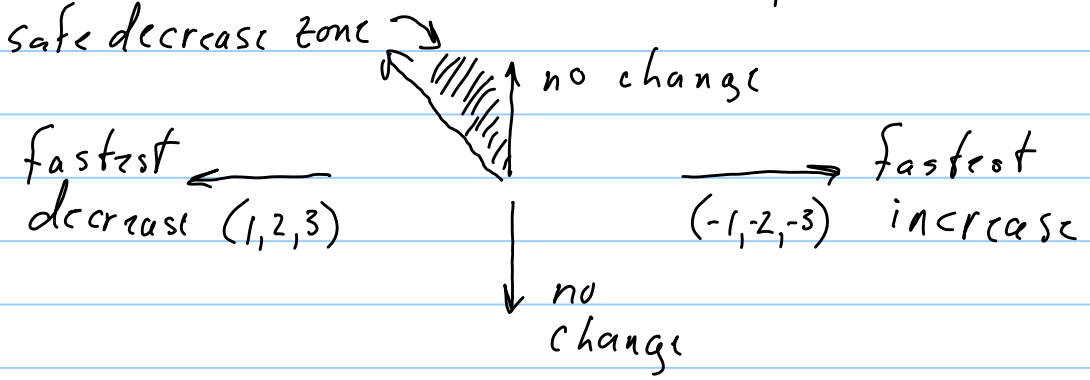
To produce a temp decrease, want $\cos\theta < 0$.

And want $|\text{rate}| \leq \sqrt{14}e^2$

$\therefore$ Want θ s.t. $|2e^2\sqrt{14}\cos\theta| \le 14e^2$,

or $|\cos\theta| \le \frac{1}{2}$, $\therefore -\frac{1}{2} \le \cos\theta \le \frac{1}{2}$,

so $-\frac{1}{2} \le \cos\theta \le 0$, $\therefore 120^\circ \le \theta \le 90^\circ$



i.e., between 90° to 120° from direction of gradient

23.

23. A function $f: \mathbb{R}^2 \to \mathbb{R}$ is said to be *independent of the second variable* if there is a function $g: \mathbb{R} \to \mathbb{R}$ such that $f(x, y) = g(x)$ for all x in $\mathbb{R}$. In this case, calculate ∇f in terms of g'.

$\nabla f(x,y) = (f_x, f_y)$. $\dfrac{\partial f(x,y)}{\partial x} = \dfrac{d}{dx} g(x) = g'(x)$

$\dfrac{\partial f(x,y)}{\partial y} = \dfrac{\partial}{\partial y} g(x) = 0$ $\therefore \nabla f(x,y) = (g'(x), 0)$

24.

24. Let f and g be functions from $\mathbb{R}^3$ to $\mathbb{R}$. Suppose f is differentiable and $\nabla f(\mathbf{x}) = g(\mathbf{x})\mathbf{x}$. Show that spheres centered at the origin are contained in the level sets for f; that is, f is constant on such spheres.

Let $\vec{c}(t)$ be a curve on a sphere centered at $\vec{0}$.

To show $f(\vec{c}(t)): \mathbb{R} \to \mathbb{R}$ is constant,

it suffices to show $\frac{d}{dt} f(\vec{c}(t))\Big|_{t=0} = 0$, where

$\vec{c}(0)$ is some point on a sphere, call it $\vec{c}(0) = \vec{x}_0$.

$\frac{d}{dt} f(\vec{c}(t))\Big|_{t=0} = \nabla f(\vec{c}(0)) \cdot \vec{c}'(0)$. Let $\vec{v} = \vec{c}'(0)$

Let $h(x,y,z) = x^2 + y^2 + z^2 = r^2$ be any sphere

centered at the origin. $\therefore \nabla h(x,y,z) = (2x, 2y, 2z)$.

$\therefore \nabla h(\vec{x}_0) = 2\vec{x}_0$. But $h(\vec{c}(t)) = r^2$,

so $\frac{d}{dt} h(\vec{c}(t)) = 0 = \nabla h(\vec{c}(t)) \cdot \vec{c}'(t)$. At $t=0$,

$\nabla h(\vec{x}_0) \cdot \vec{v} = 0$, so $(2\vec{x}_0) \cdot \vec{v} = 0$, so $\vec{x} \cdot \vec{v} = 0$.

$\therefore \nabla f(\vec{x}_0) \cdot \vec{v} = g(\vec{x}_0)\, \vec{x}_0 \cdot \vec{v} = 0$

$\therefore \frac{d}{dt} f(\vec{c}(t))\Big|_{t=0} = \nabla f(\vec{c}(0)) \cdot \vec{v} = 0$,

Since $\vec{c}(0)$ was an arbitrary point on a sphere,

f is constant on the arbitrary sphere

centered at the origin.

25. **25.** A function $f: \mathbb{R}^n \to \mathbb{R}$ is called an *even* function if $f(\mathbf{x}) = f(-\mathbf{x})$ for every $\mathbf{x}$ in $\mathbb{R}^n$. If f is differentiable and even, find $\mathbf{D}f$ at the origin.

$Df(x) = Df(-x)$

Let $g(\vec{x}) = -\vec{x}$. $\therefore D(f \circ g)(\vec{x}) = Df(g(\vec{x})) \cdot Dg(\vec{x})$

But $Dg(\vec{x}) = -I$, I the identity matrix.

$\therefore Df(g(\vec{x})) \cdot Dg(\vec{x}) = Df(-\vec{x}) \cdot (-I) = -Df(-\vec{x})$

$\therefore Df(\vec{x}) = Df(-\vec{x}) = -Df(-\vec{x})$

$\therefore$ For $Df(\vec{x}) = -Df(-\vec{x})$, let $\vec{x} =$ origin $= \vec{0}$.

$\therefore Df(\vec{0}) = -Df(\vec{0})$, so $2Df(\vec{0}) = \vec{0}$,

$\therefore Df(\vec{0}) = \vec{0} = [0 \dots 0]$, a $1 \times n$ matrix.

26. **26.** Suppose that a mountain has the shape of an elliptic paraboloid $z = c - ax^2 - by^2$, where a, b, and c are positive constants, x and y are the east–west and north–south map coordinates, and z is the altitude above sea level (x, y, z are all measured in meters). At the point (1, 1), in what direction is the altitude increasing most rapidly? If a marble were released at (1, 1), in what direction would it begin to roll?

(a) $z = f(x,y) = c - ax^2 - by^2$. $\therefore \nabla f = (-2ax, -2by)$

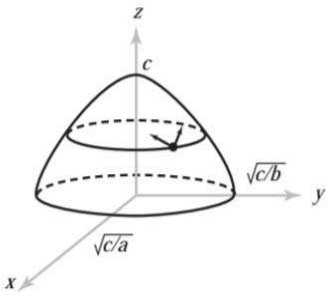
$\therefore$ At $(x,y) = (1,1)$, $f(x,y)$ increases most in

the direction of $(-2a, -2b)$ [i.e., toward origin].

(b) In the opposite direction of (a), so in the direction of $(2a, 2b)$, or (a, b). This is the direction of fastest decrease.

Note also: consider tangent plane at $(1,1)$, as it should have no lateral forces at the point. For $F(x,y,z) = c - ax^2 - by^2 - z$, normal to tangent plane is $\pm(-2ax, -2by, -1)$. So at $(x,y) = (1,1)$, normal is $\pm(-2a, -2b, -1)$.
∴ It will move along (x,y) vector of $\pm(a,b)$. From (a), $(-a,b)$ is uphill. ∴ (a,b) is downhill with the only force of gravity on it.

27. **27.** An engineer wishes to build a railroad up the mountain of Exercise 26. Straight up the mountain is much too steep for the power of the engines. At the point (1, 1), in what directions may the track be laid so that it will be climbing with a 3% grade—that is, an angle whose tangent is 0.03? (There are two possibilities.) Make a sketch of the situation indicating the two possible directions for a 3% grade at (1, 1).



$\nabla f(x,y) \cdot \vec{v}$ = rate of change of $f(x,y)$ in direction of unit vector $\vec{v}$. Let r = grade = 0.03

Let $\vec{v} = (x,y)$ be the unit vector in xy-plane. $x^2 + y^2 = 1$

$\nabla f(1,1) = (-2a, -2b)$. Assume $a > 0$, $b > 0$.

$\therefore \nabla f(1,1)\cdot(x,y) = -2ax - 2by = r$

Or, $ax + by = -\frac{r}{2}$ [1]

$\therefore ax + \frac{r}{2} = by$

$a^2x^2 + arx + \frac{r^2}{4} = b^2y^2 = b^2(1-x^2)$

$\therefore (a^2+b^2)x^2 + arx + \frac{r^2}{4} - b^2 = 0$ [2]

Checking the discriminant,

$a^2r^2 - 4(a^2+b^2)\left(\frac{r^2}{4} - b^2\right) > 0 \iff$

$a^2r^2 > (a^2+b^2)(r^2 - 4b^2) \iff$

$a^2r^2 > a^2r^2 - 4a^2b^2 + b^2r^2 - 4b^4 \iff$

$4a^2 + 4b^2 > r^2$, or $a^2+b^2 > \frac{r^2}{4}$

For $r = 0.03$ and any reasonable mountain (i.e., values for a, b), this will be true

$\therefore$ For [2] with $r = 0.03$,

$(a^2+b^2)x^2 + (0.03)ax + \frac{(0.03)^2}{4} - b^2 = 0$ [3]

has two real solutions, call them x_1, x_2.

From [1], $ax + by = -\frac{r}{2}$, or $y = \frac{-\frac{r}{2} - ax}{b}$, or

$$y = -\frac{(r + 2ax)}{2b}.$$

$$\therefore \left[x_1, \ -\frac{(0.03 + 2ax_1)}{2b}\right] \text{ and } \left[x_2, -\frac{(0.03 + 2ax_2)}{2b}\right] \quad [4]$$

where x_1, x_2 are solutions to [3]:

$$(a^2 + b^2)x^2 + (0.03)ax + \frac{(0.03)^2}{4} - b^2 = 0$$

As a concrete example, let $a = 40$, $b = 50$, $c = 100$, so

$z = 100 - 40x^2 - 50y^2$. $\therefore \nabla f(1,1) = (-80, -100)$

[3] becomes: $4100x^2 + 1.2x - 2500 = 0$,

$x = 0.781, -0.781$

From [4], $(x,y) = [0.781, -0.625]$ and $[-0.781, 0.625]$

28.

28. In electrostatics, the force **P** of attraction between two particles of opposite charge is given by $\mathbf{P} = k(\mathbf{r}/\|\mathbf{r}\|^3)$ (*Coulomb's law*), where k is a constant and $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Show that **P** is the gradient of $f = -k/\|\mathbf{r}\|$.

$$f(x,y,z) = \frac{-k}{\sqrt{x^2+y^2+z^2}} = -k(x^2+y^2+z^2)^{-\frac{1}{2}}$$

$$\therefore f_x = -\frac{1}{2}k(x^2+y^2+z^2)^{-\frac{3}{2}}(2x) = -kx(x^2+y^2+z^2)^{-3/2}$$

$$= -\frac{Kx}{(\sqrt{x^2+y^2+z^2})^3} = -\frac{Kx}{\|\vec{r}\|^3}$$

Similarly, $f_y = -\frac{Ky}{\|\vec{r}\|^3}$, $f_z = \frac{-Kz}{\|\vec{r}\|^3}$

$$\therefore \nabla f(x,y,z) = -\frac{k}{\|\vec{r}\|^3}(x,y,z) = -\frac{K\vec{r}}{\|\vec{r}\|^3} = \vec{F}$$

29. **29.** The electrostatic potential V due to two infinite parallel filaments with linear charge densities λ and $-\lambda$ is $V = (\lambda/2\pi\varepsilon_0)\ln(r_2/r_1)$, where $r_1^2 = (x-x_0)^2 + y^2$ and $r_2^2 = (x+x_0)^2 + y^2$. We think of the filaments as being in the z-direction, passing through the xy plane at $(-x_0, 0)$ and $(x_0, 0)$. Find $\nabla V(x, y)$.

$$V(x,y) = \frac{\lambda}{2\pi\epsilon_0}\cdot\frac{1}{2}\ln\left(\frac{r_2}{r_1}\right)^2 = \frac{\lambda}{4\pi\epsilon_0}\ln\frac{r_2^2}{r_1^2}$$

$$= \frac{\lambda}{4\pi\epsilon_0}\ln\left[\frac{(x+x_0)^2+y^2}{(x-x_0)^2+y^2}\right]$$

$$\therefore \frac{\partial V}{\partial x} = \frac{\lambda}{4\pi\epsilon_0}\cdot\left[\frac{(x-x_0)^2+y^2}{(x+x_0)^2+y^2}\right]\left[\frac{[(x-x_0)^2+y^2]\cdot[2(x+x_0)] - [(x+x_0)^2+y^2][2(x-x_0)]}{[(x-x_0)^2+y^2]^2}\right]$$

$$= \frac{\lambda}{2\pi\epsilon_0}\left[\frac{r_1^2}{r_2^2}\right]\left[\frac{r_1^2(x+x_0) - r_2^2(x-x_0)}{r_1^4}\right]$$

$$= \frac{\lambda}{2\pi\epsilon_0}\left[\frac{r_1^2(x+x_0) - r_2^2(x-x_0)}{r_1^2 r_2^2}\right] = \frac{\lambda}{2\pi\epsilon_0}\left[\frac{(x+x_0)}{r_2^2} - \frac{(x-x_0)}{r_1^2}\right]$$

$$\frac{\partial V}{\partial y} = \frac{\lambda}{4\pi\epsilon_0}\left[\frac{(x-x_0)^2+y^2}{(x+x_0)^2+y^2}\right]\left[\frac{[(x-x_0)^2+y^2][2y]-[(x+x_0)^2+y^2][2y]}{[(x-x_0)^2+y^2]^2}\right]$$

$$= \frac{\lambda}{2\pi\epsilon_0} y\left[\frac{r_1^2}{r_2^2}\right]\left[\frac{r_1^2-r_2^2}{r_1^4}\right] = \frac{\lambda y}{2\pi\epsilon_0}\left[\frac{r_1^2-r_2^2}{r_1^2 r_2^2}\right]$$

$$= \frac{\lambda}{2\pi\epsilon_0} y\left[\frac{1}{r_2^2}-\frac{1}{r_1^2}\right]$$

$$\therefore \nabla V(x,y) = \frac{\lambda}{2\pi\epsilon_0}\left[\frac{(x+x_0)}{r_2^2}-\frac{(x-x_0)}{r_1^2}, y\left(\frac{1}{r_2^2}-\frac{1}{r_1^2}\right)\right]$$

30.

30. For each of the following, find the maximum and minimum values attained by the function f along the path $\mathbf{c}(t)$:

(a) $f(x, y) = xy$; $\mathbf{c}(t) = (\cos t, \sin t)$; $0 \le t \le 2\pi$

(b) $f(x, y) = x^2 + y^2$; $\mathbf{c}(t) = (\cos t, 2\sin t)$; $0 \le t \le 2\pi$

(a) $(f \circ c)(t) = (\cos t)(\sin t) = \frac{1}{2}\sin 2t$

$\therefore (f \circ c)'(t) = \cos 2t, \quad 0 \le t \le 2\pi$

$(f \circ c)''(t) = -2\sin 2t$

$(f \circ c)'(t) = 0 \Leftrightarrow \cos 2t = 0, \quad 0 \le t \le 2\pi$

$\Leftrightarrow 2t = \frac{\pi}{2}, \frac{3}{2}\pi, \frac{5}{2}\pi, \ldots$

$\Leftrightarrow t = \frac{\pi}{4}, \frac{3}{4}\pi, \frac{5}{4}\pi, \frac{7}{4}\pi \quad (0 \le t \le 2\pi)$

$(f\circ c)''\left(\frac{\pi}{4}\right) = -2\sin\left(2\cdot\frac{\pi}{4}\right) = -2 < 0$ $\therefore$ a max

$(f\circ c)''\left(\frac{3}{4}\pi\right) = -2\sin\left(2\cdot 3\pi/4\right) = 2 > 0$ $\therefore$ a min

$(f\circ c)''(5\pi/4) = -2\sin\left(2\cdot 5\pi/4\right) = 2 < 0$ $\therefore$ a max

$(f\circ c)''(7\pi/4) = -2\sin\left(2\cdot 7\pi/4\right) = 2 > 0$ $\therefore$ a min

$\therefore (f\circ c)\left(\frac{\pi}{4}\right) = \cos\frac{\pi}{4}\sin\frac{\pi}{4} = \left(\frac{\sqrt{2}}{2}\right)^2 = \frac{1}{2}$

$(f\circ c)\left(\frac{5\pi}{4}\right) = \cos\frac{5\pi}{4}\sin\frac{5\pi}{4} = \left(-\frac{\sqrt{2}}{2}\right)\left(-\frac{\sqrt{2}}{2}\right) = \frac{1}{2}$

$(f\circ c)\left(\frac{3\pi}{4}\right) = \cos\frac{3\pi}{4}\sin\frac{3\pi}{4} = \left(-\frac{\sqrt{2}}{2}\right)\left(\frac{\sqrt{2}}{2}\right) = -\frac{1}{2}$

$(f\circ c)(7\pi/4) = \cos\frac{7\pi}{4}\sin\frac{7\pi}{4} = \left(\frac{\sqrt{2}}{2}\right)\left(-\frac{\sqrt{2}}{2}\right) = -\frac{1}{2}$

$\therefore (f\circ c)(t) = \frac{1}{2}$, a max, at $t = \frac{\pi}{4}, \frac{5\pi}{4}$

$(f\circ c)(t) = -\frac{1}{2}$, a min, at $t = \frac{3\pi}{4}, \frac{7\pi}{4}$

(6) (b) $f(x, y) = x^2 + y^2; \mathbf{c}(t) = (\cos t, 2\sin t); 0 \le t \le 2\pi$

$(f\circ c)(t) = \cos^2 t + 4\sin^2 t, \quad 0 \le t \le 2\pi$

Let $h(t) = (f\circ c)t)$.

$\therefore h'(t) = -2\cos t\sin t + 8\sin t\cos t$

$= 6\sin t\cos t$

$= 3\sin 2t$

$\therefore h''(t) = (3\cos 2t)(2) = 6\cos 2t$

$\therefore h'(t) = 3\sin 2t = 0 \iff 2t = 0, \pi, 2\pi, 3\pi, \ldots$

$\Leftrightarrow t = 0, \frac{\pi}{2}, \pi, \frac{3}{2}\pi, 2\pi$

$h''(0) = 6\cos(0) = 6 > 0 \quad \therefore$ min

$h''(\frac{\pi}{2}) = 6\cos(2 \cdot \pi/2) = -6 < 0 \quad \therefore$ max

$h''(\pi) = 6\cos(2\pi) = 6 > 0 \quad \therefore$ min

$h''(3\pi/2) = 6\cos(2 \cdot 3\pi/2) = -6 < 0 \quad \therefore$ max

$h''(2\pi) = 6\cos(2 \cdot 2\pi) = 6 > 0 \quad \therefore$ min

$\therefore (f \circ c)(t) = 1$, a min, at $t = 0, \pi, 2\pi$

$(f \circ c)(t) = 4$, a max, at $t = \frac{\pi}{2}, \frac{3\pi}{2}$

31.

31. Suppose that a particle is ejected from the surface $x^2 + y^2 - z^2 = -1$ at the point $(1, 1, \sqrt{3})$ along the normal directed toward the xy plane to the surface at time $t = 0$ with a speed of 10 units per second. When and where does it cross the xy plane?

should read, "... normal to the surface directed toward the xy plane at..."

Find line from $(1,1,\sqrt{3})$ to xy plane, along path of normal to surface.

$\therefore$ Normal to surface: $\nabla f(x,y,z) = (2x, 2y, -2z)$

$\therefore \nabla f(1,1,\sqrt{3}) = (2, 2, -2\sqrt{3})$.

$\therefore$ line is $(1,1,\sqrt{3}) + s(2,2,-2\sqrt{3})$.

This intersects xy plane when z coord. $= 0$.

$\therefore \sqrt{3} - s2\sqrt{3} = 0, \quad s = \frac{1}{2}$

$\therefore$ intersects xy plane at $(1,1,\sqrt{3}) + \frac{1}{2}(2,2,-2\sqrt{3}) = (2,2,0)$

∴ Distance from $(1,1,\sqrt{3})$ to $(2,2,0)$ =

$$\sqrt{(2-1)^2+(2-1)^2+(0-\sqrt{3})^2} = \sqrt{5}$$

∴ Takes $\frac{\sqrt{5}\text{ units}}{10\text{ units/sec}} = \frac{\sqrt{5}}{10}$ secs.

∴ $\frac{\sqrt{5}}{10}$ secs to reach $(2,2,0)$.

32. **32.** Let $f: \mathbb{R}^3 \to \mathbb{R}$ and regard $\mathbf{D}f(x, y, z)$ as a linear map of $\mathbb{R}^3$ to $\mathbb{R}$. Show that the kernel (that is, the set of vectors mapped to zero) of $\mathbf{D}f$ is the plane in $\mathbb{R}^3$ orthogonal to ∇f.

$$Df(x,y,z) = \begin{bmatrix} f_x & f_y & f_z \end{bmatrix}$$

Let $\vec{x} \in \mathbb{R}^3$, so $\vec{x} = (x,y,z)$ or $\vec{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$

$$\therefore [Df(x,y,z)]\,\vec{x} = \begin{bmatrix} f_x & f_y & f_z \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$= f_x(x,y,z)\cdot x + f_y(x,y,z)\cdot y + f_z(x,y,z)\cdot z$$

$$= \nabla f(x,y,z)\cdot \vec{x} \quad \text{(a dot product)}.$$

∴ Kernal of Df is all $\vec{x} \in \mathbb{R}^3$ s.t. $\nabla f \cdot \vec{x} = 0$.

i.e., Kernal of Df is all $\vec{x} \in \mathbb{R}^3$ perpendicular to $\nabla f(x,y,z)$.

But this is just the plane in $\mathbb{R}^3$ orthogonal to $\nabla f(x,y,z)$.

Review Exercises for Chapter 2

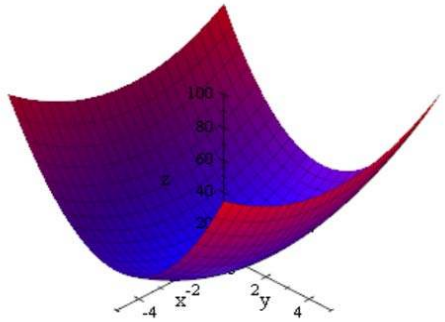
Note Title 3/9/2016

1.

1. Describe the graphs of:

(a) $f(x, y) = 3x^2 + y^2$

(b) $f(x, y) = xy + 3x$

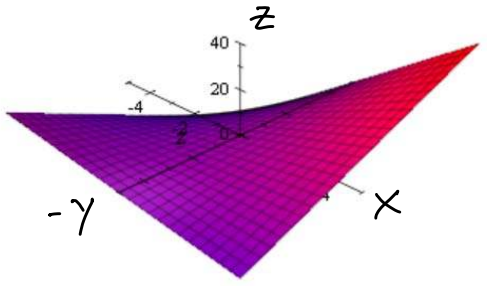


(a) for $y = 0$, $z = 3x^2$, a parabola
for $x = 0$, $z = y^2$, a parabola

for level set $K = 3x^2 + y^2 \iff \frac{K}{3} = x^2 + \frac{y^2}{3}$, an ellipse.

$\therefore$ Elliptical paraboloid.

(b) for $y = 0$, $z = 3x$
for $y = 5$, $z = 8x$
for $y = -5$, $z = -2x$



This is a "twisted" sheet, that rises (from neg. x to pos. x) for large pos. y, and decreases for large negative y. The sheet is smooth.

3.

3. Compute the derivative $\mathbf{D}f(\mathbf{x})$ of each of the following functions:

(a) $f(x, y) = (x^2 y, e^{-xy})$

(b) $f(x) = (x, x)$

(c) $f(x, y, z) = e^x + e^y + e^z$

(d) $f(x, y, z) = (x, y, z)$

(a) $f: \mathbb{R}^2 \to \mathbb{R}^2$, so $Df(\vec{x})$ is 2×2 matrix

$$\frac{\partial x^2 y}{\partial x} = 2xy, \quad \frac{\partial x^2 y}{\partial y} = x^2$$

$$\frac{\partial e^{-xy}}{\partial x} = -y e^{-xy} \qquad \frac{\partial e^{-xy}}{\partial y} = -x e^{-xy}$$

$$\therefore Df(\vec{x}) = \begin{bmatrix} 2xy & x^2 \\ -ye^{-xy} & -xe^{-xy} \end{bmatrix}$$

(b) $f: \mathbb{R} \to \mathbb{R}^2$, $\therefore$ a 2×1 matrix $\quad \frac{\partial x}{\partial x} = 1$

$$\therefore \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(c) $f: \mathbb{R}^3 \to \mathbb{R}^1$, $\therefore$ a 1×3 matrix

$$\frac{\partial f}{\partial x} = e^x \qquad \frac{\partial f}{\partial y} = e^y \qquad \frac{\partial f}{\partial z} = e^z$$

$$\therefore \begin{bmatrix} e^x & e^y & e^z \end{bmatrix}$$

(d) $f: \mathbb{R}^3 \to \mathbb{R}^3$, $\therefore$ a 3×3 matrix

$$\left.\begin{array}{lll} \frac{\partial x}{\partial x} = 1 & \frac{\partial x}{\partial y} = 0 & \frac{\partial x}{\partial z} = 0 \\ \frac{\partial y}{\partial x} = 0 & \frac{\partial y}{\partial y} = 1 & \frac{\partial y}{\partial z} = 0 \\ \frac{\partial z}{\partial x} = 0 & \frac{\partial z}{\partial y} = 0 & \frac{\partial z}{\partial z} = 1 \end{array}\right\} \therefore \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

4.

4. Suppose $f(x, y) = f(y, x)$ for all (x, y). Prove that

$$(\partial f/\partial x)(a, b) = (\partial f/\partial y)(b, a).$$

$$\frac{\partial f}{\partial x}(a,b) = \lim_{h\to 0} \frac{f(a+h,b) - f(a,b)}{h}$$

$$= \lim_{h\to 0} \frac{f(b,a+h) - f(b,a)}{h} = \frac{\partial f}{\partial y}(b,a)$$

5.

5. Let $f(u, v) = (\cos u, v + \sin u)$ and $g(x, y, z) = (x^2 + \pi y^2, xz)$. Compute $D(f \circ g)$ at $(0, 1, 1)$ using the chain rule.

$$D(f\circ g)(0,1,1) = Df(g(0,1,1)) \cdot Dg(0,1,1)$$

Note $g(0,1,1) = (\pi, 0)$

$$Df(u,v) = \begin{bmatrix} -\sin u & 0 \\ \cos u & 1 \end{bmatrix} \quad \therefore Df(\pi,0) = \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix}$$

$$Dg(x,y,z) = \begin{bmatrix} 2x & 2\pi y & 0 \\ z & 0 & x \end{bmatrix} \quad \therefore Dg(0,1,1) = \begin{bmatrix} 0 & 2\pi & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

$$\therefore Df(g(0,1,1)) \cdot Dg(0,1,1) = \begin{bmatrix} 0 & 0 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 0 & 2\pi & 0 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 1 & -2\pi & 0 \end{bmatrix}$$

6.

6. Use the chain rule to find $D(f \circ g)(-2, 1)$ for $f(u, v, w) = (v^2 + uw, u^2 + w^2, u^2 v - w^3)$ and $g(x, y) = (xy^3, x^2 - y^2, 3x + 5y)$.

$$D(f\circ g)(-2,1) = Df(g(-2,1)) \cdot Dg(-2,1)$$

$$g(-2,1) = (-2, 3, -1)$$

$$Df(u,v,w) = \begin{bmatrix} w & 2v & u \\ 2u & 0 & 2w \\ 2uv & u^2 & -3w^2 \end{bmatrix} \quad \therefore Df(-2,3,-1) = \begin{bmatrix} -1 & 6 & -2 \\ -4 & 0 & -2 \\ -12 & 4 & -3 \end{bmatrix}$$

$$Dg(x,y) = \begin{bmatrix} y^3 & 3xy^2 \\ 2x & -2y \\ 3 & 5 \end{bmatrix} \quad \therefore Dg(-2,1) = \begin{bmatrix} 1 & -6 \\ -4 & -2 \\ 3 & 5 \end{bmatrix}$$

$$\therefore Df(g(-2,1)) \cdot Dg(-2,1) = \begin{bmatrix} -1 & 6 & -2 \\ -4 & 0 & -2 \\ -12 & 4 & -3 \end{bmatrix} \begin{bmatrix} 1 & -6 \\ -4 & -2 \\ 3 & 5 \end{bmatrix} = \begin{bmatrix} -31 & -16 \\ -10 & 14 \\ -37 & 49 \end{bmatrix}$$

7.

7. Use the chain rule to find $D(f \circ g)(-1, 2)$ for $f(u, v, w) = (v^2 + w^2, u^3 - vw, u^2v + w)$ and $g(x, y) = (3x + 2y, x^3y, y^2 - x^2)$.

$$D(f \circ g)(-1,2) = Df(g(-1,2)) \cdot Dg(-1,2)$$

$$g(-1,2) = (1, -2, 3)$$

$$Df(u,v,w) = \begin{bmatrix} 0 & 2v & 2w \\ 3u^2 & -w & -v \\ 2uv & u^2 & 1 \end{bmatrix} \quad \therefore Df(1,-2,3) = \begin{bmatrix} 0 & -4 & 6 \\ 3 & -3 & 2 \\ -4 & 1 & 1 \end{bmatrix}$$

$$Dg(x,y) = \begin{bmatrix} 3 & 2 \\ 3x^2y & x^3 \\ -2x & 2y \end{bmatrix} \quad \therefore Dg(-1,2) = \begin{bmatrix} 3 & 2 \\ 6 & -1 \\ 2 & 4 \end{bmatrix}$$

$$\therefore Df(g(-1,2)) \cdot Dg(-1,2) = \begin{bmatrix} 0 & -4 & 6 \\ 3 & -3 & 2 \\ -4 & 1 & 1 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 6 & -1 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} -12 & 28 \\ -5 & 17 \\ -4 & -5 \end{bmatrix}$$

8. **8.** Let $f(x, y) = (xy, \frac{x}{y}, x + y)$ and $g(w, s, t) = (we^s, se^{wt})$. Find $D(f \circ g)(3, 1, 0)$.

$$D(f\circ g)(3,1,0) = Df(g(3,1,0)) \cdot Dg(3,1,0)$$

Note: $g(3,1,0) = (3e, 1)$

$$Df(x,y) = \begin{bmatrix} y & x \\ \frac{1}{y} & -\frac{x}{y^2} \\ 1 & 1 \end{bmatrix} \therefore Df(3e,1) = \begin{bmatrix} 1 & 3e \\ 1 & -3e \\ 1 & 1 \end{bmatrix}$$

$$Dg(w,s,t) = \begin{bmatrix} e^s & we^s & 0 \\ tse^{wt} & e^{wt} & wse^{wt} \end{bmatrix}$$

$$\therefore Dg(3,1,0) = \begin{bmatrix} e & 3e & 0 \\ 0 & 1 & 3 \end{bmatrix}$$

$$\therefore D(fog)(3,1,0) = \begin{bmatrix} 1 & 3e \\ 1 & -3e \\ 1 & 1 \end{bmatrix} \begin{bmatrix} e & 3e & 0 \\ 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} e & 6e & 9e \\ e & 0 & -9e \\ e & 3e+1 & 3 \end{bmatrix}$$

9. **9.** Let $\mathbf{r}(t) = (t\cos(\pi t), t\sin(\pi t), t)$ be a path. Where will the tangent line to $\mathbf{r}$ at $t = 5$ intersect the xy plane?

$$\vec{r}'(t) = [\cos(\pi t) - \pi t \sin(\pi t), \sin(\pi t) + \pi t \cos(\pi t), 1]$$

$$\therefore \vec{r}'(5) = [-1-0, 0-5\pi, 1] = [-1, -5\pi, 1]$$

$$\vec{r}(5) = [-5, 0, 5].$$

$\therefore$ Tangent line : $(-5,0,5) + a(-1,-5\pi,1)$

At xy plane intersection, z-component $= 0$.

$5 + a(1) = 0, \quad a = -5$

$\therefore$ At $(-5,0,5) - 5(-1,-5\pi,1) = \underline{(0, 25\pi, 0)}$

10. **10.** Let $f(x, y) = x^2 e^{-xy}$.

(a) Find a vector normal to the graph of f at (1, 2).

(b) Find the equation of the tangent plane to the graph of f at (1, 2).

(c) What point on the surface given by $z = x^2 - y^2$ has a tangent plane parallel to the plane found in (b)?

(a) $z = x^2 e^{-xy}$. Let $F(x,y,z) = x^2 e^{-xy} - z = 0$. $f(1,2) = e^{-2}$

$\nabla F(x,y,z) = (F_x, F_y, F_z) = (2xe^{-xy} - yx^2 e^{-xy}, -x^3 e^{-xy}, -1)$

$\therefore \nabla F(1,2,e^{-2}) = (2e^{-2} - 2e^{-2}, -e^{-2}, -1) = \underline{(0, e^{-2}, -1)}$

(b) Equation of plane: $\vec{n} \cdot (\vec{x} - \vec{x_0}) = 0$, $\vec{n}$ = normal.

$\therefore (0, e^{-2}, -1) \cdot [(x,y,z) - (1,2,e^{-2})]$

$= (0, e^{-2}, -1) \cdot (x-1, y-2, z-e^{-2})$

$= e^{-2}(y-2) - (z-e^{-2}) = 0$, or

$\underline{e^{-2}y - z - e^{-2} = 0}$

(c) The planes will have the same normal vectors.

Let $F(x,y,z) = x^2 - y^2 - z = 0$.

normal $= \nabla F = (2x, -2y, -1)$

$\therefore (2x, -2y-1) = (0, e^{-2}, -1)$, from normal in (a).

$\therefore x = 0, \; -2y = e^{-2}, \; y = -\frac{1}{2}e^{-2}$.

$\therefore z = x^2 - y^2 = 0 - (-\frac{1}{2}e^{-2}) = \frac{1}{4}e^{-4}$

$\therefore$ At the point: $\underline{(0, -\frac{1}{2}e^{-2}, \frac{1}{4}e^{-4})}$

11. **11.** Let $f(x, y) = (1 - x^2 - y^2)^{1/2}$. Show that the plane tangent to the graph of f at $(x_0, y_0, f(x_0, y_0))$ is orthogonal to the vector $(x_0, y_0, f(x_0, y_0))$. Interpret geometrically.

$z = (1-x^2-y^2)^{1/2}$, $\therefore$ Let $F(x,y,z) = (1-x^2-y^2)^{\frac{1}{2}} - z = 0$.

$\therefore$ Normal for tangent plane $= \nabla F(x,y,z)$

$F_x = \frac{1}{2}(1-x^2-y^2)^{-\frac{1}{2}}(-2x) = -\frac{x}{\sqrt{1-x^2-y^2}}$

$F_y = -\frac{y}{\sqrt{1-x^2-y^2}} \qquad F_z = -1$

$\therefore \nabla F(x_0, y_0, f(x_0,y_0)) = \left[-\frac{x_0}{\sqrt{1-x_0^2-y_0^2}}, -\frac{y_0}{\sqrt{1-x_0^2-y_0^2}}, -1 \right]$

$$= \left[\frac{-x_0}{f(x_0,y_0)}, \frac{-y_0}{f(x_0,y_0)}, -1 \right]$$

Let $k = -f(x_0,y_0)$.

$$\therefore k\nabla F(x_0,y_0,f(x_0,y_0)) = -f(x_0,y_0)\left[\frac{-x_0}{f(x_0,y_0)}, \frac{-y_0}{f(x_0,y_0)}, -1 \right]$$

$$= (x_0, y_0, f(x_0,y_0))$$

$\therefore$ normal to tangent plane is parallel to $(x_0,y_0,f(x_0,y_0))$.

$\therefore$ Tangent plane is perpendicular to $(x_0,y_0,f(x_0,y_0))$.

Geometrically, $z = (1-x^2-y^2)^{\frac{1}{2}}$ is a hemisphere, as $z^2 = 1-x^2-y^2$, or $x^2+y^2+z^2 = 1$. $\therefore$ Plane tangent to a sphere is $\perp$ vector from the origin: $(x_0,y_0,f(x_0,y_0))$.

12. **12.** Let $F(u, v)$ and $u = h(x, y, z)$, $v = k(x, y, z)$ be given (differentiable) real-valued functions and let $f(x, y, z)$ be defined by $f(x, y, z) = F(h(x, y, z), k(x, y, z))$. Write a formula for the gradient of f in terms of the partial derivatives of F, h, and k.

$$\nabla f(x,y,z) = [f_x, f_y, f_z]$$

$$\frac{\partial f}{\partial x} = \frac{\partial F}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial F}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$= \frac{\partial F}{\partial h}\cdot\frac{\partial h}{\partial x} + \frac{\partial F}{\partial k}\cdot\frac{\partial k}{\partial x}$$

Similarly, $\frac{\partial f}{\partial y} = \frac{\partial F}{\partial h}\cdot\frac{\partial h}{\partial y} + \frac{\partial F}{\partial k}\cdot\frac{\partial k}{\partial y}$

$\frac{\partial f}{\partial z} = \frac{\partial F}{\partial h}\cdot\frac{\partial h}{\partial z} + \frac{\partial F}{\partial k}\cdot\frac{\partial k}{\partial z}$

13. **13.** Find an equation for the tangent plane of the graph of f at the point $(x_0, y_0, f(x_0, y_0))$ for:

(d) $f(x, y) = \log(x + y) + x\cos y + \arctan(x + y)$, $(x_0, y_0) = (1, 0)$

$F(x,y,z) = \log(x+y) + x\cos y + \arctan(x+y) - z = 0.$

$f(1,0) = 0 + 1(1) + \arctan 1 = 1 + \frac{\pi}{4}$

$F_x = \frac{1}{x+y} + \cos y + \frac{1}{1+(x+y)^2}$

$F_y = \frac{1}{x+y} - x\sin y + \frac{1}{1+(x+y)^2}$

$F_z = -1.$

$\therefore \nabla F\left(1, 0, 1+\frac{\pi}{4}\right) = \left(1+1+\frac{1}{2}, 1-0+\frac{1}{2}, -1\right) = \left(\frac{5}{2}, \frac{3}{2}, -1\right)$

$\therefore \vec{n}\cdot(\vec{x}-\vec{x_0}) = 0 : \left(\frac{5}{2}, \frac{3}{2}, -1\right)\cdot\left(x-1, y-0, z-\left(1+\frac{\pi}{4}\right)\right) = 0$

or $\frac{5}{2}(x-1) + \frac{3}{2}y - \left(z-1-\frac{\pi}{4}\right) = 0$

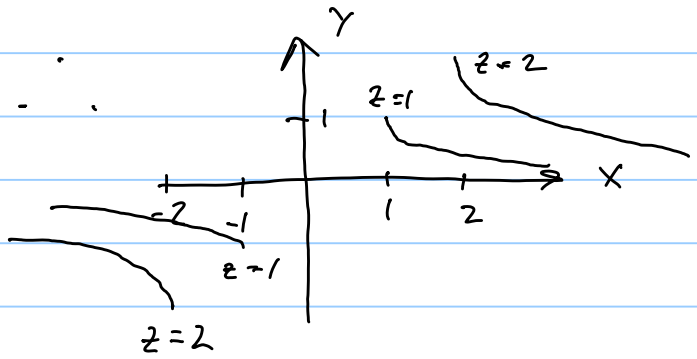
or $5(x-1) + 3y - 2z + 2 + \frac{\pi}{2} = 0$

or $10x + 6y - 4z = 6 - \pi$

16. **16.** Consider a temperature function $T(x, y) = x \sin y$. Plot a few level curves. Compute ∇T and explain its meaning.

For any x, $z = x \sin y$ is just the sine function with amplitude x.

For level curves, $\frac{z}{x} = \sin y$, or $y = \arcsin\left(\frac{z}{x}\right)$. Note arcsin is defined only for $-1 \le \frac{z}{x} \le 1$, where z is fixed.



$\nabla T = (\sin y, x \cos y)$. This gives the direction for which the temperature changes the greatest.

17. **17.** Find the following limits if they exist:

(a) $\lim_{(x,y)\to(0,0)} \frac{\cos xy - 1}{x}$

(b) $\lim_{(x,y)\to(0,0)} \sqrt{|(x+y)/(x-y)|}$, $x \neq y$

(a) Using L'Hopital's just for h, $\lim_{h\to 0} \frac{\cos h - 1}{h} = \lim_{h\to 0} (-\sin h) = 0$

$\therefore$ Given any $\epsilon > 0$, $\exists \delta > 0$ s.t. if $0 < |h| < \delta$, then $\left| \frac{\cos h - 1}{h} \right| < \epsilon$.

$\therefore$ If $0 < |xy| < \delta$, then $\left| \frac{\cos xy - 1}{xy} \right| < \epsilon$.

$\therefore |\cos xy - 1| < \epsilon |xy|$ if $0 < |xy| < \delta$. [1]

If $0 < \sqrt{(x-0)^2 + (y-0)^2} < \delta$, then $|x| < \sqrt{x^2+y^2} < \delta$ and $|y| < \sqrt{x^2+y^2} < \delta$.

$\therefore$ if $\delta < 1$, then $|x| < 1, |y| < 1$, so $|xy| < |x| < \delta$.

$\therefore$ Given any $\epsilon > 0$, choose the $\delta > 0$ from $\lim_{h \to 0} \frac{\cos h - 1}{h} = 0$. $\therefore$ Let $\delta' = \min\{1, \delta\}$.

$\therefore$ if $0 < \sqrt{x^2+y^2} < \delta'$, then $0 < |xy| < \delta'$ as shown above. And from [1],

$|\cos xy - 1| < \epsilon |xy| < \epsilon |x|$, since $|y| < 1$.

$\therefore \left| \frac{\cos xy - 1}{x} - 0 \right| < \epsilon$ if $0 < \sqrt{x^2+y^2} < \delta'$

$\therefore \lim_{(x,y) \to (0,0)} \frac{\cos xy - 1}{x} = 0$

(6) (b) $\lim_{(x,y) \to (0,0)} \sqrt{|(x+y)/(x-y)|}$, $x \neq y$

Along the line $y = 2x$, $\sqrt{\left|\frac{x+y}{x-y}\right|} = \sqrt{\left|\frac{3x}{-x}\right|} = \sqrt{3}$

Along the line $y = 3x$, $\sqrt{\left|\frac{x+y}{x-y}\right|} = \sqrt{\left|\frac{4x}{-2x}\right|} = \sqrt{2}$

$\therefore$ limit doesn't exist

18.

18. Compute the first partial derivatives and gradients of the following functions:

(a) $f(x, y, z) = xe^z + y\cos x$

(b) $f(x, y, z) = (x + y + z)^{10}$

(c) $f(x, y, z) = (x^2 + y)/z$

(a) $f_x = e^z - y\sin x$

$f_y = \cos x$

$f_z = xe^z$ $\quad \therefore \nabla f(x,y,z) = (e^z - y\sin x, \cos x, xe^z)$

(b) $f_x = f_y = f_z = 10(x+y+z)^9$

$\therefore \nabla f(x,y,z) = 10\left[(x+y+z)^9, (x+y+z)^9, (x+y+z)^9\right]$

(c) $f_x = \frac{2x}{z}$ $\quad f_y = \frac{1}{z}$ $\quad f_z = -\frac{(x^2+y)}{z^2}$

$\therefore \nabla f(x,y,z) = \left(\frac{2x}{z}, \frac{1}{z}, -\frac{(x^2+y)}{z^2}\right)$.

19.

19. Compute $\frac{\partial}{\partial x}[x\exp(1 + x^2 + y^2)]$

$\frac{\partial}{\partial x} = e^{1+x^2+y^2} + xe^{1+x^2+y^2}(2x)$

$= (1 + 2x^2)e^{1+x^2+y^2}$

20. **20.** Let $f: \mathbb{R}^2 \to \mathbb{R}^4$ and $g: \mathbb{R}^2 \to \mathbb{R}^2$ be given by $f(x, z) = (x^2 - y^2, 0, \sin(xy), 1)$ and $g(x, y) = (ye^{x^2}, xe^{y^2})$. Compute $D(f \circ g)(1, 2)$.

$$D(F \circ g)(1,2) = Df(g(1,2)) \cdot Dg(1,2)$$

$$g(1,2) = (2e, e^4)$$

$$Df(x,y) = \begin{bmatrix} 2x & -2y \\ 0 & 0 \\ y\cos(xy) & x\cos(xy) \\ 0 & 0 \end{bmatrix}$$

$$\therefore Df(g(1,2)) = Df(2e, e^4) = \begin{bmatrix} 4e & -2e^4 \\ 0 & 0 \\ e^4\cos(2e^5) & 2e\cos(2e^5) \\ 0 & 0 \end{bmatrix}$$

$$Dg(x,y) = \begin{bmatrix} 2xye^{x^2} & e^{x^2} \\ e^{y^2} & 2xye^{y^2} \end{bmatrix} \quad \therefore Dg(1,2) = \begin{bmatrix} 4e & e \\ e^4 & 4e^4 \end{bmatrix}$$

$$\therefore Df(g(1,2)) \cdot Dg(1,2) = \begin{bmatrix} 4e & -2e^4 \\ 0 & 0 \\ e^4\cos(2e^5) & 2e\cos(2e^5) \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 4e & e \\ e^4 & 4e^4 \end{bmatrix}$$

$$= \begin{bmatrix} 16e^2 - 2e^8 & 4e^2 - 8e^8 \\ 0 & 0 \\ 6e^5\cos(2e^5) & 9e^5\cos(2e^5) \\ 0 & 0 \end{bmatrix}$$

21. **21.** Let $f(x, y) = (x^2 + y^2)e^{-(x^2+y^2+10)}$. Find the rate of change of f at (2, 1) in the direction pointing toward the origin.

"Toward the origin" from (2,1) is $(0,0)-(2,1)=(-2,-1)$.
Unit vector is $-\frac{1}{\sqrt{5}}(2,1)$.

Rate of change in direction toward origin: $\nabla f \cdot \vec{v}$

$\nabla f(x,y) = (f_x, f_y)$

$f_x = 2x e^{-(x^2+y^2+10)} + (x^2+y^2)e^{-(x^2+y^2+10)}(-2x)$

$f_y = 2y e^{-(x^2+y^2+10)} + (x^2+y^2)e^{-(x^2+y^2+10)}(-2y)$

$\therefore f_x(2,1) = 4e^{-15} + (-4)(5)e^{-15} = -16e^{-15}$

$f_y(2,1) = 2e^{-15} + (-2)(5)e^{-15} = -8e^{-15}$

$\therefore \nabla f(2,1)\cdot\vec{v} = (-16e^{-15}, -8e^{-15})\cdot\left(-\frac{2}{\sqrt{5}}, -\frac{1}{\sqrt{5}}\right)$

$= \frac{32}{\sqrt{5}}e^{-15} + \frac{8}{\sqrt{5}}e^{-15} = \frac{40}{\sqrt{5}}e^{-15}$

$= \underline{\underline{8\sqrt{5}e^{-15}}}$

22.

22. Let $y(x)$ be a differentiable function defined implicitly by $F(x, y(x)) = 0$. From Exercise 19(a), Section 2.5, we know that

$$\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y}.$$

Consider the surface $z = F(x, y)$, and suppose F is increasing as a function of x and as a function of y; that is, $\partial F/\partial x > 0$ and $\partial F/\partial y > 0$. By considering the graph and the plane $z = 0$, show that for z fixed at $z = 0$, y should *decrease* as x increases and x should *decrease* as y increases. Does this agree with the minus sign in the formula for dy/dx?

Since $\frac{\partial F}{\partial x} > 0$, $\frac{\partial F}{\partial y} > 0$, then $\frac{\partial F/\partial x}{\partial F/\partial y} > 0$.

$\therefore -\frac{\partial F/\partial x}{\partial F/\partial y} < 0$, so $\frac{dy}{dx} < 0$.

$\therefore$ y decreases as x increases, for any fixed z.

Note also, $\frac{dx}{dy} = -\frac{\partial F/\partial y}{\partial F/\partial x}$, and since

$\frac{\partial F/\partial y}{\partial F/\partial x} > 0$, so $-\frac{\partial F/\partial y}{\partial F/\partial x} < 0$, so $\frac{dx}{dy} < 0$.

$\therefore$ x decreases as y increases, for any fixed z.

This obviously agrees w/ the minus sign in the formula for dy/dx.

Considering $z \approx F(x,y) + \frac{\partial F}{\partial x}\Delta x + \frac{\partial F}{\partial y}\Delta y$, for $z = 0$,

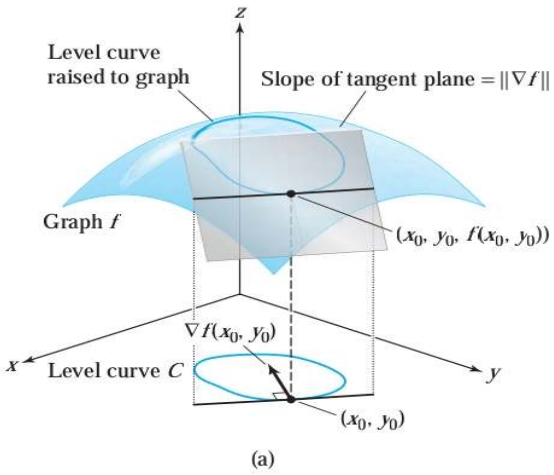
to keep balance, Δx and Δy must work in opposite ways since $f_x > 0$ and $f_y > 0$.

23.

23. (a) Consider the graph of a function $f(x, y)$ [Figure 2.R.1(a)]. Let (x_0, y_0) lie on a level curve C, so $\nabla f(x_0, y_0)$ is perpendicular to this curve. Show that the tangent plane of the graph is the plane that (i) contains the line perpendicular to $\nabla f(x_0, y_0)$ and lying in the horizontal plane $z = f(x_0, y_0)$, and (ii) has slope $\|\nabla f(x_0, y_0)\|$ relative to the xy plane.

(By the *slope* of a plane P relative to the xy plane we mean the tangent of the angle θ, $0 \le \theta \le \pi$, between the upward-pointing normal $\mathbf{p}$ to P and the unit vector $\mathbf{k}$.)

(b) Use this method to show that the tangent plane of the graph of $f(x, y) = (x + \cos y)x^2$ at $(1, 0, 2)$ is as sketched in Figure 2.R.1(b).



(a)

(a)(i) Tangent plane at $(x_0, y_0, f(x_0, y_0))$ is

$$z = f(x_0, y_0) + \frac{\partial f}{\partial x}(x_0, y_0)(x - x_0) + \frac{\partial f}{\partial y}(x_0, y_0)(y - y_0) \quad [1]$$

$$\nabla f(x_0, y_0) = \left(\frac{\partial f}{\partial x}(x_0, y_0), \frac{\partial f}{\partial y}(x_0, y_0)\right).$$

A vector, in the xy-plane, perpendicular to $\nabla f(x_0, y_0)$ is of the form $\left(-\frac{\partial f}{\partial y}(x_0, y_0), \frac{\partial f}{\partial x}(x_0, y_0)\right)$, since its dot product with $\nabla f(x_0, y_0)$ is 0.

$\therefore$ a line parallel to this vector, through $(x_0, y_0, f(x_0, y_0))$ is:

$$\ell(t) = (x_0, y_0, f(x_0, y_0)) + t\left[-f_y(x_0, y_0), f_x(x_0, y_0), 0\right]$$

∴ if, for every t, if $\ell(t)$ satisfies [1], the tangent plane, then $\ell(t)$ lies in the plane.

Every point of $\ell(t)$ is of the form:

$$x = x_0 - t f_x(x_0, y_0)$$

$$y = y_0 + t f_x(x_0, y_0)$$

$$z = f(x_0, y_0)$$

Substituting into [1],

$$f(x_0, y_0) = f(x_0, y_0) + f_x(x_0, y_0)\left[x_0 - t f_y(x_0, y_0) - x_0\right] + f_y(x_0, y_0)\left[y_0 + t f_x(x_0, y_0) - y_0\right]$$

$$\therefore f(x_0, y_0) = f(x_0, y_0) + f_x(x_0, y_0)\left(-t f_y(x_0, y_0)\right) + f_y(x_0, y_0)\left(t f_x(x_0, y_0)\right),$$

or $$f(x_0, y_0) = f(x_0, y_0) - t f_x f_y + t f_y f_x$$

so $$f(x_0, y_0) = f(x_0, y_0)$$

∴ $\ell(t)$ satisfies the equation of the plane for all t, so the line lies in the plane.

(ii) The normal to the tangent plane at $(x_0, y_0, f(x_0, y_0))$ is: $(f_x(x_0,y_0), f_y(x_0,y_0), -1)$, so the "upward" normal is: $(-f_x(x_0,y_0), -f_y(x_0,y_0), 1) = \vec{n}$

$\therefore \vec{n} \cdot \hat{k} = \|\vec{n}\| \|\hat{k}\| \cos\theta$

Since $\vec{n} \cdot \hat{k} = 1$, $\|\hat{k}\| = 1$, then

$1 = \sqrt{f_x^2 + f_y^2 + 1} \cos\theta$, or

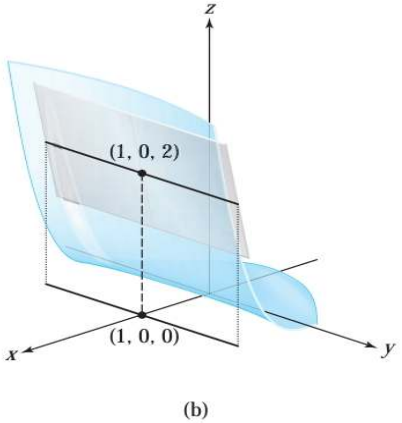
$\sec\theta = \sqrt{f_x^2 + f_y^2 + 1}$, $\sec^2\theta = f_x^2 + f_y^2 + 1$

But $1 + \tan^2\theta = \sec^2\theta$, so

$\sec^2\theta - 1 = \tan^2\theta = f_x^2 + f_y^2$

$\therefore \underline{\tan\theta = \sqrt{f_x^2 + f_y^2}} = \underline{\|\nabla f(x_0, y_0)\|}$

(b) (b) Use this method to show that the tangent plane of the graph of $f(x, y) = (x + \cos y)x^2$ at (1, 0, 2) is as sketched in Figure 2.R.1(b).



(b)

$z = f(x,y) = x^3 + x^2\cos y$

$\nabla f(x,y) = (3x^2 + 2x\cos y, -x^2 \sin y)$

$\therefore \nabla f(1,0) = (5, 0)$. A vector perpendicular to

This is $(0,-5)$. $\therefore \ell(t) = (1,0) + t(0,-5)$ in the xy-plane, and through $(1,0,2)$,

$$\ell(t) = (1,0,2) + t(0,-5,0).$$

Tangent plane at $(1,0,2)$ is: $z = 2 + 5(x-1) + 0(y-0)$

or: $z = 5x - 3$

Does $z = 5x - 3$ contain $\ell(t)$?

For $\ell(t)$: $x = 1 + t\cdot 0 = 1$

$y = 0 - t\cdot 5 = -5t$

$z = 2 + t\cdot 0 = 2$

$\therefore 2 \stackrel{?}{=} 5(1) - 3 = 2$. yes

24. **24.** Find the plane tangent to the surface $z = x^2 + y^2$ at the point $(1, -2, 5)$. Explain the geometric significance, for this surface, of the gradient of $f(x, y) = x^2 + y^2$ (see Exercise 23).

$f_x = 2x$, $f_y = 2y$. $\therefore f_x(1,-2) = 2$, $f_y(1,-2) = -4$

$\therefore$ Tangent plane: $z = 5 + 2(x-1) - 4(y+2)$

or, $\underline{z = 2x - 4y - 5}$

$\nabla f(x,y)$ = gradient, gives projection into the xy-plane of the normal to the plane.

25. **25.** In which direction is the directional derivative of $f(x, y) = (x^2 - y^2)/(x^2 + y^2)$ at $(1, 1)$ equal to zero?

$$f_x = \frac{(x^2+y^2)(2x) - (x^2-y^2)(2x)}{(x^2+y^2)^2} = \frac{4xy^2}{(x^2+y^2)^2}$$

$$f_y = \frac{(x^2+y^2)(-2y) - (x^2-y^2)(2y)}{(x^2+y^2)^2} = \frac{-4yx^2}{(x^2+y^2)^2}$$

$\therefore f_x(1,1) = 1 \qquad f_y(1,1) = -1$

Directional derivative $= \nabla f(1,1) \cdot \vec{v}$, $\vec{v}$ a unit vector.

$\therefore (1,-1) \cdot (a,b) = 0$, $a - b = 0 \Rightarrow a = b$. (e.g., $a = b = 1$).

Unit vector for $(1,1)$ is: $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$. $\therefore$ along $\pm\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$

26. **26.** Find the directional derivative of the given function at the given point and in the direction of the given vector.

(a) $f(x, y, z) = e^x \cos(yz)$, $p_0 = (0, 0, 0)$, $\mathbf{v} = (2, 1, -2)$

(b) $f(x, y, z) = xy + yz + zx$, $p_0 = (1, 1, 2)$, $\mathbf{v} = (10, -1, 2)$

(a) $f_x = e^x \cos(yz)$ $\quad f_y = -ze^x \sin(yz)$ $\quad f_z = -ye^x \sin(yz)$

$f_x(0,0,0) = 1$ $\quad f_y(0,0,0) = 0$ $\quad f_z(0,0,0) = 0$

$\|\vec{v}\| = 3$. $\therefore \vec{u} = \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right)$.

$\therefore \nabla f(0,0,0) \cdot \vec{u} = (1,0,0) \cdot \left(\frac{2}{3}, \frac{1}{3}, -\frac{2}{3}\right) = \frac{2}{3}$

(5) $f_x = y+z \quad f_y = x+z \quad f_z = y+x$

$f_x(1,1,2) = 3 \quad f_y(1,1,2) = 3 \quad f_z(1,1,2) = 2$

$\|\vec{v}\| = \sqrt{105} \quad \therefore \vec{u} = \frac{1}{\sqrt{105}}(10,-1,2)$

$\nabla f(1,1,2) \cdot \vec{u} = (3,3,2) \cdot \frac{1}{\sqrt{105}}(10,-1,2) = \frac{30-3+4}{\sqrt{105}}$

$= \frac{31}{\sqrt{105}}$

27. **27.** Find the tangent plane and normal to the hyperboloid $x^2 + y^2 - z^2 = 18$ at $(3, 5, -4)$.

$f_x = 2x \quad f_y = 2y \quad f_z = -2z \quad \nabla f(3,5,-4) = (6,10,8)$

$\therefore \nabla f(3,5,-4) \cdot (\vec{x} - (3,5,-4)) = 0 \Rightarrow$

$6(x-3) + 10(y-5) + 8(z+4) = 0$, or

$3x + 5y + 4z = 18 \quad$ Normal: $(3,5,4)$

28. **28.** Let $(x(t), y(t))$ be a path in the plane, $0 \le t \le 1$, and let $f(x, y)$ be a C^1 function of two variables. Assume that $(dx/dt)\, f_x + (dy/dt)\, f_y \le 0$. Show that $f(x(1), y(1)) \le f(x(0), y(0))$.

Let $h(t) = f(x(t), y(t))$, $0 \le t \le 1$.

$\therefore h'(t) = (f_x)(dx/dt) + (f_y)(dy/dt)$ by chain rule.

$\therefore h'(t) \le 0$, since $(f_x)\left(\frac{dx}{dt}\right) + (f_y)\left(\frac{dy}{dt}\right) \le 0$.

$\therefore h(t)$ is a decreasing function, and $\therefore h(0) \ge h(1)$.

$\therefore h(0) = f(x(0), y(0)) \ge f(x(1), y(1)) = h(1)$.

29. **29.** A bug finds itself in a toxic environment. The toxicity level is given by $T(x, y) = 2x^2 - 4y^2$. The bug is at $(-1, 2)$. In what direction should it move to lower the toxicity the fastest?

The bug should move in the $-\nabla T$ direction, which is opposite ∇T, the direction of fastest increase.

$\nabla T(x,y) = (T_x, T_y) = (4x, -8y)$.

$\therefore -\nabla T(x,y) = (-4x, 8y)$. $\therefore -\nabla T(-1,2) = \underline{(4, 16)}$.

30. **30.** Find the direction in which the function $w = x^2 + xy$ increases most rapidly at the point $(-1, 1)$. What is the magnitude of ∇w at this point? Interpret this magnitude geometrically.

$\nabla w(x,y) = (2x+y, x)$ $\therefore \underline{\nabla w(-1,1) = (-1, -1)}$.

$\|\nabla w(-1,1)\| = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$. Thus, $w(x,y)$ changes most rapidly by $\sqrt{2}$/unit.

31.

31. Let f be defined on an open set S in $\mathbb{R}^n$. We say that f is ***homogeneous of degree*** p over S if $f(\lambda \mathbf{x}) = \lambda^p f(\mathbf{x})$ for every real λ and for every $\mathbf{x}$ in S for which $\lambda \mathbf{x} \in S$.

(a) If such a function is differentiable at $\mathbf{x}$, show that $\mathbf{x} \cdot \nabla f(\mathbf{x}) = p f(\mathbf{x})$. This is known as ***Euler's theorem*** for homogeneous functions. [HINT: For fixed $\mathbf{x}$, define $g(\lambda) = f(\lambda \mathbf{x})$ and compute $g'(1)$.]

(b) Find p and check Euler's theorem for the function $f(x, y, z) = x - 2y - \sqrt{xz}$, on the region where $xz > 0$.

Note: should state $f: S \to \mathbb{R}^1$

(a) Let $\vec{x}$ be any "fixed" point in $\mathbb{R}^n$, and define

$g(\lambda) = f(\lambda \vec{x}) = \lambda^p f(\vec{x})$, $h(\lambda) = \lambda \vec{x}$

$\therefore g'(\lambda) = p\lambda^{p-1} f(\vec{x})$ (here, $f(\vec{x})$ acts as a constant).

and as $g(\lambda) = f(h(\lambda)) = f(\lambda \vec{x})$, then $g'(\lambda) = D(f \circ h)(\lambda)$

$= Df(h(\lambda)) \cdot Dh(\lambda) = \underset{1\times n}{Df(\lambda \vec{x})} \cdot \underset{n\times 1}{D(\lambda \vec{x})}$ ← matrix multiplication

$= \nabla f(\lambda \vec{x}) \cdot \vec{x}$ (vector dot product)

$\therefore p\lambda^{p-1} f(\vec{x}) = \nabla f(\lambda \vec{x}) \cdot \vec{x}$

Letting $\lambda = 1$, $p f(\vec{x}) = \nabla f(\vec{x}) \cdot \vec{x}$.

Note: if $h(\lambda) = (\lambda x_1, \lambda x_2, \ldots, \lambda x_n)$, $\vec{x} \in \mathbb{R}^n$,

where $h: \mathbb{R}^1 \to \mathbb{R}^n$, and $h(\lambda) = (h_1(\lambda), \ldots, h_n(\lambda))$,

then $Dh(\lambda) = \begin{bmatrix} \partial h_1/\partial \lambda \\ \vdots \\ \partial h_n/\partial \lambda \end{bmatrix} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$, the x_i act as constants

(6) Note $f(\lambda\vec{x}) = f(\lambda x, \lambda y, \lambda z) = (\lambda x) - 2(\lambda y) - \sqrt{(\lambda x)(\lambda z)}$

$= \lambda x - 2\lambda y - \lambda\sqrt{xz}$

$= \lambda(x - 2y - \sqrt{xz})$

$= \lambda f(x, y, z)$.

$\therefore$ f is homogeneous of degree 1.

$\therefore$ Check for $\nabla f \cdot \vec{x} = f(\vec{x})$

$f_x = 1 - \frac{1}{2}(xz)^{-\frac{1}{2}}(z) = 1 - \frac{z}{2\sqrt{xz}}$

$f_y = -2$

$f_z = -\frac{1}{2}(xz)^{-\frac{1}{2}}(x) = -\frac{x}{2\sqrt{xz}}$

$\therefore \nabla f \cdot \vec{x} = (f_x, f_y, f_z) \cdot (x, y, z)$

$= x f_x + y f_y + z f_z$

$= \left(x - \frac{xz}{2\sqrt{xz}}\right) + (-2y) + \left(-\frac{xz}{2\sqrt{xz}}\right)$

$= x - 2y - \frac{xz}{\sqrt{xz}} = x - 2y - \sqrt{xz} = f(x, y, z)$

$\therefore \nabla f \cdot \vec{x} = f(\vec{x})$

32.

32. If $z = [f(x - y)]/y$ (where f is differentiable and $y \neq 0$), show that the identity $z + y(\partial z/\partial x) + y(\partial z/\partial y) = 0$ holds.

$$\frac{\partial z}{\partial x} = Df(x-y)\cdot\frac{1}{y} \qquad \frac{\partial z}{\partial y} = \frac{y[Df(x-y)](-1) - f(x-y)(1)}{y^2}$$

$$= -\frac{Df(x-y)}{y} - \frac{f(x-y)}{y^2}$$

$$\therefore z + y\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} =$$

$$\frac{f(x-y)}{y} + y\left(\frac{Df(x-y)}{y}\right) + y\left(-\frac{Df(x-y)}{y} - \frac{f(x-y)}{y^2}\right) =$$

$$\frac{f(x-y)}{y} + Df(x-y) - Df(x-y) - \frac{f(x-y)}{y} = 0$$

33. **33.** Given $z = f((x+y)/(x-y))$ for f a C^1 function, show that

$$x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = 0.$$

$f: R^1 \to R^1$. Let $h: R^2 \to R^1$, $h(x,y) = \frac{x+y}{x-y}$.

$\therefore z = g(x,y) = (f \circ h)(x,y)$ $\therefore g: R^2 \to R^1$, Dg a 1×2 matrix.

$$Dg = [\partial g/\partial x \quad \partial g/\partial y] = \left[\frac{\partial z}{\partial x} \quad \frac{\partial z}{\partial y}\right].$$

$$\frac{\partial z}{\partial x} = f'(h(x,y))\cdot\frac{\partial h}{\partial x} \qquad \frac{\partial h}{\partial x} = \frac{(x-y)(1) - (x+y)(1)}{(x-y)^2} = \frac{-2y}{(x-y)^2}$$

$$\frac{\partial z}{\partial y} = f'(h(x,y)\cdot\frac{\partial h}{\partial y} \qquad \frac{\partial h}{\partial y} = \frac{(x-y)(1) - (x+y)(-1)}{(x-y)^2} = \frac{2x}{(x-y)^2}$$

$$\therefore x\frac{\partial z}{\partial x} + y\frac{\partial z}{\partial y} = x\left[f'(h(x,y))\left(\frac{-2y}{(x-y)^2}\right)\right] + y\left[f'(h(x,y))\left(\frac{2x}{(x-y)^2}\right)\right]$$

$$= \frac{f'(h(x,y))}{(x-y)^2}\left[-2xy + 2xy\right] = 0$$

34. **34.** Let f have partial derivatives $\partial f(\mathbf{x})/\partial x_i$, where $i = 1, 2, \ldots, n$, at each point $\mathbf{x}$ of an open set U in $\mathbb{R}^n$. If f has a local maximum or a local minimum at the point $\mathbf{x}_0$ in U, show that $\partial f(\mathbf{x}_0)/\partial x_i = 0$ for each i.

Since f has a local max. or local min. at $\vec{x}_0$, $\exists \delta > 0$ s.t. $f(\vec{x}_0) \geq f(\vec{x})$ $\forall \vec{x} \in D_\delta(\vec{x}_0)$, or $f(\vec{x}_0) \leq f(\vec{x})$ $\forall \vec{x} \in D_\delta(\vec{x}_0)$.

Let $g_{\vec{x}}(x_i) = f(\vec{x})$, for $\vec{x} \in D\delta(x_0)$

(i.e., holding all components of $\vec{x}$ constant except the "i"th)

$\therefore \frac{d}{dx_i} g_{\vec{x}}(x_i) = \frac{\partial f(\vec{x})}{\partial x_i}$, for all $x \in D\delta(\vec{x}_0)$

and $g_{\vec{x}}((\vec{x}_0)_i)$ is a local max or local min on $((\vec{x}_0)_i - \delta, (x_0)_i + \delta)$, since $g_{\vec{x}}((\vec{x}_0)_i) = f(\vec{x}_0)$.

$\therefore g'_{\vec{x}}((\vec{x}_0)_i) = 0$ from calculus of one-variable.

$\therefore \frac{\partial f}{\partial x_i}(\vec{x}_0) = g'_{\vec{x}}((\vec{x}_0)_i) = 0$

35.

35. Consider the functions defined in $\mathbb{R}^2$ by the following formulas:

(i) $f(x, y) = xy/(x^2 + y^2)$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$

(ii) $f(x, y) = x^2y^2/(x^2 + y^4)$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$

(a) In each case, show that the partial derivatives $\partial f(x, y)/\partial x$ and $\partial f(x, y)/\partial y$ exist for every (x, y) in $\mathbb{R}^2$, and evaluate these derivatives explicitly in terms of x and y.

(b) Explain why the functions described in (i) and (ii) are or are not differentiable at (0, 0).

(a) for (i) and (ii) f_x and f_y clearly exist for $(x,y) \neq (0,0)$ since they are the product, sum, and quotient of differentiable scalar-valued functions.

For $(x,y) = (0,0)$, must refer to limit definitions.

(i) $f_x(0,0) = \lim_{h\to 0} \frac{f(0+h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{f(0+h,0) - 0}{h}$

$= \lim_{h\to 0} \frac{\frac{h\cdot 0}{h^2+0^2}}{h} = \lim_{h\to 0} \frac{0}{h^3} = 0$

$f_y(0,0) = \lim_{h\to 0} \frac{f(0,0+h) - f(0,0)}{h} = \lim_{h\to 0} \frac{f(0,0+h) - 0}{h}$

$= \lim_{h\to 0} \frac{\frac{0\cdot h}{0^2+h^2}}{h} = \lim_{h\to 0} \frac{0}{h^3} = 0$

(ii) $f_x(0,0) = \lim_{h\to 0} \frac{f(0+h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{f(h,0) - 0}{h}$

$= \lim_{h\to 0} \frac{\frac{h^2\cdot 0^2}{h^2+0^4}}{h} = \lim_{h\to 0} \frac{0}{h^3} = 0$

$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,0+h)-f(0,0)}{h} = \lim_{h\to 0} \frac{f(0,h)-0}{h}$$

$$= \lim_{h\to 0} \frac{0^2\cdot h^2}{\frac{0^2+h^4}{h}} = \lim_{h\to 0} \frac{0}{h^5} = 0.$$

(6) (i) is not differentiable at $(0,0)$. If it were, it would be continuous at $(0,0)$. But along $y=x$ path, (i) becomes $\lim_{(x,y)\to(0,0)} \frac{x^2}{x^2+x^2} = \frac{1}{2} \neq 0 = f(0,0)$.

For (ii), use definition: f is differentiable at $\vec{x}_0$ if

$$\lim_{\vec{x}\to\vec{x}_0} \frac{\| f(\vec{x}) - f(\vec{x}_0) - T(\vec{x}-\vec{x}_0)\|}{\|\vec{x}-\vec{x}_0\|} = 0, \qquad [1]$$

where $\vec{x}_0 = (0,0)$, $f(\vec{x}_0) = 0$, and

$$Df(\vec{x}_0) = T = \left[\frac{\partial f}{\partial x}(0,0) \quad \frac{\partial f}{\partial y}(0,0)\right] = [0\ 0] \text{ from (a)}$$

$\therefore T(\vec{x}-\vec{x}_0) = 0.$

$\therefore$ [1] reduces to $\lim_{\vec{x}\to(0,0)} \frac{\|f(\vec{x})\|}{\|x\|} = 0,$

or $\lim_{(x,y)\to(0,0)} \frac{\frac{x^2y^2}{x^2+y^4}}{\sqrt{x^2+y^2}} = \lim_{(x,y)\to(0,0)} \frac{x^2y^2}{(x^2+y^4)\sqrt{x^2+y^2}} = 0$

If switch to polar coordinates,

$x = r\cos\theta$, $y = r\sin\theta$, then $0 < \sqrt{x^2+y^2} < \delta$ becomes $0 < r < \delta$, so $(x,y) \to (0,0)$ becomes $r \to 0$.

$$\therefore \lim_{r\to 0} \frac{r^4\cos^2\theta\sin^2\theta}{(r^2\cos^2\theta + r^4\sin^2\theta)(r)} = \lim_{r\to 0} \frac{r\cos^2\theta\sin^2\theta}{\cos^2\theta + r^2\sin^2\theta} \quad [2]$$

if $\cos\theta = 0$, then [2] becomes ($\sin^2\theta = 1$):

$$\lim_{r\to 0} \frac{0}{0 + r^2} = 0, \text{ so } f \text{ is differentiable.}$$

if $\cos\theta \neq 0$, then $\lim_{r\to 0} \frac{r\cos^2\theta\sin^2\theta}{\cos^2 + r^2\sin^2\theta} = \frac{0}{\cos^2\theta + 0} = 0$

so f is differentiable.

$\therefore$ For (i), f is differentiable at $(0,0)$

36.

36. Compute the gradient vector $\nabla f(x, y)$ at all points (x, y) in $\mathbb{R}^2$ for each of the following functions:

(a) $f(x, y) = x^2 y^2 \log(x^2 + y^2)$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$

(b) $f(x, y) = xy \sin[1/(x^2 + y^2)]$ if $(x, y) \neq (0, 0)$, $f(0, 0) = 0$

(a) If $(x,y) \neq (0,0)$: $f_x = 2xy^2\log(x^2+y^2) + x^2y^2 \cdot \frac{1}{x^2+y^2} \cdot 2x$

$f_y = 2yx^2\log(x^2+y^2) + x^2y^2 \cdot \frac{1}{x^2+y^2} \cdot 2y$

$$\therefore \nabla f(x,y) = \left(2xy^2\log(x^2+y^2) + \frac{2x^3y^2}{x^2+y^2},\ 2yx^2\log(x^2+y^2) + \frac{2x^2y^3}{x^2+y^2}\right)$$

If $(x,y) = (0,0)$, use definition.

$$f_x = \lim_{h\to 0} \frac{f(0+h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{(0+h)^2 0^2 \log((0+h)^2+0) - 0}{h}$$

$$= \lim_{h\to 0} \frac{0}{h} = 0$$

$$f_y = \lim_{h\to 0} \frac{f(0,0+h) - f(0,0)}{h} = \lim_{h\to 0} \frac{0^2 (0+h)^2 \log(0^2+(0+h)^2) - 0}{h}$$

$$= \lim_{h\to 0} \frac{0}{h} = 0.$$

$\therefore \nabla f(0,0) = (0,0)$

(5) For $(x,y) \neq (0,0)$ $\quad f_x = y\sin\left(\frac{1}{x^2+y^2}\right) + xy\cos\left(\frac{1}{x^2+y^2}\right)\left(\frac{-2x}{(x^2+y^2)^2}\right)$

$$f_y = x\sin\left(\frac{1}{x^2+y^2}\right) + xy\cos\left(\frac{1}{x^2+y^2}\right)\left(\frac{-2y}{(x^2+y^2)^2}\right)$$

$$\therefore \nabla f(x,y) = \left[y\sin\left(\frac{1}{x^2+y^2}\right) - \frac{2x^2y}{(x^2+y^2)^2}\cos\left(\frac{1}{x^2+y^2}\right), \right.$$
$$\left. \left(x\sin\left(\frac{1}{x^2+y^2}\right) - \frac{2xy^2}{(x^2+y^2)^2}\cos\left(\frac{1}{x^2+y^2}\right)\right)\right]$$

For $(x,y) = (0,0)$, use definition

$$f_x = \lim_{h\to 0} \frac{f(0+h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{(0+h)0\sin\left(\frac{1}{(0+h)^2+0}\right) - 0}{h}$$

$$= \lim_{h\to 0} \frac{0}{h} = 0$$

$$f_y = \lim_{h\to 0} \frac{f(0,0+h)-f(0,0)}{h} = \lim_{h\to 0} \frac{0(0+h)\sin\left(\frac{1}{0+(0+h)^2}\right)-0}{h}$$

$$= \lim_{h\to 0} \frac{0}{h} = 0$$

$$\therefore \nabla f(0,0) = (0,0)$$

37.

37. Find the directional derivatives of the following functions at the point $(1, 1)$ in the direction $(\mathbf{i}+\mathbf{j})/\sqrt{2}$:

(a) $f(x, y) = x\tan^{-1}(x/y)$

(b) $f(x, y) = \cos(\sqrt{x^2+y^2})$

(c) $f(x, y) = \exp(-x^2-y^2)$

(a) $f_x = \tan^{-1}\left(\frac{x}{y}\right) + x\left(\frac{1}{1+\left(\frac{x}{y}\right)^2}\right)\left(\frac{1}{y}\right)$

$\therefore f_x(1,1) = \tan^{-1}(1) + \frac{1}{2} = \frac{\pi}{4}+\frac{1}{2}$

$f_y = x\left(\frac{1}{1+\left(\frac{x}{y}\right)^2}\right)\cdot\left(-\frac{x}{y^2}\right) \quad \therefore f_y(1,1) = -\frac{1}{2}$

$\therefore \nabla f(1,1)\cdot\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right) = \left(\frac{\pi}{4}+\frac{1}{2}\right)\frac{1}{\sqrt{2}} + \left(-\frac{1}{2}\right)\left(\frac{1}{\sqrt{2}}\right)$

$$= \frac{\pi}{4\sqrt{2}} = \underline{\frac{\pi\sqrt{2}}{8}}$$

(b) $f_x = -\sin\left(\sqrt{x^2+y^2}\right)\left(\frac{1}{2}(x^2+y^2)^{-\frac{1}{2}}(2x)\right) \quad \therefore f_x(1,1) = -\frac{1}{\sqrt{2}}\sin(\sqrt{2})$

$f_y = -\sin\left(\sqrt{x^2+y^2}\right)\left(\frac{1}{2}(x^2+y^2)^{-\frac{1}{2}}(2y)\right) \quad \therefore f_y(1,1) = -\frac{1}{\sqrt{2}}\sin(\sqrt{2})$

$\therefore \nabla f(1,1)\cdot\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right) = \left(-\frac{1}{\sqrt{2}}\sin\sqrt{2}\right)\frac{1}{\sqrt{2}} + \left(-\frac{1}{\sqrt{2}}\sin\sqrt{2}\right)\left(\frac{1}{\sqrt{2}}\right)$

$$= \underline{-\sin\sqrt{2}}$$

(c) $f_x = e^{-x^2-y^2}(-2x) \quad \therefore f_x(1,1) = -2e^{-2}$

$f_y = e^{-x^2-y^2}(-2y) \quad \therefore f_y(1,1) = -2e^{-2}$

$\therefore \nabla f(1,1) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = (-2e^{-2})\frac{1}{\sqrt{2}} + (-2e^{-2})\frac{1}{\sqrt{2}}$

$= -2\sqrt{2}e^{-2}$

38.

38. (a) Let $\mathbf{u} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}$. Find: $\|\mathbf{u}\|$, $\mathbf{u} \cdot \mathbf{v}$, $\mathbf{u} \times \mathbf{v}$, and a vector in the same direction as $\mathbf{u}$, but of unit length.

(b) Find the rate of change of $e^{xy}\sin(xyz)$ in the direction $\mathbf{u}$ at (0, 1, 1).

(a) $\|\vec{u}\| = \sqrt{1^2+2^2+2^2} = \sqrt{9} = 3$

$\vec{u} \cdot \vec{v} = (1,-2,2)\cdot(2,1,-3) = 2-2-6 = -6$

$\vec{u} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 2 \\ 2 & 1 & -3 \end{vmatrix} = (6-2, -(-3-4), 1+4) =$

$(4, 7, 5)$

(b) $f_x = ye^{xy}\sin(xyz) + e^{xy}\cos(xyz)(yz)$

$f_y = xe^{xy}\sin(xyz) + e^{xy}\cos(xyz)(xz)$

$f_z = e^{xy}\cos(xyz)(xy)$

$\therefore f_x(0,1,1) = 0 + 1 = 1$

$f_y(0,1,1) = 0 + 0 = 0$

$f_z(0,1,1) = 0$

$\therefore \nabla f(0,1,1) = (1,0,0)$

$$\therefore \nabla f(0,1,1) \cdot \frac{\vec{u}}{\|\vec{u}\|} = (1,0,0) \cdot \left(\frac{1}{3}, -\frac{2}{3}, \frac{2}{3}\right) = \frac{1}{3}$$

39.

39. Let $h(x, y) = 2e^{-x^2} + e^{-3y^2}$ denote the height on a mountain at position (x, y). In what direction from $(1, 0)$ should one begin walking in order to climb the fastest?

$h_x = 2e^{-x^2}(-2x) \quad \therefore f_x(1,0) = -4e^{-1}$

$h_y = e^{-3y^2}(-6y) \quad \therefore f_y(1,0) = 0$

$\therefore \nabla h(1,0) = \left(-\frac{4}{e}, 0\right)$

40.

40. Compute an equation for the plane tangent to the graph of

$$f(x, y) = \frac{e^x}{x^2 + y^2}$$

at $x = 1$, $y = 2$.

$z = f(x,y) = \frac{e^x}{x^2+y^2}$. At $f(1,2)$, $z = \frac{e}{1+2^2} = \frac{e}{5}$

$\therefore$ tangent plane at $\left(1, 2, \frac{e}{5}\right)$.

$f_x = \frac{(x^2+y^2)e^x - e^x(2x)}{(x^2+y^2)^2} \qquad f_x(1,2) = \frac{5e-2e}{5^2} = \frac{3e}{25}$

$f_y = -\frac{e^x}{(x^2+y^2)^2}(2y) = -\frac{2ye^x}{(x^2+y^2)^2} \qquad f_y(1,2) = -\frac{4e}{25}$

$\therefore$ Tangent plane: $z = f(1,2) + f_x(1,2)(x-1) + f_y(1,2)(y-2)$

$$\therefore z = \frac{e}{5} + \frac{3e}{25}(x-1) - \frac{4e}{25}(y-2), \text{ or}$$

$$25z = 5e + 3ex - 3e - 4ey + 8e, \text{ or}$$

$$\underline{3ex - 4ey - 25z + 10e = 0}$$

41.

41. (a) Give a careful statement of the general form of the chain rule.

(b) Let $f(x, y) = x^2 + y$ and $\mathbf{h}(u) = (\sin 3u, \cos 8u)$. Let $g(u) = f(\mathbf{h}(u))$. Compute dg/du at $u = 0$ both directly *and* by using the chain rule.

(a) The derivative of a composite function, $f \circ g$, at a point $\vec{x}_0$, is the Jacobian matrix of partial derivatives of f at $g(\vec{x}_0)$ times the Jacobian matrix of partial derivatives of g at $\vec{x}_0$.

(b) Directly: $f(\vec{h}(u)) = f(\sin 3u, \cos 8u)$

$$= \sin^2 3u + \cos 8u$$

$$\therefore \frac{dg}{du}(0) = 2(\sin 3u)(\cos 3u)(3) - 8\sin 8u \Big|_{u=0}$$

$$= 6\sin 3u \cos 3u - 8\sin 8u \Big|_{u=0}$$

$$= 0 - 0 = \underline{0}$$

Chain Rule: $D\vec{h}(u) = \begin{bmatrix} 3\cos 3u \\ -8\sin 3u \end{bmatrix} \quad \therefore D\vec{h}(0) = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$

$$Df(x,y) = \begin{bmatrix} 2x & 1 \end{bmatrix}.$$

$$\therefore Df(h(0)) = Df(0,1) = \begin{bmatrix} 0 & 1 \end{bmatrix}.$$

$$\therefore Df(h(0)) \cdot Dh(0) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \end{bmatrix} = \underline{0}$$

42.

42. (a) Sketch the level curves of $f(x, y) = -x^2 - 9y^2$ for $c = 0, -1, -10$.

(b) On your sketch, draw in ∇f at $(1, 1)$. Discuss.

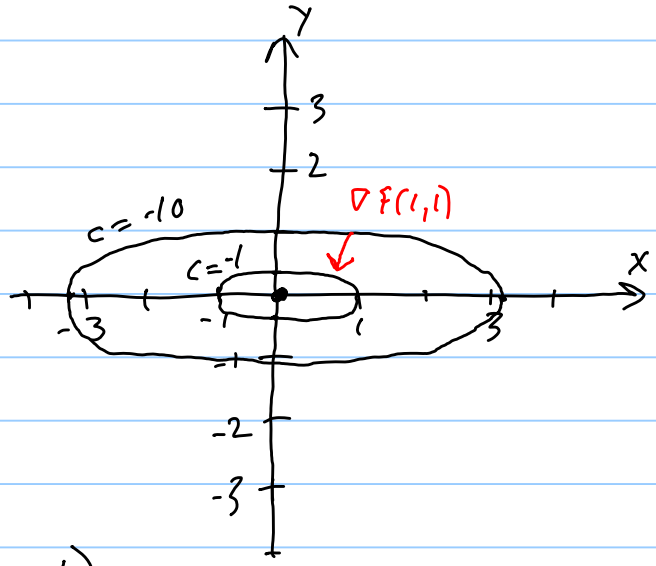
(a) Note $f(x,y) \le 0, \forall x,y$

For $c=0$, $x=0, y=0$ is only solution.

For $c=-1$, $x^2 + 9y^2 = 1$,

or $\frac{x^2}{1} + \frac{y^2}{1/9} = 1.$

$\therefore$ an ellipse with vertices at $(\pm 1, 0), (0, \pm \frac{1}{3})$.



For $c=-10$, $x^2 + 9y^2 = 10$, or

$\frac{x^2}{10} + \frac{y^2}{10/9} = 1$. $\therefore$ An ellipse with vertices at $(\pm\sqrt{10}, 0), (0, \pm\sqrt{10}/3)$

(b) $f_x = -2x$ $\quad f_x(1,1) = -2$ $\quad \therefore \nabla f(1,1) = (-2, -18)$

$f_y = -18y$ $\quad f_y(1,1) = -18$

$\nabla f(1,1)$ indicates the direction (in the xy-plane) of steepest ascent.

The 3D shape is an elliptical paraboloid, with peak at $(x,y,z) = (0,0,0)$.

At $(x,y) = (1,1)$, $(x,y,z) = (1,1,-10)$, and at $(1,1,-10)$ on paraboloid, heading toward $(-2,-18)$ gives the direction of maximum increase in elevation (or Δz).

43.

43. At time $t = 0$, a particle is ejected from the surface $x^2 + 2y^2 + 3z^2 = 6$ at the point (1, 1, 1) in a direction normal to the surface at a speed of 10 units per second. At what time does it cross the sphere $x^2 + y^2 + z^2 = 103$?

Let $f(x,y,z) = x^2 + 2y^2 + 3z^2 = C$. Normal to surface is: $\nabla f(x,y,z) = (2x, 4y, 6z)$. $\therefore \nabla f(1,1,1) = (2,4,6)$.

$\therefore \ell(t) = (1,1,1) + t(s\vec{v})$, t = time, $s\vec{v} = s(2,4,6)$, where $s\vec{v}$ is a vector in the direction of $\vec{v}$.

Note: $\left|\frac{d\ell(t)}{dt}\right| = 10 = \left|s(2,4,6)\right| = s\sqrt{2^2+4^2+6^2} = s\sqrt{56}$

$\therefore s = \frac{10}{\sqrt{56}} = \frac{5}{\sqrt{14}}$

$\therefore \ell(t) = (1,1,1) + t\frac{5}{\sqrt{14}}(2,4,6)$

or, $\ell(t) = \left[1 + \frac{10}{\sqrt{14}}t,\ 1 + \frac{20}{\sqrt{14}}t,\ 1 + \frac{30}{\sqrt{14}}t\right]$

$\therefore$ If $g(x,y,z) = x^2+y^2+z^2 = 103$, then solve for $(g \circ l)(t) = 103$.

$\therefore \left(1+\frac{10}{\sqrt{14}}t\right)^2 + \left(1+\frac{20}{\sqrt{14}}t\right)^2 + \left(1+\frac{30}{\sqrt{14}}t\right)^2 = 103$

$\left(1+\frac{20}{\sqrt{14}}t+\frac{100}{14}t^2\right)+\left(1+\frac{40}{\sqrt{14}}t+\frac{400}{14}t^2\right)+\left(1+\frac{60}{\sqrt{14}}t+\frac{900}{14}t^2\right)=103$

$\therefore \frac{120}{\sqrt{14}}t + \frac{1400}{14}t^2 = 100$, or $\frac{120}{\sqrt{14}}t + 100t^2 = 100$

or $10t^2 + \frac{12}{\sqrt{14}}t - 10 = 0$

$\therefore t = \dfrac{-\frac{12}{\sqrt{14}} \pm \sqrt{\frac{144}{14}+400}}{20}$. Reject negative root.

$\therefore t = -\frac{12}{20\sqrt{14}} + \frac{1}{20}\sqrt{\frac{5744}{14}} = -\frac{3}{5\sqrt{14}} + \frac{1}{20}\sqrt{\frac{16\times 359}{14}}$

$= -\frac{3}{70}\sqrt{14} + \frac{1}{5}\sqrt{\frac{359}{14}} = -\frac{3}{70}\sqrt{14} + \frac{\sqrt{14}}{70}\sqrt{359}$

$= \underline{\underline{\frac{\sqrt{14}}{70}\left(-3+\sqrt{359}\right)}}$

44. **44.** At what point(s) on the surface in Exercise 43 is the normal vector parallel to the line $x = y = z$?

From #43, $\nabla f(x,y,z) = (2x, 4y, 6z) =$ normal to surface.

A vector parallel to line $\Rightarrow 2x = 4y = 6z$.

$\therefore x = 3z, \; y = \frac{3}{2}z.$

$\therefore (3z)^2 + 2\left(\frac{3}{2}z\right)^2 + 3z^2 = 6$, or

$9z^2 + \frac{9}{2}z^2 + 3z^2 = 6$, or $33z^2 = 12$, or

$z^2 = \frac{4}{11}$, $\therefore z = \pm\frac{2}{\sqrt{11}}$

$\therefore x = \pm\frac{6}{\sqrt{11}}$, $y = \pm\frac{3}{\sqrt{11}}$

$\therefore \pm\left(\frac{6}{\sqrt{11}}, \frac{3}{\sqrt{11}}, \frac{2}{\sqrt{11}}\right)$

45. **45.** Compute $\partial z/\partial x$ and $\partial z/\partial y$ if

$$z = \frac{u^2 + v^2}{u^2 - v^2}, \qquad u = e^{-x-y}, \qquad v = e^{xy}$$

(a) by substitution and direct calculation, and (b) by the chain rule.

(a) $z = \dfrac{\left(e^{-x-y}\right)^2 + \left(e^{xy}\right)^2}{\left(e^{-x-y}\right)^2 - \left(e^{xy}\right)^2} = \dfrac{e^{-2x-2y} + e^{2xy}}{e^{-2x-2y} - e^{2xy}}$

$$z_x = \frac{\left(e^{-2x-2y} - e^{2xy}\right)\left(-2e^{-2x-2y} + 2ye^{2xy}\right) - \left(e^{-2x-2y} + e^{2xy}\right)\left(-2e^{-2x-2y} - 2ye^{2xy}\right)}{\left(e^{-2x-2y} - e^{2xy}\right)^2}$$

$$= \frac{4e^{2xy-2x-2y} + 4ye^{2xy-2x-2y}}{\left(e^{-2x-2y} - e^{2xy}\right)^2}$$

$$z_y = \frac{\left(e^{-2x-2y} - e^{2xy}\right)\left(-2e^{-2x-2y} + 2xe^{2xy}\right) - \left(e^{-2x-2y} + e^{2xy}\right)\left(-2e^{-2x-2y} - 2xe^{2xy}\right)}{\left(e^{-2x-2y} - e^{2xy}\right)^2}$$

$$= \frac{4e^{2xy-2x-2y} + 4xe^{2xy-2x-2y}}{\left(e^{-2x-2y} - e^{2xy}\right)^2}$$

(5) $\frac{\partial z}{\partial x} = \frac{\partial z}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \cdot \frac{\partial v}{\partial x}$

$$\frac{\partial z}{\partial u} = \left[\frac{(u^2-v^2)2u - (u^2+v^2)(2u)}{(u^2-v^2)^2}\right] = \left[\frac{-4uv^2}{(u^2-v^2)^2}\right]$$

$$\frac{\partial z}{\partial v} = \left[\frac{(u^2-v^2)(2v) - (u^2+v^2)(-2v)}{(u^2-v^2)^2}\right] = \left[\frac{4vu^2}{(u^2-v^2)^2}\right]$$

$$\therefore \frac{\partial z}{\partial x} = \left[\frac{-4uv^2}{(u^2-v^2)^2}\right]\cdot\left(-e^{-x-y}\right) + \left[\frac{4vu^2}{(u^2-v^2)^2}\right]\left(ye^{xy}\right)$$

$$= \frac{4e^{-x-y}uv^2 + 4ye^{xy}vu^2}{(u^2-v^2)^2}$$

$$= \frac{4e^{-x-y}(e^{-x-y})(e^{xy})^2 + 4ye^{xy}(e^{xy})(e^{-x-y})^2}{(e^{-2x-2y} - e^{2xy})^2}$$

$$= \frac{4e^{2xy-2x-2y} + 4ye^{2xy-2x-2y}}{(e^{-2x-2y} - e^{2xy})^2}$$

$$\frac{\partial z}{\partial y} = \frac{\partial z}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial z}{\partial v}\cdot\frac{\partial v}{\partial y}$$

$$= \left[\frac{-4uv^2}{(u^2-v^2)^2}\right]\left(-e^{-x-y}\right) + \left[\frac{4vu^2}{(u^2-v^2)^2}\right]\left(xe^{xy}\right)$$

$$= \frac{4e^{-x-y}(e^{-x-y})(e^{xy})^2 + 4xe^{xy}(e^{xy})(e^{-x-y})^2}{(e^{-2x-2y} - e^{2xy})^2}$$

$$= \frac{4e^{2xy-2x-2y} + 4xe^{2xy-2x-2y}}{(e^{-2x-2y} - e^{2xy})^2}$$

46. **46.** Compute the partial derivatives as in Exercise 45 if $z = uv$, $u = x + y$, and $v = x - y$.

(a) $z = (x+y)(x-y) = x^2 - y^2$

$\therefore \dfrac{\partial z}{\partial x} = 2x \qquad \dfrac{\partial z}{\partial y} = -2y$

(b) $\dfrac{\partial z}{\partial x} = \dfrac{\partial z}{\partial u}\cdot\dfrac{\partial u}{\partial x} + \dfrac{\partial z}{\partial v}\dfrac{\partial v}{\partial x} = (v)(1) + (u)(1) = v + u$

$= (x-y) + (x+y) = 2x$

$\dfrac{\partial z}{\partial y} = \dfrac{\partial z}{\partial u}\cdot\dfrac{\partial u}{\partial y} + \dfrac{\partial z}{\partial v}\cdot\dfrac{\partial v}{\partial y} = (v)(1) + (u)(-1) = v - u$

$= (x-y) - (x+y) = -2y$

47. **47.** What is wrong with the following argument? Suppose that $w = f(x, y)$ and $y = x^2$. By the chain rule,

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial x}\frac{\partial x}{\partial x} + \frac{\partial w}{\partial y}\frac{\partial y}{\partial x} = \frac{\partial w}{\partial x} + 2x\frac{\partial w}{\partial y}.$$

Hence, $0 = 2x(\partial w/\partial y)$, and so $\partial w/\partial y = 0$. Choose an explicit example to really see that this is incorrect.

$\dfrac{\partial w}{\partial x}$ on the left side of the equation is ambiguous.

It is unclear if this means derivative with respect to first independent variable of f, or with respect to "x" of the second, composite function.

Assuming the latter, let $w = f(u, v)$, $u(x) = x$, $v(x) = x^2$.

Then $\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial w}{\partial v}\cdot\frac{\partial v}{\partial x}$

$= \frac{\partial w}{\partial u} + \frac{\partial w}{\partial v}(2x)$

Example: Let $w = f(u,v) = uv$. And $u(x) = x$, $v(x) = x^2$.

$\therefore \frac{\partial w}{\partial x} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}$

$= (v)(1) + (u)(2x) = x^2 + 2x^2 = 3x^2$

if allowed the confusion, and allowed

$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial x} + 2x\frac{\partial w}{\partial y}$, then $\frac{\partial w}{\partial y} = \frac{\partial w}{\partial v} = u = x$.

so $\frac{\partial w}{\partial y} \neq 0$.

48. **48.** A boat is sailing northeast at 20 km/h. Assuming that the temperature drops at a rate of 0.2°C/km in the northerly direction and 0.3°C/km in the easterly direction, what is the time rate of change of temperature as observed on the boat?

Let temp vector $\vec{T} = (-0.2°C/km)\,\hat{j} + (-0.3°C/km)\,\hat{i}$

velocity vector: Northeast is 45°,
and $\cos 45° = \sin 45° = \sqrt{2}/2$.

$\therefore (20\ km/h)\left(\frac{\sqrt{2}}{2}\right) = 10\sqrt{2}\ km/hr.$

$\therefore \vec{V} = (10\sqrt{2}\ km/hr)\,\hat{i} + (10\sqrt{2}\ km/hr)\,\hat{j}$.

$$\therefore \vec{T} \cdot \vec{v} \equiv \nabla T \cdot \vec{v}, \text{ directional derivative}$$

$$= (-0.2, -0.3) \cdot (10\sqrt{2}, 10\sqrt{2})$$

$$= -2\sqrt{2} - 3\sqrt{2} = \underline{-5\sqrt{2}\ ^\circ C/hr}$$

49. **49.** Use the chain rule to find a formula for $(d/dt) \exp[f(t)g(t)]$.

$$\frac{d}{dt} e^{f(t)g(t)} = e^{f(t)g(t)} \left[f'(t)g(t) + f(t)g'(t) \right]$$

50. **50.** Use the chain rule to find a formula for $(d/dt)(f(t)^{g(t)})$.

Let $h(t) = f(t)^{g(t)}$ $\therefore \log h(t) = g(t) \log f(t)$

$$\therefore h(t) = e^{\log h(t)} = e^{g(t) \log f(t)}$$

$$\therefore \frac{d}{dt} f(t)^{g(t)} = \frac{d}{dt} e^{g(t) \log f(t)}$$

$$= e^{g(t) \log f(t)} \left[g'(t) \log f(t) + g(t) \cdot \frac{1}{f(t)} \cdot f'(t) \right]$$

$$= f(t)^{g(t)} \left[g'(t) \log f(t) + \frac{g(t)}{f(t)} \cdot f'(t) \right]$$

$$= f(t)^{g(t)} \left[g'(t) \log f(t) \right] + f(t)^{g(t)-1} \left[g(t) \cdot f'(t) \right]$$

51.

51. Verify the chain rule for the function $f(x, y, z) = [\ln(1 + x^2 + 2z^2)]/(1 + y^2)$ and the path $\mathbf{c}(t) = (t, 1 - t^2, \cos t)$.

(a) $(f \circ \vec{c})(t) = \dfrac{\ln(1 + t^2 + 2\cos^2 t)}{1 + (1 - t^2)^2}$

$\therefore (f \circ \vec{c})'(t) =$

$$\frac{\left[1 + (1-t^2)^2\right] \cdot \dfrac{2t - 4\cos t \sin t}{1 + t^2 + 2\cos^2 t} - \left[\ln(1 + t^2 + 2\cos^2 t)\right] 2(1-t^2)(-2t)}{\left[1 + (1-t^2)^2\right]^2}$$

$$= \frac{\left[1 + (1-t^2)^2\right](2t - 4\cos t \sin t) + 4t(1-t^2)\left[\ln(1 + t^2 + 2\cos^2 t)\right](1 + t^2 + 2\cos^2 t)}{\left[1 + (1-t^2)^2\right]^2 (1 + t^2 + 2\cos^2 t)}$$

(b) Using chain rule, $D(f \circ \vec{c})(t) = \nabla f(\vec{c}(t)) \cdot D\vec{c}(t)$

$$D\vec{c}(t) = \begin{bmatrix} 1 \\ -2t \\ -\sin t \end{bmatrix} \qquad f(x, y, z) = \frac{\ln(1 + x^2 + 2z^2)}{1 + y^2}$$

$$f_x = \frac{1}{1+y^2} \cdot \frac{1}{1 + x^2 + 2z^2} \cdot 2x = \frac{2x}{(1+y^2)(1 + x^2 + 2z^2)}$$

$$f_y = -\frac{\ln(1 + x^2 + 2z^2)}{(1+y^2)^2}(2y)$$

$$f_z = \frac{1}{1+y^2} \cdot \frac{1}{1 + x^2 + 2z^2} \cdot 4z = \frac{4z}{(1+y^2)(1 + x^2 + 2z^2)}$$

$\therefore \nabla f(\vec{c}(t)) \cdot D\vec{c}(t) = f_x - 2t f_y - \sin t \, f_z$

$$= \frac{2x}{(1+y^2)(1+x^2+2z^2)} + \frac{4ty\ln(1+x^2+2z^2)}{(1+y^2)^2} - \frac{4z\sin t}{(1+y^2)(1+x^2+2z^2)}$$

$$= \frac{2x(1+y^2) + 4ty\ln(1+x^2+2z^2)[1+x^2+2z^2] - 4z\sin t(1+y^2)}{(1+y^2)^2(1+x^2+2z^2)}$$

$$= \frac{[1+y^2](2x-4z\sin t) + 4ty[\ln(1+x^2+2z^2)](1+x^2+2z^2)}{(1+y^2)^2(1+x^2+2z^2)}$$

$$= \frac{[1+(1-t^2)^2](2t-4\cos t\sin t) + 4t(1-t^2)[\ln(1+t^2+2\cos^2 t)](1+t^2+2\cos^2 t)}{[1+(1-t^2)^2]^2(1+t^2+2\cos^2 t)}$$

Note (a) = (b)

52. **52.** Verify the chain rule for the function $f(x, y) = x^2/(2 + \cos y)$ and the path $x = e^t$, $y = e^{-t}$.

$$f(x,y) = \frac{x^2}{2+\cos y} \qquad \vec{c}(t) = (e^t, e^{-t})$$

(a) $$(f \circ \vec{c})(t) = \frac{e^{2t}}{2+\cos(e^{-t})}$$

$$\therefore (f\circ\vec{c})'(t) = \frac{[2+\cos(e^{-t})]2e^{2t} - e^{2t}[-\sin(e^{-t})(-e^{-t})]}{[2+\cos(e^{-t})]^2}$$

$$= \frac{2e^{2t}[2+\cos(e^{-t})] - e^t\sin(e^{-t})}{[2+\cos(e^{-t})]^2}$$

(6) Chain rule: $D(f \circ \vec{c})(t) = Df(\vec{c}(t)) \cdot D\vec{c}(t)$

$$D\vec{c}(t) = \begin{bmatrix} e^t \\ -e^{-t} \end{bmatrix} \qquad Df(x,y) = \begin{bmatrix} f_x & f_y \end{bmatrix}$$

$$f_x = \frac{2x}{2+\cos y} \qquad f_y = -\frac{x^2}{(2+\cos y)^2} \cdot (-\sin y) = \frac{x^2 \sin y}{(2+\cos y)^2}$$

$$\therefore \nabla f \cdot D\vec{c}(t) = \begin{bmatrix} \frac{2x}{2+\cos y} & \frac{x^2 \sin y}{(2+\cos y)^2} \end{bmatrix} \begin{bmatrix} e^t \\ -e^{-t} \end{bmatrix}$$

$$= \frac{2x e^t}{2+\cos y} - \frac{x^2 e^{-t} \sin y}{(2+\cos y)^2} = \frac{2x e^t (2+\cos y) - x^2 e^{-t} \sin y}{(2+\cos y)^2}$$

$$= \frac{2e^t e^t \left[2 + \cos(e^{-t})\right] - e^{2t} e^{-t} \sin(e^{-t})}{(2+\cos(e^{-t}))^2}$$

$$= \frac{2e^{2t} \left[2 + \cos(e^{-t})\right] - e^t \sin(e^{-t})}{\left[2+\cos(e^{-t})\right]^2}$$

(a) = (6)

53. **53.** Suppose that $u(x, t)$ satisfies the differential equation $u_t + uu_x = 0$ and that x, as a function $x = f(t)$ of t, satisfies $dx/dt = u(x, t)$. Prove that $u(f(t), t)$ is constant in t.

Let $y(t) = t$, $h(t) = u(f(t), t) = u(x(t), y(t))$

By chain rule, $h'(t) = u_x \cdot x'(t) + u_y y'(t)$

$= u_x u + u_y \cdot 1 = u_x u + u_y$.

But $u_y = u_t$ since this is the derivative of u with respect to the second variable.

$\therefore h'(t) = u_x u + u_t = 0$, so $h(t) = C$, a constant.

$\therefore h(t) = u(f(x), t) = C$, a constant.

54. **54.** The displacement at time t and horizontal position on a line x of a certain violin string is given by $u = \sin(x - 6t) + \sin(x + 6t)$. Calculate the velocity of the string at $x = 1$ when $t = \frac{1}{3}$.

For $x = 1$, $u(t) = \sin(1 - 6t) + \sin(1 + 6t)$

$\therefore u'(t) = -6\cos(1 - 6t) + 6\cos(1 + 6t)$

$\therefore u'\left(\frac{1}{3}\right) = -6\cos\left(1 - 6 \cdot \frac{1}{3}\right) + 6\cos\left(1 + 6 \cdot \frac{1}{3}\right)$

$= -6\cos(-1) + 6\cos(3)$

$= 6(\cos 3 - \cos 1)$

55. **55.** The ***ideal gas law*** $PV = nRT$ involves a constant R, the number n of moles of the gas, the volume V, the Kelvin temperature T, and the pressure P.

(a) Show that each of n, P, T, V is a function of the remaining variables, and determine explicitly the defining equations.

(b) Calculate $\partial V/\partial T$, $\partial T/\partial P$, $\partial P/\partial V$ and show that their product equals -1.

(a) $P = \frac{nRT}{V}$, $V = \frac{nRT}{P}$, $n = \frac{PV}{RT}$, $T = \frac{PV}{nR}$

(b) $\frac{\partial V}{\partial T} = \frac{nR}{P}$, $\frac{\partial T}{\partial P} = \frac{V}{nR}$, $\frac{\partial P}{\partial V} = -\frac{nRT}{V^2}$

$$\therefore \frac{\partial V}{\partial T} \cdot \frac{\partial T}{\partial P} \cdot \frac{\partial P}{\partial V} = \left(\frac{nR}{P}\right)\left(\frac{V}{nR}\right)\left(-\frac{nRT}{V^2}\right)$$

$$= -\frac{nRT}{PV} = -\frac{PV}{PV} = -1$$

56.

56. The ***potential temperature*** θ is defined in terms of temperature T and pressure p by

$$\theta = T\left(\frac{1000}{p}\right)^{0.286}.$$

The temperature and pressure may be thought of as functions of position (x, y, z) in the atmosphere and also of time t.

(a) Find formulas for $\partial\theta/\partial x$, $\partial\theta/\partial y$, $\partial\theta/\partial z$, $\partial\theta/\partial t$ in terms of partial derivatives of T and p.

(b) The condition $\partial\theta/\partial z < 0$ is regarded as an unstable atmosphere, for it leads to large vertical excursions of air parcels from a single upward or downward impetus. Meteorologists use the formula

$$\frac{\partial\theta}{\partial z} = \frac{\theta}{T}\left(\frac{\partial T}{\partial z} + \frac{g}{C_p}\right),$$

where $g = 32.2$ and C_p is a positive constant. How does the temperature change in the upward direction for an unstable atmosphere?

$\theta(T, P) =$

$\theta(T(x,y,z,t), P(x,y,z,t))$

$\theta(T, P) = (1000)^{0.286} T P^{-0.286}$

(a) $\frac{\partial\theta}{\partial x} = \frac{\partial\theta}{\partial T} \cdot \frac{\partial T}{\partial x} + \frac{\partial\theta}{\partial P} \cdot \frac{\partial P}{\partial x}$

$$= \left(\frac{1000}{P}\right)^{0.286} \cdot T_x - 0.286(1000)^{0.286} T P^{-1.286} \cdot P_x$$

Similarly, $\frac{\partial \theta}{\partial y} = \left(\frac{1000}{p}\right)^{0.286} \cdot T_y - 0.286(1000)^{0.286} T p^{-1.286} \cdot p_y$

$\frac{\partial \theta}{\partial z} = \left(\frac{1000}{p}\right)^{0.286} T_z - 0.286(1000)^{0.286} T p^{-1.286} p_z$

$\frac{\partial \theta}{\partial t} = \left(\frac{1000}{p}\right)^{0.286} T_t - 0.286(1000)^{0.286} T p^{-1.286} p_t$

This follows from chain rule:

Let $H(x,y,z,t) = [T(x,y,z,t), p(x,y,z,t)]$ $H: \mathbb{R}^4 \to \mathbb{R}^2$

$\theta(T,p): \mathbb{R}^2 \to \mathbb{R}^1$

$\therefore$ Pot Temp $= \theta \circ H : \mathbb{R}^4 \to \mathbb{R}^1$

$\therefore D(\theta \circ H) = D\theta \cdot DH$

$$= \begin{bmatrix} \theta_T & \theta_p \end{bmatrix} \begin{bmatrix} T_x & T_y & T_z & T_t \\ p_x & p_y & p_z & p_t \end{bmatrix}$$

$$= \begin{bmatrix} \theta_T T_x + \theta_p p_x & \theta_T T_y + \theta_p p_y & \theta_T T_z + \theta_p p_z & \theta_T T_t + \theta_p p_t \end{bmatrix}$$

The above formulae can be simplified using:

$\frac{\theta}{T} = \left(\frac{1000}{p}\right)^{0.286}$ and $(1000)^{0.286} T p^{-1.286} = \left(\frac{1000}{p}\right)^{0.286} T p^{-1}$

$= \frac{\theta}{T} \cdot T p^{-1} = \frac{\theta}{p}$

$$\therefore \frac{\partial \theta}{\partial x} = \frac{\theta}{T} T_x - 0.286 \frac{\theta}{\rho} \rho_x$$

$$\frac{\partial \theta}{\partial y} = \frac{\theta}{T} T_y - 0.286 \frac{\theta}{\rho} \rho_y$$

$$\frac{\partial \theta}{\partial z} = \frac{\theta}{T} T_z - 0.286 \frac{\theta}{\rho} \rho_z$$

$$\frac{\partial \theta}{\partial t} = \frac{\theta}{T} T_t - 0.286 \frac{\theta}{\rho} \rho_t$$

(b) Given $\frac{\partial \theta}{\partial z} = \frac{\theta}{T}\left(\frac{\partial T}{\partial z} + \frac{g}{c_p}\right)$, then

$$\frac{\partial T}{\partial z} = \frac{T}{\theta}\frac{\partial \theta}{\partial z} - \frac{g}{c_p}, \text{ and } \frac{\theta}{T} = \left(\frac{1000}{p}\right)^{0.286}$$

$$\text{so } \frac{T}{\theta} = \left(\frac{p}{1000}\right)^{0.286} > 0$$

$$\therefore \frac{T}{\theta}\left(\frac{\partial \theta}{\partial z}\right) < 0, \text{ and since } c_p > 0, \frac{-g}{c_p} < 0$$

$\therefore \frac{\partial T}{\partial z} < 0$, i.e., temperature decreases in the upward direction.

57.

57. The specific volume V, pressure P, and temperature T of a van der Waals gas are related by $P = RT/(V - \beta) - \alpha/V^2$, where α, β, and R are constants.

(a) Explain why any two of V, P, and T can be considered independent variables that determine the third variable.

(b) Find $\partial T/\partial P$, $\partial P/\partial V$, $\partial V/\partial T$. Identify which variables are constant, and interpret each partial derivative physically.

(c) Verify that $(\partial T/\partial P)(\partial P/\partial V)(\partial V/\partial T) = -1$(not $+1$!).

(a) From the formula, given the value of any two of V, P, T, the third can be found.

(b) From $P = \frac{RT}{V-\beta} - \frac{\alpha}{V^2}$, $\left(P + \frac{\alpha}{V^2}\right)\frac{V-\beta}{R} = T$

$\therefore T_P = \frac{V-\beta}{R}$ β, R constants

$P_V = -\frac{RT}{(V-\beta)^2} + \frac{2\alpha}{V^3}$ α, β, R constants

From $PV - \beta P + \alpha\left(\frac{V-\beta}{V^2}\right) = RT$

$\frac{d}{dT}\left(PV - \beta P + \alpha\left(\frac{V-\beta}{V^2}\right)\right) = \frac{d}{dT}(RT)$

$\therefore P V_T + \alpha\left[\frac{V^2(V_T) - (V-\beta)2VV_T}{V^4}\right] = R$

$\therefore PV^4 V_T + \alpha V^2 V_T - \alpha(V-\beta)2VV_T = RV^4$

$$\therefore V_T = \frac{RV^4}{PV^4 + \alpha V^2 - 2\alpha(V-\beta)V} = \frac{RV^3}{PV^3 + \alpha V - 2\alpha(V-\beta)}$$

$$= \frac{R}{P + \frac{\alpha}{V^2} - \frac{2\alpha(V-\beta)}{V^3}} \qquad \text{But } P = \frac{RT}{V-\beta} - \frac{\alpha}{V^2}$$

$$= \frac{R}{\frac{RT}{(V-\beta)} - \frac{2\alpha(V-\beta)}{V^3}} = \frac{R(V-\beta)V^3}{RTV^3 - 2\alpha(V-\beta)^2}$$

$$(C)\ T_P \cdot P_V \cdot V_T = \left(\frac{V-\beta}{R}\right)\left(\frac{-RT}{(V-\beta)^2} + \frac{2\alpha}{V^3}\right)\left(\frac{R(V-\beta)V^3}{RTV^3 - 2\alpha(V-\beta)^2}\right)$$

$$= \left(\frac{V-\beta}{R}\right)\left(\frac{-RTV^3 + 2\alpha(V-\beta)^2}{(V-\beta)^2 V^3}\right)\left(\frac{R(V-\beta)V^3}{RTV^3 - 2\alpha(V-\beta)^2}\right)$$

$$= \frac{-RTV^3 + 2\alpha(V-\beta)^2}{RTV^3 - 2\alpha(V-\beta)^2}$$

$$= -1$$

58.

58. The height h of the Hawaiian volcano Mauna Loa is (roughly) described by the function $h(x, y) = 2.59 - 0.00024y^2 - 0.00065x^2$, where h is the height above sea level in miles and x and y measure east–west and north–south distances in miles from the top of the mountain. At $(x, y) = (-2, -4)$:

(a) How fast is the height increasing in the direction $(1, 1)$ (that is, northeastward)? Express your answer in miles of height per mile of horizontal distance traveled.

(b) In what direction is the steepest upward path?

(a) $\vec{v} = \frac{1}{\sqrt{2}}(1,1)$, a unit vector.

Need to find $\nabla h(-2,-4) \cdot \vec{v}$, a directional derivative

$\nabla h = [-0.0013x, -0.00048y]$

$\therefore \nabla h(-2,-4) = (0.0026, 0.00192)$

$\therefore \nabla h(-2,-4) \cdot \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right) = \frac{.0026}{\sqrt{2}} + \frac{.00192}{\sqrt{2}} = \frac{.00452}{\sqrt{2}}$

$\therefore$ Increasing at .0032 miles/mile

(b) $\nabla h(-2,-4) = (0.0026, 0.00192)$, in xy-plane.

59.

59. (a) In what direction is the directional derivative of $f(x, y) = (x^2 - y^2)/(x^2 + y^2)$ at $(1, 1)$ equal to zero?

(b) How about at an arbitrary point (x_0, y_0) in the first quadrant?

(c) Describe the level curves of f. In particular, discuss them in terms of the result of part (b).

(a) $f_x = \frac{(x^2+y^2)(2x) - (x^2-y^2)(2x)}{(x^2+y^2)^2} = \frac{4xy^2}{(x^2+y^2)^2}$

$f_y = \frac{(x^2+y^2)(-2y) - (x^2-y^2)(2y)}{(x^2+y^2)^2} = \frac{-4yx^2}{(x^2+y^2)^2}$

$\therefore \nabla f(1,1) = (1,-1)$. Let $\vec{v}$ = direction.

$\therefore (1,-1) \cdot (v_1, v_2) = 0$, or $v_1 - v_2 = 0$, $v_1 = v_2$.

$\therefore$ parallel to vector $(1,1)$, or $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ as a unit vector.

(b) In first quadrant, $x_0 > 0$, $y_0 > 0$.

$$\therefore \left(\frac{4x_0 y_0^2}{(x_0^2+y_0^2)^2}, \frac{-4y_0 x_0^2}{(x_0^2+y_0^2)^2} \right) \cdot (v_1, v_2) = 0 \Rightarrow$$

$$4x_0 y_0^2 v_1 - 4y_0 x_0^2 v_2 = 0, \quad x_0 y_0^2 v_1 = y_0 x_0^2 v_2$$

$$y_0 v_1 = x_0 v_2, \text{ so } \left(v_1, v_1 \frac{y_0}{x_0}\right)$$

or parallel to $v_1\left(1, \frac{y_0}{x_0}\right)$, or (x_0, y_0).

Making it a unit vector, $\frac{1}{\sqrt{x_0^2+y_0^2}}(x_0, y_0)$

(c) The directional derivative tangent to a level curve is 0, since the change in value is 0.

From (b), tangents to any level curve at (x,y) are parallel to a line from $(0,0)$ to (x,y).

Looking at $\frac{x^2-y^2}{x^2+y^2} = c$, if $c \neq 0$, then $(0,0)$ is not part of level curve.

$\therefore$ Level curves are half lines from origin

Example: if $\frac{x^2-y^2}{x^2+y^2} = c$, then if (x_0, y_0) is

a point on the curve, so is (ax_0, ay_0), $a \neq 0$.

$\therefore$ The level curve for $f(x,y) = c$ is the line $a(x_0, y_0)$, a line through the origin but not containing the origin, so, 2 half-lines.

60.

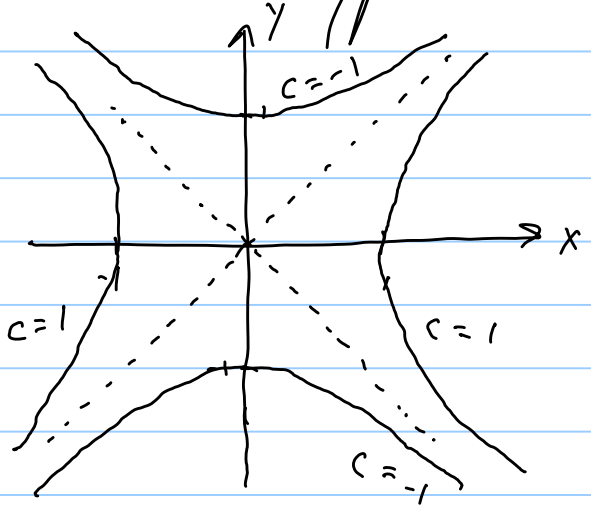
60. (a) Show that the curve $x^2 - y^2 = c$, for *any* value of c, satisfies the differential equation $dy/dx = x/y$.

(b) Draw in a few of the curves $x^2 - y^2 = c$, say for $c = \pm 1$. At several points (x, y) along each of these curves, draw a short segment of slope x/y; check that these segments appear to be tangent to the curve. What happens when $y = 0$? What happens when $c = 0$?

(a) $\frac{d}{dx}\left(x^2 - y(x)^2\right) = 2x - 2y(x)\cdot y' = \frac{d}{dx}(c) = 0$

$\therefore 2x = 2y(x)\cdot y'$, $y' = \frac{x}{y(x)}$, or $\frac{dy}{dx} = \frac{x}{y}$

(b) These are hyperbolas.



When $y=0$, slope is vertical line.

When $c=0$, level curves are $y=x$, $y=-x$, asymptotes to the hyperbolas.

61. **61.** Suppose that f is a differentiable function of one variable and that a function $u = g(x, y)$ is defined by

$$u = g(x, y) = xyf\left(\frac{x+y}{xy}\right).$$

Show that u satisfies a (partial) differential equation of the form

$$x^2\frac{\partial u}{\partial x} - y^2\frac{\partial u}{\partial y} = G(x, y)\,u$$

and find the function $G(x, y)$.

Let $h(x,y): \mathbb{R}^2 \to \mathbb{R}^1$ be $h(x,y) = \frac{x+y}{xy}$

$\therefore$ h is differentiable for $x \neq 0, y \neq 0$.

Let $j(x,y) = xy$, so j is differentiable.

$\therefore$ $u = g(x,y) = j(x,y) \cdot (f \circ h)(x,y)$

$\therefore$ $\frac{\partial u}{\partial x} = \frac{\partial j}{\partial x}(x,y) \cdot (f \circ h)(x,y) + j(x,y) \cdot \frac{\partial}{\partial x}(f \circ h)(x,y)$

$\frac{\partial j}{\partial x} = y$ $\quad \frac{\partial}{\partial x}(f \circ h)(x,y) = f'(h(x,y)) \cdot \frac{\partial h}{\partial x}(x,y)$

$= f'(h(x,y)) \cdot \frac{(xy) - y}{(xy)^2}$

$\therefore$ $\frac{\partial u}{\partial x} = y f\left(\frac{x+y}{xy}\right) + xy f'\left(\frac{x+y}{xy}\right)\left[\frac{x-1}{x^2y}\right]$

$= y f\left(\frac{x+y}{xy}\right) + \left(\frac{x-1}{x}\right) f'\left(\frac{x+y}{xy}\right)$

Similarly, $\frac{\partial u}{\partial y} = \frac{\partial}{\partial y} \cdot f\left(\frac{x+y}{xy}\right) + xy f'\left(\frac{x+y}{xy}\right) \cdot \frac{xy - x}{(xy)^2}$

$= x f\left(\frac{x+y}{xy}\right) + \left(\frac{y-1}{y}\right) f'\left(\frac{x+y}{xy}\right)$

$\therefore x^2 \frac{\partial u}{\partial x} = x^2 y f\left(\frac{x+y}{xy}\right) + x(x-1) f'\left(\frac{x+y}{xy}\right)$

$y^2 \frac{\partial u}{\partial y} = xy^2 f\left(\frac{x+y}{xy}\right) + y(y-1) f'\left(\frac{x+y}{xy}\right)$

$\therefore x^2 \frac{\partial u}{\partial x} - y^2 \frac{\partial u}{\partial y} = (x^2 y - xy^2) f\left(\frac{x+y}{xy}\right) + \left[x^2 - x - y^2 + y\right] f'\left(\frac{x+y}{xy}\right)$

$= (x-y)\, xy\, f\left(\frac{x+y}{xy}\right) + \left[x^2 - x - y^2 + y\right] f'\left(\frac{x+y}{xy}\right)$

$= (x-y)\, u + \left[x^2 - x - y^2 + y\right] f'\left(\frac{x+y}{xy}\right)$

$= u \left[(x-y) + \frac{\left[x^2 - x - y^2 + y\right] f'\left(\frac{x+y}{xy}\right)}{xy f\left(\frac{x+y}{xy}\right)} \right]$

$= u\, G(x, y)$

where $G(x,y) = \left[(x-y) + \frac{(x^2 - x - y^2 + y)}{xy} \frac{f'\left(\frac{x+y}{xy}\right)}{f\left(\frac{x+y}{xy}\right)} \right]$

62.

62. (a) Let F be a function of one variable and f a function of two variables. Show that the gradient vector of $g(x, y) = F(f(x, y))$ is parallel to the gradient vector of $f(x, y)$.

(b) Let $f(x, y)$ and $g(x, y)$ be functions such that $\nabla f = \lambda \nabla g$ for some function $\lambda(x, y)$. What is the relation between the level curves of f and g? Explain why there might be a function F such that $g(x, y) = F(f(x, y))$.

(a) $\dfrac{\partial g}{\partial x} = F'(f(x,y)) \cdot \dfrac{\partial f}{\partial x}$

$\dfrac{\partial g}{\partial y} = F'(f(x,y)) \cdot \dfrac{\partial f}{\partial y}$

$\therefore \nabla g = F'(f(x,y)) \left(\dfrac{\partial f}{\partial x}, \dfrac{\partial f}{\partial x} \right) = F'(f(x,y)) \nabla f$

$\therefore \nabla g$ is a multiple of ∇f, and so is parallel.

(b) The level curves have a similar shape, in that the tangents at any point (x,y), are parallel. (tangents of a level curve are where directional derivative is 0).

From (a), if $g(x,y) = F(f(x,y))$, then gradients are parallel, so directional derivatives will be parallel.

$\therefore$ Look for $\lambda(x,y) = \dfrac{1}{F'(f(x,y))}$

3.1 Iterated Partial Derivatives

Note Title 4/6/2016

In Exercises 1 to 6, compute the second partial derivatives $\partial^2 f/\partial x^2$, $\partial^2 f/\partial x\,\partial y$, $\partial^2 f/\partial y\,\partial x$, $\partial^2 f/\partial y^2$ for each of the following functions. Verify Theorem 1 in each case.

1.

1. $f(x, y) = 2xy/(x^2 + y^2)^2$, on the region where $(x, y) \neq (0, 0)$

$$f_x = \frac{(x^2+y^2)^2(2y) - 2(x^2+y^2)(2x)(2xy)}{(x^2+y^2)^4} = \frac{2y^3 - 6yx^2}{(x^2+y^2)^3}$$

$$f_y = \frac{(x^2+y^2)^2(2x) - 2(x^2+y^2)(2y)(2xy)}{(x^2+y^2)^4} = \frac{2x^3 - 6xy^2}{(x^2+y^2)^3}$$

$$f_{xx} = \frac{(x^2+y^2)^3(-12xy) - (2y^3 - 6yx^2)3(x^2+y^2)^2(2x)}{(x^2+y^2)^6}$$

$$= \frac{-12x^3y - 12xy^3 - 12xy^3 + 36yx^3}{(x^2+y^2)^4} = \frac{24(x^3y - xy^3)}{(x^2+y^2)^4}$$

$$f_{yy} = \frac{(x^2+y^2)^3(-12xy) - (2x^3 - 6xy^2)3(x^2+y^2)^2(2y)}{(x^2+y^2)^6} =$$

$$= \frac{-12x^3y - 12xy^3 - 12x^3y + 36xy^3}{(x^2+y^2)^4} = \frac{24(xy^3 - x^3y)}{(x^2+y^2)^4}$$

$$f_{xy} = \frac{(x^2+y^2)^3(6y^2 - 6x^2) - (2y^3 - 6yx^2)3(x^2+y^2)^2(2y)}{(x^2+y^2)^6} =$$

$$\frac{(x^2+y^2)6(y^2 - x^2) - 12y^2(y^2 - 3x^2)}{(x^2+y^2)^4} =$$

$$\frac{6y^4 - 6x^4 - 12y^4 + 36x^2y^2}{(x^2+y^2)^4} = \frac{36x^2y^2 - 6x^4 - 6y^4}{(x^2+y^2)^4}$$

$$f_{yx} = \frac{(x^2+y^2)^3(6x^2-6y^2) - (2x^3-6xy^2)3(x^2+y^2)^2(2x)}{(x^2+y^2)^6} =$$

$$\frac{(x^2+y^2)(6(x^2-y^2) - 12x^4 + 36x^2y^2}{(x^2+y^2)^4} = \frac{36x^2y^2 - 6x^4 - 6y^4}{(x^2+y^2)^4}$$

4. **4.** $f(x, y) = e^{-xy^2} + y^3x^4$

$f_x = -y^2e^{-xy^2} + 4x^3y^3$ $\quad f_y = -2xye^{-xy^2} + 3y^2x^4$

$f_{xx} = y^4e^{-xy^2} + 12x^2y^3$ $\quad f_{yy} = -2xe^{-xy^2} + 4x^2y^2e^{-xy^2} + 6yx^4$

$f_{xy} = -2ye^{-xy^2} + 2xy^3e^{-xy^2} + 12x^3y^2$

$f_{yx} = -2ye^{-xy^2} + 2xy^3e^{-xy^2} + 12y^2x^3$

$\therefore f_{xy} = f_{yx}$

7. **7.** Find all second partial derivatives of the following functions at the point $\mathbf{x}_0$.

(a) $f(x, y) = \sin(xy)$; $\mathbf{x}_0 = (\pi, 1)$

(b) $f(x, y) = xy^8 + x^2 + y^4$; $\mathbf{x}_0 = (2, -1)$

(c) $f(x, y, z) = e^{xyz}$; $\mathbf{x}_0 = (0, 0, 0)$

(a) $f_x = y\cos(xy)$ $\quad f_y = x\cos(xy)$

$f_{xx} = -y^2 \sin(xy)$ $\quad f_{xx}(\pi, 1) = \underline{0}$

$f_{yy} = -x^2 \sin(xy)$ $\quad f_{yy}(\pi, 1) = \underline{0}$

$f_{xy} = \cos(xy) - xy\sin(xy)$ $\quad f_{xy}(\pi, 1) = -1 - 0 = \underline{-1}$

$f_{yx} = \cos(xy) - xy\sin(xy)$ $\quad f_{yx}(\pi, 1) = -1 - 0 = \underline{-1}$

(b) $f_x = y^8 + 2x$ $\quad f_y = 8xy^7 + 4y^3$

$f_{xx} = 2$ $\quad f_{xx}(2, -1) = \underline{2}$

$f_{yy} = 56xy^6 + 12y^2$ $\quad f_{yy}(2, -1) = 112 + 12 = \underline{124}$

$f_{xy} = 8y^7$ $\quad f_{xy}(2, -1) = \underline{-8}$

$f_{yx} = 8y^7$ $\quad f_{yx}(2, -1) = \underline{-8}$

(c) $f_x = yz^2 e^{xyz^2}$ $\quad f_y = xz^2 e^{xyz^2}$ $\quad f_z = xye^{xyz^2}$

$f_{xx} = y^2 z^4 e^{xyz^2}$ $\quad f_{xx}(0, 0, 0) = \underline{0}$

$f_{yy} = x^2 z^4 e^{xyz^2}$ $\quad f_{yy}(0, 0, 0) = \underline{0}$

$f_{zz} = x^2 y^2 e^{xyz^2}$ $\quad f_{zz}(0, 0, 0) = \underline{0}$

$f_{xy} = f_{yx} = ze^{xyz^2} + xyz^3 e^{xyz^2}$ $\quad f_{xy}(0, 0, 0) = f_{yx}(0, 0, 0) = \underline{0}$

$f_{xz} = f_{zx} = ye^{xyz^2} + xy^2 z^2 e^{xyz^2}$ $\quad f_{xz}(0, 0, 0) = f_{zx}(0, 0, 0) = \underline{0}$

$f_{yz} = f_{zy} = xe^{xyz} + x^2yze^{xyz}$ $f_{xz}(0,0,0) = f_{zx}(0,0,0) = \underline{0}$

8. **8.** Find all second partial derivatives of $f(x, y) = \sec^3(4y - 3x)$.

$f_x = 3\sec^2(4y-3x)\left[\sec(4y-3x)\tan(4y-3x)\right](-3)$

$f_y = 3\sec^2(4y-3x)\left[\sec(4y-3x)\tan(4y-3x)\right](4)$

$f_{xx} = 54\sec(4y-3x)\left[\sec^2(4y-3x)\tan^2(4y-3x)\right]$

$\quad -9\sec^2(4y-3x)\left[-3\tan^3(4y-3x) - 3\sec^3(4y-3x)\right]$

$= 27\sec^2(4y-3x)\left[2\sec(4y-3x)\tan^2(4y-3x) + \tan^3(4y-3x) + \sec^3(4y-3x)\right]$

$f_{yy} = 96\sec(4y-3x)\left[\sec^2(4y-3x)\tan^2(4y-3x)\right]$

$\quad + 12\sec^2(4y-3x)\left[4\tan^3(4y-3x) + 4\sec^3(4y-3x)\right]$

$= 48\sec^2(4y-3x)\left[2\sec(4y-3x)\tan^2(4y-3x) + \tan^3(4y-3x) + \sec^3(4y-3x)\right]$

$f_{xy} = -72\sec(4y-3x)\left[\sec^2(4y-3x)\tan^2(4y-3x)\right]$

$\quad -9\sec^2(4y-3x)\left[4\tan^3(4y-3x) + 4\sec^3(4y-3x)\right]$

$= -36\sec^2(4y-3x)\left[2\sec(4y-3x)\tan^2(4y-3x) + \tan^3(4y-3x) + \sec^3(4y-3x)\right]$

f_{yx} should be same as f_{xy}

9. **9.** Can there exist a C^2 function $f(x, y)$ with $f_x = 2x - 5y$ and $f_y = 4x + y$?

Finding antiderivatives, from f_x, $f = x^2 - 5xy + C(y)$

from f_y, $f = 4xy + \frac{y^2}{2} + k(x)$

$C(y)$ could be $\frac{y^2}{2}$ and $k(x)$ could be x^2

But $-5xy \neq 4xy$, so can't make this work.

i.e., $f_{xy} = -5$, $f_{yx} = 4$, so $f_{xy} \neq f_{yx}$. $\therefore$ No

10. **10.** The heat conduction equation is $u_t = ku_{xx}$. Determine whether $u(x, t) = e^{-kt}\sin(x)$ is a solution.

$u_t = -ke^{-kt}\sin(x)$ $\quad u_x = e^{-kt}\cos x$ $\quad u_{xx} = -e^{-kt}\sin x$

$\therefore k u_{xx} = -ke^{-kt}\sin(x) = u_t$

$\therefore$ yes, $u(x,t) = e^{-kt}\sin(x)$ is a solution.

11. **11.** Show that the following functions satisfy the one-dimensional wave equation

$$\frac{\partial^2 f}{\partial x^2} = \frac{1}{c^2}\frac{\partial^2 f}{\partial t^2}.$$

(a) $f(x, t) = \sin(x - ct)$

(b) $f(x, t) = \sin(x)\sin(ct)$

(c) $f(x, t) = (x - ct)^6 + (x + ct)^6$

(a) $f_x = \cos(x-ct)$ $\quad f_{xx} = -\sin(x-ct)$

$f_t = (-c)\cos(x-ct)$ $\quad f_{tt} = -c^2\sin(x-ct)$

$$\therefore \frac{1}{c^2} f_{tt} = -\sin(x-ct) = f_{xx}$$

(b) $f_x = \cos(x)\sin(ct) \quad f_{xx} = -\sin(x)\sin(ct)$

$f_t = (c)\sin(x)\cos(ct) \quad f_{tt} = -c^2 \sin(x)\sin(ct)$

$$\therefore \frac{1}{c^2} f_{tt} = -\sin(x)\sin(ct) = f_{xx}$$

(c) $f_x = 6(x-ct)^5 + 6(x+ct)^5$

$f_{xx} = 30(x-ct)^4 + 30(x+ct)^4$

$f_t = 6(x-ct)^5(-c) + 6(x+ct)^5(c)$

$f_{tt} = 30c^2(x-ct)^4 + 30c^2(x+ct)^4$

$$\therefore \frac{1}{c^2} f_{tt} = 30(x-ct)^4 + 30(x+ct)^4 = f_{xx}$$

12. **12.** (a) Show that $T(x, t) = e^{-kt}\cos x$ satisfies the one-dimensional heat equation

$$k\frac{\partial^2 T}{\partial x^2} = \frac{\partial T}{\partial t}.$$

(a) $T_x = -e^{-kt}\sin(x) \quad T_{xx} = -e^{-kt}\cos(x)$

$T_t = -ke^{-kt}\cos(x) \quad \therefore KT_{xx} = -ke^{-kt}\cos(x) = T_t$

(b) Show that $T(x, y, t) = e^{-kt}(\cos x + \cos y)$ satisfies the two-dimensional heat equation

$$k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) = \frac{\partial T}{\partial t}.$$

$T_{xx} = -e^{-kt}\cos(x) \qquad T_{yy} = -e^{-kt}\cos(y)$

$T_t = -ke^{-kt}(\cos(x) + \cos(y)) = k(T_{xx} + T_{yy})$

(c) Show that $T(x, y, z, t) = e^{-kt}(\cos x + \cos y + \cos z)$ satisfies the three-dimensional heat equation

$$k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2}\right) = \frac{\partial T}{\partial t}.$$

$T_{xx} = -e^{-kt}\cos(x) \qquad T_{yy} = -e^{-kt}\cos(y) \qquad T_{zz} = -e^{-kt}\cos(z)$

$T_t = -ke^{-kt}(\cos(x) + \cos(y) + \cos(z)) = k(T_{xx} + T_{yy} + T_{zz})$

16. Let $z = x^4y^3 - x^8 + y^4$.

(a) Compute $\partial^3 z/\partial y\,\partial x\,\partial x$, $\partial^3 z/\partial x\,\partial y\,\partial x$, and $\partial^3 z/\partial x\,\partial x\,\partial y$ (also denoted $\partial^3 z/\partial x^2\partial y$).

(b) Compute $\partial^3 z/\partial x\,\partial y\,\partial y$, $\partial^3 z/\partial y\,\partial x\,\partial y$, and $\partial^3 z/\partial y\,\partial y\,\partial x$ (also denoted $\partial^3 z/\partial y^2\partial x$).

(a) $z_x = 4x^3y^3 - 8x^7 \qquad z_y = 3x^4y^2 + 4y^3$

$z_{xx} = 12x^2y^3 - 56x^6$

$z_{xy} = 12x^3y^2 \qquad z_{yx} = 12x^3y^2$

$z_{xyx} = 36x^2y^2 \qquad z_{yxx} = 36x^2y^2$

(b) Use z_x, z_y from (a)

$z_{xy} = 12x^3y^2 \qquad z_{xyy} = 24x^3y$

$z_{yx} = 12x^3y^2$ $\quad z_{yxy} = 24x^3y$

$z_{yy} = 6x^4y + 12y^2$ $\quad z_{yyx} = 24x^3y$

17. **17.** Use Theorem 1 to show that if $f(x, y, z)$ is of class C^3, then

$$\frac{\partial^3 f}{\partial x \partial y \partial z} = \frac{\partial^3 f}{\partial y \partial z \partial x}.$$

Using easier symbols, need to prove $f_{zyx} = f_{xzy}$

$f_{xz} = f_{zx} \Rightarrow (f_{xz})_y = (f_{zx})_y$, or $f_{xzy} = f_{zxy}$ [1]

$(f_z)_{xy} = (f_z)_{yx}$, so $f_{zxy} = f_{zyx}$ [2]

[1] & [2] $\Rightarrow$ $f_{xzy} = f_{zyx}$

$\therefore$ Using ∂ symbol, $\dfrac{\partial^3 f}{\partial y \partial z \partial x} = \dfrac{\partial^3 f}{\partial x \partial y \partial z}$

20. **20.** If $f(x, y, z, w)$ is of class C^3, show that $f_{xzw} = f_{zwx}$.

$f_{zx} = f_{xz} \Rightarrow f_{zxw} = f_{xzw}$ [1]

But $(f_z)_{xw} = (f_z)_{wx}$, or $f_{zxw} = f_{zwx}$ [2]

$\therefore$ [1] & [2] $\Rightarrow$ $f_{xzw} = f_{zwx}$

22. **22.** Let $w = f(x, y)$ be a function of two variables and let $x = u + v$, $y = u - v$. Show that

$$\frac{\partial^2 w}{\partial u\, \partial v} = \frac{\partial^2 w}{\partial x^2} - \frac{\partial^2 w}{\partial y^2}.$$

$w = f(u+v, u-v)$

$\therefore w_v = f_x x_v + f_y y_v = f_x(1) + f_y(-1) = f_x - f_y$

$w_{vu} = f_{xx} \cdot x_u + f_{xy} y_u - (f_{yx} \cdot x_u + f_{yy} \cdot y_u)$

$= f_{xx}(1) + f_{xy}(1) - (f_{yx}(1) + f_{yy}(1))$

$= f_{xx} + f_{xy} - f_{yx} - f_{yy}$

$= f_{xx} - f_{yy}$, since $f_{xy} = f_{yx}$

$= w_{xx} - w_{yy}$

This all assume $f(x,y)$ is C^2 on its domain.

23. **23.** Let $f: \mathbb{R}^2 \to \mathbb{R}$ be a C^2 function and let $\mathbf{c}(t)$ be a C^2 curve in $\mathbb{R}^2$. Write a formula for the second derivative $(d^2/dt^2)((f \circ \mathbf{c})(t))$ using the chain rule twice.

Let f be $f(x,y)$, $\vec{c}(t) = (x(t), y(t))$

$\therefore \frac{d}{dt}(f \circ \vec{c})(t) = \frac{d}{dt} f(x(t), y(t))$

$= f_x \cdot x'(t) + f_y \cdot y'(t)$

$= f_x(x(t), y(t)) \cdot x'(t) + f_y(x(t), y(t)) \cdot y'(t)$

$$\therefore \frac{d^2}{dt^2}(f\circ\vec{c})(t) = \left[f_{xx}\cdot x' + f_{xy}\cdot y'\right]\cdot x' + f_x\cdot x''$$
$$+ \left[f_{yx}\cdot x' + f_{yy}\cdot y'\right]\cdot y' + f_y\cdot y''$$
$$= (f_{xx})\left(\frac{dx}{dt}\right)^2 + 2(f_{xy})\left(\frac{dx}{dt}\right)\left(\frac{dy}{dt}\right) + (f_{yy})\left(\frac{dy}{dt}\right)^2$$
$$+ \nabla f\cdot\vec{c}''(t)$$

24.

24. Let $f(x, y, z) = e^{xz}\tan(yz)$ and let $x = g(s, t)$, $y = h(s, t)$, $z = k(s, t)$, and define the function $m(s, t) = f(g(s, t), h(s, t), k(s, t))$. Find a formula for m_{st} using the chain rule and verify that your answer is symmetric in s and t.

$$m_s = f_x\cdot g_s + f_y\cdot h_s + f_z\cdot k_s$$
$$\therefore m_{st} = \left[f_{xx}\cdot g_t + f_{xy}\cdot h_t + f_{xz}\cdot k_t\right]\cdot g_s + f_x\cdot g_{st}$$
$$+ \left[f_{yx}\cdot g_t + f_{yy}\cdot h_t + f_{yz}\cdot k_t\right]\cdot h_s + f_y\cdot h_{st}$$
$$+ \left[f_{zx}\cdot g_t + f_{zy}\cdot h_t + f_{zz}\cdot k_t\right]\cdot k_s + f_z\cdot k_{st}$$

Using $f_{xy} = f_{yx}$, $f_{xz} = f_{zx}$, $f_{xz} = f_{zx}$

$$m_{st} = f_{xx}\,g_t g_s + f_{yy}\,h_t h_s + f_{zz}\,k_t k_s +$$
$$f_{xy}(h_t g_s + g_t h_s) + f_{xz}(k_t g_s + g_t k_s) + f_{yz}(k_t h_s + h_t k_s)$$
$$+ f_x\,g_{st} + f_y\,h_{st} + f_z\,k_{st}$$

From commutivity ($g_t g_s = g_s g_t$, $h_t h_s = h_s h_t$, etc.), equality of mixed partials ($g_{st} = g_{ts}$, etc.),

The formula is symmetric in s and t (i.e., $m_{st} = m_{ts}$)

This all assumes g, h, k are class C^2.
f is C^2 from definition.

25.

25. A function $u = f(x, y)$ with continuous second partial derivatives satisfying Laplace's equation

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

is called a ***harmonic function***. Show that the function $u(x, y) = x^3 - 3xy^2$ is harmonic.

$u_x = 3x^2 - 3y^2 \quad u_{xx} = 6x$

$u_y = -6xy, \quad u_{yy} = -6x \quad 6x - 6x = 0, \therefore u_{xx} + u_{yy} = 0$

26.

26. Which of the following functions are harmonic? (See Exercise 25.)

(a) $f(x, y) = x^2 - y^2$
(b) $f(x, y) = x^2 + y^2$
(c) $f(x, y) = xy$
(d) $f(x, y) = y^3 + 3x^2 y$
(e) $f(x, y) = \sin x \cosh y$
(f) $f(x, y) = e^x \sin y$

(a) $f_{xx} = 2 \quad f_{yy} = -2, \; 2 + (-2) = 0 \quad \therefore$ harmonic

(b) $f_{xx} = 2 \quad f_{yy} = 2, \; 2 + 2 \neq 0 \quad \therefore$ not harmonic

(c) $f_{xx} = 0 \quad f_{yy} = 0 \quad \therefore$ harmonic

(d) $f_{xx} = 6y \quad f_{yy} = 6y, \; 6y + 6y \neq 0$ for $y \neq 0$. $\therefore$ not harmonic

(e) $f_{xx} = -\sin x \cosh y \quad f_{yy} = \sin x \cosh y$

$f_{xx} + f_{yy} = 0 \quad \therefore$ harmonic

(f) $f_{xx} = e^x \sin y \quad f_{yy} = -e^x \sin y$

$f_{xx} + f_{yy} = 0 \quad \therefore$ harmonic

27.

27. (a) Is the function $f(x, y, z) = x^2 - 2y^2 + z^2$ harmonic? What about $f(x, y, z) = x^2 + y^2 - z^2$?

(b) Laplace's equation for functions of n variables is

$$\frac{\partial^2 f}{\partial x_1^2} + \frac{\partial^2 f}{\partial x_2^2} + \cdots + \frac{\partial^2 f}{\partial x_n^2} = 0.$$

Find an example of a function of n variables that is harmonic, and show that your example is harmonic.

(a) $f_{xx} = 2 \quad f_{yy} = -4 \quad f_{zz} = 2. \quad \therefore f_{xx} + f_{yy} + f_{zz} = 0$

$\therefore x^2 - 2y^2 + z^2$ is harmonic

for $x^2 + y^2 - z^2$: $f_{xx} = 2, \ f_{yy} = 2, \ f_{zz} = -2.$

$\therefore$ not harmonic

(b) Examples: polynomials of degree 1 (or 0) will have all 2nd partials 0. $\therefore$ harmonic

Polynomials of degree 2: If n is even, and all coefficients are the same, and there are the same number of positive coefficients as negative coefficients, then harmonic.
for example: $2x^2 - 2y^2 + 2z^2 - 2w^2$

29.

29. Let f and g be C^2 functions of one variable. Set $\phi = f(x-t) + g(x+t)$.

(a) Prove that ϕ satisfies the wave equation: $\partial^2\phi/\partial t^2 = \partial^2\phi/\partial x^2$.

(b) Sketch the graph of ϕ against t and x if $f(x) = x^2$ and $g(x) = 0$.

(a) $\phi_x = f'(x-t)\cdot(1) + g'(x+t)\cdot(1) = f'(x-t) + g'(x+t)$

$\phi_{xx} = f''(x-t)\cdot(1) + g''(x+t)\cdot(1) = f''(x-t) + g''(x+t)$

$\phi_t = f'(x-t)\cdot(-1) + g'(x+t)\cdot(1) = -f'(x-t) + g'(x+t)$

$\phi_{tt} = -f''(x-t)\cdot(-1) + g''(x+t)\cdot(1) = f''(x-t) + g''(x+t)$

$\therefore \phi_{xx} = f''(x-t) + g''(x+t) = \phi_{tt}$

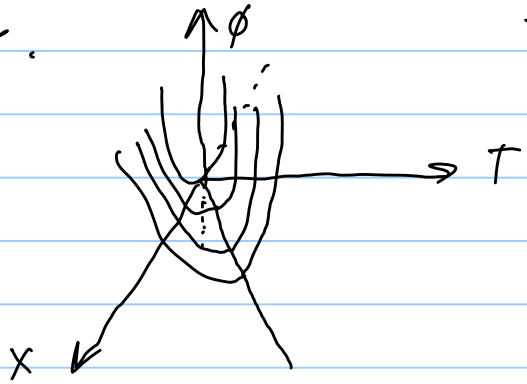
(b) $f(x-t) = (x-t)^2 \quad g(x+t) = 0$

$\therefore \phi(x,t) = (x-t)^2$

For $t = 0$, $\phi = x^2$,
$t = 1$, $\phi = (x-1)^2$
$x = t$, $\phi = 0$.

} parabolas

$\therefore$ a one-sheet paraboloid oriented along 45° axis of xt plane.



30.

30. (a) Show that function $g(x, t) = 2 + e^{-t} \sin x$ satisfies the heat equation: $g_t = g_{xx}$. [Here $g(x, t)$ represents the temperature in a metal rod at position x and time t.]

(b) Sketch the graph of g for $t \geq 0$. (HINT: Look at sections by the planes $t = 0$, $t = 1$, and $t = 2$.)

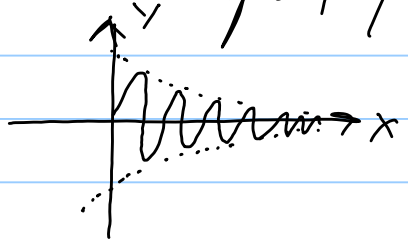
(c) What happens to $g(x, t)$ as $t \to \infty$? Interpret this limit in terms of the behavior of heat in the rod.

(a) $g_t = -e^{-t}\sin x$

$g_x = e^{-t}\cos x$, $g_{xx} = -e^{-t}\sin x$. $\therefore$ $g_t = g_{xx}$

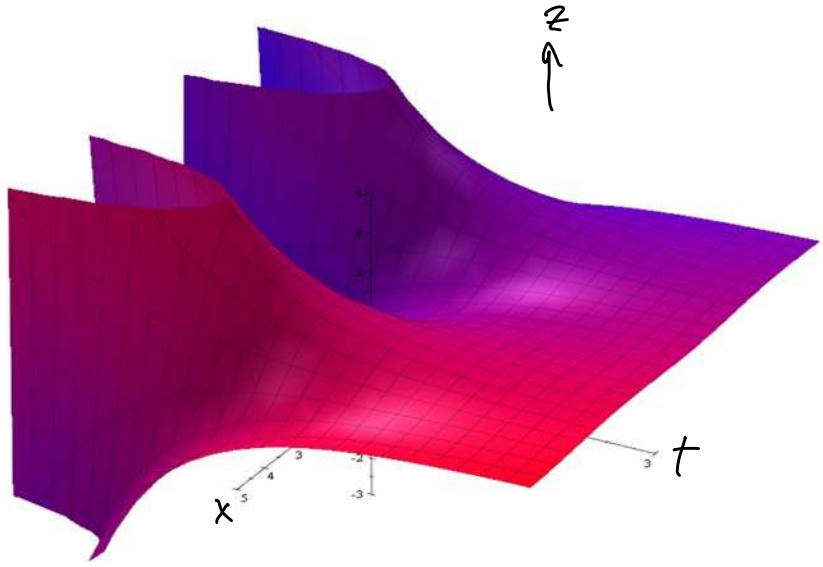
(b) e^{-t} acts as an "envelope" for $\sin x$

For example, $y = e^{-x}\sin x$ would look like:



If you just had $z = \sin x$, then you would have a one-sheeted sinusoid.

The e^{-t} term damps the oscillation as $t \to \infty$, and expands the oscillation as $t \to -\infty$.



(c) as $t \to \infty$, $g(x,t) \to 2$ as $e^{-t} \to 0$

As time goes on, the rod equilibrates, becomes uniform in temp.

31.

31. Show that Newton's potential $V = -GmM/r$ satisfies Laplace's equation

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2} = 0 \quad \text{for} \quad (x, y, z) \neq (0, 0, 0).$$

Let $r = \sqrt{x^2+y^2+z^2}$ $\therefore V = -GMm(x^2+y^2+z^2)^{-\frac{1}{2}}$

$\therefore V_x = \frac{1}{2} GMm(x^2+y^2+z^2)^{-\frac{3}{2}}(2x) = GMm\,x(x^2+y^2+z^2)^{-\frac{3}{2}}$

$V_{xx} = GMm(x^2+y^2+z^2)^{-\frac{3}{2}} + GMmx\left(-\frac{3}{2}\right)(x^2+y^2+z^2)^{-\frac{5}{2}}(2x)$

$= GMm(x^2+y^2+z^2)^{-\frac{3}{2}} - 3GMmx^2(x^2+y^2+z^2)^{-\frac{5}{2}}$

$= GMm\left(\frac{1}{r^3} - \frac{3x^2}{r^5}\right)$

Similarly, $V_{yy} = GMm\left(\frac{1}{r^3} - \frac{3y^2}{r^5}\right)$

$V_{zz} = GMm\left(\frac{1}{r^3} - \frac{3z^2}{r^5}\right)$

$\therefore V_{xx} + V_{yy} + V_{zz} = GMm\left(\frac{3}{r^3} - \frac{3(x^2+y^2+z^2)}{r^5}\right)$

$= GMm\left[\frac{3}{r^3} - \frac{3(x^2+y^2+z^2)}{r^3(x^2+y^2+z^2)}\right]$

using $r^2 = x^2+y^2+z^2$

$= GMm\left[\frac{3}{r^3} - \frac{3}{r^3}\right] = \underline{0}$

32.

32. Let

$$f(x, y) = \begin{cases} xy(x^2 - y^2)/(x^2 + y^2), & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

(see Figure 3.1.4).

(a) If $(x, y) \neq (0, 0)$, calculate $\partial f/\partial x$ and $\partial f/\partial y$.

(b) Show that $(\partial f/\partial x)(0, 0) = 0 = (\partial f/\partial y)(0, 0)$.

(c) Show that $(\partial^2 f/\partial x\, \partial y)(0, 0) = 1$, $(\partial^2 f/\partial y\, \partial x)(0, 0) = -1$.

(d) What went wrong? Why are the mixed partials not equal?

$$f(x,y) = \frac{xy(x^2-y^2)}{x^2+y^2}$$

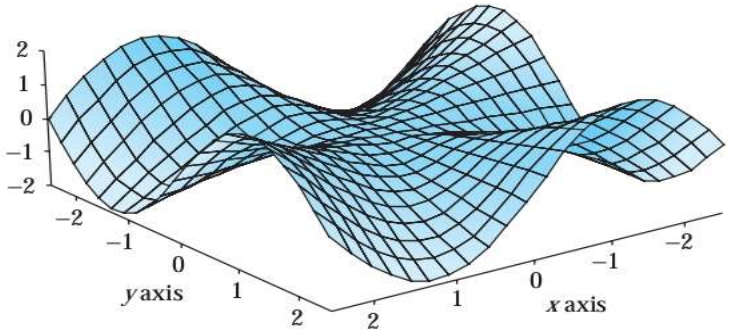


figure **3.1.4** The graph of the function in Exercise 32.

(a) $$f_x = \frac{(x^2+y^2)\left[y(x^2-y^2) + xy(2x)\right] - xy(x^2-y^2)[2x]}{(x^2+y^2)^2}$$

$$= \frac{y(x^4-y^4) + 2x^4y + 2x^2y^3 - 2x^4y + 2x^2y^3}{(x^2+y^2)^2}$$

$$= \frac{y(x^4-y^4) + 4x^2y^3}{(x^2+y^2)^2} = \frac{x^4y - y^5 + 4x^2y^3}{(x^2+y^2)^2}$$

$$f_y = \frac{(x^2+y^2)\left[x(x^2-y^2) + xy(-2y)\right] - xy(x^2-y^2)[2y]}{(x^2+y^2)^2}$$

$$= \frac{x(x^4-y^4) - 2x^3y^2 - 2xy^4 - 2x^3y^2 + 2xy^4}{(x^2+y^2)^2}$$

$$= \frac{x(x^4 - y^4) - 4x^3y^2}{(x^2+y^2)^2} = \frac{x^5 - xy^4 - 4x^3y^2}{(x^2+y^2)^2}$$

(b) Can't use formulas in (a), so try direct definition.

$$f_x(0,0) = \lim_{h\to 0} \frac{f(0+h,0) - f(0,0)}{h}$$

$$= \lim_{h\to 0} \frac{\dfrac{(0+h)(0)\left[(0+h)^2 - 0^2\right]}{\left[(0+h)^2 + 0^2\right]} - 0}{h} = \lim_{h\to 0} \frac{(h)(0)(h^2)}{h^3} = 0$$

$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,0+h) - f(0,0)}{h}$$

$$\lim_{h\to 0} \frac{\dfrac{0(0+h)\left[0^2 - (0+h)^2\right]}{\left[0^2 + (0+h)^2\right]} - 0}{h} = \lim_{h\to 0} \frac{0(h)(-h^2)}{h^3} = 0$$

(c) As in (b), use definition

$$f_{yx}(0,0) = \lim_{h\to 0} \frac{f_y(0+h,0) - f_y(0,0)}{h}$$

$$= \lim_{h\to 0} \frac{\dfrac{(0+h)^5 - (0+h)(0)^4 - 4(0+h)^3(0)^2}{\left[(0+h)^2 + 0^2\right]^2} - 0}{h}$$

using $f_y(0,0) = 0$ from (b)

$$= \lim_{h\to 0} \frac{h^5 - 0 - 0}{h^5} = \underline{1}$$

$$f_{xy}(0,0) = \lim_{h\to 0} \frac{f_x(0,0+h) - f_x(0,0)}{h}$$

$$= \lim_{h\to 0} \frac{\dfrac{0^4(0+h) - (0+h)^5 + 4(0)^2(0+h)^2}{[0^2+(0+h)^2]^2} - 0}{h}$$

using $f_x(0,0) = 0$ from (b)

$$= \lim_{h\to 0} \frac{0 - h^5 + 0}{h^5} = \underline{-1}$$

(d) $f(x,y)$ is not class C^2 since f_{xy} and f_{yx} are not continuous at $(0,0)$.

For $(x,y) \neq (0,0)$, $f_x(x,y) = \dfrac{x^4y - y^5 + 4x^2y^3}{(x^2+y^2)^2}$

$$\therefore f_{xy} = \frac{(x^2+y^2)^2\left[x^4 - 5y^4 + 12x^2y^2\right] - (x^4y - y^5 + 4x^2y^3)\left[2(x^2+y^2)(2y)\right]}{(x^2+y^2)^4}$$

From (c), $f_{xy}(0,0) = -1$, but along the $y=x$ path, $f_{xy}(x,x) = \dfrac{4x^4(x^4-5x^4+12x^4) - (x^5-x^5+4x^5)[8x^3]}{16x^8}$

$$= \frac{4x^8 - 20x^8 + 48x^8 - 32x^8}{16x^8} = 0$$

So can't define $f_{xy}(0,0)$ so that $\lim_{(\Delta x,\Delta y)\to(0,0)} f_{xy} = -1$

3.2 Taylor's Theorem

Note Title 4/26/2016

1.

1. Let $f(x, z) = e^{x+y}$.

(a) Find the first-order Taylor formula for f at (0, 0).

(b) Find the second-order Taylor formula for f at (0, 0).

(a) $f(0,0) = e^0 = 1$

$f_x = e^{x+y}$, $f_x(0,0) = 1$ $\quad f_y = e^{x+y}$, $f_y(0,0) = 1$

$\therefore \quad f(0,0) + f_x(0,0)(x-0) + f_y(0,0)(y-0)$

or, $f(x,y) = 1 + x + y + R_1((0,0),(x,y))$

(b) $f_{xx} = e^{x+y}$, $f_{xx}(0,0) = 1$ $\quad f_{yy} = e^{x+y}$, $f_{yy}(0,0) = 1$

$f_{xy} = e^{x+y}$, $f_{xy}(0,0) = 1$

$\therefore f(\vec{x_0} + \vec{h}) = f(\vec{x_0}) + \nabla f(x_0) \cdot \vec{h} + \frac{1}{2} \vec{h}^T H f(x_0) \vec{h}$

Here, $\vec{x_0} = (0,0)$, $\vec{h} = (h_1, h_2)$

$\therefore f(h_1, h_2) = 1 + (1,1) \cdot (h_1, h_2) + \frac{1}{2} [h_1, h_2] \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$

$= 1 + h_1 + h_2 + \frac{1}{2} h_1^2 + h_1 h_2 + \frac{1}{2} h_2^2 + R_2((0,0),(h_1,h_2))$

2.

2. Suppose $L: \mathbb{R}^2 \to \mathbb{R}$ is linear, so that L has the form $L(x, y) = ax + by$.

(a) Find the first-order Taylor approximation for L.

(b) Find the second-order Taylor approximation for L.

(c) What will higher-order approximations look like?

(a) Let $\vec{h} = (h_1, h_2)$, $\vec{x_0} = (x_0, y_0)$

$$\therefore L(\vec{x_0} + \vec{h}) = L(\vec{x_0}) + \nabla L(\vec{x_0}) \cdot \vec{h}$$

$$= ax_0 + by_0 + \begin{bmatrix} a & b \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$$

$$= ax_0 + by_0 + ah_1 + bh_2 + R_1(\vec{x_0}, \vec{h})$$

$$= a(x_0 + h_1) + b(y_0 + h_2) + R_1(\vec{x_0}, \vec{h})$$

$$R_1(\vec{x_0}, \vec{h}) = \sum_{i,j=1}^{2} \int_0^1 (t-1) f_{x_i x_j}(\vec{x_0} + t\vec{h}) h_i h_j \, dt$$

But $f_{x_i x_j} = 0 \quad \therefore R_1(\vec{x_0}, \vec{h}) = 0$

$$\therefore L(\vec{x_0} + \vec{h}) = ax_0 + by_0 + ah_1 + bh_2 = \underline{L(\vec{x_0} + \vec{h})}$$

(b) $L(\vec{x_0} + \vec{h}) = L(\vec{x_0}) + \nabla L(x_0) \cdot \vec{h} + \frac{1}{2} \vec{h}^T \cdot H L(x_0) \cdot \vec{h}$

Since $L_x = a$, $L_y = b$, $L_{xx} = 0$, $L_{yy} = 0$, $L_{xy} = 0$

$$\therefore H L(\vec{x_0}) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad \therefore \frac{1}{2} \vec{h}^T H L(\vec{x_0}) \vec{h} = 0$$

$$\therefore L(\vec{x_0} + \vec{h}) = ax_0 + by_0 + ah_1 + bh_2 + R_2(\vec{x_0}, \vec{h})$$

$$R_2(\vec{x_0}, \vec{h}) = 0 \text{ as } L_{xxx} = L_{yyy} = L_{xxy} = L_{xyy} = 0$$

$$\therefore L(\vec{x_0} + \vec{h}) = ax_0 + by_0 + ah_1 + bh_2 = \underline{L(\vec{x_0} + \vec{h})}$$

(c) Higher order approximations will look the same as first order = the linear function.

In each of Exercises 3 to 8, determine the second-order Taylor formula for the given function about the given point (x_0, y_0).

3. **3.** $f(x, y) = (x + y)^2$, where $x_0 = 0$, $y_0 = 0$

$f(0,0) = 0$ $\quad f_x = 2(x+y),\ f_x(0,0) = 0$

$f_y = 2(x+y),\ f_y(0,0) = 0$

$f_{xx} = 2 \quad f_{xy} = 2 \quad f_{yy} = 2$

$$\therefore f(h_1, h_2) = 0 + [0,0]\cdot\vec{h} + \frac{1}{2}\vec{h}^T \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \vec{h}$$

$$= \underline{h_1^2 + 2h_1h_2 + h_2^2}$$

$R_2(\vec{0}, \vec{h}) = 0$ as 3rd order partial derivatives $= 0$.

4. **4.** $f(x, y) = 1/(x^2 + y^2 + 1)$, where $x_0 = 0$, $y_0 = 0$

$f(0,0) = 1 \quad f_x = -\frac{1}{(x^2+y^2+1)^2}\cdot 2x = -\frac{2x}{(x^2+y^2+1)^2} \quad f_x(0,0) = 0$

$f_y = \frac{-2y}{(x^2+y^2+1)^2} \quad f_y(0,0) = 0$

$f_{xx} = \underline{\frac{(x^2+y^2+1)^2(-2) - (-2x)[2(x^2+y^2+1)2x]}{(x^2+y^2+1)^4}} \quad f_{xx}(0,0) = -2$

$f_{yy} = \underline{\frac{(x^2+y^2+1)^2(-2) - (-2y)[2(x^2+y^2+1)2y]}{(x^2+y^2+1)^4}} \quad f_{yy}(0,0) = -2$

$$f_{xy} = \frac{(x^2+y^2+1)^2(0) - (-2x)\left[2(x^2+y^2+1)\,2y\right]}{(x^2+y^2+1)^4} \qquad f_{xy}(0,0) = 0$$

$$\therefore f(h_1,h_2) = 1 + [0\ 0]\vec{h} + \frac{1}{2}\vec{h}^T\begin{bmatrix} -2 & 0 \\ 0 & -2 \end{bmatrix}\vec{h} + R_2(\vec{0},\vec{h})$$

$$= 1 + \frac{1}{2}(-2)h_1^2 + 0\cdot h_1 h_2 + \frac{1}{2}(-2)h_2^2 + R_2(\vec{0},\vec{h})$$

$$= \underline{1 - h_1^2 - h_2^2 + R_2(\vec{0},\vec{h})}$$

5. **5.** $f(x, y) = e^{x+y}$, where $x_0 = 0$, $y_0 = 0$

Same as problem #1

6. **6.** $f(x, y) = e^{-x^2-y^2}\cos(xy)$, where $x_0 = 0$, $y_0 = 0$

$$f(0,0) = e^0\cos(0) = 1$$

$$f_x = (-2x)e^{-x^2-y^2}\cos(xy) - ye^{-x^2-y^2}\sin(xy) \qquad f_x(0,0) = 0$$

$$f_y = (-2y)e^{-x^2-y^2}\cos(xy) - xe^{-x^2-y^2}\sin(xy) \qquad f_y(0,0) = 0$$

$$f_{xx} = (-2)e^{-x^2-y^2}\cos(xy) + (-2x)^2e^{-x^2-y^2}\cos(xy) + 2xye^{-x^2-y^2}\sin(xy)$$

$$+ 2xye^{-x^2-y^2}\sin(xy) - y^2e^{-x^2-y^2}\cos(xy) \qquad f_{xx}(0,0) = -2$$

$$f_{yy} = (-2)e^{-x^2-y^2}\cos(xy) + (-2y)^2e^{-x^2-y^2}\cos(xy) + 2xye^{-x^2-y^2}\sin(xy)$$

$$+ 2xye^{-x^2-y^2} - x^2e^{-x^2-y^2}\cos(xy) \qquad f_{yy}(0,0) = -2$$

$$f_{xy} = 4xye^{-x^2-y^2}\cos(xy) + 2x^2e^{-x^2-y^2}\sin(xy)$$
$$- e^{-x^2-y^2}\sin(xy) + 2y^2e^{-x^2-y^2}\sin(xy) - xye^{-x^2-y^2}\cos(xy)$$

$$f_{xy}(0,0) = 0$$

$$\therefore f(h_1,h_2) = 1 + 0 + \tfrac{1}{2}(-2)h_1^2 + 0\cdot h_1h_2 + \tfrac{1}{2}(-2)h_2^2 + R_2(\vec{0},\vec{h})$$
$$= \underline{1 - h_1^2 - h_2^2 + R_2(\vec{0},\vec{h})}$$

8. **8.** $f(x, y) = e^{(x-1)^2}\cos y$, where $x_0 = 1$, $y_0 = 0$

$f(1,0) = 1$ $\quad f_x = 2(x-1)e^{(x-1)^2}\cos y$ $\quad f_x(1,0) = 0$

$f_y = -e^{(x-1)^2}\sin y$ $\quad f_y(1,0) = 0$

$f_{xx} = 2e^{(x-1)^2}\cos y + 4(x-1)^2e^{(x-1)^2}\cos y$ $\quad f_{xx}(1,0) = 2$

$f_{yy} = -e^{(x-1)^2}\cos y$ $\quad f_{yy}(1,0) = -1$

$f_{xy} = -2(x-1)e^{(x-1)^2}\sin y$ $\quad f_{xy}(1,0) = 0$

$$\therefore f(h_1,h_2) = 1 + [0\ 0]\cdot\vec{h} + \tfrac{1}{2}(2)h_1^2 + 0\cdot h_1h_2 + \tfrac{1}{2}(-1)h_2^2 + R_2(\vec{0},\vec{h})$$
$$= \underline{1 + h_1^2 - h_2^2 + R_2(\vec{0},\vec{h})}$$

9. **9.** Calculate the second-order Taylor approximation to $f(x, y) = \cos x \sin y$ at the point $(\pi, \pi/2)$.

$f(\pi, \frac{\pi}{2}) = (-1)(1) = -1$ Let $\vec{h} = (\vec{x} - \vec{x_0}) = \begin{bmatrix} x - \pi \\ y - \frac{\pi}{2} \end{bmatrix}$

$f_x = -\sin x \sin y$ $f_x(\pi, \frac{\pi}{2}) = 0$

$f_y = \cos x \cos y$ $f_y(\pi, \frac{\pi}{2}) = 0$

$f_{xx} = -\cos x \sin y$ $f_{xx}(\pi, \frac{\pi}{2}) = 1$

$f_{yy} = -\cos x \sin y$ $f_{yy}(\pi, \frac{\pi}{2}) = 1$ $H = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$f_{xy} = -\sin x \cos y$ $f_{xy}(\pi, \frac{\pi}{2}) = 0$

$\therefore f(x,y) \approx f(\pi, \frac{\pi}{2}) + \nabla f(\pi, \frac{\pi}{2}) \cdot \vec{h} + \frac{1}{2} \vec{h}^T \cdot Hf(\pi, \frac{\pi}{2}) \cdot \vec{h}$

$-1 + 0 + \frac{1}{2}(1)(x - \pi)^2 + 0 + \frac{1}{2}(1)(y - \frac{\pi}{2})^2$

$= -1 + \frac{1}{2}(x - \pi)^2 + \frac{1}{2}(y - \frac{\pi}{2})^2$

10. **10.** Let $f(x, y) = x\cos(\pi y) - y\sin(\pi x)$. Find the second-order Taylor approximation for f at the point $(1, 2)$.

$f(1,2) = (1)(1) - 2(0) = 1$ Let $\vec{h} = \begin{bmatrix} x - 1 \\ y - 2 \end{bmatrix}$

$f_x = \cos(\pi y) - \pi y \cos(\pi x)$ $f_x(1,2) = 1 - 2\pi(-1) = 1 + 2\pi$

$f_y = -\pi x \sin(\pi y) - \sin(\pi x)$ $f_y(1,2) = 0$

$f_{xy} = -\pi \sin(\pi y) - \pi \cos(\pi x)$ $\quad f_{xy}(1,2) = \pi$

$f_{xx} = \pi^2 y \sin(\pi x)$ $\quad f_{xx}(1,2) = 0$

$f_{yy} = -\pi^2 x \cos(\pi y)$ $\quad f_{yy}(1,2) = -\pi^2$

$$\therefore f(x,y) \approx 1 + \nabla f(1,2) \cdot \vec{h} + \frac{1}{2} \vec{h}^T Hf(1,2) \vec{h}$$

$$= 1 + \begin{bmatrix} 1+2\pi & 0 \end{bmatrix} \begin{bmatrix} x-1 \\ y-2 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} x-1 & y-2 \end{bmatrix} \begin{bmatrix} 0 & \pi \\ \pi & -\pi^2 \end{bmatrix} \begin{bmatrix} x-1 \\ y-2 \end{bmatrix}$$

$$= 1 + (1+2\pi)(x-1) + \frac{1}{2}(0)(x-1)^2 + \pi(x-1)(y-2) + \frac{1}{2}(-\pi^2)(y-2)^2$$

$$= 1 + (1+2\pi)(x-1) + \pi(x-1)(y-2) - \frac{\pi^2}{2}(y-2)^2$$

11. **11.** Let $g(x, y) = \sin(xy) - 3x^2 \log y + 1$. Find the degree 2 polynomial which best approximates g near the point $(\pi/2, 1)$.

$g\left(\frac{\pi}{2}, 1\right) = 2$ $\quad g_x = y\cos(xy) - 6x \log y$ $\quad g_x\left(\frac{\pi}{2}, 1\right) = 0$

$g_y = x\cos(xy) - \frac{3x^2}{y}$ $\quad g_y\left(\frac{\pi}{2}, 1\right) = -\frac{3}{4}\pi^2$

$g_{xx} = -y^2 \sin(xy) - 6 \log y$ $\quad g_{xx}\left(\frac{\pi}{2}, 1\right) = -1$

$g_{yy} = -x^2 \sin(xy) + \frac{3x^2}{y^2}$ $\quad g_{yy}\left(\frac{\pi}{2}, 1\right) = -\frac{\pi^2}{4} + \frac{3}{4}\pi^2 = \frac{\pi^2}{2}$

$g_{xy} = \cos(xy) - xy\sin(xy) - \frac{6x}{y}$ $\quad g_{xy}\left(\frac{\pi}{2}, 1\right) = -\frac{\pi}{2} - 3\pi = -\frac{7}{2}\pi$

$$\therefore g(x,y) \approx g(\vec{x_0}) + \nabla g(\vec{x_0}) \cdot \vec{h} + \frac{1}{2}\vec{h}^T Hg(\vec{x_0})\vec{h}$$

$$= 2 + \begin{bmatrix} 0 & -\frac{3}{4}\pi^2 \end{bmatrix} \begin{bmatrix} x - \frac{\pi}{2} \\ y - 1 \end{bmatrix} + \frac{g_{xx}(\vec{x_0})}{2}\left(x - \frac{\pi}{2}\right)^2 + g_{xy}(\vec{x_0})\left(x - \frac{\pi}{2}\right)(y-1)$$

$$+ \frac{g_{yy}(\vec{x_0})}{2}(y-1)^2$$

$$= 2 - \frac{3}{4}\pi^2(y-1) - \frac{1}{2}\left(x - \frac{\pi}{2}\right)^2 - \frac{\pi}{2}\left(x - \frac{\pi}{2}\right)(y-1) + \frac{\pi^2}{4}(y-1)^2$$

12. **12.** For each of the functions in Exercises 3 to 7, use the second-order Taylor formula to approximate $f(-1, -1)$. Compare your approximation to the exact value using a calculator.

3. **3.** $f(x, y) = (x + y)^2$, where $x_0 = 0$, $y_0 = 0$

From #3, $f(h_1, h_2) = h_1^2 + 2h_1h_2 + h_2^2$

For $\vec{x} = (-1, -1)$, $h_1 = -1$, $h_2 = -1$.

$\therefore f(-1, -1) \approx (-1)^2 + 2(-1)(-1) + (-1)^2 = 4$

Actual: $f(x,y) = (x+y)^2$, $\therefore f(-1,-1) = (-2)^2 = 4$

4. **4.** $f(x, y) = 1/(x^2 + y^2 + 1)$, where $x_0 = 0$, $y_0 = 0$

From #4, $f(h_1, h_2) = 1 - h_1^2 - h_2^2$

For $\vec{h} = (-1, -1)$, $h_1 = -1$, $h_2 = -1$.

$\therefore f(-1, -1) \approx 1 - (-1)^2 - (-1)^2 = -1$ Actual: $f(-1,-1) = \frac{1}{3}$

5. **5.** $f(x, y) = e^{x+y}$, where $x_0 = 0$, $y_0 = 0$

From #1, $f(h_1, h_2) \approx 1 + h_1 + h_2 + \frac{1}{2}h_1^2 + h_1h_2 + \frac{1}{2}h_2^2$

$\therefore f(-1,-1) \approx 1 + (-1) + (-1) + \frac{1}{2}(-1)^2 + (-1)(-1) + \frac{1}{2}(-1)^2$

$= \frac{1}{2} + \frac{1}{2} = \underline{1}$

Actual: $e^{-1-1} = e^{-2} = 0.135$

6. **6.** $f(x, y) = e^{-x^2-y^2}\cos(xy)$, where $x_0 = 0$, $y_0 = 0$

From #6, $f(h_1, h_2) \approx 1 - h_1^2 - h_2^2$

$\therefore f(-1,-1) \approx 1 - (-1)^2 - (-1)^2 = \underline{-1}$

Actual $e^{-1-1}\cos(1) = e^{-2}\cos(1) = 0.073$

7. **7.** $f(x, y) = \sin(xy) + \cos(xy)$, where $x_0 = 0$, $y_0 = 0$

From answers in back of text, $f(h_1, h_2) \approx 1 + h_1 + h_2$

$\therefore f(-1,-1) \approx 1 + (-1) + (-1) = \underline{-1}$

Actual: $\sin(1) + \cos(1) = 1.38$

In all of these, the approximation is not close to actual since $\|\vec{h}\|$ is not small.

13.

13. (Challenging) A function $f: \mathbb{R} \to \mathbb{R}$ is called an ***analytic*** function provided

$$f(x+h) = f(x) + f'(x)\,h + \cdots + \frac{f^{(k)}(x)}{k!}h^k + \cdots$$

[i.e., the series on the right-hand side converges and equals $f(x+h)$].

(a) Suppose f satisfies the following condition: On any closed interval $[a, b]$, there is a constant M such that for all $k = 1, 2, 3, \ldots$, $|f^{(k)}(x)| \le M^k$ for all $x \in [a, b]$. Prove that f is analytic.

(b) Let $f(x) = \begin{cases} e^{-1/x} & x > 0 \\ 0 & x \le 0. \end{cases}$
Show that f is a C^∞ function, but f is not analytic.

(c) Give a definition of analytic functions from $\mathbb{R}^n$ to $\mathbb{R}$. Generalize the proof of part (a) to this class of functions.

(d) Develop $f(x, y) = e^{x+y}$ in a power series about $x_0 = 0$, $y_0 = 0$.

(a) Look at the remainder:

$$R_k(x_0, h) = \frac{1}{k!}\int_{x_0}^{x_0+h} (x_0+h-t)^k f^{(k+1)}(t)\,dt$$

Since $|f^{(k+1)}(t)| \le M^{k+1}$ for all $t \in [x_0, x_0+h]$

then $$|R_k(x_0,h)| = \left|\frac{1}{k!}\int_{x_0}^{x_0+h} (x_0+h-t)^k f^{(k+1)}(t)\,dt\right|$$

$$\le \left|\frac{1}{k!}\int_{x_0}^{x_0+h} (x_0+h-t)^k M^{k+1}\,dt\right|$$

$$= \frac{|M^{k+1}|}{k!}\left|\int_{x_0}^{x_0+h} (x_0+h-t)^k\,dt\right| = \frac{|M^{k+1}|}{k!}\left|-\frac{1}{k+1}(x_0+h-t)^{k+1}\Big|_{x_0}^{x_0+h}\right|$$

$$= \frac{|M^{k+1}|}{k!}\left|\frac{h^{k+1}}{k+1}\right| = \frac{|M^{k+1}|\,|h^{k+1}|}{(k+1)!}$$

$\therefore |R_k(x_0,h)| \le \dfrac{|M^{k+1}||h^{k+1}|}{(k+1)!}$

But $\lim_{k\to\infty} \dfrac{|M^k|}{k!} = 0$, so $\lim_{k\to\infty} \dfrac{|M^{k+1}||h^{k+1}|}{(k+1)!} = 0$

$\therefore \lim_{k\to\infty} |R_k(x_0,h)| = 0$, so $\lim_{k\to\infty} R_k(x_0,h) = 0$.

$\therefore$ The series converges.

(6) (b) Let $f(x) = \begin{cases} e^{-1/x} & x > 0 \\ 0 & x \le 0. \end{cases}$

Show that f is a C^∞ function, but f is not analytic.

$g(x) = -\frac{1}{x}$ is infinitely differentiable for $x > 0$.

$h(x) = e^x$ is infinitely differentiable.

$\therefore (h \circ g)(x)$ is infinitely differentiable for $x > 0$ by chain rule.

$f(x) = 0$ is infinitely differentiable for $x \le 0$.

$\therefore f(x)$ is C^∞ for all x.

Look at first Taylor expansion.

$f'(x) = \frac{1}{x^2} e^{-\frac{1}{x}}$. Can't compute $f'(0)$ using formula.

Try definition.

$$\lim_{x\to 0^+} \frac{f(x)-f(0)}{x} = \lim_{x\to 0^+} \frac{e^{-\frac{1}{x}}}{x} = \lim_{x\to 0^+} \frac{\frac{1}{x}}{e^{\frac{1}{x}}} =$$

$$\lim_{x\to 0^+} \frac{-\frac{1}{x^2}}{-\frac{1}{x^2}e^{\frac{1}{x}}} = \lim_{x\to 0^+} e^{-\frac{1}{x}} = 0.$$

$\therefore$ Can define $f'(0) = 0$.

Looking at $f'(x) = x^{-2}e^{-1/x}$, all higher derivatives will have $e^{-1/x}$ in every term, and every term will have a term of $c_n x^{-(n+1)}$:

$f^{(n)}(x) = p_n\left(\frac{1}{x}\right)e^{-\frac{1}{x}}$, where $p_n(x)$ is a polynomial in x, and so $p_n(1/x)$ is a polynomial in $1/x$.

Both $p_n\left(\frac{1}{x}\right)$ and $e^{-\frac{1}{x}}$ are C^∞, and so by the chain rule and product rule, $f(x)$ is C^∞.

To find Taylor series of $f(x)$, look at coefficients.

For $x \le 0$, $f'(0) = 0$, as $\lim_{h\to 0^-} \frac{f(0+h)-f(0)}{h} =$

$$\lim_{h\to 0^-} \frac{0-0}{h} = 0$$

For $x > 0$, from above, $f'(0) = 0$

For higher orders, using induction, assume $f^{(k)}(0) = 0$. Now look at

$$f^{(k+1)}(0) = \lim_{h\to 0^+} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} = \lim_{h\to 0^+} \frac{f^{(k)}(h)}{h}$$

$$= \lim_{h\to 0^+} \frac{p_k\left(\frac{1}{h}\right) e^{-\frac{1}{h}}}{h}$$. Substitute $x = \frac{1}{h}$, so this becomes $\lim_{x\to +\infty} \frac{x\, p_k(x)}{e^x}$. Using L'Hopital's rule a finite number of times, this becomes

$\lim_{x\to +\infty} \frac{c}{e^x} = 0$, c a constant, and actually the coefficient of highest order of $p_k(x)$.

$\therefore f^{(k+1)}(0) = 0$, so $f^{(n)}(0) = 0$ for all n.

$\therefore$ The Taylor series about 0, having terms $\frac{f^{(n)}(0)}{n!} x^n$ will have 0 for all coefficients.

$\therefore$ There is no radius of convergence, since for any $x > 0$, $e^{-\frac{1}{x}}$ is not 0.

$\therefore f(x)$ is C^∞ but not analytic since its Taylor series about 0 does not represent the function.

(c) (c) Give a definition of analytic functions from $\mathbb{R}^n$ to $\mathbb{R}$. Generalize the proof of part (a) to this class of functions.

If the Taylor expansion of $f: \mathbb{R}^n \to \mathbb{R}$ about some open interval around $\vec{x}_0 \in \mathbb{R}^n$ is equal to the value of the function in that open interval, then f is analytic on the open interval.

(d) (d) Develop $f(x, y) = e^{x+y}$ in a power series about $x_0 = 0, y_0 = 0$.

This was done in problem #1.

3.3 Extrema of Real-Valued Functions

Note Title 5/5/2016

In Exercises 1 to 16, find the critical points of the given function and then determine whether they are local maxima, local minima, or saddle points.

1. **1.** $f(x, y) = x^2 - y^2 + xy$

$f_x = 2x + y$
$f_y = x - 2y$
} critical point : $(0,0)$

$f_{xx}(0,0) = 2 \quad f_{yy}(0,0) = -2 \quad f_{xy}(0,0) = 1$

$\begin{vmatrix} 2 & 1 \\ 1 & -2 \end{vmatrix} = -5$ $\therefore$ Discriminant < 0

$\therefore (0,0)$ is a saddle point

2. **2.** $f(x, y) = x^2 + y^2 - xy$

$f_x = 2x - y$
$f_y = -x + 2y$
} Critical point : $(0,0)$

$f_{xx}(0,0) = 2 \quad f_{yy}(0,0) = 2 \quad f_{xy}(0,0) = -1$

$\therefore \begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} = 3$ $\quad f_{xx} > 0$ $\quad \therefore (0,0)$ a local minimum

3. **3.** $f(x, y) = x^2 + y^2 + 2xy$

$f_x = 2x + 2y$
$f_y = 2x + 2y$
} Critical points : $x = -y$

$f_{xx} = 2 \quad f_{yy} = 2 \quad f_{xy} = 2$

$\begin{vmatrix} 2 & 2 \\ 2 & 2 \end{vmatrix} = 0$ $\therefore$ Usual test for 2 variables fails.

$f(x,-x) = x^2 + (-x)^2 + 2x(-x) = 0$. So $f=0$ at critical pts.

$f(x,y)$ can be rewritten as $f(x,y) = (x+y)^2 \geq 0$.

$\therefore$ At critical points $x=-y$, $f(x,y)$ is an absolute minimum ($f(x,y)=0$), and $\therefore$ are also local minima.

4. **4.** $f(x, y) = x^2 + y^2 + 3xy$

$f_x = 2x + 3y$
$f_y = 3x + 2y$
} Critical point: $(0,0)$

$f_{xx}(0,0) = 2 \quad f_{yy}(0,0) = 2 \quad f_{xy}(0,0) = 3$

$\begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = -5 < 0$. $\therefore$ $(0,0)$ a saddle point

5. **5.** $f(x, y) = e^{1+x^2-y^2}$

$f_x = 2x e^{1+x^2-y^2}$
$f_y = -2y e^{1+x^2-y^2}$
Since $e^{1+x^2-y^2} > 0 \ \forall x,y$, critical point: $(0,0)$.

$f_{xx} = 2e^{1+x^2-y^2} + 4x^2 e^{1+x^2-y^2}$ $\quad f_{xx}(0,0) = 2e$
$f_{yy} = -2e^{1+x^2-y^2} + 4y^2 e^{1+x^2-y^2}$ $\quad f_{yy}(0,0) = -2e$
$f_{xy} = -4xy e^{1+x^2-y^2}$ $\quad f_{xy}(0,0) = 0$

$\begin{vmatrix} 2e & 0 \\ 0 & -2e \end{vmatrix} = -4e^2 < 0$. $\therefore$ $(0,0)$ a saddle point

6. **6.** $f(x, y) = x^2 - 3xy + 5x - 2y + 6y^2 + 8$

$$\left.\begin{aligned} f_x &= 2x - 3y + 5 = 0 \\ f_y &= -3x + 12y - 2 = 0 \end{aligned}\right\} \begin{aligned} &5x + 18 = 0,\ x = -\tfrac{18}{5} \\ &15y + 11 = 0,\ y = -\tfrac{11}{15} \end{aligned} \Big\} \text{ Critical point}$$

$$f_{xx} = 2 \quad f_{yy} = 12 \quad f_{xy} = -3 \quad \begin{vmatrix} 2 & -3 \\ -3 & 12 \end{vmatrix} = 15 > 0$$

$f_{xx} > 0$, $\therefore \left(-\frac{18}{5}, -\frac{11}{5}\right)$ a local minimum

7. **7.** $f(x, y) = 3x^2 + 2xy + 2x + y^2 + y + 4$

$$\left.\begin{aligned} f_x &= 6x + 2y + 2 = 0 \\ f_y &= 2x + 2y + 1 = 0 \end{aligned}\right\} \begin{aligned} &4x + 1 = 0,\ x = -\tfrac{1}{4} \\ &2(-\tfrac{1}{4}) + 2y + 1 = 0,\ y = -\tfrac{1}{4} \end{aligned} \Big\} \text{ critical point}$$

$$f_{xx} = 6 \quad f_{yy} = 2 \quad f_{xy} = 2 \quad \begin{vmatrix} 6 & 2 \\ 2 & 2 \end{vmatrix} = 8$$

$f_{xx} > 0$, $\therefore \left(-\frac{1}{4}, -\frac{1}{4}\right)$ a local minimum.

8. **8.** $f(x, y) = \sin(x^2 + y^2)$ [consider only the critical point $(0, 0)$]

$f_x = 2x \cos(x^2+y^2)$ Use $(0,0)$ as critical point
$f_y = 2y \cos(x^2+y^2)$

$$f_{xx} = 2\cos(x^2+y^2) - 4x^2 \sin(x^2+y^2) \qquad f_{xx}(0,0) = 2$$
$$f_{yy} = 2\cos(x^2+y^2) - 4y^2 \sin(x^2+y^2) \qquad f_{yy}(0,0) = 2$$
$$f_{xy} = -4xy \sin(x^2+y^2) \qquad f_{xy}(0,0) = 0$$

$\therefore \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4 > 0 \quad f_{xx}(0,0) > 0.$

$\therefore$ $(0,0)$ is a local minimum

9. **9.** $f(x, y) = \cos(x^2 + y^2)$ [consider only the three critical points $(0, 0)$, $(\sqrt{\pi/2}, \sqrt{\pi/2})$, and $(0, \sqrt{\pi})$]

$f_x = -2x \sin(x^2+y^2)$

$f_y = -2y \sin(x^2+y^2)$

$f_{xx} = -2\sin(x^2+y^2) - 4x^2\cos(x^2+y^2)$

$f_{yy} = -2\sin(x^2+y^2) - 4y^2\cos(x^2+y^2)$

$f_{xy} = -4xy\cos(x^2+y^2)$

$(0,0)$: $f_{xx}(0,0) = 0 \quad f_{yy}(0,0) = 0 \quad f_{xy}(0,0) = 0$

$\therefore$ Test fails

But since $f(0,0) = 1$, $(0,0)$ is a local maximum

$\left(\sqrt{\frac{\pi}{2}}, \sqrt{\frac{\pi}{2}}\right)$: $f_{xx} = 2\pi \quad f_{yy} = 2\pi \quad f_{xy} = 2\pi$

$\begin{vmatrix} 2\pi & 2\pi \\ 2\pi & 2\pi \end{vmatrix} = 0$ $\therefore$ Test fails.

But $f(\sqrt{\pi/2}, \sqrt{\pi/2}) = \cos\left(\frac{\pi}{2} + \frac{\pi}{2}\right) = -1.$

$\therefore$ $(\sqrt{\pi/2}, \sqrt{\pi/2})$ is a local minimum

$(0, \sqrt{\pi})$: $f_{xx} = 0 \quad f_{yy} = 4\pi \quad f_{xy} = 0$

$\begin{vmatrix} 0 & 0 \\ 0 & 4\pi \end{vmatrix} = 0$ $\therefore$ Test fails.

But $f(0, \sqrt{\pi}) = \cos(\pi) = -1.$

$\therefore$ $(0, \sqrt{\pi})$ is a local minimum

10. **10.** $f(x, y) = y + x \sin y$

$f_x = \sin y = 0 \Rightarrow y = n\pi \quad (n = 0, 1, 2, 3, \ldots)$

$f_y = 1 + x\cos y = 0 \quad \therefore$ use $y = n\pi \ (\pm 1, \pm 3, \pm 5, \ldots)$ $y = n\pi \ (0, \pm 2, \pm 4, \ldots)$

n odd: $1 - x = 0 \Rightarrow x = 1$

n even: $1 + x = 0 \Rightarrow x = -1$

$\therefore$ Critical points: $(1, n\pi)$ $n = \pm 1, \pm 3, \pm 5, \ldots$

$(-1, n\pi)$ $n = 0, \pm 2, \pm 4, \pm 6, \ldots$

$f_{xx} = 0 \quad f_{yy} = -x\sin y \quad f_{xy} = \cos y$

$\therefore (-1, 0): \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1. \quad \therefore (-1, n\pi)$, n even, a saddle point

$(1, \pi): \begin{vmatrix} 0 & -1 \\ -1 & 0 \end{vmatrix} = -1 \quad \therefore (1, n\pi)$, n odd, a saddle point

11. **11.** $f(x, y) = e^x \cos y$

$f_x = e^x \cos y \quad \} \quad f_x = 0 \Rightarrow \cos y = 0 \Rightarrow y = n\frac{\pi}{2}, \ n = \pm 1, \pm 3, \ldots$

$f_y = -e^x \sin y \quad \} \quad f_y = 0 \Rightarrow \sin y = 0 \Rightarrow y = n\pi, \ n = 0, \pm 1, \pm 2, \ldots$

$\therefore$ $\sin y$ & $\cos y$ can't both be 0 for any y.

$\therefore$ no critical points

12. **12.** $f(x, y) = (x - y)(xy - 1) = x^2y - x - xy^2 + y$

$f_x = 2xy - 1 - y^2$

$f_y = x^2 - 2xy + 1$

Add together: $f_x + f_y = 0 = x^2 - y^2 \therefore x = \pm y$

$y = x$: $f_x = 0 \Rightarrow 2x(x) - 1 - (x)^2 = x^2 - 1 = 0$, or $x = \pm 1$

$\therefore (1,1), (-1,-1)$

$y = -x$: $f_x = 0 \Rightarrow 2x(-x) - 1 - (-x)^2 = -3x^2 - 1 = 0$, $x^2 = -\frac{1}{3}$, $\therefore$ not real

$\therefore$ Critical points: $(1,1), (-1,-1)$

$f_{xx} = 2y \quad f_{yy} = -2x \quad f_{xy} = 2x - 2y$

$(1,1)$: $\begin{vmatrix} 2 & 0 \\ 0 & -2 \end{vmatrix} = -4 \quad \therefore$ $(1,1)$ a saddle point

$(-1,-1)$: $\begin{vmatrix} -2 & 0 \\ 0 & 2 \end{vmatrix} = -4 \quad \therefore$ $(-1,-1)$ a saddle point

13. **13.** $f(x, y) = xy + \frac{1}{x} + \frac{1}{y}$

$f_x = y - \frac{1}{x^2}$ $\quad f_x$ not defined at $x = 0$

$f_y = x - \frac{1}{y^2}$ $\quad f_y$ not defined at $y = 0$

But f also not defined at $x = 0$, or $y = 0$.

$\therefore f(x,y)$ is C^2 for $x \neq 0, y \neq 0$

$f_x = 0 \Rightarrow x^2 y = 1$

$f_y = 0 \Rightarrow xy^2 = 1$

$x^2 y = xy^2$, or $x = y$, $\therefore x = y = 1$

$\therefore (1,1)$ is only critical point.

$f_{xx} = 2x^{-3} \quad f_{yy} = 2y^{-3} \quad f_{xy} = 1$

$f_{xx}(1,1) = 2 \quad f_{yy}(1,1) = 2 \quad f_{xy}(1,1) = 1$

$\therefore \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 3 . \quad \therefore$ (1,1) is a local minimum.

14. **14.** $f(x, y) = \log(2 + \sin xy)$ Note $2 + \sin(x,y) \geq 1 \quad \forall x, y$

$f_x = \dfrac{y\cos(xy)}{2 + \sin(xy)} = 0 \Rightarrow y\cos(xy) = 0 \therefore y = 0$ or $\cos(xy) = 0$

$f_y = \dfrac{x\cos(xy)}{2 + \sin(xy)} = 0 \Rightarrow x\cos(xy) = 0 \therefore x = 0$ or $\cos(xy) = 0$

$\therefore y = 0$ or $xy = (2n+1)\dfrac{\pi}{2}$, n = any integer

$x = 0$ or $xy = (2n+1)\dfrac{\pi}{2}$, n = any integer

$\therefore (0,0), \left(x, \dfrac{(2n+1)\pi/2}{x}\right), \left(\dfrac{(2n+1)\pi/2}{y}, y\right)$ critical points.

$$f_{xx} = \frac{[2 + \sin(xy)][-y^2\sin(xy)] - y\cos(xy)[y\cos(xy)]}{[2 + \sin(xy)]^2}$$

$$= \frac{-y^2(2\sin(xy) + 1)}{[2 + \sin(xy)]^2}$$

$$f_{yy} = \frac{[2 + \sin(xy)][-x^2\sin(xy)] - x\cos(xy)[x\cos(xy)]}{[2 + \sin(xy)]^2}$$

$$= \frac{-x^2(2\sin(xy)+1)}{[2+\sin(xy)]^2}$$

$$f_{xy} = \frac{[2+\sin(xy)][\cos(xy) - xy\sin(xy)] - y\cos(xy)[x\cos(xy)]}{[2+\sin(xy)]^2}$$

$$= \frac{2\cos(xy) + \sin(xy)\cos(xy) - 2xy\sin(xy) - xy}{[2+\sin(xy)]^2}$$

$(0,0)$: $f_{xx} = 0 \quad f_{yy} = 0 \quad f_{xy} = \frac{2}{4} = \frac{1}{2}$

$\therefore \begin{vmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{vmatrix} = -\frac{1}{4}$. $\therefore$ $(0,0)$ a saddle point

$\left(x, \frac{(2n+1)\pi/2}{x}\right)$: i.e., $xy = (2n+1)\frac{\pi}{2}$ $\therefore \cos(xy) = 0 \quad \sin(xy) = \pm 1$

$$f_{xy} = 0 + 0 - (2n+1)\pi[\pm 1] - (2n+1)\frac{\pi}{2}$$

$$= -(2n+1)\pi - (2n+1)\frac{\pi}{2}, \quad (2n+1)\pi - (2n+1)\frac{\pi}{2}$$

$$= -\frac{3}{2}(2n+1)\pi, \quad \frac{(2n+1)\pi}{2}$$

$\therefore$ when $\sin(xy) = 1$, $f_{xy} = -\frac{3}{2}(2n+1)\pi$

$\sin(xy) = -1$, $f_{xy} = \frac{(2n+1)\pi}{2}$

f_{xx}: $\sin(xy) = 1$, $f_{xx} = -\frac{1}{3}y^2$, $y \neq 0$

$\sin(xy) = -1$, $f_{xx} = y^2$, $y \neq 0$

f_{yy} : $\sin(xy) = 1$, $f_{yy} = -\frac{1}{3}x^2$, $x \neq 0$

$\sin(xy) = -1$, $f_{yy} = x^2$, $x \neq 0$

$\therefore \sin(xy) = 1$, $\begin{vmatrix} -\frac{1}{3}y^2 & -\frac{3}{2}(2n+1)\pi \\ -\frac{3}{2}(2n+1)\pi & -\frac{1}{3}x^2 \end{vmatrix} = \frac{1}{9}x^2y^2 - \frac{9}{4}(2n+1)^2\pi^2$

$y = \frac{(2n+1)\frac{\pi}{2}}{x}$, $\therefore y^2 = \frac{(2n+1)^2\pi^2}{4x^2}$

$\therefore \frac{(2n+1)^2\pi}{36} - \frac{9}{9}\cdot\frac{9}{4}(2n+1)^2\pi^2 < 0$

$\therefore f_{xx} = -\frac{1}{3}y^2 < 0$, $f_{xx}f_{yy} - (f_{xy})^2 < 0$

$\therefore$ when $\sin(xy) = 1$, a local max

$\sin(xy) = -1$, $\begin{vmatrix} y^2 & \frac{(2n+1)\pi}{2} \\ \frac{(2n+1)\pi}{2} & x^2 \end{vmatrix} = x^2y^2 - \frac{(2n+1)^2\pi^2}{4}$

$y = \frac{(2n+1)\frac{\pi}{2}}{x}$, $\therefore y^2 = \frac{(2n+1)^2\pi^2}{4x^2}$

$\therefore x^2y^2 = \frac{(2n+1)^2\pi^2}{4}$. $\therefore$ det $= 0$

$\therefore$ test in this case fails.

But looking at $f(x,y) = \log(2 + \sin(xy))$,

when $\sin(xy) = -1$, $f(x,y)$ is a local min.

Summary : $(0,0)$ a saddle point

For $\sin(xy) = 1$ and $xy = \frac{(2n+1)\pi}{2}$, or

$xy = \frac{\pi}{2} + 2n\pi$, a local maximum

For $\sin(xy) = -1$ and $xy = \frac{(2n+1)\pi}{2}$, or

$xy = -\frac{\pi}{2} + 2n\pi$, a local minimum

15. **15.** $f(x, y) = x \sin y$

$f_x = \sin y \quad f_y = x\cos y$ when $\sin y = 0$, $\cos y \neq 0$

Critical points : $(0, n\pi)$

$f_{xx} = 0 \quad f_{yy} = -x\sin y \quad f_{xy} = \cos y$

$\begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1$ or $\begin{vmatrix} 0 & -1 \\ -1 & 0 \end{vmatrix} = -1$. $\therefore (0, n\pi)$ saddle points.

16. **16.** $f(x, y) = (x + y)(xy + 1) = x^2y + x + xy^2 + y$

$f_x = 2xy + 1 + y^2 \quad f_y = x^2 + 2xy + 1$

$f_x = 0$ and $f_y = 0$, so $f_x - f_y = 0 \Rightarrow x^2 - y^2 = 0$, $x = \pm y$

$\therefore$ $x=y$: $f_x = 3x^2 - 1 = 0$ no real solutions

$x=-y$: $f_x = -x^2 - 1 = 0$, $x = \pm 1$, so $y = \mp 1$

$\therefore$ Critical points: $(1,-1)$, $(-1,1)$.

$f_{xx} = 2y \quad f_{yy} = 2x \quad f_{xy} = 2x - 2y$

$\therefore$ at $(1,-1)$: $\begin{vmatrix} 2 & 0 \\ 0 & -2 \end{vmatrix} = -4$

at $(-1,1)$: $\begin{vmatrix} -2 & 0 \\ 0 & 2 \end{vmatrix} = -4$

$\therefore$ Both $(1,-1)$ and $(-1,1)$ are saddle points

17. **17.** Find all local extrema for $f(x, y) = 8y^3 + 12x^2 - 24xy$.

$f_x = 24x - 24y \quad f_y = 24y^2 - 24x$

$\therefore$ Critical points: $f_y = 0 \Rightarrow x = y^2$, $f_x = 0 \Rightarrow 24y^2 - 24y = 0$

$y = 0$ or $y = 1$

$\therefore$ $(0,0)$ or $(1,1)$

$f_{xx} = 24 \quad f_{yy} = 48y \quad f_{xy} = -24$

$\therefore$ at $(0,0)$ $\begin{vmatrix} 24 & -24 \\ -24 & 0 \end{vmatrix} = -24^2 < 0$ $\therefore$ $(0,0)$ a saddle point

at $(1,1)$ $\begin{vmatrix} 24 & -24 \\ -24 & 48 \end{vmatrix} = 24(48) - (24)^2 > 0$, and $f_{xx} > 0$

$\therefore$ $(1,1)$ a local minimum

18. **18.** Let $f(x, y, z) = x^2 + y^2 + z^2 + kyz$.

(a) Verify that $(0, 0, 0)$ is a critical point for f.

(b) Find all values of k such that f has a local minimum at $(0, 0, 0)$.

(a) $f_x = 2x \quad f_y = 2y + kz \quad f_z = 2z + ky$

$\therefore f_x(0,0,0) = 0, \; f_y(0,0,0) = 0, \; f_z(0,0,0) = 0$

(b) $f_{xx} = 2 \quad f_{xy} = 0 \quad f_{xz} = 0$

$f_{yy} = 2 \quad f_{yz} = k$

$f_{zz} = 2$

$\therefore$ Hessian matrix: $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & k \\ 0 & k & 2 \end{bmatrix}$

$|2| > 0, \; \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} > 0, \; \begin{vmatrix} 2 & 0 & 0 \\ 0 & 2 & k \\ 0 & k & 2 \end{vmatrix} = 2(4 - k^2) > 0$ if $4 > k^2$

$\therefore (0,0,0)$ a local minimum if $|k| < 2$

20. **20.** Suppose $(4, 2)$ is a critical point for the C^2 function $f(x, y)$. In each case, determine whether $(4, 2)$ is a local maximum, a local minimum, or a saddle point.

(a) $f_{xx}(4, 2) = 1, \quad f_{xy}(4, 2) = 3, \quad f_{yy} = 5$

(b) $f_{xx}(4, 2) = 2, \quad f_{yx}(4, 2) = -1, \quad f_{yy} = 4$

(c) $f_{xx}(4, 2) = -2, \quad f_{xy}(4, 2) = 1, \quad f_{yy} = 3$

(a) $\begin{vmatrix} 1 & 3 \\ 3 & 5 \end{vmatrix} = -1 < 0$. $\therefore$ a saddle point

(b) $\begin{vmatrix} 2 & -1 \\ -1 & 4 \end{vmatrix} = 7 > 0$ and $f_{xx} > 0$. $\therefore$ local minimum

(c) $\begin{vmatrix} -2 & 1 \\ 1 & 3 \end{vmatrix} = -7 < 0$ $\therefore$ a saddle point

22.

22. Let $f(x, y) = x^2 + y^2 + kxy$. If you imagine the graph changing as k increases, at what values of k does the shape of the graph change qualitatively?

$f_x = 2x + ky$ $f_y = 2y + kx$

$\therefore$ $\begin{matrix} 2x + ky = 0 \\ kx + 2y = 0 \end{matrix}$ $\Rightarrow$ $\begin{matrix} 2kx + k^2y = 0 \\ 2kx + 4y = 0 \end{matrix}$ $\Rightarrow (4 - k^2)y = 0$

$y = 0$ or $k = \pm 2$

$\therefore$ Critical points: $(0,0)$ for $k \neq \pm 2$, $y = \pm x$ for $k = \pm 2$

$f_{xx} = 2$ $f_{xy} = k$ $f_{yy} = 2$

$\begin{vmatrix} 2 & k \\ k & 2 \end{vmatrix} = 4 - k^2$

$4 - k^2 < 0 \Rightarrow |k| > 2$

$4 - k^2 = 0 \Rightarrow k = \pm 2$

$4 - k^2 > 0 \Rightarrow |k| < 2$

$\therefore$ For $|k| < 2$, $f_{xx} > 0$, so $(0,0)$ a local minimum.

For $|k| > 2$, $\begin{vmatrix} 2 & k \\ k & 2 \end{vmatrix} < 0$, so $(0,0)$ a saddle point.

For $K=2$, $f(x,y)=(x+y)^2$, so $y=-x$ local minima

$K=-2$, $f(x,y)=(x-y)^2$, $y=x$ local minima.

23.

23. An examination of the function $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto (y - 3x^2)(y - x^2)$ will give an idea of the difficulty of finding conditions that guarantee that a critical point is a relative extremum when Theorem 6 fails.[8] Show that

(a) the origin is a critical point of f;

(b) f has a relative minimum at $(0, 0)$ on every straight line through $(0, 0)$; that is, if $g(t) = (at, bt)$, then

$f \circ g: \mathbb{R} \to \mathbb{R}$ has a relative minimum at 0, for every choice of a and b;

(c) the origin is not a relative minimum of f.

(a) $f(x,y) = y^2 - 3x^2y - x^2y + 3x^4 = 3x^4 - 4x^2y + y^2$

$\therefore f_x = -6xy - 2xy + 12x^3 = 12x^3 - 8xy$

$f_y = 2y - 3x^2 - x^2 = 2y - 4x^2$

$(0,0) \Rightarrow f_x = 0, f_y = 0.$

For other points: $12x^3 - 8xy = 0 \Rightarrow y = \frac{3}{2}x^2, x \neq 0$

substituting into $f_y = 0$, $3x^2 - 4x^2 = 0$ for $x \neq 0$.

$\therefore (0,0)$ is the only critical point.

(b) $f_{xx} = 36x^2 - 8y \quad f_{xy} = -8x \quad f_{yy} = 2$

$\therefore$ at $(0,0)$, $\begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} = 0$, $\therefore$ test fails.

Look at $h(t) = (f \circ g)(t) = 3(at)^4 - 4(at)^2(bt) + (bt)^2$

$= 3a^4t^4 - 4a^2bt^3 + b^2t^2$

$\therefore h'(t) = 12a^4t^3 - 12a^2bt^2 + 2b^2t$

$= 2t\left[6a^4t^2 - 6a^2bt + b^2\right]$

$\therefore t = 0$ is the critical point corresponding to $(x,y) = (0,0)$.

$h''(t) = 36a^4t^2 - 24a^2bt + 2b^2$

$\therefore h''(0) = 2b^2 > 0$, for $b \neq 0$.

$\therefore t = 0$ is a local minimum $\forall a$ if $b \neq 0$.

For $b = 0$, $h(t) = 3at^4$, so $t = 0$ is a local minimum.

$\therefore (0,0)$ is a local minimum along (at, bt) $\forall a, b$.

(c) To show $(0,0)$ not a local minimum for $f(x,y)$, need to show $f(x,y) < 0$ for some (x,y) close to $(0,0)$.

Consider points along curve $y = 2x^2$ as $(x,y) \to (0,0)$

$\therefore (y - 3x^2) < 0$ and $y - x^2 > 0$.

$\therefore f(x,y) = (y-3x^2)(y-x^2) < 0$

24. **24.** Let $f(x, y) = Ax^2 + E$, where A and E are constants. What are the critical points of f? Are they local maxima or local minima?

$f_x = 2Ax \quad f_y = 0 \quad \therefore (0, y)$ are all critical points

$f_{xx} = 2A \quad f_{xy} = 0 \quad f_{yy} = 0 \quad \therefore f_{xx} f_{yy} - f_{xy}^2 = 0$, test fails.

$f(x,y)$ is a parabolic cylinder

For $A > 0$, $(0, y)$ are all local minima

For $A < 0$, $(0, y)$ are all local maxima

25. **25.** Let $f(x, y) = x^2 - 2xy + y^2$. Here $D = 0$. Can you say whether the critical points are local minima, local maxima, or saddle points?

$f_x = 2x - 2y \quad f_y = 2y - 2x \quad$ Critical points: $x = y$

$f_{xx} = 2 \quad f_{xy} = -2 \quad f_{yy} = 2 \quad \begin{vmatrix} 2 & -2 \\ -2 & 2 \end{vmatrix} = 0.$

But $f(x,y) = (x-y)^2 \geq 0 \ \forall x, y$

$\therefore$ Points along $y = x$ are all local minima.

27. **27.** Suppose $f: \mathbb{R}^3 \to \mathbb{R}$ is C^2, and that $\mathbf{x}_0$ is a critical point for f. Suppose $Hf(\mathbf{x}_0)(\mathbf{h}) = h_1^2 + h_2^2 + h_3^2 + 4h_2h_3$. Does f have a local maximum, minimum, or saddle at $\mathbf{x}_0$?

for $\vec{h} = (0, 1, -1)$, $Hf(\vec{x_0})\vec{h} = 2 - 4 = -2 < 0$

for $\vec{h} = (1,0,0)$, $H f(\vec{x}_0)\vec{h} = 1 > 0$.

$\therefore$ Not pos-def, not neg-def.

Now try to construct Hessian matrix.

$$[h_1\ h_2\ h_3]\begin{bmatrix} f_{xx} & f_{xy} & f_{xz} \\ f_{yx} & f_{yy} & f_{yz} \\ f_{zx} & f_{zy} & f_{zz} \end{bmatrix}\begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix} = h_1^2 + h_2^2 + h_3^2 + 4h_2h_3$$

With $\vec{h} = (1,0,0)$, $f_{xx}(\vec{x}_0) = 1$ $\therefore$ $\begin{bmatrix} 1 & f_{xy} & f_{xz} \\ f_{xy} & 1 & f_{yz} \\ f_{xz} & f_{yz} & 1 \end{bmatrix}$

$\vec{h} = (0,1,0)$, $f_{yy}(\vec{x}_0) = 1$

$\vec{h} = (0,0,1)$, $f_{zz}(\vec{x}_0) = 1$

$\vec{h} = (1,1,0)$, $\begin{bmatrix} 1 & 1 & 0 \end{bmatrix}\begin{bmatrix} 1+f_{xy} \\ f_{xy}+1 \\ f_{xz}+f_{yz} \end{bmatrix} = 2$, $2 + 2f_{xy} = 2$,

$\therefore f_{xy} = 0$

$\vec{h} = (1,0,1)$, $\begin{bmatrix} 1 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1+f_{xz} \\ f_{yz} \\ f_{xz}+1 \end{bmatrix} = 2$, $2 + 2f_{xz} = 2$,

$f_{xz} = 0$

So far, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & f_{yz} \\ 0 & f_{yz} & 1 \end{bmatrix}$

$\vec{h} = (0,1,1)$, $\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} 0 \\ 1+f_{yz} \\ f_{yz}+1 \end{bmatrix} = 6$, $2 + 2f_{yz} = 6$

$\therefore f_{yz} = 2$

$\therefore Hf(\vec{x}_0) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix}$, $\det Hf(\vec{x}_0) = -3$

$\therefore$ Not pos-def, not neg-def, and $\det \neq 0 \Rightarrow$ $\vec{x}_0$ is a saddle point.

28. **28.** Find the point on the plane $2x - y + 2z = 20$ nearest the origin.

$2z = 20 - 2x + y$

Each point on plane is of: $(x, y, 10 - x + \frac{1}{2}y)$

$\therefore$ Distance from $(0,0,0) = d = \sqrt{x^2 + y^2 + (10 - x + \frac{1}{2}y)^2}$

Or, $d^2 = f(x,y) = x^2 + y^2 + (10 - x + \frac{1}{2}y)^2$

$\therefore f_x = 2x + 2(10 - x + \frac{1}{2}y)(-1) = 2x - 20 + 2x - y = 4x - y - 20$

$f_y = 2y + 2(10 - x + \frac{1}{2}y)(\frac{1}{2}) = \frac{5}{2}y - x + 10$

Critical point: $f_x + 4f_y = 9y + 20 = 0$, $y = \frac{20}{9}$

$\therefore 4x = 20 + \frac{20}{9}$, $36x = 200$, $9x = 50$, $x = \frac{50}{9}$

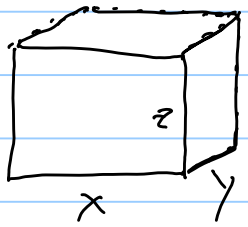
$\therefore z = 10 - \frac{50}{9} + \frac{1}{2}(\frac{20}{9}) = 10 - \frac{40}{9} = \frac{50}{9}$

$\therefore \left(\frac{50}{9}, \frac{20}{9}, \frac{50}{9}\right)$

29. **29.** Show that a rectangular box of given volume has minimum surface area when the box is a cube.

Let V = volume, let sides be of length x, y, z

$\therefore V = xyz, \quad z = \frac{V}{xy}$



Surface area $= 2xy + 2yz + 2xz$

$\therefore S(x,y) = 2xy + 2y\left(\frac{V}{xy}\right) + 2x\left(\frac{V}{xy}\right)$

$= 2xy + \frac{2V}{x} + \frac{2V}{y}$

$\therefore S_x = 2y - \frac{2V}{x^2} \qquad S_y = 2x - \frac{2V}{y^2}$

$\therefore 2yx^2 - 2V = 0 \qquad 2xy^2 - 2V = 0$

$\therefore yx^2 = xy^2$, or $x = y$.

$\therefore$ Critical point : $y = x$

$S_{xx} = 4Vx^{-3} \qquad S_{yy} = 4Vy^{-3} \qquad S_{xy} = 2$

At $y = x$, $\begin{vmatrix} 4Vx^{-3} & 2 \\ 2 & 4Vx^{-3} \end{vmatrix} = 16V^2x^{-6} > 0$, for all $x > 0$.

And $S_{xx} > 0 \quad \forall x > 0$.

$\therefore$ At $y=x$, $S(x,y)$ is a local minimum.

Using $S_x = 2yx^2 - 2V = 0$, and $y=x$,

you get $x^3 = V$, so $x = y = V^{\frac{1}{3}}$

But $V = xyz$, so $V = (V^{\frac{1}{3}})(V^{\frac{1}{3}})z$, $\Rightarrow z = V^{\frac{1}{3}}$

$\therefore x = y = z$, and box is a cube.

30. **30.** Show that the rectangular parallelepiped with fixed surface area and maximum volume is a cube.

Let $S = 2xy + 2yz + 2xz$ = surface area of box

$= 2xy + z(2y + 2x)$

$\therefore z = \dfrac{S - 2xy}{2x + 2y}$, and $2x + 2y \neq 0$ [0]

$\therefore V(x,y) = xy\left(\dfrac{S - 2xy}{2x + 2y}\right) = xyz$

$\therefore V_x = y\left(\dfrac{S - 2xy}{2x + 2y}\right) + xy\left[\dfrac{(2x+2y)(-2y) - (S-2xy)(2)}{(2x+2y)^2}\right]$

$V_y = x\left(\dfrac{S - 2xy}{2x + 2y}\right) + xy\left[\dfrac{(2x+2y)(-2x) - (S-2xy)(2)}{(2x+2y)^2}\right]$

$\therefore V_x = \dfrac{y(2x+2y)(S-2xy) - 2xy^2(2x+2y) - 2xy(S-2xy)}{(2x+2y)^2} = 0$

or, $(2x+2y)(5-2xy) - 2xy(2x+2y) - 2x(5-2xy) = 0$ [1]

$V_y = x(2x+2y)(5-2xy) - 2x^2y(2x+2y) - 2xy(5-2xy) = 0$

or, $(2x+2y)(5-2xy) - 2xy(2x+2y) - 2y(5-2xy) = 0$ [2]

Subtracting [1] - [2] gives:

$-2x(5-2xy) + 2y(5-2xy) = 0$, so $y = x$

This is the only critical point.

It must be a local max (without using 2nd derivative test) as a "box" with top and bottom of surface area $\frac{1}{2}S$ and height near 0 has a volume $(\frac{1}{2}S)(h) \approx 0$.

Substituting $y = x$ into [1] to solve for S gives:

$4x(5-2x^2) - 2x^2(4x) - 2x(5-2x^2) = 0$,

or, $(5-2x^2)(2x) = 8x^3$, or $5 - 2x^2 = 4x^2 \Rightarrow S = 6x^2$

Now substitute $y = x$ and $S = 6x^2$ into [0] to solve for z:

$$z = \frac{5-2xy}{2x+2y} = \frac{6x^2-2x^2}{4x} = \frac{4x^2}{4x} = x$$

$\therefore y = x = z$ $\therefore$ A cube

31. **31.** Write the number 120 as a sum of three numbers so that the sum of the products taken two at a time is a maximum.

$x + y + z = 120, \quad z = 120 - x - y$

$P = xy + xz + yz = xy + x(120 - x - y) + y(120 - x - y)$

$\therefore P(x,y) = xy + 120x - x^2 - xy + 120y - xy - y^2$

$= 120x + 120y - x^2 - y^2 - xy$

$\therefore P_x = 120 - 2x - y \quad P_y = 120 - 2y - x$

$P_x - P_y = -2x + 2y - y + x = 0 \Rightarrow x = y$

$\therefore P_x = 120 - 3x = 0 \Rightarrow x = 40, \; y = 40$

$\therefore (40,40)$ a critical point

$P_{xx} = -2 \quad P_{yy} = -2 \quad P_{xy} = -1 \quad \begin{vmatrix} -2 & -1 \\ -1 & -2 \end{vmatrix} = 3 > 0.$

Since $P_{xx}(40,40) = -2 < 0$, Then $(40,40)$ a local max.

From $x + y + z = 120, \; z = 40.$

$\therefore \underline{x = y = z = 40}$

32.

32. Show that if (x_0, y_0) is a critical point of a quadratic function $f(x, y)$ and $D < 0$, then there are points (x, y) near (x_0, y_0) at which $f(x, y) > f(x_0, y_0)$ and, similarly, points for which $f(x, y) < f(x_0, y_0)$.

Let $\vec{x_0} = (x_0, y_0)$ Let $a = f_{xx}(\vec{x_0})$, $b = f_{xy}(\vec{x_0})$, $c = f_{yy}(\vec{x_0})$

$D = ac - b^2 < 0$. Let $\vec{x} = (x, y)$, $\vec{h} = \vec{x} - \vec{x_0}$, $\vec{x}$ near $\vec{x_0}$.

Let $H(\vec{h}) = \frac{1}{2}ah_1^2 + bh_1h_2 + \frac{1}{2}ch_2^2$, $\vec{h} = (h_1, h_2)$

(i) if $a = 0$, $H(\vec{h}) = h_2(bh_1 + \frac{1}{2}ch_2)$

$\therefore$ if $h_2 = 0$ and $h_1 \neq 0$, $\vec{h}$ is still $\neq \vec{0}$ ($\vec{x}$ near $\vec{x_0}$)

Then $H(\vec{h}) = 0$, so $H(\vec{h})$ is not pos-def. and not neg-def., as $\vec{h} \neq \vec{0}$.

(ii) if $a \neq 0$, Then fix h_2 and solve for h_1 ($\vec{x}$ near $\vec{x_0}$)

$$ah_1^2 + 2bh_2h_1 + ch_2^2 = 0$$

$$h_1 = \frac{-2bh_2 \pm \sqrt{4b^2h_2^2 - 4ach_2^2}}{2a}$$

$$= \frac{bh_2 \pm |h_2|\sqrt{b^2 - ac}}{a}$$

Since $D = ac - b^2 < 0$, Then $b^2 - ac > 0$, so for $h_2 \neq 0$, there is at least one solution for h_1. $\therefore \vec{h} \neq 0$ and $H(\vec{h}) = 0$.

$\therefore H(\vec{h})$ is not pos-def. and not neg-def.

$\therefore$ $\vec{x}$ near $\vec{x}_0$ and $D<0$ means $H(\vec{h})$ is not pos-def and not neg-def.

(iii) Not pos-def. $\Rightarrow$ $H(\vec{h}) \leq 0$ for all $\vec{h}$ near $\vec{0}$.

Suppose $H(\vec{h}) = 0$ $\forall \vec{h}$ near $\vec{0}$.

$\therefore \begin{bmatrix} h_1 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} h_1 \\ 0 \end{bmatrix} = a h_1^2 = 0 \Rightarrow a = 0,$ $h_1 \neq 0$

$\begin{bmatrix} 0 & h_2 \end{bmatrix} \begin{bmatrix} 0 & b \\ b & c \end{bmatrix} \begin{bmatrix} 0 \\ h_2 \end{bmatrix} = c h_2^2 = 0 \Rightarrow c = 0$ $h_2 \neq 0$.

$\therefore \begin{bmatrix} h_1 & h_2 \end{bmatrix} \begin{bmatrix} 0 & b \\ b & 0 \end{bmatrix} \begin{bmatrix} h_1 \\ h_2 \end{bmatrix} = 2 b h_1 h_2 = 0 \Rightarrow b = 0$ for $h_1, h_2 \neq 0$

$\therefore ac - b^2 = 0$, contradicting $D = ac - b^2 < 0$.

$\therefore$ There is some $\vec{h}$ near $\vec{0}$ ($\vec{x}$ near $\vec{x}_0$) s.t.

$H(\vec{h}) < 0$

(iv) Not neg-def. $\Rightarrow$ $H(\vec{h}) \geq 0$ for all $\vec{h}$ near $\vec{0}$.

Suppose $H(\vec{h}) = 0$ $\forall \vec{h}$ near $\vec{0}$.

As in (iii), using $\begin{bmatrix} h_1 \\ 0 \end{bmatrix}$, $\begin{bmatrix} 0 \\ h_2 \end{bmatrix}$, then $\begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$,

This implies $a = 0$, $b = 0$, $c = 0$, so

$ac - b^2 = 0$ contradicting $D = ac - b^2 < 0$

$\therefore H(\vec{h})$ can't be 0 $\forall \vec{h}$ near $\vec{0}$, so

there is some $\vec{h}$ near $\vec{0}$ ($\vec{x}$ near $\vec{x_0}$) s.t.

$H(\vec{h}) > 0$.

$\therefore$ For (x,y) near (x_0,y_0), $D<0 \Rightarrow H(\vec{x}-\vec{x_0})$ is

not pos-def. and not neg.-def. $\Rightarrow$ from (iii)+(iv)

that there is an (x,y) near (x_0,y_0) s.t. $f(x,y) < f(x_0,y_0)$

and another (x,y) near (x_0,y_0) s.t. $f(x,y) > f(x_0,y_0)$.

33.

33. Let $f(x, y) = x^6 + x^2 + y^6$, $g(x, y) = -x^6 - x^2 - y^6$, $h(x, y) = x^6 - x^4 + y^6$.

(a) Show that (0, 0) is a degenerate critical point for f, g, and h.

(b) Show that (0, 0) is a local minimum for f, a local maximum for g, and a saddle for h.

(a) $f_x = 6x^5 + 2x$ $\quad f_{yy} = 6y^5$ $\quad \therefore (0,0)$ is a critical point

$f_{xx} = 30x^4 + 2$ $\quad f_{xy} = 0$ $\quad f_{yy} = 30y^4$

$\therefore \begin{vmatrix} 2 & 0 \\ 0 & 0 \end{vmatrix} = 0$ $\quad$ degenerate

$g_x = -6x^5 - 2x$ $\quad g_y = -6y^5$ $\quad \therefore (0,0)$ a critical point

$g_{xx} = -30x^4 - 2$ $\quad g_{xy} = 0$ $\quad g_{yy} = -30y^4$

$\therefore \begin{vmatrix} -2 & 0 \\ 0 & 0 \end{vmatrix} = 0$ degenerate

$h_x = 6x^5 - 4x^3 \quad h_y = 6y^5 \quad \therefore (0,0)$ a critical point

$h_{xx} = 30x^4 - 12x^2 \quad h_{xy} = 0 \quad h_{yy} = 30y^4$

$\therefore \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} = 0$ degenerate

(6) $f(0,0) = 0$, $f(x,y) \geq 0 \ \forall (x,y)$. as exponents are even

$\therefore f(0,0)$ is a local minimum

$g(0,0) = 0$, $g(x,y) \leq 0 \ \forall (x,y)$ as exponents are even

$\therefore g(0,0)$ is a local maximum.

h: for $0 < |x| < 1$, $x^6 < x^4$ so $x^6 - x^4 < 0$.

$\therefore$ along $(x,0)$ $h(x,y) < 0$.

along $(0,y)$, $h(x,y) > 0$ as $h(0,y) = y^6$

$\therefore$ near $(0,0)$, a saddle point

Not a good example because $a = f_{xx}(0,0)$ is different in each case!

34. **34.** Let $f(x, y) = 5ye^x - e^{5x} - y^5$.

(a) Show that f has a unique critical point and that this point is a local maximum for f.

(b) Show that f is unbounded on the y axis, and thus has no global maximum. [Note that for a function $g(x)$ of a single variable, a unique critical point which is a local extremum is necessarily a global extremum. This example shows that this is not the case for functions of several variables.]

(a) $f_x = 5ye^x - 5e^{5x}$ $\quad f_y = 5e^x - 5y^4$

$f_x = 0 \Rightarrow y = e^{4x}$ $\quad f_y = 0 \Rightarrow 5e^x = 5(e^{4x})^4$, $e^x = e^{16x}$,

$e^{15x} = 1$, $x = 0$.

$\therefore y = e^0 = 1$.

$\therefore (0,1)$ is a unique critical point as it is the only solution to $f_x = 0$, $f_y = 0$

$f_{xx} = 5ye^x - 25e^{5x}$ $\quad f_{xy} = 5e^x$ $\quad f_{yy} = -20y^3$

$\therefore f_{xx}(0,1) = -20$ $\quad f_{xy}(0,1) = 5$ $\quad f_{yy}(0,1) = -20$

$\therefore \begin{vmatrix} -20 & 5 \\ 5 & -20 \end{vmatrix} > 0.$

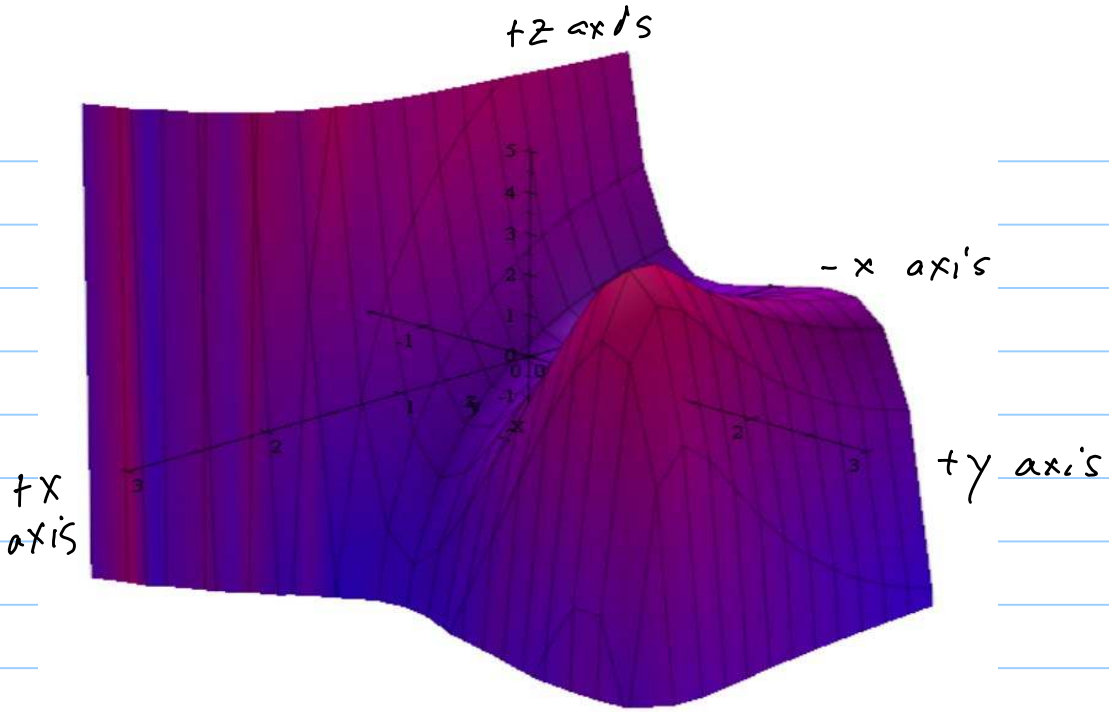
$\therefore$ Since $f_{xx}(0,1) < 0$ and $f_{xx}f_{yy} - f_{xy}^2\big|_{(0,1)} > 0$,

the critical point $(0,1)$ is a local maximum.

(b) for $x = 0$, $f(0,y) = 5y - 1 - y^5$

$\therefore$ as $y \to -\infty$, $-y^5 \to +\infty$, and $\therefore 5y - 1 - y^5 \to +\infty$

A plot is shown on next page



To go from local max to area of global max, the function dips in a saddle fashion to avoid another local extremum.

38. **38.** Find the absolute maximum and minimum values of the function $f(x, y) = (x^2 + y^2)^4$ on the disc $x^2 + y^2 \leq 1$. (You do not *have* to use calculus.)

$f(x,y)$ acts like a paraboloid, so $(0,0)$ is a global minimum and $f(0,0) = 0$

The global max will be on the edge of the disc, since $x^2+y^2 < 1$ $\forall (x,y)$ inside disc.
$\therefore$ global max is $f(x,y) = 1$ for all $x^2+y^2 = 1$.

39. **39.** Repeat Exercise 38 for the function $f(x, y) = x^2 + xy + y^2$.

$f_x = 2x + y \quad f_y = x + 2y \quad f_x - 2f_y = -3y = 0, \; y = 0.$
$\therefore x = 0.$

$\therefore (0,0)$ is a critical point.

$f_{xx} = 2 \quad f_{xy} = 1 \quad f_{yy} = 2. \quad \begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 3 > 0.$

as $f_{xx} > 0$, $(0,0)$ is a local min.

Now look at $f(x,y)$ on edge of disc: $x^2+y^2=1$.

$\therefore f(x,y) = 1 + xy$. Using $y = \pm\sqrt{1-x^2}$,

$g(x) = 1 + x\sqrt{1-x^2} \qquad h(x) = 1 - x\sqrt{1-x^2}$

$g'(x) = \sqrt{1-x^2} + \frac{1}{2}x(1-x^2)^{-\frac{1}{2}}(-2x)$

$= \sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}} = 0, \quad \therefore 1-x^2 = x^2, \quad x = \pm\frac{\sqrt{2}}{2}$

$h'(x) = -\sqrt{1-x^2} - \frac{1}{2}x(1-x^2)^{-\frac{1}{2}}(-2x)$

$= -\sqrt{1-x^2} + \frac{x^2}{\sqrt{1-x^2}} = 0, \quad \therefore x^2 = 1-x^2, \quad x = \pm\frac{\sqrt{2}}{2}$

$\therefore$ Critical points $\left(\frac{\sqrt{2}}{2}, \pm\frac{\sqrt{2}}{2}\right), \left(-\frac{\sqrt{2}}{2}, \pm\frac{\sqrt{2}}{2}\right)$

$\therefore$ On $x^2+y^2=1$, $f(x,y) = 1+xy$, and at critical points, $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ and $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ is a local max, and $1+xy = 1+\frac{1}{2} = \frac{3}{2}$

$\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ and $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ a local min and $1+xy = 1-\frac{1}{2} = \frac{1}{2}$.

$\therefore (0,0)$ a global minimum, $\underline{\underline{f(x,y) = 0}}$

$\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ and $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ global maxima,

and $f(x,y) = \frac{3}{2}$

40.

40. A curve C in space is defined *implicitly* on the cylinder $x^2 + y^2 = 1$ by the additional equation $x^2 - xy + y^2 - z^2 = 1$. Find the point or points on C closest to the origin.

Any point, (x,y,z), on the cylinder will be of distance $d = \sqrt{x^2+y^2+z^2}$ from the origin, or $d^2 = x^2+y^2+z^2$. Since $x^2+y^2=1$ on cylinder, then $d^2 = 1+z^2$. This is clearly a minimum when $z=0$. $\therefore$ Only points under consideration are $(x,y,0)$, where $x^2+y^2=1$, $x^2-xy+y^2-z^2=1$

$\therefore x^2-xy+y^2-z^2=1$ becomes $-xy=0$.

$\therefore x=0$ or $y=0$ (not both, since $x^2+y^2=1$).

$\therefore x=0: y^2=1 \Rightarrow y=\pm 1$

$y=0: x^2=1 \rightarrow x=\pm 1$.

$\therefore$ 4 points: $(1,0,0), (-1,0,0), (0,1,0), (0,-1,0)$.

All are of distance 1 from origin.

41. **41.** Find the absolute maximum and minimum values for $f(x, y) = \sin x + \cos y$ on the rectangle R defined by $0 \le x \le 2\pi, 0 \le y \le 2\pi$.

$\frac{3}{2}\pi, \pi$

The obvious max is $1+1=2$, min is $-1-1=-2$.
Use calculus to corroborate this.

$f_x = \cos x \quad f_y = -\sin y \quad f_x = 0$ for $x = \frac{\pi}{2}, \frac{3}{2}\pi$
$f_y = 0$ for $y = \pi$

$f(\frac{\pi}{2}, \pi) = 0 \quad f(\frac{3\pi}{2}, \pi) = -2$

(a) X-axis's boundary: $0 \le x \le 2\pi, y = 0$

$\therefore f(x, 0) = g(x) = \sin x + 1$

$g'(x) = \cos x, \; g'(x) = 0$ at $x = \frac{\pi}{2}, \frac{3}{2}\pi$

$g''(x) = -\sin x, \; g''(\frac{\pi}{2}) = 0, \; g''(\frac{3}{2}\pi) = 0,$

so, 2nd derivative test fails at critical pts.

$\therefore g(0) = 1, \; g(\frac{\pi}{2}) = 1+1 = 2, \; g(\frac{3}{2}\pi) = 0, \; g(2\pi) = 1$

$\therefore$ So far, $f(\frac{\pi}{2}, 0) = 2, \quad f(\frac{3}{2}\pi, \pi) = -2$

(b) y-axis's boundary: $x = 0, \; 0 \le y \le 2\pi$

$\therefore f(0, y) = g(y) = \cos y$

$g'(y) = -\sin y = 0$ at $y = 0, \pi, 2\pi$

$g''(y) = -\cos y \quad g''(0) = -1,\ g''(\pi) = 1,\ g''(2\pi) = -1$

$\therefore g''(\pi) > 0$, so $g(y)$ a local min.

$f(0,0) = 1,\ f(0,\pi) = -1,\ f(0,2\pi) = 1$

$\therefore$ So far, $f(\frac{\pi}{2}, 0) = 2,\ f(\frac{3}{2}\pi, \pi) = -2$

(c) $x = 2\pi,\ 0 \le y \le 2\pi$

$\therefore f(2\pi, y) = g(y) = \cos y$

This is exactly the same as (b)

(d) $0 \le x \le 2\pi,\ y = 2\pi$

$\therefore f(x, 2\pi) = \sin x + 1$

This is identical to (a)

$\therefore$ Global max at $(\frac{\pi}{2}, 0)$, $f(x,y) = 2$ (on boundary)

Global min at $(\frac{3}{2}\pi, \pi)$, $f(x,y) = -2$ (on interior)

42. **42.** Find the absolute maximum and minimum values for the function $f(x, y) = xy$ on the rectangle R defined by $-1 \le x \le 1, -1 \le y \le 1$.

$f_x = y \quad f_y = x \quad$ Critical point: $(0,0)$

$f(0,0) = 0.$

Boundary $x = -1,\ -1 \le y \le 1$ $\therefore f(x,y) = f(-1,y) = -y.$

$\therefore$ Min: -1 at $(-1,1)$
Max: 1 at $(-1,-1)$

Boundary $x = 1,\ -1 \le y \le 1$ $\therefore f(x,y) = f(1,y) = y$

$\therefore$ Min: -1 at $(1,-1)$
Max: 1 at $(1,1)$

Boundary $-1 \le x \le 1,\ y = -1$ $\therefore f(x,y) = f(x,-1) = -x$

$\therefore$ Min: -1 at $(1,-1)$
Max: 1 at $(-1,-1)$

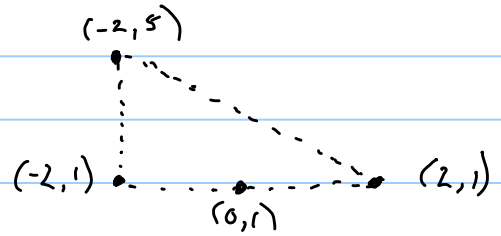
Boundary $-1 \le x \le 1,\ y = 1$ $\therefore f(x,y) = f(x,1) = x$

$\therefore$ Min: -1 at $(-1,1)$
Max: 1 at $(1,1)$

$\therefore$ Global max $= 1$ at $(1,1)$ or $(-1,-1)$
Global min $= -1$ at $(-1,1)$ or $(1,-1)$

43.

43. Let $f(x, y) = 1 + xy - 2x + y$ and let D be the triangular region in $\mathbb{R}^2$ with vertices $(-2, 1)$, $(-2, 5)$, and $(2, 1)$. Find the absolute maximum and minimum values of f on D. Give all points where these extreme values occur.



$f_x = y - 2 \quad f_y = x + 1$ $\therefore$ Critical point: $(-1, 2)$

Line connecting $(-2,5)$ and $(2,1)$: $\frac{1-5}{2-(-2)} = \frac{-4}{4} = -1$.

$\therefore y = -x + 3$

For $x = -1$ of critical point, $y = 4$, and $1 < 2 < 4$, so critical point "inside" triangle

$\underline{f(-1,2) = 1 + (-1)(2) - 2(-1) + (2) = 3}$

Endpoints: $f(-2,5) = 1 + (-2)(5) - 2(-2) + 5 = 6$

$f(2,1) = 1 + (2)(1) - 2(2) + 1 = 0$

$f(-2,1) = 1 + (-2)(1) - 2(-2) + 1 = 4$

Boundary $y = -x + 3$ for $-2 < x < 2$

$\therefore f(x, -x+3) = 1 + x(-x+3) - 2x + (-x+3) = -x^2 + 4$

for $g(x) = -x^2 + 4$, $g'(x) = -2x$,

$\therefore x = 0$ a critical point. $\therefore y = 3$

$\underline{f(0,3) = 4}$

Boundary $x = -2$, $1 < y < 5$

$\therefore f(-2,y) = 1 - 2y + 4 + y = 5 - y$

for $g(y) = 5 - y$, $g'(y) = -1$, so no critical points

Boundary $y = 1$, $-2 < x < 2$

$\therefore f(x,1) = 1 + x - 2x + 1 = 2 - x$

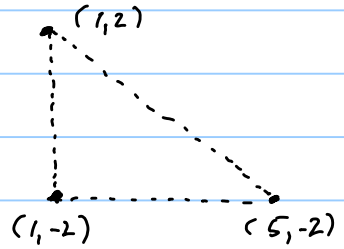
for $g(x) = 2 - x$, $g'(x) = -1$, so no critical pts.

$\therefore$ Global min = 0 at $(-2,5)$ and $(2,1)$

Global max = 4 at $(-2,1)$ and $(0,3)$

44.

44. Let $f(x, y) = 1 + xy + x - 2y$ and let D be the triangular region in $\mathbb{R}^2$ with vertices $(1, -2)$, $(5, -2)$, and $(1, 2)$. Find the absolute maximum and minimum values of f on D. Give all points where these extreme values occur.



$f_x = y + 1$ $\quad f_y = x - 2$ $\quad \therefore$ Critical point: $(2,-1)$

Line connecting $(1,2)$ and $(5,-2)$: $\frac{-2-2}{5-1} = \frac{-4}{4} = -1$

$\therefore y = -x + 3$

Since $1 < 2 < 5$ for $(2,-1)$

and $y = -(2) + 3 = 1$, so $-2 < -1 < 1$ for $(2,-1)$

Then critical point $(2,-1)$ is "inside" triangle.

$\therefore f(2,-1) = 1 + (-2) + (2) - (-2) = 3$ "inside" triangle.

Endpoints: $f(1,2) = 1 + (2) + (1) - (4) = 0$

$f(1,-2) = 1 + (-2) + (1) - (-4) = 4$

$f(5,-2) = 1 + (-10) + (5) - (-4) = 0$

Boundary, $y = -x + 3$, $1 < x < 5$

$\therefore f(x, -x+3) = 1 + (-x^2 + 3x) + x - (-2x + 6) = -x^2 + 6x - 5$

for $g(x) = -x^2 + 6x - 7$, $g'(x) = -2x + 6$,

$x = 3$ a critical point. $\therefore y = 0$

$f(3,0) = 1 + (0) + (3) - (0) = 4$

Boundary $x = 1$, $-2 < y < 2$

$\therefore f(1,y) = 1 + y + 1 - 2y = 2 - y$

for $g(y) = 2 - y$, $g'(y) = -1$, so no critical pts.

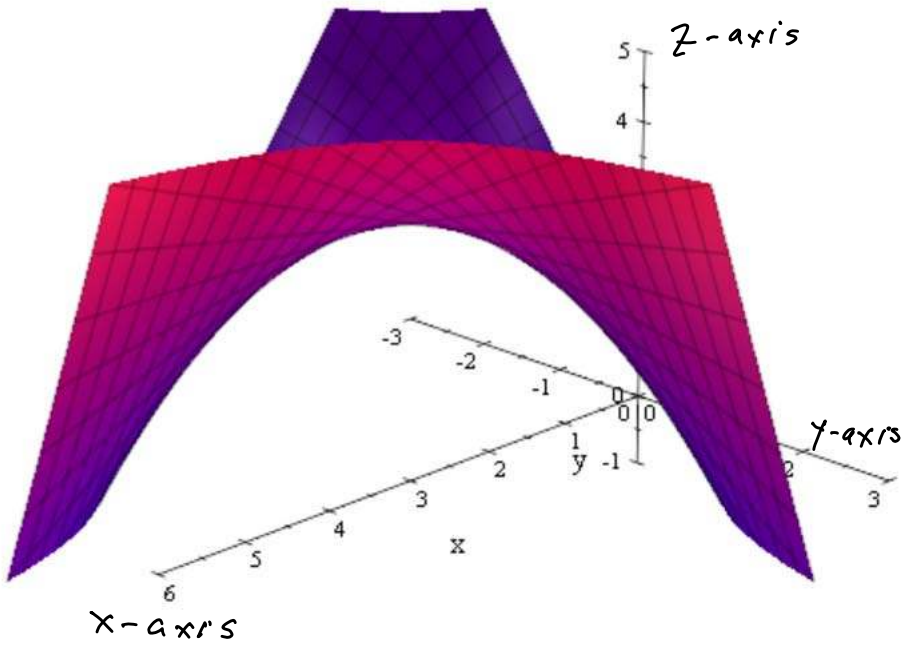
Boundary $y = -2$, $1 < x < 5$

$\therefore f(x,-2) = 1 + (-2x) + x - (-4) = 5 - x$

for $g(x) = 5 - x$, $g'(x) = -1$, so no critical points.

$\therefore$ Global minimum = 0 at $(1,2)$ and $(5,-2)$

Global maximum = 4 at $(1,-2)$ and $(3,0)$



plot of $f(x,y) = 1 + xy + x - 2y$

Note the extremum of $(2,-1)$ "inside" triangle domain is a saddle pt.

In Exercises 46 through 50, D denotes the unit disc.

4C.

46. Let u be a C^2 function on D which is "strictly subharmonic"; that is, the following inequality holds: $\nabla^2 u = (\partial^2 u/\partial x^2) + (\partial^2 u/\partial y^2) > 0$. Show that u cannot have a maximum point in $D \backslash \partial D$ (the set of points in D, but not in ∂D).

Assume $u : \mathbb{R}^2 \to \mathbb{R}$

Suppose u has a max in $D \backslash \partial D$ ($\therefore$ a local max) at $\vec{x}_0$

$\therefore u_x(\vec{x}_0) = u_y(\vec{x}_0) = 0$ Let $\vec{h} = (h_1, h_2)$

Consider $g(t) = u(\vec{x}_0 + t\vec{h})$, $\vec{x}_0 + t\vec{h} \in D \backslash \partial D$

for $t \in (-a, a) = I$, some $a \neq 0 \in \mathbb{R}$.

Then $g(0)$ is a local max on I

$g(t)$ is C^2 since u is C^2 and $\vec{x}_0 + t\vec{h}$ is C^2

$\therefore g'(t) = Du(\vec{x}_0 + t\vec{h})\vec{h}$, by chain rule

$= u_x(\vec{x}_0 + t\vec{h})h_1 + u_y(\vec{x}_0 + \vec{h})h_2$

and $g''(t) = h_1[u_{xx}(\vec{x}_0 + t\vec{h})h_1 + u_{xy}(\vec{x}_0 + t\vec{h})h_2]$

$+ h_2[u_{yx}(\vec{x}_0 + t\vec{h})h_1 + u_{yy}(\vec{x}_0 + t\vec{h})h_2]$

$= h_1^2 u_{xx}(\vec{x}_0 + t\vec{h}) + 2h_1h_2 u_{xy}(\vec{x}_0 + t\vec{h}) + h_2^2 u_{yy}(\vec{x}_0 + t\vec{h})$ [1]

since $u_{xy} = u_{yx}$

Since $u_{xx}(\vec{x}) + u_{yy}(\vec{x}) > 0$, $\forall \vec{x} \in D$, then one of u_{xx} or u_{yy} must be positive at $\vec{x}_0$

(i) Suppose it is $u_{xx}(\vec{x}_0) > 0$

Consider $g(t)$ for $\vec{h} = (h_1, 0)$.

$\therefore g''(0) = h_1^2 u_{xx}(\vec{x}_0) > 0$, from [1]

$\therefore g(0)$ is a local minimum on I, by 2nd derivative test for $g(t)$,

contradicting $g(0)$ is a local max.

$\therefore$ Original assumption that $u(x,y)$ has a local max on $D \setminus \partial D$ must be false.

(ii) Suppose it is $u_{yy}(\vec{x}_0) > 0$

Consider $g(t)$ for $\vec{h} = (0, h_2)$

$\therefore g''(0) = h_2^2 u_{yy}(\vec{x}_0) > 0$, from [1]

$\therefore g(0)$ is a local minimum on I, by 2nd derivative test for $g(t)$,

contradicting $g(0)$ is a local max

$\therefore$ Original assumption that $u(x,y)$ has a local max. on $D \setminus \partial D$ must be false.

$\therefore$ (i) & (ii) $\Rightarrow$ $u(x,y)$ cannot have a max. on $D \setminus \partial D$

Note: same reasoning $\Rightarrow$ if $u_{xx} + u_{yy} < 0$ then $u(x,y)$ cannot have a local min. on $D \setminus \partial D$.

47.

47. Let u be a harmonic function on D—that is, $\nabla^2 u = 0$ on $D \setminus \partial D$—and be continuous on D. Show that if u achieves its maximum value in $D \setminus \partial D$, it also achieves it on ∂D. This is sometimes called the "weak maximum principle" for harmonic functions. [HINT: Consider $\nabla^2(u + \varepsilon e^x)$, $\varepsilon > 0$. You can use the following fact, which is proved in more advanced texts: Given a sequence $\{\mathbf{p}_n\}$, $n = 1, 2, \ldots$, of points in a closed bounded set A in $\mathbb{R}^2$ or $\mathbb{R}^3$, there exists a point $\mathbf{q}$ such that every neighborhood of $\mathbf{q}$ contains at least one member of $\{\mathbf{p}_n\}$.]

Assume $u : \mathbb{R}^2 \to \mathbb{R}$

Have to show there is a point $\vec{q}$ on ∂D s.t. $u(\vec{q}) \geq u(\vec{x})$, $\forall \vec{x} \in D$!

D can be viewed as $D \setminus \partial D \cup \partial D$, i.e. the union of the open disc D with its border.

If $\vec{q}$ is a max for $u(x,y)$ on ∂D, then if can show $u(\vec{q}) \geq u(\vec{x})$, where $\vec{x}$ is any point of $D \setminus \partial D$, then assertion is proved.

The idea is to use the hint, so that $\vec{q}$ is the limit of a sequence $\vec{p}_n$ of points on ∂D, and each $\vec{p}_n$ is where a max occurs for some constructed sequence of functions involving

$u(x,y)$. Then use continuity to say $u(x,y)$ has a max at the limit point $\vec{q}$.

The constructed sequence of functions will have to be s.t. each function of the sequence can't have a max on $D \setminus \partial D$ (using # 4C above), and so must have a max on ∂D.

The hint suggests constructing $u(\vec{x}) + \epsilon e^x$

$= u(x,y) + \epsilon e^x$, where $\vec{x} = (x,y) \in D$, $\epsilon > 0$

Then $\nabla^2 (u(x,y) + \epsilon e^x) = u_{xx} + u_{yy} + \epsilon e^x$

$= \epsilon e^x$ on $D \setminus \partial D$.

By # 4b above, this constructed function is C^2 on D, and so cannot have a max on $D \setminus \partial D$, and so can only have a max on ∂D, the boundary.

If we let $\epsilon = \frac{1}{n}$, n an integer $\neq 0$, then

$u_n(x,y) = u(x,y) + \frac{1}{n} e^x$ is such a function,

and so each $u_n(x,y)$ has its max, call it $\vec{p}_n$, on ∂D. Note also, $u_n(x,y) > u(x,y)$

From the hint, for this sequence $\vec{p}_n$, there is a point $\vec{q}$ s.t. every neighborhood of $\vec{q}$ contains a least one $\vec{p}_n$, and so contains infinitely many such $\vec{p}_n$ since $\frac{1}{n+1} < \frac{1}{n}$,

so that a neighborhood of radius $\frac{1}{r}$, contains all points with radius $\frac{1}{s}$, $s > r$.

Note, since for each $n > 0$, u_n has its max at $\vec{p}_n \in \partial D$, and letting $\vec{p}_n = (x_n, y_n)$, then

$$u_n(\vec{p}_n) = u(\vec{p}_n) + \frac{1}{n} e^{x_n} \leq u(\vec{p}_n) + \frac{1}{n} e$$

since $x_n \leq 1$ on ∂D (so $e^{x_n} \leq e^1$)

$$\therefore u_n(\vec{p}_n) - \frac{e}{n} \leq u(\vec{p}_n) \qquad [1]$$

Now let $\vec{x}$ be any point in D. As u_n is max at $\vec{p}_n$,

$$\therefore u_n(\vec{x}) \leq u_n(\vec{p}_n) \qquad [2]$$

As above from the definition of $u_n(x,y)$

$$u(\vec{x}) < u_n(\vec{x}) \qquad [3]$$

$\therefore$ From [2], [3],

$$u(\vec{x}) < u_n(\vec{p}_n), \text{ and so}$$

$$u(\vec{x}) - \frac{e}{n} < u_n(\vec{p}_n) - \frac{e}{n} \qquad [4]$$

$\therefore$ From [1] & [4],

$$u(\vec{x}) - \frac{e}{n} < u(\vec{p}_n)$$

Now look at $\vec{q}$. Given any $\delta > 0$,

$\exists N > 0$ s.t. if $n > N$, then $\vec{p}_n \in D_\delta(\vec{q})$

i.e., $u(\vec{x}) = \lim_{n\to\infty} \left[u(\vec{x}) - \frac{e}{n}\right] \le \lim_{n\to\infty} u(\vec{p}_n) = u(\vec{q})$

The latter limit using continuity of u.

$\therefore u(\vec{x}) \le u(\vec{q})$ for any $\vec{x} \in D$.

$\therefore u(\vec{q})$ is a max for u, and $\vec{q} \in \partial D$

48. **48.** Define the notion of a strict superharmonic function u on D by mimicking Exercise 46. Show that u cannot have a minimum in $D \setminus \partial D$.

Let $u(x,y)$ be a C^2 function on D s.t.

$u_{xx}(x,y) + u_{yy}(x,y) = \nabla^2 u(x,y) < 0$. Then $u(x,y)$ cannot have a minimum in $D \setminus \partial D$ (i.e., open D).

Pf: This is analogous to proof of #46.

Suppose $u(x,y)$ has a min on $D \setminus \partial D$ (and $\therefore$ a local min).

at $\vec{x}_0 = (x_0, y_0)$. By 1st derivate test,

$u_x(\vec{x}_0) = 0$, $u_y(\vec{x}_0) = 0$. Let $\vec{h} = (h_x, h_y)$

Consider $g(t) = u(\vec{x}_0 + t\vec{h})$, $\vec{x}_0 + t\vec{h} \in D \setminus \partial D$

for $t \in (-a, a) = I$, some $a \in \mathbb{R}$

Then $g(0)$ has a local min on I.

$g(t)$ is C^2 since u is C^2, and $\vec{x}_0 + t\vec{h}$ is C^2

$\therefore g'(t) = \nabla u(\vec{x}_0 + t\vec{h})\,\vec{h}$, by chain rule

$= u_x(\vec{x}_0 + t\vec{h})\,h_x + u_y(\vec{x}_0 + t\vec{h})\,h_y$

and $g''(t) = h_x\left[u_{xx}(\vec{x}_0 + t\vec{h})\,h_x + u_{xy}(\vec{x}_0 + t\vec{h})\,h_y\right]$

$+ h_y\left[u_{yx}(\vec{x}_0 + t\vec{h})\,h_x + u_{yy}(\vec{x}_0 + t\vec{h})\,h_y\right]$

$= h_x^2 u_{xx}(\vec{x}_0 + t\vec{h}) + 2h_x h_y u_{xy}(\vec{x}_0 + t\vec{h}) + h_y^2 u_{yy}(\vec{x}_0 + t\vec{h})$ [1]

since $u_{xy} = u_{yx}$

Since $u_{xx}(\vec{x}) + u_{yy}(\vec{x}) < 0$, $\forall \vec{x} \in D$, Then

one of u_{xx} or u_{yy} must be negative at $\vec{x}_0$

(i) Suppose it is $u_{xx}(\vec{x}_0) < 0$

Consider $g(t)$ for $\vec{h} = (h_x, 0)$.

$\therefore g''(0) = h_x^2 u_{xx}(\vec{x}_0) < 0$, from [1] above

$\therefore$ $g(0)$ is a local maximum on I, by 2nd derivative test for $g(t)$,

contradicting $g(0)$ is a local min.

$\therefore$ Original assumption that $u(x,y)$ has a local min on $D \setminus \partial D$ must be false.

(ii) Suppose it is $u_{yy}(\vec{x}_0) < 0$

Consider $g(t)$ for $\vec{h} = (0, h_y)$

$\therefore g''(0) = h_y^2 u_{yy}(\vec{x}_0) < 0$, from [i] above

$\therefore$ $g(0)$ is a local maximum on I, by 2nd derivative test for $g(t)$,

contradicting $g(0)$ is a local min.

(i) & (ii) $\Rightarrow$ $u(x,y)$ cannot have a minimum on $D \setminus \partial D$

49. **49.** Let u be harmonic in D as in Exercise 47. Show that if u achieves its minimum value in $D \setminus \partial D$, it also achieves it on ∂D. This is sometimes called the "weak minimum principle" for harmonic functions.

Pf: Analogous to #47

The idea is to construct functions based on $u(x,y)$, whose minimums are on ∂D, and the limit

of the sequence of minimums is on ∂D, and so u(sequence of minimums) = u(limit) on ∂D, and u(limit) $\leq u$(sequence of minimums).

Use hint to construct $u_n(\vec{x}) = u(\vec{x}) - \frac{1}{n}e^x$, $\vec{x} = (x,y) \in D$, n a positive integer.

$\therefore \nabla^2 u_n = u_{xx}(x,y) + u_{yy}(x,y) - \frac{1}{n}e^x$, $(x,y) \in D$

$= -\frac{1}{n}e^x < 0$ on $D \setminus \partial D$

since $u_{xx} + u_{yy} = 0$ on $D \setminus \partial D$, by assumption.

By #48, u_n cannot have a minimum on $D \setminus \partial D$, and so must have a minimum on ∂D, as u_n is C^2 on D.

Let the min of u_n be located at $\vec{p}_n \in \partial D$

Note since $u_n = u - \frac{1}{n}e^x$, $u_n < u$ $\forall \vec{x} \in D$.

From the hint in #47, since $\vec{p}_n$ is a sequence on closed, bounded ∂D, $\exists \vec{q} \in \partial D$ s.t. for every $\delta > 0$,

$\vec{p}_n \in D_\delta(\vec{q})$, some integer $n > 0$.

Since $D_{\delta'}(\vec{q}) \subseteq D_\delta(\vec{q})$, $\delta' < \delta$, then

$D_\delta(\vec{q})$ contains infinitely many $\vec{p}_n$

so that given any $\delta > 0$, $\exists\, N > 0$ s.t. if $n > N$, then

$$\vec{\rho}_n \in D_\delta(\vec{q})$$

$$\therefore \lim_{n\to\infty} \vec{\rho}_n = \vec{q}$$

Note, since for each $n > 0$, u_n has its min at $\vec{\rho}_n \in \partial D$, and letting $\vec{\rho}_n = (x_n, y_n)$, then

$$u_n(\vec{\rho}_n) = u(\vec{\rho}_n) - \frac{1}{n} e^{x_n} \geq u(\vec{\rho}_n) - \frac{1}{n} e$$

since $x_n \leq 1$ on ∂D (so $e^{x_n} \leq e^1 \Rightarrow -e^{x_n} \geq -e$)

$$\therefore u(\vec{\rho}_n) \leq u_n(\vec{\rho}_n) + \frac{e}{n} \qquad [1]$$

Now let $\vec{x}$ be any point in D. As u_n is min at $\vec{\rho}_n$,

$$\therefore u_n(\vec{\rho}_n) \leq u_n(\vec{x}) \qquad [2]$$

As above from the definition of $u_n(x,y)$

$$u_n(\vec{x}) < u(\vec{x}) \qquad [3]$$

$\therefore$ From [2], [3],

$$u_n(\vec{\rho}_n) < u(\vec{x}), \quad \text{and so}$$

$$u_n(\vec{\rho}_n) + \frac{e}{n} < u(\vec{x}) + \frac{e}{n} \qquad [4]$$

$\therefore$ From [1] and [4],

$$u(\vec{p}_n) < u(\vec{x}) + \frac{e}{n} \qquad [5]$$

Since u is continuous on D,

$$u(\vec{q}) = \lim_{n \to \infty} u(\vec{p}_n)$$

and from [5] $\lim_{n \to \infty} u(\vec{p}_n) \leq \lim_{n \to \infty} \left[u(\vec{x}) + \frac{e}{n} \right] = u(\vec{x})$

$\therefore u(\vec{q}) \leq u(\vec{x})$, where $\vec{x} \in D$.

$\therefore u(\vec{q})$ is a minimum, $\vec{q} \in \partial D$

50.

50. Let ϕ: $\partial D \to \mathbb{R}$ be continuous and let T be a solution on D to $\nabla^2 T = 0$, continuous on D and $T = \phi$ on ∂D.

(a) Use Exercises 46 to 49 to show that such a solution, if it exists, must be unique.

(b) Suppose that $T(x, y)$ represents a temperature function that is independent of time, with ϕ representing the temperature of a circular plate at its boundary. Can you give a physical interpretation of the principle stated in part (a)?

(a) Suppose $T_1(\vec{x})$ and $T_2(\vec{x})$ are two different solutions.

Consider $T(\vec{x}) = T_1(\vec{x}) - T_2(\vec{x})$.

Since $T_1 = \phi$ on ∂D, $T_2 = \phi$ on ∂D, then

$T = T_1 - T_2 = \phi - \phi = 0$ on ∂D.

Since $\nabla^2 T_1 = 0$, $\nabla^2 T_2 = 0$, then $\nabla^2 T = 0$ on D.

By #47 and #49, $T(\vec{x})$ must achieve its maximum and minimum on ∂D.

But since $T(\vec{x}) = 0$ for all points on ∂D, 0 must be the max and min $\forall \vec{x} \in D$.

$\therefore T(\vec{x}) = 0 \quad \forall \vec{x} \in D \Rightarrow T_1 = T_2 \quad \forall \vec{x} \in D$.

$\therefore$ There is only one solution.

(b) Since max & min achieved at boundary, measuring temp just at boundary gives idea of temp. distribution in disc.

51. **51.** (a) Let f be a C^1 function on the real line $\mathbb{R}$. Suppose that f has exactly one critical point x_0 that is a strict local minimum for f. Show that x_0 is also an absolute minimum for f; that is, that $f(x) \geq f(x_0)$ for all x.

(a) x_0 a local minimum $\Rightarrow \exists \delta > 0$ s.t. $\forall x \in D\delta(x_0)$, $f(x_0) < f(x)$. $\therefore$ Let $x_1 \in D\delta(x_0)$ s.t. $x_1 < x_0$, and let $x_2 \in D\delta(x_0)$ s.t. $x_0 < x_2$.

$\therefore f(x_1) > f(x_0)$, and $f(x_0) < f(x_2)$

$\therefore \dfrac{f(x_1) - f(x_0)}{x_1 - x_0} < 0$ and $\dfrac{f(x_2) - f(x_0)}{x_2 - x_0} > 0$

By mean value Theorem, $\exists\ z_1, z_2$ s.t.

$f'(z_1) = \frac{f(x_1)-f(x_0)}{x_1-x_0} < 0$ and $f'(z_2) = \frac{f(x_2)-f(x_0)}{x_2-x_0} > 0$.

(i) Suppose there is a point $y < x_1$ s.t. $f(y) < f(x_0)$.

$\therefore f(y) < f(x_1)$ and $\frac{f(y)-f(x_1)}{y-x_1} > 0$.

By mean value Theorem, $\exists\ z'$ s.t. $f'(z') = \frac{f(y)-f(x_1)}{y-x_1}$

$\therefore f'(z') > 0$ and $f'(z_1) < 0$ and $z' < z_1$

$\therefore$ as f is C^1, and f' changes sign from z' to z_1, then there must be a point w s.t. $z' < w < z_1$ s.t. $f'(w) = 0$

making w another critical point, a contradiction

(ii) Suppose there is another point $y > x_2$ s.t. $f(y) < f(x_0)$

$\therefore f(y) < f(x_2)$, and $\frac{f(y)-f(x_2)}{y-x_2} < 0$

By mean value Theorem, $\exists\ z'$ s.t. $f'(z') = \frac{f(y)-f(x_2)}{y-x_2}$

$\therefore f'(z') < 0$ and $f'(z_2) > 0$ and $z_2 < z'$

As f is C^1, and f' changes sign from z_2 to z',

Then there must be a point w s.t. $z_2 < w < z'$

s.t. $f'(w) = 0$, so w is another critical

point, a contradiction

(i) & (ii) $\Rightarrow$ There is no point $y < x_0$ or $y > x_0$

s.t. $f(y) < f(x_0)$.

$\therefore f(x_0)$ is an absolute minimum.

(5) (b) The next example shows that the conclusion of part (a) does not hold for functions of more than one variable. Let $f: \mathbb{R}^2 \to \mathbb{R}$ be defined by

$$f(x, y) = -y^4 - e^{-x^2} + 2y^2\sqrt{e^x + e^{-x^2}}.$$

(i) Show that (0, 0) is the only critical point for f and that it is a local minimum.
(ii) Argue informally that f has no absolute minimum.

(i) $f_x = 2xe^{-x^2} + 2y^2\left(\frac{1}{2}\right)\left(e^x + e^{-x^2}\right)^{-\frac{1}{2}}\left(e^x - 2xe^{-x^2}\right)$

$$= 2xe^{-x^2} + \frac{y^2\left(e^x - 2xe^{-x^2}\right)}{\sqrt{e^x + e^{-x^2}}}$$

$$f_y = -4y^3 + 4y\sqrt{e^x + e^{-x^2}}$$

$$f_y = 0 : 4y\sqrt{e^x + e^{-x^2}} = 4y^3$$

$$\Rightarrow y = 0, \text{ or } y^2 = (e^x + e^{-x^2})^{\frac{1}{2}}$$

$$\therefore y = 0 \text{ or } y = \pm (e^x + e^{-x^2})^{\frac{1}{4}}$$

Substituting above values of y into

$$f_x = 0 : y = 0 : f_x = 2xe^{-x^2} \Rightarrow x = 0.$$

$$y^2 = (e^x + e^{-x^2})^{\frac{1}{2}} : f_x = 2xe^{-x^2} + (e^x - 2xe^{-x^2})$$

$$= e^x$$

$$\therefore f_x \neq 0 \text{ for } y^2 = (e^x + e^{-x^2})^{\frac{1}{2}}$$

$\therefore$ Only solution for $f_x = f_y = 0$ is $(0,0)$.

$\therefore$ $(0,0)$ only critical point

$$f_{yy} = -12y^2 + 4\sqrt{e^x + e^{-x^2}} \qquad \therefore f_{yy}(0,0) = 4\sqrt{2}$$

$$f_{yx} = 4y\left(\tfrac{1}{2}\right)(e^x + e^{-x^2})^{-\frac{1}{2}}(e^x - 2xe^{-x^2}) \qquad \therefore f_{yx}(0,0) = 0$$

$$f_{xx} = 2e^{-x^2} - 4x^2e^{-x^2} + y^2\left[\frac{\partial}{\partial x}\left(\frac{e^x - 2xe^{-x^2}}{\sqrt{e^x + e^{-x^2}}}\right)\right]$$

$$\therefore f_{xx}(0,0) = 2$$

$$\therefore \begin{vmatrix} 2 & 0 \\ 0 & 4\sqrt{2} \end{vmatrix} = 8\sqrt{2} > 0.$$

$\therefore f_{xx} > 0$ and $Hf(0,0) > 0 \Rightarrow (0,0)$ local min.

(ii) as $y \to \infty$, at $x = 0$, $-y^4$ dominates y^2,

so $f(x,y) \to -\infty$

52. Suppose that a pentagon is composed of a rectangle topped by an isosceles triangle (see Figure 3.3.8). If the length of the perimeter is fixed, find the maximum possible area.

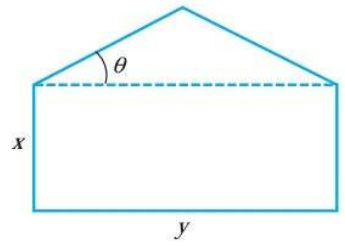


figure 3.3.8 Maximize the area for fixed perimeter.

Let z = length of one leg of the isosceles triangle.

Let P = perimeter of pentagon $= 2x + y + 2z$

$\therefore z\cos\theta = \frac{1}{2}y$, $\therefore z = \frac{y}{2\cos\theta}$, $2z = \frac{y}{\cos\theta}$

Area $= A = xy + \frac{1}{2}yz\sin\theta$

$= xy + \frac{y^2\sin\theta}{4\cos\theta}$ $\quad 0 < x,\ 0 < y,\ 0 < \theta < 90^\circ$

$P = 2x + y + \frac{y}{\cos\theta}$, $\therefore x = \frac{1}{2}\left(P - y - \frac{y}{\cos\theta}\right)$

$\therefore A(y,\theta) = \frac{1}{2}Py - \frac{1}{2}y^2 - \frac{y^2}{2\cos\theta} + \frac{y^2\sin\theta}{4\cos\theta}$

$\therefore A_y = 0$ $A_\theta = 0$ to find critical points.

$A_y = \frac{P}{2} - y - \frac{y}{\cos\theta} + \frac{y\sin\theta}{2\cos\theta} = 0$

$$\text{or } \frac{\rho}{2} = y\left(1 + \frac{1}{\cos\theta} - \frac{\sin\theta}{2\cos\theta}\right)$$

$$= y\left(\frac{2 + 2\cos\theta - \sin\theta}{2\cos\theta}\right)$$

$$\therefore y = \frac{\rho\cos\theta}{2 + 2\cos\theta - \sin\theta}$$

$$A_\theta = -\frac{y^2}{2}\cdot\frac{\sin\theta}{\cos^2\theta} + \frac{y^2}{4}\left(\frac{1}{\cos^2\theta}\right) = 0$$

$$\text{Or, } \frac{y^2}{2\cos^2\theta}(\sin\theta) = \frac{y^2}{2\cos^2\theta}\left(\frac{1}{2}\right)$$

$$\therefore \sin\theta = \frac{1}{2}, \quad \underline{\theta = 30^\circ} \quad \therefore \cos\theta = \frac{\sqrt{3}}{2}$$

$$\text{From } A_y = 0: \quad y = \frac{\rho\left(\frac{\sqrt{3}}{2}\right)}{2 + 2\left(\frac{\sqrt{3}}{2}\right) - \left(\frac{1}{2}\right)} = \underline{\frac{\rho\sqrt{3}}{3 + 2\sqrt{3}}}$$

$$\text{From } x = \frac{1}{2}\left(\rho - y - \frac{y}{\cos\theta}\right) = \frac{1}{2}\left(\rho - \frac{\rho\sqrt{3}}{3+2\sqrt{3}} - \frac{2\rho}{3+2\sqrt{3}}\right)$$

$$= \frac{\rho}{2}\left(\frac{3 + 2\sqrt{3} - \sqrt{3} - 2}{3 + 2\sqrt{3}}\right)$$

$$\therefore x = \underline{\underline{\frac{\rho(1+\sqrt{3})}{2(3+2\sqrt{3})}}}$$

$$\therefore \text{Area} = xy + \frac{y^2\sin\theta}{4\cos\theta}$$

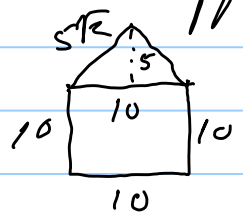
$$= \frac{\rho(1+\sqrt{3})}{2(3+2\sqrt{3})}\left(\frac{\rho\sqrt{3}}{3+2\sqrt{3}}\right) + \frac{1}{4}\left(\frac{\rho\sqrt{3}}{3+2\sqrt{3}}\right)^2\left(\frac{\sqrt{3}}{3}\right)$$

$$= \frac{\rho^2(3+\sqrt{3})}{2(9+12\sqrt{3}+12)} + \frac{\sqrt{3}}{12}\left(\frac{3\rho^2}{9+12\sqrt{3}+12}\right)$$

$$= \rho^2\left[\frac{18+6\sqrt{3}+3\sqrt{3}}{12(21+12\sqrt{3})}\right]$$

$$= \rho^2\frac{9(2+\sqrt{3})}{12\cdot 3(7+4\sqrt{3})} = \frac{\rho^2}{4}\left(\frac{2+\sqrt{3}}{7+4\sqrt{3}}\right)$$

Test: Suppose a $45°$ Δ on top of a square of side 10.



$\therefore \rho = 30 + 10\sqrt{2} \approx 44.142$

$A = 125$

From above formula, $\frac{(44.142)^2}{4}\frac{(2+\sqrt{3})}{(7+4\sqrt{3})} \approx 130.52$

$130.52 > 125$

3.4 Constrained Extrema and Lagrange Multipliers

Note Title 5/20/2016

1.

1. Let $f(x, y) = x^2 + 3y^2$. Find the maximum and minimum values of f subject to the given constraint.

(a) $x^2 + y^2 = 1$

(b) $x^2 + y^2 \le 1$

$f_x = 2x \quad f_y = 6y$

(a) Let $g(x,y) = x^2+y^2 \quad \therefore g_x = 2x, \; g_y = 2y$

$\therefore (2x, 6y) = \lambda(2x, 2y)$ and $x^2+y^2=1$

$2x = \lambda 2x : x \neq 0 : \lambda = 1, \; 6y = 2y \Rightarrow y = 0, \; x = \pm 1$

$x = 0 : \therefore y = \pm 1 \; (\therefore \lambda = \pm 3)$

$\therefore$ Critical points: $(\pm 1, 0), (0, \pm 1)$

$f(1,0) = 1$, $f(-1,0) = 1$ } min $\quad$ $f(0,1) = 3$, $f(0,-1) = 3$ } max

(b) $f_x = 0 \Rightarrow 2x = 0 \quad f_y = 0 \Rightarrow 6y = 0$

$\therefore (0,0)$ a critical point on $x^2+y^2 < 1$

$f_{xx} = 2 \quad f_{yy} = 6 \quad f_{xy} = 0 \quad \therefore \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4$

$\therefore f_{xx} > 0, \; ac - b^2 > 0 \Rightarrow$ Hessian pos-def. $\Rightarrow$

$(0,0)$ a minimum on $x^2+y^2 < 1$

and $f(0,0) = 0$

At the boundary, from (a) min = 1, max = 3.

$\therefore$ On closed disk: $x^2+y^2 \le 1$,

Min: 0, at (0,0)

Max: 3, at (0,1) and (0,-1)

2. **2.** Consider all rectangles with fixed perimeter p. Use Lagrange multipliers to show that the rectangle with maximal area is a square.

Let x, y be lengths of sides. $\therefore p = 2x + 2y$

Let $A(x,y) = xy$, the area of the rectangle.

$\therefore A_x = y \quad A_y = x$

Let $g(x,y) = 2x + 2y - p = 0$. $\therefore g_x = 2 \quad g_y = 2$

$\therefore [y \; x] = \lambda [2 \; 2]$, or $y = 2\lambda \quad x = 2\lambda$

$\therefore p = 2(2\lambda) + 2(2\lambda) = 8\lambda. \quad \lambda = \frac{p}{8}$

$\therefore y = 2\left(\frac{p}{8}\right) = \frac{p}{4}, \quad x = 2\left(\frac{p}{8}\right) = \frac{p}{4}.$

$\therefore x = y \Rightarrow$ a square, and Area $= \frac{p^2}{16}$

In Exercises 3 to 7, find the extrema of f subject to the stated constraints.

3. **3.** $f(x, y, z) = x - y + z$, subject to $x^2 + y^2 + z^2 = 2$

$f_x = 1 \quad f_y = -1 \quad f_z = 1 \quad g(x,y) = x^2 + y^2 + z^2$

$\therefore \nabla g(x,y) = [2x \; 2y \; 2z]$

$\therefore [1 \; -1 \; 1] = \lambda [2x \; 2y \; 2z] \quad \therefore \lambda \neq 0$

$\therefore \quad 1 = 2\lambda x \quad x = \frac{1}{2\lambda} \quad \therefore x = z = -y$

$-1 = 2\lambda y \quad y = -1/2\lambda$

$1 = 2\lambda z \quad z = \frac{1}{2\lambda}$

$\therefore x^2 + (-x)^2 + (x^2) = 2,$

$x^2 = 2/3, \; x = \pm \frac{\sqrt{6}}{3}$

$\therefore$ Critical points: $\left(\frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}\right)$ and $\left(-\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}\right)$

$\therefore f\left(\frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}\right) = \sqrt{6} = \max$

$f\left(-\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}\right) = -\sqrt{6} = \min$

4. **4.** $f(x, y) = x - y$, subject to $x^2 - y^2 = 2$

$\nabla f = [1 \; -1] \quad \nabla g = [2x \; -2y] \quad g(x,y) = x^2 - y^2$

$\therefore \nabla f = \lambda \nabla g \; : \; [1 \; -1] = \lambda [2x \; -2y] \Rightarrow \lambda \neq 0$

$\therefore \quad x = \frac{1}{2\lambda} = y,$ but from $x^2 - y^2 = 2$, $x \neq y$.

Note $\{(x,y) \mid x^2 - y^2 = 2\}$ is not bounded.

This test fails to find maxima & minima

Also note, asymptotes of $x^2-y^2=2$ are $y=x$, $y=-x$

So as $x \to +\infty$, from $y=-x$, $y \to -\infty$, so $x-y \to +\infty$ so no max.

As $x \to -\infty$, from $y=-x$, $y \to +\infty$, so $x-y \to -\infty$, so no min.

5. **5.** $f(x, y) = x$, subject to $x^2 + 2y^2 = 3$

Note $S = \{(x,y) \mid x^2+2y^2=3\}$ is bounded, so continuous f has a max & min on S.

$f_x = 1 \quad f_y = 0 \quad \nabla g = [2x \;\; 4y]$

$\therefore [1 \;\; 0] = \lambda [2x \;\; 4y] \Rightarrow \lambda \neq 0$ since $1 = 2\lambda x$.

$\therefore 0 = 4\lambda y \Rightarrow y = 0$. $\therefore x^2 = 3$, $x = \pm\sqrt{3}$

$\therefore (\pm\sqrt{3}, 0)$ are critical points

$f(-\sqrt{3}, 0) = -\sqrt{3}$ a min

$f(\sqrt{3}, 0) = \sqrt{3}$ a max

6. **6.** $f(x, y, z) = x + y + z$, subject to $x^2 - y^2 = 1$, $2x + z = 1$

Let $g_1(x,y,z) = x^2 - y^2$, $g_2(x,y,z) = 2x + z$

$\nabla f = [1\ 1\ 1]$ $\nabla g_1 = [2x\ -2y\ 0]$ $\nabla g_2 = [2\ 0\ 1]$

$\therefore [1\ 1\ 1] = \lambda_1 [2x\ -2y\ 0] + \lambda_2 [2\ 0\ 1]$

$\therefore 1 = 2\lambda_1 x + 2\lambda_2$ [1]

$1 = -2\lambda_1 y$ [2] $\Rightarrow \lambda_1 \neq 0$

$1 = \lambda_2$ [3]

$\therefore -1 = 2\lambda_1 x$ [1] + [3]

$\therefore x = -\frac{1}{2\lambda_1}$ and [2] $\Rightarrow y = -\frac{1}{2\lambda_1}$

$\therefore x = y$ But $x \neq y$ from $x^2 - y^2 = 1$.

$\therefore$ Test fails to find a max or min.

And note $\{(x,y) \mid x^2 - y^2 = 1\}$ is not bounded, and for $x = \frac{1-z}{2}$, $\{(y,z) \mid \frac{(1-z)^2}{4} - y^2\}$ is not bounded.

Also, $y = \pm x$ are asymptotes for $x^2 - y^2 = 1$.

As $x \to \infty$, $y \to \infty$, $z \to 0$ from $2x + z = 1$, but $f(x,y,z) \to \infty$ $\therefore$ No max

As $x \to -\infty$, $y \to -\infty$, $z \to 0$, $f \to -\infty$. $\therefore$ No min

7. **7.** $f(x, y) = 3x + 2y$, subject to $2x^2 + 3y^2 = 3$

$\nabla f = [3 \; 2] \quad \nabla g = [4x \; 6y]$

$\therefore \nabla f = \lambda \nabla g : [3 \; 2] = \lambda [4x \; 6y] \Rightarrow \lambda \neq 0$

$\therefore 3 = 4\lambda x \;, \; x = \frac{3}{4\lambda}$

$2 = 6\lambda y \;, \; y = \frac{1}{3\lambda}$

$\therefore 2\left(\frac{3}{4\lambda}\right)^2 + 3\left(\frac{1}{3\lambda}\right)^2 = 3 \;, \; \frac{9}{8\lambda^2} + \frac{1}{3\lambda^2} = 3$

$27 + 8 = 72\lambda^2, \; \lambda = \pm \frac{1}{6}\sqrt{\frac{35}{2}} = \pm \frac{\sqrt{70}}{12}$

$\therefore (x, y) = \pm\left(\frac{3}{4}\frac{12}{\sqrt{70}}, \frac{12}{3}\frac{1}{\sqrt{70}}\right) = \pm\left(\frac{9}{\sqrt{70}}, \frac{4}{\sqrt{70}}\right)$

$\therefore f\left(\frac{9}{\sqrt{70}}, \frac{4}{\sqrt{70}}\right) = \underline{\max} = \frac{35}{\sqrt{70}} = \frac{\sqrt{70}}{2}$

$f\left(-\frac{9}{\sqrt{70}}, -\frac{4}{\sqrt{70}}\right) = \underline{\min} = -\frac{\sqrt{70}}{2}$

Find the relative extrema of $f|S$ in Exercises 8 to 11.

8. **8.** $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2, S = \{(x, 2) \mid x \in \mathbb{R}\}$ Note: S not bounded

$\nabla f = [2x \; 2y] \quad g(x,y) = y \;, \; g(x,y) = 2 \; \therefore \nabla g = [0 \; 1]$

$\therefore \nabla f = \lambda \nabla g : [2x \; 2y] = \lambda [0 \; 1]$

$\therefore$ $2x = 0 \Rightarrow x = 0$ $\therefore$ $(0,2)$ a critical point
$2y = \lambda$
and $y = 2$, so $\lambda = 4$

$(0,2)$ only critical point $f(0,2) = 4$

Note for $y = 2$, $f(x,y) = x^2 + 4$, so no max.

$\therefore$ $(0,2)$ a min and $f(0,2) = 4$

Using second-derivative test,

Let $h(x,y) = f - \lambda g = x^2 + y^2 - 4(2)$

$h_{xx} = 2$ $h_{xy} = 0$ $h_{yy} = 2$

$$\therefore |\bar{H}| = \begin{vmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{vmatrix} = (-1)(0 - (-2)) = -2$$

Since $|\bar{H}| < 0$, $(0,2)$ is a local min.

9. **9.** $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2, S = \{(x, y) \mid y \geq 2\}$ Note: S not bounded

Let $U = \{(x,y) \mid y > 2\}$, $\partial U = \{(x,y) \mid y = 2\}$

(a) U: $f_x = 2x$ $f_y = 2y$ $(0,0)$ a critical point in R^2,
but $(0,0) \notin U$. $\therefore$ No critical point in U.

(b) ∂U: $g(x,y) = y$, where ∂U is $g(x,y) = 2$.

$\nabla f = \lambda \nabla g \Rightarrow [2x \;\; 2y] = \lambda [0 \;\; 1] \Rightarrow x = 0,$

$y = 2 \Rightarrow \lambda = 4$

$\therefore (0,2)$ a critical point on ∂U.

$f(0,2) = 4$

2nd derivative test: let $h = f - \lambda g$

$h_{xx} = 2 \quad h_{xy} = 0 \quad h_{yy} = 2$

$$\therefore |\bar{H}| = \begin{vmatrix} 0 & 0 & -1 \\ 0 & 2 & 0 \\ -1 & 0 & 2 \end{vmatrix} = (-1)(0 - (-2)) = -2$$

$\therefore |\bar{H}| < 0 \Rightarrow (0,2)$ a local min.

10. **10.** $f: \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 - y^2, S = \{(x, \cos x) \mid x \in \mathbb{R}\}$

$y = \cos x$, or $y - \cos x = 0$. $\therefore$ Let $g(x,y) = y - \cos x$

$\therefore$ For $g(x,y) = 0$, $\nabla g = [\sin x \;\; 1] \neq \vec{0}$.

$\therefore \nabla f = \lambda \nabla g \Rightarrow [2x \;\; -2y] = \lambda [\sin x \;\; 1]$

$\lambda = 0 \therefore \Rightarrow (x,y) = (0,0)$, but $(0,0) \notin S$ as $\cos(0) = 1$.

$\lambda \neq 0$: $2x = \lambda \sin x$

$-2y = \lambda$

and $y = \cos x$

$\therefore -2\cos x = \lambda$, and $x \neq n\frac{\pi}{2}$, which makes $\lambda = 0$.

$\therefore 2x = -2\cos x \sin x = -\sin 2x \Rightarrow x = 0$

For $s(x) = 2x + \sin 2x$, $s'(x) = 2 + 2\cos x \geq 0$

$\therefore s(x)$ is increasing $\forall x$, $s(0) = 0$, and for $-\frac{\pi}{4} < x < \frac{\pi}{4}$, $s'(x) > 0$, so $x = 0$ is the only value at which $s(x) = 0$.

$\therefore y = \cos(0) = 1$.

$\therefore (0,1)$ the only critical point for $\lambda \neq 0$.

$\therefore (0,1)$ the only critical point for $f|S$.

$f(0,1) = -1$.

Note for $f(x,y) = x^2 - y^2$ and $y = \cos x$, $f(x,y) = x^2 - \cos^2 x$.

As $x^2 \geq 0$ $\forall x$, and $-1 \leq -\cos^2 x$ $\forall x$,

then $-1 \leq x^2 - \cos^2 x$, and $x^2 - \cos^2 x \to \infty$ as $x \to \infty$

$\therefore$ $(0,1)$ is an absolute minimum on $f|S$, no max.

11. **11.** $f: \mathbb{R}^3 \to \mathbb{R}, (x, y, z) \mapsto x^2 + y^2 + z^2, S = \{(x, y, z) \mid z \geq 2 + x^2 + y^2\}$

Let $g(x,y,z) = z - x^2 - y^2 - 2$

$U = \{(x,y,z) \mid g(x,y,z) > 0\}$, $\partial U = \{(x,y,z) \mid g(x,y,z) = 0\}$.

Note f is not bounded on U or ∂U.

U: $f_x = 2x$ $f_y = 2y$ $f_z = 2z$ $\therefore (0,0,0)$ a possible critical point, but $(0,0,0) \notin U$.

∂U: $\nabla f = \lambda \nabla g \Rightarrow \langle 2x\ 2y\ 2z \rangle = \lambda \langle -2x\ -2y\ 1 \rangle$

Since $(0,0,0) \notin S$, $\lambda \neq 0$.

$\therefore$ $\left.\begin{aligned} 2x &= -2\lambda x \\ 2y &= -2\lambda y \\ 2z &= \lambda \Rightarrow z \neq 0. \end{aligned}\right\}$ $\begin{aligned} 2x &= -4z x \quad \text{or} \quad x = -2z x \\ 2y &= -4z y \qquad\quad y = -2z y \end{aligned}$

and $z = 2 + x^2 + y^2$

$x = 0, y = 0$: $z = 2$ $(\lambda = 4)$ $\therefore$ $(0,0,2)$ possible

$x = 0, y \neq 0$: $\therefore 2 = -2\lambda \Rightarrow \lambda = -1$,
$2z = \lambda \Rightarrow z = -\frac{1}{2}$
Not possible since $z = 2 + x^2 + y^2 > 0$

$x \neq 0, y = 0$: Same as above, $\lambda = -1$, $z = -\frac{1}{2}$.

$x \neq 0, y \neq 0$: Same as above, $\lambda = -1$, $z = -\frac{1}{2}$

$\therefore$ $(0,0,2)$ only critical point

Note for S, $z \geq x^2+y^2+2 \Rightarrow z \geq 2$

$\therefore x^2+y^2+z^2 \geq 4$

$\therefore$ $f(0,0,2)=4$ an absolute minimum for $f|_S$

12. **12.** Use the method of Lagrange multipliers to find the absolute maximum and minimum values of

$f(x, y) = x^2 + y^2 - x - y + 1$ on the unit disc (see Example 10 of Section 3.3).

Let $U = \{(x,y) \mid x^2+y^2 < 1\}$ $\partial U = \{(x,y) \mid x^2+y^2 = 1\}$.

Both U and ∂U are bounded.

U: $f_x = 2x-1$ $f_y = 2y-1$ $\therefore \left(\frac{1}{2}, \frac{1}{2}\right)$ a critical pt.

$f_{xx} = 2$ $f_{xy} = 0$ $f_{yy} = 2$

Hessian: $\begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4 > 0$, $f_{xx} > 0$

$\therefore \left(\frac{1}{2}, \frac{1}{2}\right)$ a local minimum, $f\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$

∂U: $g(x,y) = x^2+y^2$ $\nabla g = [2x \; 2y]$.

$\therefore [2x-1 \;\; 2y-1] = \lambda [2x \; 2y]$

$\lambda \neq 0$, as $\left(\frac{1}{2}, \frac{1}{2}\right)$ is not on ∂U.

$\therefore \begin{cases} 2x-1 = 2\lambda x \\ 2y-1 = 2\lambda y \end{cases}$ so $\lambda \neq 1$, otherwise $-1=0$.

$\therefore \begin{cases} 2x(1-\lambda)=1 \\ 2y(1-\lambda)=1 \end{cases} \Rightarrow x=y. \quad \therefore x^2+y^2=1 \Rightarrow x=\pm\sqrt{2}/2$
$y=\pm\sqrt{2}/2$

$\therefore \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ critical points.

$\therefore f\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = \frac{1}{2}+\frac{1}{2}-\frac{\sqrt{2}}{2}-\frac{\sqrt{2}}{2}+1 = 2-\sqrt{2} > \frac{1}{2}$

$f\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = \frac{1}{2}+\frac{1}{2}+\frac{\sqrt{2}}{2}+\frac{\sqrt{2}}{2}+1 = 2+\sqrt{2}$

$\therefore$ Absolute minimum at $\left(\frac{1}{2}, \frac{1}{2}\right)$, $f\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2}$

Absolute maximum at $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$, $f\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = 2+\sqrt{2}$

13.

13. Consider the function $f(x, y) = x^2 + xy + y^2$ defined on the unit disc, namely, $D = \{(x, y) \mid x^2 + y^2 \leq 1\}$. Use the method of Lagrange multipliers to locate the maximum and minimum points for f on the unit circle. Use this to determine the absolute maximum and minimum values for f on D.

Let $U = \{(x,y) \mid x^2+y^2 < 1\}$, $\partial U = \{(x,y) \mid x^2+y^2 = 1\}$

Let $g(x,y) = x^2+y^2$ U is open, ∂U is closed, bounded.

U: $f_x = 2x+y$ $f_y = x+2y$ $\therefore \begin{matrix} 2x+y=0 \\ x+2y=0 \end{matrix} \Rightarrow (x,y)=(0,0)$.

Hessian: $f_{xx}=2$, $f_{xy}=1$, $f_{yy}=2$ $\begin{vmatrix} 2 & 1 \\ 1 & 2 \end{vmatrix} = 3 > 0$

As $f_{xx} > 0$, $(0,0)$ is a local minimum

$f(0,0) = 0$

∂U: $\nabla g = [2x \;\; 2y]$

$\therefore \nabla f = \lambda \nabla g \Rightarrow [2x+y \;\; x+2y] = \lambda [2x \;\; 2y]$

$\lambda \neq 0$ as $(x,y) = (0,0) \notin \partial U$.

$\left.\begin{aligned} 2x+y &= 2\lambda x \\ x+2y &= 2\lambda y \\ \text{and } x^2+y^2 &= 1. \end{aligned}\right\} \Rightarrow \begin{aligned} &\text{if } x=0 \text{ then } y=0 \\ &\text{if } y=0 \text{ then } x=0. \\ &\text{and } \therefore \lambda \neq 1. \end{aligned}$

$\left.\begin{aligned} y &= 2x(\lambda-1) \\ x &= 2y(\lambda-1) \end{aligned}\right\} \; y = 4y(\lambda-1)^2, \; (\lambda-1)^2 = \frac{1}{4}, \; \lambda = 1 \pm \frac{1}{2}$

$\therefore \lambda = \frac{1}{2}, \frac{3}{2}$

$\lambda = \frac{1}{2}$: $\left.\begin{aligned} 2x+y &= x \\ x+2y &= y \end{aligned}\right\} \Rightarrow x = -y, \; \therefore x^2+y^2 = 1 \Rightarrow$

$(x,y) = \left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right), \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

$f\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = \frac{1}{2} - \frac{1}{2} + \frac{1}{2} = \frac{1}{2}$

$f\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = \frac{1}{2} - \frac{1}{2} + \frac{1}{2} = \frac{1}{2}$

$\lambda = \frac{3}{2}$: $\left.\begin{aligned} 2x+y &= 3x \\ x+2y &= 3y \end{aligned}\right\} \; x = y, \; x^2+y^2 = 1 \Rightarrow$

$(x,y) = \pm\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

$$f\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right) = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

$$f\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2}$$

$\therefore$ Absolute min : 0 at $(0,0)$

Absolute max : $\frac{3}{2}$ at $\pm\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

14. **14.** Find the absolute maximum and minimum values of $f(x, y, z) = 2x + y$, subject to the constraint $x + y + z = 1$.

Intuitively, two planes intersecting in a line.

Let $g(x,y,z) = x+y+z$, $g(x,y,z) = 1$. $\nabla g = [1\ 1\ 1]$

$\nabla f = [2\ 1\ 0]$

$\therefore \nabla f = \lambda \nabla g \Rightarrow [2\ 1\ 0] = \lambda [1\ 1\ 1] \Rightarrow \lambda = 0$.

The test is not helpful.

Note: for $x+y+z=1$, $x+y = z-1$

$\therefore f(x,y,z) = x+z-1$, and this is not bounded.

$\therefore$ There is no absolute max or min.

15. **15.** Find the extrema of $f(x, y) = 4x + 2y$, subject to the constraint $2x^2 + 3y^2 = 21$.

Let $g(x,y) = 2x^2 + 3y^2$ $\therefore \nabla g = [4x \;\; 6y]$

$\nabla f = [4 \;\; 2]$ $\therefore [4 \;\; 2] = \lambda [4x \;\; 6y] \Rightarrow \lambda \neq 0$.

$$\left.\begin{aligned} 4 &= 4\lambda x \\ 2 &= 6\lambda y \end{aligned}\right\} \quad 4\lambda x = 12\lambda y, \; x = 3y$$

$\therefore 2x^2 + 3y^2 = 2(3y)^2 + 3y^2 = 21y^2 = 21, \; y = \pm 1$

$\therefore (x,y) = \pm(3,1)$

Note: $S = g(x,y) = 21$ is closed, bounded.

$\therefore f$ (continuous) assumes a max, min on S.

$f(3,1) = 4(3) + 2(1) = 14$, a max

$f(-3,-1) = -14$, a min

Note: $(0,0)$, our answer in back of textbook is not on the constraint.

16. **16.** Use Lagrange multipliers to find the distance from the point $(2, 0, -1)$ to the plane $3x - 2y + 8z + 1 = 0$. Compare your answer to Example 12 in Section 1.3.

Let $f(x,y,z) = (x-2)^2 + y^2 + (z+1)^2$ (distance function^2)

Let constraint $g(x,y,z) = 3x - 2y + 8z = -1$

$$\therefore \nabla g = [3 \;\; -2 \;\; 8]$$

$$\nabla f = [2x-4 \;\; 2y \;\; 2z+2] = \lambda [3 \;\; -2 \;\; 8]$$

$$\begin{aligned} \therefore \; 2x - 4 &= 3\lambda \\ 2y &= -2\lambda \\ 2z + 2 &= 8\lambda \\ 3x - 2y + 8z &= -1 \end{aligned} \qquad \begin{bmatrix} 2 & 0 & 0 & -3 \\ 0 & 2 & 0 & 2 \\ 0 & 0 & 2 & -8 \\ 3 & -2 & 8 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ \lambda \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -2 \\ -1 \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} 2 & 0 & 0 & -3 & 4 \\ 0 & 2 & 0 & 2 & 0 \\ 0 & 0 & 2 & -8 & -2 \\ 0 & -2 & 8 & 9/2 & -7 \end{bmatrix} \Leftrightarrow \begin{bmatrix} 2 & 0 & 0 & -3 & 4 \\ 0 & 2 & 0 & 2 & 0 \\ 0 & 0 & 2 & -8 & -2 \\ 0 & 0 & 8 & 13/2 & -7 \end{bmatrix} \Leftrightarrow$$

$$\begin{bmatrix} 2 & 0 & 0 & -3 & 4 \\ 0 & 2 & 0 & 2 & 0 \\ 0 & 0 & 2 & -8 & -2 \\ 0 & 0 & 0 & \frac{77}{2} & 1 \end{bmatrix}$$

$$\therefore \frac{77}{2}\lambda = 1, \quad \lambda = \frac{2}{77}$$

$$2z - 8\left(\frac{2}{77}\right) = -2, \quad z = -\frac{69}{77}$$

$$2y + 2\left(\frac{2}{77}\right) = 0, \quad y = -\frac{2}{77}$$

$$2x - 3\left(\frac{2}{77}\right) = 4, \quad x = \frac{157}{77}$$

$$\therefore \text{Only critical point is: } (x,y,z) = \left(\frac{157}{77}, -\frac{2}{77}, -\frac{69}{77}\right)$$

$$f\left(\frac{157}{77}, -\frac{2}{77}, -\frac{69}{77}\right) = \left(\frac{157}{77} - 2\right)^2 + \left(-\frac{2}{77}\right)^2 + \left(-\frac{69}{77} + 1\right)^2$$

$$= \frac{9}{77^2} + \frac{4}{77^2} + \frac{64}{77^2} = \frac{77}{77^2} = \frac{1}{77} = \text{distance}^2$$

$\therefore$ Distance at critical point $= \underline{\underline{\frac{1}{\sqrt{77}}}}$.

From geometric considerations, this is a minimum.

Compared to Example 12, section 1.3, the distance is: $\frac{|Ax_1 + By_1 + Cz_1 + D|}{\sqrt{A^2+B^2+C^2}} = \frac{|(3,-2,8)\cdot(2,0,-1)+1|}{\sqrt{3^2+(-2)^2+8^2}} =$

$$\frac{|6+0-8+1|}{\sqrt{9+4+64}} = \frac{1}{\sqrt{77}}$$

17. **17.** Find the maximum and minimum values attained by $f(x, y, z) = xyz$ on the unit ball $x^2 + y^2 + z^2 \leq 1$.

On $x^2+y^2+z^2<1$: $f_x = yz$ $f_y = xz$ $f_z = xy$

$f_x = 0$: if $y \neq 0$, then $x=0, z=0$
$\therefore$ $(0, y, 0)$ critical points.

Similarly for $f_y = 0$, $f_z = 0$, $(x,0,0)$, $(0,0,z)$ critical points.

For all these critical points, $f(x,y,z) = xyz = 0$.
$\therefore$ 0 is clearly a minimum.

On $x^2+y^2+z^2=1$: $\nabla f = [yz, xz, xy] = \lambda \nabla g = \lambda [2x, 2y, 2z]$

If $\lambda = 0$, then $(\pm 1, 0, 0)$, $(0, \pm 1, 0)$, $(0, 0, \pm 1)$ possible.

These all gives $f(x,y,z)=0$

If $\lambda \neq 0$:

[1] $yz = 2\lambda x$ $\quad z(y-x) = 2\lambda(x-y)$, $(z+2\lambda)(y-x)=0$

[2] $xz = 2\lambda y$

[3] $xy = 2\lambda z$ $\quad x(z-y) = 2\lambda(y-z)$, $(x+2\lambda)(z-y)=0$

[4] $x^2+y^2+z^2=1$ $\quad y(z-x) = 2\lambda(x-z)$, $(y+2\lambda)(z-x)=0$

$\therefore (z+2\lambda)(x-y)=0.$

If $x=y$, assume $x,y \neq 0$ as $f(x,y,z)=0$.

$\therefore$ [1] $\Rightarrow z=2\lambda$, $\therefore$ [3] $\Rightarrow x^2 = 4\lambda^2$

and [4] $\Rightarrow 2x^2+4\lambda^2=1$. $\therefore 12\lambda^2=1$

$\lambda = \pm\sqrt{\frac{1}{12}} = \pm\frac{1}{2\sqrt{3}}$

$\therefore z = \pm\frac{1}{\sqrt{3}}$, $\therefore$ [4] $\Rightarrow 2x^2 = \frac{2}{3}$, $x = \pm\frac{1}{\sqrt{3}}$

$\therefore (x,y) = \pm\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$.

$\therefore$ Possibilities: $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}\right), \left(-\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}\right)$

Similarly for $x = \pm\frac{1}{\sqrt{3}}$, $y = \pm\frac{1}{\sqrt{3}}$

$\therefore f(x,y,z) = \pm\left|\frac{1}{\sqrt{3}}\cdot\frac{1}{\sqrt{3}}\cdot\frac{1}{\sqrt{3}}\right| = \pm\frac{1}{3\sqrt{3}}$. These are greater than and less than 0.

$\therefore$ On $x^2+y^2+z^2 \le 1$, $f(x,y,z)$ $\underline{\max} = 3\sqrt{3}$

$\underline{\min} = -\frac{1}{3\sqrt{3}}$

18.

18. Let S be the sphere of radius 1 centered at (1, 2, 3). Find the distance from S to the plane $x + y + z = 0$. (HINT: Use Lagrange multipliers to find the distance from the plane to the center of the sphere.)

$S: (x-1)^2+(y-2)^2+(z-3)^2$, centered at $(1,2,3)$

The minimum distance between S and $P = x+y+z=0$ will be on a line perpendicular to S and P, and this line will include center of S.

$\therefore$ Find (distance)^2 from plane to $(1,2,3)$ and minimize it.

$d^2 = (x-1)^2+(y-2)^2+(z-3)^2$, constraint $g(x,y,z) = x+y+z=0$.

$\therefore \nabla d^2 = [2x-2, 2y-4, 2z-6] = \lambda \nabla g = \lambda[1,1,1]$

$\therefore$ $2x-2=\lambda$, $2y-4=\lambda$, $2z-6=\lambda$ and $x+y+z=0$

$$\therefore \begin{bmatrix} 2 & 0 & 0 & -1 \\ 0 & 2 & 0 & -1 \\ 0 & 0 & 2 & -1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ \lambda \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 6 \\ 0 \end{bmatrix} \Leftrightarrow$$

$$\begin{bmatrix} 2 & 0 & 0 & -1 & 2 \\ 0 & 2 & 0 & -1 & 4 \\ 0 & 0 & 2 & -1 & 6 \\ 0 & 1 & 1 & \frac{1}{2} & -1 \end{bmatrix} \Leftrightarrow \begin{bmatrix} 2 & 0 & 0 & -1 & 2 \\ 0 & 2 & 0 & -1 & 4 \\ 0 & 0 & 2 & -1 & 6 \\ 0 & 0 & 1 & 1 & -3 \end{bmatrix} \Leftrightarrow$$

$$\begin{bmatrix} 2 & 0 & 0 & -1 & 2 \\ 0 & 2 & 0 & -1 & 4 \\ 0 & 0 & 2 & -1 & 6 \\ 0 & 0 & 0 & \frac{3}{2} & -6 \end{bmatrix} \Rightarrow \tfrac{3}{2}\lambda = -6,\ \lambda = -4$$

$\therefore 2z + 4 = 6,\ z = 1$

$2y + 4 = 4,\ y = 0$

$2x + 4 = 2,\ x = -1$

$\therefore (1, 0, -1)$ on P and $d^2 = (1-1)^2 + (0-2)^2 + (-1-3)^2 = 20$

$\therefore$ Distance from P to center of $S = \sqrt{20}$

$\therefore$ Distance from P to $S = \sqrt{20} - 1 = \underline{2\sqrt{5} - 1}$

19.

19. (a) Find three numbers whose product is 27 and whose sum is minimal.

(b) Find three numbers whose sum is 27 and whose product is maximal.

(a) $f(x, y, z) = x + y + z$ subject to $g(x, y, z) = xyz = 27$.

$\therefore \nabla f = [1\ 1\ 1] = \lambda \nabla g = \lambda [yz\ \ xz\ \ xy]$. $\therefore \lambda \neq 0$

Also, $x, y, z \neq 0$.

$$\left.\begin{aligned} 1 &= \lambda yz \\ 1 &= \lambda xz \\ 1 &= \lambda xy \end{aligned}\right\} \therefore x = y = z = 3$$

$\therefore \underline{3, 3, 3}$ Sum $= 9$

(6) $f(x,y,z) = xyz$ subject to $g(x,y,z) = x+y+z = 27$

$\therefore \nabla f = [yz \;\; xz \;\; xy] = \lambda \nabla g = \lambda [1 \;\; 1 \;\; 1]$

Note, $x,y,z \neq 0$ as product would be 0.
$\therefore \lambda \neq 0$.

$$\left.\begin{aligned} yz &= \lambda \\ xz &= \lambda \\ xy &= \lambda \end{aligned}\right\} \therefore x = y = z = 9$$

$\therefore$ 9, 9, 9 product = 729

20. **20.** A rectangular box with no top is to have a surface area of 16 m^2. Find the dimensions that maximize its volume.

Assume "surface area" means surfaces exposed, so each side has 2 surfaces exposed (inside & outside).

Assume dimensions of w = width, d = depth, h = height.

$\therefore$ Volume = $V(w,d,h) = wdh$

Surface area = $S(w,d,h) = 2wd + 4wh + 4dh = 16$

$V(w,d,h)$ unbounded above on R^3 domain.
($V = 0$ a clear minimum).

$\therefore$ Look at "boundary", or constraint, and assume $w,d,h \neq 0$

$\nabla V = [dh, \; wh, \; wd] = \lambda \nabla S = \lambda [2d+4h, \; 2w+4h, \; 4w+4d]$

$\therefore \lambda \neq 0$

$\therefore$ [1] $dh = 2\lambda d + 4\lambda h$

[2] $wh = 2\lambda w + 4\lambda h$

[3] $wd = 4\lambda w + 4\lambda d$

[1] + [2] $\Rightarrow h(d-w) = 2\lambda(d-w)$

If $d \neq w$, then $h = 2\lambda$. Substituting in [1],

$2\lambda d = 2\lambda d + 8\lambda^2 \Rightarrow \lambda = 0$, a contradiction.

$\therefore d = w$

From [3], $d^2 = 8\lambda d$, $d = 8\lambda$, $\therefore 4\lambda = \frac{d}{2}$, $2\lambda = \frac{d}{4}$

$S(d, d, h) = 2(d)d + 4(d)h + 4dh = 2d^2 + 8dh = 16$,

[4] $\therefore d^2 + 4dh = 8$

From [2], $(w - 4\lambda)h = 2\lambda w$

$\therefore (d - \frac{d}{2})h = (\frac{d}{4})d$, as $w = d$, $4\lambda = \frac{d}{2}$, $2\lambda = \frac{d}{4}$

$\therefore h = \frac{2}{d} \cdot \frac{d}{4} d = \frac{d}{2}$.

$\therefore$ From [4], $d^2 + 4d(\frac{d}{2}) = 8$, $3d^2 = 8$

$\therefore d = \sqrt{\frac{8}{3}} = \frac{\sqrt{24}}{3} = \frac{2}{3}\sqrt{6}$.

$\therefore V = wdh = (\frac{2}{3}\sqrt{6})(\frac{2}{3}\sqrt{6})(\frac{\sqrt{6}}{2}) = \underline{\underline{\frac{24}{27}\sqrt{6}\ m^3}}$

21. **21.** Design a cylindrical can (with a lid) to contain 1 liter ($= 1000\ \text{cm}^3$) of water, using the minimum amount of metal.

Let dimensions be r = radius of lid
h = height of can.

Volume = $V(r,h) = \pi r^2 h = 1000\ \text{cm}^3$ $\quad \therefore r, h \neq 0$

Outside Surface area = $S(r,h) = 2\pi r^2 + 2\pi r h$
and S is a measure of amount of metal used.

$\therefore$ Minimize S subject to $V = 1000$

$S(r,h)$ not bounded above on R^2. $\therefore$ Look at "boundary" of constraint.

$$\therefore \nabla S(r,h) = [4\pi r + 2\pi h,\ 2\pi r] = \lambda \nabla V(r,h) = \lambda [2\pi r h,\ \pi r^2]$$

$$\therefore [1]\ 4\pi r + 2\pi h = 2\pi\lambda r h, \text{ or } 2r + h = \lambda r h$$
$$[2]\ \ 2\pi r = \pi \lambda r^2 \qquad\qquad 2 = \lambda r$$

$\therefore 2r + h = 2h$, $2r = h$ (diameter = height).

$$\therefore V = \pi r^2 h = \pi \left(\frac{d}{2}\right)^2 d = 1000,\ d^3 = \frac{4000}{\pi}$$

$\therefore$ Diameter = height, diameter $= 10\sqrt[3]{\frac{4}{\pi}}$

Note $10\sqrt[3]{\frac{4}{\pi}} = 10\sqrt[3]{\frac{8}{2\pi}} = \frac{20}{\sqrt[3]{2\pi}}$

22.

22. Show that solutions of equations (4) and (5) are in one-to-one correspondence with the critical points of

$$h(x_1, \ldots, x_n, \lambda_1, \ldots, \lambda_k) = f(x_1, \ldots, x_n) - \lambda_1[g_1(x_1, \ldots, x_n) - c_1] - \cdots - \lambda_k[g_k(x_1, \ldots, x_n) - c_k].$$

(4) & (5) are:

$$\left.\begin{aligned} g_1(x_1, \ldots, x_n) &= c_1 \\ g_2(x_1, \ldots, x_n) &= c_2 \\ &\vdots \\ g_k(x_1, \ldots, x_n) &= c_k \end{aligned}\right\}, \tag{4}$$

$$\nabla f(\mathbf{x}_0) = \lambda_1 \nabla g_1(\mathbf{x}_0) + \cdots + \lambda_k \nabla g_k(\mathbf{x}_0). \tag{5}$$

The critical points of $h()$ are where the partial derivatives are 0; i.e., $\nabla h = \vec{0}$

But $\frac{\partial h}{\partial x_i} = \frac{\partial f}{\partial x_i} - \lambda_1 \frac{\partial g_1}{\partial x_i} - \ldots - \lambda_k \frac{\partial g_k}{\partial x_i}$ [1]

and $\frac{\partial h}{\partial \lambda_j} = \frac{\partial f}{\partial \lambda_j} - [g_j - c_j] = -[g_j - c_j]$, as $\frac{\partial f}{\partial \lambda_j} = 0$ [2]

Looking at right side of [1] set to 0,

$\frac{\partial f}{\partial x_i} = \nabla f(\vec{x}_0)_i$, the "$i$"th component, $\vec{x}_0 = (x_1, \ldots, x_n)$

Similarly, $-\lambda_1 \frac{\partial g_1}{\partial x_i} - \ldots - \lambda_k \frac{\partial g_k}{\partial x_i} = [\lambda_1 \nabla g_1(\vec{x}_0) + \ldots + \lambda_k \nabla g_k(\vec{x}_0)]_i$

The "i"th component

$\therefore \forall i = 1, \ldots, n$, [1] is $\nabla h_i = [\nabla f - \lambda_1 g_1 - \ldots - \lambda_k g_k]_i = 0$

$\Leftrightarrow \nabla f_i = [\lambda_1 g_1 + \ldots + \lambda_k g_k]_i$

which is [5]

Looking at [2], $\frac{\partial h}{\partial \lambda_j} = 0 = -[g_j - c_j]$, or $g_j = c_j$

for $j = 1, \ldots, k$

For $\forall j = 1, \ldots, k$, $g_j = c_j$ is [4]

$\therefore$ For $j = 1, \ldots, k$, $\nabla h_j(\vec{x_0}, \lambda_1, \ldots, \lambda_k) = 0 \iff g_j(\vec{x_0}) = c_j$

which is [4]

$\therefore \nabla h = \vec{0} \iff$ [4] and [5]

23. **23.** Find the absolute maximum and minimum for the function $f(x, y, z) = x + y - z$ on the ball $B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$.

For $x^2 + y^2 + z^2 < 1$, $\nabla f = [1, 1, -1] \neq 0$ for any x, y, z.

$\therefore$ No critical points at interior of ball

For $x^2 + y^2 + z^2 = 1$, $\nabla f = [1, 1, -1] = \lambda [2x, 2y, 2z]$

so $\lambda \neq 0$.

$\therefore$ $2\lambda x = 1$ [1]

$2\lambda y = 1$ [2] [1] $\div$ [2] $\Rightarrow x = y$

$2\lambda z = -1$ [3] [2] $\div$ [3] $\Rightarrow y = -z$

$\therefore x^2 + x^2 + (-x)^2 = 1$, $3x^2 = 1$, $x = \pm \frac{\sqrt{3}}{3}$

$\therefore$ Critical points: $\left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right)$, $\left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right)$

$f\left(\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}\right) = \underline{\sqrt{3}}$ = absolute max.

$f\left(-\frac{\sqrt{3}}{3}, -\frac{\sqrt{3}}{3}, \frac{\sqrt{3}}{3}\right) = \underline{-\sqrt{3}}$ = absolute min

24. **24.** Repeat Exercise 23 for $f(x, y, z) = x + yz$.

$\nabla f = [1, z, y]$

For $x^2+y^2+z^2<1$, no critical points since $\nabla f \neq \vec{0}$ for any $(x,0,0)$

For $x^2+y^2+z^2=1$, $[1, z, y] = \lambda[2x, 2y, 2z]$, $\therefore \lambda \neq 0$

$\therefore$ $2\lambda x = 1$ [1]
$2\lambda y = z$ [2]
$2\lambda z = y$ [3]
and $x^2+y^2+z^2=1$ [4]

[2] & [3] $\Rightarrow z^2 = y^2$ If $z=y=0$, then $x^2+0+0=1$, $x=\pm 1$

$\therefore$ Possible critical points: $(1,0,0)$, $(-1,0,0)$

$\therefore$ $\underline{f(1,0,0)=1 \quad f(-1,0,0)=-1}$

Assume $y \neq 0$, $z \neq 0$.

$x=0$ $\therefore$ [4] becomes $2y^2=1$, $y = \pm\frac{\sqrt{2}}{2}$

$\therefore \left(0, \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(0, \frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right), \left(0, -\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right), \left(0, -\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

are possible critical points.

$\therefore$ $f(x,y,z) = \pm\frac{1}{2}$

$x \neq 0$: [1] $\Rightarrow$ $2\lambda = \frac{1}{x}$

$\therefore$ [2] $\Rightarrow$ $y = zx$

[3] $\Rightarrow$ $z = yx$

$\Rightarrow y = yx^2,\ x^2 = 1$

$x = \pm 1.$

$\therefore$ [4] becomes $1 + 2y^2 = 1,\ y = z = 0,$

but not a possibility as assumed $y \neq 0,\ z \neq 0.$

$\therefore$ Absolute max at $(1,0,0)$ and $f(1,0,0) = 1$

Absolute min at $(-1,0,0)$ and $f(-1,0,0) = -1$

25. **25.** A rectangular mirror with area A square feet is to have trim along the edges. If the trim along the horizontal edges costs p cents per foot and that for the vertical edges costs q cents per foot, find the dimensions that will minimize the total cost.

Let h = horiz. length, v = vertical length.

$\therefore$ Area $= A(h,v) = hv = A$

Trim $= T(h,v) = 2ph + 2qv$, assume $p \neq 0,\ q \neq 0.$

$\therefore$ Minimize $T(h,v)$ subject to $Area(h,v) = A$

Assume $h \neq 0, v \neq 0$ since $hv = A \neq 0$.

On open R^2, $T_h = 2p$, $T_v = 2q$.

$\therefore$ No critical points s.t. $\nabla T = \vec{0}$.

Boundary: $\nabla T = [2p \;\; 2q] = \lambda \nabla Area = \lambda [v \;\; h]$. $\therefore \lambda \neq 0$

$\therefore$ $2p = \lambda v$ [1]

$2q = \lambda h$ [2] and $hv = A$ [3]

[1] & [2] $\Rightarrow \frac{2p}{v} = \frac{2q}{h}$, $\therefore ph = qv$, $h = \frac{qv}{p}$, $v = \frac{ph}{q}$

$\therefore$ [3] $\Rightarrow \left(\frac{qv}{p}\right) v = A$, $v^2 = \frac{pA}{q}$, $v = \underline{\sqrt{\frac{pA}{q}}}$

and $h\left(\frac{ph}{q}\right) = A$, or $h^2 = \frac{qA}{p}$, $h = \underline{\sqrt{\frac{qA}{p}}}$

But "boundary" $Area(h,v) = hv = A$ is not a bounded domain. So is critical point a max or min? Look at bordered Hessian.

Let $H(h,v) = T(h,v) - \lambda Area(h,v)$

From above, $2p = \lambda \sqrt{\frac{pA}{q}}$, $\lambda = \frac{2p}{\sqrt{\frac{pA}{q}}}$, or

$$\lambda = 2\sqrt{\frac{pq}{A}} \quad \therefore H_h = T_h - \lambda\, Area_h = 2p - \lambda v$$

$$H_{hh} = 0$$

$$H_v = T_v - \lambda\, Area_v = 2q - \lambda h \quad H_{vv} = 0$$

$$H_{hv} = -\lambda$$

$$\therefore |H| = \begin{vmatrix} 0 & -v & -h \\ -v & 0 & -\lambda \\ -h & -\lambda & 0 \end{vmatrix} = 0 + v(0 - h\lambda) + (-h)(\lambda v)$$

$$= -2\lambda h v < 0.$$

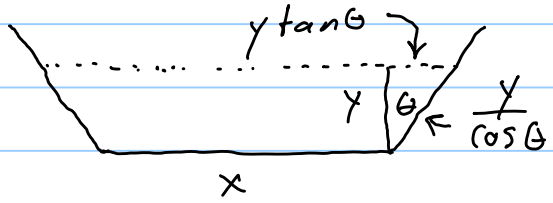
$\therefore \left(\sqrt{\frac{qA}{p}}, \sqrt{\frac{pA}{q}}\right)$ is a local min on $Area(h,v) = A$.

$\therefore h = \sqrt{\frac{qA}{p}}, \; v = \sqrt{\frac{pA}{q}}$ minimizes cost.

26.

26. An irrigation canal in Arizona has concrete sides and bottom with trapezoidal cross section of area $A = y(x + y\tan\theta)$ and wetted perimeter $P = x + 2y/\cos\theta$, where x = bottom width, y = water depth, and θ = side inclination, measured from vertical.

The best design for a fixed inclination θ is found by solving P = minimum subject to the condition A = constant. Show that $y^2 = (A\cos\theta)/(2 - \sin\theta)$.



Area of trapezoid is (ave of bases)(height)

$$= \left[\frac{x + (x + 2y\tan\theta)}{2}\right](y) = y(x + y\tan\theta)$$

Wetted perimeter $= x + 2\left(\frac{y}{\cos\theta}\right) = x + \frac{2y}{\cos\theta}$

$\therefore$ Minimize $P(x,y) = x + \frac{2y}{\cos\theta}$

with constraint $Area(x,y) = xy + y^2\tan\theta = A$

Assume $x, y \neq 0$. $P(x,y)$ not bounded above on R^2

$\therefore$ Look at "boundary" for possible critical points.

$\nabla P = \left[1, \frac{2}{\cos\theta}\right] = \lambda \nabla Area = \lambda\left[y, x + 2y\tan\theta\right]$

$\therefore \lambda \neq 0$ as $P_x = 1$.

$\therefore$ [1] $1 = \lambda y$, or $\lambda = \frac{1}{y}$

[2] $\frac{2}{\cos\theta} = \lambda x + (2\lambda\tan\theta)y$

[3] $A = xy + y^2\tan\theta$

[1] & [2] $\Rightarrow \frac{2}{\cos\theta} = \frac{x}{y} + 2\tan\theta$

or $2y = x\cos\theta + 2y\sin\theta$

or $\frac{2y(1-\sin\theta)}{\cos\theta} = x$ [4]

[3] & [4] $\Rightarrow A = \frac{2y^2(1-\sin\theta)}{\cos\theta} + y^2\tan\theta$

or $A\cos\theta = 2y^2(1-\sin\theta) + y^2\sin\theta$

$$\therefore A\cos\theta = y^2[2 - 2\sin\theta + \sin\theta]$$

$$\therefore y^2 = \frac{A\cos\theta}{2 - \sin\theta}$$

Note: This is only possible critical point. From physical considerations, if it is a minimum, as perimeter (mainly x dimension) can be made as large as possible given any A.

27. **27.** Apply the second-derivative test to study the nature of the extrema in Exercises 3 and 7.

3. $f(x, y, z) = x - y + z$, subject to $x^2 + y^2 + z^2 = 2$

Let $h(x,y,z) = f(x,y,z) - \lambda g(x,y,z)$

$f_x = 1 \quad f_y = -1 \quad f_z = 1 \qquad g_x = 2x \quad g_y = 2y \quad g_z = 2z$

$\therefore h_{xx} = -2\lambda \quad h_{yy} = -2\lambda \quad h_{zz} = -2\lambda$

$h_{xy} = 0 \quad h_{xz} = 0 \quad h_{yz} = 0$

$\left(\frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}\right)$ was a max

$\left(-\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}\right)$ was a min

$\lambda = \frac{1}{2x}$

$$\therefore |\bar{H}| = \begin{vmatrix} 0 & -2x & -2y & -2z \\ -2x & -2\lambda & 0 & 0 \\ -2y & 0 & -2\lambda & 0 \\ -2z & 0 & 0 & -2\lambda \end{vmatrix}$$

For $\left(\frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}\right)$, a max: $\lambda = \frac{3}{2\sqrt{6}}$

$$|\bar{H}_3| = \begin{vmatrix} 0 & -\frac{2\sqrt{6}}{3} & \frac{2\sqrt{6}}{3} & -\frac{2\sqrt{6}}{3} \\ -\frac{2\sqrt{6}}{3} & -\frac{3}{\sqrt{6}} & 0 & 0 \\ \frac{2\sqrt{6}}{3} & 0 & -\frac{3}{\sqrt{6}} & 0 \\ -\frac{2\sqrt{6}}{3} & 0 & 0 & -\frac{3}{\sqrt{6}} \end{vmatrix}$$

$$= 0 + \frac{2\sqrt{6}}{3}\left[-\frac{2\sqrt{6}}{3}\cdot\frac{9}{6}\right] + \frac{2\sqrt{6}}{3}\left[\frac{3}{\sqrt{6}}\cdot\left(\frac{2\sqrt{6}}{3}\cdot\frac{-3}{\sqrt{6}}\right)\right] + \frac{2\sqrt{6}}{3}\left[\frac{3}{\sqrt{6}}\cdot\left(\frac{3}{\sqrt{6}}\cdot\frac{3}{\sqrt{6}}\cdot\frac{2\sqrt{6}}{3}\right)\right]$$

$$= -4 + (-4) + (\sqrt{6}) < 0$$

$$|\bar{H}_2| = \begin{vmatrix} 0 & -\frac{2\sqrt{6}}{3} & \frac{2\sqrt{6}}{3} \\ -\frac{2\sqrt{6}}{3} & -\frac{3}{\sqrt{6}} & 0 \\ \frac{2\sqrt{6}}{3} & 0 & -\frac{3}{\sqrt{6}} \end{vmatrix} = 0 + \frac{2\sqrt{6}}{3}\left(\frac{9}{6}\right) + \frac{2\sqrt{6}}{3}(0+2)$$

$$> 0$$

$\therefore |\bar{H}_2|, |\bar{H}_3| : >0, <0$

$\therefore \left(\frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}\right)$ a local max

For $\left(-\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}\right)$, a min: $\lambda = -\frac{3}{2\sqrt{6}}$

$$|\bar{H}_3| = \begin{vmatrix} 0 & -\frac{2\sqrt{6}}{3} & \frac{2\sqrt{6}}{3} & -\frac{2\sqrt{6}}{3} \\ -\frac{2\sqrt{6}}{3} & \frac{3}{\sqrt{6}} & 0 & 0 \\ \frac{2\sqrt{6}}{3} & 0 & \frac{3}{\sqrt{6}} & 0 \\ -\frac{2\sqrt{6}}{3} & 0 & 0 & \frac{3}{\sqrt{6}} \end{vmatrix} = 0$$

$$+ \frac{2\sqrt{6}}{3}\left[-\frac{2\sqrt{6}}{3} \cdot \frac{3}{\sqrt{6}} \cdot \frac{3}{\sqrt{6}}\right]$$

$$+ \frac{2\sqrt{6}}{3}\left[-\frac{3}{\sqrt{6}}\left(\frac{2\sqrt{6}}{3} \cdot \frac{3}{\sqrt{6}}\right)\right]$$

$$+ \frac{2\sqrt{6}}{3}\left[-\frac{3}{\sqrt{6}}\left(\frac{2\sqrt{6}}{3} \cdot \frac{3}{\sqrt{6}}\right)\right]$$

$$= (-) + (-) + (-) < 0$$

$$|\bar{H}_2| = \begin{vmatrix} 0 & -\frac{2\sqrt{6}}{3} & \frac{2\sqrt{6}}{3} \\ -\frac{2\sqrt{6}}{3} & \frac{3}{\sqrt{6}} & 0 \\ \frac{2\sqrt{6}}{3} & 0 & \frac{3}{\sqrt{6}} \end{vmatrix} = 0$$

$$+ \frac{2\sqrt{6}}{3}\left(-\frac{2\sqrt{6}}{3} \cdot \frac{3}{\sqrt{6}}\right)$$

$$+ \frac{2\sqrt{6}}{3}\left(-\frac{2\sqrt{6}}{3} \cdot \frac{3}{\sqrt{6}}\right) < 0$$

$\therefore |\bar{H}_2|, |\bar{H}_3|: <0, <0$

$\therefore \left(-\frac{\sqrt{6}}{3}, \frac{\sqrt{6}}{3}, -\frac{\sqrt{6}}{3}\right)$ a local min.

7. $f(x, y) = 3x + 2y$, subject to $2x^2 + 3y^2 = 3$

Let $L(x,y) = f(x,y) - \lambda g(x,y)$

$f_x = 3 \quad f_y = 2 \quad g_x = 2x \quad g_y = 6y$

$$h_{xx} = -2\lambda \quad h_{yy} = -6\lambda \quad h_{xy} = 0$$

$$\therefore |\bar{H}| = \begin{vmatrix} 0 & -2x & -6y \\ -2x & -2\lambda & 0 \\ -6y & 0 & -6\lambda \end{vmatrix}$$

For $\left(\frac{9}{\sqrt{70}}, \frac{4}{\sqrt{70}}\right)$, a max: $\lambda = \frac{3}{4x} = \frac{3}{36/\sqrt{70}} = \frac{\sqrt{70}}{12}$

$$|H_2| = \begin{vmatrix} 0 & -\frac{18}{\sqrt{70}} & -\frac{24}{\sqrt{70}} \\ -\frac{18}{\sqrt{70}} & -\frac{\sqrt{70}}{6} & 0 \\ -\frac{24}{\sqrt{70}} & 0 & -\frac{\sqrt{70}}{2} \end{vmatrix} = 0 + \frac{18}{\sqrt{70}}\left(\frac{18}{\sqrt{70}} \cdot \frac{\sqrt{70}}{2}\right) + \left(-\frac{24}{\sqrt{70}}\right)\left(0 - \frac{24}{\sqrt{70}} \cdot \frac{\sqrt{70}}{6}\right)$$

$$= \frac{162}{\sqrt{70}} + \frac{96}{\sqrt{70}} > 0$$

$$\therefore |H_2| > 0 \Rightarrow \left(\frac{9}{\sqrt{70}}, \frac{4}{\sqrt{70}}\right) \text{ a } \underline{\text{local max.}}$$

For $\left(-\frac{9}{\sqrt{70}}, -\frac{4}{\sqrt{70}}\right)$, a min: $\lambda = \frac{3}{4x} = -\frac{\sqrt{70}}{12}$

$$|H_2| = \begin{vmatrix} 0 & -\frac{18}{\sqrt{70}} & -\frac{24}{\sqrt{70}} \\ -\frac{18}{\sqrt{70}} & \frac{\sqrt{70}}{6} & 0 \\ -\frac{24}{\sqrt{70}} & 0 & \frac{\sqrt{70}}{2} \end{vmatrix} = 0 + \frac{18}{\sqrt{70}}\left(-\frac{18}{\sqrt{70}} \cdot \frac{\sqrt{70}}{2}\right) + \left(-\frac{24}{\sqrt{70}}\right)\left[0 - \left(-\frac{24}{\sqrt{70}} \cdot \frac{\sqrt{70}}{6}\right)\right]$$

$$= -\frac{162}{\sqrt{70}} - \frac{96}{\sqrt{70}} < 0$$

$\therefore |\bar{H}_2| < 0 \Rightarrow \left(-\frac{9}{\sqrt{70}}, -\frac{4}{\sqrt{70}}\right)$ a local min.

28.

28. A light ray travels from point A to point B crossing a boundary between two media (see Figure 3.4.7). In the first medium its speed is v_1, and in the second it is v_2. Show that the trip is made in minimum time when *Snell's law* holds:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}.$$

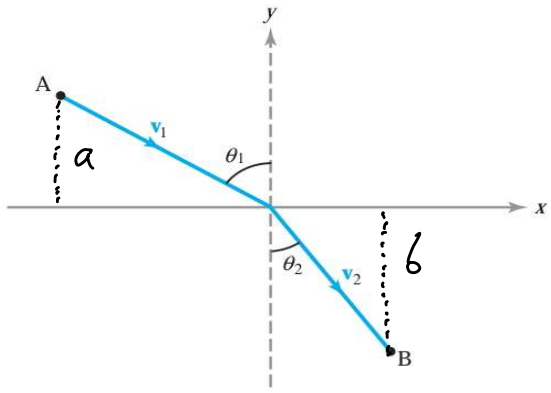


figure **3.4.7** Snell's law of refraction.

Note: interface is x-axis. θ's measured from $\perp$ to interface.

Let a = distance from A to interface parallel to y-axis

b = distance from B to interface parallel to y-axis

Since points A & B are fixed, their separation (y-coordinate) is fixed: a + b

It is the refraction point on the x-axis that may vary, and thus θ_1 and θ_2 vary.

The x-axis separation between A and B is also fixed, but each component may vary, along with θ_1 and θ_2

This x-axis separation = $a\tan\theta_1 + b\tan\theta_2$,

and is a constant. Assume $\theta_1 \neq 0$, $\theta_2 \neq 0$

$\therefore$ Let $g(\theta_1, \theta_2) = a\tan\theta_1 + b\tan\theta_2$

The distance for the V_1 trip is $\frac{a}{\cos\theta_1} = D_1$

The distance for the V_2 trip is $\frac{b}{\cos\theta_2} = D_2$

$\therefore$ Total time of light trip is $\frac{D_1}{V_1} + \frac{D_2}{V_2}$

$\therefore$ Total time of light trip is:

$$T(\theta_1, \theta_2) = \frac{a}{V_1\cos\theta_1} + \frac{b}{V_2\cos\theta_2}$$

$\therefore$ Minimize $T(\theta_1, \theta_2)$ subject to constraint

$$g(\theta_1, \theta_2) = a\tan\theta_1 + b\tan\theta_2 = \text{constant}$$

and a, b constants, V_1, V_2 constants

$\therefore$ Set $\nabla T = \lambda \nabla g$

$$T_{\theta_1} = -\frac{a}{V_1\cos^2\theta_1}\cdot -\sin\theta_1 = \frac{a\sin\theta_1}{V_1\cos^2\theta_1}$$

$$T_{\theta_2} = \frac{b\sin\theta_2}{V_2\cos^2\theta_2}$$

$$g_{\theta_1} = a\sec^2\theta_1 = \frac{a}{\cos^2\theta_1} \qquad g_{\theta_2} = \frac{b}{\cos^2\theta_2}$$

Note $\nabla g = \left[\frac{a}{\cos^2\theta_1}, \frac{b}{\cos^2\theta_2}\right] \neq \vec{0}$

$$\therefore \left[\frac{a\sin\theta_1}{v_1\cos^2\theta_1}, \frac{b\sin\theta_2}{v_2\cos^2\theta_2}\right] = \lambda\left[\frac{a}{\cos^2\theta_1}, \frac{b}{\cos^2\theta_2}\right]$$

$$\therefore \frac{a\sin\theta_1}{v_1\cos^2\theta_1} = \frac{\lambda a}{\cos^2\theta_1} \qquad \frac{b\sin\theta_2}{v_2\cos^2\theta_2} = \frac{\lambda b}{\cos^2\theta_2}$$

$$\text{or,} \quad \frac{\sin\theta_1}{v_1} = \lambda \qquad \frac{\sin\theta_2}{v_2} = \lambda$$

$$\therefore \frac{\sin\theta_1}{v_1} = \frac{\sin\theta_2}{v_2}, \text{ or } \underline{\frac{v_2}{v_1} = \frac{\sin\theta_2}{\sin\theta_1}}$$

This is a critical point. To show it is a minimum, look at bordered Hessian.

Let $h(\theta_1,\theta_2) = T(\theta_1,\theta_2) - \lambda g(\theta_1,\theta_2)$

From above: $T_{\theta_1} = \frac{a\sin\theta_1}{v_1\cos^2\theta_1}$ $T_{\theta_2} = \frac{b\sin\theta_2}{v_2\cos^2\theta_2}$

$$g_{\theta_1} = \frac{a}{\cos^2\theta_1} \qquad g_{\theta_2} = \frac{b}{\cos^2\theta_2}$$

$$\therefore T_{\theta_1\theta_1} = \frac{v_1\cos^2\theta_1(a\cos\theta_1) - (a\sin\theta_1)(-2v_1\cos\theta_1\sin\theta_1)}{v_1^2\cos^4\theta_1}$$

$$= \frac{av_1\cos\theta_1\left[\cos^2\theta_1 + 2\sin^2\theta_1\right]}{v_1^2\cos^4\theta_1} = \frac{a\left[1+\sin^2\theta_1\right]}{v_1\cos^3\theta_1}$$

$$T_{\theta_1\theta_2} = 0$$

$$T_{\theta_2\theta_2} = \frac{V_2\cos^2\theta_2\,(b\cos\theta_2) - (b\sin\theta_2)(-2V_2\cos\theta_2\sin\theta_2)}{V_2^2\cos^4\theta_2}$$

$$= \frac{b\left[1+\sin^2\theta_2\right]}{V_2\cos^3\theta_2}$$

$$g_{\theta_1\theta_1} = -\frac{a}{\cos^4\theta_1}(-2\cos\theta_1\sin\theta_1) = \frac{2a\sin\theta_1}{\cos^3\theta_1}$$

$$g_{\theta_2\theta_2} = \frac{2b\sin\theta_2}{\cos^3\theta_2}$$

$$g_{\theta_1\theta_2} = 0$$

Note $\lambda = \frac{\sin\theta_1}{V_1} = \frac{\sin\theta_2}{V_2}$

$$\therefore |\bar{H}_2| = \begin{vmatrix} 0 & -\frac{a}{\cos^2\theta_1} & \frac{-b}{\cos^2\theta_2} \\ -\frac{a}{\cos^2\theta_1} & \frac{a\left[1+\sin^2\theta_1\right]}{V_1\cos^3\theta_1} - \lambda\frac{2a\sin\theta_1}{\cos^3\theta_1} & 0 \\ -\frac{b}{\cos^2\theta_2} & 0 & \frac{b\left[1+\sin^2\theta_2\right]}{V_2\cos^3\theta_2} - \lambda\frac{2b\sin\theta_2}{\cos^3\theta_2} \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -\frac{a}{\cos^2\theta_1} & \frac{-b}{\cos^2\theta_2} \\ -\frac{a}{\cos^2\theta_1} & \frac{a\left[1-\sin^2\theta_1\right]}{V_1\cos^3\theta_1} & 0 \\ -\frac{b}{\cos^2\theta_2} & 0 & \frac{b\left[1-\sin^2\theta_2\right]}{V_2\cos^3\theta_2} \end{vmatrix}$$

$$= \begin{vmatrix} 0 & -\frac{a}{\cos^2\theta_1} & -\frac{b}{\cos^2\theta_2} \\ -\frac{a}{\cos^2\theta_1} & \frac{a}{v_1\cos\theta_1} & 0 \\ -\frac{b}{\cos^2\theta_2} & 0 & \frac{b}{v_2\cos\theta_2} \end{vmatrix}$$

$$= 0 + \frac{a}{\cos^2\theta_1}\left(\frac{-ab}{v_2\cos^2\theta_1\cos\theta_2}\right) - \frac{b}{\cos^2\theta_2}\left(\frac{ab}{v_1\cos^2\theta_2\cos\theta_1}\right)$$

$$= \frac{-a^2b}{v_2\cos^4\theta_1\cos\theta_2} - \frac{ab^2}{v_1\cos^4\theta_2\cos\theta_1} < 0$$

since all $a, b, v_1, v_2, \cos\theta_1, \cos\theta_2 > 0$

$\therefore |H_2| < 0 \Rightarrow$ critical point a minimum

30.

30. Let P be a point on a surface S in $\mathbb{R}^3$ defined by the equation $f(x, y, z) = 1$, where f is of class C^1. Suppose that P is a point where the distance from the origin to S is maximized. Show that the vector emanating from the origin and ending at P is perpendicular to S.

This follows the proof to Theorem 8 on p. 186 of the text.

Let the distance function of any point to the origin be:

$$D = \sqrt{x^2+y^2+z^2} \quad \therefore \nabla D = -\frac{1}{D}[x, y, z]$$

Let $P = (x_0, y_0, z_0)$

Let $\vec{c}(t) : R \to R^3$ be any path in S, s.t. $\vec{c}(0) = P$

Since $D(x,y,z)$ is a maximum at P, then

$D(\vec{c}(t))$ is a maximum at $t=0$, and so

$$\frac{d}{dt} D(\vec{c}(t)) = 0 = \nabla D(P) \cdot \vec{c}\,'(0)$$

Note that $\vec{c}\,'(0)$ is tangent to S at P.

$\therefore$ $\nabla D(P)$ is perpendicular to every $\vec{c}\,'(0)$, and so $\nabla D(P)$ is perpendicular to S at P.

But $\nabla D(P) = -\frac{1}{\sqrt{x_0^2+y_0^2+z_0^2}}[x_0, y_0, z_0]$, which is a vector from P pointing to the origin. A vector from the origin to P is parallel to $\nabla D(P)$.

$\therefore$ Vector from origin to P is $\perp$ S at P.

31.

31. Let A be a nonzero symmetric 3×3 matrix. Thus, its entries satisfy $a_{ij} = a_{ji}$. Consider the function $f(\mathbf{x}) = \frac{1}{2}(A\mathbf{x}) \cdot \mathbf{x}$.

(a) What is ∇f?

(b) Consider the restriction of f to the unit sphere $S = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$ in $\mathbb{R}^3$. By Theorem 7 we know that f must have a maximum and a minimum on S. Show that there must be an $\mathbf{x} \in S$ and a $\lambda \neq 0$ such that $A\mathbf{x} = \lambda\mathbf{x}$. (The vector $\mathbf{x}$ is called an ***eigenvector***, while the scalar λ is called an ***eigenvalue***.)

(c) What are the maxima and minima for f on $B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$?

(a) Let $\vec{x} = (x,y,z)$, $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}$

$$\therefore A\vec{x} = \begin{bmatrix} a_{11} \\ a_{12} \\ a_{13} \end{bmatrix} x + \begin{bmatrix} a_{12} \\ a_{22} \\ a_{23} \end{bmatrix} y + \begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \end{bmatrix} z$$

$$\therefore A\vec{x}\cdot\vec{x} = a_{11}x^2 + a_{12}yx + a_{13}zx$$
$$+ a_{12}xy + a_{22}y^2 + a_{23}zy$$
$$+ a_{13}xz + a_{23}yz + a_{33}z^2$$

$$\therefore \frac{\partial}{\partial x} A\vec{x}\cdot\vec{x} = a_{11}(2x) + a_{12}(y) + a_{13}(z)$$
$$+ a_{12}(y) + a_{22}(0) + a_{23}(0)$$
$$+ a_{13}(z) + a_{23}(0) + a_{33}(0)$$
$$= a_{11}(2x) + a_{12}(2y) + a_{13}(2z)$$
$$= 2\begin{bmatrix} a_{11} & a_{12} & a_{13} \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\frac{\partial}{\partial y} A\vec{x}\cdot\vec{x} = a_{11}(0) + a_{12}(x) + a_{13}(0)$$
$$+ a_{12}(x) + a_{22}(2y) + a_{23}(z)$$
$$+ a_{13}(0) + a_{23}(z) + a_{33}(0)$$
$$= a_{12}(2x) + a_{22}(2y) + a_{23}(2z)$$
$$= 2\begin{bmatrix} a_{12} & a_{22} & a_{23} \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\frac{\partial}{\partial z} A\vec{x}\cdot\vec{x} = \begin{aligned} &0 + 0 + a_{13}(x) \\ &+ 0 + 0 + a_{23}(y) \\ &+ a_{13}(x) + a_{23}(y) + a_{33}(2z) \end{aligned}$$

$$= a_{13}(2x) + a_{23}(2y) + a_{33}(2z)$$

$$= 2\begin{bmatrix} a_{13} & a_{23} & a_{33} \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\therefore \nabla f = \frac{1}{2}\left[\frac{\partial}{\partial x} A\vec{x}\cdot\vec{x} \quad \frac{\partial}{\partial y} A\vec{x}\cdot\vec{x} \quad \frac{\partial}{\partial z} A\vec{x}\cdot\vec{x}\right]$$

$$= \left(\frac{1}{2}\right)(2)\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = A\vec{x}$$

$$\therefore \underline{\nabla f = A\vec{x}}$$

(6) Let $g(x,y,z) = x^2 + y^2 + z^2$ so that S is the level set for $g(\vec{x}) = 1$. $\nabla g = [2x \; 2y \; 2z]$, and

$\therefore \nabla g \neq \vec{0}$ for $\vec{x} \in S$, since $\vec{x} \neq \vec{0}$ on S.

Let $\vec{x} = (x,y,z)$ be the local max for $f|_S$

$\therefore$ Theorem 8 (p. 186 of text) says there is a real number λ, s.t. $\nabla f(\vec{x}) = \lambda \nabla g(\vec{x})$

From (a), $\nabla f(\vec{x}) = A\vec{x}$ and $\nabla g(\vec{x}) = [2x \; 2y \; 2z]$,

Then $A\vec{x} = \lambda_1 [2x \; 2y \; 2z] = 2\lambda_1 \vec{x}$

$\therefore$ Let $\lambda = 2\lambda_1$. $\therefore A\vec{x} = \lambda\vec{x}$.

Note that even though $\vec{x} \neq \vec{0}$ and A is non-zero, $A\vec{x}$ could still be $\vec{0}$ so that λ could be 0.

(c) (c) What are the maxima and minima for f on $B = \{(x, y, z) \mid x^2 + y^2 + z^2 \leq 1\}$?

B is closed and bounded, and so f, continuous on B, must have a max and min in B.

Inside B, $\nabla f = A\vec{x}$ and so critical points are in nullspace of A. On surface of B, critical points are eigenvectors of A.

From $\frac{1}{2} A\vec{x} \cdot \vec{x} \leq \frac{1}{2} \|A\vec{x}\| \|\vec{x}\| \leq \frac{1}{2} \|A\vec{x}\|$, since $\|\vec{x}\| \leq 1$,

and $\|A\vec{x}\| \leq k\|x\| \leq k$, where $k = \sqrt{\sum_{i=1}^{} \sum_{j=1}^{} a_{ij}^2}$

Then $-\frac{1}{2}k \leq \frac{1}{2} A\vec{x} \cdot \vec{x} \leq \frac{1}{2}k$.

32.

32. Suppose that A in the function f defined in Exercise 31 is not necessarily symmetric.

(a) What is ∇f?

(b) Can we conclude the existence of an eigenvector and eigenvalues as in Exercise 31?

(a) From #31, $A\vec{x}\cdot\vec{x} = a_{11}x^2 + a_{12}yx + a_{13}zx$

$+ a_{21}xy + a_{22}y^2 + a_{23}zy$

$+ a_{31}xz + a_{32}yz + a_{33}z^2$

$\therefore \frac{\partial f}{\partial x} = \frac{1}{2}\left[a_{11}(2x) + (a_{12}+a_{21})y + (a_{13}+a_{31})z\right]$

$\frac{\partial f}{\partial y} = \frac{1}{2}\left[(a_{12}+a_{21})x + a_{22}(2y) + (a_{23}+a_{32})z\right]$

$\frac{\partial f}{\partial z} = \frac{1}{2}\left[(a_{13}+a_{31})x + (a_{23}+a_{32})y + a_{33}(2z)\right]$

$\nabla f = \frac{1}{2}\left[\left[(A+A^T)\vec{x}\right]_{row_1}, \left[(A+A^T)\vec{x}\right]_{row_2}, \left[(A+A^T)\vec{x}\right]_{row_3}\right]$

$= \frac{1}{2}(A+A^T)\vec{x}$

(b) $f(\vec{x}) = \frac{1}{2}A\vec{x}\cdot\vec{x}$ is continuous on R^3, whether A is symmetric or not. If f is defined on a level set and the conditions of Theorem 8

(p. 186 of text) are satisfied, then you can guarantee an eigenvalue.

33. **33.** (a) Find the critical points of $x + y^2$, subject to the constraint $2x^2 + y^2 = 1$.

(b) Use the bordered Hessian to classify the critical points.

(a) Let $f(x,y) = x + y^2$, $g(x,y) = 2x^2 + y^2$

$\therefore \nabla g = [4x, 2y] \neq \vec{0}$ if $(x,y) \neq (0,0)$.

$\nabla f = [1, 2y]$.

$\therefore$ Set $[1, 2y] = \lambda [4x, 2y]$, $\therefore \lambda \neq 0$ since $f_x = 1 \neq 0$.

$\therefore 2y = \lambda 2y \Rightarrow \lambda = 1$, assuming $y \neq 0$

$\therefore 1 = \lambda 4x \Rightarrow x = \frac{1}{4}$

From $2x^2 + y^2 = 1$, $2\left(\frac{1}{4}\right)^2 + y^2 = 1$, $y^2 = \frac{7}{8}$, $y = \pm\sqrt{\frac{7}{8}}$

$\therefore$ For $y \neq 0$: $\left(\frac{1}{4}, \pm\sqrt{\frac{7}{8}}\right)$ and $\lambda = 1$

For $y = 0$: $2x^2 + y^2 = 1 \Rightarrow x = \pm\sqrt{\frac{1}{2}}$, so

$$\lambda = \frac{1}{4x} = \pm\frac{\sqrt{2}}{4}$$

$\therefore \left(\pm\sqrt{\frac{1}{2}}, 0\right)$

(6) Let $h(x,y) = f(x,y) - \lambda g(x,y)$

$\therefore h_x = 1 - \lambda 4x \quad h_y = 2y - \lambda 2y$

$h_{xx} = -4\lambda \quad h_{xy} = 0 \quad h_{yy} = 2 - 2\lambda$

$$\therefore |H_2| = \begin{vmatrix} 0 & -4x & -2y \\ -4x & -4\lambda & 0 \\ -2y & 0 & 2-2\lambda \end{vmatrix}$$

For $(x, y, \lambda) = \left(\frac{1}{4}, \pm\sqrt{\frac{7}{8}}, 1\right)$,

$$|H_2| = \begin{vmatrix} 0 & -1 & \pm 2\sqrt{\frac{7}{8}} \\ -1 & -4 & 0 \\ \pm 2\sqrt{\frac{7}{8}} & 0 & 0 \end{vmatrix} = 0 + (1)(0) + 2\sqrt{\frac{7}{8}}\left(8\sqrt{\frac{7}{8}}\right) = 14$$

or $0 + (1)(0) - 2\sqrt{\frac{7}{8}}\left(-8\sqrt{\frac{7}{8}}\right) = 14$

$\therefore \left(\frac{1}{4}, \pm\sqrt{\frac{7}{8}}\right)$ are relative maxs. as $|H_2| > 0$

For $(x, y, \lambda) = \left(\sqrt{\frac{1}{2}}, 0, \frac{\sqrt{2}}{4}\right), \left(-\sqrt{\frac{1}{2}}, 0, -\frac{\sqrt{2}}{4}\right)$

$$|H_2| = \begin{vmatrix} 0 & -4\sqrt{\frac{1}{2}} & 0 \\ -4\sqrt{\frac{1}{2}} & -\sqrt{2} & 0 \\ 0 & 0 & 2 - \frac{\sqrt{2}}{2} \end{vmatrix} = 0 + 4\sqrt{\frac{1}{2}}\left(-8\sqrt{\frac{1}{2}} + 2\right) < 0$$

$\therefore$ Rel. min.

$$|H_2| = \begin{vmatrix} 0 & 4\sqrt{\frac{1}{2}} & 0 \\ 4\sqrt{\frac{1}{2}} & \sqrt{2} & 0 \\ 0 & 0 & 2+\frac{\sqrt{2}}{2} \end{vmatrix} = 0 - 4\sqrt{\tfrac{1}{2}}\left(8\sqrt{\tfrac{1}{2}}+2\right) < 0$$

$\therefore$ Rel. min.

$\therefore \left(\pm\sqrt{\frac{1}{2}}, 0\right)$ are both relative mins.

$$f\left(\frac{1}{4}, \sqrt{\frac{7}{8}}\right) = \frac{1}{4} + \frac{7}{8} = \frac{9}{8}$$

$$f\left(\frac{1}{4}, -\sqrt{\frac{7}{8}}\right) = \frac{1}{4} + \frac{7}{8} = \frac{9}{8}$$

$$f\left(\sqrt{\frac{1}{2}}, 0\right) = \sqrt{\frac{1}{2}}$$

$$f\left(-\sqrt{\frac{1}{2}}, 0\right) = -\sqrt{\frac{1}{2}}$$

$\therefore \left(\frac{1}{4}, \pm\sqrt{\frac{7}{8}}\right)$ are absolute maxs.

$\left(-\sqrt{\frac{1}{2}}, 0\right)$ is an absolute min.

$\left(\sqrt{\frac{1}{2}}, 0\right)$ is a relative min.

34.

34. Answer the question posed in the last line of Example 9.

distance from the point (x, y) to the origin. Therefore, if (x_1, y_1) and (x_2, y_2) denote the nontrivial solutions to equation (9) corresponding to t_1 and t_2, and if t_1 and t_2 are positive, we get $\sqrt{x_2^2 + y_2^2} = 1/\sqrt{t_2}$ and $\sqrt{x_1^2 + y_1^2} = 1/\sqrt{t_1}$. Consequently, if $t_1 > t_2$, the lengths of the semiminor and semimajor axes are $1/\sqrt{t_1}$ and $1/\sqrt{t_2}$, respectively. If the curve is an ellipse, both t_1 and t_2 are, in fact, real and positive. What happens with a hyperbola or a parabola? ▲

For parabolas & hyperbolas, there is only a

minimum distance. These curves are not bounded from above.

35. **35.** Try to find the extrema of $xy + yz$ among points satisfying $xz = 1$.

Let $f(x,y,z) = xy + yz$, $g(x,y,z) = xz$

$\therefore \nabla f = [y, x+z, y]$ $\nabla g = [z, 0, x]$.

$\therefore$ if f has max/min on $g(x,y,z) = 1$, then $\exists \lambda$ s.t.

$$[y, x+z, y] = \lambda [z, 0, x]$$

$$\therefore \quad y = \lambda z \quad [1]$$
$$x + z = 0 \quad [2]$$
$$y = \lambda x \quad [3]$$

Assume $\lambda \neq 0$.

$\therefore$ [1], [3] $\Rightarrow x = z$. $\therefore$ [2] $\Rightarrow x = z = 0$

& $\therefore y = 0$.

But $(0,0,0)$ not a point on $g(x,y,z) = xz = 1$.

$\therefore$ No critical points for $\lambda \neq 0$.

Assume $\lambda = 0$. $\therefore y = 0$ and $f(x,y,z) = 0$

But [2] $\Rightarrow z = -x \Rightarrow g(x,y,z) = -x^2 = 1$.

There is no such x.

∴ No critical points

36. **36.** A company's production function is $Q(x, y) = xy$. The cost of production is $C(x, y) = 2x + 3y$. If this company can spend $C(x, y) = 10$, what is the maximum quantity that can be produced?

Maximize Q given constraint $C = 10$.

$$\nabla Q(x,y) = [y, x] = \lambda \nabla C(x,y) = \lambda [2, 3]$$

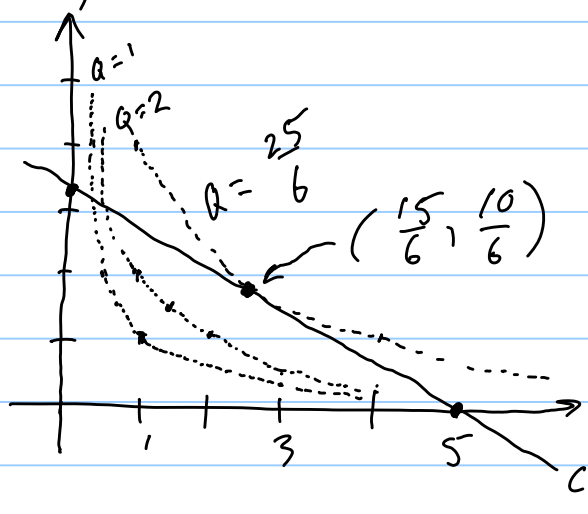
$$\therefore \quad y = 2\lambda \quad [1]$$
$$x = 3\lambda \quad [2]$$
$$2x + 3y = 10 \quad [3]$$

$$\therefore 2(3\lambda) + 3(2\lambda) = 12\lambda = 10, \ \lambda = \frac{5}{6}$$

$$\therefore \quad x = 3\left(\frac{5}{6}\right) = \frac{15}{6} \qquad y = 2\left(\frac{5}{6}\right) = \frac{10}{6}$$

$$\therefore Q\left(\frac{15}{6}, \frac{10}{6}\right) = \frac{15}{6} \cdot \frac{10}{6} = \frac{150}{36} = \underline{\frac{25}{6}}$$

Graphically, $C(x,y) = \frac{x}{5} + \frac{y}{10/3} = 1$



Each $Q = xy = c$ represents a curve of different $Q = c$. The max Q just touches $C = 2x + 3y = 10$.

37. **37.** Find the point on the curve $(\cos t, \sin t, \sin(t/2))$ that is farthest from the origin.

Note: The curve is in $\mathbb{R}^3$, actually on a cylinder of radius 1, perpendicular to xy plane, centred at origin.

Let distance function be $d(t)$:

$$d(t) = \sqrt{(\cos t - 0)^2 + (\sin t - 0)^2 + (\sin(\tfrac{t}{2}) - 0)^2}$$

$$= \sqrt{1 + \sin^2(\tfrac{t}{2})}$$

The max of $\sin^2()$ is when $\sin^2() = 1$, or $\sin() = \pm 1$

$\therefore \frac{t}{2} = \pm\frac{\pi}{2}$, $t = \pm\pi$.

$\therefore t = \pi$: $(\cos\pi, \sin\pi, \sin\frac{\pi}{2}) = \underline{(-1, 0, 1)}$

$t = -\pi$: $(\cos(-\pi), \sin(-\pi), \sin\frac{-\pi}{2}) = \underline{(-1, 0, -1)}$

To use Lagrange multipliers, consider distance function $d(x,y,z) = x^2 + y^2 + z^2$, subject to

constraints: $x^2 + y^2 = 1$ [1]

$z = \sin\left(\frac{\arcsin x}{2}\right)$ [2]

Since $y = \sin t$, $t = \arcsin y$, $\therefore \sin\frac{t}{2} = \sin\left(\frac{\arcsin y}{2}\right)$

$\therefore$ let $g_1(x,y,z) = x^2 + y^2 = 1$

$$g_2(x,y,z) = z - \sin\left(\frac{\arcsin y}{2}\right) = 0$$

$\therefore \nabla d = \lambda_1 \nabla g_1 + \lambda_2 \nabla g_2$

$$\nabla g_1 = [2x, 2y, 0]$$

$$\nabla g_2 = \left[0, -\cos\left(\frac{\arcsin y}{2}\right)\cdot\frac{1}{2}\frac{1}{\sqrt{1-y^2}}, 1\right]$$

$$\nabla d = [2x, 2y, 2z].$$

$$\therefore [2x, 2y, 2z] = \lambda_1 [2x, 2y, 0] + \lambda_2 \left[0, -\cos\left(\frac{\arcsin y}{2}\right)\cdot\frac{1}{2\sqrt{1-y^2}}, 1\right]$$

$$\therefore \quad 2x = \lambda_1 2x \qquad [3]$$

$$2y = \lambda_1 2y - \lambda_2 \cos\left(\frac{\arcsin y}{2}\right)\cdot\frac{1}{2\sqrt{1-y^2}} \qquad [4]$$

$$2z = \lambda_2 \qquad [5]$$

If $x = 0$, [1] $\Rightarrow y = \pm 1$, so [2] $\Rightarrow z = \sin\left(\pm\frac{\pi}{4}\right) = \pm\sqrt{2}/2$

$\therefore$ possibilities: $(0, 1, \sqrt{2}/2)$, $(0, -1, -\sqrt{2}/2)$

$\therefore$ distance $= \sqrt{0 + 1 + \frac{1}{2}} = \sqrt{1\frac{1}{2}}$

If $x \neq 0$, [3] $\Rightarrow \lambda_1 = 1$

$$\lambda_2 \cos\left(\frac{\arcsin y}{2}\right)\cdot\frac{1}{2\sqrt{1-y^2}} = 0 \qquad [4']$$

$$2z = \lambda_2 \qquad [5]$$

Note is $x \neq 0$, $y \neq 1$ from [1].

$\therefore$ [2], [4'], and [5] $\Rightarrow$

$$2\sin\left(\frac{\arcsin y}{2}\right)\cos\left(\frac{\arcsin y}{2}\right) = 0$$

Or, using $2\sin\theta\cos\theta = \sin 2\theta$,

$$\sin\left(2 \cdot \frac{\arcsin y}{2}\right) = y = 0$$

$\therefore$ $x \neq 0$, $y = 0$, and [1] $\Rightarrow$ $x = \pm 1$

with $y = 0$, [2]: $z = \sin\left(\frac{\arcsin(y)}{2}\right)$

$\arcsin(0) = 0, \pi, -\pi$

$\therefore z = 0, 1, -1.$

$\therefore$ $(1,0,0)$, $(1,0,1)$, $(1,0,-1)$

$(-1,0,0)$, $(-1,0,1)$, $(-1,0,-1)$

But from $(x,y,z) = (\cos t, \sin t, \sin\frac{t}{2})$,

no such t value for $(1,0,1)$, $(1,0,-1)$, $(-1,0,0)$

$\therefore$ possibilities: $(1,0,0)$, $(-1,0,1)$, $(-1,0,-1)$

Corresponding distances: $1, \sqrt{2}, \sqrt{2}$

$\therefore$ Absolute max at $\underline{(-1,0,1), (-1,0,-1)}$

38. **38.** A firm uses wool and cotton fiber to produce cloth. The amount of cloth produced is given by $Q(x, y) = xy - x - y + 1$, where x is the number of pounds of wool, y the number of pounds of cotton, $x > 1$, and $y > 1$. If wool costs p dollars per pound, cotton costs q dollars per pound, and the firm can spend B dollars on material, what should the ratio of cotton and wool be to produce the most cloth?

Cost = $C(x,y) = px + qy = B$, the constraint. with $x > 1$, $y > 1$.

$\therefore \nabla Q(x,y) = [y-1, x-1] = \lambda \nabla C(x,y) = \lambda [p, q]$

$\therefore$ $y - 1 = \lambda p$, $p \neq 0$, $y > 1$ [1]

$x - 1 = \lambda q$, $q \neq 0$, $x > 1$ [2]

$px + qy = B$ [3]

$\therefore$ [1] $\Rightarrow y = \lambda p + 1$ [2] $\Rightarrow x = \lambda q + 1$

$\therefore$ [3] $\Rightarrow p(\lambda q + 1) + q(\lambda p + 1) = B$

$\lambda pq + p + \lambda pq + q = B$

$\lambda = \dfrac{B - p - q}{2pq}$

Note: from $px + qy = B$, x, y, p, q all > 0, $B > p$, $B > q$

$\therefore x = 1 + \left(\dfrac{B-p-q}{2pq}\right) q = 1 + \dfrac{B-p-q}{2p}$

$= \dfrac{B + p - q}{2p}$

$$y = 1 + \left(\frac{\beta - \rho - q}{2\rho q}\right)\rho = 1 + \frac{\beta - \rho - q}{2q}$$

$$= \frac{\beta - \rho + q}{2q}$$

$$\text{Ratio } \frac{\text{cotton}}{\text{wool}} = \frac{Y}{X} = \frac{L}{q}\,\frac{(\beta - \rho + q)}{(\beta + \rho - q)}$$

39. **39.** Carry out the analysis of Example 10 for the production function $Q(K, L) = AK^{\alpha} L^{1-\alpha}$, where A and α are positive constants and $0 < \alpha < 1$. This is called a ***Cobb–Douglas production function*** and is sometimes used as a simple model for the national economy. Q is then the aggregate output of the economy for a given input of capital and labor.

Maximize $Q(K, L) = AK^{\alpha} L^{1-\alpha}$

under constraint $g(K, L) = \rho L + qK = \beta$

$\therefore \nabla Q = [A\alpha K^{\alpha-1} L^{1-\alpha},\ A(1-\alpha) K^{\alpha} L^{-\alpha}] = \lambda \nabla g = \lambda [q, \rho]$

$$A\alpha K^{\alpha-1} L^{1-\alpha} = \lambda q \qquad [1]$$

$$A(1-\alpha) K^{\alpha} L^{-\alpha} = \lambda \rho \qquad [2]$$

$$\rho L + qK = \beta \qquad [3]$$

$[1] \Rightarrow q = \dfrac{A\alpha K^{\alpha-1} L^{1-\alpha}}{\lambda}$ $\quad [2] \Rightarrow \rho = \dfrac{A(1-\alpha) K^{\alpha} L^{-\alpha}}{\lambda}$

$$\therefore [3] \Rightarrow \frac{A(1-\alpha)K^{\alpha}L^{-\alpha}}{\lambda}(L) + \frac{A\alpha K^{\alpha-1}L^{1-\alpha}}{\lambda}(K) = B$$

$$\text{or, } \frac{A(1-\alpha)K^{\alpha}L^{1-\alpha} + A\alpha K^{\alpha}L^{1-\alpha}}{\lambda} = B$$

$$\text{or, } \frac{AK^{\alpha}L^{1-\alpha}(1-\alpha+\alpha)}{\lambda} = B$$

$$\text{or, } \lambda = \frac{AK^{\alpha}L^{1-\alpha}}{B}$$

From [1], $A\alpha K^{\alpha-1}L^{1-\alpha} = \left(\frac{AK^{\alpha}L^{1-\alpha}}{B}\right)q$

$$\text{or } B\alpha K^{-1} = q, \text{ or } K = \frac{\alpha B}{q}$$

From [2], $A(1-\alpha)K^{\alpha}L^{-\alpha} = \left(\frac{AK^{\alpha}L^{1-\alpha}}{B}\right)p$

$$\text{or, } \frac{B(1-\alpha)}{L} = p, \text{ or } L = \frac{(1-\alpha)B}{p}$$

$$\therefore (K, L) = \left(\frac{\alpha B}{q}, \frac{(1-\alpha)B}{p}\right) \text{ will optimize } Q()$$

3.5 The Implicit Function Theorem

Note Title 6/16/2016

1.

1. Show that the equation $x + y - z + \cos(xyz) = 0$ can be solved for $z = g(x, y)$ near the origin. Find $\frac{\partial g}{\partial x}$ and $\frac{\partial g}{\partial y}$ at $(0, 0)$.

By "the origin" must mean $(x,y) = (0,0)$.

$F(x,y,z) = x + y - z + \cos(xyz)$. $F(0,0,1) = 0$.

F is C^1, and $F_z = -1 - xy\sin(xyz)$

$\therefore F_z(0,0,1) = -1 \neq 0$. Here $\vec{x_0} = (0,0)$, $z_0 = 1$.

$\therefore$ By implicit function theorem, $\exists$ an open ball U about $\vec{x_0} = (0,0)$ and V around $z_0 = 1$

s.t. $g(x,y) = z$ and $F(x,y,g(x,y)) = 0$.

$$g_x(\vec{x_0}) = -\frac{F_x(\vec{x_0}, z_0)}{F_z(\vec{x_0}, z_0)}, \quad g_y(\vec{x_0}) = -\frac{F_y(\vec{x_0}, z_0)}{F_z(\vec{x_0}, z_0)}$$

$$\therefore g_x(0,0) = -\frac{1 - yz\sin(xyz)}{-1 - xy\sin(xyz)}\Bigg|_{(x,y,z)=(0,0,1)}$$

$$= -\frac{1-0}{-1-0} = 1$$

$$g_y(0,0) = -\frac{1 - xz\sin(xyz)}{-1 - xy\sin(xyz)}\Bigg|_{(x,y,z)=(0,0,1)}$$

$$= 1$$

2. **2.** Show that $xy + z + 3xz^5 = 4$ is solvable for z as a function of (x, y) near $(1, 0, 1)$. Compute $\partial z/\partial x$ and $\partial z/\partial y$ at $(1, 0)$.

Let $F(x,y,z) = xy + z + 3xz^5 - 4 = 0$

$F(1,0,1) = 0 + 1 + 3 - 4 = 0$

F is C^1, and $F_z(1,0,1) = 1 + 15xz^4\big|_{(1,0,1)} = 16 \neq 0$

$\therefore$ $\exists$ a function $g(x,y) = z$ s.t. $F(x,y,g(x,y)) = 0$

near $\vec{x}_0 = (1,0)$ and $z_0 = 1$

$$g_x(1,0) = z_x(1,0) = -\frac{F_x(1,0,1)}{F_z(1,0,1)} = -\frac{y + 3z^5}{16}\bigg|_{(1,0,1)} = -\frac{3}{16}$$

$$g_y(1,0) = z_y(1,0) = -\frac{F_y(1,0,1)}{F_z(1,0,1)} = -\frac{x}{16}\bigg|_{(1,0,1)} = -\frac{1}{16}$$

3. **3.** (a) Check directly (i.e., without using Theorem 11) where we can solve the equation $F(x, y) = y^2 + y + 3x + 1 = 0$ for y in terms of x.

(b) Check that your answer in part (a) agrees with the answer you expect from the implicit function theorem. Compute dy/dx.

(a) Using quadratic formula ($c = 3x+1$),

$$y = \frac{-1 \pm \sqrt{1 - 4(3x+1)}}{2} = -\frac{1}{2} \pm \frac{1}{2}\sqrt{-3-12x}$$

$\therefore -3 - 12x \geq 0$, $12x \leq -3$, $x \leq -\frac{1}{4}$.

For $x < -\frac{1}{4}$, $y = -\frac{1}{2} + \frac{1}{2}\sqrt{-3-12x}$ or $y = -\frac{1}{2} - \frac{1}{2}\sqrt{-3-12x}$

$x = -\frac{1}{4}$, $y = -\frac{1}{2}$, just a point

(6) $F(x,y)$ is C^1. $F_y = 2y+1$, $F_y \neq 0$ for $y \neq -\frac{1}{2}$

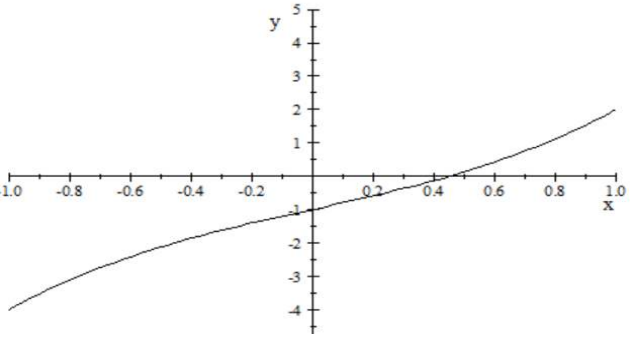
$\therefore$ For $y \neq -\frac{1}{2}$, can find a $g(x) = y$.

$$\frac{dy}{dx} = g'(x) = -\frac{F_x}{F_y} = -\frac{3}{2y+1}$$

4. **4.** Repeat Exercise 3 with $F(x, y) = xy^2 - 2y + x^2 + 2 = 0$.

(a) $$y = \frac{2 \pm \sqrt{4 - 4(x)(x^2+2)}}{2x} = \frac{1}{x} \pm \frac{\sqrt{1 - x^3 - 2x}}{x}$$

$\therefore x \neq 0$ and $1 - x^3 - 2x \geq 0$, or $x^3 + 2x - 1 \leq 0$



(6) $F(x,y)$ is C^1,

$F_y = 2xy - 2 \neq 0$

for $xy \neq 1$, or $y \neq \frac{1}{x}$

$\therefore$ For $y \neq \frac{1}{x}$, $\exists$ a $g(x) = y$ around a point (x,y) s.t. $F(x,y) = 0$. $(0,1)$ is one such point, and $F_y(0,1) = -2 \neq 0$.

5. **5.** Let $F(x, y) = 0$ define a curve in the xy plane through the point (x_0, y_0), where F is C^1. Assume that $(\partial F/\partial y)(x_0, y_0) \neq 0$. Show that this curve can be locally represented by the graph of a function $y = g(x)$. Show that (i) the line orthogonal to $\nabla F(x_0, y_0)$ agrees with (ii) the tangent line to the graph of $y = g(x)$.

By the Implicit Function Theorem, since F is C^1 and $F_y(x_0, y_0) \neq 0$, $\exists$ a function $y = g(x)$ s.t. near (x_0, y_0), $F(x, g(x)) = 0$.

(i) Gradients (i.e., ∇F) are perpendicular to level sets. $\therefore$ $\nabla F(x_0, y_0)$ is $\perp$ to the tangent to $F(x,y)$ at (x_0, y_0).

$\nabla F(x_0, y_0) = [F_x(x_0, y_0), F_y(x_0, y_0)]$.

$\therefore$ The line orthogonal to $\nabla F(x_0, y_0)$ is parallel to $[F_y(x_0, y_0), -F_x(x_0, y_0)]$

(ii) Tangent line to $y = g(x)$ is parallel to $g'(x)$.

By Special Implicit Function Theorem,

$g'(x_0) = -\dfrac{F_x(x_0, y_0)}{F_y(x_0, y_0)}$. $\therefore$ a vector with this slope would be $\left[1, -\dfrac{F_x(x_0, y_0)}{F_y(x_0, y_0)}\right]$,

which is parallel to $F_y(x_0,y_0)\left[1, -\frac{f_x(x_0,y_0)}{f_y(x_0,y_0)}\right]$

$= \left[F_y(x_0,y_0), -F_x(x_0,y_0)\right]$

This vector is the same as (i).

6. **6.** Consider the surface S given by $3y^2z^2 - 3x = 0$.

(a) Using the implicit function theorem, verify that we can solve for x as a function of y and z near any point on S. Explicitly write x as a function of y and z.

(b) Show that near $(1, 1, -1)$ we can solve for either y or z, and give explicit expressions for these variables in terms of the other two.

(a) Let $F(y,z,x) = 3y^2z^2 - 3x = 0$.

F is C^1 and $F_x = -3 \neq 0$.

$\therefore$ By Special Implicit Function Theorem (Theorem 11 in text), $\exists$ a function $x = g(y,z): \mathbb{R}^2 \to \mathbb{R}$

s.t. $F(y,z,g(y,z)) = 0$.

Note: $3y^2z^2 - 3x = 0 \Rightarrow y^2z^2 = x$

(b) Now let $F(x,y,z) = 3y^2z^2 - 3x = 0$. $\therefore F_y = 6yz^2$

$\therefore F_y(1,1,-1) = 6 \neq 0$ $\therefore$ near $(1,1,-1)$ can solve $y = g(x,z)$.

$F_z = 6y^2z$ so $F_z(1,1,-1) = -6 \neq 0$. $\therefore$ Can solve $z = g(x,y)$ near $(1,1,-1)$.

7. **7.** Show that $x^3z^2 - z^3yx = 0$ is solvable for z as a function of (x, y) near $(1, 1, 1)$, but not near the origin. Compute $\partial z/\partial x$ and $\partial z/\partial y$ at $(1, 1)$.

Let $F(x,y,z) = x^3z^2 - z^3yx = 0$. F is C^1.

(a) $F_z = 2x^3z - 3z^2yx$. $\therefore F_z(1,1,1) = -1 \neq 0$.

$\therefore$ $\exists$ a function $z = g(x,y)$ near $(1,1,1) = (x,y,z)$.

(b) $F_z(0,0,0) = 0$. $\therefore$ Can't guarantee a function near $\vec{0}$.

However, $x^3z^2 - z^3yx = 0 \iff x^3z^2 = z^3yx$

For $(x,y) \neq 0$, $\frac{x^2z^2}{y} = z^3$, so $z=0$ is a solution.

$\therefore$ $(x,y,0)$ are solutions.

But $x=y=z$ is also a solution, even at origin.

$\therefore$ For $x=y$ and $(x,y) \neq (0,0)$ near origin,

we have two possible z values.

$\therefore$ No unique z for $x=y$, $(x,y) \neq (0,0)$ near origin.

$\therefore$ No function of $z = g(x,y)$ possible near $\vec{0}$.

(c) $\frac{\partial z}{\partial x} = -\frac{F_x(1,1,1)}{F_z(1,1,1)} = -\left.\frac{3x^2z^2 - z^3y}{2x^3z - 3z^2yx}\right|_{(1,1,1)} = -\frac{2}{-1} = 2$

(d) $\frac{\partial z}{\partial y} = -\frac{F_y(1,1,1)}{F_z(1,1,1)}$. $F_y = -z^3 x$. $\therefore F_y(1,1,1) = -1$.

$\therefore \frac{\partial z}{\partial y} = -\frac{-1}{-1} = \underline{-1}$

8.

8. Discuss the solvability in the system

$$\begin{aligned} 3x + 2y + z^2 + u + v^2 &= 0 \\ 4x + 3y + z + u^2 + v + w + 2 &= 0 \\ x + z + w + u^2 + 2 &= 0 \end{aligned}$$

for u, v, w in terms of x, y, z near $x = y = z = 0$, $u = v = 0$, and $w = -2$.

Let $F_1(x,y,z,u,v,w) = 3x + 2y + z^2 + u + v^2 = 0$

$F_2(x,y,z,u,v,w) = 4x + 3y + z + u^2 + v + w + 2 = 0$

$F_3(x,y,z,u,v,w) = x + z + w + u^2 + 2 = 0$

Each F_i is C^1

$$\Delta = \begin{vmatrix} F_{1u} & F_{1v} & F_{1w} \\ F_{2u} & F_{2v} & F_{2w} \\ F_{3u} & F_{3v} & F_{3w} \end{vmatrix} = \begin{vmatrix} 1 & 2v & 0 \\ 2u & 1 & 1 \\ 2u & 0 & 1 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{vmatrix} = 1 \neq 0 \text{ at } (x,y,z,u,v,w) = (0,0,0,0,0,-2)$$

$\therefore$ can solve u, v, w in terms of x, y, z

9. **9.** Discuss the solvability of

$$y + x + uv = 0$$
$$uxy + v = 0$$

for u, v in terms of x, y near $x = y = u = v = 0$ and check directly.

Let $F_1(x,y,u,v) = y + x + uv = 0$ [1] F_1 is C^1

$F_2(x,y,u,v) = uxy + v = 0$ [2] F_2 is C^1

$$\Delta = \begin{vmatrix} F_{1u} & F_{1v} \\ F_{2u} & F_{2v} \end{vmatrix} = \begin{vmatrix} v & u \\ xy & 1 \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ 0 & 1 \end{vmatrix} = 0 \quad \text{at } (x,y,u,v) = \vec{0}$$

$\therefore$ Can't guarantee a function of (u,v) in terms of (x,y).

Check: From [2], $v = uxy$

$\therefore$ From [1], $y + x + u(uxy) = 0$, $u^2xy = -x - y$

$\therefore$ at $(x,y) = (0,0)$, u can have any value, so not uniquely determined.

And for $x < 0$, $y < 0$ near $(0,0)$, u can have two values.

$\therefore$ No function exists $u(x,y)$.

10. **10.** Investigate whether or not the system

$$u(x, y, z) = x + xyz$$
$$v(x, y, z) = y + xy$$
$$w(x, y, z) = z + 2x + 3z^2$$

can be solved for x, y, z in terms of u, v, w near $(x, y, z) = (0, 0, 0)$.

This involves inverse function Theorem

u, v, w are all C^1. Look at Jacobian at $(0,0,0)$

$$\begin{vmatrix} u_x & u_y & u_z \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{vmatrix} = \begin{vmatrix} 1+yz & xz & xy \\ y & 1+x & 0 \\ 2 & 0 & 1+6z \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 0 & 1 \end{vmatrix} \text{ at } (x,y,z) = (0,0,0)$$

$= 1 \neq 0$

$\therefore$ Near $(x, y, z, u, v, w) = (0,0,0,0,0,0)$ $\exists$ functions

$x = g_1(u, v, w)$ i.e., can be solved.
$y = g_2(u, v, w)$
$z = g_3(u, v, w)$

11. **11.** Consider $f(x, y) = ((x^2 - y^2)/(x^2 + y^2), xy/(x^2 + y^2))$. Does this map of $\mathbb{R}^2 \backslash (0, 0)$ to $\mathbb{R}^2$ have a local inverse near $(x, y) = (0, 1)$?

Let $f_1(x,y) = \dfrac{x^2-y^2}{x^2+y^2}$ $\quad f_2(x,y) = \dfrac{xy}{x^2+y^2}$

$\therefore f_{1x} = \dfrac{(x^2+y^2)(2x) - (x^2-y^2)(2x)}{(x^2+y^2)^2}$ $\quad f_{1x}(0,1) = 0$

$f_{1y} = \dfrac{(x^2+y^2)(-2y) - (x^2-y^2)(2y)}{(x^2+y^2)^2}$ $\quad f_{1y}(0,1) = 0$

Don't need to calculate f_{2x}, f_{2y}

$$\text{Jacobian}(0,1) = \begin{vmatrix} f_{1x} & f_{1y} \\ f_{2x} & f_{2y} \end{vmatrix} = \begin{vmatrix} 0 & 0 \\ - & - \end{vmatrix} = 0$$

∴ Inverse function Theorem doesn't guarantee an inverse near $(x,y) = (0,1)$.

Note that at $(x,y) = (0,1)$, $f(0,1) = (-1,0)$

In fact, $f(0,y) = (-1,0)$, $\forall y \neq 0$.

∴ Multiple points map to the same point, $(-1,0)$.

An inverse could not exist because there is no unique (x,y) which could be mapped from $(-1,0)$.

12.

12. (a) Define $x: \mathbb{R}^2 \to \mathbb{R}$ by $x(r,\theta) = r\cos\theta$ and define $y: \mathbb{R}^2 \to \mathbb{R}$ by $y(r,\theta) = r\sin\theta$. Show that

$$\left.\frac{\partial(x,y)}{\partial(r,\theta)}\right|_{(r_0,\theta_0)} = r_0.$$

(b) When can we form a smooth inverse function $(r(x,y), \theta(x,y))$? Check directly and with the inverse function theorem.

(c) Consider the following transformations for spherical coordinates (see Section 1.4):

$$\begin{aligned} x(\rho,\phi,\theta) &= \rho\sin\phi\cos\theta \\ x(\rho,\phi,\theta) &= \rho\sin\phi\sin\theta \\ z(\rho,\phi,\theta) &= \rho\cos\phi. \end{aligned}$$

Show that the Jacobian determinant is given by

$$\frac{\partial(x,y,z)}{\partial(\rho,\phi,\theta)} = \rho^2\sin\phi.$$

(d) When can we solve for (ρ,ϕ,θ) in terms of (x,y,z)?

(a) $$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix}_{(r_0,\theta_0)} = \begin{vmatrix} \cos\theta_0 & -r_0\sin\theta_0 \\ \sin\theta_0 & r_0\cos\theta_0 \end{vmatrix}$$

$$= r_0\cos^2\theta_0 + r_0\sin^2\theta_0 = r_0(\cos^2\theta_0 + \sin^2\theta_0) = r_0$$

(b) If $x = r\cos\theta$ Then $x - r\cos\theta = 0$
$y = r\sin\theta$ $y - r\sin\theta = 0$

Let $F_1(x,y,r,\theta) = x - r\cos\theta$
$F_2(x,y,r,\theta) = y - r\sin\theta$

By Implicit Function Theorem (Th. 12 in text),
(which is the inverse function Theorem in this case),

if $\Delta = \begin{vmatrix} \frac{\partial F_1}{\partial r} & \frac{\partial F_1}{\partial \theta} \\ \frac{\partial f_2}{\partial r} & \frac{\partial F}{\partial \theta} \end{vmatrix} \neq 0$ near (x_0,y_0,r_0,θ_0)

Then there exists functions $r = g_1(x,y)$ near (x_0,y_0,r_0,θ_0)
$\theta = g_2(x,y)$
and g_1, g_2 are C^1.

Here, $\Delta = \begin{vmatrix} \frac{\partial F_1}{\partial r} & \frac{\partial F_1}{\partial \theta} \\ \frac{\partial f_2}{\partial r} & \frac{\partial F}{\partial \theta} \end{vmatrix} = \begin{vmatrix} -\cos\theta & r\sin\theta \\ -\sin\theta & -r\cos\theta \end{vmatrix} = r$

So when $r \neq 0$, such inverse functions exist.

As a check, $x^2 = r^2\cos^2\theta$ and $\frac{y}{x} = \frac{r\sin\theta}{r\cos\theta} = \tan\theta$ $r \neq 0$

$y^2 = r^2\sin^2\theta$

$\therefore x^2 + y^2 = r^2(\cos^2\theta + \sin^2\theta) = r^2$

$\therefore r = \pm\sqrt{x^2+y^2}$, $\theta = \arctan\left(\frac{y}{x}\right)$

From $x = r\cos\theta$, assuming $r \neq 0$, and for $0 \leq \theta < 2\pi$, for $x = 0$, $\theta = \frac{\pi}{2}, \frac{3}{2}\pi$

and since $\cos\theta = \cos(-\theta)$, define domain as $r > 0$, $0 \leq \theta < \frac{\pi}{2}$. Then can have a 1-1 correspondence and $r = \sqrt{x^2+y^2}$, $\theta = \arctan\left(\frac{y}{x}\right)$.

(c)

(c) Consider the following transformations for spherical coordinates (see Section 1.4):

$$x(\rho, \phi, \theta) = \rho\sin\phi\cos\theta$$
$$x(\rho, \phi, \theta) = \rho\sin\phi\sin\theta$$
$$z(\rho, \phi, \theta) = \rho\cos\phi.$$

should be $y(\rho,\phi,\theta) = \rho\sin\phi\sin\theta$

Show that the Jacobian determinant is given by

$$\frac{\partial(x, y, z)}{\partial(\rho, \phi, \theta)} = \rho^2\sin\phi.$$

$x_\rho = \sin\phi\cos\theta$ $x_\phi = \rho\cos\phi\cos\theta$ $x_\theta = -\rho\sin\phi\sin\theta$

$y_\rho = \sin\phi\sin\theta$ $y_\phi = \rho\cos\phi\sin\theta$ $y_\theta = \rho\sin\phi\cos\theta$

$z_\rho = \cos\phi$ $z_\phi = -\rho\sin\phi$ $z_\theta = 0$

$$\therefore \frac{\partial(x,y,z)}{\partial(\rho,\phi,\theta)} = \begin{vmatrix} \sin\phi\cos\theta & \rho\cos\phi\cos\theta & -\rho\sin\phi\sin\theta \\ \sin\phi\sin\theta & \rho\cos\phi\sin\theta & \rho\sin\phi\cos\theta \\ \cos\phi & -\rho\sin\phi & 0 \end{vmatrix}$$

$$= \cos\phi\left[\rho^2\sin\phi\cos\phi\cos^2\theta + \rho^2\sin\phi\cos\phi\sin^2\theta\right]$$
$$-(-\rho\sin\phi)\left[\rho\sin^2\phi\cos^2\theta + \rho\sin^2\phi\sin^2\theta\right]$$
$$+0$$
$$= \cos\phi\left[\rho^2\sin\phi\cos\phi\left(\cos^2\theta+\sin^2\theta\right)\right]$$
$$+\rho\sin\phi\left[\rho\sin^2\phi\left(\cos^2\theta+\sin^2\theta\right)\right]$$
$$= \rho^2\sin\phi\cos^2\phi + \rho^2\sin\phi\sin^2\phi$$
$$= \rho^2\sin\phi\left(\cos^2\phi+\sin^2\phi\right)$$
$$= \underline{\rho^2\sin\phi}$$

(d) (d) When can we solve for (ρ, ϕ, θ) in terms of (x, y, z)?

From (c), when $\rho^2\sin\phi = 0$, so when $\rho \neq 0$ or $\phi \neq 0$. $\therefore$ To solve, $\rho \neq 0$ and $\phi \neq 0, \frac{\pi}{2}$

Makes sense, since if $\rho = 0$, ϕ and θ could have any value. If $\phi = 0, \frac{\pi}{2}$, then $x = y = 0$ for any value of θ.

13.

13. Let (x_0, y_0, z_0) be a point of the locus defined by $z^2 + xy - a = 0$, $z^2 + x^2 - y^2 - b = 0$, where a and b are constants.

(a) Under what conditions may the part of the locus near (x_0, y_0, z_0) be represented in the form $x = f(z)$, $y = g(z)$?

(b) Compute $f'(z)$ and $g'(z)$.

(a) Consider $F_1(z,x,y) = z^2 + xy - a = 0$ $\quad F_1$ is C^1

$F_2(z,x,y) = z^2 + x^2 - y^2 - b = 0$ $\quad F_2$ is C^1

Using General Implicit Theorem, $n=1$, $m=2$, and we've flipped the order of x, y, z to make it analogous to Theorem in text.

If $\begin{vmatrix} \frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} \\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} \end{vmatrix} \neq 0$, then can solve for

$x = f(z)$

$y = g(z)$.

$\therefore \frac{\partial F_1}{\partial x} = y \quad \frac{\partial F_1}{\partial y} = x \quad \frac{\partial F_2}{\partial x} = 2x \quad \frac{\partial F_2}{\partial y} = -2y$

$\therefore \begin{vmatrix} y & x \\ 2x & -2y \end{vmatrix} = -2y^2 - 2x^2 \neq 0 \iff -y^2 \neq x^2$

Since $x^2 \geq 0$, $-y^2 \leq 0$, $-y^2 \neq x^2$ everywhere

except $(x,y) = (0,0)$.

(b) Using implicit differentiation,

$\frac{\partial}{\partial z}(z^2 + xy - a) = 2z + \frac{\partial x}{\partial z} y + x \frac{\partial y}{\partial z} = 0$ [1]

$$\frac{\partial}{\partial z}\left(z^2 + x^2 - y^2 - 6\right) = 2z + 2x\frac{\partial x}{\partial z} - 2y\frac{\partial y}{\partial z} = 0 \quad [2]$$

Multiply [1] by $2y$, [2] by x and add:

$$4zy + 2y^2\frac{\partial x}{\partial z} + 2xy\frac{\partial y}{\partial z} = 0 \qquad [1]'$$

$$2xz + 2x^2\frac{\partial x}{\partial z} - 2xy\frac{\partial y}{\partial z} = 0 \qquad [2]'$$

$$\therefore 4zy + 2xz + (2y^2 + 2x^2)\frac{\partial x}{\partial z} = 0$$

From (a), $-y^2 \neq x^2$, or $x^2 + y^2 \neq 0$, then

$$f'(z) = \frac{\partial x}{\partial z} = \frac{-2zy - xz}{x^2 + y^2}$$

Now multiply [1] by $2x$, [2] by y, and subtract

$$4xz + 2xy\frac{\partial x}{\partial z} + 2x^2\frac{\partial y}{\partial z} = 0 \qquad [1]''$$

$$2yz + 2xy\frac{\partial x}{\partial z} - 2y^2\frac{\partial y}{\partial z} = 0 \qquad [2]''$$

$$\therefore 4xz - 2yz + (2x^2 + 2y^2)\frac{\partial y}{\partial z} = 0, \text{ and since } x^2 + y^2 \neq 0,$$

$$g'(z) = \frac{\partial y}{\partial z} = \frac{yz - 2xz}{x^2 + y^2}$$

14. **14.** Consider the unit sphere S given by $x^2 + y^2 + z^2 = 1$. S intersects the x axis at two points. Which variables can we solve for at these points? What about the points of intersection of S with the y and z axes?

We need 1-1 functions at points $(x,y,z) = \pm(1,0,0)$.

$\therefore$ Look at hemispheres where $\pm(1,0,0)$ are not on the edge.

$\therefore$ $x = f(y,z)$ or $x = -f(y,z)$, when $\pm(1,0,0)$ is at top or bottom of hemisphere.

i.e., $x(y,z) = \pm\sqrt{1-y^2-z^2}$

For $\pm(0,1,0)$, the y-coordinate is at top/bottom of hemisphere: $y = \pm\sqrt{1-x^2-z^2}$

For $\pm(0,0,1)$, the z-coordinate is at the top/bottom of hemisphere: $z = \pm\sqrt{1-x^2-y^2}$

15. **15.** Let $F(x, y) = x^3 - y^2$ and let C denote the level curve given by $F(x, y) = 0$.

(a) Without using the implicit function theorem, show that we can describe C as the graph of x as a function of y near any point.

(b) Show that $F_x(0, 0) = 0$. Does this contradict the implicit function theorem?

(a) $x^3 - y^2 = 0 \iff x^3 = y^2$. $\therefore$ $x = y^{2/3}$

Since $y^{\frac{1}{3}}$ is defined for negative & positive y,

then $y^{2/3}$ is defined for all y.

(5) $\frac{\partial F}{\partial x} = 3x^2 \quad \therefore F_x(0,0) = 3(0)^2 = 0$.

The Implicit Function Theorem says if $\frac{\partial F}{\partial x} \neq 0$, then we can solve x in terms of y.

If $F_x = 0$, may still be able to solve. There is just no guarantee.

16. **16.** Consider the system of equations

$$x^5v^2 + 2y^3u = 3$$
$$3yu - xuv^3 = 2.$$

Show that near the point $(x, y, u, v) = (1, 1, 1, 1)$, this system defines u and v implicitly as functions of x and y. For such local functions u and v, define the local function f by $f(x, y) = (u(x, y), v(x, y))$. Find $Df(1, 1)$.

(a) Let $F_1 = x^5v^2 + 2y^3u - 3 = 0$

$F_2 = 3yu - xuv^3 - 2 = 0$

$\frac{\partial F_1}{\partial u} = 2y^3 \quad \frac{\partial F_1}{\partial v} = 2x^5v$

$\frac{\partial F_2}{\partial u} = 3y - xv^3 \quad \frac{\partial F_2}{\partial v} = -3xuv^2$

$$\therefore \begin{vmatrix} 2y^3 & 2x^5v \\ 3y - xv^3 & -3xuv^2 \end{vmatrix}_{(x,y,u,v) = (1,1,1,1)} = \begin{vmatrix} 2 & 2 \\ 2 & -3 \end{vmatrix} = -10 \neq 0$$

$\therefore \Delta \neq 0$, so can solve $u(x,y)$, $v(x,y)$.

(6) If $f(x,y) = (u(x,y), v(x,y))$, Then

$$Df(1,1) = \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix}_{(x,y)=(1,1)}$$

Note, when $(x,y) = (1,1)$, $(u,v) = (1,1)$.

Use implicit differentiation to find u_x, u_y, v_x, v_y.

$$\frac{\partial}{\partial x}(x^5v^2 + 2y^3u - 3) = 5x^4v^2 + 2x^5v\frac{\partial v}{\partial x} + 2y^3\frac{\partial u}{\partial x} = 0 \quad [1]$$

$$\frac{\partial (3yu - xuv^3 - 2)}{\partial x} = 3y\frac{\partial u}{\partial x} - uv^3 - x\frac{\partial u}{\partial x}v^3 - 3xuv^2\frac{\partial v}{\partial x} = 0$$

$$= -uv^3 + (3y - xv^3)\frac{\partial u}{\partial x} - 3xuv^2\frac{\partial v}{\partial x} = 0 \quad [2]$$

Using $(x,y,u,v) = (1,1,1,1)$,

$$5 + 2u_x + 2v_x = 0 \quad [1]'$$
$$-1 + 2u_x - 3v_x = 0 \quad [2]'$$

$\therefore 6 = 5v_x$, $v_x = \frac{6}{5}$, $13 + 10u_x = 0$, $u_x = \frac{-13}{10}$

$$\frac{\partial}{\partial y}(x^5v^2 + 2y^3u - 3) = 2x^5vv_y + 6y^2u + 2y^3u_y = 0 \quad [3]$$

$$\frac{\partial}{\partial y}(3yu - xuv^3 - 2) \quad 3u + 3yu_y - xu_yv^3 - 3xuv^2v_y = 0 \quad [4]$$

Using $(x,y,u,v)=(1,1,1,1)$,

$$6 + 2v_y + 2u_y = 0 \quad [3]'$$

$$3 - 3v_y + 2u_y = 0 \quad [4]'$$

$$\therefore\ 3 + 5v_y = 0,\ v_y = -\frac{3}{5},\quad 24 + 10u_y = 0,\ u_y = -\frac{12}{5}$$

$$\therefore\ D f(1,1) = \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix}_{(x,y)=(1,1)} = \begin{bmatrix} -\frac{13}{10} & -\frac{12}{5} \\ \frac{6}{5} & -3/5 \end{bmatrix}$$

17.

17. Consider the equations

$$x^2 - y^2 - u^3 + v^2 + 4 = 0$$
$$2xy + y^2 - 2u^2 + 3v^4 + 8 = 0.$$

(a) Show that these equations determine functions $u(x, y)$ and $v(x, y)$ near the point $(x, y, u, v) = (2, -1, 2, 1)$.

(b) Compute $\frac{\partial u}{\partial x}$ at $(x, y) = (2, -1)$.

(a) Let $F_1(x,y,u,v) = x^2 - y^2 - u^3 + v^2 + 4 = 0$ $\quad F_1$ is C^1

$F_2(x,y,u,v) = 2xy + y^2 - 2u^2 + 3v^4 + 8 = 0$ $\quad F_2$ is C^1

$$\frac{\partial F_1}{\partial u} = -3u^2 \quad \frac{\partial F_1}{\partial v} = 2v \quad \frac{\partial F_2}{\partial u} = -4u \quad \frac{\partial F_2}{\partial v} = 12v^3$$

$$\Delta = \begin{vmatrix} -3u^2 & 2v \\ -4u & 12v^3 \end{vmatrix} = -36u^2v^3 + 8uv$$

$$\therefore\ \Delta(2,1) = -36(4)(1) + 8(2)(1) = -128 \neq 0$$

$\therefore$ By Implicit Function Theorem, near $(x,y,u,v)=(2,-1,2,1)$,

There exist unique C^1 functions

$u = K_1(x,y)$ and $v = K_2(x,y)$ s.t.

$$F_1[x,y,K_1(x,y),K_2(x,y)] = 0$$

$$F_2[x,y,K_1(x,y),K_2(x,y)] = 0$$

(6) $$\frac{\partial F_1}{\partial x} = 2x - 3u^2u_x + 2vv_x = 0 \quad [1]$$

$$\frac{\partial F_2}{\partial x} = 2y - 4uu_x + 12v^3v_x = 0 \quad [2]$$

Multiply [1] by $6v^2$ and subtract [2]

$$12xv^2 - 18u^2v^2u_x + 12v^3v_x = 0 \quad [1]'$$

$$2y - 4uu_x + 12v^3v_x = 0 \quad [2]$$

$$12xv^2 - 2y - (18u^2v^2 - 4u)u_x = 0 \quad [3]$$

Using $(x,y,u,v)=(2,-1,2,1)$ [3] becomes:

$$12(2)(1) - 2(-1) = [18(4)(1) - 4(2)]u_x, \text{ or}$$

$$26 = 64u_x, \quad u_x = \frac{13}{32}$$

$\therefore$ $\underline{u_x(2,-1) = \frac{13}{32}}$

18. **18.** Is it possible to solve the system of equations

$$xy^2 + xzu + yv^2 = 3$$
$$u^3yz + 2xv - u^2v^2 = 2$$

for $u(x, y, z)$, $v(x, y, z)$ near $(x, y, z) = (1, 1, 1)$, $(u, v) = (1, 1)$? Compute $\partial v/\partial y$ at $(x, y, z) = (1, 1, 1)$.

(a) Let $F_1(x,y,z,u,v) = xy^2 + xzu + yv^2 - 3 = 0$ F_1 is C^1

$F_2(x,y,z,u,v) = u^3yz + 2xv - u^2v^2 - 2 = 0$ F_2 is C^1.

Note $F_1(1,1,1,1,1) = 0$, $F_2(1,1,1,1,1) = 0$

$\frac{\partial F_1}{\partial u} = xz$ $\frac{\partial F_1}{\partial v} = 2yv$

$\frac{\partial F_2}{\partial u} = 3u^2yz - 2uv^2$ $\frac{\partial F_2}{\partial v} = 2x - 2u^2v$

$$\therefore \Delta = \begin{vmatrix} xz & 2yv \\ 3u^2yz - 2uv^2 & 2x - 2u^2v \end{vmatrix}$$

$$= 2x^2z - 2xzu^2v - 6y^2u^2zv + 4yuv^3$$

$\Delta(1,1,1,1,1) = 2 - 2 - 6 + 4 = -2 \neq 0.$

$\therefore$ yes, near $(x,y,z,u,v) = (1,1,1,1,1)$, can solve for $u(x,y,z)$, $v(x,y,z)$.

(b) $\frac{\partial F_1}{\partial y} = 2xy + xzu_y + v^2 + 2yvv_y = 0$ [1]

$$\frac{\partial F_2}{\partial y} = a^3 z + 3u^2 y z u_y + 2x v_y - 2u u_y v^2 - 2u^2 v v_y = 0 \quad [2]$$

Using $(x,y,z,u,v) = (1,1,1,1,1)$, [1] and [2] become

$$3 + u_y + 2v_y = 0 \quad [1]'$$

$$1 + u_y = 0 \quad [2]'$$

$$\therefore 2 + 2v_y = 0, \quad \therefore \underline{v_y(1,1,1) = -1}$$

19.

19. The problem of factoring a polynomial $x^n + a_{n-1}x^{n-1} + \cdots + a_0$ into linear factors is, in a sense, an "inverse function" problem. The coefficients a_i may be thought of as functions of the n roots r_j. We would like to find the roots as functions of the coefficients in some region. With $n = 3$, apply the inverse function theorem to this problem and state what it tells you about the possibility of doing this.

For $n = 3$, $x^3 + a_2 x^2 + a_1 x + a_0 = 0$

Let r_1, r_2, r_3 be the roots.

$$\therefore (x - r_1)(x - r_2)(x - r_3) = x^3 + a_2 x^2 + a_1 x + a_0$$

$$\text{or } r_1 \left[x^2 - (r_1 + r_2)x + r_1 r_2\right](x - r_3) =$$

$$x^3 - (r_1 + r_2)x^2 + r_1 r_2 x$$

$$- r_3 x^2 + r_3(r_1 + r_2)x - r_1 r_2 r_3$$

$$= x^3 - (r_1 + r_2 + r_3)x^2 + (r_1 r_2 + r_1 r_3 + r_2 r_3)x - r_1 r_2 r_3 = 0$$

$$\therefore \quad a_2(r_1, r_2, r_3) = -(r_1 + r_2 + r_3)$$

$$a_1(r_1, r_2, r_3) = (r_1 r_2 + r_1 r_3 + r_2 r_3)$$

$$a_0(r_1, r_2, r_3) = -r_1 r_2 r_3$$

$$\therefore \quad \frac{\partial a_2}{\partial r_1} = -1 \quad \frac{\partial a_2}{\partial r_2} = -1 \quad \frac{\partial a_2}{\partial r_3} = -1$$

$$\frac{\partial a_1}{\partial r_1} = r_2 + r_3 \quad \frac{\partial a_1}{\partial r_2} = r_1 + r_3 \quad \frac{\partial a_1}{\partial r_3} = r_1 + r_2$$

$$\frac{\partial a_0}{\partial r_1} = -r_2 r_3 \quad \frac{\partial a_0}{\partial r_2} = -r_1 r_3 \quad \frac{\partial a_0}{\partial r_3} = -r_1 r_2$$

From Inverse Function Theorem, can we find $r_1(a_2, a_1, a_0)$, $r_2(a_2, a_1, a_0)$, $r_3(a_2, a_1, a_0)$?

Look at above Jacobian determinant:

$$\frac{\partial(a_2, a_1, a_0)}{\partial(r_1, r_2, r_3)} = \begin{vmatrix} -1 & -1 & -1 \\ r_2 + r_3 & r_1 + r_3 & r_1 + r_2 \\ -r_2 r_3 & -r_1 r_3 & -r_1 r_2 \end{vmatrix}$$

$$= -1\left[-r_1 r_2 (r_1 + r_3) + r_1 r_3 (r_1 + r_2)\right]$$

$$+ 1\left[-r_1 r_2 (r_2 + r_3) + r_2 r_3 (r_1 + r_2)\right]$$

$$-1\left[-r_1 r_3 (r_2 + r_3) + r_2 r_3 (r_1 + r_3)\right]$$

$$= r_1^2 r_2 + r_1 r_2 r_3 - r_1^2 r_3 - r_1 r_2 r_3$$

$$- r_1 r_2^2 - r_1 r_2 r_3 + r_1 r_2 r_3 + r_2^2 r_3$$

$$+ r_1 r_2 r_3 + r_1 r_3^2 - r_1 r_2 r_3 - r_2 r_3^2$$

$$= r_1^2 r_2 - r_1^2 r_3 + r_2^2 r_3 - r_1 r_2^2 + r_1 r_3^2 - r_2 r_3^2$$

$$= r_1 (r_1 r_2 - r_1 r_3 - r_2^2) + r_3 (r_2^2 + r_1 r_3 - r_2 r_3)$$

$$= r_1 (r_1 r_2 - r_1 r_3 - r_2^2) + r_1 r_2 r_3 - r_1 r_2 r_3 + r_3 (r_2^2 + r_1 r_3 - r_2 r_3)$$

$$= r_1 (r_1 r_2 - r_1 r_3 - r_2^2 + r_2 r_3) + r_3 ($$

$$= r_1 (r_1 - r_2)(r_2 - r_3) + r_3 (r_2 - r_3)(r_2 - r_1)$$

$$= r_1 (r_1 - r_2)(r_2 - r_3) - r_3 (r_2 - r_3)(r_1 - r_2)$$

$$= (r_1 - r_3)(r_1 - r_2)(r_2 - r_3)$$

$$\therefore \frac{\partial (a_2, a_1, a_0)}{\partial (r_1, r_2, r_3)} = (r_1 - r_3)(r_1 - r_2)(r_2 - r_3)$$

This will be non-zero when all roots are different.

If so, then there are open balls U of (a_2, a_1, a_0) and V of (r_1, r_2, r_3), s.t. there are C^1 functions of the roots in terms of the coefficients (by inverse function theorem).

Review Exercises for Chapter 3

Note Title 7/21/2016

1. **1.** Let f be any differentiable function. Show that $u = f(y - kx)$ is a solution to the partial differential equation $\frac{\partial u}{\partial x} + k\frac{\partial u}{\partial y} = 0$.

Assume $f: \mathbb{R} \to \mathbb{R}$. Let $g(x,y): \mathbb{R}^2 \to \mathbb{R}$ be $g(x,y) = y - kx$.

$\therefore u(x,y) = (f \circ g)(x,y)$, where $u: \mathbb{R}^2 \to \mathbb{R}$

By chain rule $D u(x,y) = D f(g(x,y)) \cdot D g(x,y)$.

size of matrices $\to$ 1×2 $\quad$ 1×1 $\quad$ 1×2

$\therefore [u_x \;\; u_y] = f'(g(x,y)) \cdot [g_x \;\; g_y]$

$\therefore u_x = f'(g(x,y)) \cdot g_x$, $u_y = f'(g(x,y)) \cdot g_y$

$$g_x = \frac{\partial (y - kx)}{\partial x} = -k$$

$$g_y = \frac{\partial (y - kx)}{\partial y} = 1$$

$\therefore [u_x \;\; u_y] = f'(y - kx)[-k \;\; 1]$

$$\therefore \frac{\partial u}{\partial x} + k\frac{\partial u}{\partial y} = f'(y-kx)(-k) + k[f'(y-kx)(1)]$$

$$= -k f'(y-kx) + k f'(y-kx)$$

$$= 0$$

2. **2.** Prove that if u and v have continuous mixed second partial derivatives and satisfy the Cauchy–Riemann equations

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$$
$$\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x},$$

then both u and v are harmonic.

u and v having continuous mixed second partial derivitafes $\Rightarrow u_{xy} = u_{yx}$ and $v_{xy} = v_{yx}$.

A function $f(x,y): \mathbb{R}^2 \to \mathbb{R}$ is harmonic if its 2nd partials are continuous and $f_{xx} + f_{yy} = 0$

$$\therefore \frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial v}{\partial y}\right) = v_{yx}$$

$$\frac{\partial^2 u}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}\right) = \frac{\partial}{\partial y}\left(-\frac{\partial v}{\partial x}\right) = -v_{xy} = -v_{yx}$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = v_{yx} + (-v_{yx}) = 0$$

$\therefore u(x,y)$ is harmonic

Similarly, $\frac{\partial^2 v}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial v}{\partial x}\right) = \frac{\partial}{\partial x}\left(-\frac{\partial u}{\partial y}\right) = -u_{yx}$

$$\frac{\partial^2 v}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial v}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial u}{\partial x}\right) = u_{xy} = u_{yx}$$

$$\therefore \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = -u_{yx} + u_{yx} = 0$$

$\therefore V(x,y)$ is harmonic

3. **3.** Let $f(x, y) = x^2 - y^2 - xy + 5$. Find all critical points of f and determine whether they are local minima, local maxima, or saddle points.

$$\left.\begin{aligned} f_x &= 2x - y = 0 \\ f_y &= -x - 2y = 0 \end{aligned}\right\} \quad \therefore 5x = 0 \Rightarrow x = 0, \quad \therefore y = 0$$

$\therefore (0,0)$ is the only critical point.

$$f_{xx} = 2 \quad f_{xy} = -1 \quad f_{yy} = -2$$

$$\therefore \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = \begin{vmatrix} 2 & -1 \\ -1 & -2 \end{vmatrix} = -4 - 1 = -5 < 0$$

$\therefore (0,0)$ is a saddle point (by Theorem 6 of text).

4. **4.** Find the absolute minimum and maximum values of the function $f(x, y) = x^2 + 3xy + y^2 + 5$ on the unit disc $D = \{(x, y) \mid x^2 + y^2 \le 1\}$.

D is closed and bounded, f is continuous on D, so absolute max & min exist.

Critical points: $\left.\begin{aligned} f_x &= 2x + 3y = 0 \\ f_y &= 3x + 2y = 0 \end{aligned}\right\} \Rightarrow (x,y) = (0,0)$

and $(0,0) \in D$.

$f(0,0) = 5$

Check boundary: $x^2 + y^2 = 1$.

Use Lagrange multiplier: Let $g(x,y) = x^2 + y^2 - 1 = 0$

$\therefore \nabla f = [2x + 3y, 3x + 2y] = \lambda [2x, 2y]$

$\therefore \begin{cases} 2x + 3y = \lambda 2x \\ 3x + 2y = \lambda 2y \end{cases}$ If $x = 0$, then $3y = 0$ which is impossible since $(0,0) \notin x^2 + y^2 - 1 = 0$

$\therefore \dfrac{2x + 3y}{2x} = \lambda$

$\therefore 3x + 2y = \left(\dfrac{2x + 3y}{2x}\right) 2y$, or

$6x^2 + 4xy = 4xy + 6y^2 \Rightarrow x^2 = y^2$.

$\therefore x = \pm y$.

From $x^2 + y^2 = 1$, $2x^2 = 1$, $x = \pm \dfrac{\sqrt{2}}{2}$

$\therefore \left(\dfrac{\sqrt{2}}{2}, \dfrac{\sqrt{2}}{2}\right), \left(\dfrac{\sqrt{2}}{2}, -\dfrac{\sqrt{2}}{2}\right), \left(-\dfrac{\sqrt{2}}{2}, \dfrac{\sqrt{2}}{2}\right), \left(-\dfrac{\sqrt{2}}{2}, -\dfrac{\sqrt{2}}{2}\right)$.

$\therefore f\left(\dfrac{\sqrt{2}}{2}, \dfrac{\sqrt{2}}{2}\right) = \left(\dfrac{\sqrt{2}}{2}\right)^2 + 3\left(\dfrac{\sqrt{2}}{2}\right)\left(\dfrac{\sqrt{2}}{2}\right) + \left(\dfrac{\sqrt{2}}{2}\right)^2 + 5 = 7\dfrac{1}{2}$

$f\left(\dfrac{\sqrt{2}}{2}, -\dfrac{\sqrt{2}}{2}\right) = \left(\dfrac{\sqrt{2}}{2}\right)^2 + 3\left(\dfrac{\sqrt{2}}{2}\right)\left(-\dfrac{\sqrt{2}}{2}\right) + \left(-\dfrac{\sqrt{2}}{2}\right)^2 + 5 = 4\dfrac{1}{2}$

$f\left(-\dfrac{\sqrt{2}}{2}, \dfrac{\sqrt{2}}{2}\right) = \left(-\dfrac{\sqrt{2}}{2}\right)^2 + 3\left(-\dfrac{\sqrt{2}}{2}\right)\left(\dfrac{\sqrt{2}}{2}\right) + \left(\dfrac{\sqrt{2}}{2}\right)^2 + 5 = 4\dfrac{1}{2}$

$$f\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right) = \left(-\frac{\sqrt{2}}{2}\right)^2 + 3\left(-\frac{\sqrt{2}}{2}\right)\left(-\frac{\sqrt{2}}{2}\right) + \left(-\frac{\sqrt{2}}{2}\right)^2 = 7\frac{1}{2}$$

$\therefore$ Absolute max $= 7\frac{1}{2}$, located at $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$ and $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

Absolute min $= 4\frac{1}{2}$, located at $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$ and $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

5.

5. Find the second-order Taylor polynomial for $f(x, y) = y^2 e^{-x^2}$ at $(1, 1)$.

$$f(\vec{x_0} + \vec{h}) = f(\vec{x_0}) + \nabla f(\vec{x_0}) \cdot \vec{h} + \frac{1}{2}\vec{h}^T Hf(\vec{x_0})\vec{h} + R_2(\vec{x_0}, \vec{h})$$

or, $$f(\vec{x}) = f(\vec{x_0}) + \nabla f(\vec{x_0})(\vec{x} - \vec{x_0}) + \frac{1}{2}(\vec{x} - \vec{x_0})^T Hf(\vec{x_0})(\vec{x} - \vec{x_0}) + R_2$$

$$f(1,1) = e^{-1}$$

$$\nabla f = \left[-2xy^2 e^{-x^2},\ 2y e^{-x^2}\right]$$

$$\therefore \nabla f(1,1) = \left[-2e^{-1},\ 2e^{-1}\right]$$

$$f_{xx} = -2y^2 e^{-x^2} + 4x^2 y^2 e^{-x^2} \qquad f_{xx}(1,1) = 2e^{-1}$$

$$f_{xy} = -4xy e^{-x^2} \qquad f_{xy}(1,1) = -4e^{-1}$$

$$f_{yy} = 2e^{-x^2} \qquad f_{yy}(1,1) = 2e^{-1}$$

$$\therefore f(x,y) = e^{-1} + \left[-2(x-1)e^{-1} + 2(y-1)e^{-1}\right] + \frac{1}{2}\left[2(x-1)^2 e^{-1} - 8(x-1)(y-1)e^{-1} + 2(y-1)^2 e^{-1}\right]$$

$$= e^{-1}\left[1 - 2(x-1) + 2(y-1) + (x-1)^2 + (y-1)^2 - 4(x-1)(y-1)\right]$$

Note: $\frac{1}{2}\vec{h}^T \begin{Bmatrix} a & b \\ b & c \end{Bmatrix} \vec{h} = \frac{1}{2}ah_1^2 + bh_1h_2 + \frac{1}{2}ch_2^2$

6. **6.** Let $f(x, y) = ax^2 + bxy + cy^2$.

(a) Find $g(x, y)$, the second-order Taylor approximation to f at $(0, 0)$.

(b) What is the relationship between g and f?

(c) Prove that $R_2(\mathbf{x}_0, \mathbf{h}) = 0$ for all $\mathbf{x}_0$, $\mathbf{h} \in \mathbb{R}^2$. (HINT: Show that f is equal to its second-order Taylor approximation at every point.)

(a) $g(x,y) = f(0,0) + \nabla f(0,0)\cdot[x\ y] + \frac{1}{2}[x\ y]\,Hf(0,0)\begin{bmatrix} x \\ y \end{bmatrix}$

$= 0 + [2ax + by \quad bx + 2cy]_{(0,0)}\cdot[x\ y]$

$+ \frac{1}{2}[x\ y]\begin{bmatrix} 2a & b \\ b & 2c \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$

$= \underline{ax^2 + bxy + cy^2}$

(b) $g(x,y) = f(x,y)$

(c) $f(\vec{x_0} + \vec{h}) = f(\vec{x_0}) + Df(\vec{x_0})\cdot\vec{h} + \frac{1}{2}\vec{h}^T Hf(\vec{x_0})\vec{h} + R_2(\vec{x_0}, \vec{h})$

Let $\vec{x_0} = (x, y)$, $\vec{h} = (h_1, h_2)$

$\therefore Df(x,y) = [2ax + by \quad bx + 2cy]$

$Df(x,y)\cdot\vec{h} = 2ah_1x + bh_1y + bh_2x + 2ch_2y$

$$\frac{1}{2}\vec{h}\,Hf(\vec{x}_0)\vec{h} = \frac{1}{2}\begin{bmatrix} h_1 & h_2 \end{bmatrix}\begin{bmatrix} 2a & b \\ b & 2c \end{bmatrix}\begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$$

$$= ah_1^2 + bh_1h_2 + ch_2^2$$

$$\therefore f(\vec{x}_0) + Df(\vec{x}_0)\cdot\vec{h} + \frac{1}{2}\vec{h}^T Hf(\vec{x}_0)\vec{h} = \quad [1]$$

$$\left[ax^2 + bxy + cy^2\right] + \left[2ah_1x + bh_1y + bh_2x + 2ch_2y\right]$$
$$+ \left[ah_1^2 + bh_1h_2 + ch_2^2\right]$$

$$= \left[ax^2 + 2ah_1x + ah_1^2\right] + \left[cy^2 + 2ch_2y + ch_2^2\right]$$
$$+ \left[bxy + bh_1y + bh_2x + bh_1h_2\right]$$

$$= a(x+h_1)^2 + c(y+h_2)^2 + b(x+h_1)y + b(x+h_1)h_2$$

$$= a(x+h_1)^2 + c(y+h_2)^2 + b(x+h_1)(y+h_2) \quad [2]$$

$$f(\vec{x}_0+\vec{h}) = f(x+h_1, y+h_2) \quad [3]$$

$$= a(x+h_1)^2 + b(x+h_1)(y+h_2) + c(y+h_2)^2 \quad [4]$$

Since [2] = [4], then [1] = [3].

i.e., $f(\vec{x}_0+\vec{h}) = f(\vec{x}_0) + Df(\vec{x}_0)\cdot\vec{h} + \frac{1}{2}\vec{h}^T Hf(\vec{x}_0)\vec{h}$

this means $R_2(\vec{x}_0, \vec{h}) = 0$ since

$$f(\vec{x}_0+\vec{h}) = f(\vec{x}_0) + Df(\vec{x}_0)\cdot\vec{h} + \frac{1}{2}\vec{h}^T Hf(\vec{x}_0)\vec{h} + R_2(\vec{x}_0, \vec{h})$$

7.

7. Analyze the behavior of the following functions at the indicated points. [Your answer in part (b) may depend on the constant C.]

(a) $z = x^2 - y^2 + 3xy, \quad (x, y) = (0, 0)$

(b) $z = x^2 - y^2 + Cxy, \quad (x, y) = (0, 0)$

(a) $z_x = 2x + 3y \quad z_x(0,0) = 0$
$z_y = -2y + 3x \quad z_y(0,0) = 0$
$\therefore (0,0)$ a critical point

$z_{xx} = 2 \quad z_{xy} = 3 \quad z_{yy} = -2$

$\begin{vmatrix} 2 & 3 \\ 3 & -2 \end{vmatrix} = -4 - 6 = -10 \quad \therefore (0,0)$ is a saddle point

(b) $z_x = 2x + Cy \quad z_x(0,0) = 0$
$z_y = -2y + Cx \quad z_y(0,0) = 0$
$\therefore (0,0)$ is a critical point

$z_{xx} = 2 \quad z_{xy} = C \quad z_{yy} = -2$

$\begin{vmatrix} 2 & C \\ C & -2 \end{vmatrix} = -4 - C^2 < 0$ independent of C.

$\therefore (0,0)$ is a saddle point

8.

8. Find and classify the extreme values (if any) of the functions on $\mathbb{R}^2$ defined by the following expressions:

(a) $y^2 - x^3$

(b) $(x - 1)^2 + (x - y)^2$

(c) $x^2 + xy^2 + y^4$

(a) $f_x = -3x^2$
$f_y = 2y$
$\therefore$ only critical point is $(0,0)$.

$f_{xx} = -6x \quad f_{xy} = 0 \quad f_{yy} = 2$

$$\begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}_{(0,0)} = \begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} = 0. \quad \therefore \text{ 2nd derivative test not helpful.}$$

But along line $x=0$, $f(x,y) = y^2$, so $(0,0)$ would appear to be a minimum.

But along line $y=0$, $f(x,y) = -x^3$, $(0,0)$ like an inflection point.

$\therefore$ $(0,0)$ acts like a saddle point, where any local neighborhood has values greater and less than 0.

(6) $f_x = 2(x-1) + 2(x-y)$

$f_y = -2(x-y)$

$f_y = 0 \Rightarrow x = y$

For f_x at $x=y$, $f_x = 2(x-1) = 0 \Rightarrow x = 1$

$\therefore$ Only critical point is $(x,y) = (1,1)$

$f_{xx} = 4 \quad f_{xy} = -2 \quad f_{yy} = 2$

$\therefore \begin{vmatrix} 4 & -2 \\ -2 & 2 \end{vmatrix} = 8 - 4 = 4 > 0$, and $f_{xx}(1,1) = 4 > 0$.

$\therefore$ $(1,1)$ a local minimum.

As domain is all of $\mathbb{R}^2$, the only extremum is $(1,1)$ and it is an absolute minimum.

(c) $f(x,y) = x^2 + xy^2 + y^4$

$\therefore f_x = 2x + y^2$ $\left.\right\}$ $f_x = 0 \Rightarrow x = -\frac{y^2}{2}$

$f_y = 2xy + 4y^3$ $\left.\right\}$ $\therefore f_y = 2\left(-\frac{y^2}{2}\right)y + 4y^3$

or $f_y = 3y^3 = 0 \Rightarrow y = 0 \quad \therefore x = 0.$

$\therefore (0,0)$ the only critical point.

$f_{xx} = 2 \quad f_{xy} = 2y \quad f_{yy} = 2x + 12y^2$

$\therefore \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}_{(0,0)} = \begin{vmatrix} 2 & 0 \\ 0 & 0 \end{vmatrix} = 0.$ $\therefore$ 2nd derivative test not helpful.

At $(0,0)$, $x^2 + xy^2 + y^4 = 0.$

Along $x = 0$, $f(x,y) = y^4$, so $f(x,y) \geq 0$

Along $y = 0$, $f(x,y) = x^2$, so $f(x,y) \geq 0$

Can $x^2 + xy^2 + y^4$ ever be < 0?

Examining terms, $x^2 \geq 0$, $y^4 \geq 0$ $\forall x, y$.

The only possible negative term is xy^2 when $x < 0$.

Consider any small ball around $(0,0)$: $B_\delta(0,0)$

where $\delta < 1$. That is, close to $(0,0)$.

$\therefore |x| < \delta$, $|y| < \delta$. $\therefore x^2 < \delta^2$, $|x|y^2 < \delta^3$ and $y^4 < \delta^4$.

Since for $\delta < 1$, $\delta^4 < \delta^3 < \delta^2$, the dominant term will be x^2, so that very near $(0,0)$, $y^4 + |x|y^2 < x^2$. $\therefore x^2 - xy^2$ will always be > 0 when (x,y) is very close to $(0,0)$.

$\therefore$ very near $(0,0)$, $x^2 + xy^2 + y^4 \geq 0$, so

$(0,0)$ is a local minimum.

A more elegant way to show this is to "square" the terms: $x^2 + xy^2 + y^4$

$(x + y^2)^2 = x^2 + 2xy^2 + y^4$.

$\therefore f(x,y) = (x+y^2)^2 - xy^2$

For $x < 0$, $xy^2 \geq 0$ and $(x+y^2)^2 \geq 0$.

$\therefore$ For all (x,y), $x^2 + xy^2 + y^4 \geq 0$, so

$(0,0)$ is an absolute minimum.

9. **9.** (a) Find the minimum distance from the origin in $\mathbb{R}^3$ to the surface $z = \sqrt{x^2 - 1}$.

(b) Repeat part (a) for the surface $z = 6xy + 7$.

(a) This can be rephrased as finding the minimum of $f(x,y,z) = x^2+y^2+z^2$ subject to the constraint $g(x,y,z) = \sqrt{x^2-1} - z = 0$, as $x^2+y^2+z^2$ is the $(\text{distance})^2$ to the origin.

Note that S as defined by $g(x,y,z)$ is not a closed bounded set, so no guarantee of a min or max of $f(x,y,z)$ existing on S.

In fact, $\nabla f = (2x, 2y, 2z)$ and

$$g_x = \frac{x}{\sqrt{x^2-1}}, \quad g_y = 0, \quad g_z = -1.$$

$$\therefore 2x = \lambda \frac{x}{\sqrt{x^2-1}} \qquad [1]$$

$$4x^2 - 4 = \lambda^2$$

$$2y = \lambda(0) \qquad [2]$$

$$2z = -\lambda \qquad [3]$$

$$\sqrt{x^2-1} - z = 0 \qquad [4]$$

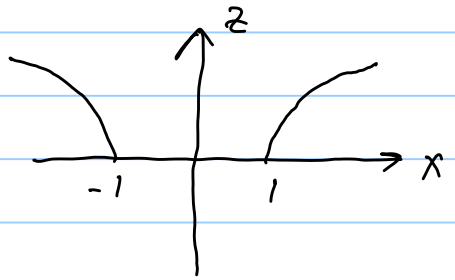
Since $x \geq 1$, $x \neq 0$, so [1] becomes $2 = \frac{\lambda}{\sqrt{x^2-1}}$

or $2\sqrt{x^2-1} = \lambda$. With [4], $2z = \lambda$

But [3] says $2z = -\lambda$. $\therefore \lambda = 0$.
But $\lambda = 0 \Rightarrow x = 0$ from [1].

$\therefore$ Contradictory results, and can't use Lagrange test since $f(x,y,z)$ has no max or min on S (for $|x| > 1$).

By inspection,



The minimum point is $(1, y, 0)$, $(-1, y, 0)$,
so minimum distance is on x-axis, and
is 1. (Note, y can be anything, and
$z = \sqrt{x^2 - 1}$ is a double "sheet").

Perhaps parameterizing S can give an analytical method for the solution.

Let $x(t) = \sec(t)$, $-\frac{\pi}{2} < t < \frac{\pi}{2}$

Then $g(x(t), y, z) = \sqrt{\sec^2(t) - 1} - z = |\tan(t)| - z$.

and note for $-\frac{\pi}{2} < t < \frac{\pi}{2}$, $\sec(t) \geq 1$.

$\therefore f(x(t), y, z) = \sec^2(t) + y^2 + z^2$.

$\therefore \frac{\partial f}{\partial t} = 2\sec(t)\cdot\sec(t)\cdot\tan(t) = 2\sec^2(t)\tan(t)$

$\frac{\partial f}{\partial y} = 2y \quad \frac{\partial f}{\partial z} = 0.$

$\therefore$ Setting $\frac{\partial f}{\partial t} = 0 \Rightarrow \tan(t) = 0 \Rightarrow t = 0.$

$\frac{\partial f}{\partial y} = 0$ and $\frac{\partial f}{\partial z} = 0 \Rightarrow y = z = 0.$

$\therefore$ a critical point is $(t, y, z) = (0, 0, 0)$,

which is, $\sec(0) = 1$, so $(x, y, z) = (1, 0, 0)$.

For $\frac{\pi}{2} < t < \frac{3\pi}{2}$, you get $(x, y, z) = (-1, 0, 0)$.

$\therefore$ Critical points are $(x, y, z) = (\pm 1, 0, 0)$.

Since $f(x, y, z) \geq 0$ and $\to \infty$ $(x, y, z) \to (\infty, \infty, \infty)$,

and the constraint requires $x \geq 1$, f is

a minimum at $(\pm 1, 0, 0)$ and its value is 1.

(6) Let $g(x, y, z) = 6xy + 7 - z = 0.$

$\nabla g = \langle 6y, 6x, -1 \rangle$

Suppose $f(x, y, z) = x^2 + y^2 + z^2$ restricted to $g(x, y, z)$

has an extremum.

$\therefore [2x, 2y, 2z] = \lambda [6y, 6x, -1]$, or

$2x = \lambda 6y \quad [1]$
$2y = \lambda 6x \quad [2]$
$2z = -\lambda \quad [3]$
$6xy + 7 - z = 0 \quad [4]$

$\lambda = 0 \Rightarrow (x, y, z) = (0, 0, 0)$, but $(0, 0, 0)$ is not on $6xy + 7 - z = 0$. $\therefore \lambda \neq 0$

Suppose $x = 0$. $\therefore [2] \Rightarrow y = 0$, $[4] \Rightarrow z = 7$

$\therefore$ One possibility is $(0, 0, 7)$. Distance $= 7$.

Suppose $x \neq 0$. $\therefore [1] \Rightarrow y \neq 0$.

$\therefore [1] \div [2]: \frac{2x}{6y} = \lambda = \frac{2y}{6x} \Rightarrow x^2 = y^2 \Rightarrow x = \pm y$

$x = y$: $[1] \Rightarrow 2x = \lambda 6x$, $\lambda = \frac{1}{3}$, $[3] \Rightarrow z = -\frac{1}{6}$

$[4] \Rightarrow 6x^2 + 7 - (-\frac{1}{6}) \neq 0$ for any x.

$\therefore x \neq y$

$x = -y$: $\therefore [1] \Rightarrow \lambda = -\frac{1}{3}$, $[3] \Rightarrow z = \frac{1}{6}$

$[4] \Rightarrow -6x^2 + 7 - \frac{1}{6} = 0$, $x^2 = \frac{41}{36}$,

$$x = \pm \frac{\sqrt{41}}{6}$$

$\therefore \left(\frac{\sqrt{41}}{6}, -\frac{\sqrt{41}}{6}, \frac{1}{6}\right)$ and $\left(-\frac{\sqrt{41}}{6}, \frac{\sqrt{41}}{6}, \frac{1}{6}\right)$

are possibilities.

$$\text{Distance} = \sqrt{\frac{41}{36} + \frac{41}{36} + \frac{1}{36}} = \frac{\sqrt{83}}{6}$$

$$\frac{\sqrt{83}}{6} < 7.$$

$\therefore$ Minimum distance $= \frac{\sqrt{83}}{6}$

10. **10.** Find the first few terms in the Taylor expansion of $f(x, y) = e^{xy} \cos x$ about $x = 0$, $y = 0$.

$f(0,0) = 1$

$f_x = ye^{xy}\cos x - e^{xy}\sin x \quad \therefore f_x(0,0) = 0$

$f_y = xe^{xy}\cos x \quad \therefore f_y(0,0) = 0$

$f_{xx} = y^2e^{xy}\cos x - ye^{xy}\sin x - ye^{xy}\sin x - e^{xy}\cos x$

$\therefore f_{xx}(0,0) = -1$

$f_{yx} = e^{xy}\cos x + x^2e^{xy}\cos x - xe^{xy}\sin x$

$\therefore f_{xy}(0,0) = 1$

$f_{yy} = x^2e^{xy}\cos x \quad \therefore f_{yy}(0,0) = 0$

$\frac{1}{2}ax^2 + bxy + \frac{1}{2}cy^2$

$\therefore$ Hessian component: $\frac{1}{2}(-1)x^2 + (1)xy + \frac{1}{2}(0)y^2$

$\therefore f(x,y) = 1 + 0 - \frac{1}{2}x^2 + xy + 0 + R_2((x,y),(0,0))$

or $f(x,y) = 1 - \frac{1}{2}x^2 + xy$ to the second order

11.

11. Prove that

$$z = \frac{3x^4 - 4x^3 - 12x^2 + 18}{12(1 + 4y^2)}$$

$z = f(x,y)$

has one local maximum, one local minimum, and one saddle point. (The graph is shown in Figure 3.R.1.)

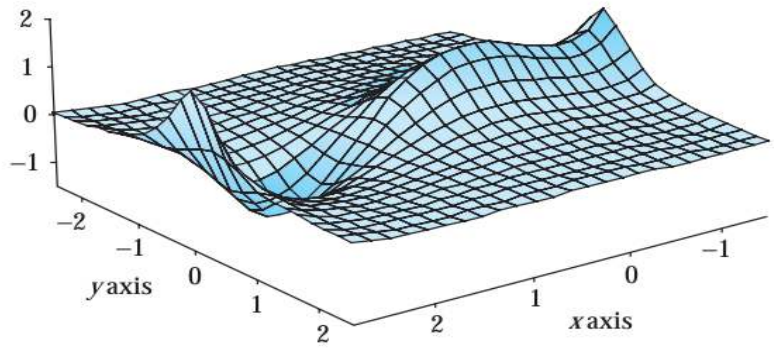


figure **3.R.1** Graph of $z = (3x^4 - 4x^3 - 12x^2 + 18)/12(1 + 4y^2)$.

$$f_x = \frac{12x^3 - 12x^2 - 24x}{12(1+4y^2)} = \frac{x^3 - x^2 - 2x}{(1+4y^2)}$$

$$f_y = -\frac{(3x^4 - 4x^3 - 12x^2 + 18)[8y]}{12(1+4y^2)^2}$$

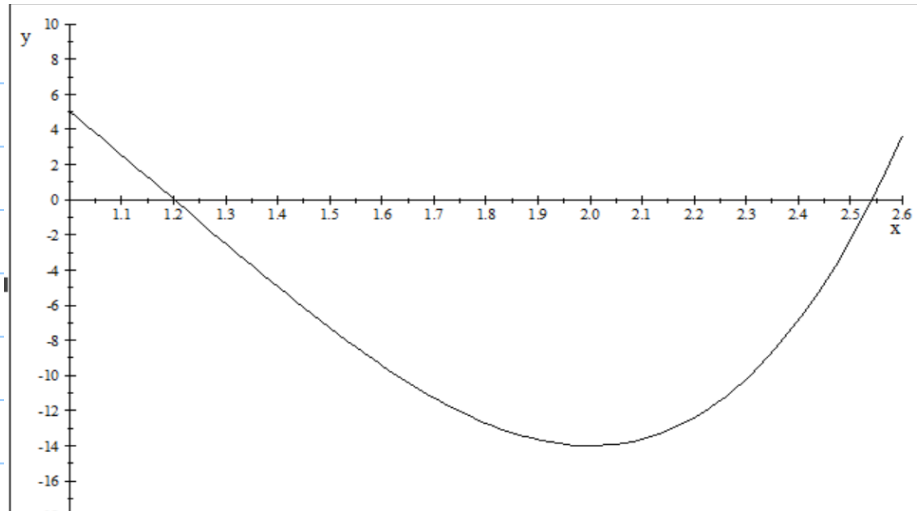
$f_x = 0 \Rightarrow x(x^2 - x - 2) = 0$

or $x(x-2)(x+1) = 0 \Rightarrow x = 0, 2, -1$

$f_y = 0 \Rightarrow (3x^4 - 4x^3 - 12x^2 + 18)y = 0 \Rightarrow y = 0$

or $3x^4 - 4x^3 - 12x^2 + 18 = 0$

Plot of $y = 3x^4 - 4x^3 - 12x^2 + 18$



$\therefore x \approx 1.2, 2.54$

But for these values, $f_x \neq 0$.

And for $x = 0, 2, -1$, y must be 0 for $f_y = 0$.

$\therefore$ Only critical points are: $(0,0), (2,0), (-1,0)$.

$$f_{xx} = \frac{3x^2 - 2x - 2}{(1+y^2)}$$

$$f_{xy} = -\frac{(x^3 - x^2 - 2x)[8y]}{(1+4y^2)^2}$$

$$f_{yy} = -(3x^4 - 4x^3 - 12x^2 + 18)\left(\frac{2}{3}\right)\left[\frac{(1+4y^2)^2 - y[2(1+4y^2)(8y)]}{(1+4y^2)^4}\right]$$

$\therefore (0,0)$: $\begin{vmatrix} -2 & 0 \\ 0 & -12 \end{vmatrix} = 24$, $f_{xx} < 0$, $\therefore$ Maximum

$(2,0)$: $\begin{vmatrix} 6 & 0 \\ 0 & 14 \end{vmatrix} = 84$, $f_{xx} > 0$, $\therefore$ Minimum

$(-1,0): \begin{vmatrix} 3 & 0 \\ 0 & -\frac{26}{3} \end{vmatrix} = -26$, $\therefore$ Saddle point

$\therefore$ Only 3 critical points:

$(-1,0)$: a saddle point
$(0,0)$: a local maximum
$(2,0)$: a local minimum

12.

12. Find the maxima, minima, and saddles of the function $z = (2 + \cos \pi x)(\sin \pi y)$, which is graphed in Figure 3.R.2.

$z = f(x,y)$

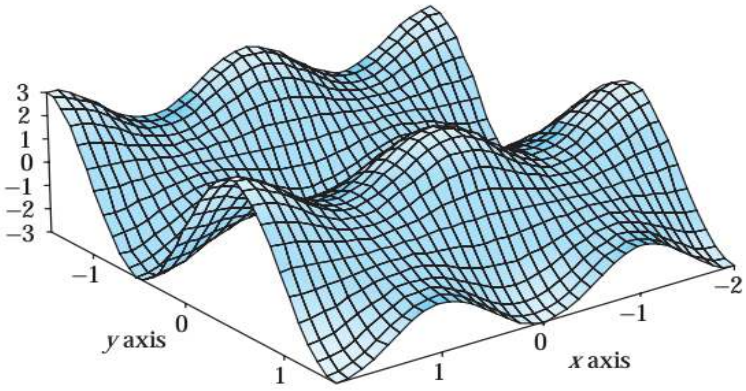


figure **3.R.2** Graph of $z = (2 + \cos \pi x)(\sin \pi y)$.

$f_x = (-\pi \sin \pi x)(\sin \pi y)$

$f_y = \pi(2 + \cos \pi x)(\cos \pi y)$

$\therefore f_x = 0$ when $x = \ldots, -2, -1, 0, 1, 2, \ldots$
or $y = \ldots, -2, -1, 0, 1, 2, \ldots$

$f_y = 0$ only when $\cos \pi y = 0 \Rightarrow y = n + \frac{1}{2}$
n any integer

$\therefore$ Critical points: $(x,y) = (m, n+\frac{1}{2})$

m, n any integer

$$f_{xx} = -\pi^2(\cos\pi x)(\sin\pi y)$$

$$f_{xy} = -\pi^2(\sin\pi x)(\cos\pi y)$$

$$f_{yy} = -\pi^2(2+\cos\pi x)(\sin\pi y)$$

m even, n even:

$\cos\pi m = 1$, $\sin\pi(n+\frac{1}{2}) = 1$ $\quad\therefore f_{xx} = -\pi^2$

$\sin\pi m = 0$, $\cos\pi(n+\frac{1}{2}) = 0$ $\quad\therefore f_{xy} = 0$

$2+\cos\pi m = 3$, $\sin\pi(n+\frac{1}{2}) = 1$ $\quad\therefore f_{yy} = -3\pi^2$

$$\therefore \begin{vmatrix} -\pi^2 & 0 \\ 0 & -3\pi^2 \end{vmatrix} = 3\pi^4,\ f_{xx} < 0 \therefore \text{rel. max.}$$

m even, n odd:

$\cos\pi m = 1$, $\sin\pi(n+\frac{1}{2}) = -1$ $\quad\therefore f_{xx} = \pi^2$

$\sin\pi m = 0$, $\cos\pi(n+\frac{1}{2}) = 0$ $\quad\therefore f_{xy} = 0$

$2+\cos\pi m = 3$, $\sin\pi(n+\frac{1}{2}) = -1$ $\quad\therefore f_{yy} = 3\pi^2$

$$\therefore \begin{vmatrix} \pi^2 & 0 \\ 0 & 3\pi^2 \end{vmatrix} = 3\pi^4,\ f_{xx} > 0 \therefore \text{rel. min.}$$

m odd, n even:

$\cos\pi m = -1$, $\sin\pi(n+\frac{1}{2}) = 1$ $\quad\therefore f_{xx} = \pi^2$

$\sin\pi m = 0$, $\cos\pi(n+\frac{1}{2}) = 0$ $\quad\therefore f_{xy} = 0$

$2 + \cos \pi m = 1$, $\sin \pi (n + \frac{1}{2}) = 1$ $\therefore f_{yy} = -\pi^2$

$\therefore \begin{vmatrix} \pi^2 & 0 \\ 0 & -\pi^2 \end{vmatrix} = -\pi^4$, $\therefore$ saddle point

m odd, n odd:

$\cos \pi m = -1$, $\sin \pi (n + \frac{1}{2}) = -1$ $\therefore f_{xx} = -\pi^2$
$\sin \pi m = 0$, $\cos \pi (n + \frac{1}{2}) = 0$ $\therefore f_{xy} = 0$
$2 + \cos \pi m = 1$, $\sin \pi (n + \frac{1}{2}) = -1$ $\therefore f_{yy} = \pi^2$

$\therefore \begin{vmatrix} -\pi^2 & 0 \\ 0 & \pi^2 \end{vmatrix} = -\pi^4$, $\therefore$ saddle point

$\therefore$ Critical points: $(x, y) = (m, n + \frac{1}{2})$, m, n any integer

When m even, n even, a relative maximum

m even, n odd, a relative minimum

m odd, a saddle point (n odd or even)

13.

13. Find and describe the critical points of $f(x, y) = y \sin(\pi x)$. (See Figure 3.R.3.)

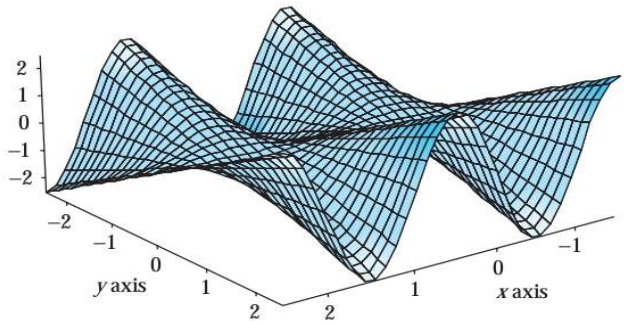


figure **3.R.3** Graph of $z = y \sin(\pi x)$.

$f_x = \pi y \cos(\pi x) = 0$ for $y = 0$ or $x = \ldots, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \ldots$

$f_y = \sin(\pi x) = 0$ for $x = \ldots, -2, -1, 0, 1, 2, \ldots$

$\therefore$ for both $f_x = 0$ and $f_y = 0$, $(x, y) = \underline{(n, 0)}$,
n any integer.

$f_{xx} = -\pi^2 y \sin(\pi x)$ $\quad f_{xx}(n, 0) = 0$

$f_{xy} = \pi \cos(\pi x)$ $\quad f_{xy}(n, 0) = \pi$ (n even), $-\pi$ (n odd)

$f_{yy} = 0$ $\quad f_{yy}(n, 0) = 0$

n even: $\begin{vmatrix} 0 & \pi \\ \pi & 0 \end{vmatrix} = -\pi^2$ $\therefore$ saddle points

n odd: $\begin{vmatrix} 0 & -\pi \\ -\pi & 0 \end{vmatrix} = -\pi^2$ $\therefore$ saddle points

$\therefore$ All critical points are of form $\underline{(n, 0)}$, where n is any integer, and are saddle points.

14.

14. A graph of the function $z = \sin(\pi x)/(1 + y^2)$ is shown in Figure 3.R.4. Verify that this function has alternating maxima and minima on the x axis, with no other critical points.

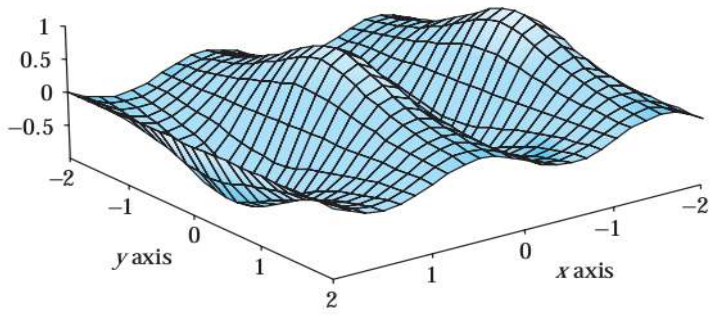


figure **3.R.4** Graph of $z = \sin(\pi x)/(1 + y^2)$.

$z = f(x,y)$

$f_x = \frac{\pi \cos(\pi x)}{1+y^2} = 0$ for $x = \ldots, -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \ldots$

$f_y = -\frac{\sin(\pi x)}{(1+y^2)^2} \cdot 2y = 0$ for $x = \ldots, -2, -1, 0, 1, 2, \ldots$ or $y = 0$

$\therefore$ Critical points when $(x,y) = (n + \frac{1}{2}, 0)$, n any integer

$f_{xx} = -\frac{\pi^2 \sin(\pi x)}{1+y^2}$ $\quad f_{xx}(n+\frac{1}{2}, 0) = -\pi^2$, n even $= \pi^2$, n odd

$f_{xy} = -\frac{\pi \cos(\pi x)}{(1+y^2)^2}(2y)$ $\quad f_{xy}(n+\frac{1}{2}, 0) = 0$, all n

$f_{yy} = -\sin(\pi x)\left[\frac{(1+y^2)^2(2) - 2y(2(1+y^2)2y)}{(1+y^2)^4}\right]$

$f_{yy}(n+\frac{1}{2}, 0) = -2$, n even $= 2$, n odd

$\therefore$ n even: $\begin{vmatrix} -\pi^2 & 0 \\ 0 & -2 \end{vmatrix} = 2\pi^2$, $f_{xx} < 0$, so max

n odd: $\begin{vmatrix} \pi^2 & 0 \\ 0 & 2 \end{vmatrix} = 2\pi^2$, $f_{xx} > 0$, so min

$\therefore$ All critical points of form $(n + \frac{1}{2}, 0)$, n any integer.

$y = 0 \Rightarrow$ critical points on x-axis.

When n is even, points are relative maxima.

When n is odd, points are relative minima.

In Exercises 15 to 20, find the extrema of the given functions subject to the given constraints.

15. **15.** $f(x, y) = x^2 - 2xy + 2y^2$, subject to $x^2 + y^2 = 1$

$S = \{(x,y) : x^2 + y^2 = 1\}$ is closed and bounded, f is continuous, so max & min exist.

Let $g(x,y) = x^2 + y^2 - 1 = 0$

$\nabla f = [2x - 2y, -2x + 4y]$ $\quad \nabla g = [2x, 2y]$

$\therefore$ $2x - 2y = \lambda 2x$ [1]

$-2x + 4y = \lambda 2y$ [2]

$x^2 + y^2 = 1$ [3]

$\lambda = 0$: [1] $\Rightarrow x = y$, [2] $\Rightarrow x = 2y$, so $x = y = 0$
But that contradicts [3].
$\therefore \lambda \neq 0$.

$\lambda \neq 0$: [1] $\Rightarrow 2x(1-\lambda) = 2y$, or $x(1-\lambda) = y$ [4]

If $\lambda = 1$, then $y = 0$, [2] $\Rightarrow x = 0$, which contradicts [3]. $\therefore \lambda \neq 1$.

[2] $\Rightarrow 2y(2-\lambda) = 2x$, or $y(2-\lambda) = x$ [5]

[4], [5] $\Rightarrow y(1-\lambda)(2-\lambda) = y$ [6]

If $y = 0$, [2] $\Rightarrow x = 0$, contradicting [3]
$\therefore y \neq 0$, and [6] becomes:

$(1-\lambda)(2-\lambda) = 1$, or $\lambda^2 - 3\lambda + 1 = 0$

$\therefore \lambda = \dfrac{3 \pm \sqrt{9-4}}{2} = \dfrac{3 \pm \sqrt{5}}{2} = 2.618, 0.382$

[4], [3] $\Rightarrow x^2 + x^2(1-\lambda)^2 = 1$, or $x^2[(\lambda-1)^2 + 1] = 1$

$\lambda = \dfrac{3+\sqrt{5}}{2} = 2.618$: $x^2[3.618] = 1$, $x = \pm 0.526$

[4] $\Rightarrow y = \pm 0.526(1-2.618) = \mp 0.851$

$\therefore$ Critical points: $(x, y) = \pm(0.526, -0.851)$

$\lambda = \frac{3-\sqrt{5}}{2} = 0.382 \therefore x^2(1.382) = 1,\ x = \pm 0.851$

$[4] \Rightarrow y = \pm 0.851(1-0.382) = \pm 0.526$

$\therefore$ Critical points: $(x,y) = \pm(0.851, 0.526)$

$\therefore f(x,y) = (x-y)^2 + y^2$

$f(0.526, -0.851) = 2.62$

$f(-0.526, 0.851) = 2.62$

$f(0.851, 0.526) = 0.382$

$f(-0.851, -0.526) = 0.382$

$\therefore$ Relative max of 2.62 at $\pm(0.526, -0.851)$

Relative min of 0.382 at $\pm(0.851, 0.526)$

16. **16.** $f(x, y) = xy - y^2$, subject to $x^2 + y^2 = 1$

$f(x,y)$ is continuous and $S = \{(x,y) : x^2+y^2 = 1\}$ is closed & bounded. $\therefore$ max & min exist.

Let $g(x,y) = x^2 + y^2 - 1 = 0 \quad \therefore \nabla g = [2x, 2y]$

$\nabla f = [y, x-2y]$

$\therefore \quad y = \lambda 2x \qquad [1]$

$x - 2y = \lambda 2y \qquad [2]$

$x^2 + y^2 = 1 \qquad [3]$

Note $\lambda = 0 \Rightarrow y = 0$ from [1], $x = 0$ from [2], which contradicts [3]. $\therefore \lambda \neq 0$.

Also, if $y = 0$, from [2] $x = 0$. $\therefore y \neq 0$.

$\therefore [2] \Rightarrow x = 2y(\lambda + 1)$

$[1] \Rightarrow y = \lambda 2(2y)(\lambda + 1)$

$\therefore 1 = 4\lambda^2 + 4\lambda$, or $4\lambda^2 + 4\lambda - 1 = 0$

$\therefore \lambda = \dfrac{-4 \pm \sqrt{16 + 16}}{8} = -\frac{1}{2} \pm \frac{\sqrt{2}}{2}$

$\therefore \lambda = 0.207, -1.207$

$\lambda = 0.207: \quad [2] \Rightarrow x = 2.414y \qquad [4]$

$[3] \Rightarrow (2.414y)^2 + y^2 = 1, \quad y = \pm 0.383$

$\therefore [4] \Rightarrow x = 2.414(\pm 0.383) = \pm 0.924$

$\therefore$ Critical points: $(x, y) = \pm (0.924, 0.383)$

$\lambda = -1.207: \quad [2] \Rightarrow x = 2y(-0.207) = -0.414y \qquad [5]$

$\therefore [3] \Rightarrow (-0.414y)^2 + y^2 = 1, \quad y = \pm 0.924$

$\therefore [5] \Rightarrow x = -0.414(\pm 0.924) = \mp (0.383)$

$\therefore$ Critical points: $(x,y) = \pm(-0.383, 0.924)$

$\therefore f(0.924, 0.383) = 0.207$

$f(-0.924, -0.383) = 0.207$

$f(-0.383, 0.924) = -1.208$

$f(0.383, -0.924) = -1.208$

$\therefore$ Max of 0.207 at $(x,y) = \pm(0.924, 0.383)$

Min of -1.208 at $(x,y) = \pm(-0.383, 0.924)$

17. **17.** $f(x, y) = \cos(x^2 - y^2)$, subject to $x^2 + y^2 = 1$

f is continuous on closed & bounded set.

Let $g(x,y) = x^2 + y^2 - 1$. $\therefore \nabla g = [2x, 2y]$.

$\nabla f = [-2x \sin(x^2-y^2), 2y \sin(x^2-y^2)]$

$\therefore$ $-2x \sin(x^2-y^2) = \lambda 2x$ [1]

$2y \sin(x^2-y^2) = \lambda 2y$ [2]

$x^2 + y^2 = 1$ [3]

Note: $(x,y) = (0, \pm 1)$ and $(x,y) = (\pm 1, 0)$ are solutions,

and so are possible critical points.

∴ Suppose $x \neq 0$

∴ [1] $\Rightarrow \sin(x^2 - y^2) = -\lambda$

∴ [2] $\Rightarrow -2y\lambda = 2y\lambda \Rightarrow y = 0$ or $\lambda = 0$

$y = 0 \Rightarrow x = \pm 1$ from [3]

$\lambda = 0 \Rightarrow \sin(x^2 - y^2) = 0$ from [1]

$\Rightarrow x^2 - y^2 = n\pi$, n some integer

or $x^2 = y^2 + n\pi$

∴ [3] $\Rightarrow 2y^2 + n\pi = 1$, $y^2 = \frac{1 - n\pi}{2}$

∴ $y^2 = \frac{1}{2}, \frac{1+\pi}{2}, \frac{1+2\pi}{2}, \ldots$

But [3] $\Rightarrow y^2 \leq 1$, ∴ $y^2 = \frac{1}{2}$, $y = \pm \frac{1}{\sqrt{2}}$

∴ From [3], $(x,y) = \left(\frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right)$ or $\left(-\frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right)$

∴ For $(x,y) = \left(\frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right)$ or $\left(-\frac{1}{\sqrt{2}}, \pm \frac{1}{\sqrt{2}}\right)$

$\cos(x^2 - y^2) = \cos(0) = 1$

For $(x,y) = (0, \pm 1)$ or $(\pm 1, 0)$, $\cos(x^2 - y^2) = \cos(1)$

$\therefore$ Max of 1 at $(x,y) = \left(\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\right)$ or $\left(-\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\right)$

Min of $\cos(1)$ at $(x,y) = (0, \pm 1)$ or $(\pm 1, 0)$

18. **18.** $f(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$, subject to $x + y = 1$

Note $S = \{(x,y) : x+y = 1\}$ is not bounded. $\therefore$ no guarantee a max & min exist.

Let $g(x,y) = x+y-1 = 0$. $\therefore \nabla g = [1, 1]$

$$f_x = \frac{(x^2+y^2)2x - (x^2-y^2)2x}{(x^2+y^2)^2} = \frac{4xy^2}{(x^2+y^2)^2}$$

$$f_y = \frac{(x^2+y^2)(-2y) - (x^2-y^2)(2y)}{(x^2+y^2)^2} = \frac{-4yx^2}{(x^2+y^2)^2}$$

$$\therefore \frac{4xy^2}{(x^2+y^2)^2} = \lambda \quad [1]$$

$$\frac{-4yx^2}{(x^2+y^2)^2} = \lambda \quad [2]$$

$x + y = 1$ [3] So $(x,y) \neq (0,0)$

$\lambda = 0$: [1] $\Rightarrow x = 0$ or $y = 0$.

[3] $\Rightarrow$ $(x,y) = (0,1)$ or $(1,0)$

critical points.

$[1] = [2] \Rightarrow 4xy^2 = -4yx^2$, or

$xy^2 + yx^2 = 0 \Rightarrow x(y^2 + yx) = 0$

if $x = 0$, then $[3] \Rightarrow y = 1$

if $x \neq 0$, $y(y+x) = 0$, and $[3] \Rightarrow y = 0$.

$\therefore [3] \Rightarrow x = 1$.

$\therefore (x,y) = (0,1)$ or $(1,0)$ only critical points.

$f(0,1) = \frac{0-1}{0+1} = -1$

$f(1,0) = \frac{1-0}{1+0} = 1$.

$\therefore$ Relative max $= 1$ at $(x,y) = (1,0)$

Relative min $= -1$ at $(x,y) = (0,1)$

Note $\frac{x^2-y^2}{x^2+y^2} = \frac{(x-y)(x+y)}{x^2+y^2} = \frac{x-y}{x^2+y^2} = \frac{x-(1-x)}{x^2+(1-x)^2} =$

$\frac{2x-1}{2x^2-2x+1}$ $\lim_{x \to \pm\infty} \frac{2x-1}{2x^2-2x+1} = 0$

Also $\frac{x^2-y^2}{x^2+y^2} = \frac{x-y}{x^2+y^2} = \frac{(1-y)-y}{(1-y)^2+y^2} = \frac{1-2y}{2y^2-2y+1}$

$\therefore \lim_{y \to \pm\infty} \frac{1-2y}{2y^2-2y+1} = 0.$

$\therefore$ The relative extrema appear to be absolute extrema.

19. **19.** $z = xy$, subject to the condition $x + y = 1$

$z_x = y$
$z_y = x$

Let $g(x,y) = x+y-1 = 0$ $\nabla g = [1, 1]$

$\therefore$ $y = \lambda$ [1]
$x = \lambda$ [2]
$x+y = 1$ [3]

[2], [3] $\Rightarrow x = y$, [3] $\Rightarrow x = y = \frac{1}{2}$

$\therefore$ One critical point: $(x,y) = (\frac{1}{2}, \frac{1}{2})$

$z(\frac{1}{2}, \frac{1}{2}) = \frac{1}{4}$.

Note: $y = 1-x$, $\therefore z = x(1-x) = x - x^2$

$\therefore \lim_{x \to \infty} x - x^2 = -\infty$ Similarly for $y \to -\infty$.

$\therefore$ No absolute max for $z(x,y)$.

Also, $z'(x) = 1-2x$, $z''(x) = -2 < 0$.

$\therefore z''\left(\frac{1}{2}\right) = -2$, so $x = \frac{1}{2}$ is a relative max

20. **20.** $z = \cos^2 x + \cos^2 y$, subject to the condition $x + y = \pi/4$

$z_x = -2\cos x \sin x$ Let $g(x,y) = x + y - \frac{\pi}{4} = 0$.

$z_y = -2\cos y \sin y$ $\therefore \nabla g = [1, 1]$.

$\therefore -2\cos x \sin x = \lambda$ [1]

$-2\cos y \sin y = \lambda$ [2]

$x + y = \frac{\pi}{4}$ [3]

[1], [2] $\Rightarrow 2\cos x \sin x = 2\cos y \sin y$

or $\sin 2x = \cos 2y$ [4]

Since $\sin^2(2x) + \cos^2(2y) = 1$,

[4] $\Rightarrow \sin 2x = \cos 2y = \pm \frac{\sqrt{2}}{2}$

$\sin 2x = \frac{\sqrt{2}}{2} \Rightarrow 2x = \frac{\pi}{4}, \frac{3}{4}\pi$, $x = \frac{\pi}{8}, \frac{3}{8}\pi$

$= -\frac{\sqrt{2}}{2} \Rightarrow x = -\frac{\pi}{8}, -\frac{3}{8}\pi$

$\therefore \cos 2y = \frac{\sqrt{2}}{2} \Rightarrow y = \pm\frac{\pi}{8}$,

$= -\frac{\sqrt{2}}{2} \Rightarrow y = \pm\frac{3}{8}\pi$

$\therefore \left(\frac{\pi}{8}, \frac{\pi}{8}\right)$ and $\left(-\frac{3}{8}\pi, -\frac{3}{8}\pi\right)$ satisfy [4]

but only $\left(\frac{\pi}{8}, \frac{\pi}{8}\right)$ satisfies [3].

$\therefore (x,y) = \left(\frac{\pi}{8} + n\pi, \frac{\pi}{8} - n\pi\right)$, n any integer

satisfies [3] and [4], since

$$\sin\left[2\left(\frac{\pi}{8} + n\pi\right)\right] = \sin\left[\frac{\pi}{4} + 2n\pi\right] = \sin\left[\frac{\pi}{4}\right]$$

and $\cos\left[2\left(\frac{\pi}{8} - n\pi\right)\right] = \cos\left[\frac{\pi}{4}\right]$

$\therefore$ for points $(x,y) = \left(\frac{\pi}{8} + n\pi, \frac{\pi}{8} - n\pi\right)$,

$$\cos^2\left(\frac{\pi}{8} + n\pi\right) = \left[\cos\left(\frac{\pi}{8} + n\pi\right)\right]^2 = \left[\pm\cos\frac{\pi}{8}\right]^2$$

$$= \cos^2\frac{\pi}{8}$$

Similarly, $\cos^2\left(\frac{\pi}{8} - n\pi\right) = \cos^2\frac{\pi}{8}$

$\therefore z(x,y) = \underline{2\cos^2\frac{\pi}{8}}$ at the extrema

of $(x,y) = \left(\frac{\pi}{8} + n\pi, \frac{\pi}{8} - n\pi\right)$,

n any integer

21.

21. Find the points on the surface $z^2 - xy = 1$ nearest to the origin.

Let $f(x,y,z) = x^2 + y^2 + z^2$, $g(x,y,z) = z^2 - xy - 1 = 0$

Need to find extrema of f subject to $g(x,y,z)$.

$\nabla f = [2x, 2y, 2z]$ $\nabla g = [-y, -x, 2z]$

$\therefore$ $2x = -\lambda y$ [1] $\quad \lambda = 0 \Rightarrow (x,y,z) = (0,0,0)$,
$2y = -\lambda x$ [2] $\quad$ incompatible with [4].
$2z = \lambda 2z$ [3] $\quad \therefore \lambda \neq 0$
$z^2 - xy = 1$ [4]

[3] $\Rightarrow \lambda = 1$ or $z = 0$

$\lambda = 1$: [1], [2] $\Rightarrow 2x = -y$, $2y = -x$,
$\therefore 4xy = xy \Rightarrow xy = 0$
$\therefore$ [1] $\Rightarrow x = 0$ and $y = 0$.

$\therefore$ [4] $\Rightarrow z^2 = 1$, $z = \pm 1$

$\therefore (x,y,z) = (0,0,\pm 1)$.

$z = 0$: [4] $\Rightarrow xy = -1$, $\therefore x \neq 0$ and $y \neq 0$.

[1], [2] $\Rightarrow -\lambda = \frac{2x}{y} = \frac{2y}{x} \Rightarrow x^2 = y^2$

$\therefore x = \pm y$, and with $xy = -1$, then
$x = -y$, so $(x,y) = \pm(1,-1)$

$\therefore (x,y,z) = \pm(1,-1,0)$.

$f(0,0,\pm 1) = 1 \qquad f(\pm(1,-1,0)) = 2$

$\therefore$ Minimum distance $= 1 = \sqrt{f(0,0,\pm 1)}$

The points are: $(0,0,1)$ and $(0,0,-1)$

22. **22.** Use the implicit function theorem to compute dy/dx for

(a) $x/y = 10$

(b) $x^3 - \sin y + y^4 = 4$

(c) $e^{x+y^2} + y^3 = 0$

(a) Let $F(x,y) = \frac{x}{y} - 10 = 0$

$F_y = -\frac{x}{y^2}$, $F_y \neq 0$ for $y \neq 0$.

Implicit function theorem says $\exists\ g(x) = y$

s.t. $F(x, g(x)) = 0$, and

$$D g(x) = -\frac{D_x F}{\frac{\partial F}{\partial y}} = -\frac{\partial F/\partial x}{\partial F/\partial y}$$

But $D g(x) = \frac{dy}{dx}$, so $\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y}$

$\frac{\partial F}{\partial x} = \frac{1}{y}$ $\quad \frac{\partial F}{\partial y} = -\frac{x}{y^2}$

$$\therefore \frac{dy}{dx} = -\frac{\frac{1}{y}}{-\frac{x}{y^2}} = \frac{y}{x} = \frac{1}{10} \quad \text{since } \frac{x}{y} = 10$$

(b) Let $F(x,y) = x^3 - \sin y + y^4 - 4 = 0$

as in (a), $\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y}$, when $\frac{\partial F}{\partial y} \neq 0$

$$\frac{\partial F}{\partial x} = 3x^2 \qquad \frac{\partial F}{\partial y} = -\cos y + 4y^3$$

$$\therefore \frac{dy}{dx} = -\frac{3x^2}{-\cos y + 4y^3} = \frac{3x^2}{\cos y - 4y^3}$$

when $\cos y - 4y^3 \neq 0$

(c) Let $F(x,y) = e^{x+y^2} + y^3 = 0$

as in (a), $\frac{dy}{dx} = -\frac{\partial F/\partial x}{\partial F/\partial y}$ when $\frac{\partial F}{\partial y} \neq 0$

$$\frac{\partial F}{\partial x} = e^{x+y^2} \qquad \frac{\partial F}{\partial y} = 2ye^{x+y^2} + 3y^2$$

$$\therefore \frac{dy}{dx} = -\frac{e^{x+y^2}}{2ye^{x+y^2} + 3y^2}, \quad y \neq 0$$

23. **23.** Find the shortest distance from the point $(0, b)$ to the parabola $x^2 - 4y = 0$. Solve this problem using the Lagrange multiplier method and also without using Lagrange's method.

Assume $b \neq 0$ as $(0,0)$ is on the parabola.

(a) Lagrange method:

Let $f(x,y) = \text{distance}^2 = x^2 + (y-b)^2$ subject to $g(x,y) = x^2 - 4y = 0$

$$\therefore \nabla f = [2x, 2y-2b] = \lambda \nabla g = \lambda[2x, -4]$$

$\therefore 2x = \lambda 2x$ [1] $\quad \lambda = 0 \Rightarrow x = 0$, and $y = b$

$2y - 2b = -4\lambda$ [2] $\quad \therefore$ [3] $\Rightarrow -4b = 0 \Rightarrow b = 0$.

$x^2 - 4y = 0$ [3] $\quad \therefore \lambda \neq 0$.

[1] $\Rightarrow \lambda = 1$ or $x = 0$

$x = 0 \therefore$ [3] $\Rightarrow y = 0$

$\therefore$ Distance is $(0, b)$ to $(0, c) = |b|$

$x \neq 0 : \therefore$ [1] $\Rightarrow \lambda = 1$

[2] $\Rightarrow 2y - 2b = -4$, $y = b - 2$

[3] $\Rightarrow x^2 = 4(b-2)$, $x = \pm 2\sqrt{b-2}$, $b \geq 2$

$\therefore f(x,y) = 4(b-2) + [b-2-b]^2 = 4b - 4$

$\therefore \sqrt{f(x,y)} = |\sqrt{4b-4}| = 2\sqrt{b-1}$, $b \geq 2$.

Note for $b \geq 2$, $2\sqrt{b-1} \leq b$, as

$2 \leq b \Rightarrow 0 \leq b - 2$

$\Rightarrow 0 \leq (b-2)^2 = b^2 - 4b + 4$

$\Rightarrow 4b - 4 \leq b^2$

$\Rightarrow 4(b-1) \leq b^2$

$\Rightarrow 2\sqrt{b-1} \leq b$

$\therefore$ for $b \geq 2$, minimum distance $= 2\sqrt{b-1}$

$b < 2$, minimum distance $= |b|$

$x\left(\frac{x^2}{4}+2-6\right)$ $x=0,\pm 2\sqrt{6-2}$ $2+\frac{3x^2}{4}-6$

$x=0$ $2\cdot 6$ $2(6-2)$

(5) Without Lagrange

Let (x,y) be a point on $x^2-4y=0$, $\therefore y\geq 0$

Distance^2 to $(0,6)$ is $x^2+(y-6)^2 = 4y+(y-6)^2$
$= 4y+y^2-26y+6^2$

Let $D(y) = 4y+y^2-26y+6^2$

$\therefore D'(y) = 4+2y-26 = 0 \Rightarrow y = 6-2$
and $6\geq 2$ as $y\geq 0$

$D''(y) = 2 > 0 \Rightarrow y = 6-2$ is a minimum

$\therefore x^2 = 4(6-2)$, $x = \pm 2\sqrt{6-2}$, $\therefore 6\geq 2$

$\therefore$ For $6\geq 2$, Distance $= \sqrt{x^2+(y-6)^2}$

$= \sqrt{4(6-2)+[(6-2)-6]^2}$

$= \sqrt{46-8+4} = 2\sqrt{6-1}$

For $6<2$, $0<2-6$, $\therefore 0<4-26$

As $y\geq 0$ since (x,y) is on $x^2-4y=0$,

then $0\leq (4-26)y = 4y-26y$

$\therefore 0\leq y^2+4y-26y$ as $0\leq y^2$

$\therefore b^2 \le y^2 + (4-2b)y + b^2 = D(y)$

$\therefore$ for $b < 2$, $|b| \le \sqrt{D(y)}$

and $\sqrt{D(y)}$ is the distance from a point on $x^2 - 4y = 0$ to $(0, b)$.

$\therefore$ For $b \ge 2$, minimum distance $= 2\sqrt{b-1}$

$b < 2$, minimum distance $= |b|$ } same as (a).

Perhaps it is better to view distance as a function of x, since x can assume all values for $x^2 - 4y = 0$, whereas when distance is a function of y, y can only be ≥ 0.

$\therefore$ Distance$^2(x) = D(x) = x^2 + \left(\frac{x^2}{4} - b\right)^2$

$\therefore D'(x) = 2x + 2\left(\frac{x^2}{4} - b\right)\left(\frac{x}{2}\right) = 2x + x\left(\frac{x^2}{4} - b\right)$

$= x\left[\frac{x^2}{4} - b + 2\right]$

Set $D'(x) = 0 \Rightarrow x = 0$ or $x^2 = 4(b-2)$

$\therefore$ Critical points: $x = 0, \pm 2\sqrt{b-2}$

$D''(x) = 2 + \frac{3}{4}x^2 - b$

$\therefore D''(0) = 2 - b > 0$, a minimum if $b < 2$

$$D''(\pm 2\sqrt{b-2}) = 2 + \frac{3}{4}(4(b-2)) - b$$
$$= 2(b-2) > 0 \quad \text{for } b > 0$$

$\therefore$ $b < 2$, $x = 0 \Rightarrow y = 0$ from $x^2 - 4y = 0$.

$\therefore$ Minimum distance from parabola to $(0, b)$ is $|b|$

$b > 2$, $x = \pm 2\sqrt{b-2}$, $\therefore x^2 = 4(b-2)$, and

from $x^2 - 4y = 0$, $y = b - 2$

$\therefore$ Distance^2 $= (\pm 2\sqrt{b-2})^2 + ((b-2) - b)^2$
$= 4b - 8 + 4 = 4b - 4$

$\therefore$ Distance $= 2\sqrt{b-1}$

If $b = 2$, Distance^2 $= x^2 + (y-2)^2 = 4y + (y-2)^2$

$= y^2 + 4$, which is a minimum when $y = 0$.
$\therefore x = 0$.

$\therefore$ Distance $= b = 2$.

24. **24.** Determine all values of k for which the function $g(x, y, z) = x^2 + kxy + kxz + ky^2 + kz^2$ has a local minimum at $(0, 0, 0)$.

First look at possible extrema: $\nabla g(0,0,0) = 0$

Then use 2nd derivative test and look at Hessian.

$g_x = 2x + ky + kz = 0$ [1] $\therefore g_x(0,0,0) = 0$

$g_y = kx + 2ky = 0$ [2] $g_y(0,0,0) = 0$

$g_z = kx + 2kz = 0$ [3] $g_z(0,0,0) = 0$

IF Hessian $Hf(0,0,0)$ is positive definite, then $(0,0,0)$ is a local minimum.

$g_{xx} = 2 \quad g_{yy} = 2k \quad g_{zz} = 2k$

$g_{xy} = k \quad g_{xz} = k \quad g_{yz} = 0$

$$\therefore Hf(0,0,0)(h) = \frac{1}{2}\begin{bmatrix} h_1 & h_2 & h_3 \end{bmatrix}\begin{bmatrix} 2 & k & k \\ k & 2k & 0 \\ k & 0 & 2k \end{bmatrix}\begin{bmatrix} h_1 \\ h_2 \\ h_3 \end{bmatrix}$$

Using the determinant test, this is positive definite if determinant of submatrices are >0.

$$|2| \qquad \begin{vmatrix} 2 & k \\ k & 2k \end{vmatrix} \qquad \begin{vmatrix} 2 & k & k \\ k & 2k & 0 \\ k & 0 & 2k \end{vmatrix}$$

$|2| > 0, \quad \begin{vmatrix} 2 & k \\ k & 2k \end{vmatrix} = 4k - k^2$

$$\begin{vmatrix} 2 & k & k \\ k & 2k & 0 \\ k & 0 & 2k \end{vmatrix} = 2(4k^2) - k(2k^2) + k(-2k^2) = 8k^2 - 4k^3$$

$\therefore$ If $4k - k^2 > 0$ and $8k^2 - 4k^3 > 0$, Then positive-definite, and $(0,0,0)$ a local

$4k - k^2 > 0 \Rightarrow 4k > k^2 > 0$ for $k > 0$

$\therefore 4 > k$

So, if $0 < k < 4$, $4k - k^2 > 0$

$8k^2 - 4k^3 > 0 \Rightarrow 8 - 4k > 0, \; 2 > k$

$\therefore$ if $0 < k < 2$, $8k^2 - 4k^3 > 0$.

$\therefore$ If $0 < k < 2$, Then Hessian determinant will be positive definite, and $\therefore (0,0,0)$ will be a local minimum.

25. **25.** Find and classify all critical points of the function $g(x, y) = \frac{1}{4}x^4 - \frac{5}{3}x^3 + y^3 + 3x^2 - \frac{3}{2}y^2 + 20$.

$\nabla g = [x^3 - 5x^2 + 6x, \; 3y^2 - 3y]$

$x(x^2 - 5x + 6) = x(x-3)(x-2) = 0$ for $x = 0, 2, 3$

$3y^2 - 3y = 3y(y-1) = 0$ for $y = 0, 1$

$g_{xx} = 3x^2 - 10x + 6 \quad g_{xy} = 0 \quad g_{yy} = 6y - 3$

$\therefore$ Critical points:

$(0,0), (0,1)$
$(2,0), (2,1)$
$(3,0), (3,1)$

$\therefore$ $(0,0)$: $g_{xx}(0,0) = 6 \quad g_{yy}(0,0) = -3$

$$\begin{vmatrix} 6 & 0 \\ 0 & -3 \end{vmatrix} = -18, \therefore \text{Saddle point}$$

$(0,1)$: $g_{xx}(0,1) = 6 \quad g_{yy}(0,1) = 3$

$$\begin{vmatrix} 6 & 0 \\ 0 & 3 \end{vmatrix} = 18 > 0, \ 6 > 0 \ \therefore \text{local min.}$$

$(2,0)$: $g_{xx}(2,0) = -2 \quad g_{yy}(2,0) = -3$

$$\begin{vmatrix} -2 & 0 \\ 0 & -3 \end{vmatrix} = 6 > 0, \ -2 < 0. \ \therefore \text{local max.}$$

$(2,1)$: $g_{xx}(2,1) = -2 \quad g_{yy}(2,1) = 3$

$$\begin{vmatrix} -2 & 0 \\ 0 & 3 \end{vmatrix} = -6 < 0 \quad \therefore \text{Saddle point}$$

$(3,0)$: $g_{xx}(3,0) = 3 \quad g_{yy}(3,0) = -3$

$$\begin{vmatrix} 3 & 0 \\ 0 & -3 \end{vmatrix} = -9 < 0 \quad \therefore \text{Saddle point}$$

$(3,1)$: $g_{xx}(3,1) = 3$ $g_{yy}(3,1) = 3$

$\begin{vmatrix} 3 & 0 \\ 0 & 3 \end{vmatrix} = 9 > 0,\ 3 > 0$ $\therefore$ local min.

$\therefore$ Local min: $(0,1), (3,1)$

Local max: $(2,0)$

Saddle point: $(0,0), (2,1), (3,0)$

26.

26. Solve the following geometric problems by Lagrange's method.

(a) Find the shortest distance from the point (a_1, a_2, a_3) in $\mathbb{R}^3$ to the plane whose equation is given by $b_1x_1 + b_2x_2 + b_3x_3 + b_0 = 0$, where $(b_1, b_2, b_3) \neq (0, 0, 0)$.

(b) Find the point on the line of intersection of the two planes $a_1x_1 + a_2x_2 + a_3x_3 = 0$ and $b_1x_1 + b_2x_2 + b_3x_3 + b_0 = 0$ that is nearest to the origin.

(c) Show that the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

is $8abc/3\sqrt{3}$.

(a) Let $D(x_1, x_2, x_3) = (x_1 - a_1)^2 + (x_2 - a_2)^2 + (x_3 - a_3)^2$

subject to $b_1x_1 + b_2x_2 + b_3x_3 + b_0 = 0$, $(b_1, b_2, b_3) \neq \vec{0}$

$\nabla D(x_1, x_2, x_3) = [2x_1 - 2a_1, 2x_2 - 2a_2, 2x_3 - 2a_3]$

Let $g(x_1, x_2, x_3) = b_1x_1 + b_2x_2 + b_3x_3 + b_0$

$$\therefore \nabla g(x_1, x_2, x_3) = [b_1, b_2, b_3] \quad \therefore \nabla g \neq 0$$

$$\therefore \begin{aligned} 2x_1 - 2a_1 &= \lambda b_1 && [1] \\ 2x_2 - 2a_2 &= \lambda b_2 && [2] \\ 2x_3 - 2a_3 &= \lambda b_3 && [3] \\ b_1 x_1 + b_2 x_2 + b_3 x_3 + b_0 &= 0 && [4] \end{aligned}$$

Use Gauss-Jordan elimination

$$\therefore \begin{bmatrix} 2 & 0 & 0 & -b_1 \\ 0 & 2 & 0 & -b_2 \\ 0 & 0 & 2 & -b_3 \\ b_1 & b_2 & b_3 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \lambda \end{bmatrix} = \begin{bmatrix} 2a_1 \\ 2a_2 \\ 2a_3 \\ -b_0 \end{bmatrix}$$

$$\Rightarrow \left[\begin{array}{cccc|c} 2 & 0 & 0 & -b_1 & 2a_1 \\ 0 & 2 & 0 & -b_2 & 2a_2 \\ 0 & 0 & 2 & -b_3 & 2a_3 \\ 0 & b_2 & b_3 & b_1^2/2 & -b_0 - b_1 a_1 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{cccc|c} 2 & 0 & 0 & -b_1 & 2a_1 \\ 0 & 2 & 0 & -b_2 & 2a_2 \\ 0 & 0 & 2 & -b_3 & 2a_3 \\ 0 & 0 & b_3 & b_1^2/2 + b_2^2/2 & -b_0 - b_1 a_1 - b_2 a_2 \end{array}\right]$$

$$\Rightarrow \left[\begin{array}{cccc} 2 & 0 & 0 & -b_1 \\ 0 & 2 & 0 & -b_2 \\ 0 & 0 & 2 & -b_3 \\ 0 & 0 & 0 & \frac{b_1^2}{2} + \frac{b_2^2}{2} + \frac{b_3^2}{2} \end{array}\right] \left[\begin{array}{c} x_1 \\ x_2 \\ x_3 \\ \lambda \end{array}\right] = \left[\begin{array}{c} 2a_1 \\ 2a_2 \\ 2a_3 \\ -b_0 - b_1 a_1 - b_2 a_2 - b_3 a_3 \end{array}\right]$$

$$\therefore \lambda = -2\frac{(b_0 + b_1 a_1 + b_2 a_2 + b_3 a_3)}{b_1^2 + b_2^2 + b_3^2}$$

$$2x_3 - b_3\lambda = 2a_3, \quad x_3 = \frac{2a_3 + b_3\lambda}{2}$$

$$2x_2 - b_2\lambda = 2a_2, \quad x_2 = \frac{2a_2 + b_2\lambda}{2}$$

$$2x_1 - b_1\lambda = 2a_1, \quad x_1 = \frac{2a_1 + b_1\lambda}{2}$$

Look at Hessian:

$$D_{x_1x_1} = 2 \quad D_{x_1x_2} = 0 \quad D_{x_1x_3} = 0$$
$$D_{x_2x_2} = 2 \quad D_{x_2x_3} = 0$$
$$D_{x_3x_3} = 2$$

$$\therefore HD() = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}, \quad \therefore H_1 = |2| > 0$$

$$H_2 = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = 4 > 0$$

$$H_3 = \begin{vmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{vmatrix} = 8 > 0$$

$\therefore$ Critical point is a local minimum.

$$D\left(\frac{2a_1 + b_1\lambda}{2}, \frac{2a_2 + b_2\lambda}{2}, \frac{2a_3 + b_3\lambda}{2}\right) =$$

$$\left(\frac{2a_1 + b_1\lambda}{2} - a_1\right)^2 + \left(\frac{2a_2 + b_2\lambda}{2} - a_2\right) + \left(\frac{2a_3 + b_3\lambda}{2} - a_3\right)^2$$

$$= \left(\frac{b_1\lambda}{2}\right)^2 + \left(\frac{b_2\lambda}{2}\right)^2 + \left(\frac{b_3\lambda}{2}\right)^2$$

$$= \frac{\lambda^2}{4}\left(b_1^2 + b_2^2 + b_3^2\right)$$

Using $\lambda = -2\dfrac{(b_0 + a_1b_1 + a_2b_2 + a_3b_3)}{b_1^2 + b_2^2 + b_3^2}$,

$$D\left(\frac{2a_1+b_1\lambda}{2}, \frac{2a_2+b_2\lambda}{2}, \frac{2a_3+b_3\lambda}{2}\right) =$$

$$\frac{(b_0 + a_1b_1 + a_2b_2 + a_3b_3)^2}{b_1^2 + b_2^2 + b_3^2}$$

$\therefore$ Minimum distance $= \sqrt{D\left(\frac{2a_1+b_1\lambda}{2}, \frac{2a_2+b_2\lambda}{2}, \frac{2a_3+b_3\lambda}{2}\right)}$

$$= \frac{|b_0 + a_1b_1 + a_2b_2 + a_3b_3|}{\sqrt{b_1^2 + b_2^2 + b_3^2}}$$

(b) (b) Find the point on the line of intersection of the two planes $a_1x_1 + a_2x_2 + a_3x_3 = 0$ and $b_1x_1 + b_2x_2 + b_3x_3 + b_0 = 0$ that is nearest to the origin.

Let $\vec{a} = (a_1, a_2, a_3)$
$\vec{b} = (b_1, b_2, b_3)$.

Distance function^2 $= D(x_1, x_2, x_3) = (x_1-0)^2 + (x_2-0)^2 + (x_3-0)^2$
$= x_1^2 + x_2^2 + x_3^2$

subject to $g_1(x_1, x_2, x_3) = a_1x_1 + a_2x_2 + a_3x_3 = 0$
$g_2(x_1, x_2, x_3) = b_1x_1 + b_2x_2 + b_3x_3 + b_0 = 0$

Assume $\nabla g_1 \neq 0$, $\nabla g_2 \neq 0$.

$$\nabla f = [2x_1, 2x_2, 2x_3] \quad \nabla g_1 = [a_1, a_2, a_3] \quad \nabla g_2 = [b_1, b_2, b_3]$$

$$\therefore \quad 2x_1 = \lambda_1 a_1 + \lambda_2 b_1 \qquad [1]$$
$$2x_2 = \lambda_1 a_2 + \lambda_2 b_2 \qquad [2]$$
$$2x_3 = \lambda_1 a_3 + \lambda_2 b_3 \qquad [3]$$

$$a_1 x_1 + a_2 x_2 + a_3 x_3 = 0 \qquad [4]$$
$$b_1 x_1 + b_2 x_2 + b_3 x_3 = -b_0 \qquad [5]$$

$$\begin{bmatrix} 2 & 0 & 0 & -a_1 & -b_1 \\ 0 & 2 & 0 & -a_2 & -b_2 \\ 0 & 0 & 2 & -a_3 & -b_3 \\ a_1 & a_2 & a_3 & 0 & 0 \\ b_1 & b_2 & b_3 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -b_0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 0 & -a_1 & -b_1 \\ 0 & 2 & 0 & -a_2 & -b_2 \\ 0 & 0 & 2 & -a_3 & -b_3 \\ 0 & a_2 & a_3 & a_1^2/2 & a_1 b_1/2 \\ 0 & b_2 & b_3 & a_1 b_1/2 & b_1^2/2 \end{bmatrix}$$

Using Gauss elimination

$$\Rightarrow \begin{bmatrix} 2 & 0 & 0 & -a_1 & -b_1 \\ 0 & 2 & 0 & -a_2 & -b_2 \\ 0 & 0 & 2 & -a_3 & -b_3 \\ 0 & 0 & a_3 & (a_1^2 + a_2^2)/2 & (a_1 b_1 + a_2 b_2)/2 \\ 0 & 0 & b_3 & (a_1 b_1 + a_2 b_2)/2 & (b_1^2 + b_2^2)/2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 0 & -a_1 & -b_1 \\ 0 & 2 & 0 & -a_2 & -b_2 \\ 0 & 0 & 2 & -a_3 & -b_3 \\ 0 & 0 & 0 & (a_1^2+a_2^2+a_3^2)/2 & (a_1b_1+a_2b_2+a_3b_3)/2 \\ 0 & 0 & 0 & (a_1b_1+a_2b_2+a_3b_3)/2 & (b_1^2+b_2^2+b_3^2)/2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2 & 0 & 0 & -a_1 & -b_1 \\ 0 & 2 & 0 & -a_2 & -b_2 \\ 0 & 0 & 2 & -a_3 & -b_3 \\ 0 & 0 & 0 & (a_1^2+a_2^2+a_3^2)/2 & (a_1b_1+a_2b_2+a_3b_3)/2 \\ 0 & 0 & 0 & 0 & \frac{(b_1^2+b_2^2+b_3^2)}{2} - \frac{(a_1b_1+a_2b_2+a_3b_3)^2}{2(a_1^2+a_2^2+a_3^2)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \lambda_1 \\ \lambda_2 \end{bmatrix}$$

$$\therefore 2x_1 - \lambda_1 a_1 - \lambda_2 b_1 = 0 \quad [1]$$

$$2x_2 - \lambda_1 a_2 - \lambda_2 b_2 = 0 \quad [2]$$

$$2x_3 - \lambda_1 a_3 - \lambda_2 b_3 = 0 \quad [3]$$

$$\frac{(a_1^2+a_2^2+a_3^2)}{2}\lambda_1 + \frac{(a_1b_1+a_2b_2+a_3b_3)}{2}\lambda_2 = 0 \quad [4]$$

$$\frac{(b_1^2+b_2^2+b_3^2)}{2}\lambda_2 - \frac{(a_1b_1+a_2b_2+a_3b_3)^2}{2(a_1^2+a_2^2+a_3^2)}\lambda_2 = -b_0 \quad [5]$$

If $b_0 = 0$, then [5] $\Rightarrow \lambda_2 = 0$ as $\vec{a} \neq \vec{0}$, $\vec{b} \neq \vec{0}$.

$\therefore$ [4] $\Rightarrow \lambda_1 = 0$, and $\therefore$ [1], [2], [3] $\Rightarrow x_1 = x_2 = x_3 = 0$

This means $(0,0,0)$ is on the intersection of the planes, and the distance to the origin is 0.

$\therefore$ Assume $b_0 \neq 0$, so that $\lambda_2 \neq 0$.

[4] and [5] can be restated, for brevity,

$$(\vec{a}\cdot\vec{a})\lambda_1 + (\vec{a}\cdot\vec{b})\lambda_2 = 0 \quad [4]$$

$$\frac{\lambda_2}{2}\left[\vec{b}\cdot\vec{b} - \frac{(\vec{a}\cdot\vec{b})^2}{\vec{a}\cdot\vec{a}}\right] = -b_0 \quad [5]$$

$$\therefore \lambda_2 = -2b_0\left[\frac{\vec{a}\cdot\vec{a}}{(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})^2}\right] \quad [6]$$

$\therefore$ [4] becomes

$$\vec{a}\cdot\vec{a}\,\lambda_1 = -(\vec{a}\cdot\vec{b})\lambda_2 = 2b_0\left[\frac{(a\cdot b)(\vec{a}\cdot\vec{a})}{(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})^2}\right]$$

$$\therefore \lambda_1 = 2b_0\left[\frac{\vec{a}\cdot\vec{b}}{(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})^2}\right] \quad [7]$$

Letting $\vec{x} = (x_1, x_2, x_3)$, [1], [2], and [3] become

$$2\vec{x} = \lambda_1\vec{a} + \lambda_2\vec{b}$$

$$\vec{x} = \frac{\lambda_1}{2}\vec{a} + \frac{\lambda_2}{2}\vec{b}$$

$$\therefore \vec{x} = b_0\left[\frac{\vec{a}\cdot\vec{b}}{(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})^2}\right]\vec{a} - b_0\left[\frac{\vec{a}\cdot\vec{a}}{(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})^2}\right]\vec{b}$$

$$\therefore \vec{x} = \frac{b_0}{(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})^2}\left[(\vec{a}\cdot\vec{b})\vec{a} - (\vec{a}\cdot\vec{a})\vec{b}\right] \quad [8]$$

Noting That $\vec{a}\cdot\vec{b} = |\vec{a}||\vec{b}|\cos\theta$, $\vec{a}\cdot\vec{a} = |\vec{a}|^2$, and $\vec{b}\cdot\vec{b} = |\vec{b}|^2$, then

$$(\vec{a}\cdot\vec{a})(\vec{b}\cdot\vec{b}) - (\vec{a}\cdot\vec{b})^2 = |\vec{a}|^2|\vec{b}|^2 - |\vec{a}|^2|\vec{b}|^2\cos^2\theta$$

$$= |\vec{a}|^2|\vec{b}|^2(1-\cos^2\theta) = |\vec{a}|^2|\vec{b}|^2\sin^2\theta$$

$$= |\vec{a}\times\vec{b}|^2$$

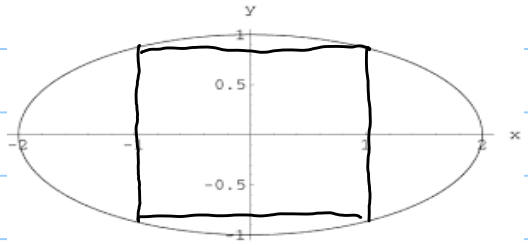
$$\therefore \vec{x} = \frac{b_0}{|\vec{a}\times\vec{b}|^2}\left[(\vec{a}\cdot\vec{b})\vec{a} - (\vec{a}\cdot\vec{a})\vec{b}\right]$$

This $\vec{x}$ is on the intersection of the two planes and is closest to the origin.

(c) Show that the volume of the largest rectangular parallelepiped that can be inscribed in the ellipsoid

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

is $8abc/3\sqrt{3}$.



cross section of ellipsoid

Assume rectangular parallelepiped is such that its edges are parallel to axes.
Unknown if a rectangle can be inscribed in an ellipse that is not parallel to axes.

∴ For each x at a corner, $-x$ is at the other corner. Similarly for y, z.

∴ Edges have length $2x$, $2y$, $2z$.

$\therefore$ Volume $= 8xyz$.

$\therefore$ Maximize $V(x,y,z) = 8xyz$ subject to constraint $g(x,y,z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 = 0$

$\therefore \nabla V = [8yz, 8xz, 8xy]$

$\nabla g = \left[\frac{2x}{a^2}, \frac{2y}{b^2}, \frac{2z}{c^2}\right]$.

Assume $(x,y,z) \neq (0,0,0)$ so $V \neq 0$, $\nabla g \neq 0$.

$\therefore$

$$8yz = \lambda \frac{2x}{a^2} \quad [1]$$

$$8xz = \lambda \frac{2y}{b^2} \quad [2]$$

$$8xy = \lambda \frac{2z}{c^2} \quad [3]$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad [4]$$

Multiply [1] by x, [2] by y, [3] by z, and divide by 2:

$$4xyz = \lambda \frac{x^2}{a^2} \quad [1]'$$

$$4xyz = \lambda \frac{y^2}{b^2} \quad [2]'$$

$$4xyz = \lambda \frac{z^2}{c^2} \quad [3]'$$

Multiply [4] by λ, and substitute [1]', [2]', [3]'

$$4xyz + 4xyz + 4xyz = \lambda, \quad \lambda = 12xyz \quad [5]$$

Substitute [5] into [1]', [2]', [3]':

$$4xyz = 12xyz\,\frac{x^2}{a^2} \quad [1]''$$

$$4xyz = 12xyz\,\frac{y^2}{b^2} \quad [2]''$$

$$4xyz = 12xyz\,\frac{z^2}{c^2} \quad [3]''$$

$$\therefore [1]'' \Rightarrow \frac{x^2}{a^2} = \frac{1}{3}, \quad x = \frac{|a|}{\sqrt{3}}$$

$$\text{Similarly, } [2]'' \Rightarrow y = \frac{|b|}{\sqrt{3}}, \quad z = \frac{|c|}{\sqrt{3}}$$

$$\therefore \text{Volume}(x,y,z) = 8xyz = \frac{8|abc|}{3\sqrt{3}}$$

27.

27. A particle moves in a potential $V(x, y) = x^3 - y^2 + x^2 + 3xy$. Determine whether $(0, 0)$ is a stable equilibrium point; that is, whether or not $(0, 0)$ is a strict local minimum of V.

$$V_x = 3x^2 + 2x + 3y \quad \therefore V_x(0,0) = 0$$
$$V_y = -2y + 3x \qquad\qquad V_y(0,0) = 0$$

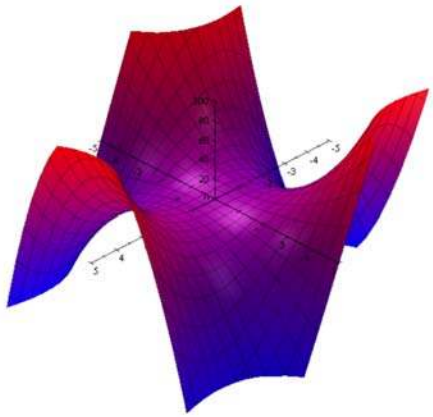
$$V_{xx} = 6x + 2 \qquad V_{xy} = 3 \qquad V_{yy} = -2$$
$$V_{xx}(0,0) = 2$$

$\therefore \begin{vmatrix} 2 & 3 \\ 3 & -2 \end{vmatrix} = -4-6 = -10 < 0$, so $(0,0)$ is a saddle point.

$\therefore$ $(0,0)$ is not a stable equilibrium point

28. **28.** Study the nature of the function $f(x, y) = x^3 - 3xy^2$ near $(0, 0)$. Show that the point $(0, 0)$ is a degenerate critical point; that is, $D = 0$. This surface is called a *monkey saddle*.

D means discriminant



$f_x = 3x^2 - 3y^2 \qquad f_x(0,0) = 0$

$f_y = -3x \qquad f_y(0,0) = 0$

$f_{xx} = 6x \qquad f_{xy} = -6y \qquad f_{yy} = 0$

$f_{xx}(0,0) = 0 \qquad f_{xy}(0,0) = 0 \qquad f_{yy}(0,0) = 0$

$\therefore \begin{vmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{vmatrix}_{(x,y)=(0,0)} = \begin{vmatrix} 0 & 0 \\ 0 & 0 \end{vmatrix} = 0$

$\therefore D = 0$, so $(0,0)$ a degenerate critical point

29. **29.** Find the maximum of $f(x, y) = xy$ on the curve $(x + 1)^2 + y^2 = 1$.

$(x+1)^2+y^2=1$ closed, bounded

Let $g(x,y) = (x+1)^2 + y^2 = 1 \qquad \nabla g = [2x+2, 2y]$

$\nabla f = [y, x]$. $\therefore$

$y = \lambda(2x+2) \quad [1]$

$x = \lambda(2y) \quad [2]$

$(x+1)^2 + y^2 = 1 \quad [3]$

If $\lambda = 0$, $x = y = 0$ from [1], [2]. And $(0,0)$ satisfies [3]. $f(0,0) = 0$

Assume $\lambda \neq 0$

$[1], [2] \Rightarrow \frac{y}{2x+2} = \frac{x}{2y}$, or $2y^2 = 2x^2 + 2x$

or $x^2 + x - y^2 = 0$

$[3] \Rightarrow x^2 + 2x + 1 + y^2 = 1$

Adding, $2x^2 + 3x = 0$, or $x(2x+3) = 0$ [4]

If $x = 0$, Then $[2] \Rightarrow y = 0$, again $f(0,0) = 0$

$x = -\frac{3}{2}$, $[1] \Rightarrow y = -\lambda$, $[2] \Rightarrow x = -2y^2$,

$\therefore -\frac{3}{2} = -2y^2$, $y^2 = \frac{3}{4}$, $y = \pm \frac{\sqrt{3}}{2}$

$f\left(-\frac{3}{2}, -\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{4}$ $\quad$ $f\left(-\frac{3}{2}, \frac{\sqrt{3}}{2}\right) = -\frac{3\sqrt{3}}{4}$

$\therefore$ Max at $\left(-\frac{3}{2}, -\frac{\sqrt{3}}{2}\right) = \frac{3\sqrt{3}}{4}$

30. **30.** Find the maximum and minimum of $f(x, y) = xy - y + x - 1$ on the set $x^2 + y^2 \leq 2$.

Let $u = x^2 + y^2 < 2$

$2u = x^2 + y^2 = 2$

$f_x = y + 1$ $\quad$ $f_x = 0 \Rightarrow y = -1$

$f_y = x - 1$ $\quad$ $f_y = 0 \Rightarrow x = 1$

$(1,-1) \notin U$. $\therefore$ No critical points in U.

At boundary, let $g(x,y) = x^2+y^2 = 2$. $\nabla g = [2x, 2y]$

$\therefore$ $y+1 = \lambda 2x$ [1]

$x-1 = \lambda 2y$ [2]

$x^2+y^2 = 2$ [3]

If $\lambda = 0$, $(x,y) = (1,-1)$ and satisfies [3].

Assume $\lambda \neq 0$:

Adding [1], [2] $\Rightarrow$ $x+y = 2\lambda(x+y)$

$\therefore$ $\lambda = \frac{1}{2}$ or $x+y = 0$, or $y = -x$

$y = -x$: From [3], $x^2+(-x)^2 = 2$, $x = \pm 1$

$\therefore (x,y) = \pm(1,-1)$

$f(1,-1) = 0$, $f(-1,1) = -4$

$\lambda = \frac{1}{2}$: [1] $\Rightarrow$ $y+1 = x$

Using [3], $(y+1)^2+y^2 = 2$, $2y^2+2y-1 = 0$,

$y = \frac{-2 \pm \sqrt{4+8}}{4} = \frac{-1 \pm \sqrt{3}}{2}$

$$\therefore x = y + 1 = \frac{1 \pm \sqrt{3}}{2}$$

$$f\left(\frac{1+\sqrt{3}}{2}, \frac{-1+\sqrt{3}}{2}\right) = \frac{3-1}{4} - \frac{-1+\sqrt{3}}{2} + \frac{1+\sqrt{3}}{2} - 1$$

$$= \frac{1}{2} + \frac{1-\sqrt{3}+1+\sqrt{3}}{2} - 1$$

$$= \frac{1}{2}$$

$$f\left(\frac{1-\sqrt{3}}{2}, \frac{-1-\sqrt{3}}{2}\right) = \frac{3-1}{4} - \frac{-1-\sqrt{3}}{2} + \frac{1-\sqrt{3}}{2} - 1$$

$$= \frac{1}{2} + \frac{1+\sqrt{3}+1-\sqrt{3}}{2} - 1$$

$$= \frac{1}{2}$$

$\therefore$ Minimum $= -4$ at $(-1, 1)$

Maximum $= \frac{1}{2}$ at $\left(\frac{1+\sqrt{3}}{2}, \frac{-1+\sqrt{3}}{2}\right)$ and $\left(\frac{1-\sqrt{3}}{2}, \frac{-1-\sqrt{3}}{2}\right)$

31.

31. The Baraboo, Wisconsin, plant of International Widget Co., Inc., uses aluminium, iron, and magnesium to produce high-quality widgets. The quantity of widgets that may be produced using x tons of aluminum, y tons of iron, and z tons of magnesium is $Q(x, y, z) = xyz$. The cost of raw materials is aluminum, \$6 per ton; iron, \$4 per ton; and magnesium, \$8 per ton. How many tons each of aluminum, iron, and magnesium should be used to manufacture 1000 widgets at the lowest possible cost?

(HINT: Find an extreme value for what function subject to what constraint.)

Cost $= C(x, y, z) = 6x + 4y + 8z$

Subject to constraint $Q(x, y, z) = xyz = 1000$

$\therefore \nabla C = [6, 4, 8] = \lambda \nabla Q = \lambda [yz, xz, xy]$

$\therefore \quad 6 = \lambda yz \quad [1] \qquad x, y, z, \lambda \neq 0$

$4 = \lambda xz \quad [2]$

$8 = \lambda xy \quad [3]$

$xyz = 1000 \quad [4]$

$[1], [2], [3] \Rightarrow \frac{6}{yz} = \frac{4}{xz} = \frac{8}{xy} = \lambda$

$\therefore 6x = 4y = 8z \qquad [5]$

$\therefore \frac{3}{2}x = y, \quad \frac{3}{4}x = z$

$\therefore [4] \Rightarrow x\left(\frac{3}{2}x\right)\left(\frac{3}{4}x\right) = 1000, \quad x^3 = \frac{8(1000)}{9} = \frac{2^3 \cdot 2^3 \cdot 5^3}{3^2}$

$x^3 = \frac{2^3 \cdot 2^3 \cdot 5^3}{3^3} \cdot 3,$

$\underline{x = \frac{20}{3}\sqrt[3]{3} = 9.6}$ tons

$\underline{y = \frac{3}{2}x = 10\sqrt[3]{3} = 14.4}$ tons

$\underline{z = \frac{3}{4}x = 5\sqrt[3]{3} = 7.2}$ tons

Note: $\underline{\lambda = \frac{8}{xy} > 0}$
$= 0.06$

To verify this is a minimum for $C(x,y,z)$, use bordered Hessian.

Form $h(x,y,z) = C(x,y,z) - \lambda Q(x,y,z)$

$= 6x + 4y + 8z - \lambda(xyz)$

$$-Q_x = -yz \qquad -Q_y = -xz \qquad -Q_z = -xy$$

$$h_x = 6 - \lambda yz \qquad h_{xx} = 0 \qquad h_{xy} = -\lambda z \qquad h_{xz} = -\lambda y$$

$$h_y = 4 - \lambda xz \qquad h_{yx} = -\lambda z \qquad h_{yy} = 0 \qquad h_{yz} = -\lambda x$$

$$h_z = 8 - \lambda xy \qquad h_{zx} = -\lambda y \qquad h_{zy} = -\lambda x \qquad h_{zz} = 0$$

$$\therefore H = \begin{bmatrix} 0 & -yz & -xz & -xy \\ -yz & 0 & -\lambda z & -\lambda y \\ -xz & -\lambda z & 0 & -\lambda x \\ -xy & -\lambda y & -\lambda x & 0 \end{bmatrix}$$

Look at diagonal submatrices ≥ 3

$$\therefore \bar{H}_3 = \begin{vmatrix} 0 & -yz & -xz \\ -yz & 0 & -\lambda z \\ -xz & -\lambda z & 0 \end{vmatrix} = 0 + yz(0 - \lambda x z^2) + -xz(\lambda y z^2 - 0)$$

$$= -2\lambda x y z^3 < 0 \quad (\text{as } \lambda, x, y, z \text{ all} > 0)$$

$$\bar{H}_4 = \begin{vmatrix} 0 & -yz & -xz & -xy \\ -yz & 0 & -\lambda z & -\lambda y \\ -xz & -\lambda z & 0 & -\lambda x \\ -xy & -\lambda y & -\lambda x & 0 \end{vmatrix} = 0 + yz(\lambda^2 x^2 yz - \lambda^2 x^2 yz - \lambda^2 x^2 yz) - xz(\lambda^2 xy^2 z + 0 + 0) + xy(-\lambda^2 xyz^2)$$

$$= -\lambda^2 x^2 y^2 z^2 - \lambda^2 x^2 y^2 z^2 - \lambda^2 x^2 y^2 z^2 = -3\lambda^2 x^2 y^2 z^2 < 0$$

$\therefore \bar{H}_3 < 0, \bar{H}_4 < 0 \Rightarrow$ The extremum is a minimum

32. **32.** Let $f\colon \mathbb{R} \to \mathbb{R}$ be of class C^1 and let

$$u = f(x)$$
$$v = -y + xf(x).$$

If $f'(x_0) \neq 0$, show that this transformation of $\mathbb{R}^2$ to $\mathbb{R}^2$ is invertible near (x_0, y_0) and its inverse is given by

$$x = f^{-1}(u)$$
$$y = -v + uf^{-1}(u).$$

Let $F_1(u, v, x, y) = f(x) - u = 0$

$F_2(u, v, x, y) = -y + xf(x) - v = 0$

$\frac{\partial F_1}{\partial x} = f'(x) \qquad \frac{\partial F_1}{\partial y} = 0$

$\frac{\partial F_2}{\partial x} = f(x) + xf'(x) \qquad \frac{\partial F_2}{\partial y} = -1$

With $F(u, v, x, y) = (F_1(u, v, x, y), F_2(u, v, x, y)) = (0, 0)$

Then $JF(x_0, y_0) = \begin{vmatrix} f'(x_0) & 0 \\ f(x_0) + x_0 f'(x_0) & -1 \end{vmatrix}$

$= -f'(x_0) \neq 0.$

$\therefore$ By Inverse function Theorem, for x near x_0, y near y_0, there exists a function

$g: \mathbb{R}^2 \to \mathbb{R}^2$ s.t. $(x, y) = g(u, v) = (g_1(u, v), g_2(u, v))$

Note that by using the Inverse function Theorem for $f: \mathbb{R} \to \mathbb{R}$ at x_0, since $f'(x_0) \neq 0$, then f^{-1} exists for x near x_0.

$\therefore$ Consider $x = g_1(u,v) = f^{-1}(u)$.

$$y = g_2(u,v) = -v + u f^{-1}(u)$$

$$\therefore F(u,v,x,y) = F(u,v,g_1(u,v), g_2(u,v))$$

$$= F[u, v, f^{-1}(u), -v + u f^{-1}(u)]$$

$$= [F_1(u,v,f^{-1}(u), -v+uf^{-1}(u)), \; F_2(u,v,f^{-1}(u), -v+uf^{-1}(u))]$$

$$= [f(f^{-1}(u)) - u \;, \; v - uf^{-1}(u) + f^{-1}(u)\, u - v]$$

since $F_1 = f(x) - u$ $\quad F_2 = -y + xf(x) - v$

$$= [u - u \;, \; v - v + uf^{-1}(u) - uf^{-1}(u)]$$

$$= [0, 0]$$

Or, $x = f^{-1}(u) = f^{-1}(f(x)) = x$

and $y = -v + uf^{-1}(u)$

$$= -(-y + xf(x)) + f(x) f^{-1}(f(x))$$

$$= y - xf(x) + f(x) \cdot x$$

$$= y$$

33.

33. Show that the pair of equations

$$x^2 - y^2 - u^3 + v^2 + 4 = 0$$
$$2xy + y^2 - 2u^2 + 3v^4 + 8 = 0$$

determine functions $u(x, y)$ and $v(x, y)$ defined for (x, y) near $x = 2$ and $y = -1$ such that $u(2, -1) = 2$ and $v(2, -1) = 1$. Compute $\partial u/\partial x$ at $(2, -1)$.

Let $F_1(x, y, u, v) = x^2 - y^2 - u^3 + v^2 + 4 = 0$ $\quad F_1(2, -1, 2, 1) = 0$

$F_2(x, y, u, v) = 2xy + y^2 - 2u^2 + 3v^4 + 8 = 0$ $\quad F_2(2, -1, 2, 1) = 0$

Look at $\begin{vmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{vmatrix}$ evaluated at $(x, y, u, v) = (2, -1, 2, 1)$

$\frac{\partial F_1}{\partial u} = -3u^2 \quad \therefore \frac{\partial F_1}{\partial u}(2, -1, 2, 1) = -3(2)^2 = -12$

$\frac{\partial F_1}{\partial v} = 2v \quad \therefore \frac{\partial F_1}{\partial v}(2, -1, 2, 1) = 2(1) = 2$

$\frac{\partial F_2}{\partial u} = -4u \quad \therefore \frac{\partial F_2}{\partial u}(2, -1, 2, 1) = -4(2) = -8$

$\frac{\partial F_2}{\partial v} = 12v^3 \quad \therefore \frac{\partial F_2}{\partial v} = 12(1)^3 = 12$

$\therefore \begin{vmatrix} -12 & 2 \\ -8 & 12 \end{vmatrix} = -12(12) - (-8)(2) = -128 \neq 0$

By Implicit Function Theorem, there exist unique functions $(x,y) \to (u,v)$ around $(x,y) = (2,-1)$

s.t. $u(x,y) = u$, $v(x,y) = v$, $u(2,-1) = 2$, $v(2,-1) = 1$

(b) By Implicit differentiation,

$$\frac{\partial F_1}{\partial x} = 2x - 3u^2 \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x} = 0 \quad [1]$$

$$\frac{\partial F_2}{\partial x} = 2y - 4u \frac{\partial u}{\partial x} + 12v^3 \frac{\partial v}{\partial x} = 0 \quad [2]$$

$$\frac{\partial F_1}{\partial x}(2,-1,2,1) = 2(2) - 3[u(2,-1)]^2 \frac{\partial u}{\partial x}(2,-1) + 2v(2,-1) \frac{\partial v}{\partial x}(2,-1) = 0$$

$$= 4 - 12 \frac{\partial u}{\partial x}(2,-1) + 2 \frac{\partial v}{\partial x}(2,-1) = 0 \quad [3]$$

$$\frac{\partial F_2}{\partial x}(2,-1,2,1) = 2(-1) - 4(2) \frac{\partial u}{\partial x}(2,-1) + 12 \frac{\partial v}{\partial x}(2,-1) = 0$$

$$= -2 - 8 \frac{\partial u}{\partial x}(2,-1) + 12 \frac{\partial v}{\partial x}(2,-1) = 0 \quad [4]$$

[3] becomes $-24 + 72 u_x(2,-1) - 12 v_x(2,-1) = 0$ [5]

Adding [4], [5] $\Rightarrow -26 + 64 u_x(2,-1) = 0$,

$$\therefore \frac{\partial u}{\partial x}(2,-1) = \frac{13}{32}$$

34.

34. Show that there are positive numbers p and q and unique functions u and v from the interval $(-1-p, -1+p)$ into the interval $(1-q, 1+q)$ satisfying

$$xe^{u(x)} + u(x)e^{v(x)} = 0 = xe^{v(x)} + v(x)\,e^{u(x)}$$

for all x in the interval $(-1-p, -1+p)$ with $u(-1) = 1 = v(-1)$.

Let $F_1(x,u,v) = xe^u + ue^v = 0$

Note $F_1(-1,1,1) = -e + e = 0$

$F_2(x,u,v) = xe^v + ve^u$

Note $F_2(-1,1,1) = -e + e = 0$

Look at $\begin{vmatrix} \frac{\partial F_1}{\partial u} & \frac{\partial F_1}{\partial v} \\ \frac{\partial F_2}{\partial u} & \frac{\partial F_2}{\partial v} \end{vmatrix}$ at $(x,u,v) = (-1,1,1)$

$\frac{\partial F_1}{\partial u} = xe^u + e^v \qquad \frac{\partial F_1}{\partial v} = ue^v$

$\frac{\partial F_2}{\partial u} = ve^u \qquad \frac{\partial F_2}{\partial v} = xe^v + e^u$

$\therefore \begin{vmatrix} xe^u + e^v & ue^v \\ ve^u & xe^v + e^u \end{vmatrix}_{(x,u,v)=(-1,1,1)} = \begin{vmatrix} 0 & e \\ e & 0 \end{vmatrix} = -e^2 \neq 0$

$\therefore$ By Implicit Function Theorem, $\exists$ unique smooth functions $u = u(x)$, $v = v(x)$, and open balls U around 1 and V around $(1,1)$, s.t.

$u(-1) = 1$, $v(-1) = 1$.

Let $U = D_p(-1)$, so $U = (-1-p, -1+p)$, p some positive number

Let $V = D_q(1,1)$, q some positive number, so $V = (1-q, 1+q) \times (1-q, 1+q)$

38.

38. Consider the surface S given by $x^2z + x\sin y + ye^{z-1} = 1$.

(a) Find the equation of the tangent plane to S at the point $(1, 0, 1)$.

(b) Is it possible to solve the equation defining S for the variable y as a function of the variables x and z near $(1, 0, 1)$? Why?

(c) Find $\frac{\partial y}{\partial x}$ at $(1, 0, 1)$.

Let $S(x,y,z) = x^2z + x\sin y + ye^{z-1} - 1 = 0$

(a) Note $S(1,0,1) = 0$, $\therefore (1,0,1)$ is on S.

∇S is normal to tangent plane of S (a level set)

$\therefore \nabla S = [2xz + \sin y, x\cos y + e^{z-1}, x^2 + ye^{z-1}]$

$\therefore \nabla S(1,0,1) = [2, 2, 1]$

$\therefore (2,2,1)\cdot\{(x,y,z)-(1,0,1)\}=0$, or

$$\underline{2x+2y+z=3}$$

(b) By Theorem 11 of text (p. 204), the Special Implicit Function Theorem,

if $\frac{\partial S}{\partial y}(1,0,1)\neq 0$, then yes, since

$$\frac{\partial S}{\partial y}=x\cos y+e^{z-1},\ \therefore \frac{\partial S}{\partial y}(1,0,1)=2\neq 0.$$

$\therefore$ <u>yes</u>, can solve y as a function of x,z

near $(x,y,z)=(1,0,1)$.

(c) Use implicit differentiation.

$$\frac{\partial}{\partial x}\left(x^2z+x\sin y+ye^{z-1}-1\right)=$$

$$2xz+\sin y+(x\cos y)\frac{\partial y}{\partial x}+e^{z-1}\frac{\partial y}{\partial x}=0$$

$$\therefore \frac{\partial y}{\partial x}\left(x\cos y+e^{z-1}\right)=-2xz-\sin y$$

$$\therefore \underline{\frac{\partial y}{\partial x}=\frac{-2xz-\sin y}{x\cos y+e^{z-1}}}$$

39.

39. Consider the system of equations

$$2xu^3v - yv = 1$$
$$y^3v + x^5u^2 = 2$$

Show that near the point $(x, y, u, v) = (1, 1, 1, 1)$, this system defines u and v implicitly as functions of x and y. For such local functions u and v, define the local function f by $f(x, y) = (u(x, y), v(x, y))$. Find $Df(1, 1)$.

Let $f_1(x,y,u,v) = 2xu^3v - yv = 1$
$f_2(x,y,u,v) = y^3v + x^5u^2 = 2$

(a) By General Implicit Function Theorem, need to

Show $\begin{vmatrix} \frac{\partial f_1}{\partial u} & \frac{\partial f_1}{\partial v} \\ \frac{\partial f_2}{\partial u} & \frac{\partial f_2}{\partial v} \end{vmatrix}_{(1,1,1,1)} \neq 0$

$\therefore f_{1u} = 6xu^2v \qquad f_{1v} = 2xu^3 - y$

$f_{2u} = 2x^5u \qquad f_{2v} = y^3$

$\therefore \begin{vmatrix} 6 & 1 \\ 2 & 1 \end{vmatrix} = 4 \neq 0$

$\therefore$ Near $(x,y,u,v) = (1,1,1,1)$, $\exists$ unique smooth functions $u(x,y)$ and $v(x,y)$

(b) Note $f: \mathbb{R}^2 \to \mathbb{R}^2$, so Df is 2×2 matrix

and $Df = \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix}_{(x,y)=(1,1)}$

Note since $(x,y,u,v) = (1,1,1,1)$, $(x,y) = (1,1) \Leftrightarrow (u,v) = (1,1)$

By implicit differentiation on f_1 and f_2,

$f_1(x,y,u,v) = 2xu^3v - yv = 1$

$f_2(x,y,u,v) = y^3v + x^5u^2 = 2$

f_{1x} : $2u^3v + 6xu^2v(u_x) + 2xu^3v_x - yv_x = 0$ [1]

f_{2x} : $y^3v_x + 5x^4u^2 + 2x^5u(u_x) = 0$ [2]

Evaluating at $(x,y,u,v) = (1,1,1,1)$, [1] & [2] become:

$$2 + 6u_x + 2v_x - v_x = 0 \quad [1']$$

$$v_x + 5 + 2u_x = 0 \quad [2']$$

or $$6u_x + v_x = -2 \quad [1'']$$

$$2u_x + v_x = -5 \quad [2'']$$

Subtracting, $4u_x = 3$, $u_x(1,1) = \frac{3}{4}$

$\therefore 6(\frac{3}{4}) + v_x = -2$, $v_x(1,1) = -\frac{13}{2}$

Now differentiate by y:

f_{1y} : $6xu^2v(u_y) + 2xu^3(v_y) - v - yv_y = 0$ [3]

$$3y^2v + y^3v_y + 2x^5u \cdot (u_y) = 0 \quad [4]$$

Or, using $(x,y,u,v) = (1,1,1,1)$

$$6u_y + 2v_y - 1 - v_y = 0 \qquad [3']$$

$$3 + v_y + 2u_y = 0 \qquad [4']$$

Or, $$6u_y + v_y = 1 \qquad [3'']$$

$$2u_y + v_y = 3 \qquad [4'']$$

Subtracting, $4u_y = -2$, $u_y(1,1) = -\frac{1}{2}$

$\therefore 2(-\frac{1}{2}) + v_y = 3$, $v_y(1,1) = 4$

$$\therefore Df(1,1) = \begin{pmatrix} \frac{3}{4} & -\frac{1}{2} \\ -\frac{13}{2} & 4 \end{pmatrix}$$

40.

40. For each set of three data points, plot the points, write down the function $f(m, b)$ from the preceding equation, find m and b to give the best straight-line fit according to the method of least squares, and plot the straight line.

(a) $(x_1, y_1) = (1, 1)$
$(x_2, y_2) = (2, 3)$
$(x_3, y_3) = (4, 3)$

(b) $(x_1, y_1) = (0, 0)$
$(x_2, y_2) = (1, 2)$
$(x_3, y_3) = (2, 3)$

For (a) $y = m(1) + 6 = m + 6$
$y = m(2) + 6 = 2m + 6$
$y = m(4) + 6 = 4m + 6$

(a) $f(m, 6) = (1 - m - 6)^2 + (3 - 2m - 6)^2 + (3 - 4m - 6)^2$

$(x_1, y_1) = (1,1)$ $(x_2, y_2) = (2,3)$ $(x_3, y_3) = (4,3)$

Here, $f: \mathbb{R}^2 \to \mathbb{R}$ and $Df = [f_m, f_6]$

If (x,y) is a local extremum, then $Df(x,y) = 0$.

$$f_m = 2(1-m-6)(-1) + 2(3-2m-6)(-2) + 2(3-4m-6)(-4)$$
$$= -2 + 2m + 26 - 12 + 8m + 46 - 24 + 32m + 86$$
$$= -38 + 42m + 146$$

$$f_6 = 2(1-m-6)(-1) + 2(3-2m-6)(-1) + 2(3-4m-6)(-1)$$
$$= -2 + 2m + 26 - 6 + 4m + 26 - 6 + 8m + 26$$
$$= -14 + 14m + 66$$

$\therefore$ f_m: $42m + 146 = 38$ [1]

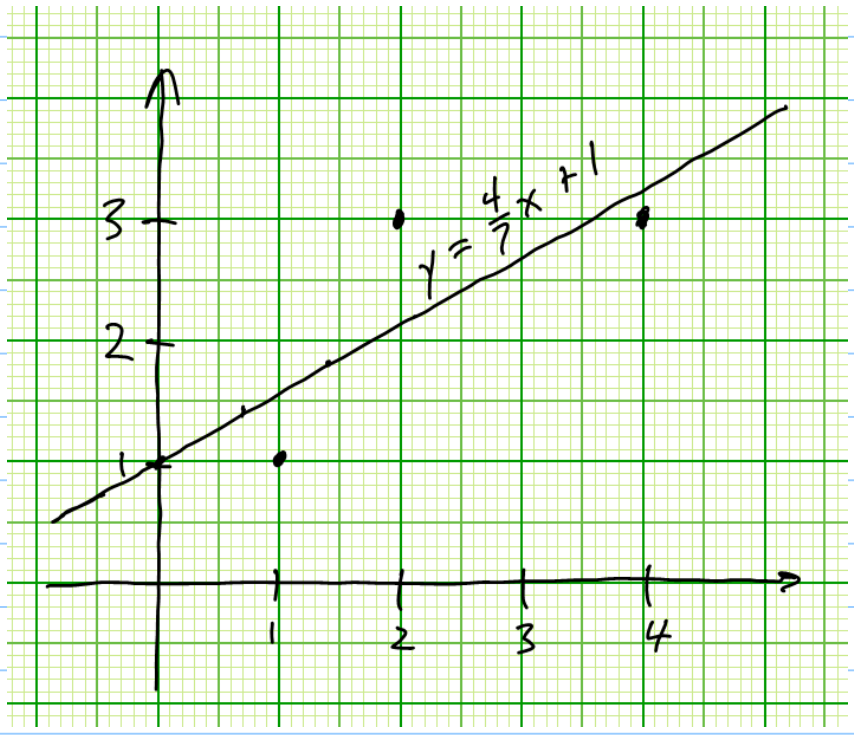
f_6: $14m + 66 = 14$ [2]

Subtracting [1] − 3[2]:

$$-46 = -4, \quad 6 = 1$$

$$\therefore m = \frac{8}{14} = \frac{4}{7}$$

$$\therefore y = \frac{4}{7}x + 1$$



Discrepancies:

$\frac{4}{7}(1) + 1 = \frac{11}{7}$

$\frac{4}{7}(2) + 1 = \frac{15}{7}$

$\frac{4}{7}(4) + 1 = \frac{23}{7}$

$$\therefore \left(\frac{11}{7}-1\right)^2+\left(\frac{15}{7}-3\right)^2+\left(\frac{23}{7}-3\right)^2=$$

$$\frac{16}{49}+\frac{36}{49}+\frac{4}{49}=\frac{56}{49}\approx 1.14$$

Note: The answer book gives $y=\frac{13}{14}x+\frac{1}{2}$ as the best curve. Checking:

$$y=\frac{13}{14}(1)+\frac{1}{2}=\frac{20}{14}$$

$$y=\frac{13}{14}(2)+\frac{1}{2}=\frac{33}{14}$$

$$y=\frac{13}{14}(4)+\frac{1}{2}=\frac{59}{14}$$

$$\therefore \left(\frac{20}{14}-1\right)^2+\left(\frac{33}{14}-3\right)^2+\left(\frac{59}{14}-3\right)^2=$$

$$\frac{36}{196}+\frac{81}{196}+\frac{289}{196}=\frac{406}{196}\approx 2.09$$

$1.14<2.09$, so $y=\frac{4}{7}x+1$ is a better fit.

(5) (b) $(x_1, y_1)=(0,0)$
$(x_2, y_2)=(1,2)$
$(x_3, y_3)=(2,3)$

$y=m(0)+b=b$
$y=m(1)+b=m+b$
$y=m(2)+b=2m+b$

$$\therefore f(m,b)=(0-0-b)^2+(2-m-b)^2+(3-2m-b)^2$$

$$f_m=2(2-m-b)(-1)+2(3-2m-b)(-2)$$
$$=-4+2m+2b-12+8m+4b$$

$= -16 + 10m + 6b$

$f_b = 2(-b)(-1) + 2(2-m-b)(-1) + 2(3-2m-b)(-1)$

$= 2b - 4 + 2m + 2b - 6 + 4m + 2b$

$= -10 + 6m + 6b$

$\therefore f_m: \quad 10m + 6b = 16 \quad [1]$

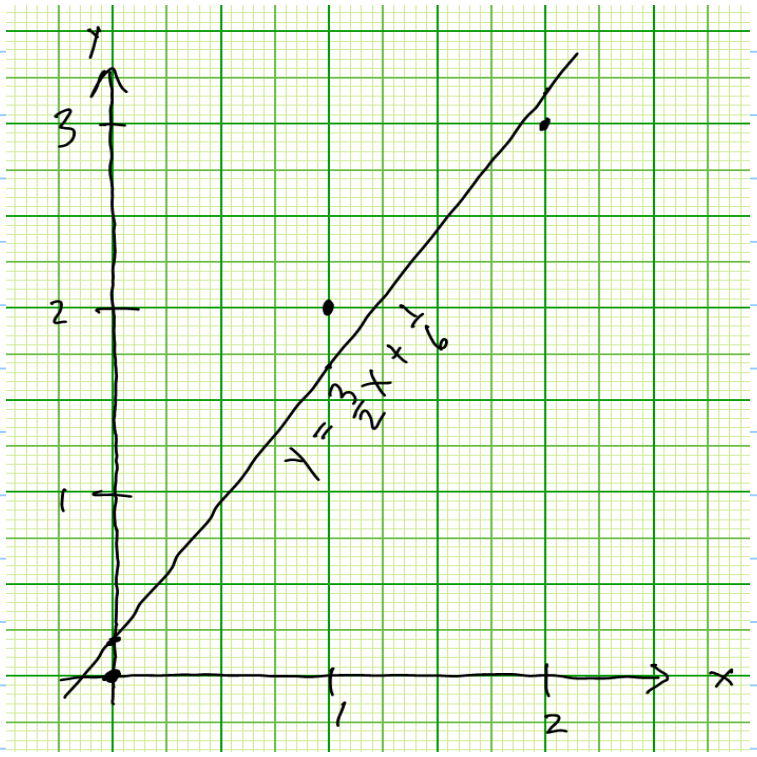
$f_b: \quad 6m + 6b = 10 \quad [2]$

Subtracting $[1] - [2]$:

$4m = 6, \quad m = \frac{3}{2}$

$\therefore 6\left(\frac{3}{2}\right) + 6b = 10, \quad b = \frac{1}{6}$

$\therefore y = \frac{3}{2}x + \frac{1}{6}$



41. **41.** Show that if only two data points (x_1, y_1) and (x_2, y_2) are given, this method produces the line through (x_1, y_1) and (x_2, y_2).

Let $y = mx + b$ be the line of best fit by least squares.

$\therefore \quad y_1' = mx_1 + b \qquad y_2' = mx_2 + b$

Let $f(m, b) = (y_1 - y_1')^2 + (y_2 - y_2')^2$

$= (y_1 - mx_1 - b)^2 + (y_2 - mx_2 - b)^2$

$\therefore f_m = 2(y_1 - mx_1 - b)(-x_1) + 2(y_2 - mx_2 - b)(-x_2)$

$= -2x_1y_1 + 2mx_1^2 + 2x_1b - 2x_2y_2 + 2mx_2^2 + 2x_2b$

$= -2(x_1y_1 + x_2y_2) + 2(x_1^2 + x_2^2)m + 2(x_1 + x_2)b$

$f_b = 2(y_1 - mx_1 - b)(-1) + 2(y_2 - mx_2 - b)(-1)$

$= -2y_1 + 2mx_1 + 2b - 2y_2 + 2mx_2 + 2b$

$= -2(y_1 + y_2) + 2(x_1 + x_2)m + 4b$

Setting to 0:

$f_m: \quad 2(x_1^2 + x_2^2)m + 2(x_1 + x_2)b = 2(x_1y_1 + x_2y_2) \quad [1]$

$f_b: \quad 2(x_1 + x_2)m + 4b = 2(y_1 + y_2) \quad [2]$

Multiply [1] by 2 and [2] by (x_1+x_2), then subtract:

$$\left\{4(x_1^2+x_2^2)-2(x_1+x_2)^2\right\}m = 4(x_1y_1+x_2y_2)-2(y_1+y_2)(x_1+x_2)$$

$$m = \frac{2(x_1y_1+x_2y_2)-(y_1+y_2)(x_1+x_2)}{2(x_1^2+x_2^2)-(x_1+x_2)^2}$$

$$= \frac{2x_1y_1+2x_2y_2-x_1y_1-x_1y_2-x_2y_1-x_2y_2}{2x_1^2+2x_2^2-x_1^2-2x_1x_2-x_2^2}$$

$$= \frac{x_1y_1+x_2y_2-x_1y_2-x_2y_1}{x_1^2-2x_1x_2+x_2^2}$$

$$= \frac{(x_1-x_2)y_1-(x_1-x_2)y_2}{(x_1-x_2)^2}, \quad \text{assuming } x_1 \neq x_2$$

$$\therefore m = \frac{y_1-y_2}{x_1-x_2} = \frac{y_2-y_1}{x_2-x_1}, \quad x_1 \neq x_2$$

Substituting this m into [2],

$$2(x_1+x_2)\frac{(y_2-y_1)}{x_2-x_1}+46 = 2(y_1+y_2)$$

$$2b = (y_1 + y_2) - \frac{(x_1 + x_2)(y_2 - y_1)}{x_2 - x_1}$$

$$= \frac{(y_1 + y_2)(x_2 - x_1) - (x_1 + x_2)(y_2 - y_1)}{x_2 - x_1}$$

$$= \frac{x_2 y_1 + x_2 y_2 - x_1 y_1 - x_1 y_2 - x_1 y_2 - x_2 y_2 + x_1 y_1 + x_2 y_1}{x_2 - x_1}$$

$$= \frac{2x_2 y_1 - 2x_1 y_2}{x_2 - x_1}$$

$$\therefore b = \frac{x_2 y_1 - x_1 y_2}{x_2 - x_1}, \quad x_2 \neq x_1$$

$$\therefore m = \frac{y_2 - y_1}{x_2 - x_1}, \quad b = \frac{x_2 y_1 - x_1 y_2}{x_2 - x_1}, \quad x_2 \neq x_1$$

This is the line through (x_1, y_1) and (x_2, y_2) with slope $m = \frac{y_2 - y_1}{x_2 - x_1}$, $x_1 \neq x_2$.

42.

42. Show that the equations for a critical point, $\partial s/\partial b = 0$ and $\partial s/\partial m = 0$, are equivalent to

$$m\left(\sum x_i\right) + nb = \left(\sum y_i\right)$$

and

$$m\left(\sum x_i^2\right) + b\left(\sum x_i\right) = \left(\sum x_i y_i\right),$$

where all the sums run from $i = 1$ to $i = n$.

Assume: $s = f(m, b) = d_1^2 + d_2^2 + \cdots + d_n^2 = \sum_{i=1}^{n} (y_i - mx_i - b)^2,$

$$\frac{\partial s}{\partial b} = 0 \iff \sum_{i=1}^{n} 2(y_i - mx_i - b)(-1) = 0$$

$$\iff \sum_{i=1}^{n} (y_i - mx_i - b) = 0$$

$$\iff \sum_{i=1}^{n} y_i - \sum_{i=1}^{n} mx_i - \sum_{i=1}^{n} b = 0$$

$$\iff \sum_{i=1}^{n} y_i = m\sum_{i=1}^{n} x_i + nb$$

$$\frac{\partial s}{\partial m} = 0 \iff \sum_{i=1}^{n} 2(y_i - mx_i - b)(-x_i) = 0$$

$$\iff \sum_{i=1}^{n} (-x_i y_i + mx_i^2 + bx_i) = 0$$

$$\iff \sum_{i=1}^{n} (-x_i y_i) + \sum_{i=1}^{n} mx_i^2 + \sum_{i=1}^{n} bx_i = 0$$

$$\Leftrightarrow \quad m\sum_{i=1}^{n} x_i^2 + b\sum_{i=1}^{n} x_i = \sum_{i=1}^{n} x_i y_i$$

43.

43. If $y = mx + b$ is the best-fitting straight line to the data points $(x_1, y_1), \ldots, (x_n, y_n)$ according to the least-square method, show that

$$\sum_{i=1}^{n} (y_i - mx_i - b) = 0;$$

that is, the positive and negative deviations cancel (see Exercise 42).

If $y = mx + b$ is the best fitting curve by least-square, then $mx_i + b$ is the expected data point for any given x_i data point. ∴ Since y_i is the measured point, $y_i - (mx_i + b)$ is the difference, and ∴ the sum of the squared differences is, for n data points, $\sum_{i=1}^{n} (y_i - mx_i - b)^2$

Let $S(m, b) = \sum_{i=1}^{n} (y_i - mx_i - b)^2$

At an extremum, $\frac{\partial S}{\partial b} = 0$.

$$\frac{\partial S}{\partial b} = \sum_{i=1}^{n} 2(y_i - mx_i - b)(-1) = 0$$

$$\Leftrightarrow \sum_{i=1}^{n} (y_i - mx_i - b) = 0$$

44. **44.** Use the second-derivative test to show that the critical point of f is a minimum.

From $S(m,b) = \sum_{i=1}^{n} (y_i - mx_i - b)^2 = 0$,

if (m,b) is a critical point (so $S_m(m,b)=0$, and $S_b(m,b)=0$), Then look at Hessian:

$$Hf\,S(m,b)\,\vec{h} = \frac{1}{2}\begin{bmatrix} h_1 & h_2 \end{bmatrix}\begin{bmatrix} S_{mm} & S_{mb} \\ S_{bm} & S_{bb} \end{bmatrix}\begin{bmatrix} h_1 \\ h_2 \end{bmatrix}$$

where $\vec{h} = (h_1, h_2)$.

$$S_m = \sum_{i=1}^{n} 2(y_i - mx_i - b)(-x_i) = -2\sum_{i=1}^{n} (x_i y_i - mx_i^2 - bx_i)$$

$$S_b = \sum_{i=1}^{n} 2(y_i - mx_i - b)(-1) = -2\sum_{i=1}^{n} (y_i - mx_i - b)$$

$$S_{mm} = -2\sum_{i=1}^{n} (-x_i^2) = 2\sum_{i=1}^{n} x_i^2$$

$$S_{mb} = -2\sum_{i=1}^{n} (-x_i) = 2\sum_{i=1}^{n} x_i$$

$$S_{66} = -2 \sum_{i=1}^{n} (-1) = 2n$$

$$\therefore (S_{mm})(S_{66}) - (S_{m6})^2 =$$

$$\left(2 \sum_{i=1}^{n} x_i^2\right)(2n) - \left(2 \sum_{i=1}^{n} x_i\right)^2 \qquad [1]$$

For $n=1$, this is just $4x_1^2 - 4x_1^2 = 0$

For $n=2$, this is $8(x_1^2 + x_2^2) - 4(x_1 + x_2)^2$

$$= 8x_1^2 + 8x_2^2 - 4x_1^2 - 8x_1x_2 - 4x_1^2$$

$$= 4(x_1 - x_2)^2 > 0 \text{ for } x_1 \neq x_2$$

For $n > 2$, [1] can be rewritten

$$\left(4 \sum_{i=1}^{n} x_i^2\right)(n-2) + 4\left[2 \sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2\right] \qquad [2]$$

The first term, $\left(4 \sum_{i=1}^{n} x_i^2\right)(n-2)$, is > 0 for $n > 2$.

Now consider, for $n > 2$, $4 \sum_{i=1}^{n} x_i^2 - 2\left(\sum_{i=1}^{n} x_i\right)^2$

$$\left(\sum_{i=1}^{n} x_i\right)^2 = x_1^2 + x_2^2 + \ldots + x_n^2 + 2x_1x_2 + 2x_1x_3 + \ldots + 2x_nx_{n-1}$$

$$= \sum_{i=1}^{n} x_i^2 + 2 \sum_{\substack{i,j=1 \\ i \neq j}}^{n} x_i x_j$$

The second term contains elements such as $2x_1x_3$, $2x_2x_3$, ..., i.e., every possible combination of x_i and x_j, $i \neq j$.

$$\therefore 2\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2 = \left(x_1 - x_2 - \ldots - x_n\right)^2$$

$$\therefore 2\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2 \geq 0 \quad \text{for } n > 2$$

$\therefore$ [2] is always positive for $n > 2$

$\therefore$ [1] is always positive for all $n \geq 2$

$$\therefore S_{mm}S_{bb} - (S_{mb})^2 > 0 \quad \text{for } n \geq 2$$

Since $S_{mm} = 2\sum_{i=1}^{n} x_i^2 > 0$ for all n,

by second derivitive test, $S(m,b)$ is a minimum. i.e., $y = mx + b$ produces the minimum sum of least squares.

45. **45.** Use the method of least squares to find the straight line that best fits the points (0, 1), (1, 3), (2, 2), (3, 4), and (4, 5). Plot the points and line.[16]

Let $mx + b$ be the line of best fit. Look at sum of least squares for the data.

$$S(m,b) = (1 - m\cdot 0 - b)^2 + (3 - m\cdot 1 - b)^2 + (2 - 2m - b)^2 + (4 - 3m - b)^2 +$$

$$(5-4m-6)^2$$

$$= (1-6)^2 + (3-m-6)^2 + (2-2m-6)^2 + (4-3m-6)^2 + (5-4m-6)^2$$

$$S_m = 2(3-m-6)(-1) + 2(2-2m-6)(-2) + 2(4-3m-6)(-3) + 2(5-4m-6)(-4)$$

$$= -6 + 2m + 26 - 8 + 8m + 46 - 24 + 18m + 66 - 40 + 32m + 86$$

$$= -78 + 60m + 20b \qquad [1]$$

$$S_b = 2(1-b)(-1) + 2(3-m-b)(-1) + 2(2-2m-b)(-1) + 2(4-3m-b)(-1) + 2(5-4m-b)(-1)$$

$$= -2 + 2b - 6 + 2m + 2b - 4 + 4m + 2b - 8 + 6m + 2b - 10 + 8m + 2b$$

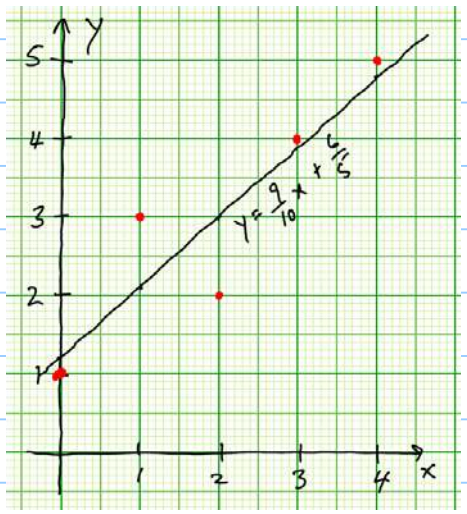
$$= -30 + 20m + 10b \qquad [2]$$

Setting [1], [2] to zero:

$$60m + 20b = 78 \text{ , or } \quad 3m + b = 3.9 \quad [1']$$
$$20m + 10b = 30 \text{ , or } \quad 2m + b = 3 \quad [2']$$

Subtracting, $m = 0.9 = \frac{9}{10}$, $b = 1.2 = \frac{6}{5}$

$$\therefore \underline{\underline{y = \frac{9}{10}x + \frac{6}{5}}}$$



46.

46. The partial differential equation

$$\frac{\partial^4 u}{\partial x^4} = -\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2},$$

where c is a constant, comes up in the study of deflections of a thin beam. Show that

$$u(x, t) = \sin(\lambda \pi x)\cos(\lambda^2 \pi^2 ct)$$

is a solution for any choice of the parameter λ.

$u_x = \lambda\pi\cos(\lambda\pi x)\cos(\lambda^2\pi^2 ct)$

$u_{xx} = -\lambda^2\pi^2\sin(\lambda\pi x)\cos(\lambda^2\pi^2 ct)$

$u_{xxx} = -\lambda^3\pi^3\cos(\lambda\pi x)\cos(\lambda^2\pi^2 ct)$

$u_{xxxx} = \frac{\partial^4 u}{\partial x^4} = \lambda^4\pi^4\sin(\lambda\pi x)\cos(\lambda^2\pi^2 ct)$ [1]

$u_t = -\lambda^2\pi^2 c\sin(\lambda\pi x)\sin(\lambda^2\pi^2 ct)$

$u_{tt} = -\lambda^4\pi^4 c^2\sin(\lambda\pi x)\cos(\lambda^2\pi^2 ct)$ [2]

Multiply [2] by $-\frac{1}{c^2}$:

$-\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2} = \lambda^4\pi^4\sin(\lambda\pi x)\cos(\lambda^2\pi^2 ct)$ [3]

$\therefore$ [1] = [3]:

$\frac{\partial^4 u}{\partial x^4} = \lambda^4\pi^4\sin(\lambda\pi x)\cos(\lambda^2\pi^2 ct) = -\frac{1}{c^2}\frac{\partial^2 u}{\partial t^2}$,

for all λ

47. **47.** The Kortweg–DeVries equation

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} = 0$$

arises in modeling shallow water waves (called **solitons**). Show that

$$u(x, t) = 12a^2\operatorname{sech}^2(ax - 4a^3 t)$$

is a solution to the Kortweg–DeVries equation (see the Internet supplement).

Note $\frac{d}{dx}\operatorname{sech}(x) = -\operatorname{sech}(x)\tanh(x)$

$\frac{d}{dx}\tanh(x) = \operatorname{sech}^2(x)$

$\therefore \frac{\partial u}{\partial t} = -96a^5\operatorname{sech}(ax-4a^3t)[-\operatorname{sech}(ax-4a^3t)\tanh(ax-4a^3t)]$

$= 96a^5\operatorname{sech}^2(ax-4a^3t)\tanh(ax-4a^3t)$

$\frac{\partial u}{\partial x} = -24a^3\operatorname{sech}^2(ax-4a^3t)\tanh(ax-4a^3t)$

$\therefore uu_x = -288a^5\operatorname{sech}^4(ax-4a^3t)\tanh(ax-4a^3t)$

$u_{xx} = 48a^4\operatorname{sech}^2(ax-4a^3t)\tanh^2(ax-4a^3t)$

$-24a^4\operatorname{sech}^4(ax-4a^3t)$

$u_{xxx} = -96a^5\operatorname{sech}^2(ax-4a^3t)\tanh^3(ax-4a^3t)$

$+96a^5\operatorname{sech}^4(ax-4a^3t)\tanh(ax-4a^3t)$

$+96a^5\operatorname{sech}^4(ax-4a^3t)\tanh(ax-4a^3t)$

$$= -96a^5 \operatorname{sech}^2(ax-4a^3t)\tanh^3(ax-4a^3t)$$

$$+192a^5 \operatorname{sech}^4(ax-4a^3t)\tanh(ax-4a^3t)$$

For simplicity, let $y = ax-4a^3t$

$$\therefore \frac{\partial u}{\partial t} = 96a^5 \operatorname{sech}^2(y)\tanh(y)$$

$$u\frac{\partial u}{\partial x} = -288a^5 \operatorname{sech}^4(y)\tanh(y)$$

$$\frac{\partial^3 u}{\partial x^3} = -96a^5 \operatorname{sech}^2(y)\tanh^3(y) + 192a^5 \operatorname{sech}^4(y)\tanh(y)$$

$$\therefore \frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} =$$

$$96a^5 \operatorname{sech}^2(y)\tanh(y) - 96a^5 \operatorname{sech}^4(y)\tanh(y)$$

$$-96a^5 \operatorname{sech}^2(y)\tanh^3(y) \qquad [1]$$

Now use $\operatorname{sech}^2(y) = 1-\tanh^2(y)$ $\therefore$ [1] becomes:

$$96a^5 \operatorname{sech}^2(y)\tanh(y) - 96a^5(1-\tanh^2(y))^2\tanh(y)$$

$$-96a^5(1-\tanh^2(y))\tanh^3(y)$$

$$= 96a^5 \operatorname{sech}^2(y)\tanh(y) - 96a^5\left[1-2\tanh^2(y)+\tanh^4(y)\right]\tanh(y)$$

$$-96a^5\tanh^3(y)+96a^5\tanh^5(y)$$

$$=96a^5\operatorname{sech}^2(y)\tanh(y)$$

$$-96a^5\tanh(y)+192a^5\tanh^3(y)-96a^5\tanh^5(y)$$

$$-96a^5\tanh^3(y)+96a^5\tanh^5(y)$$

$$=96a^5\operatorname{sech}^2(y)\tanh(y)-96a^5\tanh(y)+96a^5\tanh^3(y)$$

$$=96a^5\left[1-\tanh^2(y)\right]\tanh(y)-96a^5\tanh(y)+96a^5\tanh^3(y)$$

$$=96a^5\tanh(y)-96a^5\tanh^3(y)-96a^5\tanh(y)+96a^5\tanh^3(y)$$

$$=0$$

$\therefore$ When $u(x,t)=12a^2\operatorname{sech}(ax-4a^3t)$,

$$\frac{\partial u}{\partial t}+u\frac{\partial u}{\partial x}+\frac{\partial^3 u}{\partial x^3}=0$$

48. **48.** The heat-conduction equation in two space dimensions is

$$k(u_{xx}+u_{yy})=u_t.$$

Assuming that $u(x, y, t)=X(x)\,Y(y)\,T(t)$, find ordinary differential equations satisfied by $X(x)$, $Y(y)$, and $T(t)$.

$$kX''YT+kXY''T=XYT$$

$$\therefore X''Y+XY''=\frac{XY}{k}$$

Let $X = f\left(\frac{x}{\sqrt{k}}\right) \therefore X' = \frac{1}{\sqrt{k}} f\left(\frac{x}{\sqrt{k}}\right), X'' = \frac{1}{k} f''\left(\frac{x}{\sqrt{k}}\right)$

Similarly, Let $Y = f\left(\frac{y}{\sqrt{k}}\right) \therefore Y'' = \frac{1}{k} f''\left(\frac{y}{\sqrt{k}}\right)$

Need to find functions whose 2nd derivative is similar in form to the function.

Possibilities: $\sin, \cos, e^x$

Let $X(x) = \sin\left(\frac{x}{\sqrt{k}}\right) \therefore X'' = -\frac{1}{k}\sin\left(\frac{x}{\sqrt{k}}\right)$.

$X(x) = -\sin\left(\frac{x}{\sqrt{k}}\right) \therefore X'' = \frac{1}{k}\sin\left(\frac{x}{\sqrt{k}}\right)$

$Y(y) = -\cos\left(\frac{y}{\sqrt{k}}\right), \quad Y'' = \frac{1}{k}\cos\left(\frac{y}{\sqrt{k}}\right)$

$\therefore$ Let $u(x,y,t) = \sin\left(\frac{x}{\sqrt{k}}\right)\sin\left(\frac{y}{\sqrt{k}}\right)e^{-t}$

$\therefore \quad u_{xx} = -\frac{1}{k}\sin\left(\frac{x}{\sqrt{k}}\right)\sin\left(\frac{y}{\sqrt{k}}\right)e^{-ct}$

$u_{yy} = -\frac{1}{k}\sin\left(\frac{x}{\sqrt{k}}\right)\sin\left(\frac{y}{\sqrt{k}}\right)e^{-ct}$

$u_t = -c\sin\left(\frac{x}{\sqrt{k}}\right)\sin\left(\frac{y}{\sqrt{k}}\right)e^{-ct}$

$\therefore k(u_{xx} + u_{yy}) = -2u = u_t$ if $c = 2$

$\therefore \quad u_1(x,y,t) = \sin\left(\frac{x}{\sqrt{k}}\right)\sin\left(\frac{y}{\sqrt{k}}\right)e^{-2t}$

Similar results occur with

$u_2(x,y,t) = \cos\left(\frac{x}{\sqrt{k}}\right)\cos\left(\frac{y}{\sqrt{k}}\right)e^{-2t}$

Also, the system is linear, so $c_1 u_1$ and $c_2 u_2$ are solutions, and so is $c_1 u_1 + c_2 u_2$

$\therefore$ A general solution would be:

$$u(x,y,t) = c_1 \sin\left(\frac{x}{\sqrt{k}}\right)\sin\left(\frac{y}{\sqrt{k}}\right)e^{-2t} + c_2 \cos\left(\frac{x}{\sqrt{k}}\right)\cos\left(\frac{y}{\sqrt{k}}\right)e^{-2t}$$

Note that $u(x,y,t) = c\sin\left(\frac{x}{\sqrt{k}}\right)\cos\left(\frac{y}{\sqrt{k}}\right)e^{-2t}$

will also work.

Since $\sin(x) = \cos\left(x - \frac{\pi}{2} + 2n\pi\right)$, $n = 0, \pm 1, \pm 2, \ldots$
a phase adjustment to the term $\frac{x}{\sqrt{k}}$ can be used to simplify the general solution.

$\therefore$ A general solution would be any linear combination of:

$$\underline{u(x,y,t) = c\sin\left(\frac{x+\phi_1}{\sqrt{k}}\right)\sin\left(\frac{x+\phi_2}{\sqrt{k}}\right)e^{-2t}}$$

where, c, ϕ_1, ϕ_2 are constants.

Note if $X(x) = e^{\frac{x}{\sqrt{k}}}$, then $X'' = \frac{1}{k}e^{\frac{x}{\sqrt{k}}} = \frac{1}{k}X$.

$\therefore$ $\underline{u(x,y,t) = ce^{\frac{x}{\sqrt{k}}}e^{\frac{y}{\sqrt{k}}}e^{kt}}$ will also be a solution.

Corresponding ordinary differential equations these

solutions would satisfy are: $ku_{xx} = u$, $ku_{yy} = u$ and $u_t = ku$

49.

49. The heat conduction equation in two space dimensions may be expressed in terms of polar coordinates as

$$k\left(u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta}\right) = u_t.$$

Assuming that $u(r, \theta, t) = R(r)\Theta(\theta)\,T(t)$, find ordinary differential equations satisfied by $R(r)$, $\Theta(\theta)$, and $T(t)$.

Write equation as:

$$k(r^2u_{rr} + ru_r + u_{\theta\theta}) - r^2u_t = 0$$

$\therefore$ Ordinary (one-variable) differential equations satisfied by $u(r,\theta,t)$ would be:

$$\frac{dT}{dt} = cT \qquad (\text{so } T_{\theta\theta} = 0,\ T_r = 0,\ T_{rr} = 0)$$

$$\frac{d^2\Theta(\theta)}{d\theta^2} = c\Theta(\theta) \qquad (\Theta_t = 0,\ \Theta_r = 0,\ \Theta_{rr} = 0)$$

$$r^2\frac{d^2R(r)}{dr^2} + r\frac{dR(r)}{dr} = cR(r) \qquad (R_{\theta\theta} = 0,\ R_t = 0)$$

where c is a constant, and could be 0.

4.1 Acceleration and Newton's Second Law

Note Title 8/16/2016

In Exercises 1 to 4, find the velocity and acceleration vectors and the equation of the tangent line for each of the following curves, at the given value of t.

1. **1.** $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin 2t)\mathbf{j}$, at $t = 0$

$\vec{r}'(t) = -\sin t\,\hat{i} + 2\cos 2t\,\hat{j}$ = velocity vector

$\therefore \vec{r}'(0) = 2\hat{j}$

$\vec{r}''(t) = -\cos t\,\hat{i} - 4\sin 2t\,\hat{j}$ = acceleration vector

$\therefore \vec{r}''(0) = -\hat{i}$

Tangent line at $t=0$: $\vec{r}(0) + s\vec{v}(0)$, $s \in \mathbb{R}$

$\therefore \vec{r}(0) = [1,0]$ $\vec{v}(0) = \vec{r}'(0) = [0,2]$

$\therefore \ell(s) = [1,0] + s[0,2] = [1, 2s] = \hat{i} + 2s\hat{j}$

2. **2.** $\mathbf{c}(t) = (t\sin t, t\cos t, \sqrt{3}t)$, at $t = 0$

$\vec{c}'(t) = [\sin t + t\cos t, \cos t - t\sin t, \sqrt{3}]$ = velocity vector

$\therefore \vec{c}'(0) = [0, 1, \sqrt{3}] = \hat{j} + \sqrt{3}\hat{k}$

$\vec{c}''(t) = [\cos t + \cos t - t\sin t, \sin t - \sin t - t\cos t, 0]$

$= [2\cos t - t\sin t, -t\cos t, 0]$ = acceleration vector

$\therefore \vec{c}''(0) = [2, 0, 0] = 2\hat{i}$

Tangent line at $t=0$: $\vec{c}(0) + s\vec{v}(0)$, $s \in \mathbb{R}$

$\vec{c}(0) = (0, 0, 0)$ $\quad \vec{v}(0) = \vec{c}'(0) = (0, 1, \sqrt{3})$

$\therefore \ell(s) = s(0, 1, \sqrt{3}) = s\hat{j} + s\sqrt{3}\hat{k}$

3. **3.** $\mathbf{r}(t) = \sqrt{2}t\mathbf{i} + e^t\mathbf{j} + e^{-t}\mathbf{k}$, at $t = 0$

$\vec{r}'(t) = (\sqrt{2}, e^t, -e^{-t})$ = velocity vector

$\therefore \vec{r}'(0) = (\sqrt{2}, 1, -1) = \sqrt{2}\hat{i} + \hat{j} - \hat{k}$

$\vec{r}''(t) = (0, e^t, e^{-t})$ = acceleration vector

$\therefore \vec{r}''(0) = (0, 1, 1) = \hat{j} + \hat{k}$

Tangent line at $t=0$: $\vec{r}(0) + s\vec{v}(0)$, $s \in \mathbb{R}$

$\vec{r}(0) = (0, 1, 1)$ $\quad \vec{v}(0) = \vec{c}'(0) = (\sqrt{2}, 1, -1)$

$\therefore \ell(s) = (0, 1, 1) + s(\sqrt{2}, 1, -1)$

$= s\sqrt{2}\hat{i} + (1+s)\hat{j} + (1-s)\hat{k}$

4. **4.** $\mathbf{c}(t) = t\mathbf{i} + t\mathbf{j} + \frac{2}{3}t^{3/2}\mathbf{k}$, at $t = 9$

$\vec{c}'(t) = (1, 1, t^{1/2})$ = velocity vector

$\therefore \vec{c}'(9) = (1, 1, 3) = \hat{i} + \hat{j} + 3\hat{k}$

$\vec{c}''(t) = (0, 0, \frac{1}{2}t^{-\frac{1}{2}})$ = acceleration vector

$\therefore \vec{c}''(9) = (0, 0, \frac{1}{6}) = \frac{1}{6}\hat{k}$

Tangent line at $t=9$: $\vec{c}(9) + s\vec{v}(9)$, $s \in \mathbb{R}$

$\vec{c}(9) = (9, 9, 18)$ $\vec{v}(9) = \vec{c}'(9) = (1, 1, 3)$

$\therefore l(s) = (9, 9, 18) + s(1, 1, 3)$

$= (9+s)\hat{i} + (9+s)\hat{j} + (18+3s)\hat{k}$

In Exercises 5 to 8, let $\mathbf{c}_1(t) = e^t\mathbf{i} + (\sin t)\mathbf{j} + t^3\mathbf{k}$ *and* $\mathbf{c}_2(t) = e^{-t}\mathbf{i} + (\cos t)\mathbf{j} - 2t^3\mathbf{k}$. *Find each of the stated derivatives in two different ways to verify the rules in the box preceding Example 1.*

5. **5.** $\frac{d}{dt}[\mathbf{c}_1(t) + \mathbf{c}_2(t)]$

$\vec{c}_1'(t) = (e^t, \cos t, 3t^2)$ $\vec{c}_2'(t) = (-e^{-t}, -\sin t, -6t^2)$

$\therefore \vec{c}_1'(t) + \vec{c}_2'(t) = (e^t - e^{-t}, \cos t - \sin t, -3t^2)$

$$\vec{c}_1(t) + \vec{c}_2(t) = (e^t + e^{-t},\ \sin t + \cos t,\ -t^3)$$

$$\therefore \frac{d}{dt}\left\{\vec{c}_1(t) + c_2(t)\right\} = (e^t - e^{-t},\ \cos t - \sin t,\ -3t^2)$$

6. **6.** $\frac{d}{dt}[\mathbf{c}_1(t) \cdot \mathbf{c}_2(t)]$

$$\vec{c}_1(t) \cdot \vec{c}_2(t) = (e^t \cdot e^{-t}) + (\sin t \cdot \cos t) + (t^3 \cdot (-2t^3))$$

$$= 1 + \sin t \cos t - 2t^6$$

$$\therefore \frac{d}{dt}\left\{\vec{c}_1(t) \cdot \vec{c}_2(t)\right\} = \cos^2 t - \sin^2 t - 12t^5$$

$$\vec{c}_1'(t) = (e^t,\ \cos t,\ 3t^2) \qquad \vec{c}_2'(t) = (-e^t,\ -\sin t,\ -6t^2)$$

$$\therefore \vec{c}_1'(t) \cdot \vec{c}_2(t) = (e^t, \cos t, 3t^2) \cdot (e^{-t}, \cos t, -2t^3)$$

$$= 1 + \cos^2 t - 6t^5$$

$$\vec{c}_1(t) \cdot \vec{c}_2'(t) = (e^t, \sin t, t^3) \cdot (-e^{-t}, -\sin t, -6t^2)$$

$$= -1 - \sin^2 t - 6t^5$$

$$\therefore \vec{c}_1'(t) \cdot \vec{c}_2(t) + \vec{c}_1(t) \cdot \vec{c}_2'(t) = \cos^2 t - \sin^2 t - 12t^5$$

7. **7.** $\frac{d}{dt}[\mathbf{c}_1(t) \times \mathbf{c}_2(t)]$

$$\vec{c}_1(t) \times \vec{c}_2(t) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ e^t & \sin t & t^3 \\ e^{-t} & \cos t & -2t^3 \end{vmatrix}$$

$$= \left[-2t^3\sin t - t^3\cos t,\ t^3e^{-t} + 2t^3e^t,\ e^t\cos t - e^{-t}\sin t\right]$$

$$\therefore \frac{d}{dt}\left\{\vec{c}_1(t) \times c_2(t)\right\} =$$

$$\left(-6t^2\sin t - 2t^3\cos t - 3t^2\cos t + t^3\sin t,\right.$$
$$3t^2e^{-t} - t^3e^{-t} + 6t^2e^t + 2t^3e^t,$$
$$\left.e^t\cos t - e^t\sin t + e^{-t}\sin t - e^{-t}\cos t\right) \quad [1]$$

$$\vec{c}_1'(t) \times \vec{c}_2(t) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ e^t & \cos t & 3t^2 \\ e^{-t} & \cos t & -2t^3 \end{vmatrix}$$

$$= \left(-2t^3\cos t - 3t^2\cos t,\ 3t^2e^{-t} + 2t^3e^t,\ e^t\cos t - e^{-t}\cos t\right)$$

$$\vec{c}_1(t) \times \vec{c}_2'(t) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ e^t & \sin t & t^3 \\ -e^{-t} & -\sin t & -6t^2 \end{vmatrix}$$

$$= \left(-6t^2\sin t + t^3\sin t,\ -e^{-t}t^3 + 6t^2e^t,\ -e^t\sin t + e^{-t}\sin t\right)$$

$$\therefore \vec{c}_1'(t) \times \vec{c}_2(t) + \vec{c}_1(t) \times \vec{c}_2'(t) =$$

$$\left(-2t^3\cos t - 3t^2\cos t - 6t^2\sin t + t^3\sin t,\right.$$
$$3t^2e^{-t} + 2t^3e^t - e^{-t}t^3 + 6t^2e^t,$$
$$\left.e^t\cos t - e^{-t}\cos t - e^t\sin t + e^{-t}\sin t\right) \quad [2]$$

$[1] = [2] \quad \therefore \frac{d}{dt}\left[\vec{c}_1 \times \vec{c}_2\right] = \vec{c}_1' \times \vec{c}_2 + \vec{c}_1 \times \vec{c}_2'$

8. $\frac{d}{dt}\{\mathbf{c}_1(t) \cdot [2\mathbf{c}_2(t) + \mathbf{c}_1(t)]\}$

(a) $2\vec{c}_2(t) = (2e^{-t}, 2\cos t, -4t^3)$

$\vec{c}_1(t) = (e^t, \sin t, t^3)$

$\therefore 2\vec{c}_2 + \vec{c}_1 = (2e^{-t} + e^t, 2\cos t + \sin t, -3t^3)$

$\therefore \vec{c}_1 \cdot (2\vec{c}_2 + \vec{c}_1) = 2 + e^{2t} + 2\sin t\cos t + \sin^2 t - 3t^6$

$\therefore \frac{d}{dt}\left[\vec{c}_1 \cdot (2\vec{c}_2 + \vec{c}_1)\right] =$

$$2e^{2t} + 2\cos^2 t - 2\sin^2 t + 2\sin t\cos t - 18t^5$$

(b) $\frac{d}{dt}\left[2\vec{c}_2(t)\right] = (-2e^{-t}, -2\sin t, -12t^2)$

$\vec{c}_1'(t) = (e^t, \cos t, 3t^2)$

$$\therefore \frac{d}{dt}\left[2\vec{c}_2(t) + \vec{c}_1(t)\right] =$$

$$(-2e^{-t} + e^{t},\ -2\sin t + \cos t,\ -9t^2)$$

$$\therefore \vec{c}_1(t) \cdot \frac{d}{dt}\left[2\vec{c}_2(t) + \vec{c}_1(t)\right] =$$

$$(e^{t},\ \sin t,\ t^3)\cdot(-2e^{-t} + e^{t},\ -2\sin t + \cos t,\ -9t^2)$$

$$= -2 + e^{2t} - 2\sin^2 t + \sin t \cos t - 9t^5 \quad [1]$$

$$\vec{c}_1'(t) \cdot \left[2\vec{c}_2(t) + \vec{c}_1(t)\right] =$$

$$(e^{t},\ \cos t,\ 3t^2)\cdot(2e^{-t} + e^{t},\ 2\cos t + \sin t,\ -3t^3)$$

$$= 2 + e^{2t} + 2\cos^2 t + \cos t \sin t - 9t^5 \quad [2]$$

$$[2] + [1] = \vec{c}_1' \cdot \left[2\vec{c}_2 + \vec{c}_1\right] + \vec{c}_1 \cdot \frac{d}{dt}\left[2\vec{c}_2 + \vec{c}_1\right]$$

$$= \underline{2e^{2t} + 2\cos^2 t - 2\sin^2 t + 2\sin t \cos t - 18t^5}$$

(a) = (b)

9. **9.** Consider the helix given by $\mathbf{c}(t) = (a\cos t,\ a\sin t,\ bt)$. Show that the acceleration vector is always parallel to the xy plane.

$$\vec{c}'(t) = (-a\sin t,\ a\cos t,\ b)$$

$$\vec{c}''(t) = (-a\cos t,\ -a\sin t,\ 0) = \text{acceleration vector}$$

z-component is 0. $\therefore \vec{c}''$ parallel to xy-plane.

10. **10.** Prove the dot product rule.

Let $\vec{b}(t) = \{b_1(t), b_2(t), b_3(t)\}$

$\vec{c}(t) = \{c_1(t), c_2(t), c_3(t)\}$

$\therefore \vec{b}(t) \cdot \vec{c}(t) = b_1(t)c_1(t) + b_2(t)c_2(t) + b_3(t)c_3(t)$

$\therefore \frac{d}{dt}\left[\vec{b}(t) \cdot \vec{c}(t)\right] = b_1'c_1 + b_1c_1' + b_2'c_2 + b_2c_2' + b_3'c_3 + b_3c_3'$ [1]

$\frac{d}{dt}\vec{b}(t) \cdot \vec{c}(t) = (b_1', b_2', b_3') \cdot (c_1, c_2, c_3)$

$= b_1'c_1 + b_2'c_2 + b_3'c_3$

$\vec{b}(t) \cdot \frac{d}{dt}\vec{c}(t) = (b_1, b_2, b_3) \cdot (c_1', c_2', c_3')$

$= b_1c_1' + b_2c_2' + b_3c_3'$

$\therefore \frac{d}{dt}\vec{b}(t) \cdot \vec{c}(t) + \vec{b}(t) \cdot \frac{d}{dt}\vec{c}(t) =$

$b_1'c_1 + b_1c_1' + b_2'c_2 + b_2c_2' + b_3'c_3 + b_3c_3'$ [2]

[1] = [2] $\therefore \frac{d}{dt}\left[\vec{b} \cdot \vec{c}\right] = \frac{d}{dt}\vec{b} \cdot \vec{c} + \vec{b} \cdot \frac{d}{dt}\vec{c}$

11. **11.** Determine which of the following paths are regular:

(a) $\mathbf{c}(t) = (\cos t, \sin t, t)$

(b) $\mathbf{c}(t) = (t^3, t^5, \cos t)$

(c) $\mathbf{c}(t) = (t^2, e^t, 3t + 1)$

(a) $\vec{c}'(t) = (-\sin t, \cos t, 1)$. $\therefore \vec{c}'(t) \neq \vec{0}$.

$\therefore \vec{c}(t)$ is regular

(b) $\vec{c}'(t) = (3t^2, 5t^4, -\sin t)$. For $t=0$, $\vec{c}'(t) = 0$.

$\therefore$ Not regular

(c) $\vec{c}'(t) = (2t, e^t, 3)$. $\therefore \vec{c}'(t) \neq \vec{0}$

$\therefore$ Regular

12. **12.** Let $\mathbf{v}$ and $\mathbf{a}$ denote the velocity and acceleration vectors of a particle moving on a path $\mathbf{c}$. Suppose the initial position of the particle is $\mathbf{c}(0) = (3, 4, 0)$, the initial velocity is $\mathbf{v}(0) = (1, 1, -2)$, and the acceleration function is $\mathbf{a}(t) = (0, 0, 6)$. Find $\mathbf{v}(t)$ and $\mathbf{c}(t)$.

$\vec{v}(t) = (c_1, c_2, 6t + c_3)$, c_1, c_2, c_3 are constants.

$\vec{v}(0) = (1, 1, -2)$, $\therefore c_1 = 1$, $c_2 = 1$, $c_3 = -2$.

$\therefore \vec{v}(t) = (1, 1, 6t - 2)$.

$\therefore \vec{c}(t) = (t + k_1, t + k_2, 3t^2 - 2t + k_3)$, k_i constants.

$\vec{c}(0) = (3, 4, 0) = (k_1, k_2, k_3)$. $\therefore k_1 = 3, k_2 = 4, k_3 = 0$.

$\therefore \vec{c}(t) = (t+3, t+4, 3t^2-2t)$

13.

13. The acceleration, initial velocity, and initial position of a particle traveling through space are given by

$$\mathbf{a}(t) = (2, -6, -4), \quad \mathbf{v}(0) = (-5, 1, 3), \quad \mathbf{r}(0) = (6, -2, 1).$$

The particle's trajectory intersects the *yz* plane exactly twice. Find these two intersection points.

$\vec{a}(t) = (2, -6, -4) \Rightarrow \vec{v}(t) = (2t + c_1, -6t + c_2, -4t + c_3)$

$\vec{v}(0) = (-5, 1, 3) = (c_1, c_2, c_3)$

$\therefore \vec{v}(t) = (2t - 5, -6t + 1, -4t + 3)$

$\therefore \vec{r}(t) = (t^2 - 5t + k_1, -3t^2 + t + k_2, -2t^2 + 3t + k_3)$,

k_i constants.

$\vec{r}(0) = (6, -2, 1) = (k_1, k_2, k_3)$

$\therefore \vec{r}(t) = (t^2 - 5t + 6, -3t^2 + t - 2, -2t^2 + 3t + 1)$

Cross yz plane when x component = 0.

$\therefore t^2 - 5t + 6 = 0 = (t-3)(t-2) \Rightarrow t = 2, 3.$

$\therefore t = 2: \ \vec{r}(2) = (0, -12, -1)$

$t = 3: \ \vec{r}(3) = (0, -26, -8)$

14.

14. The acceleration, initial velocity, and initial position of a particle traveling through space are given by

$$\mathbf{a}(t) = (-6, 2, 4), \quad \mathbf{v}(0) = (2, -5, 1), \quad \mathbf{r}(0) = (-3, 6, 2).$$

The particle's trajectory intersects the yz plane exactly twice. Find these two intersection points.

$\vec{a}(t) = (-6, 2, 4) \Rightarrow \vec{v}(t) = (-6t + c_1, 2t + c_2, 4t + c_3)$, c_i constants

$\vec{v}(0) = (2, -5, 1) = (c_1, c_2, c_3)$

$\therefore \vec{v}(t) = (-6t + 2, 2t - 5, 4t + 1)$

$\therefore \vec{r}(t) = \int \vec{v}(t)\,dt = (-3t^2 + 2t + k_1, t^2 - 5t + k_2, 2t^2 + t + k_3)$, k_i constants

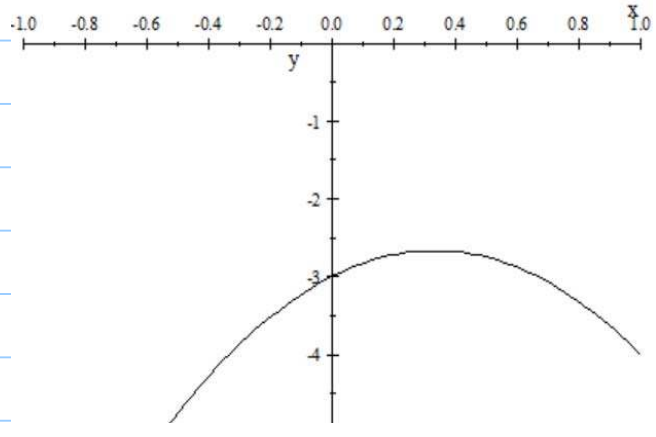
$\vec{r}(0) = (-3, 6, 2) = (k_1, k_2, k_3)$

$\therefore \vec{r}(t) = (-3t^2 + 2t - 3, t^2 - 5t + 6, 2t^2 + t + 2)$

Cross yz plane when x component $= 0$.

$\therefore -3t^2 + 2t - 3 = 0$

$t = \dfrac{-2 \pm \sqrt{4 - 36}}{-6}$. No real solutions.



Problem probably meant crossing xz plane

(y component $= 0$, at $t = 2, 3$).

15. **15.** If $\mathbf{r}(t) = 6t\mathbf{i} + 3t^2\mathbf{j} + t^3\mathbf{k}$, what force acts on a particle of mass m moving along $\mathbf{r}$ at $t = 0$?

$\vec{r}'(t) = (6, 6t, 3t^2)$, $\vec{a}(t) = r''(t) = (0, 6, 6t)$

at $t = 0$, $\vec{a}(0) = (0, 6, 0)$.

$\therefore \vec{F} = m\vec{a}(0) = (0, 6m, 0) = 6m\hat{j}$

16. **16.** Let a particle of mass 1 gram (g) follow the path in Exercise 1, with units in seconds and centimeters. What force acts on it at $t = 0$? (Give the units in your answer.)

From #1,

$\vec{r}''(t) = -\cos t\,\hat{i} - 4\sin 2t\,\hat{j}$ = acceleration vector

$\therefore \vec{r}''(0) = (-1, 0)$

$\vec{F}(0) = m\vec{r}''(0) = 1g(-1, 0) = -\hat{i}\ \frac{gm\text{-}cm}{sec^2}$

17. **17.** A body of mass 2 kilograms moves on a circle of radius 3 meters, making one revolution every 5 seconds. Find the centripetal force acting on the body.

Circumference $= 2\pi(3\text{ meter}) = 6\pi$ meters

Speed $= \frac{6\pi}{5}$ meters/sec.

Acceleration $= \frac{V^2}{r} = \frac{(6\pi/5)^2}{(3\text{ meters})} = \frac{36\pi^2}{75}$

$$\therefore F = ma = (2\,kg)\left(\frac{36\pi^2}{75}\right) = \underline{\frac{72}{75}\pi^2 \text{ Newtons}}$$

Using text $\vec{r}(t) = r_0\left(\cos\frac{5t}{r_0}, \sin\frac{5t}{r_0}\right)$,

assuming $\vec{r}(0) = (r_0, 0)$, here $r_0 = 3$ meters

$$\omega = \frac{5}{r_0} = \frac{2\pi r_0 / 5\,sec}{r_0} = \frac{2\pi}{5}\,sec^{-1}$$

$$\therefore \vec{r}(t) = 3\left(\cos\tfrac{2\pi}{5}t, \sin\tfrac{2\pi}{5}t\right)$$

$$\therefore \vec{r}'(t) = \frac{6\pi}{5}\left(-\sin\tfrac{2\pi}{5}t, \cos\tfrac{2\pi}{5}t\right)$$

$$\vec{r}''(t) = \frac{12\pi^2}{25}\left(-\cos\tfrac{2\pi}{5}t, -\sin\tfrac{2\pi}{5}t\right)$$

$$\therefore \vec{F}(t) = (2\,kg)\,\vec{r}''(t)$$

$$= \underline{-\frac{24}{25}\pi^2\left(\cos\tfrac{2\pi}{5}t, \sin\tfrac{2\pi}{5}t\right)}$$

18. **18.** Find the centripetal force acting on a body of mass 4 kilograms (kg), moving on a circle of radius 10 meters (m) with a frequency of 2 revolutions per second (rps).

Let $\vec{r}(t) = r_0(\cos\omega t, \sin\omega t)$

Here, $r_0 = 10\,m$, $\omega = \frac{5}{r_0} = \frac{2(2\pi r_0)}{r_0} = 4\pi\,sec^{-1}$

$$\therefore \vec{r}(t) = 10(\cos 4\pi t, \sin 4\pi t)$$

$\therefore \vec{r}'(t) = 40\pi(-\sin 4\pi t, \cos 4\pi t)$

$\vec{r}''(t) = -160\pi^2(\cos 4\pi t, \sin 4\pi t)$

$\vec{F}(t) = m\vec{a}(t) = (4\text{kg})(-160\pi^2)(\cos 4\pi t, \sin 4\pi t)$

$= -640\pi^2(\cos 4\pi t, \sin 4\pi t)$ Newtons

19. **19.** Show that if the acceleration of an object is always perpendicular to the velocity, then the speed of the object is constant. (HINT: See Example 1.)

Let $\vec{r}(t)$ = position of object at time t

$\therefore \vec{r}'(t)$ = velocity, $\vec{r}''(t)$ = acceleration.

Given: $\vec{r}'(t) \cdot \vec{r}''(t) = 0$

$\therefore \vec{r}'(t) \cdot \vec{r}''(t) + \vec{r}''(t) \cdot \vec{r}'(t) = 0$

$\therefore \frac{d}{dt}\left[\vec{r}'(t) \cdot \vec{r}'(t)\right] = 0$

$\therefore \vec{r}'(t) \cdot \vec{r}'(t) = c$, a constant

$\therefore \|\vec{r}'(t)\|^2 = c$, so $\|\vec{r}'(t)\| = \sqrt{c}$, a constant.

$\therefore$ Speed $= \|\vec{r}'(t)\|$ is a constant

20. **20.** Show that, at a local maximum or minimum of $\|\mathbf{r}(t)\|$, the vector $\mathbf{r}'(t)$ is perpendicular to $\mathbf{r}(t)$.

A local maximum or minimum for $\|\vec{r}(t)\|$ will also be a local max or min for $\|\vec{r}(t)\|^2$, since both are >0.

At a local max or min, $\frac{d}{dt}\left(\|\vec{r}(t)\|^2\right) = 0$

$\therefore \frac{d}{dt}\left(\vec{r}(t)\cdot\vec{r}(t)\right) = 0 \Rightarrow$

$2\,\vec{r}(t)\cdot\vec{r}'(t) = 0 \Rightarrow \vec{r}(t)\cdot\vec{r}'(t) = 0$

$\therefore \vec{r}(t) \perp \vec{r}'(t)$ at a local max or min of $\|\vec{r}(t)\|$.

21. **21.** A satellite is in a circular orbit 500 miles above the surface of the earth. What is the period of the orbit? (You may take the radius of the earth to be 4000 miles, or 6.436×10^6 meters.)

$R_{satellite} = 4500 \text{ miles} = 6.436\times10^6\left(\frac{4500}{4000}\right) = 7.24\times10^6 \text{ m}$

$T^2 = R^3\left(\frac{4\pi^2}{GM}\right) = \frac{(7.24\times10^6)^3\,4\pi^2}{(6.67\times10^{-11})(5.98\times10^{24})} = 375.6\times10^5$

$\therefore T = \sqrt{37.56\times10^6} = 6.129\times10^3 = 6{,}129 \text{ secs}$

$= 102.1 \text{ min.}$

22. **22.** What is the acceleration of the satellite in Exercise 21? The centripetal force?

(a) Acceleration $= \frac{V^2}{R} = \frac{(2\pi R/T)^2}{R} = \frac{4\pi^2 R}{T^2}$

$= \frac{4\pi^2 (7.24 \times 10^6 \text{ m})}{(6.129 \times 10^3 \text{ sec})^2} = \underline{7.61 \text{ m/sec}^2}$

(b) $F = ma$. Need to know mass of satellite.

23. **23.** Find the path $\mathbf{c}$ such that $\mathbf{c}(0) = (0, -5, 1)$ and $\mathbf{c}'(t) = (t, e^t, t^2)$.

$\int \vec{c}'(t) = \int (t, e^t, t^2) = \left(\frac{1}{2}t^2 + c_1, e^t + c_2, \frac{1}{3}t^3 + c_3\right)$,

where c_i are constants.

$\vec{c}(0) = (0, -5, 1) = \left(\frac{1}{2}(0)^2 + c_1, e^0 + c_2, \frac{1}{3}(0)^3 + c_3\right)$

$= (c_1, 1 + c_2, c_3)$

$\therefore c_1 = 0, \; c_2 = -6, \; c_3 = 1$

$\therefore \vec{c}(t) = \left(\frac{t^2}{2}, e^t - 6, \frac{t^3}{3} + 1\right)$

24. **24.** Let $\mathbf{c}$ be a path in $\mathbb{R}^3$ with zero acceleration. Prove that $\mathbf{c}$ is a straight line or a point.

$\vec{c}''(t) = (0,0,0) \quad \therefore \vec{c}'(t) = \int \vec{c}''(t) = \{c_1, c_2, c_3\}$,

where c_i are constants.

$\therefore \vec{c}(t) = \int \vec{c}'(t) = [c_1 t + k_1, c_2 t + k_2, c_3 t + k_3]$,

where c_i and k_i are constants.

If $c_1 = c_2 = c_3 = 0$, then $\vec{c}(t) = [k_1, k_2, k_3]$,

so $\vec{c}(t)$ is a point (no motion over time).

If just one of $c_i \neq 0$, then $[c_1 t + k_1, c_2 t + k_2, c_3 t + k_3]$

describes a line.

25. **25.** Find paths $\mathbf{c}(t)$ that represent the following curves or trajectories.

(a) $\{(x, y) \mid y = e^x\}$

(b) $\{(x, y) \mid 4x^2 + y^2 = 1\}$

(c) A straight line in $\mathbb{R}^3$ passing through the origin and the point (a, b, c)

(d) $\{(x, y) \mid 9x^2 + 16y^2 = 4\}$

(a) Let $\vec{c}(t) = (t, e^t)$ for $t \in \mathbb{R}$ (or $-\infty < t < \infty$)

(b) $y^2 = 1 - 4x^2$, so $y = \pm\sqrt{1-4x^2}$. Can't neatly describe all the points with one $\vec{c}(t)$.

$\therefore$ Try polar coordinates.

Let $x = \frac{\cos\theta}{2}$, $y = \sin\theta$, $0 \le \theta < 2\pi$

$\therefore 4x^2 + y^2 = 4\frac{\cos^2\theta}{4} + \sin^2\theta = 1.$

$\therefore \vec{C}(\theta) = \left(\frac{\cos\theta}{2}, \sin\theta\right)$, $0 \le \theta < 2\pi$

(c) $(0,0,0) + t[(a,b,c) - (0,0,0)] = t(a,b,c).$

$\therefore \vec{C}(t) = t(a,b,c)$, $t \in \mathbb{R}$

(d) $9x^2 + 16y^2 = 4 \iff \frac{9}{4}x^2 + 4y^2 = 1$

$\therefore x = \frac{2}{3}\cos\theta \qquad y = \frac{1}{2}\sin\theta$

$\therefore \frac{9}{4}\left(\frac{2}{3}\cos\theta\right)^2 + 4\left(\frac{1}{2}\sin\theta\right)^2 = \cos^2\theta + \sin^2\theta = 1$

$\therefore \vec{C}(\theta) = \left(\frac{2}{3}\cos\theta, \frac{1}{2}\sin\theta\right)$, $0 \le \theta < 2\pi$

26.

26. Let $\mathbf{c}(t)$ be a path, $\mathbf{v}(t)$ its velocity, and $\mathbf{a}(t)$ the acceleration. Suppose $\mathbf{F}$ is a C^1 mapping of $\mathbb{R}^3$ to $\mathbb{R}^3$, $m > 0$, and $\mathbf{F}(\mathbf{c}(t)) = m\mathbf{a}(t)$ (Newton's second law).

Prove that

$$\frac{d}{dt}[m\mathbf{c}(t) \times \mathbf{v}(t)] = \mathbf{c}(t) \times \mathbf{F}(\mathbf{c}(t))$$

(i.e., "rate of change of angular momentum = torque"). What can you conclude if $\mathbf{F}(\mathbf{c}(t))$ is parallel to $\mathbf{c}(t)$? Is this the case in planetary motion?

(a) $\frac{d}{dt}\left[m\vec{c}(t)\times\vec{v}(t)\right] = m\frac{d}{dt}\vec{c}(t)\times\vec{v}(t) + m\vec{c}(t)\times\frac{d}{dt}\vec{v}(t)$

$= m\,\vec{v}(t)\times\vec{v}(t) + m\,\vec{c}(t)\times\vec{a}(t)$

$= 0 + \vec{c}(t)\times m\,\vec{a}(t)$ since $\vec{v}\times\vec{v} = 0$

$= \vec{c}(t)\times F(\vec{c}(t))$

(b) if $\vec{F}(\vec{c}(t))$ is parallel to $\vec{c}(t)$, then

$\vec{c}(t)\times F(\vec{c}(t)) = 0$, so

$\frac{d}{dt}\left[m\vec{c}(t)\times\vec{v}(t)\right] = 0$, $\therefore$ $m\vec{c}(t)\times\vec{v}(t) =$ constant

(c) $\vec{F}(\vec{c}(t))$ is parallel to $\vec{c}(t)$ in planetary motion as gravity is a radial force.

$\therefore$ angular momentum of a planet is constant over time.

27. **27.** Continue the investigations in Exercise 26 to prove Kepler's law that a planet moving under the influence of gravity about the sun does so in a fixed plane.

From #26, $m\vec{c}(t)\times\vec{v}(t)$ = a constant vector, one that doesn't change size or direction over time.

Let $\vec{L} = m\vec{c}(t) \times \vec{v}(t)$.

Note that $\vec{L} \perp \vec{c}(t)$ for all t, as

$(\vec{a} \times \vec{b}) \perp \vec{a}$ and $(\vec{a} \times \vec{b}) \perp \vec{b}$ for any $\vec{a}, \vec{b} \neq 0$.

$\therefore \vec{L}$ is the normal to a plane, and since

$\vec{c}(t) \perp \vec{L}$, $\vec{c}(t)$ stays in the plane for all t.

4.2 Arc Length

Note Title 8/23/2016

Find the arc length of the given curve on the specified interval in Exercises 1 to 6.[4]

[4]Several of these problems make use of the formula

$$\int \sqrt{x^2 + a^2}\, dx = \tfrac{1}{2}\left[x\sqrt{x^2 + a^2} + a^2 \log\left(x + \sqrt{x^2 + a^2}\right)\right] + C$$

from the table of integrals in the back of the book.

1.

1. $(2\cos t, 2\sin t, t)$, for $0 \le t \le 2\pi$

$$\int_0^{2\pi} \sqrt{(-2\sin t)^2 + (2\cos t)^2 + 1}\, dt$$

$$= \int_0^{2\pi} \sqrt{4(\sin^2 t + \cos^2 t) + 1}\, dt = \int_0^{2\pi} \sqrt{5}\, dt = \sqrt{5}\, t \Big|_0^{2\pi}$$

$$= 2\sqrt{5}\,\pi$$

2.

2. $(1, 3t^2, t^3)$, for $0 \le t \le 1$

$$\int_0^1 \sqrt{0 + (6t)^2 + (3t^2)^2}\, dt = \int_0^1 \sqrt{36t^2 + 9t^4}\, dt$$

$$= \int_0^1 3t\sqrt{t^2 + 4}\, dt \quad \text{since } \sqrt{t^2} = t \text{ for } 0 \le t \le 1$$

$$= \frac{3}{2}\int_0^1 2t\sqrt{t^2 + 4}\, dt = \frac{3}{2}\left[\frac{2}{3}(t^2 + 4)^{3/2}\right]\Big|_0^1$$

$$= \frac{3}{2}\left[\frac{2}{3}(t^2+4)^{3/2}\right]\Big|_0^1 \qquad \text{since } \frac{d}{dt}(t^2+4) = 2t$$

$$= (5)^{3/2} - (4)^{3/2} = \underline{5^{3/2} - 8} \simeq 3.18$$

3. **3.** $(\sin 3t,\ \cos 3t,\ 2t^{3/2})$, for $0 \le t \le 1$

$$\int_0^1 \sqrt{(3\cos 3t)^2 + (-3\sin 3t)^2 + (3t^{\frac{1}{2}})^2}\ dt$$

$$= \int_0^1 \sqrt{9(\cos^2 3t + \sin^2 3t) + 9t}\ dt$$

$$= 3\int_0^1 \sqrt{t+1}\ dt = 3\left[\frac{2}{3}(t+1)^{3/2}\right]\Big|_0^1$$

$$= 2\left[2^{3/2} - 1^{3/2}\right] = 2(2\sqrt{2} - 1) = \underline{4\sqrt{2} - 2}$$

4. **4.** $\left(t+1,\ \frac{2\sqrt{2}}{3}t^{3/2} + 7,\ \frac{1}{2}t^2\right)$, for $1 \le t \le 2$

$$\int_1^2 \sqrt{1 + (\sqrt{2}t^{\frac{1}{2}})^2 + t^2}\ dt = \int_1^2 \sqrt{t^2 + 2t + 1}\ dt$$

$$\int_1^2 \sqrt{(t+1)^2}\,dt = \int_1^2 (t+1)\,dt \quad \text{since } \sqrt{(t+1)^2} = t+1 \text{ for } 1 \le t \le 2$$

$$= \frac{1}{2}(t+1)^2 \Big|_1^2 = \frac{9}{2} - \frac{4}{2} = \underline{\frac{5}{2}}$$

5. **5.** (t, t, t^2), for $1 \le t \le 2$

$$\int_1^2 \sqrt{1 + 1 + (2t)^2}\,dt = \int_1^2 \sqrt{2 + 4t^2}\,dt$$

$$= 2\int_1^2 \sqrt{t^2 + \frac{1}{2}}\,dt \quad \left(\text{using } \tfrac{1}{2}\left[x\sqrt{x^2+a^2} + a^2 \log\left(x + \sqrt{x^2+a^2}\right)\right] + C\right)$$

$$= \left[t\sqrt{t^2 + \frac{1}{2}} + \frac{1}{2}\log\left(t + \sqrt{t^2 + \frac{1}{2}}\right)\right]_1^2$$

$$= 2\sqrt{4 + \frac{1}{2}} + \frac{1}{2}\log\left(2 + \sqrt{4 + \frac{1}{2}}\right) - \left[\sqrt{1 + \frac{1}{2}} + \frac{1}{2}\log\left(1 + \sqrt{1 + \frac{1}{2}}\right)\right]$$

$$= 2\frac{3}{\sqrt{2}} + \frac{1}{2}\log\left(2 + \frac{3}{\sqrt{2}}\right) - \sqrt{\frac{3}{2}} - \frac{1}{2}\log\left(1 + \sqrt{\frac{3}{2}}\right)$$

$$= \frac{6}{\sqrt{2}} - \frac{\sqrt{3}}{\sqrt{2}} + \frac{1}{2}\log\left[\frac{2 + \frac{3}{\sqrt{2}}}{1 + \frac{\sqrt{3}}{\sqrt{2}}}\right]$$

$$= \frac{6-\sqrt{3}}{\sqrt{2}} + \frac{1}{2}\log\left(\frac{2\sqrt{2}+3}{\sqrt{2}+\sqrt{3}}\right)$$

6. **6.** $(t, t\sin t, t\cos t)$, for $0 \le t \le \pi$

$$\vec{c}\,'(t) = (1, \sin t + t\cos t, \cos t - t\sin t)$$

$$\therefore \int_0^{\pi} \sqrt{1 + (\sin t + t\cos t)^2 + (\cos t - t\sin t)^2}\, dt$$

$$= \int_0^{\pi} \sqrt{1 + (\sin^2 t + 2t\sin t\cos t + t^2\cos^2 t) + (\cos^2 t - 2t\sin t\cos t + t^2\sin^2 t)}\, dt$$

$$= \int_0^{\pi} \sqrt{1 + (\sin^2 t + \cos^2 t) + t^2(\cos^2 t + \sin^2 t)}\, dt$$

$$= \int_0^{\pi} \sqrt{t^2 + 2}\, dt$$

using: $\int \sqrt{x^2+a^2}\, dx = \frac{1}{2}\left[x\sqrt{x^2+a^2} + a^2\log\left(x+\sqrt{x^2+a^2}\right)\right] + C$

$$= \frac{1}{2}\left[t\sqrt{t^2+2} + 2\log\left(t+\sqrt{t^2+2}\right)\right]\Bigg|_0^{\pi}$$

$$= \frac{1}{2}\left[\pi\sqrt{\pi^2+2} + 2\log\left(\pi+\sqrt{\pi^2+2}\right) - 2\log\sqrt{2}\right]$$

7. **7.** Find the arc length of $\mathbf{c}(t) = (t, |t|)$ for $-1 \le t \le 1$.

Break up into $-1 \le t \le 0$ and $0 \le t \le 1$

$-1 \le t \le 0$: $\vec{c}(t) = (t, -t)$ $\therefore \vec{c}'(t) = (1, -1)$

$0 \le t \le 1$: $\vec{c}(t) = (t, t)$ $\therefore \vec{c}'(t) = (1, 1)$

$$\therefore \int_{-1}^{0} \sqrt{1^2 + (-1)^2}\, dt + \int_{0}^{1} \sqrt{1^2 + 1^2}\, dt$$

$$= \sqrt{2}\, t \Big|_{-1}^{0} + \sqrt{2}\, t \Big|_{0}^{1} = \sqrt{2} + \sqrt{2} = \underline{2\sqrt{2}}$$

8. **8.** Recall from Section 2.4 that a rolling circle of radius R traces out a cycloid, which can be parametrized by $\mathbf{c}(t) = (Rt - R\sin t,\ R - R\cos t)$. One arch of the cycloid is completed from $t = 0$ to $t = 2\pi$. Show that the length of this arch is always 4 times the diameter of the rolling circle.

$$\vec{c}'(t) = (R - R\cos t,\ R\sin t)$$

$$\therefore \int_{0}^{2\pi} \sqrt{(R - R\cos t)^2 + (R\sin t)^2}\, dt$$

$$= \int_{0}^{2\pi} R\sqrt{(1 - 2\cos t + \cos^2 t + \sin^2 t)}\, dt$$

$$= R\int_0^{2\pi} \sqrt{2-2\cos t}\, dt = \sqrt{2}R\int_0^{2\pi} \sqrt{1-\cos t}\, dt$$

$$= \sqrt{2}R\int_0^{2\pi} \sqrt{2}\sqrt{\frac{1-\cos t}{2}}\, dt = 2R\int_0^{2\pi} \sqrt{\frac{1-\cos t}{2}}\, dt$$

$$= 2R\int_0^{2\pi} \sin\frac{t}{2}\, dt \qquad \text{since } \sin\frac{\theta}{2} = \sqrt{\frac{1-\cos\theta}{2}} \text{ for } 0 \le \theta \le 2\pi$$

$$= 2R\left[-2\cos\frac{t}{2}\right]\Big|_0^{2\pi} = 2R\left[2-(-2)\right] = 8R$$

∴ Length of one arch = $8R$

Diameter of rolling circle = $2R$

∴ Length of one arch = 4 × Diameter of rolling circle

9.

9. Let C be the line segment connecting the point $\mathbf{p} = (1, 2, 0)$ to the point $\mathbf{q} = (0, 1, -1)$.

(a) Find a curve $\mathbf{c}(t)\colon [a, b] \to \mathbb{R}^3$ that traces out C.

(b) Find the arc length of $\mathbf{c}(t)$.

(c) Find $||\mathbf{p} - \mathbf{q}||$.

(a) $\vec{c}(t) = \vec{p} + t(\vec{q} - \vec{p})$, $0 \le t \le 1$

$= (1,2,0) + t[(0,1,-1) - (1,2,0)]$

$= (1,2,0) + t(-1,-1,-1) = (1-t, 2-t, -t)$, $0 \le t \le 1$

(5) $\vec{c}\,'(t) = (-1,-1,-1)$

$$\therefore \int_0^1 \sqrt{(-1)^2+(-1)^2+(-1)^2}\,dt = \sqrt{3}\int_0^1 dt = \underline{\sqrt{3}}$$

(c) $\vec{p}-\vec{q} = (1,2,0)-(0,1,-1) = (1,1,1)$

$$\therefore \|\vec{p}-\vec{q}\| = \sqrt{1^2+1^2+1^2} = \underline{\sqrt{3}}$$

10. **10.** Compute the length of the curve $\mathbf{c}(t) = (\log(\sqrt{t}),\ \sqrt{3}t,\ \frac{3}{2}t^2)$ for $1 \le t \le 2$.

$$\vec{c}\,'(t) = \left(\frac{1}{\sqrt{t}}\cdot\frac{1}{2}\frac{1}{\sqrt{t}},\ \sqrt{3},\ 3t\right) = \left(\frac{1}{2t},\ \sqrt{3},\ 3t\right)$$

$$\therefore \int_1^2 \sqrt{\left(\frac{1}{2t}\right)^2+(\sqrt{3})^2+(3t)^2}\,dt$$

$$= \int_1^2 \sqrt{\frac{1}{4t^2}+3+9t^2}\,dt = \int_1^2 \sqrt{\frac{1+12t^2+36t^4}{4t^2}}\,dt$$

$$= \int_1^2 \sqrt{\left(\frac{6t^2+1}{2t}\right)^2}\,dt = \int_1^2 \frac{6t^2+1}{2t}\,dt$$

Since $\frac{6t^2+1}{2t} > 0$ for $1 \le t \le 2$

$$= \int_1^2 3t\,dt + \int_1^2 \frac{1}{2t}\,dt$$

$$= \frac{3}{2}t^2\Big|_1^2 + \frac{1}{2}\log t\Big|_1^2$$

$$= \left(6 - \frac{3}{2}\right) + \frac{1}{2}(\log 2 - 0)$$

$$= \frac{9}{2} + \frac{1}{2}\log 2$$

11. **11.** Find the length of the path $\mathbf{c}(t)$, defined by $\mathbf{c}(t) = (2\cos t, 2\sin t, t)$, if $0 \le t \le 2\pi$ and $\mathbf{c}(t) = (2, t - 2\pi, t)$, if $2\pi \le t \le 4\pi$.

Note: The curves are connected at $t = 2\pi$

$$\vec{c}_1'(t) = (-2\sin t, 2\cos t, 1)$$

$$\vec{c}_2'(t) = (0, 1, 1)$$

$$\therefore \int_0^{2\pi} \sqrt{(-2\sin t)^2 + (2\cos t)^2 + 1}\, dt + \int_{2\pi}^{4\pi} \sqrt{(0^2 + 1^2 + 1^2)}\, dt$$

$$= \int_0^{2\pi} \sqrt{4\sin^2 + 4\cos^2 + 1}\, dt + \int_{2\pi}^{4\pi} \sqrt{2}\, dt$$

$$= \int_0^{2\pi} \sqrt{5}\, dt + \sqrt{2}t\Big|_{2\pi}^{4\pi}$$

$$= \sqrt{5}t\Big|_0^{2\pi} + (4\sqrt{2}\pi - 2\sqrt{2}\pi)$$

$= 2\sqrt{5}\pi + 2\sqrt{2}\pi = 2\pi(\sqrt{5} + \sqrt{2})$

12. **12.** Let $\mathbf{c}$ be the path $\mathbf{c}(t) = (t, t\sin t, t\cos t)$. Find the arc length of $\mathbf{c}$ between the two points $(0, 0, 0)$ and $(\pi, 0, -\pi)$.

$(0,0,0)$ corresponds to $t=0$, $(\pi,0,-\pi)$ to $t=\pi$

$\vec{c}'(t) = (1, \sin t + t\cos t, \cos t - t\sin t)$

$$\therefore \int_0^{\pi} \sqrt{1^2 + (\sin t + t\cos t)^2 + (\cos t - t\sin t)^2}\, dt$$

$$= \int_0^{\pi} \sqrt{1 + (\sin^2 t + 2t\sin t\cos t + t^2\cos^2 t) + (\cos^2 t - 2t\sin t\cos t + t^2\sin^2 t}\, dt$$

$$= \int_0^{\pi} \sqrt{1 + \sin^2 t + \cos^2 t + t^2(\cos^2 t + \sin^2 t)}\, dt$$

$$= \int_0^{\pi} \sqrt{t^2 + 2}\, dt$$

using: $\int \sqrt{x^2 + a^2}\, dx = \frac{1}{2}\left[x\sqrt{x^2 + a^2} + a^2 \log\left(x + \sqrt{x^2 + a^2}\right)\right] + C$

$$= \frac{1}{2}\left[t\sqrt{t^2+2} + 2\log\left(t + \sqrt{t^2+2}\right)\right]\Big|_0^{\pi}$$

$$= \frac{1}{2}\left[\pi\sqrt{\pi^2+2} + 2\log\left(\pi+\sqrt{\pi^2+2}\right) - 2\log\sqrt{2}\right]$$

Note: same as problem # 6

13. **13.** Let $\mathbf{c}$ be the path $\mathbf{c}(t) = (2t, t^2, \log t)$, defined for $t > 0$. Find the arc length of $\mathbf{c}$ between the points $(2, 1, 0)$ and $(4, 4, \log 2)$.

$(2,1,0)$ corresponds to $t=1$
$(4,4,\log 2)$ corresponds to $t=2$

$\vec{c}\,'(t) = \left(2, 2t, \frac{1}{t}\right)$.

$$\therefore \int_1^2 \sqrt{2^2 + (2t)^2 + \left(\frac{1}{t}\right)^2}\, dt$$

$$= \int_1^2 \sqrt{\frac{4t^2 + 4t^4 + 1}{t^2}}\, dt = \int_1^2 \frac{\sqrt{(2t^2+1)^2}}{t}\, dt$$

$$= \int_1^2 \frac{2t^2+1}{t}\, dt = \int_1^2 2t\, dt + \int_1^2 \frac{1}{t}\, dt$$

$$= t^2\Big|_1^2 + \log t\Big|_1^2 = (4-1) + (\log 2 - \log 1)$$

$$= \underline{3 + \log 2}$$

14.

14. The arc-length function $s(t)$ for a given path $\mathbf{c}(t)$, defined by $s(t) = \int_a^t \|\mathbf{c}'(\tau)\|\, d\tau$, represents the distance a particle traversing the trajectory of $\mathbf{c}$ will have traveled by time t if it starts out at time a; that is, it gives the length of $\mathbf{c}$ between $\mathbf{c}(a)$ and $\mathbf{c}(t)$. Find the arc-length functions for the curves $\boldsymbol{\alpha}(t) = (\cosh t,\ \sinh t,\ t)$ and $\boldsymbol{\beta}(t) = (\cos t,\ \sin t,\ t)$, with $a = 0$.

(a) $\vec{\alpha}'(x) = (\sinh(x), \cosh(x), 1)$

$$\therefore s(t) = \int_0^t \sqrt{\sinh^2(x) + \cosh^2(x) + 1}\, dx$$

$$= \int_0^t \sqrt{\sinh^2(x) + \cosh^2(x) + [\cosh^2 x - \sinh^2 x]}\, dx$$

$$= \int_0^t \sqrt{2\cosh^2 x}\, dx = \int_0^t \sqrt{2}\cosh x\, dx$$ using $\cosh x > 0$ for all x.

$$= \sqrt{2}\sinh x \Big|_0^t = \sqrt{2}\sinh t$$

$$\therefore s(t) = \sqrt{2}\sinh t$$

(b) $\vec{\beta}'(x) = (-\sin x, \cos x, 1)$

$$\therefore s(t) = \int_0^t \sqrt{(-\sin x)^2 + \cos^2 x + 1}\, dx$$

$$= \int_0^t \sqrt{2}\, dx = \sqrt{2}\, t$$

$$\therefore\ s(t) = \sqrt{2}\, t$$

15.

15. Let $\mathbf{c}(t)$ be a given path, $a \le t \le b$. Let $s = \alpha(t)$ be a new variable, where α is a strictly increasing C^1 function given on $[a, b]$. For each s in $[\alpha(a), \alpha(b)]$ there is a unique t with $\alpha(t) = s$. Define the function $\mathbf{d}\colon [\alpha(a), \alpha(b)] \to \mathbb{R}^3$ by $\mathbf{d}(s) = \mathbf{c}(t)$.

(a) Argue that the image curves of $\mathbf{c}$ and $\mathbf{d}$ are the same.

(a) For all $t \in [a,b]$, $\vec{d}(s) = \vec{d}(\alpha(t)) = \vec{c}(t)$

$\therefore$ On $[a,b]$, for each $t \in [a,b]$, $\vec{d} = \vec{c}$

(b) (b) Show that $\mathbf{c}$ and $\mathbf{d}$ have the same arc length.

Let l = the arc length

$$\therefore\ l = \int_{\alpha(a)}^{\alpha(b)} \|\vec{d}'(s)\|\, ds = \int_a^b \|\vec{d}'(\alpha(t))\,\alpha'(t)\|\, dt$$

$$= \int_a^b \|\vec{c}'(t)\|\, dt$$

since $\vec{c}'(t) = \frac{d}{dt}\vec{d}(\alpha(t))$

$= \vec{d}'(\alpha(t)) \cdot \alpha'(t)$

(c) Let $s = \alpha(t) = \int_a^t \|\mathbf{c}'(\tau)\|\, d\tau$. Define $\mathbf{d}$ as above by $\mathbf{d}(s) = \mathbf{c}(t)$. Show that

$$\left\| \frac{d}{ds}\mathbf{d}(s) \right\| = 1.$$

The path $s \mapsto \mathbf{d}(s)$ is said to be an ***arc-length reparametrization*** of $\mathbf{c}$ (see also Exercise 17).

(C)

$$\frac{d}{dt}\vec{d}(s) = \frac{d}{ds}\vec{d}(s) \cdot \frac{d}{dt}s(t) = \frac{d}{ds}\vec{d}(s) \cdot \alpha'(t)$$

$$\therefore \frac{\frac{d}{dt}\vec{d}(s)}{\alpha'(t)} = \frac{d}{ds}\vec{d}(s)$$

But $\vec{d}(s) = \vec{c}(t)$. $\therefore \frac{d}{dt}\vec{d}(s) = \vec{c}'(t)$

$$\therefore \frac{\vec{c}'(t)}{\alpha'(t)} = \frac{d}{ds}\vec{d}(s) \qquad [1]$$

But $\frac{d}{dt}\alpha(t) = \frac{d}{dt}\int_a^t \|\vec{c}(\tau)\|\, d\tau = \|\vec{c}'(t)\|$

$\therefore$ [1] becomes $\frac{\vec{c}'(t)}{\|\vec{c}'(t)\|} = \frac{d}{ds}\vec{d}(s)$ [2]

$\frac{\vec{c}'(t)}{\|\vec{c}'(t)\|}$ is a unit vector, so $\left\| \frac{\vec{c}'(t)}{\|\vec{c}'(t)\|} \right\| = 1$

$\therefore$ From [2], $\underline{\left\| \frac{d}{ds}\vec{d}(s) \right\| = 1}$

16.

16. Let $\mathbf{c}\colon [a, b] \to \mathbb{R}^3$ be an infinitely differentiable path (derivatives of all orders exist). Assume $\mathbf{c}'(t) \neq \mathbf{0}$ for any t. The vector $\mathbf{c}'(t)/\|\mathbf{c}'(t)\| = \mathbf{T}(t)$ is tangent to $\mathbf{c}$ at $\mathbf{c}(t)$, and, because $\|\mathbf{T}(t)\| = 1$, $\mathbf{T}$ is called the ***unit tangent*** to $\mathbf{c}$.

(a) Show that $\mathbf{T}'(t) \cdot \mathbf{T}(t) = 0$. [HINT: Differentiate $\mathbf{T}(t) \cdot \mathbf{T}(t) = 1$.]

(b) Write down a formula for $\mathbf{T}'(t)$ in terms of $\mathbf{c}$.

(a) Since $\|\vec{T}(t)\| = 1$, then $\|\vec{T}(t)\|^2 = 1$,

so $\vec{T}(t) \cdot \vec{T}(t) = 1$.

$$\therefore \frac{d}{dt}\left\{\vec{T}(t) \cdot \vec{T}(t)\right\} = 0$$

$$\therefore \vec{T}'(t) \cdot \vec{T}(t) + \vec{T}(t) \cdot \vec{T}'(t) = 2\vec{T}'(t) \cdot \vec{T}(t) = 0$$

$$\therefore \vec{T}'(t) \cdot \vec{T}(t) = 0$$

(b)
$$\vec{T}'(t) = \frac{d}{dt}\left[\frac{\vec{c}'(t)}{\|\vec{c}'(t)\|}\right]$$

$$= \frac{\|\vec{c}'(t)\|\,\vec{c}''(t) - \vec{c}'(t) \cdot \frac{d}{dt}\left[\sqrt{\vec{c}'(t) \cdot \vec{c}'(t)}\right]}{\|\vec{c}'(t)\|^2}$$

$$= \frac{\|\vec{c}'(t)\|\,\vec{c}''(t) - \vec{c}'(t)\left[\frac{1}{2}\,\frac{2[\vec{c}'(t) \cdot \vec{c}''(t)]}{\|\vec{c}'(t)\|}\right]}{\|\vec{c}'(t)\|^2}$$

$$= \frac{\|\vec{c}'(t)\|^2\,\vec{c}''(t) - [\vec{c}'(t) \cdot \vec{c}''(t)]\,\vec{c}'(t)}{\|\vec{c}'(t)\|^3}$$

$$\therefore \vec{T}'(t) = \frac{1}{\|\vec{c}'(t)\|}\left[\vec{c}''(t) - \frac{\vec{c}'(t)\cdot\vec{c}''(t)}{\|\vec{c}'(t)\|^2}\vec{c}'(t)\right] \quad [1]$$

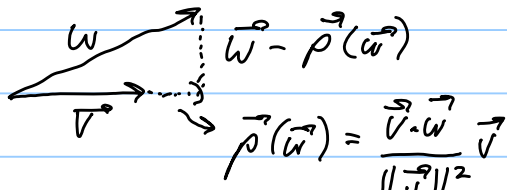
Note: $\frac{\vec{c}'(t)\cdot\vec{c}''(t)}{\|\vec{c}'(t)\|^2}\vec{c}'(t) = \frac{\vec{v}\cdot\vec{w}}{\|\vec{v}\|^2}\vec{v}$, The projection of $\vec{w}$ onto $\vec{v}$.

$$\therefore \vec{T}'(t) = \frac{1}{\|\vec{c}'(t)\|}\left[\vec{c}''(t) - \text{projection of } \vec{c}''(t) \text{ onto } \vec{c}'(t)\right]$$

Note: $\vec{w} - \frac{\vec{v}\cdot\vec{w}}{\|\vec{v}\|^2}\vec{v}$ is $\perp$ to $\vec{v}$, since

$$\vec{v}\cdot\left(\vec{w} - \frac{\vec{v}\cdot\vec{w}}{\|\vec{v}\|^2}\vec{v}\right) = \vec{v}\cdot\vec{w} - \frac{(\vec{v}\cdot\vec{w})(\vec{v}\cdot\vec{v})}{\|\vec{v}\|^2}$$

$$= \vec{v}\cdot\vec{w} - \vec{v}\cdot\vec{w} = 0$$



$$\therefore \|\vec{v}\times\vec{w}\| = \|\vec{v}\|\,\|\vec{w}\|\sin\theta$$

$$\therefore \|\vec{v}\times\vec{w}\| = \|\vec{v}\|\,\|\vec{w} - \vec{p}(\vec{w})\|$$

$$\therefore \frac{\|\vec{v}\times\vec{w}\|}{\|\vec{v}\|^2} = \frac{1}{\|\vec{v}\|}\|\vec{w} - \vec{p}(\vec{w})\| \quad [2]$$

Letting $\vec{v} = \vec{c}'(t)$, $\vec{w} = \vec{c}''(t)$, [1], [2] become

$$\|\vec{T}'(t)\| = \frac{\|\vec{c}'(t)\times\vec{c}''(t)\|}{\|\vec{c}'(t)\|^2} \quad [3]$$

17.

17. (a) A path $\mathbf{c}(s)$ is said to be ***parametrized by arc length*** or, what is the same thing, to have ***unit speed*** if $\|\mathbf{c}'(s)\| = 1$. For a path parametrized by arc length on $[a, b]$, show that $l(\mathbf{c}) = b - a$.

(b) The ***curvature*** at a point $\mathbf{c}(s)$ on a path is defined by $k = \|\mathbf{T}'(s)\|$ when the path is parametrized by arc length. Show that $k = \|\mathbf{c}''(s)\|$.

(c) If $\mathbf{c}$ is given in terms of some other parameter t and $\mathbf{c}'(t)$ is never $\mathbf{0}$, show that $k = \|\mathbf{c}'(t) \times \mathbf{c}''(t)\| / \|\mathbf{c}'(t)\|^3$.

(d) Calculate the curvature of the helix $\mathbf{c}(t) = (1/\sqrt{2})(\cos t,\ \sin t,\ t)$. (This $\mathbf{c}$ is a scalar multiple of the right-circular helix.)

(a) $$\int_a^b \|\vec{c}'(s)\|\, ds = \int_a^b 1\, ds = \underline{b-a}$$

(b) Not mentioned, $\vec{T}$ is defined as the unit tangent vector. $\therefore \vec{T}(t) = \dfrac{\vec{c}'(t)}{\|\vec{c}'(t)\|}$, $\vec{T}(s) = \dfrac{\vec{c}'(s)}{\|\vec{c}'(s)\|}$

But $\|\vec{c}'(s)\| = 1$, from (a)

$\therefore \vec{T}(s) = \vec{c}'(s)$ $\therefore \dfrac{d}{ds}\vec{T}(s) = \dfrac{d}{ds}\vec{c}'(s) = \vec{c}''(s)$

$\therefore K = \|T'(s)\| = \|\vec{c}''(s)\|$

(c) From 16(b) above, using [3],

$$\frac{\|\vec{T}'(t)\|}{\|\vec{c}'(t)\|} = \frac{\|\vec{c}'(t) \times \vec{c}''(t)\|}{\|\vec{c}'(t)\|^3}$$

But $\|\vec{T}'(t)\| = \left\|\dfrac{d}{ds}\vec{T}(s) \cdot \dfrac{ds}{dt}\right\|$

$$= \| \vec{T}'(s) \| \, \| \vec{c}'(t) \| = k \, \| \vec{c}'(t) \|$$

$$\therefore k = \frac{\| \vec{T}'(t) \|}{\| \vec{c}'(t) \|} = \frac{\| \vec{c}'(t) \times \vec{c}''(t) \|}{\| \vec{c}'(t) \|^3}$$

(d) (d) Calculate the curvature of the helix $\mathbf{c}(t) = (1/\sqrt{2})(\cos t,\ \sin t,\ t)$. (This $\mathbf{c}$ is a scalar multiple of the right-circular helix.)

Using formula in (c),

$$\vec{c}'(t) = \frac{1}{\sqrt{2}}(-\sin t, \cos t, 1) \quad \therefore \| \vec{c}'(t) \| = \frac{1}{\sqrt{2}} \sqrt{1+1} = 1$$

$$\vec{c}''(t) = \frac{1}{\sqrt{2}}(-\cos t, -\sin t, 0) \quad \therefore \| \vec{c}''(t) \| = \frac{1}{\sqrt{2}}$$

$$\vec{c}'(t) \times \vec{c}''(t) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\frac{\sin t}{\sqrt{2}} & \frac{\cos t}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{\cos t}{\sqrt{2}} & -\frac{\sin t}{\sqrt{2}} & 0 \end{vmatrix}$$

$$= \left(\frac{\sin t}{2}, -\frac{\cos t}{2}, \frac{\sin^2 t + \cos^2 t}{2} \right) = \frac{1}{2}(\sin t, -\cos t, 1)$$

$$\therefore \| \vec{c}'(t) \times \vec{c}''(t) \| = \frac{1}{2} \sqrt{1+1} = \frac{\sqrt{2}}{2}$$

$$\therefore k = \frac{\frac{\sqrt{2}}{2}}{1} = \frac{\sqrt{2}}{2}$$

18. **18.** Show that any line $\mathbf{l}(t) = \mathbf{x}_0 + t\mathbf{v}$, where $\mathbf{v}$ is a unit vector, has zero curvature.

$$\vec{l}'(t) = \vec{v}, \quad \vec{l}''(t) = \vec{0} \quad \therefore \| \vec{l}'(t) \| = \| \vec{v} \| = 1.$$

From 17(c), $K = \dfrac{\|\vec{l}'(t) \times \vec{l}''(t)\|}{\|\vec{l}'(t)\|^3} = \dfrac{\|\vec{l}'(t) \times \vec{l}''(t)\|}{1}$

$= \|\vec{v} \times \vec{0}\| = \underline{0}$

19. **19.** Consider the parametrization of the unit circle given by $\mathbf{c}(t) = (\cos t, \sin t)$.

(a) Verify that $\mathbf{c}$ is parametrized by arc length.

(b) Show that the curvature k of $\mathbf{c}$ is constant.

(a) $\vec{c}'(t) = [-\sin(t), \cos(t)]$

$\therefore \|\vec{c}'(t)\| = \sqrt{\sin^2 t + \cos^2 t} = 1$

$\therefore$ By definition given in 17(a), $\vec{c}(t)$ is parametrized by arc length.

(b) Write $\vec{c}(t) = (\cos t, \sin t, 0)$, so $\vec{c}(t) \in \mathbb{R}^3$

By 17(c), $K = \dfrac{\|\vec{c}'(t) \times \vec{c}''(t)\|}{\|\vec{c}'(t)\|} = \|\vec{c}'(t) \times \vec{c}''(t)\|$

From (a), $\|\vec{c}'(t)\| = 1$.

$\vec{c}''(t) = [-\cos(t), -\sin(t), 0]$

$\therefore \vec{c}'(t) \times \vec{c}''(t) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\sin t & \cos t & 0 \\ -\cos t & -\sin t & 0 \end{vmatrix}$

$= (0, 0, \sin^2 t + \cos^2 t) = (0, 0, 1)$

$\therefore K = \frac{\|(0,0,1)\|}{1} = 1$

20.

20. If $\mathbf{T}'(t) \neq \mathbf{0}$, it follows from Exercise 16 that $\mathbf{N}(t) = \mathbf{T}'(t)/\|\mathbf{T}'(t)\|$ is normal (i.e., perpendicular) to $\mathbf{T}(t)$; $\mathbf{N}$ is called the ***principal normal vector***. Let a third unit vector that is perpendicular to both $\mathbf{T}$ and $\mathbf{N}$ be defined by $\mathbf{B} = \mathbf{T} \times \mathbf{N}$; $\mathbf{B}$ is called the ***binormal vector***. Together, $\mathbf{T}$, $\mathbf{N}$, and $\mathbf{B}$ form a right-handed system of mutually orthogonal vectors that may be thought of as moving along the path (Figure 4.2.5). Show that

(a) $\frac{d\mathbf{B}}{dt} \cdot \mathbf{B} = 0$.

(b) $\frac{d\mathbf{B}}{dt} \cdot \mathbf{T} = 0$.

(c) $d\mathbf{B}/dt$ is a scalar multiple of $\mathbf{N}$.

(a) $\|\vec{B}\| = \|\vec{T} \times \vec{N}\| = \|\vec{T}\| \|\vec{N}\| \sin\theta$

But $\|\vec{T}\| = 1$, $\|\vec{N}\| = 1$, and $\vec{N} \perp \vec{T} \Rightarrow \sin\theta = 1$.

$\therefore \|\vec{B}\| = 1 \Rightarrow \|\vec{B}\|^2 = 1 \Rightarrow \vec{B} \cdot \vec{B} = 1$

$\Rightarrow \frac{d}{dt}(\vec{B} \cdot \vec{B}) = 0 \Rightarrow 2\frac{d\vec{B}}{dt} \cdot B = 0$

$\Rightarrow \frac{d\vec{B}}{dt} \cdot \vec{B} = 0.$

(b) Since $\vec{B} = \vec{T} \times \vec{N}$, $\vec{B} \perp \vec{T}$. $\therefore \vec{B} \cdot \vec{T} = 0$.

$\therefore \frac{d}{dt}(\vec{B} \cdot \vec{T}) = 0 \Rightarrow \frac{d\vec{B}}{dt} \cdot \vec{T} + \vec{B} \cdot \frac{d\vec{T}}{dt} = 0$ [1]

But $\frac{d\vec{T}}{dt} = \|\vec{T}'\|\vec{N}$ by definition of $\vec{N}$,

and $\vec{B} = \vec{T} \times \vec{N} \Rightarrow \vec{B} \perp \vec{N} \Rightarrow \vec{B} \perp \|\vec{T}'\|\vec{N}$

$\therefore \vec{B} \cdot \frac{d\vec{T}}{dt} = \vec{B} \cdot (\|\vec{T}'\|\vec{N}) = 0.$

$\therefore$ [1] becomes $\frac{d}{dt}(\vec{B} \cdot \vec{T}) = \frac{d\vec{B}}{dt} \cdot \vec{T} + 0 = 0$

$\therefore \frac{d\vec{B}}{dt} \cdot \vec{T} = 0$

(c) (a) + (b) $\Rightarrow \frac{d\vec{B}}{dt} \perp \vec{B}$ and $\frac{d\vec{B}}{dt} \perp \vec{T}$

But $\vec{N} \perp \vec{B}$ and $\vec{N} \perp \vec{T}$.

$\therefore \frac{d\vec{B}}{dt}$ and $\vec{N}$ are parallel.

$\therefore \frac{d\vec{B}}{dt} = s\vec{N}, \; s \in \mathbb{R}$

21.

21. If $\mathbf{c}(s)$ is parametrized by arc length, we use the result of Exercise 20(c) to define a scalar-valued function τ, called the ***torsion***, by

$$\frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}.$$

(a) Show that $\tau = [\mathbf{c}'(s) \times \mathbf{c}''(s)] \cdot \mathbf{c}'''(s)/\|\mathbf{c}''(s)\|^2$.

(b) Show that if $\mathbf{c}$ is given in terms of some other parameter t,

$$\tau = \frac{[\mathbf{c}'(t) \times \mathbf{c}''(t)] \cdot \mathbf{c}'''(t)}{\|\mathbf{c}'(t) \times \mathbf{c}''(t)\|^2}.$$

Compare with Exercise 17(c).

(c) Compute the torsion of the helix $\mathbf{c}(t) = (1/\sqrt{2})(\cos t, \sin t, t)$.

(a) $\vec{c}(s)$ parametrized by arc length $\Rightarrow \|\vec{c}'(s)\| = 1$ by definition given in 17(a)

Note that the definitions of $\vec{T}, \vec{N}, \vec{B}$ are irrelevant to the letter of the parameterization.

i.e., $T(s) = \dfrac{\vec{c}'(s)}{\|\vec{c}'(s)\|}$ or $\vec{T}(x) = \dfrac{\vec{c}'(x)}{\|\vec{c}'(x)\|}$

So, all the properties of problems #16, 17, 20 hold irrespective to the letter of the parameter.

Here, since $\|\vec{c}'(s)\| = 1$, $\vec{T}(s) = \vec{c}'(s)$ and

$\vec{T}'(s) = \vec{c}''(s)$. $\therefore \vec{N} = \dfrac{\vec{T}'}{\|\vec{T}'\|} = \dfrac{\vec{c}''(s)}{\|\vec{c}''(s)\|}$

$\therefore \vec{B} = \vec{T} \times \vec{N} \Rightarrow \vec{B}' = \vec{T}' \times \vec{N} + \vec{T} \times \vec{N}'$

$\therefore \vec{B}'(s) = \vec{T}'(s) \times \vec{N}(s) + \vec{T}(s) \times \vec{N}'(s)$

$$= \vec{c}''(s) \times \frac{\vec{c}''(s)}{\|\vec{c}''(s)\|} + \vec{c}'(s) \times \frac{d}{ds}\left[\frac{\vec{c}''(s)}{\|\vec{c}''(s)\|}\right]$$

$$= 0 + \vec{c}'(s) \times \frac{d}{ds}\left[\frac{\vec{c}''(s)}{\|\vec{c}''(s)\|}\right]$$

as $\vec{c}'' \times \vec{c}'' = 0$

$$\therefore \frac{d\vec{B}(s)}{ds} = \vec{c}'(s) \times \frac{d}{ds}\left[\frac{\vec{c}''(s)}{\|\vec{c}''(s)\|}\right]$$

But $$\frac{d}{ds}\left(\frac{\vec{c}''(s)}{\|\vec{c}''(s)\|}\right) = \frac{\|\vec{c}''(s)\|\vec{c}'''(s) - \vec{c}''(s)\frac{d}{ds}\|\vec{c}''(s)\|}{\|\vec{c}''(s)\|^2}$$

and $$\frac{d}{ds}\|\vec{c}''(s)\| = \frac{d}{ds}\sqrt{\vec{c}''(s)\cdot\vec{c}''(s)}$$

$$= \frac{1}{2}\frac{2[\vec{c}'''(s)\cdot\vec{c}''(s)]}{\sqrt{\vec{c}''(s)\cdot\vec{c}''(s)}} = \frac{\vec{c}'''(s)\cdot\vec{c}''(s)}{\|\vec{c}''(s)\|}$$

$$\therefore \frac{d}{ds}\left(\frac{\vec{c}''(s)}{\|\vec{c}''(s)\|}\right) = \frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} - \left[\frac{\vec{c}'''(s)\cdot\vec{c}''(s)}{\|\vec{c}''(s)\|^3}\right]\vec{c}''(s)$$

$$\therefore \frac{d\vec{B}(s)}{ds} = \vec{c}'(s) \times \left[\frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} - \left(\frac{\vec{c}'''(s)\cdot\vec{c}''(s)}{\|\vec{c}''(s)\|^3}\right)\vec{c}''(s)\right]$$

$$= \vec{T} \times \left[\frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} - \left(\frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} \cdot \frac{\vec{c}''(s)}{\|\vec{c}''(s)\|}\right)\frac{\vec{c}''(s)}{\|\vec{c}''(s)\|}\right]$$

$$= \vec{T} \times \left[\frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} - \left(\frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} \cdot \vec{N} \right) \vec{N} \right]$$

$$\therefore \frac{d\vec{B}(s)}{ds} = \vec{T} \times \frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} - \left(\frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} \cdot \vec{N} \right) \vec{T} \times \vec{N} \quad [1]$$

Since $\frac{d\vec{B}}{ds} = -\tau \vec{N}$, $\frac{d\vec{B}}{ds} \cdot \vec{N} = -\tau \vec{N} \cdot \vec{N} = -\tau$

since $\vec{N} \cdot \vec{N} = 1$.

$$\therefore -\frac{d\vec{B}}{ds} \cdot \vec{N} = \tau$$

$\therefore$ Using [1],

$$-\frac{d\vec{B}}{ds} \cdot \vec{N} = -\left[\vec{T} \times \frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} \right] \cdot \vec{N} + \left(\frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} \cdot \vec{N} \right) (\vec{T} \times \vec{N}) \cdot \vec{N}$$

But $(\vec{T} \times \vec{N}) \cdot \vec{N} = \vec{N} \cdot (\vec{T} \times \vec{N}) = \begin{vmatrix} n_1 & n_2 & n_3 \\ T_1 & T_2 & T_3 \\ n_1 & n_2 & n_3 \end{vmatrix} = 0$

as 2 rows of the determinant are equal.

$$\therefore \tau = -\frac{d\vec{B}}{ds} \cdot \vec{N} = -\left[\vec{T} \times \frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} \right] \cdot \vec{N}$$

$$= -\left[\vec{c}'(s) \times \frac{\vec{c}'''(s)}{\|\vec{c}''(s)\|} \right] \cdot \frac{\vec{c}''(s)}{\|\vec{c}''(s)\|}$$

$$= -\frac{1}{\|\vec{c}''(s)\|^2} \left[\vec{c}'(s) \times \vec{c}'''(s) \right] \cdot \vec{c}''(s)$$

$$\therefore \quad \tau = \frac{1}{\|\vec{c}''(s)\|^2} \left[\vec{c}'(s) \times \vec{c}''(s) \right] \cdot \vec{c}'''(s)$$

Using $-(\vec{a} \times \vec{c}) \cdot \vec{b} = (\vec{a} \times \vec{b}) \cdot \vec{c}$

$$\text{as} \quad - \begin{vmatrix} a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \quad \text{row exchange}$$

(6) (b) Show that if $\mathbf{c}$ is given in terms of some other parameter t,

$$\tau = \frac{[\mathbf{c}'(t) \times \mathbf{c}''(t)] \cdot \mathbf{c}'''(t)}{\|\mathbf{c}'(t) \times \mathbf{c}''(t)\|^2}.$$

Compare with Exercise 17(c).

From $\frac{d\vec{B}(s)}{ds} = -\tau \vec{N}(s)$ and $\therefore$ since $\vec{N}(s) \cdot \vec{N}(s) = 1$,

$$\tau = -\frac{d\vec{B}(s)}{ds} \cdot \vec{N}(s)$$

Let $\vec{c}(t)$ be another parametrization s.t.

$$s = \int_a^s \|\vec{c}'(t)\| \, dt, \text{ assuming } \vec{c}(t) \text{ is } C^1.$$

$\therefore \frac{ds}{dt} = \|\vec{c}'(t)\|$ and $\frac{d\vec{B}}{dt} = \frac{d\vec{B}}{ds} \cdot \frac{ds}{dt}$ by chain rule

$$\therefore \quad \tau = -\frac{d\vec{B}/dt}{\|\vec{c}'(t)\|} \cdot \vec{N}(s(t)) \qquad [0]$$

$\therefore$ Need to find $\vec{B}$ as a function of t.

Note: $\vec{N}$ is defined as $\dfrac{\vec{T}'(x)}{\|\vec{T}'(x)\|}$, irrespective

of the parametrization of $\vec{T}$ (and $\therefore$ of $\vec{c}$).

$\vec{B} = \vec{T} \times \vec{N}$ From 16(6),

$$\vec{T}'(t) = \frac{\|\vec{c}'(t)\|^2 \vec{c}''(t) - [\vec{c}'(t) \cdot \vec{c}''(t)]\vec{c}'(t)}{\|\vec{c}'(t)\|^3}$$

and from [3] of 16(6),

$$\|\vec{T}'(t)\| = \frac{\|\vec{c}'(t) \times \vec{c}''(t)\|}{\|\vec{c}'(t)\|^2}$$

$$\therefore \vec{N}(t) = \frac{\vec{T}'(t)}{\|\vec{T}'(t)\|} = \frac{\|\vec{c}'(t)\|^2 \vec{c}''(t) - [\vec{c}'(t) \cdot \vec{c}''(t)]\vec{c}'(t)}{\|\vec{c}'(t)\|^3} \cdot \frac{\|\vec{c}'(t)\|^2}{\|\vec{c}'(t) \times \vec{c}''(t)\|}$$

$$\frac{\|\vec{c}'(t)\|}{\|\vec{c}'(t) \times \vec{c}''(t)\|}\vec{c}''(t) - \left(\frac{\vec{c}'(t) \cdot \vec{c}''(t)}{\|\vec{c}'(t) \times \vec{c}''(t)\|}\right)\frac{\vec{c}'(t)}{\|\vec{c}'(t)\|} \quad [1]$$

$$\therefore \vec{B}(t) = \vec{T}(t) \times \vec{N}(t) = \frac{\vec{c}'(t)}{\|\vec{c}'(t)\|} \times \vec{N}(t)$$

$$= \frac{\vec{c}'(t)}{\|\vec{c}'(t)\|} \times \left[\frac{\|\vec{c}'(t)\|\vec{c}''(t)}{\|\vec{c}'(t) \times \vec{c}''(t)\|} - \left(\frac{\vec{c}'(t) \cdot \vec{c}''(t)}{\|\vec{c}'(t) \times \vec{c}''(t)\|}\right)\frac{\vec{c}'(t)}{\|\vec{c}'(t)\|}\right]$$

$$\therefore \vec{B}(t) = \frac{\vec{c}'(t) \times \vec{c}''(t)}{\|\vec{c}'(t) \times \vec{c}''(t)\|}, \quad \text{as } \vec{c}'(t) \times \vec{c}'(t) = 0.$$

$$= \vec{c}' \times \frac{\vec{c}''}{\|\vec{c}' \times \vec{c}''\|}$$

$$\therefore \frac{d\vec{B}}{dt} = \frac{\vec{c}'' \times \vec{c}''}{\|\vec{c}' \times \vec{c}''\|} + \vec{c}' \times \frac{d}{dt}\left(\frac{\vec{c}''}{\|\vec{c}' \times \vec{c}''\|}\right)$$

$$= \vec{c}' \times \frac{d}{dt}\left(\frac{\vec{c}''}{\|\vec{c}' \times \vec{c}''\|}\right) \quad \text{as } \vec{c}'' \times \vec{c}'' = 0$$

$$= \vec{c}' \times \left[\frac{\|\vec{c}' \times \vec{c}''\| \vec{c}''' - \vec{c}'' \cdot \frac{d}{dt}(\|\vec{c}' \times \vec{c}''\|}{\|\vec{c}' \times \vec{c}''\|^2}\right]$$

$$= \frac{\vec{c}' \times \vec{c}'''}{\|\vec{c}' \times \vec{c}''\|} - \frac{\vec{c}' \times \vec{c}''}{\|\vec{c}' \times \vec{c}''\|^2} \cdot \frac{d}{dt}\left[(\vec{c}' \times \vec{c}'') \cdot (\vec{c}' \times \vec{c}'')\right]^{\frac{1}{2}}$$

$$= \frac{\vec{c}' \times \vec{c}'''}{\|\vec{c}' \times \vec{c}''\|} - \frac{\vec{c}' \times \vec{c}''}{\|\vec{c}' \times \vec{c}''\|^2}\left[\frac{2(\vec{c}' \times \vec{c}'') \cdot \frac{d}{dt}(\vec{c}' \times \vec{c}'')}{2\|\vec{c}' \times \vec{c}''\|}\right]$$

$$\therefore \frac{d\vec{B}}{dt} = \frac{\vec{c}' \times \vec{c}'''}{\|\vec{c}' \times \vec{c}''\|} - \frac{\vec{c}' \times \vec{c}''}{\|\vec{c}' \times \vec{c}''\|^3}\left[(\vec{c}' \times \vec{c}'') \cdot (\vec{c}' \times \vec{c}''')\right] \quad [2]$$

as $\vec{c}'' \times \vec{c}'' = 0$

From [1],

$$\vec{N}(t) = \frac{\|\vec{c}'(t)\|}{\|\vec{c}'(t) \times \vec{c}''(t)\|} \vec{c}''(t) - \left(\frac{\vec{c}'(t) \cdot \vec{c}''(t)}{\|\vec{c}'(t) \times \vec{c}''(t)\|}\right) \frac{\vec{c}'(t)}{\|\vec{c}'(t)\|}$$

Now for the horrible task of computing $\frac{d\vec{B}(t)}{dt} \cdot \vec{N}(t)$

Using [1], [2], computing the 2 terms of $\vec{N}$ separately,

$$\frac{d\vec{B}}{dt} \cdot \frac{\|\vec{c}'(t)\|}{\|\vec{c}'(t) \times \vec{c}''(t)\|} \vec{c}''(t) =$$

$$\left[\frac{\vec{c}' \times \vec{c}'''}{\|\vec{c}' \times \vec{c}''\|} - \frac{\vec{c}' \times \vec{c}''}{\|\vec{c}' \times \vec{c}''\|^3} \left[(\vec{c}' \times \vec{c}'') \cdot (\vec{c}' \times \vec{c}''') \right] \right] \cdot \frac{\|\vec{c}'\| \vec{c}''}{\|\vec{c}' \times \vec{c}''\|}$$

$$= \frac{\|\vec{c}'\|}{\|\vec{c}' \times \vec{c}''\|^2} (\vec{c}' \times \vec{c}''') \cdot \vec{c}'' - 0, \text{ as } [\vec{c}' \times \vec{c}''] \cdot \vec{c}'' = 0$$

$$= -\frac{\|\vec{c}'\|}{\|\vec{c}' \times \vec{c}''\|^2} (\vec{c}' \times \vec{c}'') \cdot \vec{c}''', \qquad [3]$$

as $(\vec{a} \times \vec{c}) \cdot \vec{b} = -(\vec{a} \times \vec{b}) \cdot \vec{c}$

$$\frac{d\vec{B}}{dt} \cdot -\left(\frac{\vec{c}' \cdot \vec{c}''}{\|\vec{c}' \times \vec{c}''\|} \right) \frac{\vec{c}'}{\|\vec{c}'\|} =$$

$$-\left[\frac{\vec{c}' \times \vec{c}'''}{\|\vec{c}' \times \vec{c}''\|} - \frac{\vec{c}' \times \vec{c}''}{\|\vec{c}' \times \vec{c}''\|^3} \left[(\vec{c}' \times \vec{c}'') \cdot (\vec{c}' \times \vec{c}''') \right] \right] \cdot \left(\frac{\vec{c}' \cdot \vec{c}''}{\|\vec{c}' \times \vec{c}''\|} \right) \frac{\vec{c}'}{\|\vec{c}'\|}$$

$= -0 + 0 = 0$, as $(\vec{c}' \times \vec{c}''') \cdot \vec{c}' = 0$ (first term)
and $(\vec{c}' \times \vec{c}'') \cdot \vec{c}' = 0$ (second term).

$= 0$ [4]

$\therefore$ From [3], [4]: $\frac{d\vec{B}}{dt} \cdot \vec{N} = -\frac{\|\vec{c}'\|}{\|\vec{c}' \times \vec{c}''\|^2} (\vec{c}' \times \vec{c}'') \cdot \vec{c}'''$ [5]

$\therefore$ From [6], [5], $\vec{\tau} = -\dfrac{d\vec{B}(t)/dt}{\|\vec{c}'(t)\|} \cdot \vec{N}(t)$

$$= -\frac{1}{\|\vec{c}'\|}\left[-\frac{\|\vec{c}'\|}{\|\vec{c}' \times \vec{c}''\|^2}(\vec{c}' \times \vec{c}'') \cdot \vec{c}'''\right]$$

$$= \frac{\left[\vec{c}'(t) \times \vec{c}''(t)\right] \cdot \vec{c}'''(t)}{\|\vec{c}'(t) \times \vec{c}''(t)\|^2}$$

(C) (c) Compute the torsion of the helix $\mathbf{c}(t) = (1/\sqrt{2})(\cos t, \sin t, t)$.

$$\vec{c}' = \frac{1}{\sqrt{2}}(-\sin(t), \cos(t), 1)$$

$$\vec{c}'' = \frac{1}{\sqrt{2}}(-\cos(t), -\sin(t), 0)$$

$$\vec{c}''' = \frac{1}{\sqrt{2}}(\sin(t), -\cos(t), 0)$$

$$\vec{c}' \times \vec{c}'' = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -\frac{\sin}{\sqrt{2}} & \frac{\cos}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{\cos}{\sqrt{2}} & -\frac{\sin}{\sqrt{2}} & 0 \end{vmatrix} = \left(0 + \frac{\sin}{2}, -\frac{\cos}{2} - 0, \frac{\sin^2}{2} + \frac{\cos^2}{2}\right)$$

$$= \frac{1}{2}(\sin(t), -\cos(t), 1)$$

$$\therefore \|\vec{c}' \times \vec{c}''\|^2 = \frac{\sin^2}{4} + \frac{\cos^2}{4} + \frac{1}{4} = \frac{1}{2}$$

$$(\vec{c}' \times \vec{c}'') \cdot \vec{c}''' = \frac{1}{2\sqrt{2}}(\sin^2 + \cos^2 + 0) = \frac{\sqrt{2}}{4}$$

From (6) above,

$$\tau = \frac{(\vec{c}' \times \vec{c}'') \cdot \vec{c}'''}{\|\vec{c}' \times \vec{c}''\|^2} = \frac{\frac{\sqrt{2}}{4}}{\frac{1}{2}} = \frac{\sqrt{2}}{2}$$

22.

22. Show that if a path lies in a plane, then the torsion is zero. Do this by demonstrating that **B** is constant and is a normal vector to the plane in which **c** lies. (If the torsion is not zero, it gives a measure of how fast the curve is twisting out of the plane of **T** and **N**.)

Let $\vec{c}(t) = [f(t), g(t), h(t)]$.

If $\vec{c}(t)$ lies in a plane, then one of the components is zero.

$\therefore$ The same component will be zero for $\vec{c}'(t)$, $\vec{c}''(t)$, and $\vec{c}'''(t)$.

From #21(6), $\tau = \dfrac{[\vec{c}'(t) \times \vec{c}''(t)] \cdot \vec{c}'''(t)}{\|\vec{c}'(t) \times \vec{c}''(t)\|^2}$

$$= \frac{1}{\|\vec{c}' \times \vec{c}''\|^2} \begin{vmatrix} f'(t) & g'(t) & h'(t) \\ f''(t) & g''(t) & h''(t) \\ f'''(t) & g'''(t) & h'''(t) \end{vmatrix}$$

The same component being $0 \Rightarrow$ one column of $(\vec{c}' \times \vec{c}'') \cdot \vec{c}'''$ will be 0, so the entire determinant evaluates to 0. $\therefore \tau = 0$

Alternatively, $\vec{B} = \vec{T} \times \vec{N}$. $\therefore \|\vec{B}\| = \|\vec{T}\| \|\vec{N}\| \sin\theta$

By definition, $\|\vec{T}\| = 1$, $\|\vec{N}\| = 1$. And if $\vec{c}$ is in a plane, $\vec{T} = \frac{\vec{c}'}{\|\vec{c}'\|}$ is in that plane, and $\therefore$ so is $\vec{N} = \frac{\vec{T}'}{\|\vec{T}'\|}$. Since $\vec{T} \perp \vec{N}$, $\sin\theta = 1$.

$\therefore \|\vec{B}\| = 1$, so magnitude is constant.

Direction of $\vec{B}$ is always $\perp \vec{T}$ and $\vec{N}$, so $\vec{B}$ is always $\perp$ to plane of $\vec{c}$.

$\therefore \vec{B}$ never changes, so $\frac{d\vec{B}}{ds} = 0 = -\tau\vec{N}$.

Assuming $\vec{N} \neq 0$, then $\tau = 0$.

Note: $\vec{N}$ is only defined when $\vec{T}' \neq 0$, and $\vec{N} = 0 \Rightarrow \vec{T}' = 0$, so torsion not defined when $\vec{N} = 0$

23.

23. (a) Use the results of Exercises 17, 20, and 21 to prove the following ***Frenet formulas*** for a unit-speed curve:

$$\frac{d\mathbf{T}}{ds} = k\mathbf{N}; \quad \frac{d\mathbf{N}}{ds} = -k\mathbf{T} + \tau\mathbf{B}; \quad \frac{d\mathbf{B}}{ds} = -\tau\mathbf{N}.$$

(b) Reexpress the results of part (a) as

$$\frac{d}{ds}\begin{pmatrix}\mathbf{T}\\ \mathbf{N}\\ \mathbf{B}\end{pmatrix} = \boldsymbol{\omega} \times \begin{pmatrix}\mathbf{T}\\ \mathbf{N}\\ \mathbf{B}\end{pmatrix}$$

for a suitable vector $\boldsymbol{\omega}$.

(a) Unit speed $\Rightarrow \|\vec{c}'(s)\| = 1$. $\therefore \vec{T}(s) = \frac{\vec{c}'(s)}{\|\vec{c}'(s)\|} = \vec{c}'(s)$

$k = \| \vec{T}'(s) \|$ by definition

$\therefore \vec{N}(s) = \dfrac{\vec{T}'(s)}{\| \vec{T}'(s) \|} = \dfrac{\vec{T}'(s)}{k}$

$\therefore \vec{T}'(s) = k\vec{N}(s)$ [1]

Since $\vec{T}, \vec{N}, \vec{B}$ are mutually orthogonal and $\vec{B} = \vec{T} \times \vec{N}$ by definition,

Then $\vec{N} \times \vec{B} = \vec{N} \times (\vec{T} \times \vec{N}) = (\vec{N} \cdot \vec{N})\vec{T} - (\vec{N} \cdot \vec{T})\vec{N}$
$= (1)\vec{T} - (0)\vec{N} = \vec{T}$

$\therefore \vec{T} = \vec{N} \times \vec{B}$

Also, $\vec{B} \times \vec{T} = (\vec{T} \times \vec{N}) \times \vec{T} = (\vec{T} \cdot \vec{T})\vec{N} - (\vec{N} \cdot \vec{T})\vec{T}$
$= (1)\vec{N} - (0)\vec{T} = \vec{N}$

$\therefore \vec{N} = \vec{B} \times \vec{T}$

$\therefore \dfrac{d\vec{N}}{ds} = \dfrac{d\vec{B}}{ds} \times \vec{T} + \vec{B} \times \dfrac{d\vec{T}}{ds}$

$= (-\tau \vec{N}) \times \vec{T} + \vec{B} \times (k\vec{N})$

using definition of τ and using [1]

$= -\tau(-\vec{B}) + k(\vec{B} \times \vec{N})$

$= \tau \vec{B} - k\vec{T}$

$$\therefore \frac{d\vec{N}}{ds} = -k\vec{T} + \tau\vec{B} \qquad [2]$$

By definition, $\frac{d\vec{B}}{ds} = -\tau\vec{N}$ [3]

(6) (b) Reexpress the results of part (a) as

$$\frac{d}{ds}\begin{pmatrix}\mathbf{T}\\ \mathbf{N}\\ \mathbf{B}\end{pmatrix} = \omega \times \begin{pmatrix}\mathbf{T}\\ \mathbf{N}\\ \mathbf{B}\end{pmatrix}$$

for a suitable vector ω.

Since $\vec{T}, \vec{N}, \vec{B}$ form a basis, There are scalars

a, b, c s.t. $\vec{\omega} = a\vec{T} + b\vec{N} + c\vec{B}$

$\vec{\omega} \times \begin{pmatrix}\vec{T}\\ \vec{N}\\ \vec{B}\end{pmatrix}$ is interpreted as $\begin{pmatrix}\vec{\omega}\times\vec{T}\\ \vec{\omega}\times\vec{N}\\ \vec{\omega}\times\vec{B}\end{pmatrix}$

Also,

$$\frac{d}{ds}\begin{pmatrix}\vec{T}\\ \vec{N}\\ \vec{B}\end{pmatrix} = \begin{pmatrix}\frac{d\vec{T}}{ds}\\ \frac{d\vec{N}}{ds}\\ \frac{d\vec{B}}{ds}\end{pmatrix} = \begin{pmatrix}k\vec{N}\\ -k\vec{T} + \tau\vec{B}\\ -\tau\vec{N}\end{pmatrix}$$

$$\therefore \vec{\omega}\times\vec{T} = (a\vec{T} + b\vec{N} + c\vec{B}) \times \vec{T}$$

$$= a\vec{T}\times\vec{T} + b\vec{N}\times\vec{T} + c\vec{B}\times\vec{T}$$

$$= \vec{0} + (-b\vec{B}) + (c\vec{N})$$

$\therefore$ For $\vec{\omega} \times \vec{T} = k\vec{N}$, let $b = 0$, $c = k$

$\therefore \vec{\omega} = a\vec{T} + k\vec{B}$

$$\vec{\omega} \times \vec{N} = (a\vec{T} + k\vec{B}) \times \vec{N} = a\vec{T} \times \vec{N} + k\vec{B} \times \vec{N}$$

$$= a\vec{B} + k(-\vec{T})$$

For $\vec{\omega} \times \vec{N} = -k\vec{T} + \tau\vec{B}$, let $a = \tau$

$\therefore \vec{\omega} = \tau\vec{T} + k\vec{B}$

$$\vec{\omega} \times \vec{B} = (\tau\vec{T} + k\vec{B}) \times \vec{B} = \tau\vec{T} \times \vec{B} + k\vec{B} \times \vec{B}$$

$$= \tau(-\vec{N}) + \vec{0}$$

$$= -\tau\vec{N}$$

This is consistent with $\frac{d\vec{B}}{ds} = -\tau\vec{N}$

$\therefore \vec{\omega} = \underline{\tau\vec{T} + k\vec{B}}$

24.

24. In special relativity, the ***proper time*** of a path $\mathbf{c}: [a, b] \to \mathbb{R}^4$ with components given by $\mathbf{c}(\lambda) = (x(\lambda), y(\lambda), z(\lambda), t(\lambda))$ is defined to be the quantity

$$\int_a^b \sqrt{-[x'(\lambda)]^2 - [y'(\lambda)]^2 - [z'(\lambda)]^2 + c^2[t'(\lambda)]^2}\, d\lambda,$$

where c is the velocity of light, a constant. In Figure 4.2.6, show that, using self-explanatory notation, the "twin paradox inequality" holds:

proper time (AB) + proper time (BC) < proper time (AC).

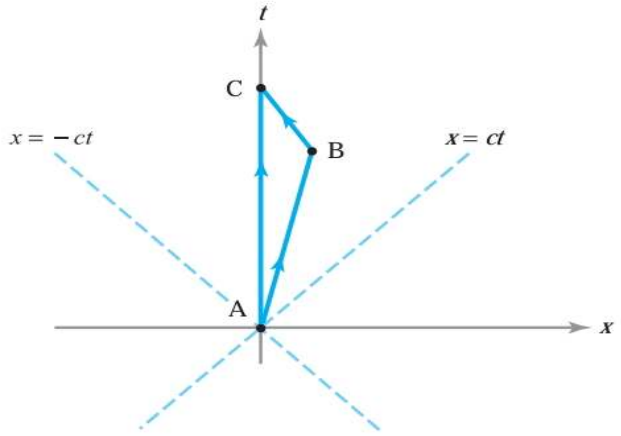


figure **4.2.6** The relativistic triangle inequality.

The "speed" along path AC is zero. i.e., $x' = 0$, $y' = 0$, $z' = 0$.

∴ path will just be

$$\sqrt{-0^2 - 0^2 - 0^2 + c^2t^2} = \sqrt{c^2t^2} = ct$$

For person going from A → B, path is:

$$\sqrt{-(x_1')^2 - (y_1')^2 - (z_1')^2 + c^2t_1^2} < \sqrt{c^2t_1^2} = ct_1$$

and path from B → C is:

$$\sqrt{-(x_2')^2 - (y_2')^2 - (z_2')^2 + c^2t_2^2} < \sqrt{c^2t_2^2} = ct_2$$

∴ proper time (AB) + proper time (BC) $< ct_1 + ct_2$

$= c(t_1 + t_2)$

$= ct$

as $t = t_1 + t_2$ from perspective of person on AC path.

∴ Proper time (AB) + Proper time (BC) < Proper time (AC)

25.

25. The early Greeks knew that a straight line was the shortest possible path between two points. Euclid, in his book *Optics*, stated the "principle of the reflection of light"—that is, light traveling in a plane travels in a straight line, and when it is reflected across a mirror, the angle of incidence equals the angle of reflection.

The Greeks could not have had a proof that straight lines provided the shortest path between two points because they, in the first place, had no definition of the length of a path. They saw this property of straight lines as more or less "obvious."

Using the justification of arc length in this section and the triangle inequality of Section 1.5, argue that if $\mathbf{c}_0$ is the straight-line path $c_0(t) = t\mathrm{P} + (1-t)\mathrm{Q}$ between P and Q in $\mathbb{R}^3$, then

$$L(\mathbf{c}_0) \leq L(\mathbf{c})$$

for any other path $\mathbf{c}$ joining P and Q.

For the first three endpoints along the path, by triangle inequality, $C_0 \to C_2 < C_0 \to C_1 + C_1 \to C_2$.

For the next point, C_3, $C_0 \to C_3 < C_0 \to C_2 + C_2 \to C_3$

$< C_0 \to C_1 + C_1 \to C_2 + C_2 \to C_3$.

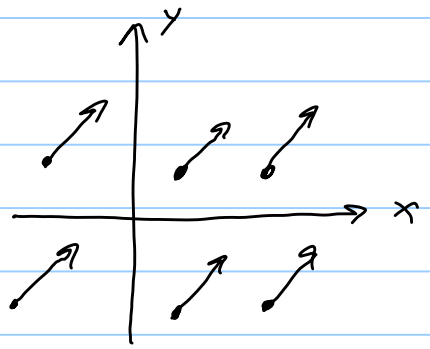
Finally, $C_0 \to C_n < C_0 \to C_1 + C_1 \to C_2 + \ldots + C_{n-1} \to C_n$

4.3 Vector Fields

Note Title 9/6/2016

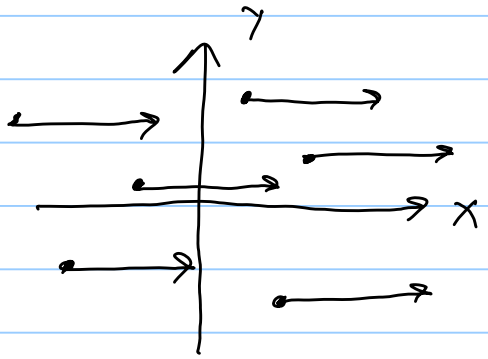
In Exercises 1 to 8, sketch the given vector field or a small multiple of it.

1. **1.** $\mathbf{F}(x, y) = (2, 2)$



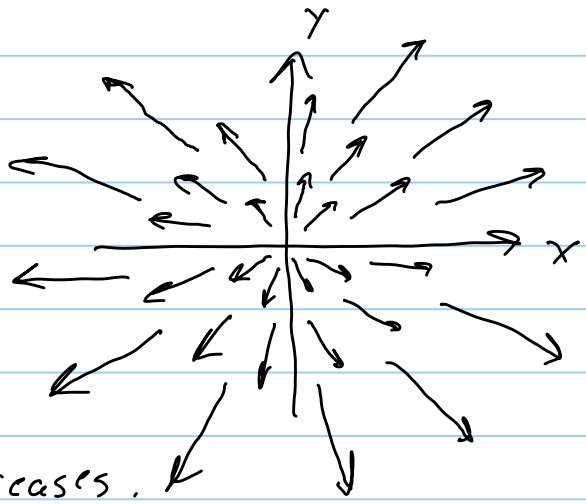
Every point in xy-plane has same vector, length of $\sqrt{2^2+2^2} = 2\sqrt{2}$, angle 45° w/ x-axis

2. **2.** $\mathbf{F}(x, y) = (4, 0)$



Every point in xy-plane has same vector, length of $\sqrt{4^2+0^2} = 4$, 0° angle with x-axis.

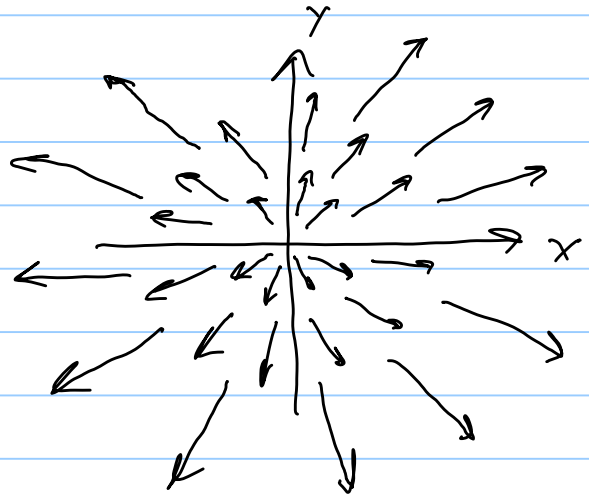
3. **3.** $\mathbf{F}(x, y) = (x, y)$



Each point radially pointed outward, magnitude increases as distance from origin increases.

Each vector is parallel to the line between the origin and the base of the vector.

4. **4.** $\mathbf{F}(x, y) = (-x, y)$



Each point is a reflection, via the y-axis. Magnitude increases with distance from origin. Vectors parallel to line through origin & base of vector. Field looks identical to #3.

5. **5.** $\mathbf{F}(x, y) = (2y, x)$

Consider axis points:

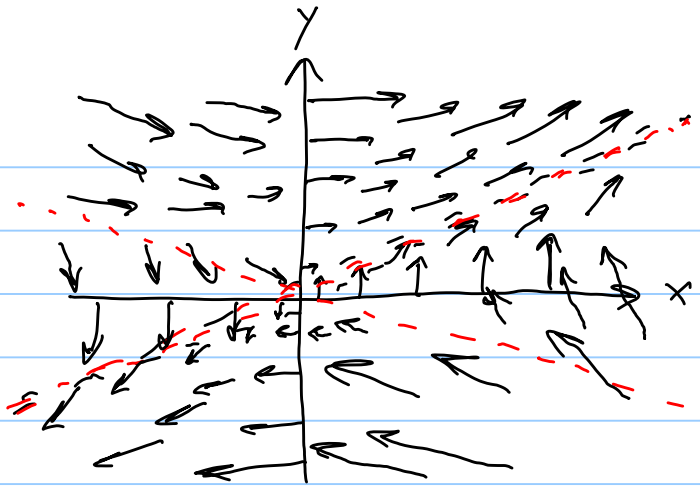
x-axis: $(2,0) \to (0,2)$, $(4,0) \to (0,4)$ i.e., vertical vectors
$(-2,0) \to (0,-2)$, $(-4,0) \to (0,-4)$

y-axis: $(0,2) \to (4,0)$, $(0,4) \to (8,0)$ i.e., horizontal vectors
$(0,-2) \to (-4,0)$, $(0,-4) \to (-8,0)$

Points on $y=x$ line become $(2y, x)$, so angle of less than 45° with x-axis.

Points on $y = \frac{1}{2}x$ line go to $y = x$ orientation.
i.e., $(x, \frac{1}{2}x) \to (x, x)$.

Magnitude increases with distance from origin



4th quadrant points:

$(1,-1) \to (-2,1)$, i.e., point toward 3rd quadrant

$(10,-1) \to (-2,10)$ $\quad (1,-10) \to (-20,1)$

3rd quadrant points: $(-5,5) \to (10,-5)$

$(-1,5) \to (10,-1)$

$(-5,1) \to (2,-10)$

$(-1,1) \to (2,-1)$

The field flows around the lines $y = \pm \frac{1}{2}x$

6. **6.** $\mathbf{F}(x, y) = (y, -2x)$

Look at axis points

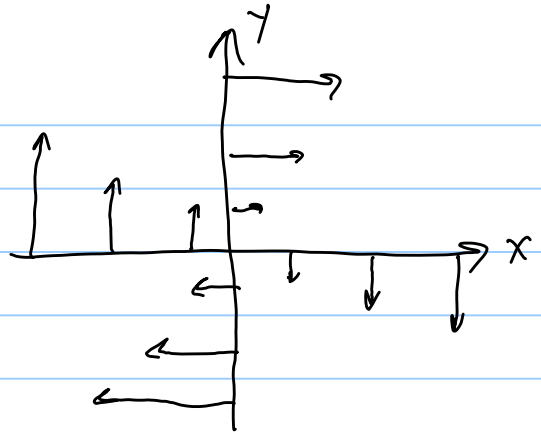
$(1,0) \to (0,-2)$ $\quad (2,0) \to (0,-4)$ $\quad (3,0) \to (0,-6)$

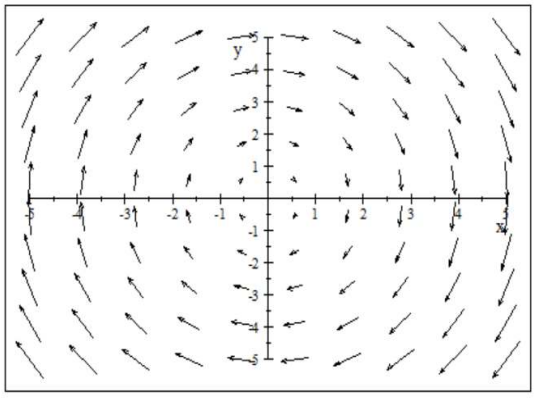
$(0,1) \to (1,0)$ $\quad (0,2) \to (2,0)$ $\quad (0,3) \to (3,0$

$(-1,0) \to (0,2$ $\quad (-2,0) \to (0,4)$ $\quad (-3,0) \to (0,6)$

$(0,-1) \to (-1,0)$ $\quad (0,-2) \to (-2,0)$ $\quad (0,-3) \to (-3,0)$

$\therefore$ Clockwise rotation, magnitude increases with distance from origin





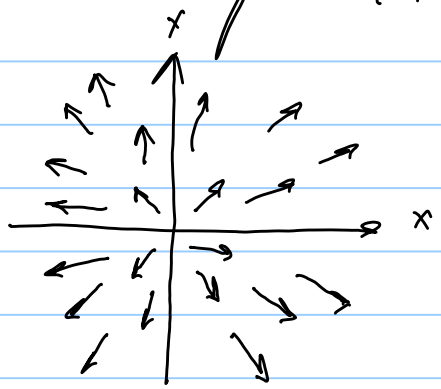
7. **7.** $\mathbf{F}(x, y) = \left(\frac{x}{\sqrt{x^2+y^2}}, \frac{y}{\sqrt{x^2+y^2}} \right)$

Note $\|\vec{F}\| = 1$, so always a unit vector.

This is much like $f(x,y) = (x,y)$, a radial vector field pointing away from origin.

All vectors parallel to line from origin.

$\therefore$

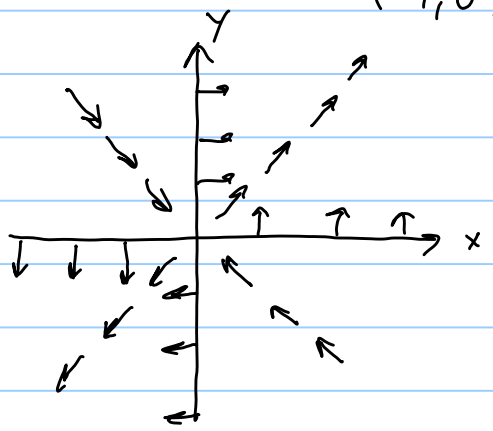


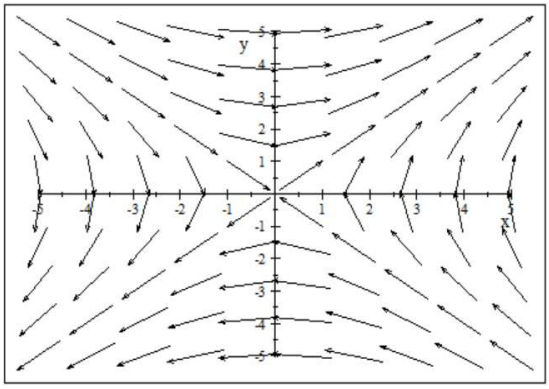
8. **8.** $\mathbf{F}(x, y) = \left(\frac{y}{\sqrt{x^2+y^2}}, \frac{x}{\sqrt{x^2+y^2}} \right)$

Like #7, $\|\vec{F}\| = 1$, so always a unit vector.

Look at axes: $(1,0) \to (0,1)$ $(0,1) \to (1,0)$

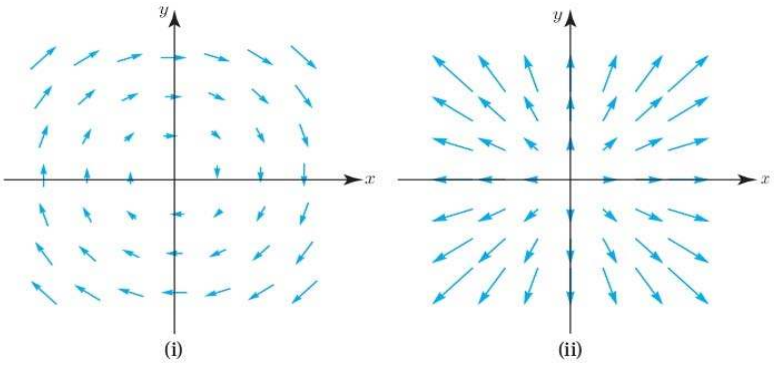
$(-1,0) \to (0,-1)$ $(0,-1) \to (-1,0)$





In the following two exercises, match the given vector field with its pictorial description (see Figures 4.3.11 and 4.3.12).

figure **4.3.11** Exercise 9.



9.

9. (a) $\mathbf{V}(x, y) = x\mathbf{i} + y\mathbf{j}$

(b) $\mathbf{V}(x, y) = y\mathbf{i} - x\mathbf{j}$

(a) a radial outward field, magnitude increasing with distance from origin.

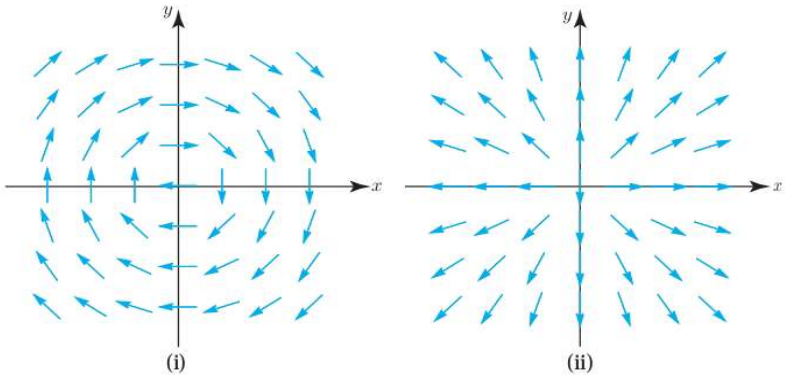
$\therefore$ (ii)

(b) At axes: $(1,0) \to (0,-1)$ $(0,1) \to (1,0)$

$(-1,0) \to (0,1)$ $(0,-1) \to (-1,0)$

$\therefore$ A clockwise rotation. $\therefore$ (i)

figure **4.3.12** Exercise 10.



10. **10.** (a) $\mathbf{V}(x, y) = \dfrac{y}{\sqrt{x^2 + y^2}}\mathbf{i} - \dfrac{x}{\sqrt{x^2 + y^2}}\mathbf{j}$

(b) $\mathbf{V}(x, y) = \dfrac{x}{\sqrt{x^2 + y^2}}\mathbf{i} + \dfrac{y}{\sqrt{x^2 + y^2}}\mathbf{j}$

Where are these vector fields *not* defined? How are these vector fields related to those in Problem 9?

(a) $\|\vec{v}\| = 1$. Like 9(b), a clockwise rotation

Not defined at origin (division by 0).

$\therefore$ (i)

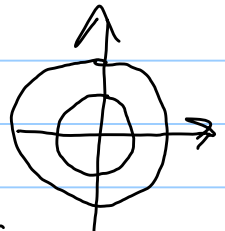
(b) $\|\vec{v}\| = 1$. Like 9(a), a radial outward field.

Not defined at origin (division by zero).

$\therefore$ (ii)

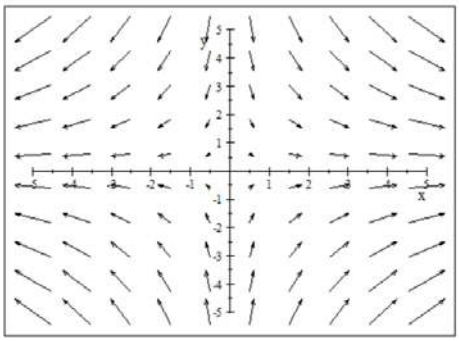
In Exercises 11 to 14, sketch a few flow lines of the given vector field.

11. **11.** $\mathbf{F}(x, y) = (y, -x)$



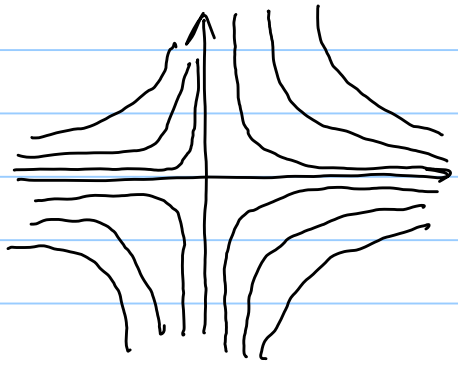
The field looks like 9(b), a clockwise rotation. $\therefore$ Concentric circles

12. **12.** $\mathbf{F}(x, y) = (x, -y)$



The field looks like:

∴ The flow lines look like hyperbolas:

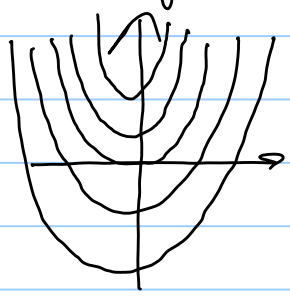


13. **13.** $\mathbf{F}(x, y) = (x, x^2)$

A series of parabolas as $y = x^2$ for (x, x^2)

Note for each y of $\vec{F}(x,y)$, the field vector is identical. So for $x = c$, vectors along the vertical line are parallel and equal in magnitude.

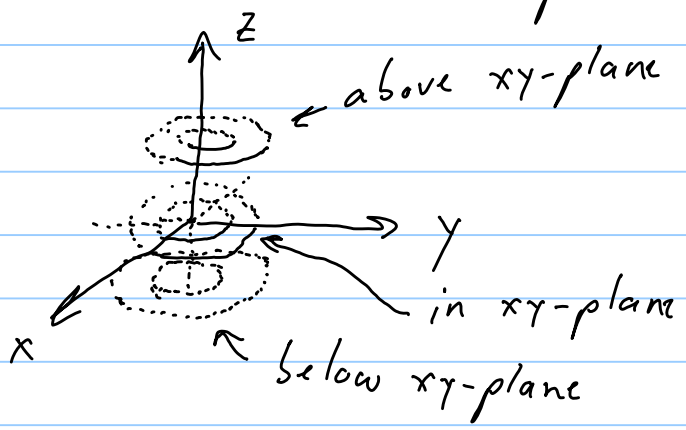
∴ Flow lines are "parallel" parabolas:



14. **14.** $\mathbf{F}(x, y, z) = (y, -x, 0)$

Field vectors are parallel to xy-plane. In each plane, the $(y, -x)$ looks like 9(6) above, a

clockwise rotation. ∴ Flow lines are concentric circles in a plane, stacked on top of one another.



In Exercises 15 to 18, show that the given curve $\mathbf{c}(t)$ *is a flow line of the given velocity vector field* $\mathbf{F}(x, y, z)$.

15.

15. $\mathbf{c}(t) = (e^{2t}, \log|t|, 1/t),\ t \neq 0; \mathbf{F}(x, y, z) = (2x, z, -z^2)$

Need to show $\vec{c}\,'(t) = \vec{F}(\vec{c}(t))$

$\vec{c}\,'(t) = \left(2e^{2t}, \frac{1}{|t|}, -\frac{1}{t^2}\right)$

$\vec{F}(\vec{c}(t)) = \vec{F}\left(e^{2t}, \log|t|, \frac{1}{t}\right) = \left(2e^{2t}, \frac{1}{t}, -\frac{1}{t^2}\right)$

∴ For $t > 0$, $\frac{1}{|t|} = \frac{1}{t}$, and $\vec{F}(\vec{c}(t)) = \vec{c}\,'(t)$

16.

16. $\mathbf{c}(t) = (t^2, 2t - 1, \sqrt{t}),\ t > 0; \mathbf{F}(x, y, z) = (y + 1, 2, 1/2z)$

$\vec{c}\,'(t) = \left(2t, 2, \frac{1}{2\sqrt{t}}\right)$

$\vec{F}(\vec{c}(t)) = \vec{F}(t^2, 2t-1, \sqrt{t}) = \left(2t-1+1, 2, \frac{1}{2\sqrt{t}}\right)$

$= (2t, 2, 2\frac{1}{\pi t})$

$\therefore \vec{c}'(t) = \vec{F}(\vec{c}(t))$

17. **17.** $\mathbf{c}(t) = (\sin t, \cos t, e^t); \mathbf{F}(x, y, z) = (y, -x, z)$

$\vec{c}'(t) = (\cos t, -\sin t, e^t)$

$\vec{F}(\vec{c}(t)) = \vec{F}(\sin t, \cos t, e^t) = (\cos t, -\sin t, e^t)$

$\therefore \vec{c}'(t) = \vec{F}(\vec{c}(t))$

18. **18.** $\mathbf{c}(t) = (\frac{1}{t^3}, e^t, \frac{1}{t}); \mathbf{F}(x, y, z) = (-3z^4, y, -z^2)$

$\vec{c}'(t) = (-3t^{-4}, e^t, -\frac{1}{t^2})$

$\vec{F}(\vec{c}(t)) = \vec{F}(\frac{1}{t^3}, e^t, \frac{1}{t}) = (-3(\frac{1}{t})^4, e^t, -(\frac{1}{t})^2)$

$= (-3t^{-4}, e^t, -\frac{1}{t^2})$

$\therefore \vec{c}'(t) = \vec{F}(\vec{c}(t))$

19. **19.** Let $\mathbf{F}(x, y, z) = (x^2, yx^2, z + zx)$ and $\mathbf{c}(t) = (\frac{1}{1-t}, 0, \frac{e^t}{1-t})$. Show $\mathbf{c}(t)$ is a flow line for $\mathbf{F}$.

$\vec{c}'(t) = \left[-\frac{1}{(1-t)^2}(-1), 0, \frac{(1-t)e^t - e^t(-1)}{(1-t)^2} \right]$

$$= \left[\frac{1}{(1-t)^2}, 0, \frac{(1-t)e^t + e^t}{(1-t)^2} \right]$$

$$\vec{F}(\vec{c}(t)) = \vec{F}\left(\frac{1}{1-t}, 0, \frac{e^t}{1-t} \right)$$

$$= \left[\frac{1}{(1-t)^2}, 0, \frac{e^t}{1-t} + \frac{e^t}{1-t}\left(\frac{1}{1-t}\right) \right]$$

$$= \left[\frac{1}{(1-t)^2}, 0, \frac{e^t(1-t) + e^t}{(1-t)^2} \right]$$

$$\therefore \vec{c}\,'(t) = \vec{F}(\vec{c}(t))$$

20.

20. Show that $\mathbf{c}(t) = (a\cos t - b\sin t,\ a\sin t + b\cos t)$ is a flow line for $\mathbf{F}(x, y) = (-y, x)$ for all real values of a and b.

$$\vec{c}\,'(t) = (-a\sin t - b\cos t,\ a\cos t - b\sin t)$$

$$\vec{F}(\vec{c}(t)) = \vec{F}(a\cos t - b\sin t,\ a\sin t + b\cos t)$$

$$= (-a\sin t - b\cos t,\ a\cos t - b\sin t)$$

$$\therefore \vec{c}\,'(t) = \vec{F}(\vec{c}(t))$$

21. **21.** (a) Let $\mathbf{F}(x, y, z) = (yz, xz, xy)$. Find a function $f: \mathbb{R}^3 \to \mathbb{R}$ such that $\mathbf{F} = \nabla f$.

(b) Let $\mathbf{F}(x, y, z) = (x, y, z)$. Find a function $f: \mathbb{R}^3 \to \mathbb{R}$ such that $\mathbf{F} = \nabla f$.

(a) $f_x = yz$, $\therefore f = xyz + g(y,z)$

$f_y = xz \Rightarrow g_y(y,z) = 0 \Rightarrow g(y,z) = h(z)$.

$\therefore f(x,y,z) = xyz + h(z)$

$f_z = xy \Rightarrow h_z = 0$, $\therefore h(z) = C$, a constant.

$\therefore \underline{f(x,y,z) = xyz + C}$, C a constant

(b) $f_x = x \Rightarrow f(x,y,z) = \frac{x^2}{2} + g(y,z)$

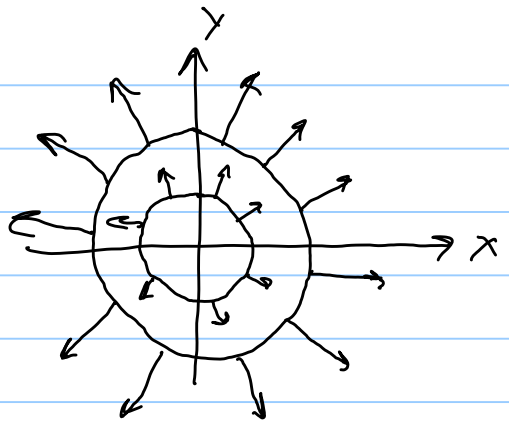
$f_y = y \Rightarrow g_y(y,z) = y \Rightarrow g(y,z) = \frac{y^2}{2} + h(z)$

$\therefore f(x,y,z) = \frac{x^2}{2} + \frac{y^2}{2} + h(z)$.

$f_z = z \to h'(z) = z \to h(z) = \frac{z^2}{2} + C$, C = constant

$\therefore \underline{f(x,y,z) = \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} + C}$, C a constant

22. **22.** Let $f(x, y) = x^2 + y^2$. Sketch the gradient vector field ∇f together with some level sets of f. How are they related?



$f: \mathbb{R}^2 \to \mathbb{R}$ $\quad \nabla f = (2x, 2y)$

$\|\nabla f\|$ increases with distance.

Level sets: $x^2 + y^2 = k$,
$\therefore$ circles

The gradient field vectors are $\perp$ to the level sets.

23. **23.** Show that it takes half as much energy to launch a satellite into an orbit just above the earth as it does to escape the earth. (Ignore the rotation of the earth.)

Let R_e = radius of earth, and so orbit radius of satellite also $= R_e$.

Let m = mass of satellite, M = mass earth.

Energy of satellite at any distance R from earth is $E = \frac{1}{2}mv^2 - \frac{GMm}{R}$

$E = 0$ for escape velocity v_e, so

$$E = 0 = \frac{1}{2}mv_e^2 - \frac{GMm}{R_e}, \quad \frac{1}{2}mv_e^2 = \frac{GMm}{R_e} \quad [1]$$

For orbit, at speed v_0, energy is

$$E = \frac{1}{2} m v_0^2 - \frac{GMm}{R_e}$$

In orbit, $F = ma = \frac{GMm}{R_e^2}$, so $a = \frac{GM}{R_e^2}$

In circular orbit, from p. 221 of text,

$$s^2 = \frac{GM}{R_0}, \text{ or } v_0^2 = \frac{GM}{R_0}.$$

$$\therefore \frac{1}{2} m v_0^2 = \frac{1}{2} \frac{GMm}{R_0} \qquad [2]$$

$\therefore$ From [1], [2] $\frac{1}{2} m v_0^2 = \frac{1}{2}\left(\frac{1}{2} m v_e^2\right)$

or, energy for orbit = $\frac{1}{2}$ energy for escape

24. **24.** Let $\mathbf{c}(t)$ be a flow line of a gradient field $\mathbf{F} = -\nabla V$. Prove that $V(\mathbf{c}(t))$ is a decreasing function of t.

A decreasing function of t means for

$t_1 < t_2$, $V(t_1) > V(t_2)$.

If $\frac{dV}{dt} \leq 0$, then V is a decreasing function.

But $\frac{dV}{dt} = \nabla V \cdot \vec{c}'(t)$ by chain rule.

And since $\vec{c}(t)$ is a flow line, $\vec{c}'(t) = \vec{F}(\vec{c}(t))$

$\therefore \frac{dV}{dt} = \nabla V \cdot \vec{c}'(t) = \nabla V \cdot \vec{F}(\vec{c}(t)) = \nabla V \cdot (-\nabla V)$

But $\|\nabla V\|^2 = \nabla V \cdot \nabla V$

$\therefore \frac{dV}{dt} = -\|\nabla V\|^2$, and $\|\vec{a}\| \geq 0$ for any $\vec{a}$.

$\therefore \frac{dV}{dt} = -\|\nabla V\|^2 \leq 0$

$\therefore$ $V(t)$ is a decreasing function of t.

25. **25.** Suppose that the isotherms in a region are all concentric spheres centered at the origin. Prove that the energy flux vector field points either toward or away from the origin.

From Example 4, p. 238 of text, $\vec{J} = -k\nabla T$,

where $\vec{J}$ = energy/heat flux vector field, k a constant, T = scalar field of temperature.

Gradient vectors, like ∇T, are perpendicular to level sets.

$\therefore$ In this case, ∇T is perpendicular to the concentric spheres, which are level sets.

$\therefore \nabla T$ points towards or away from origin.

26. **26.** Sketch the gradient field $-\nabla V$ for $V(x, y) = (x + y)/(x^2 + y^2)$ and the equipotential surface $V = 1$.

a) Compute $\nabla V = (V_x, V_y)$.

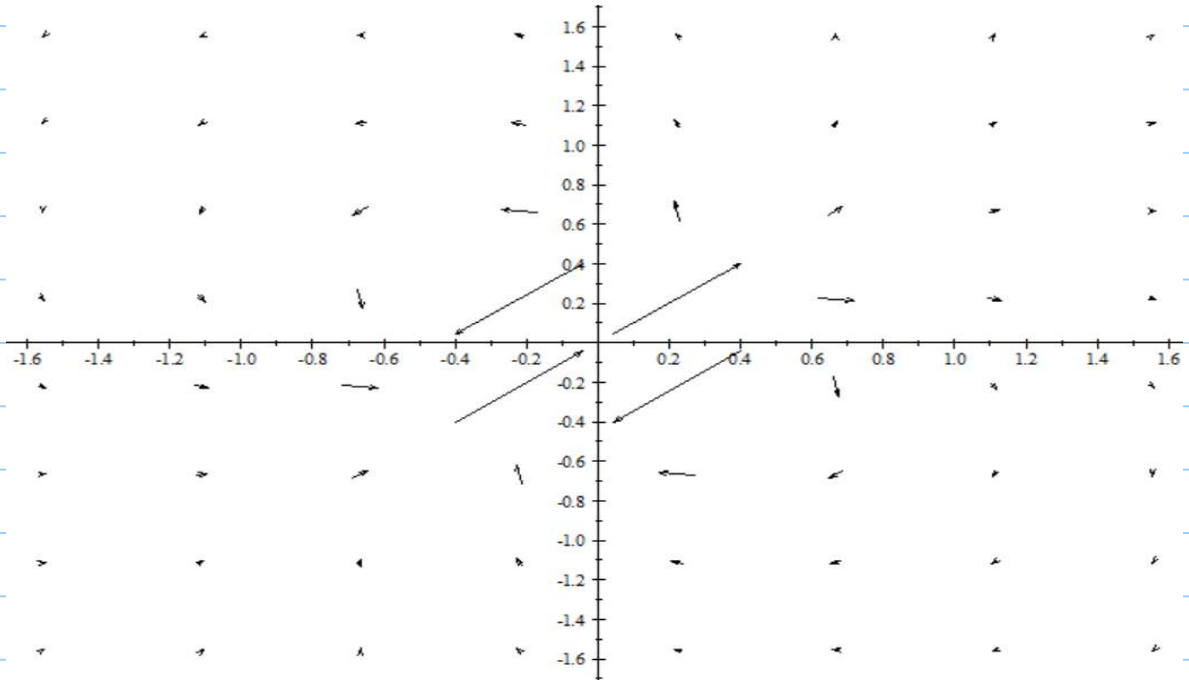
$$V_x = \frac{(x^2+y^2)(1) - (x+y)(2x)}{(x^2+y^2)^2} = \frac{-x^2 - 2xy + y^2}{(x^2+y^2)^2}$$

$$V_y = \frac{(x^2+y^2)(1) - (x+y)(2y)}{(x^2+y^2)^2} = \frac{x^2 - 2xy - y^2}{(x^2+y^2)^2}$$

$$\therefore -\nabla V = \left[\frac{x^2 + 2xy - y^2}{(x^2+y^2)^2}, \frac{y^2 + 2xy - x^2}{(x^2+y^2)^2} \right]$$

So, as $(x,y) \to \infty$, magnitude of $-\nabla V$ is very small.

as $(x,y) \to (0,0)$, magnitude of $-\nabla V$ is very large.

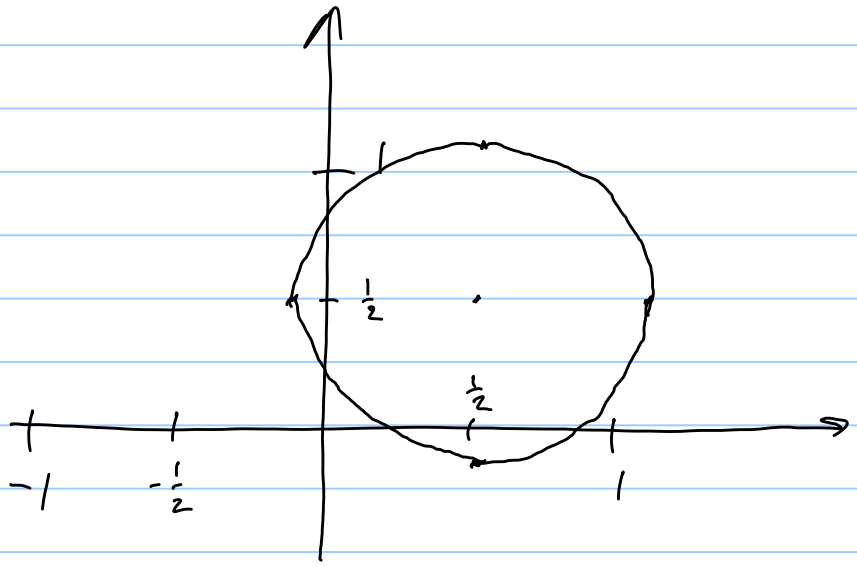


(5) For $V = 1 = \frac{x+y}{x^2+y^2}$, or $x^2+y^2 = x+y$,

$\therefore x^2 - x + y^2 - y = 0$, $(x-\frac{1}{2})^2 + (y-\frac{1}{2})^2 = \frac{1}{2}$

$\therefore$ a circle of radius $\frac{1}{\sqrt{2}}$, centred at $(\frac{1}{2}, \frac{1}{2})$

$\frac{1}{\sqrt{2}} = 0.7$



27. **27.** Let $\mathbf{F}(x, y, z) = (xe^y, y^2z^2, xyz)$ and suppose $\mathbf{c}(t) = \big(x(t), y(t), z(t)\big)$ is a flow line for $\mathbf{F}$. Find the system of differential equations that the functions $x(t)$, $y(t)$, and $z(t)$ must satisfy.

Using $\vec{c}'(t) = F(\vec{c}(t))$, $(x', y', z') = (xe^y, y^2z^2, xyz)$.

$\therefore \frac{dx}{dt} = xe^y \quad \frac{dy}{dt} = y^2z^2 \quad \frac{dz}{dt} = xyz$

4.4 Divergence and Curl

Note Title 9/12/2016

Find the divergence of the vector fields in Exercises 1 to 4.

1. **1.** $\mathbf{V}(x, y, z) = e^{xy}\mathbf{i} - e^{xy}\mathbf{j} + e^{yz}\mathbf{k}$

$\frac{\partial}{\partial x} e^{xy} = ye^{xy} \quad \frac{\partial}{\partial y}(-e^{xy}) = -xe^{xy} \quad \frac{\partial}{\partial z} e^{yz} = ye^{yz}$

$\therefore \nabla \cdot \vec{V} = ye^{xy} - xe^{xy} + ye^{yz}$

2. **2.** $\mathbf{V}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$

$\frac{\partial}{\partial x}(yz) = 0 \quad \frac{\partial}{\partial y}(xz) = 0 \quad \frac{\partial}{\partial z}(xy) = 0$

$\therefore \nabla \cdot \vec{V} = 0$

3. **3.** $\mathbf{V}(x, y, z) = x\mathbf{i} + (y + \cos x)\mathbf{j} + (z + e^{xy})\mathbf{k}$

$\frac{\partial}{\partial x}(x) = 1 \quad \frac{\partial}{\partial y}(y + \cos x) = 1 \quad \frac{\partial}{\partial z}(z + e^{xy}) = 1$

$\therefore \nabla \cdot \vec{V} = 3$

4. **4.** $\mathbf{V}(x, y, z) = x^2\mathbf{i} + (x + y)^2\mathbf{j} + (x + y + z)^2\mathbf{k}$

$\frac{\partial}{\partial x}(x^2) = 2x \quad \frac{\partial}{\partial y}(x+y)^2 = 2(x+y) \quad \frac{\partial}{\partial z}(x+y+z)^2 = 2(x+y+z)$

$\therefore \nabla \cdot \vec{V} = 2x + 2(x+y) + 2(x+y+z)$

$= 6x + 4y + 2z$

5.

5. Figure 4.4.11 shows some flow lines and moving regions for a fluid moving in the plane-field velocity field **V**. Where is div **V** > 0, and also where is div **V** < 0?

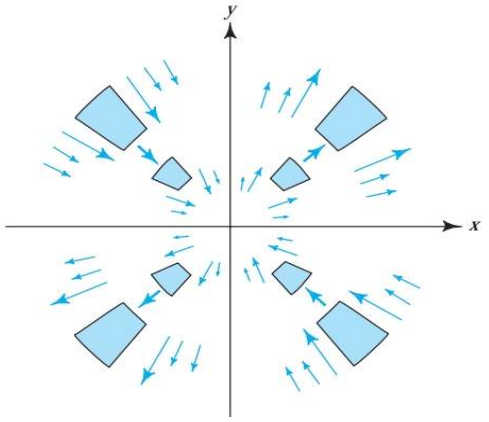


figure **4.4.11** The flow lines of a fluid moving in the plane.

$\text{div } \vec{V} > 0$ where there is expansion

so Quadrants I, III

$\text{div } \vec{V} < 0$ where there is compression.

$\therefore$ Quadrants II, IV

6.

6. Let $V(x, y, z) = x\mathbf{i}$ be the velocity field of a fluid in space. Relate the sign of the divergence with the rate of change of volume under the flow.

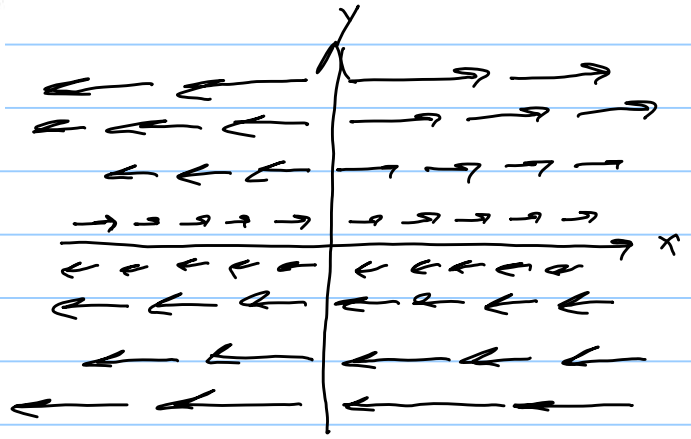
Let $V(x,y,z) = (V_1(x,y,z), V_2(x,y,z), V_3(x,y,z)) = (x, 0, 0)$

$\nabla \cdot \vec{V} = V_{1x} + V_{2y} + V_{3z} = 1 + 0 + 0 = 1 > 0$

$\therefore$ Rate of change of fluid volume = expansion

7.

7. Sketch a few flow lines for $\mathbf{F}(x, y) = y\mathbf{i}$. Calculate $\nabla \cdot \mathbf{F}$ and explain why your answer is consistent with your sketch.



$\vec{F}(x,y) = (y, 0)$

$\therefore \nabla \cdot \vec{F} = \frac{\partial}{\partial x}(y) + \frac{\partial}{\partial x}(0) = 0$

A tiny rectangle around a given point does not change in area as the flow is constant for any given y : $(y, 0)$

A tiny rectangle centered at a given (x, y) moves to the right ($y > 0$) or left ($y < 0$).

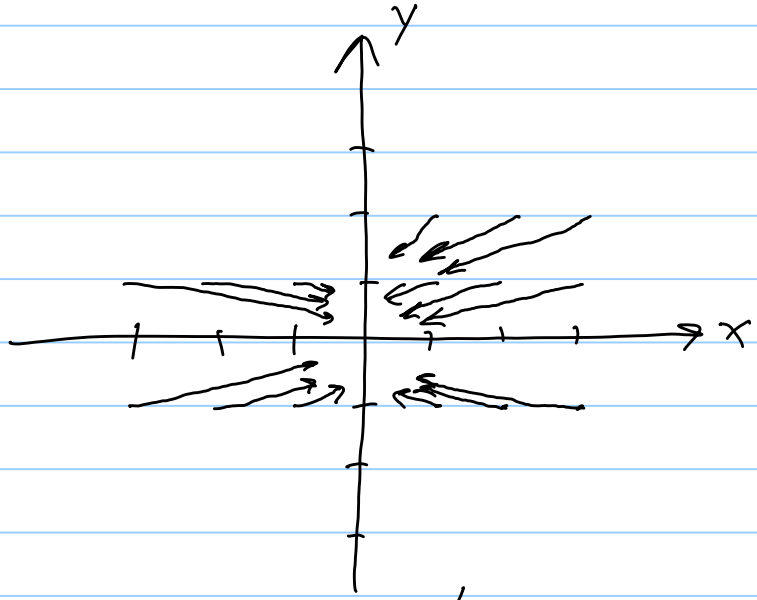
The "shear" effect (flow at $y + \Delta y$ slightly greater than at $y - \Delta y$) becomes negligible as $\Delta y \to 0$.

8. **8.** Sketch a few flow lines for $\mathbf{F}(x, y) = -3x\mathbf{i} - y\mathbf{j}$. Calculate $\nabla \cdot \mathbf{F}$ and explain why your answer is consistent with your sketch.

$\vec{F}(x, y) = (-3x, -y)$

$\therefore \nabla \cdot \vec{F} = -3 + (-1) =$

$= -4 < 0$



$\therefore$ Fluid shows compression, graph shows compression toward origin.

Calculate the divergence of the vector fields in Exercises 9 to 12.

9. **9.** $\mathbf{F}(x, y) = x^3\mathbf{i} - x\sin(xy)\mathbf{j}$

Let $\vec{F} = (F_1, F_2) = (x^3, -x\sin(xy))$

$\therefore \nabla \cdot \vec{F} = 3x^2 + (-x^2\cos(xy)) = \frac{\partial}{\partial x}x^3 + \frac{\partial}{\partial y}(-x\sin(xy))$

$= \underline{3x^2 - x^2\cos(xy)}$

10. **10.** $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$

$\vec{F} = (y, -x) \quad \therefore \nabla \cdot \vec{F} = \frac{\partial}{\partial x}y + \frac{\partial}{\partial y}(-x) = 0 + 0 = \underline{0}$

11. **11.** $\mathbf{F}(x, y) = \sin(xy)\mathbf{i} - \cos(x^2y)\mathbf{j}$

$\vec{F} = [\sin(xy), -\cos(x^2y)]$

$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(\sin(xy)) + \frac{\partial}{\partial y}(-\cos(x^2y))$

$= \underline{y\cos(xy) + x^2\sin(x^2y)}$

12. **12.** $\mathbf{F}(x, y) = xe^y\mathbf{i} - [y/(x + y)]\mathbf{j}$

$\vec{F} = (xe^y, -\frac{y}{x+y})$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(xe^y) + \frac{\partial}{\partial y}\left(\frac{-y}{x+y}\right)$$

$$= e^y + \left[\frac{(x+y)(-1) - (-y)(1)}{(x+y)^2}\right]$$

$$= e^y - \frac{x}{(x+y)^2}$$

Compute the curl, $\nabla \times \mathbf{F}$, of the vector fields in Exercises 13 to 16.

13. **13.** $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix} = (0-0, 0-0, 0-0) = \vec{0}$$

14. **14.** $\mathbf{F}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ yz & xz & xy \end{vmatrix} = (x-x, y-y, z-z) = \vec{0}$$

15. **15.** $\mathbf{F}(x, y, z) = (x^2 + y^2 + z^2)(3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k})$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3(x^2+y^2+z^2) & 4(x^2+y^2+z^2) & 5(x^2+y^2+z^2) \end{vmatrix}$$

$$= \underline{(10y - 8z,\ 6z - 10x,\ 8x - 6y)}$$

16. **16.** $\mathbf{F}(x, y, z) = \dfrac{yz\mathbf{i} - xz\mathbf{j} + xy\mathbf{k}}{x^2 + y^2 + z^2}$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \dfrac{yz}{(x^2+y^2+z^2)} & \dfrac{-xz}{(x^2+y^2+z^2)} & \dfrac{xy}{(x^2+y^2+z^2)} \end{vmatrix} =$$

$$\frac{1}{(x^2+y^2+z^2)^2}\Big[(x^2+y^2+z^2)(x) - (xy)(2y) - \big[(x^2+y^2+z^2)(-x) - (-xz)(2z)\big],$$

$$(x^2+y^2+z^2)(y) - (yz)(2z) - \big[(x^2+y^2+z^2)(y) - (xy)(2x)\big],$$

$$(x^2+y^2+z^2)(-z) - (-xz)(2x) - \big[(x^2+y^2+z^2)(z) - (yz)(2y)\big]\Big]$$

$$= \frac{1}{(x^2+y^2+z^2)^2}\Big[2x(x^2+y^2+z^2) + 2xz^2 - 2xy^2,$$

$$-2yz^2 - 2yx^2,$$

$$-2z(x^2+y^2+z^2) + 2zx^2 + 2zy^2\Big]$$

$$= \underline{\Big[2x(x^2+2z^2),\ -2yz^2 - 2yx^2,\ -2z^3\Big] \Big/ (x^2+y^2+z^2)^2}$$

Calculate the scalar curl of each of the vector fields in Exercises 17 to 20.

17. **17.** $\mathbf{F}(x, y) = \sin x\mathbf{i} + \cos x\mathbf{j}$

$$\nabla \times \vec{F} = \left[\frac{\partial}{\partial x}(\cos x) - \frac{\partial}{\partial y}(\sin x)\right]\hat{k} = -\sin x\,\hat{k}$$

$\therefore$ Scalar curl $= -\sin x$

18. **18.** $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$

$$\nabla \times \vec{F} = \left[\frac{\partial}{\partial x}(-x) - \frac{\partial}{\partial y}(y)\right]\hat{k} = -2\hat{k}$$

$\therefore$ Scalar curl $= -2$

19. **19.** $\mathbf{F}(x, y) = xy\mathbf{i} + (x^2 - y^2)\mathbf{j}$

$$\nabla \times \vec{F} = \left[\frac{\partial}{\partial x}(x^2 - y^2) - \frac{\partial}{\partial y}(xy)\right]\hat{k} = (2x - x)\hat{k} = x\hat{k}$$

$\therefore$ Scalar curl $= x$

20. **20.** $\mathbf{F}(x, y) = x\mathbf{i} + y\mathbf{j}$

$$\nabla \times \vec{F} = \left[\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(x)\right]\hat{k} = [0 - 0]\hat{k} = \vec{0}$$

$\therefore$ scalar curl $= 0$

21.

21. Let $\mathbf{F}(x, y, z) = (x^2, x^2y, z + zx)$.

(a) Verify that $\nabla \cdot (\nabla \times \mathbf{F}) = 0$.

(b) Can there exist a function $f: \mathbb{R}^3 \to \mathbb{R}$ such that $\mathbf{F} = \nabla f$? Explain.

(a) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 & x^2y & z + zx \end{vmatrix}$

$= (0 - 0, 0 - z, 2yx - 0) = (0, -z, 2xy)$

$\nabla \cdot (0, -z, 2xy) = \frac{\partial}{\partial x}(0) + \frac{\partial}{\partial y}(-z) + \frac{\partial}{\partial z}(2xy)$

$\therefore \nabla \cdot (\nabla \times \vec{F}) = 0 + 0 + 0 = \underline{0}$

(b) If $\vec{F}$ is the gradient of f, then

Curl $\vec{F} = \nabla \times \vec{F} = 0 \Rightarrow (0, -z, 2xy)$

But $\nabla \times \vec{F} \neq \vec{0}$ for all x, y, z.

$\therefore$ There is no such function $f: \mathbb{R}^3 \to \mathbb{R}$.

22.

22. (a) Which of the vector fields in Exercises 13–16 could be gradient fields?

(b) Which of the vector fields in Exercises 9–12 could be the curl of some vector field $\mathbf{V}: \mathbb{R}^3 \to \mathbb{R}^3$?

(a) If the fields are gradients, then the curl of the gradient should be zero.

13. $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ $\nabla \times \vec{F} = \vec{0}$ as shown, so this could be a gradient.

14. $\mathbf{F}(x, y, z) = yz\mathbf{i} + xz\mathbf{j} + xy\mathbf{k}$ $\nabla \times \vec{F} = \vec{0}$ as shown, so this could be a gradient

15. $\mathbf{F}(x, y, z) = (x^2 + y^2 + z^2)(3\mathbf{i} + 4\mathbf{j} + 5\mathbf{k})$ $\nabla \times \vec{F} \neq \vec{0}$, so could not be a gradient.

16. $\mathbf{F}(x, y, z) = \dfrac{yz\mathbf{i} - xz\mathbf{j} + xy\mathbf{k}}{x^2 + y^2 + z^2}$ $\nabla \times \vec{F} \neq \vec{0}$, so could not be a gradient.

$\therefore$ #13, #14 could be gradients

#15, #16 could not

(6)

(b) Which of the vector fields in Exercises 9–12 could be the curl of some vector field $\mathbf{V}: \mathbb{R}^3 \to \mathbb{R}^3$?

div (curl $\vec{F}$) = 0. $\therefore$ Look at $\nabla \cdot \vec{F}$

9. $\mathbf{F}(x, y) = x^3\mathbf{i} - x\sin(xy)\mathbf{j}$ $\nabla \cdot \vec{F} \neq 0$ as shown, so can not be the curl of some $\vec{V}$

10. $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$ $\nabla \cdot \vec{F} = 0$ as shown, so could be the curl of some $\vec{V}$

11. $\mathbf{F}(x, y) = \sin(xy)\mathbf{i} - \cos(x^2 y)\mathbf{j}$ $\nabla \cdot \vec{F} \neq 0$, so can not be the curl of some $\vec{V}$

12. $\mathbf{F}(x, y) = xe^y\mathbf{i} - [y/(x+y)]\mathbf{j}$ $\nabla \cdot \vec{F} \neq 0$, so can not be the curl of some $\vec{V}$.

∴ #10 could be

#9, 11, 12 can not

23.

23. Let $\mathbf{F}(x, y, z) = (e^{xz}, \sin(xy), x^5y^3z^2)$.

(a) Find the divergence of $\mathbf{F}$.

(b) Find the curl of $\mathbf{F}$.

(a) $\frac{\partial}{\partial x}(e^{xz}) + \frac{\partial}{\partial y}(\sin(xy)) + \frac{\partial}{\partial z}(x^5y^3z^2)$

$= \underline{ze^{xz} + x\cos(xy) + 2x^5y^3z}$

(b) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^{xz} & \sin(xy) & x^5y^3z^2 \end{vmatrix}$

$= (3x^5y^2z^2 - 0,\ xe^{xz} - 5x^4y^3z^2,\ y\cos(xy) - 0)$

$= \underline{(3x^5y^2z^2,\ xe^{xz} - 5x^4y^3z^2,\ y\cos(xy))}$

24.

24. Suppose $f: \mathbb{R}^3 \to \mathbb{R}$ is a C^2 scalar function. Which of the following expressions are meaningful, and which are nonsense? For those which are meaningful, decide whether the expression defines a scalar function or a vector field.

(a) curl(grad f)

(b) grad(curl f))

(c) div(grad f)

(d) grad(div f)

(e) curl (div f)

(f) div(curl f)

(a) grad f creates a vector in $\mathbb{R}^3$

curl operates on a vector in $\mathbb{R}^3$

so curl(grad f) makes sense, and curl generates a vector in $\mathbb{R}^3$

(b) Does not make sense

f is a scalar in $\mathbb{R}^1$, and curl operates on a vector on $\mathbb{R}^3$, and generates a vector in $\mathbb{R}^3$. Grad does not operate on a vector in $\mathbb{R}^3$.

(c) Makes sense.

grad f generates a vector in $\mathbb{R}^3$ (in this case) while div operates on the resultant vector.

(d) Does not make sense

div operates on a vector, not a scalar.

$\therefore$ div f makes no sense.

If $f: \mathbb{R}^1 \to \mathbb{R}^1$, then div f could make sense.

$\text{div } f = \frac{df}{dx}$, and $\text{grad}(\text{div } f) = \frac{d^2 f}{dx^2}$, a vector in $\mathbb{R}^1$. But here, $f: \mathbb{R}^3 \to \mathbb{R}^1$, not $\mathbb{R}^1 \to \mathbb{R}^1$.

(e) Does not make sense.

div operates on a vector, not a scalar. See (d)

(f) Does not make sense

curl operates on a vector in $\mathbb{R}^3$, not a scalar in $\mathbb{R}^1$.

25.

25. Suppose $\mathbf{F}: \mathbb{R}^3 \to \mathbb{R}^3$ is a C^2 vector field. Which of the following expressions are meaningful, and which are nonsense? For those which are meaningful, decide whether the expression defines a scalar function or a vector field.

(a) curl(grad **F**)
(b) grad(curl **F**))
(c) div(grad **F**)
(d) grad(div **F**)
(e) curl (div **F**)
(f) div(curl **F**)

(a) Nonsense. grad operates on a scalar in $\mathbb{R}^1$

(b) Nonsense. curl $\vec{F}$ is a vector in $\mathbb{R}^3$, grad requires a scalar in $\mathbb{R}^1$.

(c) Nonsense. grad operates on a scalar in R^1

(d) Makes sense. div $\vec{F}$ creates a scalar in R^1 and grad operates on a scalar in R^1 and generates a vector in R^3 (in this case).

(e) Nonsense. div $\vec{F}$ is a scalar in R^1, curl requires a vector in R^3.

(f) Makes sense. curl $\vec{F}$ generates a vector in R^3. div operates on a vector and generates a scalar.

26. **26.** Suppose $f, g, h: \mathbb{R} \to \mathbb{R}$ are differentiable. Show that the vector field $\mathbf{F}(x, y, z) = \big(f(x), g(y), h(z)\big)$ is irrotational.

Need to show $\nabla \times \vec{F} = \vec{0}$.

By the notation, f, g, h are only dependent on one variable.

$$\therefore \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ f(x) & g(y) & h(z) \end{vmatrix}$$

$$= \left(\frac{\partial}{\partial y} h(z) - \frac{\partial}{\partial z} g(y), \frac{\partial}{\partial z} f(x) - \frac{\partial}{\partial x} h(z), \frac{\partial}{\partial x} h(z) - \frac{\partial}{\partial y} f(x) \right)$$

$$= (0-0, 0-0, 0-0) = \vec{0}$$

27. **27.** Suppose $f, g, h: \mathbb{R}^2 \to \mathbb{R}$ are differentiable. Show that the vector field $\mathbf{F}(x, y, z) = \big(f(y, z), g(x, z), h(x, y) \big)$ has zero divergence.

$$\text{div } \vec{F} = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial y} + \frac{\partial h}{\partial z}$$

$$= 0 + 0 + 0 = 0$$

since f is not dependent on x
g is not dependent on y
h is not dependent on z.

28. **28.** Prove identity 13 in the list of vector identities.

13. $\text{div } (\nabla f \times \nabla g) = 0$

Let $\nabla f = \vec{F}$, $\nabla g = \vec{G}$.

$\therefore \text{div}(\nabla f \times \nabla g) = \text{div}(\vec{F} \times \vec{G})$

By identity #8, $\text{div}(\vec{F} \times \vec{G}) = \vec{G} \cdot (\nabla \times \vec{F}) - \vec{F} \cdot (\nabla \times \vec{G})$

But $\text{curl } \vec{F} = \nabla \times \vec{F} = \nabla \times (\nabla f) = \vec{0}$

since the curl of a gradient is $\vec{0}$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} & \frac{\partial f}{\partial z} \end{vmatrix} = \vec{0}$$ since mixed partials are equal.

Also, 2 rows of determinant are equal.

Similarly, $\nabla \times \vec{G} = \nabla \times (\nabla g) = \vec{0}$.

$\therefore \operatorname{div}(\vec{F} \times \vec{G}) = \vec{G} \cdot (\vec{0}) - \vec{F} \cdot (\vec{0}) = 0 + 0 = 0$

$\therefore \operatorname{div}(\nabla f \times \nabla g) = 0$

Verify that $\nabla \times (\nabla f) = \mathbf{0}$ *for the functions in Exercises 29 to 32.*

29. **29.** $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$

$$\nabla f = \left[\frac{x}{\sqrt{x^2+y^2+z^2}}, \frac{y}{\sqrt{x^2+y^2+z^2}}, \frac{z}{\sqrt{x^2+y^2+z^2}} \right]$$

$$\nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \frac{x}{\sqrt{x^2+y^2+z^2}} & \frac{y}{\sqrt{x^2+y^2+z^2}} & \frac{z}{\sqrt{x^2+y^2+z^2}} \end{vmatrix}$$

$$= \Big[-z(x^2+y^2+z^2)^{-\frac{3}{2}} y - \left(-y(x^2+y^2+z^2)^{-\frac{3}{2}} z\right),$$

$$-z(x^2+y^2+z^2)^{-\frac{3}{2}} x - \left(-x(x^2+y^2+z^2)^{-\frac{3}{2}} z\right),$$

$$-y(x^2+y^2+z^2)^{-3/2} x - \left(x(x^2+y^2+z^2)^{-3/2} y\right) \Big]$$

$= \{0, 0, 0\} = \vec{0}$

30. **30.** $f(x, y, z) = xy + yz + xz$

$\nabla f = \{y+z, x+z, y+x\}$

$\therefore \nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y+z & x+z & y+x \end{vmatrix}$

$= \{1-1, 1-1, 1-1\} = \{0, 0, 0\} = \vec{0}$

31. **31.** $f(x, y, z) = 1/(x^2 + y^2 + z^2)$

$\nabla f = \left\{-(x^2+y^2+z^2)^{-2} 2x, -(x^2+y^2+z^2)^{-2} 2y, -(x^2+y^2+z^2)^{-2} 2z\right\}$

$\therefore \nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial x & \partial/\partial z \\ \frac{-2x}{(x^2+y^2+z^2)^2} & \frac{-2y}{(x^2+y^2+z^2)^2} & \frac{-2z}{(x^2+y^2+z^2)^2} \end{vmatrix}$

$= \{4z(x^2+y^2+z^2)^{-3} 2y - (4y(x^2+y^2+z^2)^{-3} 2z),$

$4x(x^2+y^2+z^2)^{-3} 2z - (4z(x^2+y^2+z^2)^{-3} 2x),$

$4y(x^2+y^2+z^2)^{-3} 2x - (4x(x^2+y^2+z^2)^{-3} 2y)\}$

$= \{0, 0, 0\} = \vec{0}$

32. **32.** $f(x, y, z) = x^2y^2 + y^2z^2$

$$\nabla f = (2xy^2,\ 2yx^2 + 2yz^2,\ 2zy^2)$$

$$\therefore \nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xy^2 & 2yx^2 + 2yz^2 & 2zy^2 \end{vmatrix}$$

$$= [\,4zy - 4yz,\ 0 - 0,\ 4yx - 4xy\,]$$

$$= [\,0, 0, 0\,] = \vec{0}$$

33. **33.** Show that $\mathbf{F} = y(\cos x)\mathbf{i} + x(\sin y)\mathbf{j}$ is *not* a gradient vector field.

If $\vec{F}$ is a gradient, then $\nabla \times \vec{F} = \vec{0}$.

Check: $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y\cos x & x\sin y & 0 \end{vmatrix}$

$$= [\,0 - 0,\ 0 - 0,\ \sin y - \cos x\,]$$

$$= [\,0, 0, \sin y - \cos x\,] \neq \vec{0}$$

$\therefore \vec{F}$ not a gradient of some $f: \mathbb{R}^3 \to \mathbb{R}^1$

34. **34.** Show that $\mathbf{F} = (x^2 + y^2)\mathbf{i} - 2xy\mathbf{j}$ is *not* a gradient field.

If $\vec{F}$ is a gradient, then $\nabla \times \vec{F} = \vec{0}$.

Check: $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2+y^2 & -2xy & 0 \end{vmatrix}$

$= [0-0,\ 0-0,\ -2y-2y] = [0,0,-4y] \neq \vec{0}$

$\therefore \vec{F}$ not a gradient of some $f: \mathbb{R}^3 \to \mathbb{R}^1$

35. **35.** Prove identity 10 in the list of vector identities.

10. $\operatorname{curl}(f\mathbf{F}) = f\operatorname{curl}\mathbf{F} + \nabla f \times \mathbf{F}$

Let $\vec{F} = (F_1, F_2, F_3)$, where $F_i: \mathbb{R}^3 \to \mathbb{R}^1$

$\therefore \nabla \times (f\vec{F}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ fF_1 & fF_2 & fF_3 \end{vmatrix}$

$= [\, f_y F_3 + f F_{3y} - (f_z F_2 + f F_{2z}),$

$f_z F_1 + f F_{1z} - (f_x F_3 + f F_{3x}),$

$f_x F_2 + f F_{2x} - (f_y F_1 + f F_{1y}) \,]$

$= [\, f F_{3y} - f F_{2z} + f_y F_3 - f_z F_2,$

$f F_{1z} - f F_{3x} + f_z F_1 - f_x F_3,$

$f F_{2x} - f F_{1y} + f_x F_2 - f_y F_1 \,]$

$$= f\begin{bmatrix} F_{3y} - F_{2z} \\ F_{1z} - F_{3x} \\ F_{2x} - F_{1y} \end{bmatrix} + \begin{bmatrix} f_y F_3 - f_z F_2 \\ f_z F_1 - f_x F_3 \\ f_x F_2 - f_y F_1 \end{bmatrix}$$

$$= f\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ F_1 & F_2 & F_3 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ f_x & f_y & f_z \\ F_1 & F_2 & F_3 \end{vmatrix}$$

$$= f(\nabla \times \vec{F}) + \nabla f \times \vec{F}$$

$$= f(\text{curl}\,\vec{F}) + \nabla f \times \vec{F}$$

36.

36. Suppose that $\nabla \cdot \mathbf{F} = 0$ and $\nabla \cdot \mathbf{G} = 0$. Which of the following necessarily have zero divergence?

(a) $\mathbf{F} + \mathbf{G}$

(b) $\mathbf{F} \times \mathbf{G}$

(a) $\nabla \cdot (\vec{F} + \vec{G}) = \nabla \cdot \vec{F} + \nabla \cdot \vec{G} = 0 + 0 = 0$

(b) $\nabla \cdot (\vec{F} \times \vec{G}) = \vec{G} \cdot (\nabla \times \vec{F}) - \vec{F} \cdot (\nabla \times \vec{G})$

No guarantee this is zero.

Example: $\vec{F} = (x, -y, 0) \therefore \nabla \cdot \vec{F} = 1 - 1 + 0 = 0$

$\vec{G} = (z, 0, x) \therefore \nabla \cdot \vec{G} = 0 + 0 + 0 = 0$

$$\vec{F} \times \vec{G} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & -y & 0 \\ z & 0 & x \end{vmatrix} = (-xy, -x^2, yz)$$

$$\therefore \nabla \cdot (\vec{F} \times \vec{G}) = (-y, 0, y) \neq 0$$

37.

37. Let $\mathbf{F} = 2xz^2\mathbf{i} + \mathbf{j} + y^3zx\mathbf{k}$ and $f = x^2y$. Compute the following quantities.

(a) ∇f

(b) $\nabla \times \mathbf{F}$

(c) $\mathbf{F} \times \nabla f$

(d) $\mathbf{F} \cdot (\nabla f)$

(a) $\nabla f = (f_x, f_y, f_z) = (2xy, x^2, 0)$, assuming $f: \mathbb{R}^3 \to \mathbb{R}^1$

(b) $$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xz^2 & 1 & y^3zx \end{vmatrix}$$

$$= (3y^2zx - 0, 4xz - y^3z, 0 - 0)$$

$$= (3y^2zx, 4xz - y^3z, 0)$$

(c) $$\vec{F} \times \nabla f = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2xz^2 & 1 & y^3zx \\ 2xy & x^2 & 0 \end{vmatrix}$$

$$= (0 - y^3zx^3, 2y^4zx^2 - 0, 2x^3z^2 - 2xy)$$

$$= (-y^3zx^3, 2y^4zx^2, 2x^3z^2 - 2xy)$$

(d) $\vec{F} \cdot (\nabla f) = (2xz^2, 1, y^3 z x) \cdot (2xy, x^2, 0)$

$= 4x^2yz^2 + x^2 + 0 = 4x^2yz^2 + x^2$

38.

38. Let $\mathbf{r}(x, y, z) = (x, y, z)$ and $r = \sqrt{x^2 + y^2 + z^2} = \|\mathbf{r}\|$. Prove the following identities.

(a) $\nabla(1/r) = -\mathbf{r}/r^3$, $r \neq 0$; and, in general, $\nabla(r^n) = nr^{n-2}\mathbf{r}$ and $\nabla(\log r) = \mathbf{r}/r^2$.

(b) $\nabla^2(1/r) = 0$, $r \neq 0$; and, in general, $\nabla^2 r^n = n(n+1)\, r^{n-2}$.

(c) $\nabla \cdot (\mathbf{r}/r^3) = 0$; and, in general, $\nabla \cdot (r^n \mathbf{r}) = (n+3)\, r^n$.

(d) $\nabla \times \mathbf{r} = \mathbf{0}$; and, in general, $\nabla \times (r^n \mathbf{r}) = \mathbf{0}$.

(a)(1) $\frac{\partial}{\partial x}\left(\frac{1}{r}\right) = \frac{\partial}{\partial x}(x^2+y^2+z^2)^{-\frac{1}{2}} = -\frac{1}{2}(x^2+y^2+z^2)^{-\frac{3}{2}}(2x)$

$= -\frac{x}{(x^2+y^2+z^2)^{3/2}} = \frac{-x}{(\sqrt{x^2+y^2+z^2})^3} = -\frac{x}{\|\vec{r}\|^3}$

Similarly, $\frac{\partial}{\partial y}\left(\frac{1}{r}\right) = \frac{-y}{\|\vec{r}\|^3}$, $\frac{\partial}{\partial z}\left(\frac{1}{r}\right) = \frac{-z}{\|\vec{r}\|^3}$

$\therefore \nabla\left(\frac{1}{r}\right) = \left(-\frac{x}{\|\vec{r}\|^3}, -\frac{y}{\|\vec{r}\|^3}, -\frac{z}{\|\vec{r}\|^3}\right)$

$= -\frac{1}{\|\vec{r}\|^3}(x, y, z) = -\frac{\vec{r}}{\|\vec{r}\|^3} = -\frac{\vec{r}}{r^3}$

(2) $\frac{\partial}{\partial x}(r^n) = \frac{\partial}{\partial x}\left[(x^2+y^2+z^2)^{\frac{1}{2}}\right]^n = \frac{\partial}{\partial x}(x^2+y^2+z^2)^{\frac{n}{2}}$

$= \frac{n}{2}(x^2+y^2+z^2)^{\frac{n}{2}-1}(2x)$

$= n(x)(x^2+y^2+z^2)^{\frac{n-2}{2}} = nx\left[(x^2+y^2+z^2)^{\frac{1}{2}}\right]^{n-2}$

$= nx(r)^{n-2}$

Similarly, $\frac{\partial}{\partial y}(r^n) = ny(r)^{n-2}$, $\frac{\partial}{\partial z}(r^n) = nz(r)^{n-2}$

$\therefore \nabla(r^n) = \left\{nxr^{n-2}, nyr^{n-2}, nzr^{n-2}\right\}$

$= nr^{n-2}\{x, y, z\} = nr^{n-2}\vec{r}$

(3) $\frac{\partial}{\partial x}(\log r) = \frac{\partial}{\partial x}\left(\log(x^2+y^2+z^2)^{\frac{1}{2}}\right)$

$= \frac{\partial}{\partial x}\left(\frac{1}{2}\log(x^2+y^2+z^2)\right)$

$= \frac{1}{2}\frac{2x}{x^2+y^2+z^2} = \frac{x}{r^2}$

Similarly, $\frac{\partial}{\partial y}(\log r) = \frac{y}{r^2}$, $\frac{\partial}{\partial z}(\log r) = \frac{z}{r^2}$

$\therefore \nabla(\log r) = \left(\frac{x}{r^2}, \frac{y}{r^2}, \frac{z}{r^2}\right) = \frac{1}{r^2}(x, y, z)$

$= \frac{1}{r^2}\vec{r}$

(6) (b) $\nabla^2(1/r) = 0$, $r \neq 0$; and, in general, $\nabla^2 r^n = n(n+1)\,r^{n-2}$.

(1) $\nabla^2\left(\frac{1}{r}\right) = \nabla\cdot\left(\nabla\frac{1}{r}\right) = \nabla\cdot\left(-\frac{\vec{r}}{r^3}\right) = \nabla\cdot\left(\frac{-x}{r^3}, \frac{-y}{r^3}, \frac{-z}{r^3}\right)$

$$\frac{\partial}{\partial x}\left(\frac{-x}{r^3}\right) = \frac{\partial}{\partial x}\left[\frac{-x}{(x^2+y^2+z^2)^{3/2}}\right]$$

$$= \frac{(x^2+y^2+z^2)^{3/2}(-1) + (x)\frac{3}{2}(x^2+y^2+z^2)^{\frac{1}{2}}(2x)}{(x^2+y^2+z^2)^3}$$

$$= \frac{3x^2(x^2+y^2+z^2)^{\frac{1}{2}} - (x^2+y^2+z^2)^{3/2}}{(x^2+y^2+z^2)^3}$$

$$= \frac{3x^2 r - r^3}{r^6}$$

Similarly, $\frac{\partial}{\partial y}\left(-\frac{y}{r^3}\right) = \frac{3y^2 r - r^3}{r^6}$

$$\frac{\partial}{\partial z}\left(-\frac{z}{r^3}\right) = \frac{3z^2 r - r^3}{r^6}$$

$$\therefore \nabla\cdot\left(-\frac{\vec{r}}{r^3}\right) = \frac{\partial}{\partial x}\left(-\frac{x}{r^3}\right) + \frac{\partial}{\partial y}\left(-\frac{y}{r^3}\right) + \frac{\partial}{\partial z}\left(-\frac{z}{r^3}\right)$$

$$= \frac{(3x^2 r - r^3) + (3y^2 r - r^3) + (3z^2 r - r^3)}{r^6}$$

$$= \frac{3r(x^2+y^2+z^2) - 3r^3}{r^6} = \frac{3r(r^2) - 3r^3}{r^6}$$

$$= 0$$

$$\therefore \nabla^2\left(\frac{1}{r}\right) = 0$$

(2) From (a), $\nabla(r^n) = nr^{n-2}\vec{r} = nr^{n-2}(x, y, z)$

$$\therefore \frac{\partial}{\partial x}\left(nr^{n-2}x\right) = \frac{\partial}{\partial x}\left(nx\left[x^2+y^2+z^2\right]^{\frac{n-2}{2}}\right)$$

$$= n\left(x^2+y^2+z^2\right)^{\frac{n-2}{2}} + nx\left(\frac{n-2}{2}\right)\left(x^2+y^2+z^2\right)^{\frac{n-4}{2}}(2x)$$

$$= n(r)^{n-2} + x^2 n(n-2)(r)^{n-4}$$

Similarly, $\frac{\partial}{\partial y}\left(nr^{n-2}y\right) = n(r)^{n-2} + y^2 n(n-2)(r)^{n-4}$

$$\frac{\partial}{\partial z}\left(nr^{n-2}z\right) = n(r)^{n-2} + z^2 n(n-2)(r)^{n-4}$$

$$\therefore \nabla^2(r^n) = \frac{\partial}{\partial x}\left(nr^{n-2}x\right) + \frac{\partial}{\partial y}\left(nr^{n-2}y\right) + \frac{\partial}{\partial z}\left(nr^{n-2}z\right)$$

$$= 3n(r)^{n-2} + \left(x^2+y^2+z^2\right)n(n-2)(r)^{n-4}$$

$$= 3n(r)^{n-2} + r^2 n(n-2)(r)^{n-4}$$

$$= 3n(r)^{n-2} + n(n-2)(r)^{n-2}$$

$$= \left(3n + n^2 - 2n\right)(r)^{n-2}$$

$$= \underline{n(n+1)r^{n-2}}$$

(c) (c) $\nabla \cdot (\mathbf{r}/r^3) = 0$; and, in general, $\nabla \cdot (r^n \mathbf{r}) = (n+3)\, r^n$.

(1) $\dfrac{\vec{r}}{r^3} = -\nabla\left(\dfrac{1}{r}\right)$ from (a).

$\therefore \nabla \cdot \left(\dfrac{\vec{r}}{r^3}\right) = \nabla \cdot \left(-\nabla \dfrac{1}{r}\right) = -\nabla \cdot \left(\nabla \dfrac{1}{r}\right)$

$= -\nabla^2 \left(\dfrac{1}{r}\right) = -(0) = 0$ from (6)

(2) From Identity #7, $\operatorname{div}(f\vec{F}) = f \operatorname{div}\vec{F} + \vec{F} \cdot \nabla f$

$\therefore \nabla \cdot (r^n \vec{r}) = r^n \nabla \cdot \vec{r} + \vec{r} \cdot (\nabla r^n)$

$\nabla \cdot \vec{r} = \dfrac{\partial}{\partial x}(x) + \dfrac{\partial}{\partial y}(y) + \dfrac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$

$\therefore r^n \nabla \cdot \vec{r} = 3 r^n$ [1]

From (a), $\nabla r^n = n r^{n-2} \vec{r}$

$\therefore \vec{r} \cdot (\nabla r^n) = \vec{r} \cdot (n r^{n-2} \vec{r})$

$= n r^{n-2}\, \vec{r} \cdot \vec{r} = n r^{n-2} (r^2)$

$= n r^n$ [2]

$\therefore$ From [1], [2],

$\nabla \cdot (r^n \vec{r}) = 3 r^n + n r^n = \underline{(n+3) r^n}$

(d) (d) $\nabla \times \mathbf{r} = \mathbf{0}$; and, in general, $\nabla \times (r^n \mathbf{r}) = \mathbf{0}$.

(1) $\nabla \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & y & z \end{vmatrix}$

$= (0-0, 0-0, 0-0) = \vec{0}$

(2) From Identity #10 (p. 255 of text)

$\nabla \times (f\vec{F}) = f \nabla \times \vec{F} + \nabla f \times \vec{F}$

$\therefore \nabla \times (r^n \vec{r}) = r^n \nabla \times \vec{r} + \nabla r^n \times \vec{r}$

But $\nabla \times \vec{r} = \vec{0}$ from (a)

And $\nabla r^n = n r^{n-2} \vec{r}$ from (a)

$\therefore \nabla \times (r^n \vec{r}) = \vec{0} + (n r^{n-2} \vec{r}) \times \vec{r}$

$= n r^{n-2} (\vec{r} \times \vec{r}) = \vec{0}$

$\therefore \nabla \times (r^n \vec{r}) = \vec{0}$

39. **39.** Does $\nabla \times \mathbf{F}$ have to be perpendicular to $\mathbf{F}$?

No. For example, let $\vec{F} = (z, x, z)$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ z & x & z \end{vmatrix} = (0, 1, 1)$$

$$\therefore \vec{F} \cdot (\nabla \times \vec{F}) = x + z$$

40.

40. Let $\mathbf{F}(x, y, z) = 3x^2 y\mathbf{i} + (x^3 + y^3)\mathbf{j}$.

(a) Verify that curl $\mathbf{F} = \mathbf{0}$.

(b) Find a function f such that $\mathbf{F} = \nabla f$. (Techniques for constructing f in general are given in Chapter 8. The one in this problem should be sought by trial and error.)

(a) $$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3x^2 y & x^3 + y^3 & 0 \end{vmatrix}$$

$$= \left(\frac{\partial}{\partial x}(0) - \frac{\partial}{\partial z}(x^3 + y^3), \frac{\partial}{\partial z}(3x^2 y) - \frac{\partial}{\partial x}(0), \frac{\partial}{\partial x}(x^3 + y^3) - \frac{\partial}{\partial y}(3x^2 y) \right)$$

$$= (0 - 0, 0 - 0, 3x^2 - 3x^2) = (0, 0, 0) = \vec{0}$$

(b) $$\left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right) = (3x^2 y, x^3 + y^3, 0)$$

$$\therefore \frac{\partial f}{\partial x} = 3x^2 y \Rightarrow f(x, y, z) = x^3 y + g(y, z)$$

$$\text{as } \frac{\partial}{\partial x} g(y, z) = 0$$

$$\therefore \frac{\partial f}{\partial y} = x^3 + \frac{\partial g}{\partial y} = x^3 + y^3 \Rightarrow g(y, z) = \frac{y^4}{4} + h(z)$$

as $\frac{\partial}{\partial y} h(z) = 0$.

$\therefore f(x,y,z) = x^3 y + \frac{y^4}{4} + h(z)$

$\frac{\partial f}{\partial z} = h'(z) = 0 \Rightarrow h(z) = C$, a constant.

$\therefore f(x,y,z) = x^3 y + \frac{y^4}{4} + C$, C a constant

41.

41. Show that the real and imaginary parts of each of the following complex functions form the components of an irrotational and incompressible vector field in the plane; here $i = \sqrt{-1}$.

(a) $(x - iy)^2$

(b) $(x - iy)^3$

(c) $e^{x-iy} = e^x(\cos y - i \sin y)$

(a) $(x-iy)^2 = x^2 - y^2 - 2xyi$

$\therefore \vec{F} = (x^2 - y^2, -2xy, 0)$

$\therefore \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 - y^2 & -2xy & 0 \end{vmatrix}$

$= (0 - 0, 0 - 0, -2y - (-2y)) = \vec{0}$

$\therefore$ irrotational

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^2 - y^2) + \frac{\partial}{\partial y}(-2xy) + \frac{\partial}{\partial z}(0)$$

$$= 2x + (-2x) = \underline{0}$$

$\therefore$ incompressible

(b) $(x - iy)^3 = x^3 + 3x^2(-iy) + 3x(-iy)^2 + (-iy)^3$

$$= x^3 - 3xy^2 + (-3x^2y + y^3)i$$

$$\therefore \vec{F} = (x^3 - 3xy^2, -3x^2y + y^3, 0)$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^3 - 3xy^2 & -3x^2y + y^3 & 0 \end{vmatrix}$$

$$= (0 - 0, 0 - 0, -6xy - (-6xy)) = \underline{\vec{0}}$$

$\therefore$ irrotational

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^3 - 3xy^2) + \frac{\partial}{\partial y}(-3x^2y + y^3) + \frac{\partial}{\partial z}(0)$$

$$= 3x^2 - 3y^2 + (-3x^2 + 3y^2) = \underline{0}$$

$\therefore$ incompressible

(c) $\vec{F} = (e^x \cos y, -e^x \sin y, 0)$

$$\therefore \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ e^x \cos y & -e^x \sin y & 0 \end{vmatrix}$$

$$= (0-0,\ 0-0,\ -e^x \sin y - (-e^x \sin y))$$

$$= (0, 0, 0) = \vec{0}$$

$\therefore$ irrotational

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(e^x \cos y) + \frac{\partial}{\partial y}(-e^x \sin y) + \frac{\partial}{\partial z}(0)$$

$$= e^x \cos y + (-e^x \cos y) = 0$$

$\therefore$ incompressible.

Review Exercises for Chapter 4

Note Title 9/19/2016

For Exercises 1 to 4, at the indicated point, compute the velocity vector, the acceleration vector, the speed, and the equation of the tangent line.

1. **1.** $\mathbf{c}(t) = (t^3 + 1,\ e^{-t},\ \cos(\pi t/2))$, at $t = 1$

$\vec{c}'(t) = \left[3t^2,\ -e^{-t},\ -\frac{\pi}{2}\sin\left(\frac{\pi t}{2}\right)\right]$

$\therefore \vec{c}'(1) = \left[3,\ -\frac{1}{e},\ -\frac{\pi}{2}\right]$ = velocity vector

$\|\vec{c}'(1)\| = \sqrt{3^2 + \frac{1}{e^2} + \frac{\pi^2}{4}}$ = speed

$\vec{c}''(t) = \left[6t,\ e^{-t},\ -\frac{\pi^2}{4}\cos\left(\frac{\pi t}{2}\right)\right]$

$\therefore \vec{c}''(1) = \left[6,\ \frac{1}{e},\ 0\right]$ = acceleration vector

Tangent line $= \vec{c}(1) + s\,\vec{c}'(1)$, $s \in \mathbb{R}$

$= \left(2,\ \frac{1}{e},\ 0\right) + s\left(3,\ -\frac{1}{e},\ -\frac{\pi}{2}\right)$

2. **2.** $\mathbf{c}(t) = (t^2 - 1,\ \cos(t^2),\ t^4)$, at $t = \sqrt{\pi}$

Velocity vector: $\vec{c}'(t) = \left[2t,\ -2t\sin(t^2),\ 4t^3\right]$

$\therefore \vec{c}'(\sqrt{\pi}) = \left[2\sqrt{\pi},\ 0,\ 4\pi^{3/2}\right]$

speed $= \|\vec{c}'(\sqrt{\pi})\| = \sqrt{4\pi + 16\pi^3} = 2\sqrt{\pi + \pi^3}$

Acceleration vector: $\vec{c}''(t) = [2, -2\sin(t^2) - 4t^2\cos(t^2), 12t^2]$

$\therefore \vec{c}''(\sqrt{\pi}) = [2, 4\pi, 12\pi]$

Tangent line: $\vec{c}(\sqrt{\pi}) + s\,\vec{c}'(\sqrt{\pi})$, $s \in \mathbb{R}$

$= (\pi - 1, -1, \pi^2) + s(2\sqrt{\pi}, 0, 4\pi^{3/2})$

3. **3.** $\mathbf{c}(t) = (e^t, \sin t, \cos t)$, at $t = 0$

Velocity: $\vec{c}'(t) = [e^t, \cos t, -\sin t]$

$\therefore \vec{c}'(0) = [1, 1, 0]$

Speed: $\|\vec{c}'(0)\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$

Acceleration: $\vec{c}''(t) = [e^t, -\sin t, -\cos t]$

$\therefore \vec{c}''(0) = [1, 0, -1]$

Tangent line: $\vec{c}(0) + s\vec{c}'(0)$, $s \in \mathbb{R}$

$\therefore (1, 0, 1) + s(1, 1, 0)$

4. **4.** $\mathbf{c}(t) = \dfrac{t^2}{1+t^2}\mathbf{i} + t\mathbf{j} + \mathbf{k}$, at $t = 2$

Velocity: $\vec{c}\,'(t) = \left\{ \dfrac{(1+t^2)2t - t^2(2t)}{(1+t^2)^2}, 1, 0 \right\}$

$= \left\{ \dfrac{2t}{(1+t^2)^2}, 1, 0 \right\}$

$\therefore \vec{c}\,'(2) = \left\{ \dfrac{4}{25}, 1, 0 \right\}$

Speed: $\|\vec{c}\,'(2)\| = \sqrt{\dfrac{16}{625} + 1 + 0} = \sqrt{\dfrac{641}{625}} \simeq 1.01$

Acceleration: $\vec{c}\,''(t) = \left[\dfrac{(1+t^2)^2 2 - 2t(2)(1+t^2)(2t)}{(1+t^2)^4}, 0, 0 \right]$

$= \left[\dfrac{2t^4 + 4t^2 + 2 - 8t^2 - 8t^4}{(1+t^2)^4}, 0, 0 \right]$

$= \left[\dfrac{-6t^4 - 4t^2 + 2}{(1+t^2)^4}, 0, 0 \right]$

$\therefore \vec{c}\,''(2) = \left\{ \dfrac{-6(16) - 4(4) + 2}{625}, 0, 0 \right\}$

$= \left[-\dfrac{110}{625}, 0, 0 \right] = \left[-\dfrac{22}{125}, 0, 0 \right]$

Tangent line: $\vec{c}(2) + s\,\vec{c}\,'(2)$, $s \in \mathbb{R}$

$= \left(\dfrac{4}{5}, 2, 1\right) + s\left(\dfrac{4}{25}, 1, 0\right)$

5. **5.** Calculate the tangent and acceleration vectors for the helix $\mathbf{c}(t) = (\cos t, \sin t, t)$ at $t = \pi/4$.

Tangent: $\vec{c}'(t) = (-\sin t, \cos t, 1)$

$\therefore \vec{c}'\left(\frac{\pi}{4}\right) = \left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 1\right)$

Acceleration: $\vec{c}''(t) = (-\cos t, -\sin t, 0)$

$\therefore \vec{c}''\left(\frac{\pi}{4}\right) = \left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}, 0\right)$

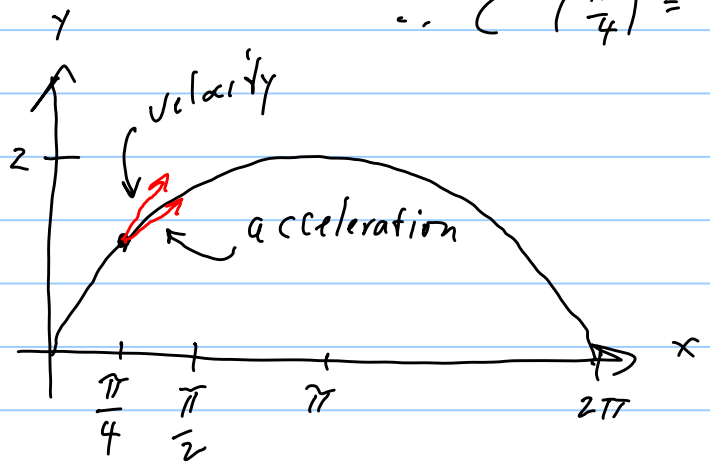
6. **6.** Calculate the tangent and acceleration vector for the cycloid $\mathbf{c}(t) = (t - \sin t, 1 - \cos t)$ at $t = \pi/4$ and sketch.

Tangent: $\vec{c}'(t) = (1 - \cos t, \sin t)$

$\therefore \vec{c}'\left(\frac{\pi}{4}\right) = \left(1 - \frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

Acceleration: $\vec{c}''(t) = (\sin t, \cos t)$

$\therefore \vec{c}''\left(\frac{\pi}{4}\right) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$



7. **7.** Let a particle of mass m move on the path $\mathbf{c}(t) = (t^2, \sin t, \cos t)$. Compute the force acting on the particle at $t = 0$.

$\vec{c}'(t) = (2t, \cos t, -\sin t)$ $\quad \vec{c}''(t) = (2, -\sin t, -\cos t)$

$\therefore \vec{c}''(0) = (2, 0, -1)$

$\therefore \vec{F} = m\vec{a} = m\vec{c}''(0) = m(2, 0, -1) = (2m, 0, -m)$

8. **8.** (a) Let $\mathbf{c}(t)$ be a path with $\|\mathbf{c}(t)\| =$ constant; that is, the curve lies on a sphere. Show that $\mathbf{c}'(t)$ is orthogonal to $\mathbf{c}(t)$.

(b) Let $\mathbf{c}$ be a path whose speed is never zero. Show that $\mathbf{c}$ has constant speed if and only if the acceleration vector $\mathbf{c}''$ is always perpendicular to the velocity vector $\mathbf{c}'$.

(a) $\|\vec{c}(t)\| = K$, a constant $\Rightarrow$

$\|\vec{c}(t)\|^2 = K^2 \Rightarrow \vec{c}(t) \cdot \vec{c}(t) = K^2$

$\therefore \frac{d}{dt}\left(\vec{c}(t) \cdot \vec{c}(t)\right) = \frac{d}{dt}(K^2) = 0$

$\therefore \vec{c}'(t) \cdot \vec{c}(t) + \vec{c}(t) \cdot \vec{c}'(t) = 2\vec{c} \cdot \vec{c}' = 0$

$\therefore \vec{c}(t) \cdot \vec{c}'(t) = 0 \Rightarrow \vec{c}(t) \perp \vec{c}'(t)$

(b) (1) Suppose $\vec{c}'(t) = (a, b, c)$, a, b, c are constants, not all zero.

$\therefore \frac{d}{dt}\vec{c}'(t) = \vec{c}''(t) = (0, 0, 0)$

$\therefore \vec{c}'(t) \cdot (0,0,0) = 0 \Rightarrow \vec{c}' \perp \vec{c}''$

(2) Suppose $\vec{c}'(t) \perp \vec{c}''(t)$ for all t.

$\therefore \vec{c}'(t) \cdot \vec{c}''(t) = 0 \Rightarrow 2\vec{c}'(t) \cdot \vec{c}''(t) = 0$

$\Rightarrow \frac{d}{dt}\left(\vec{c}'(t) \cdot \vec{c}'(t)\right) = 0$

$\Rightarrow \vec{c}'(t) \cdot \vec{c}'(t) = K$, a constant

$\Rightarrow \|\vec{c}'(t)\|^2 = K$

$\Rightarrow \|\vec{c}'(t)\|$ = speed of $\vec{c}(t) = \sqrt{K}$, a constant

9.

9. Let $\mathbf{c}(t) = (\cos t, \sin t, \sqrt{3}t)$ be a path in $\mathbb{R}^3$.

(a) Find the velocity and acceleration of this path.

(b) Find a parametrization for the tangent line to this path at $t = 0$.

(c) Find the arc length of this path for $t \in [0, 2\pi]$.

(a) $\vec{c}'(t) = (-\sin t, \cos t, \sqrt{3})$ = velocity

$\vec{c}''(t) = (-\cos t, -\sin t, 0)$ = acceleration

(b) $\vec{l}(s) = \vec{c}(t) + s\vec{c}'(t)$, $s \in \mathbb{R}$

$\therefore \vec{l}(s) = \vec{c}(0) + s\vec{c}'(0)$

$= (1,0,0) + s(0,1,\sqrt{3})$

(c) Arc length $= \int_0^{2\pi} \|\vec{c}\,'(t)\|\,dt$

$= \int_0^{2\pi} \sqrt{\sin^2 t + \cos^2 t + 3}\,dt = \int_0^{2\pi} \sqrt{4}\,dt = 2t\Big|_0^{2\pi}$

$= 4\pi$

10. **10.** Let $\mathbf{F}(x, y, z) = (\sin(xz), e^{xy}, x^2y^3z^5)$.

(a) Find the divergence of $\mathbf{F}$.

(b) Find the curl of $\mathbf{F}$.

(a) $\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = z\cos(xz) + ye^{xy} + 5x^2y^3z^4$

(b) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \sin(xz) & e^{xy} & x^2y^3z^5 \end{vmatrix}$

$= [3x^2y^2z^5 - 0,\ x\cos(xz) - 2xy^3z^5,\ ye^{xy} - 0]$

$= (3x^2y^2z^5,\ x\cos(xz) - 2xy^3z^5,\ ye^{xy})$

11. **11.** Verify that the gravitational force field $\mathbf{F}(x, y, z) = -A\frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}}$, where A is some constant, is curl free away from the origin.

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \dfrac{-Ax}{(x^2+y^2+z^2)^{3/2}} & \dfrac{-Ay}{(x^2+y^2+z^2)^{3/2}} & \dfrac{-Az}{(x^2+y^2+z^2)^{5/2}} \end{vmatrix}$$

$$= \Bigg[\frac{3}{2}\frac{(-Az)(2y)}{(x^2+y^2+z^2)^{5/2}} - \frac{3}{2}\frac{(-Ay)(2z)}{(x^2+y^2+z^2)^{5/2}}, \quad -Azy + Ayz$$

$$\frac{3}{2}\frac{(-Ax)(2z)}{(x^2+y^2+z^2)^{5/2}} - \frac{3}{2}\frac{(-Az)(2x)}{(x^2+y^2+z^2)^{5/2}}, \quad -Axz + Azx$$

$$\frac{3}{2}\frac{(-Ay)(2x)}{(x^2+y^2+z^2)^{5/2}} - \frac{3}{2}\frac{(-Ax)(2y)}{(x^2+y^2+z^2)^{5/2}} \Bigg] \quad -Ayx + Axy$$

$$= [0, 0, 0]$$

12. **12.** Show that the vector field $\mathbf{V}(x, y, z) = 2x\mathbf{i} - 3y\mathbf{j} + 4z\mathbf{k}$ is not the curl of any vector field.

If $\vec{V} = \nabla \times \vec{F}$, some $\vec{F}$, Then $\nabla \cdot (\nabla \times \vec{F}) = \nabla \cdot \vec{V} = 0$

But $\nabla \cdot \vec{V} = \frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(-3y) + \frac{\partial}{\partial z}(4z)$

$$= 2 + (-3) + 4 = 3 \neq 0$$

$\therefore \vec{V}$ is not the curl of some $\vec{F}$.

13. **13.** Express the arc length of the curve $x^2 = y^3 = z^5$ between $x = 1$ and $x = 4$ as an integral, using a suitable parametrization.

Let $x = t$. $\therefore y = t^{2/3}$ as $y^3 = t^2$

$z = t^{2/5}$ as $z^5 = t^2$

$\therefore \vec{c}(t) = (t, t^{2/3}, t^{2/5})$

$\therefore \vec{c}'(t) = (1, \frac{2}{3}t^{-\frac{1}{3}}, \frac{2}{5}t^{-\frac{3}{5}})$

Between $x=1$ and $x=4$, $t=1$, $t=4$

$\therefore$ Arc length $= \int_1^4 \sqrt{1 + \frac{4}{9}t^{-\frac{2}{3}} + \frac{4}{25}t^{-\frac{6}{5}}}\, dt$

14. **14.** Find the arc length of $\mathbf{c}(t) = t\mathbf{i} + (\log t)\mathbf{j} + 2\sqrt{2t}\mathbf{k}$ for $1 \le t \le 2$.

$\vec{c}'(t) = (1, \frac{1}{t}, 2(2t)^{-\frac{1}{2}}) = (1, \frac{1}{t}, \frac{2}{\sqrt{2t}})$

$\therefore \int_1^2 \sqrt{1 + \frac{1}{t^2} + \frac{4}{2t}}\, dt = \int_1^2 \sqrt{\frac{2t^2 + 2 + 4t}{2t^2}}\, dt$

$= \int_1^2 \sqrt{\frac{(t+1)^2}{t^2}}\, dt = \int_1^2 \frac{t+1}{t}\, dt$ since $1 \le t \le 2$

$$= \int_1^2 1 + \frac{1}{t}\,dt = \int_1^2 dt + \int_1^2 \frac{dt}{t} = t\Big|_1^2 + \ln t\Big|_1^2$$

$$= 1 + \ln 2 - \ln 1 = \underline{1 + \ln 2}$$

15.

15. A particle is constrained to move around the unit circle in the xy plane according to the formula $(x, y, z) = (\cos(t^2), \sin(t^2), 0)$, $t \geq 0$.

(a) What are the velocity vector and speed of the particle as functions of t?

(b) At what point on the circle should the particle be released to hit a target at (2, 0, 0)? (Be careful about which direction the particle is moving around the circle.)

(c) At what time t should the release take place? (Use the smallest $t > 0$ that will work.)

(d) What are the velocity and speed at the time of release?

Let $\vec{c}(t) = [\cos(t^2), \sin(t^2), 0]$

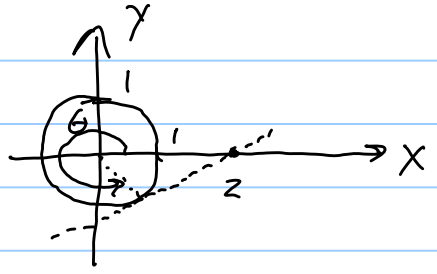
(a) Velocity $= \vec{c}\,'(t) = [-2t\sin(t^2), 2t\cos(t^2), 0]$

Speed $= \|\vec{c}\,'(t)\| = \sqrt{4t^2\sin^2(t^2) + 4t^2\cos^2(t^2)}$

$= \sqrt{4t^2} = \underline{2t}$ as $t \geq 0$

(b) Tangent line is the path the particle will follow when released.

$\therefore$ Find tangent line from unit circle to (2, 0)



This will give θ, then set $\theta = t^2$

$x^2 + y^2 = 1$, $y = -\sqrt{1-x^2}$, $\frac{dy}{dx} = -\frac{1}{2}(1-x^2)^{-\frac{1}{2}}(-2x)$

$\therefore$ slope of line $= \frac{x}{\sqrt{1-x^2}}$ at point $(x, -\sqrt{1-x^2})$

$\therefore$ Two points: $(2,0)$ and $(x, -\sqrt{1-x^2})$

$\therefore \frac{0-(-\sqrt{1-x^2})}{2-x} = \frac{x}{\sqrt{1-x^2}} = \frac{y_1 - y_2}{x_1 - x_2} = m$

$\therefore 1-x^2 = x(2-x) = 2x - x^2$, $x = \frac{1}{2}$

$\therefore$ release at point $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right)$

(c) $\vec{c}(t) = [\cos(t^2), \sin(t^2), 0] = \left(\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right)$

$\therefore t^2 = 2\pi - \frac{\pi}{3}$ since $\cos(-\frac{\pi}{3}) = \frac{1}{2}$, $\sin(-\frac{\pi}{3}) = -\frac{\sqrt{3}}{2}$

$= \frac{5\pi}{3}$

$\therefore$ Release at $t = \sqrt{\frac{5\pi}{3}}$

(d) $\vec{c}'(\sqrt{5\pi/3}) = \left(-2\sqrt{\frac{5\pi}{3}}\sin\left(\frac{5\pi}{3}\right), 2\sqrt{\frac{5\pi}{3}}\cos\left(\frac{5\pi}{3}\right), 0\right)$

$= \left\{-2\sqrt{\frac{5\pi}{3}}\left(-\frac{\sqrt{3}}{2}\right), 2\sqrt{\frac{5\pi}{3}}\left(\frac{1}{2}\right), 0\right\}$

Velocity $= \left(\sqrt{5\pi}, \sqrt{5\pi/3}, 0\right)$

Speed $= \sqrt{5\pi + \frac{5\pi}{3} + 0} = \sqrt{\frac{20\pi}{3}} = 2\sqrt{\frac{5\pi}{3}}$

(e) (e) At what time is the target hit?

Released a time $t = \sqrt{\frac{5\pi}{3}}$.

$$\frac{\text{Distance to travel}}{\text{Speed}} = \frac{\left\|\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}, 0\right) - (2,0,0)\right\|}{2\sqrt{5\pi/3}}$$

$$= \frac{\sqrt{\left(\frac{3}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2 + 0}}{2\sqrt{\frac{5\pi}{3}}} = \frac{\sqrt{\frac{9}{4} + \frac{3}{4}}}{2\sqrt{5\pi/3}}$$

$$= \frac{\sqrt{3}}{2\sqrt{\frac{5\pi}{3}}} = \frac{3}{2\sqrt{5\pi}}$$

$$\therefore t = \sqrt{\frac{5\pi}{3}} + \frac{3}{2\sqrt{5\pi}} = \frac{5\pi}{3\sqrt{5\pi}} + \frac{3}{2\sqrt{5\pi}} = \left(\frac{5\pi}{3} + \frac{3}{2}\right)\frac{1}{\sqrt{5\pi}}$$

16. **16.** A particle of mass m moves under the influence of a force $\mathbf{F} = -k\mathbf{r}$, where k is a constant and $\mathbf{r}(t)$ is the position of the particle at time t.

(a) Write down differential equations for the components of $\mathbf{r}(t)$.

(b) Solve the equations in part (a) subject to the initial conditions $\mathbf{r}(0) = \mathbf{0}$, $\mathbf{r}'(0) = 2\mathbf{j} + \mathbf{k}$.

Assume $\vec{F}, \vec{r} \in \mathbb{R}^3$

(c) $\vec{F} = m\vec{a} = m\vec{r}''(t) = -k\vec{r}(t)$

Let $\vec{r}(t) = (r_x(t), r_y(t), r_z(t))$

$\therefore m\frac{d^2 r_x(t)}{dt^2} = -k r_x(t)$, or $r_x'' = -\frac{k}{m} r$

$m\frac{d^2 r_y(t)}{dt^2} = -k r_y(t)$, or $r_y'' = -\frac{k}{m} r$

$m\frac{d^2}{dt^2} r_z(t) = -k r_z(t)$, or $r_z'' = -\frac{k}{m} r$

(b) Let $r_i(t) = a_i \cos\left(\sqrt{\frac{k}{m}}\, t\right) + b_i \sin\left(\sqrt{\frac{k}{m}}\, t\right)$

$\therefore r_i''(t) = -a_i \frac{k}{m} \cos\left(\sqrt{\frac{k}{m}}\, t\right) - b_i \frac{k}{m} \sin\left(\sqrt{\frac{k}{m}}\, t\right)$

$= -\frac{k}{m} r_i(t)$

$\therefore$ if $\vec{r}(0) = \vec{0}$, then $r_i(0) = a_i \cos(0) + b_i \sin(0)$

$\therefore 0 = a_i$

$\therefore r_i(t) = b_i \sin\left(\sqrt{\frac{k}{m}}\, t\right)$

$\vec{r}'(0) = 2\hat{j} + \hat{k} \Rightarrow r_x'(0) = 0 = b_x \sqrt{\frac{k}{m}} \cos(0)$

$\Rightarrow b_x = 0$

$r_y'(0) = 2 = b_y \sqrt{\frac{k}{m}} \cos(0)$

$\Rightarrow b_y = 2\sqrt{\frac{m}{k}}$

$$r_2'(0) = 1 = b_2\sqrt{\frac{k}{m}}\cos(0)$$

$$\Rightarrow b_2 = \sqrt{\frac{m}{k}}$$

$$\therefore \vec{r}(t) = \left\{0,\ 2\sqrt{\frac{m}{k}}\sin\left(\sqrt{\frac{k}{m}}\,t\right),\ \sqrt{\frac{m}{k}}\sin\left(\sqrt{\frac{k}{m}}\,t\right)\right\}$$

17. **17.** Write the curve described by the equations $x - 1 = 2y + 1 = 3z + 2$ in parametric form.

Let $t = x-1 = 2y+1 = 3z+2$

$$\therefore x = t+1$$
$$y = (t-1)/2$$
$$z = (t-2)/3$$

$$\therefore \vec{c}(t) = \left\{t+1,\ \frac{t-1}{2},\ \frac{t-2}{3}\right\}$$

18. **18.** Write the curve $x = y^3 = z^2 + 1$ in parametric form.

Let $t = x = y^3 = z^2+1$

$$\therefore x = t$$
$$y = t^{1/3}$$
$$z = (t-1)^{\frac{1}{2}},\ t \geq 1$$

$$\therefore \vec{c}(t) = \left\{t,\ t^{1/3},\ \sqrt{t-1}\right\},\ t \geq 1$$

19. **19.** Show that $\mathbf{c}(t) = (1/(1-t),\ 0,\ e^t/(1-t))$ is a flow line of the vector field defined by $\mathbf{F}(x, y, z) = (x^2,\ 0,\ z(1+x))$.

To show a flow line, must show $\vec{c}'(t) = \vec{F}(\vec{c}(t))$, i.e., the force vector is the same as the velocity vector, at the given point.

$$\therefore \vec{F}\left(\frac{1}{1-t}, 0, \frac{e^t}{1-t}\right) = \left[\left(\frac{1}{1-t}\right)^2, 0, \frac{e^t}{1-t}\left(1+\frac{1}{1-t}\right)\right]$$

$$= \left[\frac{1}{(1-t)^2}, 0, \frac{e^t(2-t)}{(1-t)^2}\right]$$

$$\vec{c}'(t) = \left[-\frac{1}{(1-t)^2}(-1), 0, \frac{(1-t)e^t - e^t(-1)}{(1-t)^2}\right]$$

$$= \left[\frac{1}{(1-t)^2}, 0, \frac{e^t(2-t)}{(1-t)^2}\right]$$

$$\therefore \vec{F}(\vec{c}(t)) = \vec{c}'(t)$$

20. **20.** Let $\mathbf{F}(x, y) = f(x^2 + y^2)[-y\mathbf{i} + x\mathbf{j}]$ for a function f of one variable. What equation must $g(t)$ satisfy for

$$\mathbf{c}(t) = [\cos g(t)]\mathbf{i} + [\sin g(t)]\mathbf{j}$$

to be a flow line for $\mathbf{F}$?

If $\vec{c}(t)$ is a path, the velocity vector $\vec{c}'(t)$ must equal the force vector at point $\vec{c}(t)$.

$$\therefore \vec{F}(\vec{c}(t)) = \vec{c}'(t).$$

$$\vec{c}'(t) = g'(t)\left[-\sin(g(t)), \cos(g(t))\right].$$

$$\vec{F}(\vec{c}(t)) = f\left(\cos^2(g(t)) + \sin^2(g(t))\right)\left[-\sin(g(t)), \cos(g(t))\right]$$

$$= f(1)\left[-\sin(g(t)), \cos(g(t))\right]$$

$$\therefore \vec{F}(\vec{c}(t)) = \vec{c}'(t) \iff g'(t) = f(1)$$

$$\therefore g'(t) = f(1)$$

Compute $\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$ for the vector fields in Exercises 21 to 24.

21. **21.** $\mathbf{F} = 2x\mathbf{i} + 3y\mathbf{j} + 4z\mathbf{k}$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(2x) + \frac{\partial}{\partial y}(3y) + \frac{\partial}{\partial z}(4z)$$

$$= 2 + 3 + 4 = 9$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2x & 3y & 4z \end{vmatrix}$$

$$= [0-0, 0-0, 0-0] = \vec{0}$$

22. **22.** $\mathbf{F} = x^2\mathbf{i} + y^2\mathbf{j} + z^2\mathbf{k}$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x^2) + \frac{\partial}{\partial y}(y^2) + \frac{\partial}{\partial z}(z^2)$$

$$= 2x + 2y + 2z$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2 & y^2 & z^2 \end{vmatrix}$$

$$= [0-0,\ 0-0,\ 0-0] = \vec{0}$$

23. **23.** $\mathbf{F} = (x+y)\mathbf{i} + (y+z)\mathbf{j} + (z+x)\mathbf{k}$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x+y) + \frac{\partial}{\partial y}(y+z) + \frac{\partial}{\partial z}(z+x)$$

$$= 1 + 1 + 1 = 3$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x+y & y+z & z+x \end{vmatrix}$$

$$= [0-1,\ 0-1,\ 0-1] = [-1, -1, -1]$$

24. **24.** $\mathbf{F} = x\mathbf{i} + 3xy\mathbf{j} + z\mathbf{k}$

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(3xy) + \frac{\partial}{\partial z}(z)$$

$$= 1 + 3x + 1 = 2 + 3x$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x & 3xy & z \end{vmatrix}$$

$$= [0-0,\ 0-0,\ 3y-0] = [0, 0, 3y]$$

Compute the divergence and curl of the vector fields in Exercises 25 and 26 at the points indicated.

25. **25.** $\mathbf{F}(x, y, z) = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$, at the point (1, 1, 1)

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(y) + \frac{\partial}{\partial y}(z) + \frac{\partial}{\partial z}(x)$$

$$= 0 + 0 + 0 = \underline{0}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y & z & x \end{vmatrix} = [0-1, 0-1, 0-1]$$

$$= \underline{[-1, -1, -1]}$$

"at the point" $(x, y, z) = (1, 1, 1)$ is irrelevant in this case.

26. **26.** $\mathbf{F}(x, y, z) = (x + y)^3\mathbf{i} + (\sin xy)\mathbf{j} + (\cos xyz)\mathbf{k}$, at the point (2, 0, 1)

$$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(x+y)^3 + \frac{\partial}{\partial y}(\sin(xy)) + \frac{\partial}{\partial z}(\cos(xyz))$$

$$= 3(x+y)^2 + x\cos(xy) - xy\sin(xyz)$$

$$\therefore \nabla \cdot \vec{F}(2, 0, 1) = 3(2)^2 + 2\cos(0) - 0$$

$$= 12 + 2 = \underline{14}$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ (x+y)^3 & \sin(xy) & \cos(xyz) \end{vmatrix}$$

$$= \left[-xz\sin(xyz) - 0,\ 0 - (-yz\sin(xyz)),\ y\cos(xy) - 3(x+y)^2\right]$$

$$= \left[-xz\sin(xyz),\ yz\sin(xyz),\ y\cos(xy) - 3(x+y)^2\right]$$

$$\therefore \nabla \times \vec{F}(2,0,1) = \left[0, 0, 0 - 3(2)^2\right] = \underline{[0, 0, -12]}$$

Calculate the gradients of the functions in Exercises 27 to 30, and verify that $\nabla \times \nabla f = \mathbf{0}$.

27. **27.** $f(x, y) = e^{xy} + \cos(xy)$

$$\nabla f = [f_x, f_y, f_z] = \underline{\left[ye^{xy} - y\sin(xy),\ xe^{xy} - x\sin(xy),\ 0\right]}$$

$$\nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ ye^{xy} - y\sin(xy) & xe^{xy} - x\sin(xy) & 0 \end{vmatrix}$$

$$= \left[0 - 0,\ 0 - 0,\ e^{xy} + xye^{xy} - \sin(xy) - xy\cos(xy) - \left(e^{xy} + xye^{xy} - \sin(xy) - xy\cos(xy)\right)\right]$$

$$= \underline{[0, 0, 0]}$$

28. **28.** $f(x, y) = \dfrac{x^2 - y^2}{x^2 + y^2}$

$$\nabla f = \left[\frac{(x^2+y^2)(2x) - (x^2-y^2)(2x)}{(x^2+y^2)^2},\ \frac{(x^2+y^2)(-2y) - (x^2-y^2)(2y)}{(x^2+y^2)^2}\right]$$

$$= \left[\frac{4xy^2}{(x^2+y^2)^2}, \frac{-4yx^2}{(x^2+y^2)^2}, 0 \right]$$

$$\nabla \times (\nabla f) = \begin{bmatrix} \vec{i} & \vec{j} & \vec{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \frac{4xy^2}{(x^2+y^2)^2} & \frac{-4yx^2}{(x^2+y^2)^2} & 0 \end{bmatrix}$$

$$= \left[0-0, 0-0, \frac{(x^2+y^2)^2(-8yx) - (-4yx^2)(2)(x^2+y^2)(2x)}{(x^2+y^2)^4} - \right.$$

$$\left. \frac{((x^2+y^2)^2(8xy) - (4xy^2)(2)(x^2+y^2)(2y))}{(x^2+y^2)^4} \right]$$

$$= \left[0, 0, \frac{(-16xy)(x^2+y^2)^2 + 16yx^3(x^2+y^2) + 16xy^3(x^2+y^2)}{(x^2+y^2)^4} \right]$$

$$= \left[0, 0, \frac{(-16xy)(x^2+y^2)^2 + 16xy(x^2+y^2)(x^2+y^2)}{(x^2+y^2)^4} \right]$$

$$= \left[0, 0, \frac{(-16xy)(x^2+y^2)^2 + 16xy(x^2+y^2)^2}{(x^2+y^2)^4} \right] = [0, 0, 0]$$

29. **29.** $f(x, y) = e^{x^2} - \cos(xy^2)$

$$\nabla F = [2xe^{x^2} + y^2 \sin(xy^2), 2yx \sin(xy^2), 0]$$

$$\nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xe^{x^2} + y^2 \sin(xy^2) & 2yx \sin(xy^2) & 0 \end{vmatrix}$$

$$= [0, 0, 2y \sin(xy^2) + 2xy^3 \cos(xy^2) - (2y \sin(xy^2) + 2xy^3 \cos(xy^2))]$$

$$= [0, 0, 0]$$

30. **30.** $f(x, y) = \tan^{-1}(x^2 + y^2)$

$$\nabla f = \left[\frac{1}{1 + (x^2+y^2)^2} \cdot 2x, \frac{1}{1 + (x^2+y^2)^2} \cdot 2y, 0 \right]$$

$$= \left[\frac{2x}{1 + (x^2+y^2)^2}, \frac{2y}{1 + (x^2+y^2)^2}, 0 \right]$$

$$\nabla \times (\nabla f) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ \dfrac{2x}{1 + (x^2+y^2)^2} & \dfrac{2y}{1 + (x^2+y^2)^2} & 0 \end{vmatrix}$$

$$= \left[0, 0, \frac{(1 + (x^2+y^2)^2)(0) - (2y)(2)(x^2+y^2)(2x)}{[1 + (x^2+y^2)^2]^2} - \right.$$

$$\left. \frac{(1 + (x^2+y^2)^2)(0) - (2x)(2)(x^2+y^2)(2y)}{[1 + (x^2+y^2)^2]^2} \right]$$

$$= \left[0, 0, -\frac{8xy(x^2+y^2) + 8xy(x^2+y^2)}{[1+(x^2+y^2)^2]^2}\right] = \underline{[0,0,0]}$$

31.

31. (a) Let $f(x, y, z) = xyz^2$; compute ∇f.

(b) Let $\mathbf{F}(x, y, z) = xy\mathbf{i} + yz\mathbf{j} + zy\mathbf{k}$; compute $\nabla \times \mathbf{F}$.

(c) Compute $\nabla \times (f\mathbf{F})$ using identity 10 of the list of vector identities. Compare with a direct computation.

(a) $\nabla f = [f_x, f_y, f_z] = [yz^2, xz^2, 2xyz]$

(b) $\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ xy & yz & zy \end{vmatrix} = [z-y, 0-0, 0-x]$

$= [z-y, 0, -x]$

(c) $\nabla \times (f\vec{F}) = f(\nabla \times \vec{F}) + \nabla f \times \vec{F}$

$= xyz^2[z-y, 0, -x] + [yz^2, xz^2, 2xyz] \times [z-y, 0, -x]$

$= [xyz^3 - xy^2z^2, 0, -x^2yz^2] + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ yz^2 & xz^2 & 2xyz \\ xy & yz & zy \end{vmatrix}$

$= [xyz^3 - xy^2z^2, 0, -x^2yz^2] + [xyz^3 - 2xy^2z^2, 2x^2yz - y^2z^3, y^2z^3 - x^2yz^2]$

$= [2xyz^3 - 3xy^2z^2, 2x^2yz - y^2z^3, y^2z^3 - 2x^2yz^2]$

$f\vec{F} = [x^2y^2z^2, xy^2z^3, xy^2z^3]$

$$\nabla \times (f\vec{F}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ x^2y^2z^2 & xy^2z^3 & xy^2z^5 \end{vmatrix}$$

$$= [2xyz^3 - 3xy^2z^2,\ 2x^2yz - y^2z^3,\ y^2z^3 - 2x^2yz^2]$$

Same as above

32. **32.** (a) Let $\mathbf{F} = 2xye^z\mathbf{i} + e^zx^2\mathbf{j} + (x^2ye^z + z^2)\mathbf{k}$. Compute $\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$.

(b) Find a function $f(x, y, z)$ such that $\mathbf{F} = \nabla f$.

(a) $$\nabla \cdot \vec{F} = \frac{\partial}{\partial x}(2xye^z) + \frac{\partial}{\partial y}(e^zx^2) + \frac{\partial}{\partial z}(x^2ye^z + z^2)$$

$$= 2ye^z + 0 + x^2ye^z + 2z$$

$$= 2z + (2 + x^2)ye^z$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2xye^z & e^zx^2 & x^2ye^z + z^2 \end{vmatrix}$$

$$= [x^2e^z - e^zx^2,\ 2xye^z - 2xye^z,\ 2xe^z - 2xe^z]$$

$$= [0, 0, 0]$$

(b) $$\nabla f = [f_x, f_y, f_z] = [2xye^z, e^zx^2, x^2ye^z + z^2] = \vec{F}$$

$$f_x = 2xye^z \Rightarrow f = x^2ye^z + g(y, z), \quad [1]$$

$g(y,z)$ some function of y and z but not x. $\therefore \frac{\partial}{\partial x} g = 0$.

Similarly, $f_y = e^z x^2 \Rightarrow f = x^2 y e^z + h(x,z)$, [2]

h some function of x, z, not y

Comparing [1], [2], $\frac{\partial g(y,z)}{\partial y} = \frac{\partial h(x,z)}{\partial x}$

which will be satisfied if $g(y,z) = h(x,z) = j(z)$,

j some function of z, not x, not y.

$\therefore$ So far, $f(x,y,z) = x^2 y e^z + j(z)$.

But $f_z = x^2 y e^z + z^2 = x^2 y e^z + j'(z)$.

$\therefore j(z) = \frac{z^3}{3} + C$, C some constant.

$\therefore \underline{f(x,y,z) = x^2 y e^z + \frac{z^3}{3} + C}$, C some constant.

33.

33. Let $\mathbf{F}(x, y) = f(x^2 + y^2)[-y\mathbf{i} + x\mathbf{j}]$, as in Exercise 20. Calculate div $\mathbf{F}$ and curl $\mathbf{F}$ and discuss your answers in view of the results of Exercise 20.

$\operatorname{div} \vec{F} = \frac{\partial}{\partial x}(-y f(x^2+y^2)) + \frac{\partial}{\partial y}(x f(x^2+y^2))$

$$= -y(f')(2x) + x(f')(2y)$$

$$= 2xy(f' - f') = 0 \qquad (\text{note } f: \mathbb{R}' \to \mathbb{R}').$$

$$\text{curl } \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ -y f(x^2+y^2) & x f(x^2+y^2) & 0 \end{vmatrix}$$

$$= [0, 0, f(x^2+y^2) + x f'(x^2+y^2)(2x) - \left(-f(x^2+y^2) - y f'(x^2+y^2)(2y)\right)]$$

$$= [0, 0, 2f(x^2+y^2) + (2x^2+2y^2) f'(x^2+y^2)]$$

$\text{div } \vec{F} = 0$ means the fluid is incompressible as it flows.

$\text{curl } \vec{F} \neq 0$ means points of the fluid rotate as the fluid flows.

34.

34. Let a particle of mass m move along the elliptical helix $\mathbf{c}(t) = (4\cos t, \sin t, t)$.

(a) Find the equation of the tangent line to the helix at $t = \pi/4$.

(b) Find the force acting on the particle at time $t = \pi/4$.

(c) Write an expression (in terms of an integral) for the arc length of the curve $\mathbf{c}(t)$ between $t = 0$ and $t = \pi/4$.

(a) $\vec{l}(t) = \vec{c}(t) + s\vec{c}'(t)$

$\therefore \vec{c}\left(\frac{\pi}{4}\right) = \left(2\sqrt{2}, \frac{\sqrt{2}}{2}, \frac{\pi}{4}\right)$

$\vec{c}'(t) = (-4\sin t, \cos t, 1) \quad \therefore \vec{c}'\left(\frac{\pi}{4}\right) = \left(-2\sqrt{2}, \frac{\sqrt{2}}{2}, 1\right)$

$\therefore \vec{l}\left(\frac{\pi}{4}\right) = \left[2\sqrt{2}, \frac{\sqrt{2}}{2}, \frac{\pi}{4}\right] + s\left[-2\sqrt{2}, \frac{\sqrt{2}}{2}, 1\right], \; s \in \mathbb{R}$

(5) $\vec{F} = m\vec{a} = m\vec{c}''\left(\frac{\pi}{4}\right)$

$\vec{c}''(t) = (-4\cos t, -\sin t, 0)$

$\vec{c}''\left(\frac{\pi}{4}\right) = \left(-2\sqrt{2}, -\frac{\sqrt{2}}{2}, 0\right)$

$\therefore \vec{F} = m\left(-2\sqrt{2}, -\frac{\sqrt{2}}{2}, 0\right)$

(c) $\int_0^{\frac{\pi}{4}} \|\vec{c}'(t)\| \, dt = \int_0^{\frac{\pi}{4}} \sqrt{16\sin^2 t + \cos^2 t + 1} \, dt$

35.

35. (a) Let $g(x, y, z) = x^3 + 5yz + z^2$ and let $h(u)$ be a function of one variable such that $h'(1) = 1/2$. Let $f = h \circ g$. Starting at $(1, 0, 0)$, in what directions is f changing at 50% of its maximum rate?

(b) For $g(x, y, z) = x^3 + 5yz + z^2$, calculate $\mathbf{F} = \nabla g$, the gradient of g, and verify directly that $\nabla \times \mathbf{F} = 0$ at each point (x, y, z).

(a) Note $g(1,0,0) = 1$. $g: \mathbb{R}^3 \to \mathbb{R}^1$, $f: \mathbb{R}^3 \to \mathbb{R}^1$

$\therefore \nabla f = h'(g)(\nabla g)$ by chain rule.

$\nabla g = (3x^2, 5z, 5y + 2z) \quad \therefore \nabla g(1,0,0) = (3, 0, 0)$

$\therefore \nabla f(1,0,0) = h'(g(1,0,0)) \cdot \nabla g(1,0,0)$

$= h'(1)(3,0,0) = \frac{1}{2}(3,0,0)$

∇f is the direction of greatest change of f.

Starting at $(1,0,0)$, this direction is $(\frac{3}{2}, 0, 0)$

If $\vec{s}$, a unit vector, is any other direction,

then we want $\nabla f \cdot \vec{s} = \|\nabla f\| \|\vec{s}\| \cos\theta$

$= \|\nabla f\| \cos\theta = 50\% \|\nabla f\|$, or $\cos\theta = \frac{1}{2}$

$\therefore \theta = 60°$, or in a $60°$ direction from $(1,0,0)$

This is a "cone" of possible directions

making an angle of $60° = \frac{\pi}{3}$ radians with $\hat{i}$

(6) As above, $\nabla g = (3x^2, 5z, 5y + 2z)$

$$\nabla \times (\nabla g) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 3x^2 & 5z & 5y+2z \end{vmatrix}$$

$= [5-5, 0-0, 0-0] = [0,0,0]$

36.

36. (a) Write in parametric form the curve that is the intersection of the surfaces $x^2 + y^2 + z^2 = 3$ and $y = 1$.

(b) Find the equation of the line tangent to this curve at $(1, 1, 1)$.

(c) Write an integral expression for the arc length of this curve. What is the value of this integral?

(a) $x^2+y^2+z^2=3$ (a sphere) and $y=1$ (a plane)

$\Rightarrow x^2+1+z^2=3$, or $x^2+z^2=2$, a circle in the $y=1$ plane.

$\therefore \frac{x^2}{2}+\frac{z^2}{2}=1$, or $\left(\frac{x}{\sqrt{2}}\right)^2+\left(\frac{z}{\sqrt{2}}\right)^2=1$

$\therefore$ use $\frac{x}{\sqrt{2}}=\cos t$, $\frac{z}{\sqrt{2}}=\sin t$

$\therefore \vec{c}(t)=(\sqrt{2}\cos t, 1, \sqrt{2}\sin t)$, $0\le t\le 2\pi$

Another method:

The gradient is $\perp$ to a surface, so the curve of intersection must be $\perp$ to each gradient. Let $f(x,y,z)=x^2+y^2+z^2=3$

$g(x,y,z)=y=1$.

$\therefore \nabla f=[2x,2y,2z]$ $\nabla g=[0,1,0]$

$\nabla f \times \nabla g$ is perpendicular to each gradient.

$$\nabla f \times \nabla g = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2x & 2y & 2z \\ 0 & 1 & 0 \end{vmatrix} = [-2z, 0, 2x]$$

$\therefore$ Velocity of curve of intersection is in direction of $[-z, 0, x]$. If $\vec{c}(x(t), y(t), z(t))$ is the curve, then $x'(t) = -z(t)$, $z'(t) = x(t)$

or $x''(t) = -x(t)$, $z''(t) = -z(t)$

$\therefore$ Let $x(t) = \cos(t)$, $y(t) = \sin(t)$.

$\therefore \vec{c}\,'(t) = [-\sin(t), 0, \cos(t)]$

Since $y = 1$ is a criterion, let

$\vec{c}(t) = [\cos(t), 1, \sin(t)]$.

Since $x^2 + y^2 + z^2 = 3$ is a criterion, let

$\vec{c}(t) = [\sqrt{2}\cos(t), 1, \sqrt{2}\sin(t)]$.

$\therefore \vec{c}\,'(t)$ has direction of $[-z, 0, x]$.

$\therefore \vec{c}(t) = [\sqrt{2}\cos(t), 1, \sqrt{2}\sin(t)]$, $0 \le t \le 2\pi$

(b) $\vec{l}(t) = \vec{c}(t) + s\,\vec{c}'(t)$

For $(1,1,1) = [\sqrt{2}\cos t, 1, \sqrt{2}\sin t]$, $t = \frac{\pi}{4}$

$\vec{c}'(t) = [-\sqrt{2}\sin t, 0, \sqrt{2}\cos t]$

$\therefore \vec{c}'(\frac{\pi}{4}) = [-1, 0, 1]$.

$\therefore \vec{l}(\frac{\pi}{4}) = (1,1,1) + s(-1,0,1)$, $s \in \mathbb{R}$.

(c)

(c) Write an integral expression for the arc length of this curve. What is the value of this integral?

$$\int_0^{2\pi} \|\vec{c}'(t)\|\,dt = \int_0^{2\pi} \sqrt{2\sin^2 t + 0 + 2\cos^2 t}\,dt$$

$$= \int_0^{2\pi} \sqrt{2}\,dt = 2\sqrt{2}\,\pi$$

37.

37. In meteorology, the ***negative pressure gradient*** **G** is a vector quantity that points from regions of high pressure to regions of low pressure, normal to the lines of constant pressure (***isobars***).

(a) In an xy coordinate system,

$$\mathbf{G} = -\frac{\partial P}{\partial x}\mathbf{i} - \frac{\partial P}{\partial y}\mathbf{j}.$$

Write a formula for the magnitude of the negative pressure gradient.

(b) If the horizontal pressure gradient provided the only horizontal force acting on the air, the wind would blow directly across the isobars in the direction of **G**, and for a given air mass, with acceleration proportional to the magnitude of **G**. Explain, using Newton's second law.

(c) Because of the rotation of the earth, the wind does not blow in the direction that part (b) would suggest. Instead, it obeys ***Buys–Ballot's law***, which states: "If in the Northern Hemisphere, you stand with your back to the wind, the high pressure is on your right and the low pressure is on your left." Draw a figure and introduce xy coordinates so that **G** points in the proper direction.

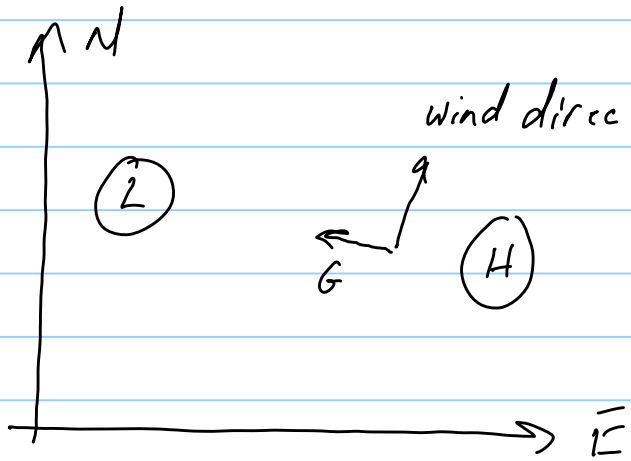
(d) State and graphically illustrate Buys–Ballot's law for the Southern Hemisphere, in which the orientation of high and low pressure is reversed.

(a) $\|\vec{G}\| = \sqrt{(-P_x)^2 + (-P_y)^2} = \sqrt{P_x^2 + P_y^2}$

(b) Let m = mass of a given pocket of air,

Then $\vec{F} = m\vec{a}$ becomes $\vec{G} = m\vec{a}$, or $\vec{a} = \frac{\vec{G}}{m}$

(c)

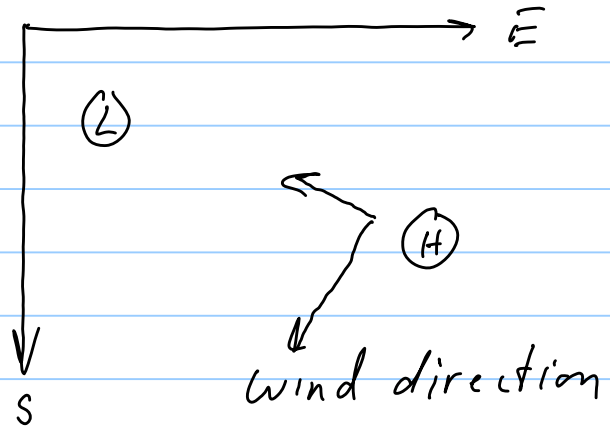


"wind at back, H on right, L on left"

H = High pressure

L = Low pressure

(d)



"wind at back, H on left, L on right"

H = High pressure

L = Low pressure

38.

38. A sphere of mass m, radius a, and uniform density has potential u and gravitational force $\mathbf{F}$, at a distance r from the center (0, 0, 0), given by

$$u = \frac{3m}{2a} - \frac{mr^2}{2a^3}, \qquad \mathbf{F} = -\frac{m}{a^3}\mathbf{r} \qquad (r \le a);$$

$$u = \frac{m}{r}, \qquad \mathbf{F} = -\frac{m}{r^3}\mathbf{r} \qquad (r > a).$$

Here, $r = \|\mathbf{r}\|$, $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$.

(a) Verify that $\mathbf{F} = \nabla u$ on the inside and outside of the sphere.

(b) Check that u satisfies Poisson's equation: $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 + \partial^2 u/\partial z^2 =$ constant inside the sphere.

(c) Show that u satisfies Laplace's equation: $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 + \partial^2 u/\partial z^2 = 0$ outside the sphere.

(a) Outside: $\nabla u = [u_x, u_y, u_z]$, $u = \frac{m}{\sqrt{x^2+y^2+z^2}}$

$$u_x = -\frac{1}{2} m (x^2+y^2+z^2)^{-\frac{3}{2}} (2x) = -mx(x^2+y^2+z^2)^{3/2}$$

$$= -\frac{mx}{r^3}, \text{ as } r = \|\vec{r}\| = \sqrt{x^2+y^2+z^2}$$

Similarly, $u_y = -\frac{my}{r^3}$, $u_z = -\frac{mz}{r^3}$

$$\therefore \nabla u = \left[-\frac{m}{r^3}x, -\frac{m}{r^3}y, -\frac{m}{r^3}z\right]$$

$$= -\frac{m}{r^3}[x,y,z] = -\frac{m}{r^3}\vec{r} = \vec{F} \quad (r \geq a)$$

Inside: $u = \frac{3m}{2a} - \frac{m(x^2+y^2+z^2)}{2a^3}$

$$\therefore u_x = -\frac{m}{2a^3}(2x) = -\frac{mx}{a^3}$$

$$\therefore u_y = -\frac{my}{a^3} \quad u_z = -\frac{mz}{a^3}$$

$$\therefore \nabla u = -\frac{m}{a^3}[x,y,z] = -\frac{m}{a^3}\vec{r} = \vec{F} \quad (r \leq a)$$

(6) (b) Check that u satisfies Poisson's equation: $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 + \partial^2 u/\partial z^2 =$ constant inside the sphere.

From (a), $\nabla u = -\frac{m}{a^3}[x, y, z]$

$\therefore u_{xx} = -\frac{m}{a^3}$, $u_{yy} = -\frac{m}{a^3}$, $u_{zz} = -\frac{m}{a^3}$

$\therefore u_{xx} + u_{yy} + u_{zz} = -\frac{3m}{a^3} =$ a constant

(c) (c) Show that u satisfies Laplace's equation: $\partial^2 u/\partial x^2 + \partial^2 u/\partial y^2 + \partial^2 u/\partial z^2 = 0$ outside the sphere.

From (a), $\nabla u = -\frac{m}{r^3}\vec{r} = -\frac{m}{(x^2+y^2+z^2)^{3/2}}[x, y, z]$

$\therefore u_{xx} = \frac{(x^2+y^2+z^2)^{3/2} - x\left(\frac{3}{2}\right)(x^2+y^2+z^2)^{\frac{1}{2}}(2x)}{(x^2+y^2+y^2)^3}$

$= \frac{(x^2+y^2+z^2)^{3/2} - 3x^2(x^2+y^2+z^2)^{\frac{1}{2}}}{(x^2+y^2+z^2)^3}$

$\therefore u_{yy} = \frac{(x^2+y^2+z^2)^{3/2} - 3y^2(x^2+y^2+z^2)^{1/2}}{(x^2+y^2+z^2)^3}$

$u_{zz} = \frac{(x^2+y^2+z^2)^{3/2} - 3z^2(x^2+y^2+z^2)^{1/2}}{(x^2+y^2+z^2)^3}$

$$\therefore u_{xx} + u_{yy} + u_{zz} = \frac{3(x^2+y^2+z^2)^{3/2} - 3(x^2+y^2+z^2)(x^2+y^2+z^2)^{\frac{1}{2}}}{(x^2+y^2+z^2)^3}$$

$$= \frac{3(x^2+y^2+z^2)^{3/2} - 3(x^2+y^2+z^2)^{3/2}}{(x^2+y^2+z^2)^3}$$

$$= 0$$

39.

39. A circular helix that lies on the cylinder $x^2 + y^2 = R^2$ with pitch ρ may be described parametrically by

$$x = R\cos\theta, \qquad y = R\sin\theta, \qquad z = \rho\theta, \qquad \theta \geq 0.$$

A particle slides under the action of gravity (which acts parallel to the z axis) without friction along the helix. If the particle starts out at the height $z_0 > 0$, then when it reaches the height z along the helix, its speed is given by

$$\frac{ds}{dt} = \sqrt{(z_0 - z)2g},$$

where s is arc length along the helix, g is the constant of gravity, t is time, and $0 \leq z \leq z_0$.

(a) Find the length of the part of the helix between the planes $z = z_0$ and $z = z_1$, $0 \leq z_1 < z_0$.

(b) Compute the time T_0 it takes the particle to reach the plane $z = 0$.

(a) $\vec{c}(\theta) = [R\cos\theta, R\sin\theta, \rho\theta]$

$\therefore \vec{c}\,'(\theta) = [-R\sin\theta, R\cos\theta, \rho]$

Height $z_0 \Rightarrow \theta_0$ from $z = \rho\theta$, and $z_1 = \rho\theta_1$

$$\text{Arc length} = \int_{\theta_1}^{\theta_0} \|\vec{c}\,'(\theta)\|\, d\theta$$

$$= \int_{\theta_1}^{\theta_0} \sqrt{R^2\sin^2\theta + R^2\cos^2\theta + \rho^2}\, d\theta$$

$$= \int_{\theta_1}^{\theta_0} \sqrt{R^2+\rho^2}\, d\theta = \sqrt{R^2+\rho^2}\, \theta \Big|_{\theta_1}^{\theta_0} =$$

$$\sqrt{R^2+\rho^2}\,(\theta_0 - \theta_1) = \sqrt{R^2+\rho^2}\left(\frac{z_0}{\rho} - \frac{z_1}{\rho}\right)$$

$$= \frac{\sqrt{R^2+\rho^2}\,(z_0 - z_1)}{\rho}$$

(6) From (a), $s(z) = \frac{\sqrt{R^2+\rho^2}}{\rho}(z_0 - z)$

$$\therefore \frac{ds}{dz} = -\frac{\sqrt{R^2+\rho^2}}{\rho}$$

Distance/speed = time.

$\therefore \left(\frac{ds}{dz}\right) / \left(\frac{ds}{dt}\right)$ = an increment of time as a function of $z = T(z)$ $= \frac{dt}{dz}$

$\therefore \int_{z_0}^{0} T(z)\, dz$ should give time from z_0 to $z = 0$.

$$\frac{ds}{dz} \Big/ \frac{ds}{dt} = -\frac{\sqrt{R^2+\rho^2}}{\rho} \cdot \frac{1}{\sqrt{2g(z_0 - z)}}$$

$$\therefore \int_{z_0}^{0} -\frac{\sqrt{R^2+\rho^2}}{\rho\sqrt{2g}} \cdot \frac{1}{\sqrt{z_0 - z}}\, dz = \left(-\frac{\sqrt{R^2+\rho^2}}{\rho\sqrt{2g}}\right)\left[-2(z_0 - z)^{\frac{1}{2}}\right]_{z_0}^{0}$$

$$= \left(-\sqrt{\frac{R^2+\rho^2}{2g\rho^2}}\right)\left[-2(z_0)^{\frac{1}{2}}+0\right]$$

$$= 2\sqrt{\frac{R^2+\rho^2}{2g\rho^2}}\sqrt{z_0} = \sqrt{\frac{4z_0(R^2+\rho^2)}{2g\rho^2}}$$

$$= \sqrt{\frac{2z_0(R^2+\rho^2)}{g\rho^2}}$$

40.

40. A sphere of radius 10 centimeters (cm) with center at (0, 0, 0) rotates about the z axis with angular velocity 4 in such a direction that the rotation looks counterclockwise from the positive z axis.

(a) Find the rotation vector ω (see Example 9, in Section 4.4).

(b) Find the velocity $\mathbf{v} = \omega \times \mathbf{r}$ when $\mathbf{r} = 5\sqrt{2}(\mathbf{i} - \mathbf{j})$ is on the "equator."

(c) Find the velocity of the point $(0, 5\sqrt{3}, 5)$ on the sphere.

(a) angular velocity 4 $\Rightarrow \|\vec{\omega}\| = 4$
counterclockwise from positive z axis $\Rightarrow \vec{\omega} = 4\hat{k}$

(b) $\vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 4 \\ 5\sqrt{2} & -5\sqrt{2} & 0 \end{vmatrix} = (0-(-20\sqrt{2}),\ 20\sqrt{2}-0,\ 0-0)$

$= (20\sqrt{2},\ 20\sqrt{2},\ 0)$

(c) $\vec{v} = \vec{\omega} \times \vec{r} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & 0 & 4 \\ 0 & 5\sqrt{3} & 5 \end{vmatrix} = (-20\sqrt{3},\ 0,\ 0)$

41.

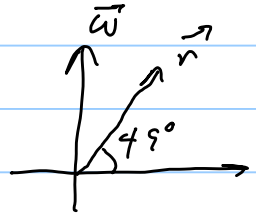
41. Find the speed of the students in a classroom located at a latitude 49°N due to the rotation of the earth. (Ignore the motion of the earth about the sun, the sun in the galaxy, etc.; the radius of the earth is 3960 miles.)

$\|\vec{v}\| = \|\vec{\omega} \times \vec{r}\|$ Here, $\vec{\omega}$ = speed / (distance from axis)

If $R = 3960$, then $2\pi R$ = circumference

$$\therefore \omega = \frac{2\pi(3960 \text{ miles})}{24 \text{ hrs}} \Big/ (3960 \text{ miles}) = \frac{2\pi}{24 \text{ hrs}}$$

At 49°N, angle between $\vec{\omega}$ and $\vec{r}$ is 41° (= 90 - 49)



$$\therefore \|\vec{v}\| = \|\vec{\omega}\| \, \|\vec{r}\| \sin(41°)$$

$$= \left(\frac{2\pi}{24}\right)(3960)\sin(41°)$$

$$= \left(\frac{2\pi}{24}\right)(3960)(0.656) = 680 \text{ miles/hr}$$

5.1 Introduction

Note Title 10/10/2016

1.

1. Evaluate the following iterated integrals:

(a) $\int_0^1 \int_0^1 (1 - x^3 + xy)\, dx\, dy$

(b) $\int_0^{\pi/2} \int_{-\pi/2}^{\pi/2} \cos x \sin y\, dx\, dy$

(c) $\int_1^2 \int_2^4 \left(\frac{x}{y} + \frac{y}{x}\right) dx\, dy$

(d) $\int_0^{\pi/4} \int_0^{\pi/4} \tan x \sec^2 y\, dx\, dy$

(a) $\int_0^1 (1 - x^3 + xy)\, dx = x - \frac{x^4}{4} + \frac{x^2}{2} y \Big|_0^1 = 1 - \frac{1}{4} + \frac{1}{2} y$

$= \frac{3}{4} + \frac{1}{2} y$

$\therefore \int_0^1 \left(\frac{3}{4} + \frac{1}{2} y\right) dy = \frac{3}{4} y + \frac{y^2}{4} \Big|_0^1 = \frac{3}{4} + \frac{1}{4} = \underline{1}$

(b) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \sin y\, dx = \sin x \sin y \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}}$

$= (1)\sin y - (-1)\sin y = 2 \sin y$

$\therefore \int_0^{\frac{\pi}{2}} 2 \sin y\, dy = -2 \cos y \Big|_0^{\frac{\pi}{2}} = 0 - (-2(1)) = \underline{2}$

(c) $\int_2^4 \left(\frac{x}{y} + \frac{y}{x}\right) dx = \frac{x^2}{2y} + y \ln x \Big|_2^4$

$= \frac{16}{2y} + y \ln 4 - \left(\frac{4}{2y} + y \ln 2\right)$

$$= \frac{6}{y} + y(\ln 4 - \ln 2) = \frac{6}{y} + y\ln 2$$

$$\therefore \int_1^2 \left(\frac{6}{y} + y\ln 2\right) dy = 6\ln y + \frac{y^2}{2}\ln 2 \Big|_1^2$$

$$= 6\ln 2 + \frac{4}{2}\ln 2 - \left(0 + \frac{1}{2}\ln 2\right)$$

$$= 8\ln 2 - \frac{1}{2}\ln 2 = \underline{\frac{15}{2}\ln 2}$$

$$= 7\ln 2 + \frac{1}{2}\ln 2 = \underline{\ln 128 + \ln\sqrt{2}}$$

(d) $$\int_0^{\pi/4} \tan x \sec^2 y \, dx = -\log|\cos x| \sec^2 y \Big|_0^{\pi/4}$$

$$= -\log\left(\frac{\sqrt{2}}{2}\right)\sec^2 y - \left[0 \sec^2 y\right]$$

$$= \log\sqrt{2}\sec^2 y$$

$$\therefore \int_0^{\pi/4} \log\sqrt{2}\sec^2 y \, dy = \log\sqrt{2}\tan y \Big|_0^{\pi/4}$$

$$= \log\sqrt{2}(1) - 0 = \underline{\ln\sqrt{2}}$$

2. **2.** Evaluate the integrals in Exercise 1 by integrating first with respect to y and then with respect to x.

(a) (a) $\int_0^1 \int_0^1 (1 - x^3 + xy)\, dx\, dy$ $\int_0^1 \int_0^1 (1 - x^3 + xy)\, dy\, dx$

$$\int_0^1 (1 - x^3 + xy)\, dy = y - x^3 y + x\frac{y^2}{2} \Big|_0^1 = 1 - x^3 + \frac{x}{2}$$

$$\therefore \int_0^1 \left(1 - x^3 + \frac{x}{2}\right) dx = x - \frac{x^4}{4} + \frac{x^2}{4}\Big|_0^1 = 1 - \frac{1}{4} + \frac{1}{4} = 1$$

(5) (b) $\displaystyle\int_0^{\pi/2}\int_{-\pi/2}^{\pi/2} \cos x \sin y \, dx\, dy$ $\displaystyle\int_{-\pi/2}^{\pi/2}\int_0^{\pi/2} \cos x \sin y \, dy\, dx$

$$\int_0^{\pi/2} \cos x \sin y \, dy = -\cos x \cos y\Big|_0^{\pi/2} = 0 - (-\cos x (1))$$

$$= \cos x$$

$$\therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x \, dx = \sin x\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = 1 - (-1) = 2$$

(c) (c) $\displaystyle\int_1^2\int_2^4 \left(\frac{x}{y} + \frac{y}{x}\right) dx\, dy$ $\displaystyle\int_2^4\int_1^2 \left(\frac{x}{y} + \frac{y}{x}\right) dy\, dx$

$$\int_1^2 \left(\frac{x}{y} + \frac{y}{x}\right) dy = x \ln y + \frac{y^2}{2x}\Big|_1^2$$

$$= x \ln 2 + \frac{2}{x} - \left(x \cdot 0 + \frac{1}{2x}\right)$$

$$= x \ln 2 + \frac{3}{2x}$$

$$\therefore \int_2^4 x \ln 2 + \frac{3}{2x} \, dx = \frac{x^2}{2} \ln 2 + \frac{3}{2} \ln x\Big|_2^4$$

$$= 8 \ln 2 + \frac{3}{2} \ln 4 - \left(2 \ln 2 + \frac{3}{2} \ln 2\right)$$

$$= 8 \ln 2 + \frac{3}{2} \ln 2^2 - 2 \ln 2 - \frac{3}{2} \ln 2$$

$= (7\frac{1}{2})\ln 2 = \frac{15}{2}\ln 2 = 7\ln 2 + \ln\sqrt{2}$

$= \ln 2^7 + \ln\sqrt{2} = \ln 128 + \ln\sqrt{2}$

(d) (d) $\int_0^{\pi/4}\int_0^{\pi/4} \tan x \sec^2 y \, dx\, dy$ $\int_0^{\pi/4}\int_0^{\pi/2} \tan x \sec^2 y \, dy\, dx$

$\int_0^{\pi/4} \tan x \sec^2 y \, dy = \tan x \tan y \Big|_0^{\pi/4}$

$= \tan x (1) - \tan x (0) = \tan x$

$\therefore \int_0^{\pi/4} \tan x \, dx = -\log|\cos x| \Big|_0^{\pi/4}$

$= -\log\left(\frac{\sqrt{2}}{2}\right) - (0) = -\log\frac{\sqrt{2}}{2}$

$= \ln\sqrt{2}$

3.

3. Evaluate the following iterated integrals:

(a) $\int_{-1}^{1}\int_{0}^{1} (x^4 y + y^2)\, dy\, dx$

(b) $\int_{0}^{\pi/2}\int_{0}^{1} (y\cos x + 2)\, dy\, dx$

(c) $\int_{0}^{1}\int_{0}^{1} (xye^{x+y})\, dy\, dx$

(d) $\int_{-1}^{0}\int_{1}^{2} (-x\log y)\, dy\, dx$

(a) $\int_0^1 (x^4 y + y^2)\, dy = x^4\frac{y^2}{2} + \frac{y^3}{3}\Big|_0^1 = \frac{x^4}{2} + \frac{1}{3}$

$$\therefore \int_{-1}^{1}\left(\frac{x^4}{2}+\frac{1}{3}\right)dx = \frac{x^5}{10}+\frac{x}{3}\Big|_{-1}^{1} = \frac{1}{10}+\frac{1}{3}-\left(\frac{-1}{10}-\frac{1}{3}\right)$$

$$= \frac{1}{5}+\frac{2}{3} = \underline{\frac{13}{15}}$$

(5) $$\int_0^1 (y\cos x+2)\,dy = \frac{y^2}{2}\cos x + 2y\Big|_0^1 = \frac{1}{2}\cos x + 2$$

$$\therefore \int_0^{\pi/2}\left(\frac{1}{2}\cos x+2\right)dx = \frac{1}{2}\sin x + 2x\Big|_0^{\frac{\pi}{2}} =$$

$$\frac{1}{2}(1)+\pi-0 = \underline{\pi+\frac{1}{2}}$$

(c) $$\int_0^1 xye^{x+y}\,dy = xye^{x+y}\Big|_0^1 - \int_0^1 xe^{x+y}\,dy$$ (integration by parts)

$$= xe^{x+1} - xe^{x+y}\Big|_0^1$$

$$= xe^{x+1} - \left[xe^{x+1} - (xe^x)\right] = xe^x$$

$$\therefore \int_0^1 xe^x\,dx = xe^x\Big|_0^1 - \int_0^1 e^x\,dx$$

$$= e-0 - e^x\Big|_0^1 = e-(e-1) = \underline{1}$$

(d) $\int_1^2 (-x\log y)\,dy = -x(y\log y - y)\Big|_1^2$

$= -xy\log y\Big|_1^2 + xy\Big|_1^2$

$= -2x\log 2 + x(0) + 2x - x$

$= x - 2x\log 2 = x(1 - 2\log 2)$

$\therefore \int_{-1}^{0} (1-2\log 2)x\,dx = (1-2\log 2)\frac{x^2}{2}\Big|_{-1}^{0}$

$= -(1-2\ln 2)\frac{1}{2} = \underline{\ln 2 - \frac{1}{2}}$

4.

4. Evaluate the integrals in Exercise 3 by integrating with respect to x and then with respect to y. [The solution to part (b) only is in the Study Guide to this text.]

(a) $\int_{-1}^{1}\int_{0}^{1} (x^4y + y^2)\,dy\,dx$ $= \int_0^1 \int_{-1}^1 (x^4y + y^2)\,dx\,dy$

$\int_{-1}^{1} (x^4y + y^2)\,dx = \frac{x^5}{5}y + xy^2\Big|_{-1}^{1} = \frac{y}{5} + y^2 - \left(-\frac{y}{5} - y^2\right)$

$= \frac{2y}{5} + 2y^2$

$\therefore \int_0^1 \left(\frac{2y}{5} + 2y^2\right)dy = \frac{y^2}{5} + \frac{2}{3}y^3\Big|_0^1 = \frac{1}{5} + \frac{2}{3} - 0$

$= \underline{\frac{13}{15}}$

(b) $\int_0^{\pi/2}\int_0^1 (y\cos x + 2)\,dy\,dx$ $= \int_0^1 \int_0^{\pi/2} (y\cos x + 2)\,dx\,dy$

$$\int_0^{\pi/2} (y\cos x + 2)\,dx = y\sin x + 2x \Big|_0^{\pi/2} = y + \pi$$

$$\therefore \int_0^1 (y+\pi)\,dy = \frac{y^2}{2} + \pi y \Big|_0^1 = \frac{1}{2} + \pi$$

(c) $\int_0^1\int_0^1 (xye^{x+y})\,dy\,dx$ $= \int_0^1 \int_0^1 (xye^{x+y})\,dx\,dy$

$$\int_0^1 xye^{x+y}\,dx = yxe^{x+y}\Big|_0^1 - \int_0^1 ye^{x+y}\,dx$$

$$= ye^{1+y} - ye^{x+y}\Big|_0^1$$

$$= ye^{1+y} - \left[ye^{1+y} - ye^{y}\right] = ye^{y}$$

$$\therefore \int_0^1 ye^{y}\,dy = ye^{y}\Big|_0^1 - \int_0^1 e^{y}\,dy$$

$$= e - 0 - e^{y}\Big|_0^1 = e - (e-1) = 1$$

(d) $\int_{-1}^{0}\int_1^2 (-x\log y)\,dy\,dx$ $= \int_1^2 \int_{-1}^{0} (-x\log y)\,dx\,dy$

$$\int_{-1}^{0} (-x \log y)\,dx = -\frac{x^2}{2} \log y \Big|_{-1}^{0} = \frac{1}{2} \log y$$

$$\therefore \int_{1}^{2} \frac{1}{2} \log y \, dy = \frac{1}{2} y \log y - \frac{1}{2} y \Big|_{1}^{2}$$

$$= \log 2 - 1 - \left[0 - \frac{1}{2}\right] = \underline{\log 2 - \frac{1}{2}}$$

5. **5.** Use Cavalieri's principle to show that the volumes of two cylinders with the same base and height are equal (see Figure 5.1.10).

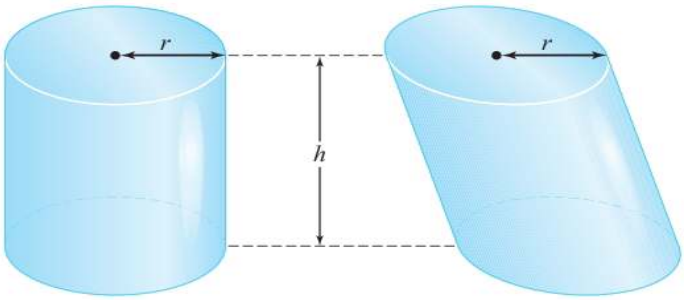


figure **5.1.10** Two cylinders with the same base and height have the same volume.

Take a cross section perpendicular to h.

Each has cross section of $\pi r^2 = A(h)$ at height h.

$\therefore$ Each has volume $\int_0^h A(h)\,dh = \int_0^h \pi r^2 \, dh$

$$= \pi r^2 h \Big|_0^h = \underline{\pi r^2 h}$$

6.

6. Using Cavalieri's principle, compute the volume of the structure shown in Figure 5.1.11; each cross section is a rectangle of length 5 and width 3.

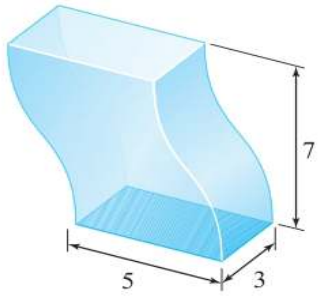


figure **5.1.11** Compute this volume.

Let h = height of solid, perpendicular to rectangular base. $\therefore A(h) = 3 \times 5 = 15$

$$\therefore \text{Volume} = \int_0^7 A(h)\,dh = \int_0^7 15\,dh = 15h\Big|_0^7$$

$$= 15(7) = \underline{105}$$

7.

7. A lumberjack cuts out a wedge-shaped piece W of a cylindrical tree of radius r obtained by making two saw cuts to the tree's center, one horizontally and one at an angle θ. Compute the volume of the wedge W using Cavalieri's principle. (See Figure 5.1.12.)

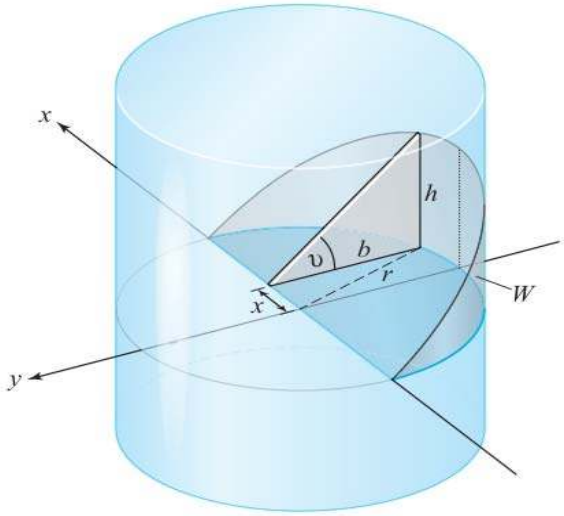


figure **5.1.12** Find the volume of W.

Using the figure, the area of the triangular cross section is $A(x) = \frac{1}{2}bh$

$b = \sqrt{r^2 - x^2}$, where $x \in [-r, r]$

$h = 6\tan\theta = \tan\theta\sqrt{r^2 - x^2}$

$\therefore A(x) = \frac{1}{2}\tan\theta\,(r^2 - x^2)$

$\therefore \text{Volume} = \int_{-r}^{r} A(x)\,dx = \int_{-r}^{r} \frac{1}{2}\tan\theta\,(r^2 - x^2)\,dx$

$= \frac{1}{2}\tan\theta\, r^2 x - \frac{1}{2}\tan\theta\,\frac{x^3}{3}\Big|_{-r}^{r}$

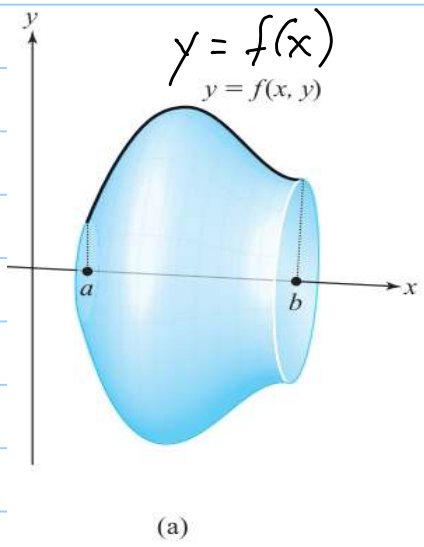
$= \frac{1}{2}\tan\theta\, r^3 - \frac{1}{2}\tan\theta\,\frac{r^3}{3} - \left[-\frac{1}{2}\tan\theta\, r^3 + \frac{1}{2}\tan\theta\,\frac{r^3}{3}\right]$

$= \tan\theta\, r^3 - \tan\theta\,\frac{r^3}{3} = \frac{2}{3}\tan\theta\, r^3$

8.

8. (a) Show that the volume of the solid of revolution shown in Figure 5.1.13(a) is

$$\pi \int_a^b [f(x)]^2\,dx.$$



(a)

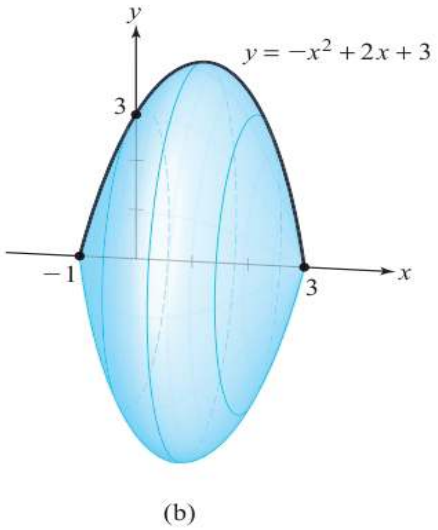
(a) Take a cross section perpendicular to the x-axis.
This is a circle of radius $f(x)$ (mistake in text).

$\therefore A(x) = \pi[f(x)]^2$

$\therefore \text{Volume} = \int_a^b \pi[f(x)]^2\,dx = \pi\int_a^b [f(x)]^2\,dx$

(6)

(b) Show that the volume of the region obtained by rotating the region under the graph of the parabola $y = -x^2 + 2x + 3$, $-1 \le x \le 3$, about the x axis is $512\pi/15$ [see Figure 5.1.13(b)].



(b)

Take a cross section perpendicular to x-axis.

This is a circle of radius y.

$\therefore A(x) = \pi y^2 = \pi(-x^2+2x+3)^2$

$\therefore$ Volume $= \int_{-1}^{3} \pi(-x^2+2x+3)^2\,dx$

$= \pi \int_{-1}^{3} (x^4 - 4x^3 - 2x^2 + 12x + 9)\,dx$

$= \pi\left(\frac{x^5}{5} - x^4 - \frac{2}{3}x^3 + 6x^2 + 9x\right)\Big|_{-1}^{3}$

$= \pi\left(\frac{243}{5} - 81 - 18 + 54 + 27\right)$

$\quad - \pi\left(-\frac{1}{5} - 1 + \frac{2}{3} + 6 - 9\right)$

$= \pi\left(\frac{153}{5}\right) - \pi\left(\frac{7}{15} - 4\right) = \pi\left(\frac{459}{15} - \frac{7}{15} + \frac{60}{15}\right)$

$= \frac{512\pi}{15}$

Evaluate the double integrals in Exercises 9 to 11, where R is the rectangle $[0, 2] \times [-1, 0]$.

9. **9.** $\iint_R (x^2y^2 + x)\, dy\, dx$ $[0,2] \times [-1,0]$ means $dx \times dy$

$$\int_{-1}^{0} (x^2y^2 + x)\, dy = x^2\frac{y^3}{3} + xy \Big|_{-1}^{0} = 0 - \left(-\frac{x^2}{3} - x\right)$$

$$= \frac{x^2}{3} + x$$

$$\therefore \int_0^2 \frac{x^2}{3} + x\, dx = \frac{x^3}{9} + \frac{x^2}{2} \Big|_0^2 = \frac{8}{9} + \frac{4}{2} - 0$$

$$= 2\frac{8}{9} = \frac{26}{9}$$

10. **10.** $\iint_R \left(|y| \cos \frac{1}{4}\pi x\right) dy\, dx$

$$\int_{-1}^{0} |y| \cos\frac{\pi x}{4}\, dy = \int_{-1}^{0} -y \cos\frac{\pi x}{4}\, dy \quad \text{since } |y| = -y \text{ for } -1 \le y \le 0$$

$$= -\frac{y^2}{2}\cos\frac{\pi x}{4} \Big|_{-1}^{0} = 0 - \left(-\frac{1}{2}\cos\frac{\pi x}{4}\right) = \frac{1}{2}\cos\frac{\pi x}{4}$$

$$\therefore \int_0^2 \frac{1}{2}\cos\frac{\pi x}{4} = \left(\frac{1}{2}\right)\left(\frac{4}{\pi}\right)\sin\frac{\pi x}{4} \Big|_0^2 = \left(\frac{2}{\pi}\right)\sin\frac{\pi}{2} - 0$$

$$= \frac{2}{\pi}$$

11. **11.** $\iint_R \left(-xe^x \sin \frac{1}{2}\pi y\right) dy\,dx$

$$\int_{-1}^{0} -xe^x \sin\frac{\pi y}{2}\, dy = xe^x\left(\frac{2}{\pi}\right)\cos\frac{\pi y}{2}\Big|_{-1}^{0}$$

$$= xe^x\left(\frac{2}{\pi}\right)(1) - xe^x\left(\frac{2}{\pi}\right)(0)$$

$$= \left(\frac{2}{\pi}\right)xe^x$$

$$\therefore \frac{2}{\pi}\int_0^2 xe^x\, dx = \frac{2}{\pi}\left[xe^x\Big|_0^2 - \int_0^2 e^x dx\right]$$

$$= \frac{2}{\pi}\left[2e^2 - 0 - e^x\Big|_0^2\right]$$

$$= \frac{2}{\pi}\left[2e^2 - (e^2 - 1)\right]$$

$$= \frac{2}{\pi}\left(e^2 + 1\right)$$

12. **12.** Evaluate the iterated integral:

$$\int_1^3 \int_1^2 \frac{xy}{\left(x^2+y^2\right)^{3/2}}\, dx\,dy.$$

$$\int_1^2 xy\left(x^2+y^2\right)^{-\frac{5}{2}} dx = \left(\frac{1}{2}\right)(-2)y\left(x^2+y^2\right)^{-\frac{1}{2}}\Big|_1^2$$

$$= -y(4+y^2)^{-\frac{1}{2}} - \left[-y(1+y^2)^{-\frac{1}{2}}\right]$$

$$= -\frac{y}{\sqrt{4+y^2}} + \frac{y}{\sqrt{1+y^2}}$$

$$\therefore \int_1^3 \left[-y(4+y^2)^{-\frac{1}{2}} + y(1+y^2)^{-\frac{1}{2}}\right] dy =$$

$$-(4+y^2)^{\frac{1}{2}} + (1+y^2)^{\frac{1}{2}} \Big|_1^3 =$$

$$\left[-\sqrt{13} + \sqrt{10}\right] - \left[-\sqrt{5} + \sqrt{2}\right] =$$

$$\underline{\sqrt{10} + \sqrt{5} - \sqrt{13} - \sqrt{2}}$$

13. **13.** Evaluate the iterated integral:

$$\int_0^1 \int_0^1 (3x+2y)^7 \, dx\,dy.$$

$$\int_0^1 (3x+2y)^7 dx = \left(\frac{1}{3}\right)\left(\frac{1}{8}\right)(3x+2y)^8 \Big|_0^1$$

$$= \frac{1}{24}(3+2y)^8 - \frac{1}{24}(2y)^8$$

$$\therefore \int_0^1 \frac{1}{24}(3+2y)^8 - \frac{1}{24}(2y)^8 \, dy =$$

$$\frac{1}{24}\left(\frac{1}{2}\right)\left(\frac{1}{9}\right)(3+2y)^9 - \frac{1}{24}\left(\frac{1}{2}\right)\left(\frac{1}{9}\right)(2y)^9 \Big|_0^1$$

$$\frac{1}{432}(5)^9 - \frac{1}{432}(2)^9 - \left[\frac{1}{432}(3)^9 - 0\right]$$

$$= \frac{5^9 - 2^9 - 3^9}{432} = \frac{1932930}{432} = \frac{2 \times 3^3 \times 5 \times 7159}{2^4 \times 3^3}$$

$$= \frac{(5)(7159)}{2^3} = \frac{35,795}{8}$$

14. **14.** Find the volume bounded by the graph of $f(x, y) = 1 + 2x + 3y$, the rectangle $[1, 2] \times [0, 1]$, and the four vertical sides of the rectangle R, as in Figure 5.1.1.

$$\text{Volume} = \int_0^1 \int_1^2 (1 + 2x + 3y)\, dx\, dy$$

Note: $1 + 2x + 3y \geq 0$ for $1 \leq x \leq 2,\ 0 \leq y \leq 1$

$$\therefore \int_1^2 (1 + 2x + 3y)\, dx = x + x^2 + 3xy \Big|_1^2$$

$$= (2 + 4 + 6y) - (1 + 1 + 3y)$$

$$= 4 + 3y$$

$$\therefore \int_0^1 (4 + 3y)\, dy = 4y + \frac{3y^2}{2} \Big|_0^1 = 4 + \frac{3}{2} = \frac{11}{2}$$

If $f(x,y) < 0$ over some portion of $[1,2] \times [0,1]$, Then we would have to break up rectangle into pieces where $f(x,y)$ is ≥ 0 and < 0, integrate over those pieces, taking the absolute value over areas where $f(x,y) < 0$.

15. **15.** Repeat Exercise 14 for the function $f(x, y) = x^4 + y^2$ and the rectangle $[-1, 1] \times [-3, -2]$.

As in #14, note $f(x,y) \geq 0$ for $[-1,1] \times [-3,-2]$

$$\therefore \text{Volume} = \int_{-1}^{1} \int_{-3}^{-2} (x^4 + y^2)\, dy\, dx$$

$$\therefore \int_{-3}^{-2} (x^4 + y^2)\, dy = x^4 y + \frac{y^3}{3} \Big|_{-3}^{-2}$$

$$= -2x^4 - \frac{8}{3} - \left[-3x^4 - \frac{27}{3}\right]$$

$$= x^4 + \frac{19}{3}$$

$$\therefore \int_{-1}^{1} \left(x^4 + \frac{19}{3}\right) dx = \frac{x^5}{5} + \frac{19}{3}x \Big|_{-1}^{1} = \frac{1}{5} + \frac{19}{3} - \left\{-\frac{1}{5} - \frac{19}{3}\right\}$$

$$= \frac{2}{5} + \frac{38}{3} = \underline{\underline{\frac{196}{15}}}$$

5.2 The Double Integral Over a Rectangle

Note Title 10/17/2016

1.

1. Evaluate each of the following integrals if $R = [0, 1] \times [0, 1]$.

(a) (a) $\iint_R (x^3 + y^2)\, dA$

$$\int_R (x^3+y^2)\,dA = \int_0^1 \int_0^1 (x^3+y^2)\,dx\,dy$$

$$\int_0^1 (x^3+y^2)\,dx = \frac{x^4}{4} + xy^2 \Big|_0^1 = \frac{1}{4} + y^2$$

$$\int_0^1 \left(\frac{1}{4}+y^2\right)dy = \frac{y}{4} + \frac{y^3}{3}\Big|_0^1 = \frac{1}{4}+\frac{1}{3} = \underline{\frac{7}{12}}$$

(b) (b) $\iint_R ye^{xy}\, dA$

$$\iint_R ye^{xy}\,dA = \int_0^1 \int_0^1 ye^{xy}\,dx\,dy$$

$$\int_0^1 ye^{xy}\,dx = e^{xy}\Big|_0^1 = e^y - 1$$

$$\int_0^1 (e^y-1)\,dy = e^y - y\Big|_0^1 = e-1-(1-0) = \underline{e-2}$$

(c) (c) $\iint_R (xy)^2 \cos x^3\, dA$

$$\iint_R (xy)^2 \cos x^3\,dA = \int_0^1 \int_0^1 (xy)^2 \cos x^3\,dx\,dy$$

$$\int_0^1 (xy)^2 \cos x^3 \, dx = \frac{1}{3} y^2 \sin(x^3) \Big|_0^1 = \frac{y^2 \sin(1)}{3}$$

$$\int_0^1 y^2 \frac{\sin(1)}{3} \, dy = \frac{\sin(1)}{3} \frac{y^3}{3} \Big|_0^1 = \underline{\frac{\sin(1)}{9}}$$

(d) (d) $\iint_R \ln[(x+1)(y+1)] \, dA$

$$\iint_R \ln[(x+1)(y+1)] \, dA = \int_0^1 \int_0^1 \ln[(x+1)(y+1)] \, dx \, dy$$

$$\int_0^1 \ln[(x+1)(y+1)] \, dx = \int_0^1 \ln(x+1) \, dx + \ln(y+1) \int_0^1 dx$$

$$= (x+1)\ln(x+1) - x \Big|_0^1 + [\ln(y+1)] x \Big|_0^1$$

$$= 2\ln 2 - 1 + \ln(y+1) = \ln 4 - 1 + \ln(y+1)$$

$$\therefore \int_0^1 [\ln 4 - 1 + \ln(y+1)] \, dy = (\ln 4 - 1) y + (y+1)\ln(y+1) - y \Big|_0^1$$

$$= \ln 4 - 1 + 2\ln 2 - 1$$

$$= \underline{2\ln 4 - 2} = \underline{\ln 16 - 2}$$

2.

2. Evaluate each of the following integrals if $R = [0, 1] \times [0, 1]$.

(a) $\iint_R (x^m y^n)\, dx\, dy$, where $m, n > 0$

$$\iint_R (x^m y^n)\, dx\, dy = \int_0^1 \int_0^1 x^m y^n \, dx\, dy$$

$$\int_0^1 x^m y^n \, dx = \frac{x^{m+1}}{m+1} y^n \Big|_0^1 = \frac{y^n}{m+1}$$

$$\therefore \int_0^1 \frac{y^n}{m+1} \, dy = \frac{y^{n+1}}{(m+1)(n+1)} \Big|_0^1 = \frac{1}{(m+1)(n+1)}$$

(b) $\iint_R (ax + by + c)\, dx\, dy$

$$\iint_R (ax+by+c)\, dx\, dy = \int_0^1 \int_0^1 (ax+by+c)\, dx\, dy$$

$$\int_0^1 (ax+by+c)\, dx = \frac{ax^2}{2} + byx + cx \Big|_0^1 = \frac{a}{2} + by + c$$

$$\int_0^1 \left(\frac{a}{2} + by + c\right) dy = \frac{a}{2} y + \frac{b}{2} y^2 + cy \Big|_0^1 = \frac{a}{2} + \frac{b}{2} + c$$

(c) $\iint_R \sin(x + y)\, dx\, dy$

$$\iint_R \sin(x+y)\, dx\, dy = \int_0^1 \int_0^1 \sin(x+y)\, dx\, dy$$

$$\int_0^1 \sin(x+y)\, dx = -\cos(x+y) \Big|_0^1 = -\cos(1+y) - [-\cos(y)]$$

$$= \cos y - \cos(1+y)$$

$$\therefore \int_0^1 \cos y - \cos(1+y)\,dy = \sin y - \sin(1+y)\Big|_0^1$$

$$= \sin(1) - \sin(2) - (\sin(0) - \sin(1))$$

$$= 2\sin(1) - \sin(2)$$

(d) (d) $\iint_R (x^2 + 2xy + y\sqrt{x})\,dx\,dy$ $= \int_0^1 \int_0^1 (x^2 + 2xy + y\sqrt{x})\,dx\,dy$

$$\int_0^1 (x^2 + 2xy + y\sqrt{x})\,dx = \frac{x^3}{3} + x^2 y + \frac{2}{3} y x^{\frac{3}{2}}\Big|_0^1$$

$$= \frac{1}{3} + y + \frac{2}{3}y = \frac{1}{3} + \frac{5}{3}y$$

$$\therefore \int_0^1 \left(\frac{1}{3} + \frac{5}{3}y\right) dy = \frac{1}{3}y + \frac{5}{6}y^2\Big|_0^1 = \frac{1}{3} + \frac{5}{6} = \frac{7}{6}$$

3. **3.** Evaluate over the region R:

$$\iint_R \frac{yx^3}{y^2+2}\,dy\,dx, \qquad R\text{: } [0, 2] \times [-1, 1].$$

$$\int_0^2 \frac{yx^3}{y^2+2}\,dx = \frac{y}{y^2+2} \cdot \frac{1}{4}x^4\Big|_0^2 = \frac{4y}{y^2+2}$$

$$\therefore \int_{-1}^1 \frac{4y}{y^2+2}\,dy = 2\ln(y^2+2)\Big|_{-1}^1 = 2\ln(3) - 2\ln(3)$$

$$= 0$$

4.

4. Evaluate over the region R:

$$\iint_R \frac{y}{1+x^2}\,dx\,dy, \qquad R\text{: } [0,1]\times[-2,2].$$

$$= \int_{-2}^{2}\int_{0}^{1} \frac{y}{1+x^2}\,dx\,dy$$

$$\int_0^1 \frac{y}{1+x^2}\,dx = y\arctan(x)\Big|_0^1 = y\arctan(1) - y\arctan(0)$$

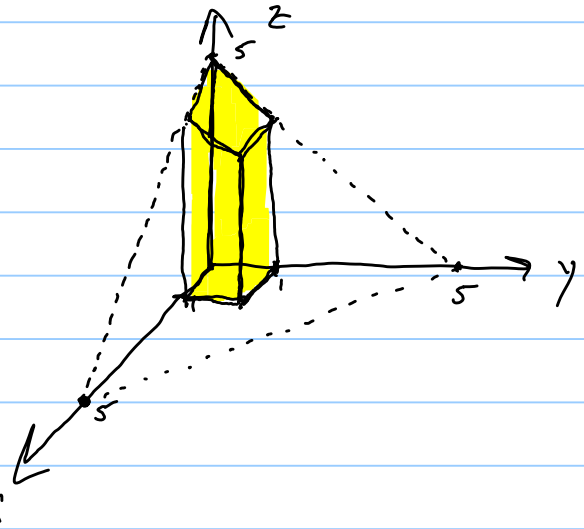
$$= y\left(\frac{\pi}{4}\right) - 0 = \frac{\pi}{4}y$$

$$\therefore \int_{-2}^{2} \frac{\pi}{4}y\,dy = \frac{\pi}{8}y^2\Big|_{-2}^{2} = \frac{\pi}{8}(4) - \frac{\pi}{8}(4) = \underline{0}$$

5.

5. Sketch the solid whose volume is given by:

$$\int_0^1\int_0^1 (5-x-y)\,dy\,dx.$$

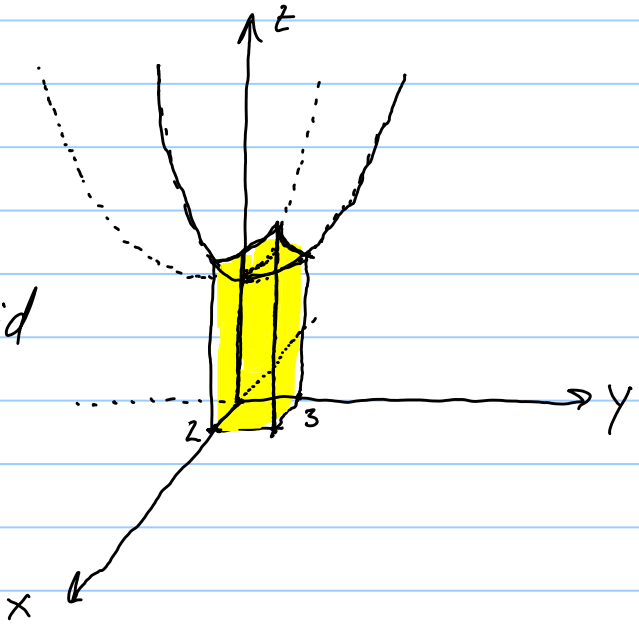


$z = 5-x-y$ is a plane intersecting axes at $(5,0,0)$, $(0,5,0)$ and $(0,0,5)$.

6.

6. Sketch the solid whose volume is given by:

$$\int_0^3\int_0^2 (9+x^2+y^2)\,dx\,dy.$$



$z = 9+x^2+y^2$ is a paraboloid whose inferior apex is 9 units above xy-plane. The solid is underneath the paraboloid.

7. **7.** Compute the volume of the region over the rectangle $[0, 1] \times [0, 1]$ and under the graph of $z = xy$.

$$\iint_R xy\,dA = \int_0^1\int_0^1 xy\,dx\,dy = \int_0^1 \left[\int_0^1 xy\,dx\right] dy$$

$$= \int_0^1 \left(\frac{x^2}{2}y\Big|_0^1\right) dy = \int_0^1 \frac{y}{2}\,dy = \frac{y^2}{4}\Big|_0^1 = \underline{\frac{1}{4}}$$

8. **8.** Compute the volume of the solid bounded by the xz plane, the yz plane, the xy plane, the planes $x = 1$ and $y = 1$, and the surface $z = x^2 + y^4$.

$R = [0,1] \times [0,1]$

$$\iint_R (x^2+y^4)\,dA = \int_0^1\int_0^1 (x^2+y^4)\,dx\,dy$$

$$\int_0^1 (x^2+y^4)\,dx = \frac{x^3}{3} + y^4 x\Big|_0^1 = \frac{1}{3} + y^4$$

$$\therefore \int_0^1 \left(\frac{1}{3}+y^4\right) dy = \frac{y}{3} + \frac{y^5}{5}\Big|_0^1 = \frac{1}{3} + \frac{1}{5} = \underline{\underline{\frac{8}{15}}}$$

9. **9.** Let f be continuous on $[a, b]$ and g continuous on $[c, d]$. Show that

$$\iint_R [f(x)g(y)]\,dx\,dy = \left[\int_a^b f(x)\,dx\right]\left[\int_c^d g(y)\,dy\right],$$

where $R = [a, b] \times [c, d]$.

$$\iint_R [f(x)\,g(y)]\,dx\,dy = \int_c^d \int_a^b [f(x)\,g(y)]\,dx\,dy$$

$$= \int_c^d \left[\int_a^b [g(y)\,f(x)]\,dx \right] dy$$

$$= \int_c^d g(y) \left[\int_a^b f(x)\,dx \right] dy \qquad \text{using } \int c\,f(x)\,dx = c\int f(x)\,dx$$

$$= \left[\int_a^b f(x)\,dx \right] \int_c^d g(y)\,dy \qquad \text{using } \int g(y)\,k\,dy = k\int g(y)\,dy$$

where $k = \int_a^b f(x)\,dx$

10. **10.** Compute the volume of the solid bounded by the surface $z = \sin y$, the planes $x = 1$, $x = 0$, $y = 0$, and $y = \pi/2$, and the xy plane.

$$R = [0,1] \times \left[0, \frac{\pi}{2}\right]$$

$$\therefore \iint_R \sin y\,dA = \int_0^1 \int_0^{\frac{\pi}{2}} \sin y\,dy\,dx$$

$$\int_0^{\frac{\pi}{2}} \sin y\,dy = -\cos y \Big|_0^{\pi/2} = 0 - (-\cos(0)) = 1$$

$$\therefore \int_0^1 1\,dx = x\Big|_0^1 = \underline{1}$$

11.

11. Compute the volume of the solid bounded by the graph $z = x^2 + y$, the rectangle $R = [0, 1] \times [1, 2]$, and the "vertical sides" of R.

$$\iint_R (x^2+y)\,dA = \int_1^2 \int_0^1 (x^2+y)\,dx\,dy$$

$$\int_0^1 (x^2+y)\,dx = \frac{x^3}{3} + yx\Big|_0^1 = \frac{1}{3} + y$$

$$\int_1^2 \left(\frac{1}{3}+y\right)dy = \frac{y}{3} + \frac{y^2}{2}\Big|_1^2 = \frac{2}{3} + 2 - \left(\frac{1}{3}+\frac{1}{2}\right)$$

$$= \frac{1}{3} + \frac{3}{2} = \underline{\frac{11}{6}}$$

12.

12. Let f be continuous on $R = [a, b] \times [c, d]$; for $a < x < b$, $c < y < d$, define

$$F(x, y) = \int_a^x \int_c^y f(u, v)\,dv\,du.$$

Show that $\partial^2 F/\partial x\,\partial y = \partial^2 F/\partial y\,\partial x = f(x, y)$. Use this example to discuss the relationship between Fubini's theorem and the equality of mixed partial derivatives.

Fubini's Theorem says,

$$\int_a^x \left[\int_c^y f(u,v)\,dv\right] du = \int_c^y \left[\int_a^x f(u,v)\,du\right] dv \quad [1]$$

Let $G(u,y) = \int_c^y f(u,v)\,dv$ $\therefore \frac{\partial G(u,y)}{\partial y} = f(u,y)$ [2]

Let $H(x,v) = \int_a^x f(u,v)\,du$ $\therefore \frac{\partial H(x,v)}{\partial x} = f(x,v)$ [3]

[2], [3] follow from the Fundamental Theorem of Calculus.

$\therefore$ [1] can be written as,

$$F(x,y) = \int_a^x G(u,y)\,du = \int_c^y H(x,v)\,dv$$

$$\therefore \frac{\partial F}{\partial x} = G(x,y) \qquad \frac{\partial F}{\partial y} = H(x,y)$$

$$\therefore \frac{\partial^2 F}{\partial y\,\partial x} = \frac{\partial}{\partial y} G(x,y) = f(x,y) \text{ from [2]}$$

$$\frac{\partial^2 F}{\partial x\,\partial y} = \frac{\partial}{\partial x} H(x,y) = f(x,y) \text{ from [3]}$$

$$\therefore f(x,y) = \frac{\partial^2 F}{\partial y\,\partial x} = \frac{\partial^2 F}{\partial x\,\partial y}$$

$\therefore$ Fubini's Theorem $\Rightarrow$ equality of mixed partial derivatives.

Note G, H are C^1, F is C^2.

13.

13. Consider the integral in 2(a) as a function of m and n; that is,

$$f(m, n) := \iint_R x^m y^n \, dx \, dy.$$

Evaluate $\lim_{m,n\to\infty} f(m, n)$.

From 2(a), $\iint_R x^m y^n \, dx \, dy = \frac{1}{(m+1)(n+1)}$, $m, n > 0$

$\therefore \lim_{m,n\to\infty} f(m,n) = \lim_{m,n\to\infty} \frac{1}{(m+1)(n+1)} = \underline{0}.$

14.

14. Let:

$$f(m, n) := \int_{-\pi}^{\pi} \int_{-\pi}^{\pi} \cos nx \sin my \, dx \, dy.$$

Show that $\lim_{m,n\to\infty} f(m, n) = 0$.

$$\int_{-\pi}^{\pi} \cos(nx)\sin(my)\,dx = \frac{\sin(nx)\sin(my)}{n}\Big|_{-\pi}^{\pi}$$

$$= \frac{1}{n}\left[\sin(n\pi)\sin(my) - \left(\sin(-n\pi)\sin(my)\right)\right]$$

$$= \frac{2}{n}\sin(n\pi)\sin(my), \quad n \neq 0.$$

If n is an integer, this is 0. $\therefore$ Assume n is not necessarily an integer, so $\sin(n\pi)$ may not be 0.

$$\therefore \int_{-\pi}^{\pi} \frac{2}{n}\sin(n\pi)\sin(my)\,dy = -\frac{2}{mn}\sin(n\pi)\cos(my)\Big|_{-\pi}^{\pi}$$

$$= -\frac{2\sin(n\pi)}{mn}\left[\cos(m\pi) - \cos(-m\pi)\right]$$

$$= -\frac{2\sin(n\pi)}{mn}[0] = 0 \quad , m \neq 0.$$

$$\therefore f(m,n) = 0. \quad \therefore \lim_{m,n\to\infty} f(m,n) = 0$$

15.

15. Let $f\colon [0,1]\times[0,1] \to \mathbb{R}$ be defined by

$$f(x,y) = \begin{cases} 1 & x \text{ rational} \\ 2y & x \text{ irrational.} \end{cases}$$

Show that the iterated integral $\int_0^1 \left[\int_0^1 f(x,y)\,dy\right]dx$ exists but that f is not integrable.

Consider $\int_0^1 f(x,y)\,dy$

For x rational: $\int_0^1 f(x,y) = \int_0^1 1\,dy = y\Big|_0^1 = 1$

For x irrational: $\int_0^1 f(x,y) = \int_0^1 2y\,dy = y^2\Big|_0^1 = 1$

$\therefore$ For all x, $\int_0^1 f(x,y)\,dy = 1.$

$$\therefore \int_0^1 \left[\int_0^1 f(x,y)\,dy \right] dx = \int_0^1 1\,dx = x\Big|_0^1 = 1.$$

f is not integrable: definition (p. 272 of text) states the sequence of sums, S_n, depends on the choice c_{jk} of the point in the rectangle R_{jk}, and the limit of S_n should be the same, S, for all choices of c_{jk}.

For every rectangle, R_{jk}, one can always find a rational or irrational coordinate for c_{jk}. The above showed the limit is the same if you choose all your coordinates to be one or the other. The trick is to choose some c_{jk} as rational coordinates, and others as irrational coordinates. Then the limit won't be the same.

For example: For R_{jk} with $0 \le y \le \frac{1}{2}$, choose

C_{jk} = (rational, rational), and for R_{jk} with $\frac{1}{2} \le y \le 1$, choose C_{jk} = (irrational, irrational).

Then $$\int_0^1 f(x,y)\,dy = \int_0^{1/2} f(x,y)\,dy + \int_{\frac{1}{2}}^1 f(x,y)\,dy$$

$$= \int_0^{\frac{1}{2}} 1\,dy + \int_{\frac{1}{2}}^1 2y\,dy$$

$$= y\Big|_0^{\frac{1}{2}} + y^2\Big|_{\frac{1}{2}}^1$$

$$= \frac{1}{2} + \left(1 - \frac{1}{4}\right) = \frac{5}{4} \neq 1.$$

$\therefore \lim_{n\to\infty} S_n$ differs depending on choice of C_{jk},

so f is not integrable.

16. **16.** Express $\iint_R \cosh xy\,dx\,dy$ as a convergent sequence, where $R = [0, 1] \times [0, 1]$.

Let $0 = x_0 < x_1 < \ldots < x_n = 1$ be a regular partition of $[0,1]$

$0 = y_0 < y_1 < \ldots < y_n = 1$ be a regular partition of $[0,1]$

Let C_{jk} be any point in $[x_j, x_{j+1}] \times [y_k, y_{k+1}]$,

$0 \le j \le n-1$, $0 \le k \le n-1$.

Let $S_n = \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} \cosh(c_{jk})(x_{j+1}-x_j)(y_{k+1}-y_k)$

Let $\Delta x = x_{j+1}-x_j$, $\Delta y = y_{k+1}-y_k$

Since $\cosh(xy)$ is continuous on $R=[0,1]\times[0,1]$,

Then $\cosh(xy)$ is integrable on R, so that

$\lim_{n\to\infty} S_n$ converges.

$$\iint_R \cosh(xy)\,dA = \lim_{n\to\infty} \sum_{j=0}^{n-1} \sum_{k=0}^{n-1} \cosh(c_{jk})\,\Delta x \Delta y$$

17.

17. Although Fubini's theorem holds for most functions met in practice, we must still exercise some caution. This exercise gives a function for which it fails. By using a substitution involving the tangent function, show that

$$\int_0^1 \int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\,dy\,dx = \frac{\pi}{4},$$

yet

$$\int_0^1 \int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\,dx\,dy = -\frac{\pi}{4}.$$

Why does this not contradict Theorem 3 or 3′?

(a) Use $1+\tan^2\theta = \sec^2\theta$. $x^2+y^2 = x^2\left(1+\left(\frac{y}{x}\right)^2\right)$

$= x^2(1+\tan^2\theta)$ using $\frac{y}{x} = \tan\theta$

$x^2-y^2 = x^2\left(1-\frac{y^2}{x^2}\right) = x^2(1-\tan^2\theta)$

$$\therefore \frac{x^2-y^2}{(x^2+y^2)^2} = \frac{x^2(1-\tan^2\theta)}{x^4(1+\tan^2\theta)^2} = \frac{1}{x^2}\frac{(1-\tan^2\theta)}{\sec^4\theta}$$

If $y = x\tan\theta$, then $dy = x\sec^2\theta\, d\theta$

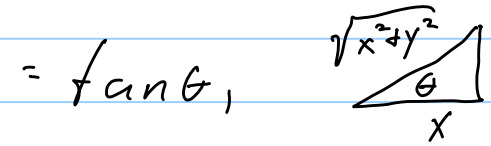
$$\therefore \int \frac{x^2-y^2}{(x^2+y^2)^2}\,dy = \int \frac{1}{x^2}\frac{(1-\tan^2\theta)}{\sec^4\theta} \times \sec^2\theta\, d\theta$$

$$= \int \frac{1}{x}\frac{(1-\tan^2\theta)}{\sec^2\theta}\,d\theta = \int \frac{1}{x}\frac{\left(1-\frac{\sin^2\theta}{\cos^2\theta}\right)}{\frac{1}{\cos^2\theta}}\,d\theta$$

$$= \int \frac{1}{x}\cos^2\theta - \sin^2\theta\,d\theta = \int \frac{1}{x}\cos(2\theta)\,d\theta$$

$$= \frac{1}{2x}\sin(2\theta) + C = \frac{\sin\theta\cos\theta}{x} + C$$

For $y = x\tan\theta$, $\frac{y}{x} = \tan\theta$,



$$\therefore \cos\theta = \frac{x}{\sqrt{x^2+y^2}},\quad \sin\theta = \frac{y}{\sqrt{x^2+y^2}}$$

$$\therefore \frac{\sin\theta\cos\theta}{x} = \frac{xy}{x(x^2+y^2)} = \frac{y}{x^2+y^2}$$

$$\therefore \int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\,dy = \frac{y}{x^2+y^2}\Big|_0^1 = \frac{1}{x^2+1} - 0 = \frac{1}{x^2+1}$$

Now, for $\int_0^1 \frac{1}{x^2+1}\,dx = \arctan x\Big|_0^1 = \frac{\pi}{4} - 0 = \frac{\pi}{4}$

$$\therefore \int_0^1 \int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\,dy\,dx = \underline{\underline{\frac{\pi}{4}}}$$

(6) For $\int \frac{x^2-y^2}{(x^2+y^2)^2}\,dx$, use $x = y\tan\theta$ so that

$$dx = y\sec^2\theta\,d\theta$$

$$\therefore x^2 - y^2 = y^2\tan^2\theta - y^2 = y^2(\tan^2\theta - 1)$$

$$(x^2+y^2)^2 = (y^2\tan^2\theta + y^2)^2 = y^4\sec^4\theta$$

$$\therefore \frac{x^2-y^2}{(x^2+y^2)^2} = \frac{\tan^2\theta - 1}{y^2\sec^4\theta}$$

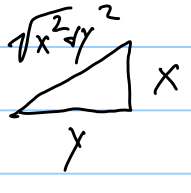
$$\therefore \int \frac{x^2-y^2}{(x^2+y^2)^2}\,dx = \int \frac{\tan^2\theta - 1}{y^2\sec^4\theta}(y\sec^2\theta)\,d\theta$$

$$= \int \frac{\tan^2\theta - 1}{y\sec^2\theta}\,d\theta = \int \frac{1}{y}\,\frac{\frac{\sin^2\theta}{\cos^2\theta} - 1}{\frac{1}{\cos^2\theta}}\,d\theta$$

$$= \int \frac{1}{y}(\sin^2\theta - \cos^2\theta)\,d\theta = -\int \frac{1}{y}\cos(2\theta)\,d\theta$$

$$= -\frac{\sin(2\theta)}{2y} + C = -\frac{\sin\theta\cos\theta}{y} + C$$

Going back, $\sin\theta = \frac{x}{\sqrt{x^2+y^2}}$, $\cos\theta = \frac{y}{\sqrt{x^2+y^2}}$



$$\therefore \int \frac{x^2-y^2}{(x^2+y^2)^2}\,dx = -\frac{xy}{y(x^2+y^2)} + C = -\frac{x}{x^2+y^2} + C$$

$$\therefore \int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\,dx = -\frac{x}{x^2+y^2}\Big|_0^1 = -\frac{1}{1+y^2}$$

$$\therefore \int_0^1 -\frac{1}{1+y^2}\,dy = -\arctan y\Big|_0^1 = -\frac{\pi}{4}$$

$$\therefore \int_0^1\int_0^1 \frac{x^2-y^2}{(x^2+y^2)^2}\,dx\,dy = \underline{\underline{-\frac{\pi}{4}}}$$

Fubini's Theorem can't be used here since $\frac{x^2-y^2}{(x^2+y^2)^2}$ is not defined at $(0,0)$, and so is not bounded on $[0,1]\times[0,1]$.

18. **18.** Let f be continuous, $f \geq 0$, on the rectangle R. If $\iint_R f\,dA = 0$, prove that $f = 0$ on R.

Suppose $f > 0$ for some $(x,y) \in R$.

Since f is continuous, $\exists$ a rectangle $R' \subset R$ s.t.

$f(x,y) > 0$ for all $(x,y) \in R'$. Let $R' = [a,b] \times [c,d]$

Let $S_n = \sum_{j}^{n-1} \sum_{k}^{n-1} f(c_{jk}) \Delta x \Delta y$.

Note that each term of S_n is ≥ 0, so $S_n \geq 0$.

Eventually, for some N, the size of the regular partition for S_N will be smaller than the size of $[a,b] \times [c,d]$.

For $n \geq N$, S_n will then always have a term $f(c_{jk}) \Delta x \Delta y > 0$

$\therefore$ If choose $\epsilon = \frac{f(x,y)(b-a)(d-c)}{2}$, for some $(x,y) \in [a,b] \times [c,d]$

Then there is an $N > 0$ s.t. for all $n > N$,

it is not true that $S_n < \epsilon$.

$\therefore \lim_{n \to \infty} S_n \neq 0$, contradicting $\iint_R f\,dA = 0$

$\therefore$ Assumption of $f > 0$ is false, so $f = 0$ on R.

5.3 The Double Integral Over More General Regions

Note Title 10/21/2016

1.

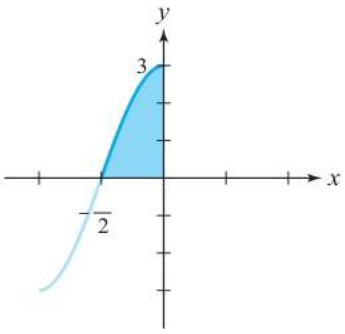
1. In parts (a) through (d) below, each iterated integral is an integral over a region D. Match the integral with the correct region of integration.

(a) $\int_1^2 \int_{\ln x}^{e^x} dy\,dx$

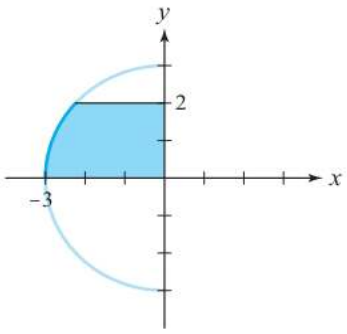
(b) $\int_0^2 \int_{(1/8)x}^{x^{1/3}} dy\,dx$

(c) $\int_0^2 \int_{-\sqrt{9-y^2}}^{0} dx\,dy$

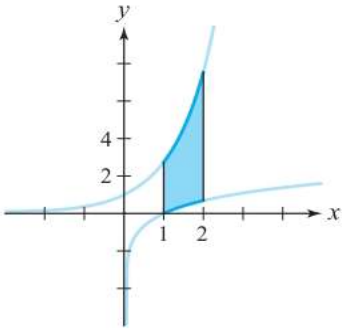
(d) $\int_0^3 \int_{\arccos y/3}^{0} dx\,dy$



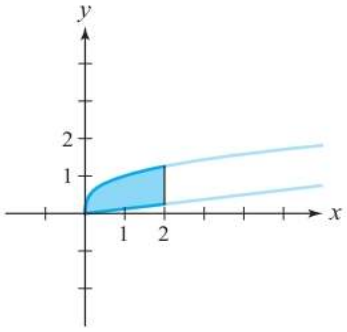
(i)



(ii)



(iii)



(iv)

(a) dy is "inside" and $y = \ln x$ to $y = e^x$ with $x = 1$ to $x = 2$ is shown in (iii).

(b) $y = \frac{x}{8}$ to $y = x^{1/3}$ are y-simple graphs, with $x = 0$ to $x = 2$. This is (iv)

(c) $x = -\sqrt{9-y^2}$ to $x = 0$ are the x-simple graphs, with $y = 0$ to $y = 2$ is shown in (ii)

(d) dx "inside" so look for $x = \arccos\frac{y}{3}$ to $x = 0$ with $y = 0$ to $y = 3$, is shown in (i)

2.

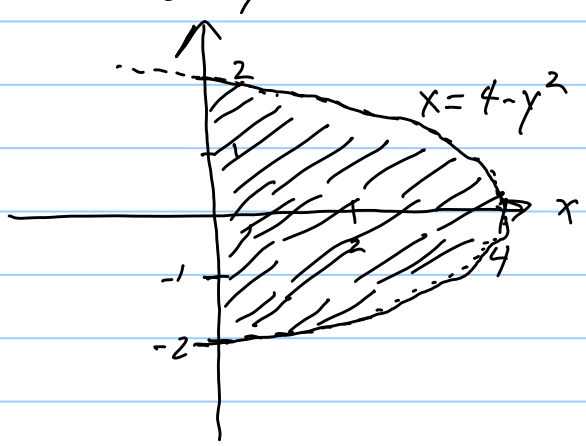
2. Sketch the region D in $\mathbb{R}^2$ that represents the region of integration:

(a) $\int_{-2}^{2} \int_{0}^{4-y^2} (4 - x)\, dx\, dy$

(b) $\int_{0}^{3} \int_{-x}^{x} (6 + y - 2x)\, dy\, dx$

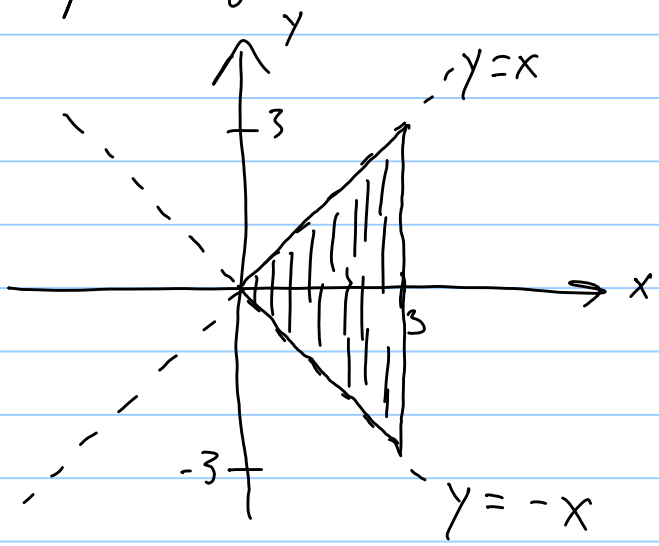
(a) dx "inside" $\Rightarrow$ look for x-simple regions, $x = 0$ to $x = 4 - y^2$ (a parabola).

The y limits are $y = -2$ to $y = 2$



(b) dy "inside" $\Rightarrow$ look for y-simple regions, from $y = -x$ to $y = x$.

The x limits are $x = 0$ to $x = 3$



3.

3. Evaluate the following iterated integrals and draw the regions D determined by the limits. State whether the regions are x-simple, y-simple, or simple.

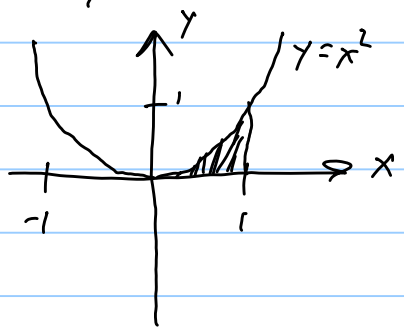
(a) $\int_{0}^{1} \int_{0}^{x^2} dy\, dx$

(b) $\int_{1}^{2} \int_{2x}^{3x+1} dy\, dx$

(c) $\int_{0}^{1} \int_{1}^{e^x} (x + y)\, dy\, dx$

(d) $\int_{0}^{1} \int_{x^3}^{x^2} y\, dy\, dx$

(a) dy "inside" ⇒ look at y-simple, $y=0$ to $y=x^2$, bounded by $x=0$ to $x=1$.

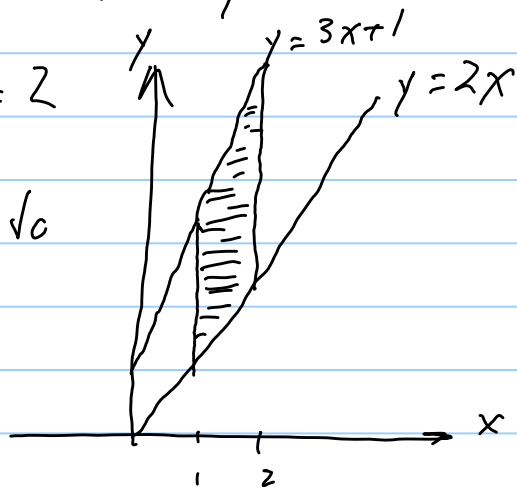


$$\int_0^{x^2} dy = y\Big|_{y=0}^{y=x^2} = x^2 - 0 = x^2$$

$$\therefore \int_0^1 x^2\,dx = \frac{x^3}{3}\Big|_0^1 = \frac{1}{3}$$

D is both y-simple and x-simple.

(b) dy "inside" ⇒ look for y-simple curves, $\therefore$ $y=2x$ to $y=3x+1$, bounded by $x=1$ to $x=2$



The figure can be broken up into either y-simple or x-simple curves.

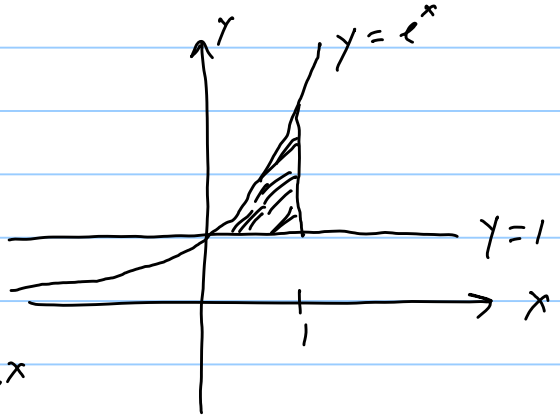
$$\int_{2x}^{3x+1} dy = y\Big|_{2x}^{3x+1} = x+1$$

$$\therefore \int_1^2 (x+1)\,dx = \frac{x^2}{2} + x\Big|_1^2 = \left(\frac{4}{2}+2\right) - \left(\frac{1}{2}+1\right)$$

$$= \frac{5}{2}$$

(c) (c) $\int_0^1 \int_1^{e^x} (x+y)\,dy\,dx$ dy "inside" ⇒ look for y-simple curves.

$\therefore$ look at $y=1$ to $y=e^x$, bounded from $x=0$ to $x=1$



Region can be viewed as x-simple or y-simple.

$$\int_1^{e^x} (x+y)\,dy = xy + \frac{y^2}{2}\Big|_{y=1}^{y=e^x}$$

$$= xe^x + \frac{e^{2x}}{2} - \left(x + \frac{1}{2}\right)$$

$$= xe^x + \frac{e^{2x}}{2} - x - \frac{1}{2}$$

$\therefore$

$$\int_0^1 \left(xe^x + \frac{e^{2x}}{2} - x - \frac{1}{2}\right)dx = xe^x - e^x + \frac{e^{2x}}{4} - \frac{x^2}{2} - \frac{x}{2}\Big|_0^1$$

$$= e - e + \frac{e^2}{4} - \frac{1}{2} - \frac{1}{2} - \left(0 - 1 + \frac{1}{4} - 0 - 0\right)$$

$$= \frac{e^2}{4} - 1 + 1 - \frac{1}{4} = \underline{\underline{\frac{e^2-1}{4}}}$$

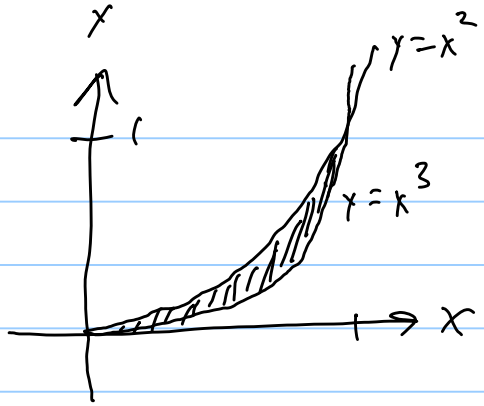
(d) (d) $\int_0^1 \int_{x^3}^{x^2} y\,dy\,dx$ dy "inside" $\Rightarrow$ look for y-simple curves.

$\therefore$ $y = x^3$ to $y = x^2$, bounded by $x=0$ to $x=1$.

$$\int_{x^3}^{x^2} y\,dy = \frac{y^2}{2}\Big|_{x^3}^{x^2} = \frac{x^4}{2} - \frac{x^6}{2}$$

$$\therefore \int_0^1 \left(\frac{x^4}{2} - \frac{x^6}{2}\right) dx =$$

$$\frac{x^5}{10} - \frac{x^7}{14}\Big|_0^1 = \frac{1}{5} - \frac{1}{14} = \frac{9}{70}$$



Domain figure can be viewed either as x-simple or y-simple.

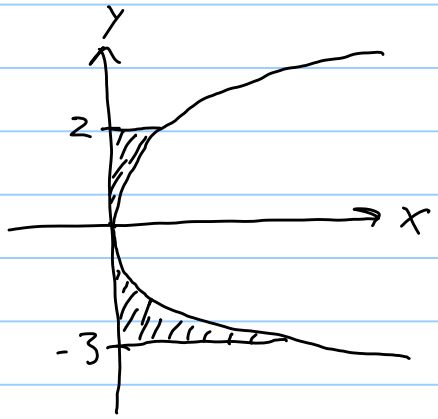
4.

4. Evaluate the following integrals and sketch the corresponding regions.

(a) $\displaystyle\int_{-3}^{2}\int_{0}^{y^2} (x^2 + y)\, dx\, dy$

$x = 0$ to $x = y^2$

$y = -3$ to $y = 2$



$$\int_0^{y^2} (x^2 + y)\, dx = \frac{x^3}{3} + xy\Big|_{x=0}^{x=y^2}$$

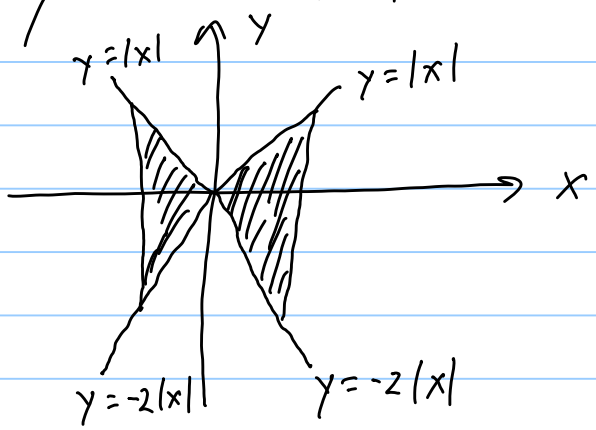
$$= \frac{y^6}{3} + y^3$$

$$\therefore \int_{-3}^{2} \left(\frac{y^6}{3} + y^3\right) dy = \frac{y^7}{21} + \frac{y^4}{4}\Big|_{-3}^{2}$$

$$= \frac{2^7}{21} + \frac{2^4}{4} - \left(\frac{(-3)^7}{21} + \frac{(-3)^4}{4}\right)$$

$$= \frac{128}{21} + \frac{16}{4} + \frac{2187}{21} - \frac{81}{4} = \frac{2059}{21} - \frac{65}{4}$$

(6) (b) $\int_{-1}^{1}\int_{-2|x|}^{|x|} e^{x+y}\,dy\,dx$ $y=-2|x|$ to $y=|x|$ bounded by $x=-1$ to $x=1$



$$\int_{-1}^{1}\int_{-2|x|}^{|x|} e^{x+y}\,dy\,dx =$$

$$\int_{-1}^{0}\int_{2x}^{-x} e^{x+y}\,dy\,dx + \int_{0}^{1}\int_{-2x}^{x} e^{x+y}\,dy\,dx$$

$$\int_{2x}^{-x} e^{x+y}\,dy = e^{x+y}\Big|_{y=2x}^{y=-x} = e^{x+(-x)} - e^{x+2x} = 1 - e^{3x}$$

and $$\int_{-2x}^{x} e^{x+y}\,dy = e^{x+y}\Big|_{y=-2x}^{y=x} = e^{2x} - e^{-x}$$

$$\therefore \int_{-1}^{0}(1-e^{3x})\,dx + \int_{0}^{1}(e^{2x}-e^{-x})\,dx =$$

$$x - \frac{e^{3x}}{3}\Big|_{-1}^{0} + \frac{e^{2x}}{2} + e^{-x}\Big|_{0}^{1} =$$

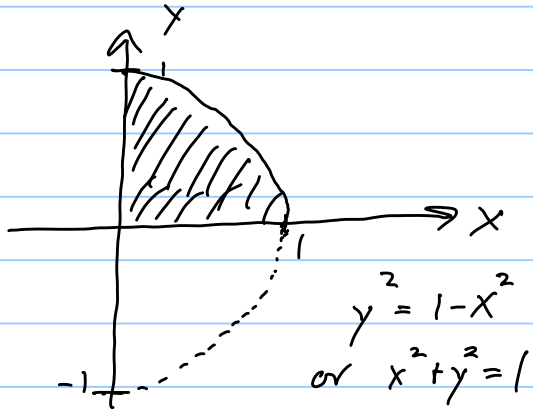
$$\left(0-\frac{1}{3}\right)-\left(-1-\frac{e^{-3}}{3}\right) + \left(\frac{e^{2}}{2}+e^{-1}\right)-\left(\frac{1}{2}+1\right) =$$

$$\frac{2}{3} + \frac{e^{-3}}{3} + \frac{e^{2}}{2} + e^{-1} - \frac{3}{2} = \frac{e^{-3}}{3} + \frac{e^{2}}{2} + e^{-1} - \frac{5}{6}$$

(c) (c) $\int_0^1 \int_0^{(1-x^2)^{1/2}} dy\,dx$ $y = 0$ to $y = (1-x^2)^{1/2}$ bounded by $x = 0$ to $x = 1$

$$\int_0^{(1-x^2)^{1/2}} dy = y\Big|_0^{(1-x^2)^{\frac{1}{2}}}$$

$$= \sqrt{1-x^2}$$



$$\therefore \int_0^1 \sqrt{1-x^2}\,dx \quad \text{Let } x = \sin\theta \therefore dx = \cos\theta\,d\theta$$

$x = 0 : \theta = 0 \quad x = 1 : \theta = \frac{\pi}{2}$

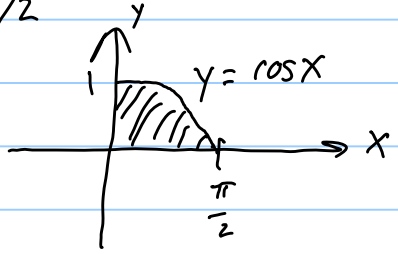
$$= \int_0^{\frac{\pi}{2}} \sqrt{1-\sin^2\theta}\,\cos\theta\,d\theta = \int_0^{\frac{\pi}{2}} \cos^2\theta\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \frac{1+\cos 2\theta}{2}\,d\theta = \frac{1}{2}\theta + \frac{\sin 2\theta}{4}\Big|_0^{\frac{\pi}{2}} = \frac{\pi}{4} + 0 - (0+0)$$

$$= \frac{\pi}{4}$$

(d) (d) $\int_0^{\pi/2} \int_0^{\cos x} y \sin x\,dy\,dx$ $y = 0$ to $y = \cos x$ bounded by $x = 0$ to $x = \pi/2$

$$\int_0^{\cos x} y \sin x\,dy =$$



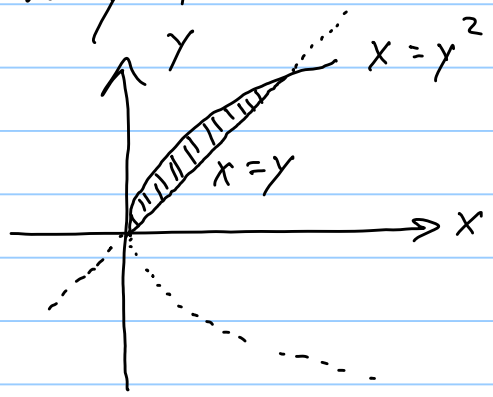
$$\frac{y^2}{2}\sin x \Big|_{y=0}^{y=\cos x} = \frac{\cos^2 x \sin x}{2}$$

$$\therefore \int_0^{\frac{\pi}{2}} \frac{\cos^2 x \sin x}{2}\, dx = -\frac{\cos^3 x}{6}\Big|_0^{\frac{\pi}{2}}$$

$$= 0 - \left(-\frac{1}{6}\right) = \frac{1}{6}$$

(e) (e) $\int_0^1 \int_{y^2}^{y} (x^n + y^m)\, dx\, dy, \quad m, n > 0$

$x = y^2$ to $x = y$ bounded by $y = 0$ to $y = 1$



$$\int_{y^2}^{y} (x^n + y^m)\, dx =$$

$$\frac{x^{n+1}}{n+1} + x y^m \Big|_{x=y^2}^{x=y}$$

$$= \frac{y^{n+1}}{n+1} + y^{m+1} - \left(\frac{y^{2n+2}}{n+1} + y^{2+m}\right)$$

$$\therefore \int_0^1 \left(\frac{y^{n+1}}{n+1} - \frac{y^{2n+2}}{n+1} + y^{m+1} - y^{m+2}\right) dy$$

$$= \frac{y^{n+2}}{(n+2)(n+1)} - \frac{y^{2n+3}}{(2n+3)(n+1)} + \frac{y^{m+2}}{m+2} - \frac{y^{m+3}}{m+3}\Big|_0^1$$

$$= \frac{1}{(n+2)(n+1)} - \frac{1}{(2n+3)(n+1)} + \frac{1}{n+2} - \frac{1}{n+3}$$

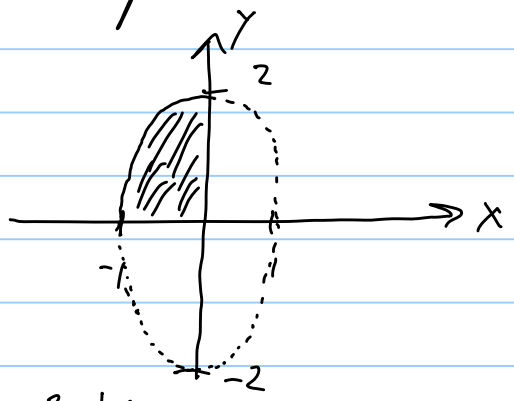
(f) (f) $\int_{-1}^{0}\int_{0}^{2(1-x^2)^{1/2}} x\,dy\,dx$

$y = 0$ to $y = 2(1-x^2)^{1/2}$ bounded by $x = -1$ to $x = 0$

Note: $y^2 = 4(1-x^2)$, $x^2 + \frac{y^2}{4} = 1$ an ellipse.

$$\int_0^{2(1-x^2)^{1/2}} x\,dy =$$

$$xy\Big|_{y=0}^{y=2(1-x^2)^{1/2}} = 2x(1-x^2)^{\frac{1}{2}}$$



$$\therefore \int_{-1}^{0} 2x(1-x^2)^{\frac{1}{2}}\,dx = -\frac{2}{3}(1-x^2)^{\frac{3}{2}}\Big|_{-1}^{0}$$

$$= -\frac{2}{3}(1-0)^{3/2} - \left(-\frac{2}{3}(1-1)^{\frac{3}{2}}\right) = -\frac{2}{3}$$

5. **5.** Use double integrals to compute the area of a circle of radius r.

Use $f(x,y) = 1$, so that $f(x,y)\,dx\,dy = dx\,dy$, and the volume effectively becomes the area.

From $x^2 + y^2 = r^2$, $y = \pm\sqrt{r^2 - x^2}$

$$\therefore \iint_R dA = \int_{-r}^{r}\int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} dy\,dx$$

$$\int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} dy = \sqrt{r^2-x^2} - \left(-\sqrt{r^2-x^2}\right) = 2\sqrt{r^2-x^2}$$

$$\therefore \int_{-r}^{r} 2\sqrt{r^2-x^2}\,dx$$

Use a substitution $x = r\cos\theta$

$\therefore dx = -r\sin\theta\,d\theta$

$\theta = -\pi \rightarrow x = -r$

$\theta = 0 \rightarrow x = r$

$$\therefore \int_{-r}^{r} 2\sqrt{r^2-x^2}\,dx = \int_{-\pi}^{0} 2\sqrt{r^2-r^2\cos^2\theta}\,(-r\sin\theta)\,d\theta$$

$$= \int_{-\pi}^{0} 2\sqrt{r^2\sin^2\theta}\,(-r\sin\theta)\,d\theta = \int_{-\pi}^{0} 2r(-\sin\theta)(-r\sin\theta)\,d\theta$$

since $\sqrt{\sin^2\theta} = -\sin\theta$, $-\pi \le \theta \le 0$

$$= \int_{-\pi}^{0} 2r^2\sin^2\theta\,d\theta$$

Now use $\cos 2\theta = \cos^2\theta - \sin^2\theta$

$\therefore \cos 2\theta = (1-\sin^2\theta) - \sin^2\theta$

$= 1 - 2\sin^2\theta$

$\therefore 2\sin^2\theta = 1 - \cos 2\theta$

$$= \int_{-\pi}^{0} r^2(1-\cos 2\theta)\,d\theta = r^2\theta - \frac{r^2}{2}\sin 2\theta \Big|_{-\pi}^{0}$$

$$= 0 - 0 - \left[-\pi r^2 - \frac{r^2}{2}\sin(-2\pi)\right] = \underline{\pi r^2}$$

6.

6. Using double integrals, determine the area of an ellipse with semiaxes of length a and b.

Let $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be the ellipse.

Rewrite as $b^2x^2 + a^2y^2 = a^2b^2$. Let $f(x,y) = 1$ be the function on the region D defined by the ellipse.

For a given x, $a^2y^2 = a^2b^2 - b^2x^2$, $y = \pm\sqrt{b^2 - \frac{b^2}{a^2}x^2}$

$\therefore \int_{-\sqrt{b^2 - \frac{b^2}{a^2}x^2}}^{\sqrt{b^2 - \frac{b^2}{a^2}x^2}} dy$ is the area of a slice at a given x, for the function $f(x,y) = 1$, from the bottom of the ellipse to the top.

$\therefore \int_{-a}^{a} \int_{-\sqrt{b^2 - \frac{b^2}{a^2}x^2}}^{\sqrt{b^2 - \frac{b^2}{a^2}x^2}} dy\,dx$ is the volume of the solid of height $1 = f(x,y)$ using the ellipse as the base. This effectively is the formula for the area of the ellipse: Volume = (height)(base) = 1(base).

$$\therefore \int_{-\sqrt{b^2-\frac{b^2}{a^2}x^2}}^{\sqrt{b^2-\frac{b^2}{a^2}x^2}} dy = 2\sqrt{b^2-\frac{b^2}{a^2}x^2}$$

$$\therefore \int_{-a}^{a} 2\sqrt{b^2-\frac{b^2}{a^2}x^2}\,dx \qquad \text{Use } x = a\cos\theta \therefore dx = -a\sin\theta\,d\theta$$
$$\theta = -\pi \to x = -a,\ \theta = 0 \to x = a$$

$$= \int_{-\pi}^{0} 2\sqrt{b^2-\frac{b^2}{a^2}(a^2\cos^2\theta)}\,(-a\sin\theta)\,d\theta$$

$$= \int_{-\pi}^{0} 2b\sqrt{\sin^2\theta}\,(-a\sin\theta)\,d\theta$$

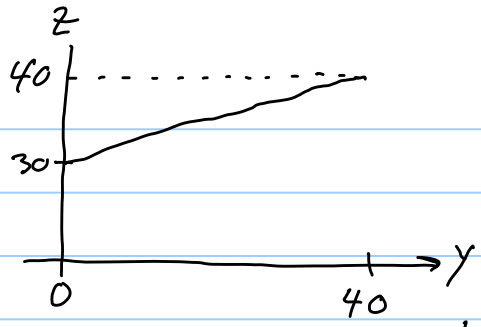
$$= \int_{-\pi}^{0} 2b(-\sin\theta)(-a\sin\theta)\,d\theta \qquad \text{as } \sqrt{\sin^2\theta} = -\sin\theta \text{ for } -\pi \le \theta \le 0$$

$$= \int_{-\pi}^{0} 2ab\sin^2\theta\,d\theta \qquad \text{Now use } 1-\cos 2\theta = 2\sin^2\theta$$

$$= \int_{-\pi}^{0} ab(1-\cos 2\theta)\,d\theta = ab\theta - \frac{ab}{2}\sin 2\theta \Big|_{-\pi}^{0}$$

$$= 0 - 0 - \left\{-ab\pi - \frac{ab}{2}\sin(-2\pi)\right\} = \underline{\pi ab}$$

7. **7.** What is the volume of a barn that has a rectangular base 20 ft by 40 ft, vertical walls 30 ft high at the front (which we assume is on the 20-ft side of the barn), and 40 ft high at the rear? The barn has a flat roof. Use double integrals to compute the volume.



Sideways view of barn gives $z = 30 + \frac{10}{40}y = 30 + \frac{1}{4}y$

∴ For a fixed y, the height is $30 + \frac{1}{4}y$.

∴ An area slice parallel to x-axis is

$$\int_0^{20} \left(30 + \frac{1}{4}y\right) dx \quad \therefore \text{ Volume} = \int_0^{40}\int_0^{20} \left(30 + \frac{y}{4}\right) dx\,dy$$

$$\int_0^{20} \left(30 + \frac{y}{4}\right) dx = 30x + \frac{y}{4}x \Big|_0^{20} = 600 + 5y$$

$$\therefore \int_0^{40} (600 + 5y)\, dy = 600y + \frac{5}{2}y^2 \Big|_0^{40}$$

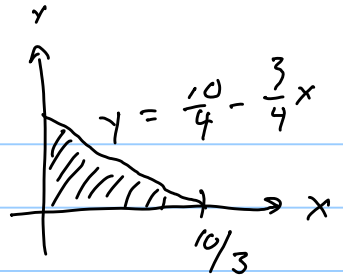
$$= 600(40) + \frac{5}{2}(40)^2 = 28{,}000 \text{ ft}^3$$

8. **8.** Let D be the region bounded by the positive x and y axes and the line $3x + 4y = 10$. Compute

$$\iint_D (x^2 + y^2)\, dA.$$

For a fixed x, $y = \frac{10 - 3x}{4}$. The line intersects the x-axis at $3x + 4(0) = 10$, $x = \frac{10}{3}$

$$\therefore \int_0^{\frac{10}{3}} \int_0^{\frac{10-3x}{4}} (x^2+y^2)\,dy\,dx$$



$$\int_0^{\frac{10}{4}-\frac{3}{4}x} (x^2+y^2)\,dy = x^2y + \frac{y^3}{3}\Bigg|_0^{\frac{10-3x}{4}}$$

$$= x^2\left(\frac{10-3x}{4}\right) + \frac{(10-3x)^3}{3\cdot 4^3}$$

$$= \frac{5x^2}{2} - \frac{3x^3}{4} + \frac{(10-3x)^3}{192}$$

$$\therefore \int_0^{\frac{10}{3}} \frac{5x^2}{2} - \frac{3x^3}{4} + \frac{(10-3x)^3}{192}\,dx$$

$$= \frac{5x^3}{6} - \frac{3}{16}x^4 + \frac{(10-3x)^4}{192(4)(-3)}\Bigg|_0^{\frac{10}{3}}$$

$$= \frac{5}{6}\left(\frac{10}{3}\right)^3 - \frac{3}{16}\left(\frac{10}{3}\right)^4 + 0 - \left[0 - 0 + \frac{10^4}{(192)(4)(-3)}\right]$$

$$= \frac{5(1000)}{6(27)} - \frac{3(10{,}000)}{16(81)} + \frac{10{,}000}{192(4)(3)}$$

$$2\cdot 3^4 \qquad 2^4\cdot 3^4 \qquad 2^8\cdot 3^2$$

$$= \frac{5(128)(1000) - 48(10{,}000) + 9(10{,}000)}{2^8\cdot 3^4}$$

$$\frac{640,000 - 480,000 + 90,000}{2^8 \cdot 3^4} = \frac{250,000}{2^8 \cdot 3^4}$$

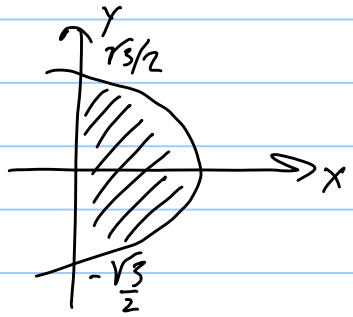
$$= \frac{5^2 \cdot 10,000}{2^8 \cdot 3^4} = \frac{5^2 \cdot (5 \cdot 2)^4}{2^8 \cdot 3^4} = \frac{5^3 \cdot 2^4}{2^8 \cdot 3^4} = \frac{5^3}{2^4 \cdot 3^4}$$

$$= \frac{125}{1296}$$

9.

9. Let D be the region bounded by the y axis and the parabola $x = -4y^2 + 3$. Compute

$$\iint_D x^3 y \, dx \, dy.$$



For $x = 0$, $4y^2 = 3$, $y = \pm \frac{\sqrt{3}}{2}$

For a fixed y, an area slice is from $x = 0$ to $x = -4y^2 + 3$

$$\therefore \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \int_0^{-4y^2+3} (x^3 y) \, dx \, dy = \iint_D x^3 y \, dx \, dy$$

$$\int_0^{-4y^2+3} (x^3 y) \, dx = \frac{x^4 y}{4} \Big|_{x=0}^{x=-4y^2+3} = \frac{(-4y^2+3)^4 y}{4}$$

$$\therefore \int_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}} \frac{(-4y^2+3)^4 y}{4} \, dy = \frac{(-4y^2+3)^5}{(5)(-8)(4)} \Big|_{-\frac{\sqrt{3}}{2}}^{\frac{\sqrt{3}}{2}}$$

$$= \frac{\left[-4\left(\frac{\sqrt{3}}{2}\right)^2 + 3\right]^5}{(5)(-8)(4)} - \frac{\left[-4\left(-\frac{\sqrt{3}}{2}\right)^2 + 3\right]^5}{(5)(-8)(4)} = 0$$

10. **10.** Evaluate $\int_0^1 \int_0^{x^2} (x^2 + xy - y^2)\, dy\, dx$. Describe this iterated integral as an integral over a certain region D in the xy plane.

dy is "inside", so the area slice is from $y = 0$ to $y = x^2$ for a given x. x varies from 0 to 1.

$\therefore$ D is area under parabola $y = x^2$, $x \in [0, 1]$.

$$\therefore \int_0^{x^2} (x^2 + xy - y^2)\, dy = x^2 y + \frac{xy^2}{2} - \frac{y^3}{3} \Big|_{y=0}^{y=x^2}$$

$$= x^4 + \frac{x^5}{2} - \frac{x^6}{3}$$

$$\therefore \int_0^1 \left(x^4 + \frac{x^5}{2} - \frac{x^6}{3}\right) dx = \frac{x^5}{5} + \frac{x^6}{12} - \frac{x^7}{21} \Big|_0^1$$

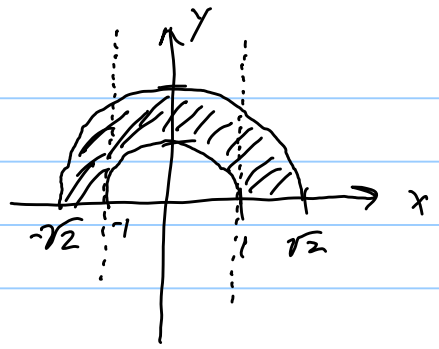
$$= \frac{1}{5} + \frac{1}{12} - \frac{1}{21} = \frac{84 + 35 - 20}{(5)(4)(3)(7)} = \frac{99}{(5)(4)(3)7}$$

$5 \times 4 \times 3 \times 7$

$$= \frac{33}{140}$$

11.

11. Let D be the region given as the set of (x, y), where $1 \le x^2 + y^2 \le 2$ and $y \ge 0$. Is D an elementary region? Evaluate $\iint_D f(x, y)\, dA$, where $f(x, y) = 1 + xy$.



(a) yes, D is 3 adjacent y-simple elementary regions. For $-\sqrt{2} \le x \le -1$ and $1 \le x \le \sqrt{2}$, use $y_2 = \sqrt{2-x^2}$ and $y_1 = 0$. For $-1 \le x \le 1$, use $y_2 = \sqrt{2-x^2}$ and $y_1 = \sqrt{x^2-1}$

(b) $$\iint_D (1+xy)\, dA = \int_{-\sqrt{2}}^{-1}\int_0^{\sqrt{2-x^2}} (1+xy)\, dy\, dx + \int_{-1}^{1}\int_{\sqrt{1-x^2}}^{\sqrt{2-x^2}} (1+xy)\, dy\, dx + \int_{1}^{\sqrt{2}}\int_0^{\sqrt{2-x^2}} (1+xy)\, dy\, dx$$

[1] $$\int_{-\sqrt{2}}^{-1}\int_0^{\sqrt{2-x^2}} (1+xy)\, dy\, dx = \int_{-\sqrt{2}}^{-1}\left[y + x\frac{y^2}{2}\Big|_0^{\sqrt{2-x^2}} \right] dx$$

$$= \int_{-\sqrt{2}}^{-1} \sqrt{2-x^2} + \frac{x(2-x^2)}{2}\, dx = \int_{-\sqrt{2}}^{-1} \sqrt{2-x^2}\, dx + \frac{x^2}{2} - \frac{x^4}{8}\Big|_{-\sqrt{2}}^{-1}$$

$$= \int_{-\sqrt{2}}^{-1} \sqrt{2-x^2}\, dx + \left(\frac{1}{2} - \frac{1}{8}\right) - \left(1 - \frac{1}{2}\right) = \int_{-\sqrt{2}}^{-1} \sqrt{2-x^2}\, dx - \frac{1}{8}$$

Now let $x = \sqrt{2}\sin\theta$ $\therefore dx = \sqrt{2}\cos\theta\, d\theta$

$\theta = -\frac{\pi}{2} \to x = -\sqrt{2}$ $\quad \theta = -\frac{\pi}{4} \to x = -1$

$$\therefore \int_{-\sqrt{2}}^{-1} \sqrt{2-x^2}\, dx = \int_{-\frac{\pi}{2}}^{-\frac{\pi}{4}} \sqrt{2-2\sin^2\theta}\,(\sqrt{2}\cos\theta)\, d\theta$$

$$\int_{-\frac{\pi}{2}}^{-\frac{\pi}{4}} 2\sqrt{\cos^2\theta}\,(\cos\theta)\, d\theta = \int_{-\frac{\pi}{2}}^{-\frac{\pi}{4}} 2\cos^2\theta\, d\theta$$

as $\cos\theta \geq 0$ for $-\frac{\pi}{2} \leq \theta \leq -\frac{\pi}{4}$

$$= \int_{-\frac{\pi}{2}}^{-\frac{\pi}{4}} (1+\cos 2\theta)\, d\theta$$

As $\cos 2\theta = \cos^2\theta - \sin^2\theta$
$= \cos^2 - (1-\cos^2)$
$= 2\cos^2\theta - 1$

$$= \theta + \frac{\sin 2\theta}{2}\Big|_{-\frac{\pi}{2}}^{-\frac{\pi}{4}} = -\frac{\pi}{4} - \frac{1}{2} - \left(-\frac{\pi}{2} + 0\right)$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

$$\therefore \int_{-\sqrt{2}}^{-1}\int_{0}^{\sqrt{2-x^2}} (1+xy)\, dy\, dx = \frac{\pi}{4} - \frac{1}{2} - \frac{1}{8} = \frac{\pi}{4} - \frac{5}{8}$$ [1]

[2] $$\int_{-1}^{1}\int_{\sqrt{1-x^2}}^{\sqrt{2-x^2}} (1+xy)\, dy\, dx = \int_{-1}^{1}\left[y + \frac{xy^2}{2}\Big|_{\sqrt{1-x^2}}^{\sqrt{2-x^2}}\right] dx$$

$$= \int_{-1}^{1} \sqrt{2-x^2} + \frac{x(2-x^2)}{2} - \sqrt{1-x^2} - \frac{x(1-x^2)}{2} \, dx$$

$$= \int_{-1}^{1} \sqrt{2-x^2}\,dx - \int_{-1}^{1} \sqrt{1-x^2}\,dx + \int_{-1}^{1} \frac{x}{2}\,dx$$

$$= \int_{-1}^{1} \sqrt{2-x^2}\,dx - \int_{-1}^{1} \sqrt{1-x^2}\,dx + x^2\Big|_{-1}^{1}$$

$$= \int_{-1}^{1} \sqrt{2-x^2}\,dx - \int_{-1}^{1} \sqrt{1-x^2}\,dx + 0$$

use $x = \sqrt{2}\sin\theta$

$dx = \sqrt{2}\cos\theta\,d\theta$

$\theta = -\frac{\pi}{4} \to x = -1$

$\theta = \frac{\pi}{4} \to x = 1$

use $x = \sin\theta$

$dx = \cos\theta\,d\theta$

$\theta = -\frac{\pi}{2} \to x = -1$

$\theta = \frac{\pi}{2} \to x = 1$

$$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \sqrt{2}\sqrt{\cos^2\theta}\,(\sqrt{2}\cos\theta)\,d\theta - \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\cos^2\theta}\,(\cos\theta)\,d\theta$$

$$= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} 2\cos^2\theta\,d\theta - \int_{-\pi/2}^{\pi/2} \cos^2\theta\,d\theta$$

Using $1+\cos 2\theta = 2\cos^2\theta$ $\qquad \cos^2\theta = \frac{1}{2} + \frac{\cos 2\theta}{2}$

$$= \int_{-\frac{\pi}{4}}^{\pi/4} (1+\cos 2\theta)\,d\theta - \int_{-\pi/2}^{\pi/2} \left(\frac{1}{2} + \frac{\cos 2\theta}{2}\right) d\theta$$

$$= \theta + \frac{\sin 2\theta}{2}\Bigg|_{-\frac{\pi}{4}}^{\frac{\pi}{4}} - \left[\frac{1}{2}\theta + \frac{\sin 2\theta}{4}\Bigg|_{-\frac{\pi}{2}}^{\pi/2}\right]$$

$$= \left(\frac{\pi}{4} + \frac{1}{2}\right) - \left(-\frac{\pi}{4} - \frac{1}{2}\right) - \left[\frac{\pi}{4} + 0 - \left(-\frac{\pi}{4} - 0\right)\right]$$

$$= \frac{\pi}{2} + 1 - \left(\frac{\pi}{2}\right) = \underline{\underline{1}} \qquad [2]$$

$$[3] \quad \int_1^{\sqrt{2}} \int_0^{\sqrt{2-x^2}} (1+xy)\,dy\,dx = \int_1^{\sqrt{2}} \left[y + \frac{xy^2}{2}\Bigg|_0^{\sqrt{2-x^2}}\right] dx$$

$$= \int_1^{\sqrt{2}} \sqrt{2-x^2} + \frac{x(2-x^2)}{2}\,dx = \int_1^{\sqrt{2}} \sqrt{2-x^2}\,dx + \frac{x^2}{2} - \frac{x^4}{8}\Bigg|_1^{\sqrt{2}}$$

$$= \int_1^{\sqrt{2}} \sqrt{2-x^2}\,dx + \left(1 - \frac{1}{2}\right) - \left(\frac{1}{2} - \frac{1}{8}\right) = \int_1^{\sqrt{2}} \sqrt{2-x^2}\,dx + \frac{1}{8}$$

Again use $x = \sqrt{2}\sin\theta \quad dx = \sqrt{2}\cos\theta\,d\theta$

$\theta = \frac{\pi}{4} \to x = 1 \qquad \theta = \frac{\pi}{2} \to x = \sqrt{2}$

$$\therefore \int_1^{\sqrt{2}} \sqrt{2-x^2}\,dx = \int_{\frac{\pi}{4}}^{\pi/2} \sqrt{2-2\sin^2\theta}\,(\sqrt{2}\cos\theta)\,d\theta$$

$$= \int_{\pi/4}^{\pi/2} 2\sqrt{\cos^2\theta}\,(\cos\theta)\,d\theta = \int_{\pi/4}^{\pi/2} 2\cos^2\theta\,d\theta$$

as $\cos\theta \geq 0$ for $\pi/4 \leq \theta \leq \pi/2$

$$= \int_{\pi/4}^{\pi/2} (1+\cos 2\theta)\,d\theta \qquad \text{using } 2\cos^2\theta = 1+\cos 2\theta$$

$$= \theta + \frac{\sin 2\theta}{2}\Bigg|_{\pi/4}^{\pi/2} = \left(\frac{\pi}{2}+0\right) - \left(\frac{\pi}{4}+\frac{1}{2}\right)$$

$$= \frac{\pi}{4} - \frac{1}{2}$$

$$\therefore \int_1^{\sqrt{2}} \int_0^{\sqrt{2-x^2}} (1+xy)\,dy\,dx = \frac{\pi}{4} - \frac{1}{2} + \frac{1}{8} = \underline{\frac{\pi}{4} - \frac{3}{8}} \quad [3]$$

$$\therefore [1] + [2] + [3] = \left(\frac{\pi}{4} - \frac{5}{8}\right) + (1) + \left(\frac{\pi}{4} - \frac{3}{8}\right)$$

$$= \underline{\frac{\pi}{2}}$$

Check: circle radius $\sqrt{2}$: $\pi(\sqrt{2})^2 = 2\pi$

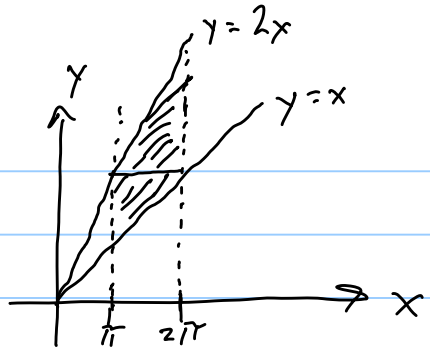
circle radius 1 : $\pi(1)^2 = \pi$

$$\frac{1}{2}(2\pi - \pi) = \frac{\pi}{2}$$

12. **12.** Evaluate the following double integral:

$$\iint_D \cos y \, dx \, dy,$$

where the region D is bounded by $y = 2x$, $y = x$, $x = \pi$, and $x = 2\pi$.



D is both a y-simple region and x-simple region

Here, dx is "inside", so use x-simple regions

Region 1: $\pi \le y \le 2\pi$, use from $x = \pi$ to $x = y$

$$\therefore \int_{\pi}^{2\pi} \int_{\pi}^{y} \cos y \, dx \, dy$$

Region 2: $2\pi \le y \le 4\pi$, use $x = \frac{y}{2}$ to $x = 2\pi$

$$\therefore \int_{2\pi}^{4\pi} \int_{\frac{y}{2}}^{2\pi} \cos y \, dx \, dy$$

$$\therefore \text{Region 1: } \int_{\pi}^{2\pi} \int_{\pi}^{y} \cos y \, dx \, dy = \int_{\pi}^{2\pi} \left[x \cos y \Big|_{\pi}^{y} \right] dy$$

$$= \int_{\pi}^{2\pi} \left[y \cos y - \pi \cos y \right] dy$$

$$= y\sin y \Big|_{\pi}^{2\pi} - \int_{\pi}^{2\pi} \sin y\, dy - \int_{\pi}^{2\pi} \pi \cos y\, dy$$

$$= 0 - 0 - (-\cos y)\Big|_{\pi}^{2\pi} - \pi \sin y \Big|_{\pi}^{2\pi}$$

$$= 0 - [-1 - (1)] - (0 - 0) = \underline{2} \quad [1]$$

Region 2: $\int_{2\pi}^{4\pi} \int_{\frac{y}{2}}^{2\pi} \cos y\, dx\, dy = \int_{2\pi}^{4\pi} \left[x \cos y \Big|_{\frac{y}{2}}^{2\pi} \right] dy$

$$= \int_{2\pi}^{4\pi} \left[2\pi \cos y - \frac{y}{2} \cos y \right] dy$$

$$= 2\pi \sin y \Big|_{2\pi}^{4\pi} - \frac{1}{2} \int_{2\pi}^{4\pi} y \cos y\, dy$$

$$= 0 - \frac{1}{2} \left[y \sin y \Big|_{2\pi}^{4\pi} - \int_{2\pi}^{4\pi} \sin y\, dy \right]$$

$$= -\frac{1}{2} \left[0 + \cos y \Big|_{2\pi}^{4\pi} \right]$$

$$= -\frac{1}{2} [1 - 1] = \underline{0} \quad [2]$$

$\therefore [1] + [2] = 2 + 0 = \underline{2}$

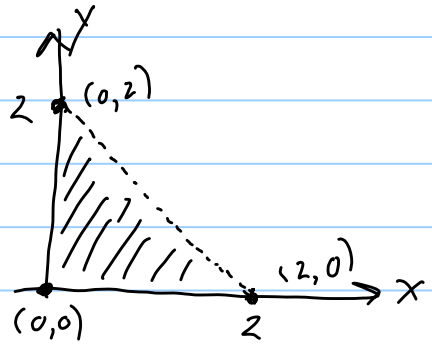
$\therefore \iint_D \cos y \, dx\,dy = \underline{2}$

13.

13. Evaluate the following double integral:

$$\iint_D xy \, dA,$$

where the region D is the triangular region whose vertices are (0, 0), (0, 2), (2, 0).



D is a y-simple region. The hypotenuse is $y = -x + 2$

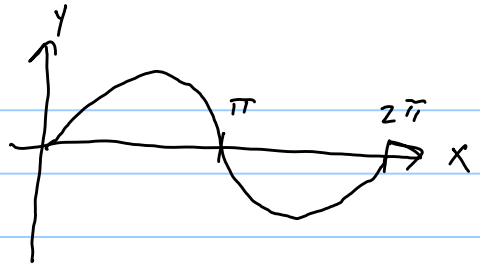
$$\therefore \int_0^2 \int_0^{-x+2} (xy)\, dy\, dx = \int_0^2 \left[\frac{xy^2}{2} \Big|_{y=0}^{y=-x+2} \right] dx$$

$$= \int_0^2 \frac{x(-x+2)^2}{2}\, dx = \int_0^2 \frac{x(x^2 - 4x + 4)}{2}\, dx$$

$$= \int_0^2 \frac{x^3}{2} - 2x^2 + 2x = \frac{x^4}{8} - \frac{2}{3}x^3 + x^2 \Big|_0^2$$

$$= 2 - \frac{16}{3} + 4 = \underline{\frac{2}{3}}$$

14. **14.** Use the formula $A(D) = \iint_D dx\,dy$ to find the area enclosed by one period of the sine function $\sin x$, for $0 \le x \le 2\pi$, and the x axis.



Two symmetrical y-simple regions: $0 \le x \le \pi$, $\pi \le x \le 2\pi$

$$\therefore \int_0^{\pi}\int_0^{\sin x} dy\,dx + \int_{\pi}^{2\pi}\int_{\sin x}^{0} dy\,dx$$

$$= \int_0^{\pi} \left[y \Big|_0^{\sin x} \right] dx + \int_{\pi}^{2\pi} \left[y \Big|_{\sin x}^{0} \right] dx$$

$$= \int_0^{\pi} \sin x\,dx + \int_{\pi}^{2\pi} (-\sin x)\,dx$$

$$= -\cos x \Big|_0^{\pi} + \cos x \Big|_{\pi}^{2\pi}$$

$$= 1 - (-1) + 1 - (-1) = 4$$

15. **15.** Find the volume of the region inside the surface $z = x^2 + y^2$ and between $z = 0$ and $z = 10$.

The domain D is a disc centred at $(x,y)=(0,0)$ with radius $\sqrt{10}$, from $10=x^2+y^2$.

$\therefore$ a y-simple region from $y=-\sqrt{10-x^2}$ to $y=\sqrt{10-x^2}$, and x ranges $-\sqrt{10}\le x\le\sqrt{10}$

The height of any one rectangular solid is $10-(x^2+y^2)$ for any $(x,y)\in dx\,dy$

$\therefore$ Volume $=\iint_D (10-x^2-y^2)\,dA$

$$=\int_{-\sqrt{10}}^{\sqrt{10}}\int_{-\sqrt{10-x^2}}^{\sqrt{10-x^2}} (10-x^2-y^2)\,dy\,dx$$

$$\int_{-\sqrt{10-x^2}}^{\sqrt{10-x^2}} (10-x^2-y^2)\,dy = 10y-x^2y-\frac{y^3}{3}\Bigg|_{y=-\sqrt{10-x^2}}^{y=\sqrt{10-x^2}}$$

$$=20\sqrt{10-x^2}-2x^2\sqrt{10-x^2}-\frac{2}{3}(10-x^2)^{3/2}$$

$\therefore$ Volume $=\int_{-\sqrt{10}}^{\sqrt{10}}\left[20\sqrt{10-x^2}-2x^2\sqrt{10-x^2}-\frac{2}{3}(10-x^2)^{3/2}\right]dx$

Using a table of integrals,

[1] $$20\int_{-\sqrt{10}}^{\sqrt{10}} \sqrt{10-x^2}\,dx = 20\left[\frac{x}{2}\sqrt{10-x^2} + \frac{10}{2}\arcsin\frac{x}{\sqrt{10}}\right]_{-\sqrt{10}}^{\sqrt{10}}$$

$$= 20\left[\frac{\sqrt{10}}{2}(0) + 5\arcsin(1) - \frac{-\sqrt{10}}{2}(0) - 5\arcsin(-1)\right]$$

$$= 20\left[5\left(\frac{\pi}{2}\right) - 5\left(-\frac{\pi}{2}\right)\right] = \underline{100\pi}$$

[2] $$-2\int_{-\sqrt{10}}^{\sqrt{10}} x^2\sqrt{10-x^2}\,dx =$$

$$-2\left[-\frac{x}{4}(10-x^2)^{\frac{3}{2}} + \frac{10}{8}x\sqrt{10-x^2} + \frac{100}{8}\arcsin\frac{x}{\sqrt{10}}\right]_{-\sqrt{10}}^{\sqrt{10}}$$

$$= -2\left[0 + 0 + \frac{100}{8}\arcsin(1) - \left(0 + 0 + \frac{100}{8}\arcsin(-1)\right)\right]$$

$$= -2\left[\frac{25}{2}\left(\frac{\pi}{2}\right) - \frac{25}{2}\left(-\frac{\pi}{2}\right)\right] = \underline{-25\pi}$$

[3] $$-\frac{2}{3}\int_{-\sqrt{10}}^{\sqrt{10}} (10-x^2)^{3/2}\,dx =$$

$$-\frac{2}{3}\left[-\frac{x}{8}\left(2x^2-5(10)\right)\sqrt{10-x^2} + \frac{3(100)}{8}\arcsin\frac{x}{\sqrt{10}}\right]_{-\sqrt{10}}^{\sqrt{10}}$$

$$= -\frac{2}{3}\left[0 + \frac{300}{8}\arcsin(1) - \left(0 + \frac{300}{8}\arcsin(-1)\right)\right]$$

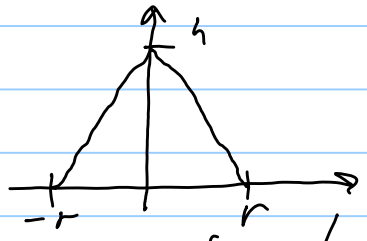
$$= -\frac{2}{3}\left[\frac{300}{8}\left(\frac{\pi}{2}\right) - \frac{300}{8}\left(-\frac{\pi}{2}\right)\right]$$

$$= -\frac{2}{3}\left(\frac{300}{8}\right)(\pi) = \underline{-25\pi}$$

$[1] + [2] + [3] = 100\pi - 25\pi - 25\pi = \underline{50\pi}$

$\therefore \iint_D (10 - x^2 - y^2)\, dA = \underline{50\pi}$

16. **16.** Set up the integral required to calculate the volume of a cone of base radius r and height h.



You can view the cone standing on a circular disc – in which case the height is a function of (x,y), using $x^2+y^2 = d^2$, d = diameter of a circle, and $0 \le d \le r$. If $d = r$, height $= 0$, if $d = 0$, height $= h$.

Let $f(x,y)$ = height function.

$\therefore f(x,y) = h - d\left(\frac{h}{r}\right) = h - \sqrt{x^2+y^2}\left(\frac{h}{r}\right)$,

$-r \le x \le r, \quad -r \le y, \le r$

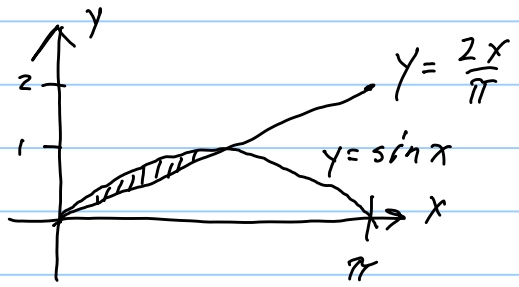
For the base, region D, can be viewed as a

y-simple region, $-\sqrt{r^2-x^2} \le y \le \sqrt{r^2-x^2}$

$\therefore$ Volume $= \iint_D f(x,y)\, dA = \int_{-r}^{r} \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} f(x,y)\, dy\, dx$

$$= \int_{-r}^{r} \int_{-\sqrt{r^2-x^2}}^{\sqrt{r^2-x^2}} \left[h - \sqrt{x^2+y^2}\left(\frac{h}{r}\right) \right] dy\,dx$$

17. **17.** Evaluate $\iint_D y\,dA$, where D is the set of points (x, y) such that $0 \le 2x/\pi \le y,\ y \le \sin x$.



First figure out what D is:

$y \le \sin x$ means values of y under $y = \sin x$ curve.

$\frac{2x}{\pi} \le y$ means those values of y above the line

$$y = \frac{2}{\pi}x$$

Note that $0 \le x$ and $y = \frac{2x}{\pi} = \sin x$

when $x = \frac{\pi}{2}$, $y = \frac{2(\frac{\pi}{2})}{\pi} = 1$, and $y = \sin\frac{\pi}{2} = 1$

$\therefore\ 0 \le x \le \frac{\pi}{2}$

View D as a y-simple region.

$$\therefore \int_0^{\frac{\pi}{2}} \int_{\frac{2x}{\pi}}^{\sin x} y\,dy\,dx = \int_0^{\frac{\pi}{2}} \left[\frac{y^2}{2} \Big|_{y=\frac{2x}{\pi}}^{y=\sin x} \right] dx$$

$$= \int_0^{\pi/2} \left[\frac{\sin^2 x}{2} - \frac{1}{2}\left(\frac{2x}{\pi}\right)^2 \right] dx$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \sin^2 x\,dx - \frac{2}{\pi^2}\int_0^{\frac{\pi}{2}} x^2\,dx$$

Use $\cos^2\theta - \sin^2\theta = \cos 2\theta$

$(1-\sin^2\theta) - \sin^2\theta = \cos 2\theta$

$\frac{1-\cos 2\theta}{2} = \sin^2\theta$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{1-\cos 2x}{2}\,dx - \frac{2}{\pi^2}\left[\frac{x^3}{3}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2}\left[\frac{x}{2} - \frac{\sin 2x}{4}\right]_0^{\frac{\pi}{2}} - \frac{2}{\pi^2}\left(\frac{\pi^3}{(3)(8)}\right)$$

$$= \frac{\pi}{8} - \frac{\sin\pi}{8} - \frac{\pi}{12} = \frac{\pi}{8} - \frac{\pi}{12} = \frac{\pi}{24}$$

18.

18. From Exercise 9, Section 5.2, $\int_a^b \int_c^d f(x)g(y)\,dy\,dx = \left(\int_a^b f(x)\,dx\right)\left(\int_c^d g(y)\,dy\right)$. Is it true that $\iint_D f(x)g(y)\,dx\,dy = \left(\int_a^b f(x)\,dx\right)\left(\int_{\phi_1(a)}^{\phi_2(b)} g(y)\,dy\right)$ for y-simple regions?

For a y-simple region, $y_1 = \phi_1(x) \le y \le \phi_2(x) = y_2$

and $a \le x \le b$

$$\therefore \iint_D f(x)\,g(y)\,dx\,dy = \iint_D f(x)\,g(y)\,dy\,dx$$

$$= \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x)\, g(y)\, dy\, dx$$

$$= \int_a^b f(x) \left[G(\phi_2(x)) - G(\phi_1(x)) \right] dx$$

assuming $G'(y) = g(y)$.

You can't factor out $[G(\phi_2(x)) - G(\phi_1(x))]$ since it is not a constant.

∴ No, The assertion is false.

19.

19. Let D be a region given as the set of (x, y) with $-\phi(x) \le y \le \phi(x)$ and $a \le x \le b$, where ϕ is a nonnegative continuous function on the interval $[a, b]$. Let $f(x, y)$ be a function on D such that $f(x, y) = -f(x, -y)$ for all $(x, y) \in D$. Argue that $\iint_D f(x, y)\, dA = 0$.

$$\iint_D f(x,y)\, dA = \int_a^b \int_{-\phi(x)}^{\phi(x)} f(x,y)\, dy\, dx$$

$$= \int_a^b \left[\int_{-\phi(x)}^{0} f(x,y)\, dy + \int_0^{\phi(x)} f(x,y)\, dy \right] dx$$

$$= \int_a^b \left[-\int_0^{-\phi(x)} f(x,y)\, dy + \int_0^{\phi(x)} f(x,y)\, dy \right] dx$$

$$= \int_a^b \left[\int_0^{-\phi(x)} -f(x,y)\,dy + \int_0^{\phi(x)} f(x,y)\,dy \right] dx$$

$$= \int_a^b \left[\int_0^{-\phi(x)} f(x,-y)\,dy + \int_0^{\phi(x)} f(x,y)\,dy \right] dx \quad [1]$$

Let $u = -y$ $\therefore du = -dy$

when $y = -\phi(x)$, $u = -y = \phi(x)$

$y = 0 \rightarrow u = 0$

$$\therefore \int_0^{-\phi(x)} f(x,-y)\,dy = \int_0^{\phi(x)} -f(x,u)\,du = -\int_0^{\phi(x)} f(x,u)\,du$$

Now use $y = u$, $dy = du$, $u = 0 \rightarrow y = 0$

$u = \phi(x) \rightarrow y = \phi(x)$

$$= -\int_0^{\phi(x)} f(x,y)\,dy$$

$\therefore$ [1] becomes

$$\int_a^b \left\{ \int_0^{-\phi(x)} f(x,-y)\,dy + \int_0^{\phi(x)} f(x,y)\,dy \right] dx =$$

$$\int_a^b \left[-\int_0^{\phi(x)} f(x,y)\,dy + \int_0^{\phi(x)} f(x,y)\,dy \right] dx =$$

$$\int_a^b 0 \, dx = \underline{0}$$

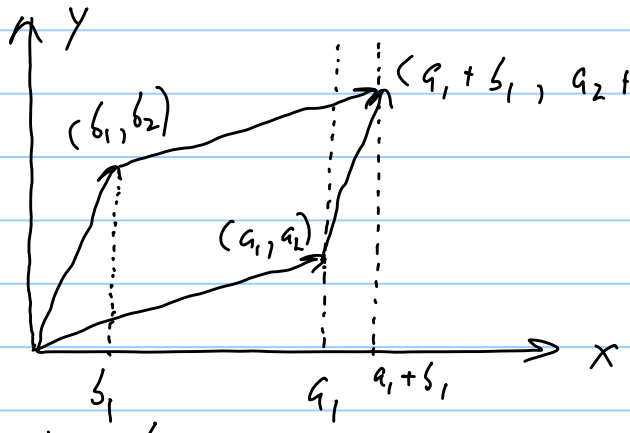
20. **20.** Use the methods of this section to show that the area of the parallelogram D determined by two planar vectors $\mathbf{a}$ and $\mathbf{b}$ is $|a_1 b_2 - a_2 b_1|$, where $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$ and $\mathbf{b} = b_1\mathbf{i} + b_2\mathbf{j}$.

Assume $\vec{a}$ and $\vec{b}$ not parallel, and not $\vec{0}$.

$\therefore \vec{a} \times \vec{b} = \|\vec{a}\| \|\vec{b}\| \sin\theta \neq 0$

so $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_1 & a_2 & 0 \\ b_1 & b_2 & 0 \end{vmatrix} = \vec{a} \times \vec{b} = a_1 b_2 - a_2 b_1 \neq 0.$

Consider



We have 3 y-simple regions

$0 \le x \le b_1 \quad y_1 = \frac{a_2}{a_1} x \quad$ to $\quad y_2 = \frac{b_2}{b_1} x$

$b_1 \le x \le a_1 \quad y_1 = \frac{a_2}{a_1} x \quad$ to $\quad y_2 = \frac{a_2}{a_1} x + \left(b_2 - \frac{a_2}{a_1} b_1\right)$

$a_1 \le x \le a_1 + b_1 \quad y_1 = \frac{b_2}{b_1} x + \left(a_2 - \frac{b_2}{b_1} a_1\right)$ to

$y_2 = \frac{a_2}{a_1} x + \left(b_2 - \frac{a_2}{a_1} b_1\right)$

Area of parallelogram $= \iint_D dy\,dx$

$$= \iint_{A_1} dy\,dx + \iint_{A_2} dy\,dx + \iint_{A_3} dy\,dx$$

$$A_1: \int_0^{b_1} \int_{\frac{a_2}{a_1}x}^{\frac{b_2}{b_1}x} dy\,dx = \int_0^{b_1} \left(\frac{b_2}{b_1}x - \frac{a_2}{a_1}x\right) dx$$

$$= \frac{b_2}{b_1}\frac{x^2}{2} - \frac{a_2}{a_1}\frac{x^2}{2}\Bigg|_0^{b_1} = \frac{b_2}{b_1}\cdot\frac{b_1^2}{2} - \frac{a_2}{a_1}\cdot\frac{b_1^2}{2}$$

$$= \frac{b_1 b_2}{2} - \frac{a_2 b_1^2}{2a_1}$$

$$A_2: \int_{b_1}^{a_1} \int_{\frac{a_2}{a_1}x}^{\frac{a_2}{a_1}x + \left(b_2 - \frac{a_2}{a_1}b_1\right)} dy\,dx =$$

$$\int_{b_1}^{a_1} \left(b_2 - \frac{a_2}{a_1}b_1\right) dx = \left(b_2 x - \frac{a_2}{a_1}b_1 x\right)\Bigg|_{b_1}^{a_1}$$

$$= a_1 b_2 - a_2 b_1 - \left(b_1 b_2 - \frac{a_2}{a_1}b_1^2\right)$$

$$= a_1 b_2 - a_2 b_1 - b_1 b_2 + \frac{a_2 b_1^2}{a_1}$$

$$A_3 : \int_{a_1}^{a_1+b_1} \int_{\frac{b_2}{b_1}x + \left(a_2 - \frac{b_2}{b_1}a_1\right)}^{\frac{a_2}{a_1}x + \left(b_2 - \frac{a_2}{a_1}b_1\right)} dy\,dx =$$

$$\int_{a_1}^{a_1+b_1} \frac{a_2}{a_1}x - \frac{b_2}{b_1}x + b_2 - a_2 - \frac{a_2}{a_1}b_1 + \frac{b_2}{b_1}a_1 \; dx$$

$$= \frac{a_2}{a_1}\frac{x^2}{2} - \frac{b_2}{b_1}\frac{x^2}{2} + b_2 x - a_2 x - \frac{a_2}{a_1}b_1 x + \frac{b_2}{b_1}a_1 x \Bigg|_{a_1}^{a_1+b_1}$$

$$= \frac{a_2}{a_1}\frac{(a_1+b_1)^2}{2} - \frac{b_2}{b_1}\frac{(a_1+b_1)^2}{2} + a_1 b_2 + b_1 b_2 - a_1 a_2 - a_2 b_1$$

$$- a_2 b_1 - \frac{a_2 b_1^2}{a_1} + \frac{b_2 a_1^2}{b_1} + a_1 b_2$$

$$- \frac{a_1 a_2}{2} + \frac{b_2}{b_1}\frac{a_1^2}{2} - a_1 b_2 + a_1 a_2 + a_2 b_1 - \frac{b_2}{b_1}a_1^2$$

$$= \frac{a_2}{a_1}\frac{(a_1+b_1)^2}{2} - \frac{b_2}{b_1}\frac{(a_1+b_1)^2}{2} + b_1 b_2 - a_2 b_1 + a_1 b_2 - \frac{a_2 b_1^2}{a_1} + \frac{b_2 a_1^2}{b_1} - \frac{a_1 a_2}{2} - \frac{a_1^2 b_2}{2b_1}$$

$$= \frac{a_2}{2a_1}\left(a_1^2 + 2a_1 b_1 + b_1^2\right) - \frac{b_2}{2b_1}\left(a_1^2 + 2a_1 b_1 + b_1^2\right)$$

$$+ b_1 b_2 - a_2 b_1 + a_1 b_2 - \frac{a_2 b_1^2}{a_1} + \frac{a_1^2 b_2}{b_1} - \frac{a_1 a_2}{2} - \frac{a_1^2 b_2}{2b_1}$$

$$= \frac{a_1 a_2}{2} + a_2 b_1 + \frac{a_2 b_1^2}{2a_1} - \frac{a_1^2 b_2}{2b_1} - a_1 b_2 - \frac{b_1 b_2}{2}$$

$$+ b_1 b_2 - a_2 b_1 + a_1 b_2 - \frac{a_2 b_1^2}{a_1} + \frac{a_1^2 b_2}{b_1} - \frac{a_1 a_2}{2} - \frac{a_1^2 b_2}{2b_1}$$

$$= -\frac{a_2 b_1^2}{2a_1} + \frac{b_1 b_2}{2}$$

$$\therefore A_1 + A_2 + A_3 =$$

$$\left(\frac{b_1 b_2}{2} - \frac{a_2 b_1^2}{2a_1}\right) + \left(a_1 b_2 - a_2 b_1 - b_1 b_2 + \frac{a_2 b_1^2}{a_1}\right) + \left(-\frac{a_2 b_1^2}{2a_1} + \frac{b_1 b_2}{2}\right)$$

$$= a_1 b_2 - a_2 b_1$$

$$\therefore \text{Area} = |A_1 + A_2 + A_3| = |a_1 b_2 - a_2 b_1|$$

Note: any parallelogram can be rotated so that the acute angle is in Quadrant I.

If a rectangle, then $(a_1, a_2) = (a_1, 0)$, $(b_1, b_2) = (0, b_2)$. $\therefore |a_1 b_2 - a_2 b_1| = |a_1 b_2|$ so formula still works.

21. **21.** Describe the area $A(D)$ of a region as a limit of areas of inscribed rectangles, as in Example 3.

This question seems to be identical to Example 3.

$$A(D) = \lim_{n\to\infty} \sum_{j,k=0}^{n-1} f(c_{jk})\,\Delta x\,\Delta y, \qquad \Delta x = x_{j+1} - x_j, \quad \Delta y = y_{j+1} - y_j$$

where $f(c_{jk}) = \begin{cases} 1, & \text{if } c_{jk} \in D \\ 0, & \text{if } c_{jk} \notin D \end{cases}$, c_{jk} any point in R_{jk}

where R_{jk} is a subrectangle of the partition R, a rectangle that contains D.

5.4 Changing the Order of Integration

Note Title 10/27/2016

1.

1. Change the order of integration, but do not evaluate, the following integrals:

(a) $\int_0^8 \int_{1/2y}^4 dx\,dy$

(b) $\int_0^9 \int_0^{\sqrt{y}} dx\,dy$

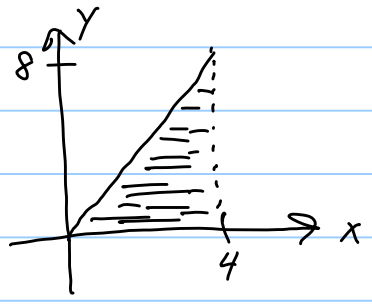
(c) $\int_0^4 \int_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}} dx\,dy$

(d) $\int_{\pi/2}^{\pi} \int_0^{\sin x} dy\,dx$

(a) The original integral must be $\int_0^8 \int_{\frac{1}{2}y}^4 dx\,dy$

as $y \neq 0$ for $\frac{1}{2y}$

$\therefore$ for $\frac{y}{2} \leq x \leq 4$ and $0 \leq y \leq 8$

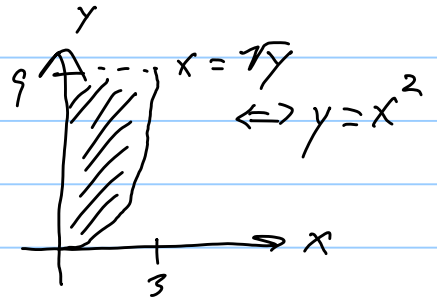


Can be also described as

$0 \leq x \leq 4$ and $0 \leq y \leq 2x$

$\therefore \int_0^4 \int_0^{2x} dy\,dx$

(b) $0 \leq y \leq 9$ and $0 \leq x \leq \sqrt{y}$

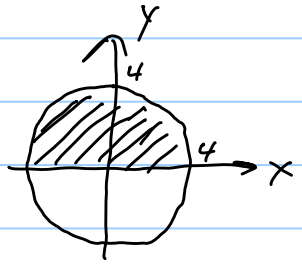


so x ranges from 0 to $\sqrt{9} = 3$

$\therefore$ $0 \leq x \leq 3$ and $x^2 \leq y \leq 9$

$$\therefore \int_0^3 \int_{x^2}^{9} dy\,dx$$

(c) (c) $\int_0^4 \int_{-\sqrt{16-y^2}}^{\sqrt{16-y^2}} dx\,dy$



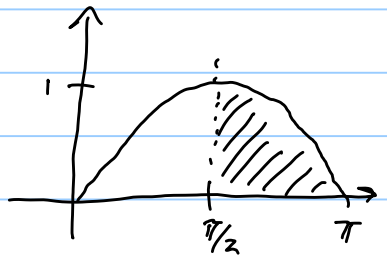
$-\sqrt{16-y^2} \le x \le \sqrt{16-y^2} \quad 0 \le y \le 4$

is the same region as $-4 \le x \le 4$

and $0 \le y \le \sqrt{16-x^2}$

$$\therefore \int_{-4}^{4} \int_{0}^{\sqrt{16-x^2}} dy\,dx$$

(d) (d) $\int_{\pi/2}^{\pi} \int_{0}^{\sin x} dy\,dx$



$0 \le y \le \sin x, \quad \frac{\pi}{2} \le x \le \pi$

The max & min values for sinx are $\sin(\frac{\pi}{2})$, $\sin(\pi)$, or, between 0 and 1.

∴ after switching, y will range $0 \le y \le 1$

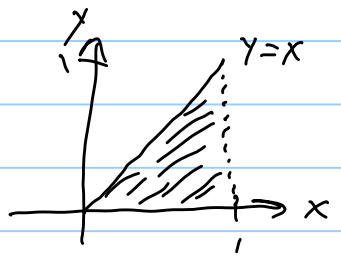
The relationship for x will be the inverse of sinx.

∴ arcsiny, so $\frac{\pi}{2} \le x \le \arcsin y$

$$\therefore \int_0^1 \int_{\frac{\pi}{2}}^{\arcsin y} dx\,dy$$

2. **2.** Change the order of integration and evaluate:

$$\int_0^1 \int_y^1 \sin(x^2)\,dx\,dy.$$



Initially, $0 \le y \le 1$, $y \le x \le 1$

$\therefore$ Same region Δ described as: $0 \le x \le 1$, $0 \le y \le x$

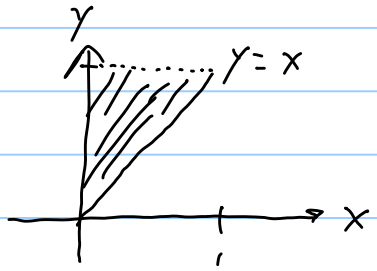
$$\therefore \int_0^1 \int_0^x \sin(x^2)\,dy\,dx = \int_0^1 \left[y \sin(x^2) \Big|_{y=0}^{y=x} \right] dx$$

$$= \int_0^1 x \sin(x^2)\,dx = -\frac{1}{2}\cos(x^2)\Big|_0^1 =$$

$$-\frac{1}{2}\cos(1) - \left(-\frac{1}{2}\cos(0)\right) = \frac{1}{2} - \frac{1}{2}\cos(1)$$

3. **3.** In the following integrals, change the order of integration, sketch the corresponding regions, and evaluate the integral both ways.

(a) (a) $\int_0^1 \int_x^1 xy\,dy\,dx$



$0 \le x \le 1$, $x \le y \le 1$

$$\int_0^1 \int_x^1 (xy)\, dy\, dx = \int_0^1 \left[\frac{xy^2}{2} \Big|_{y=x}^{y=1} \right] dx$$

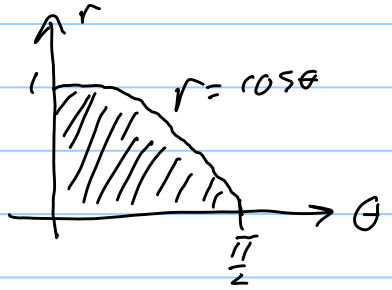
$$= \int_0^1 \left(\frac{x}{2} - \frac{x^3}{2} \right) dx = \frac{x^2}{4} - \frac{x^4}{8} \Big|_0^1 = \frac{1}{4} - \frac{1}{8} = \frac{1}{8}$$

Change order: $0 \le x \le y$, $0 \le y \le 1$

$$\therefore \int_0^1 \int_0^y (xy)\, dx\, dy = \int_0^1 \left[\frac{x^2}{2} y \Big|_{x=0}^{x=y} \right] dx$$

$$= \int_0^1 \frac{y^3}{2}\, dy = \frac{y^4}{8} \Big|_0^1 = \frac{1}{8}$$

(5) (b) $\displaystyle\int_0^{\pi/2} \int_0^{\cos\theta} \cos\theta \, dr\, d\theta$



Initially, $0 \le r \le \cos\theta$

$0 \le \theta \le \pi/2$

$$\int_0^{\frac{\pi}{2}} \int_0^{\cos\theta} \cos\theta \, dr\, d\theta = \int_0^{\frac{\pi}{2}} \left[r\cos\theta \Big|_{r=0}^{r=\cos\theta} \right] d\theta$$

$$= \int_0^{\frac{\pi}{2}} \cos^2\theta \, d\theta = \int_0^{\frac{\pi}{2}} \frac{\cos 2\theta + 1}{2}\, d\theta$$

using $\cos 2\theta = \cos^2\theta - \sin^2\theta$

$= \cos^2\theta - (1 - \cos^2\theta)$

$= 2\cos^2\theta - 1$

$$= \frac{\sin 2\theta}{4} + \frac{\theta}{2}\Big|_0^{\frac{\pi}{2}} = 0 + \frac{\pi}{4} - (0+0) = \underline{\frac{\pi}{4}}$$

The inverse of $r = \cos\theta$ is $\theta = \arccos r$

Initially, r varied between 0 and $\max(\cos\theta)$

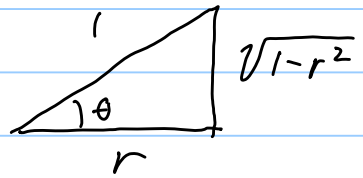
Here, $\max(\cos\theta) = \cos(0) = 1$.

For the inverse, θ varies between 0 and $\arccos(r)$ and $0 \le r \le 1$.

$$\therefore \int_0^1 \int_0^{\arccos(r)} \cos\theta \, d\theta \, dr = \int_0^1 \left[\sin\theta \Big|_{\theta=0}^{\theta=\arccos(r)} \right] dr$$

$$= \int_0^1 \sin(\arccos(r)) \, dr$$

For $\theta = \arccos(r)$, $r = \cos\theta$. $\therefore$



$$\therefore \sin\theta = \sqrt{1-r^2} = \sin(\arccos(r))$$

$$\therefore \int_0^1 \sin(\arccos(r)) \, dr = \int_0^1 \sqrt{1-r^2} \, dr$$

$$= \frac{r}{2}\sqrt{1-r^2} + \frac{1}{2}\arcsin(r)\Big|_0^1 \quad \text{using table of integrals}$$

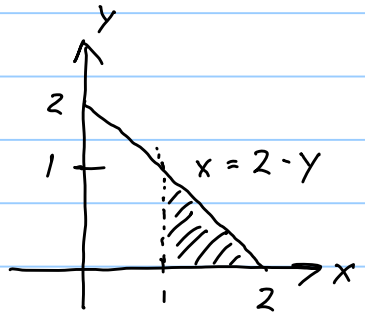
$$= \frac{1}{2}\arcsin(1) = \frac{1}{2}\left(\frac{\pi}{2}\right) = \underline{\frac{\pi}{4}}$$

(c) (c) $\int_0^1 \int_1^{2-y} (x+y)^2 \, dx\, dy$

$$\int_0^1 \left[\frac{(x+y)^3}{3} \Big|_{x=1}^{x=2-y} \right] dy = \int_0^1 \left[\frac{8}{3} - \frac{(1+y)^3}{3} \right] dy$$

$$= \frac{8}{3}y - \frac{(1+y)^4}{12} \Big|_0^1 = \frac{8}{3} - \frac{16}{12} - \left(0 - \frac{1}{12}\right) = \underline{\frac{17}{12}}$$

To change,



$x = f(y) = 2 - y$

$y = f^{-1}(x) = 2 - x$

$f(0) = 2 - 0 = 2$
$f(1) = 2 - 1 = 1$
$\Big\}$ $\therefore$ 1 is min.

$\therefore \; 0 \le y \le f^{-1}(x) = 2 - x$

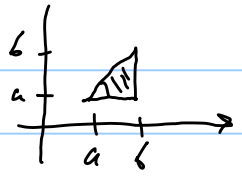
$1 \le x \le 2$ or $1 \le x \le f(y_{min})$

$$\therefore \int_1^2 \int_0^{2-x} (x+y)^2 \, dy\, dx = \int_1^2 \left[\frac{(x+y)^3}{3} \Big|_{y=0}^{y=2-x} \right] dx$$

$$= \int_1^2 \left[\frac{8}{3} - \frac{x^3}{3} \right] dx = \frac{8}{3}x - \frac{x^4}{12} \Big|_1^2 = \frac{16}{3} - \frac{16}{12} - \left(\frac{8}{3} - \frac{1}{12}\right)$$

$$= \underline{\frac{17}{12}}$$

(d) (d) $\int_a^b \int_a^y f(x, y)\, dx\, dy$ (express your answer in terms of antiderivatives).



Let $\frac{\partial G}{\partial x}(x,y) = f(x,y)$. $\therefore \int_a^y f(x,y)\,dx = G(y,y) - G(a,y)$

Now let $F(y)$ be s.t. $F'(y) = G(y,y) - G(a,y)$

$\therefore \int_a^b [G(y,y) - G(a,y)]\,dy = F(b) - F(a)$

Switching, $\int_a^b \int_a^x f(x,y)\,dy\,dx$

Let $H(x,y)$ be s.t. $\frac{\partial H}{\partial y}(x,y) = f(x,y)$

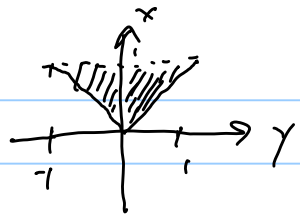
$\therefore \int_a^x f(x,y)\,dy = H(x,x) - H(x,a)$

Now let $J(x)$ be s.t. $J'(x) = H(x,x) - H(x,a)$

$\therefore \int_a^b [H(x,x) - H(x,a)]\,dx = J(b) - J(a)$

4. **4.** Find

(a) $\int_{-1}^{1}\int_{|y|}^{1}(x+y)^2\,dx\,dy$



(a) The "inner" variable limit is $x = |y|$ for $-1 \le y \le 1$,
or $x = -y$, $-1 \le y \le 0$, $x = y$, $0 \le y \le 1$

$$\therefore \int_{-1}^{1}\int_{|y|}^{1}(x+y)^2\,dx\,dy = \int_{-1}^{0}\int_{-y}^{1}(x+y)^2\,dx\,dy + \int_{0}^{1}\int_{y}^{1}(x+y)^2\,dx\,dy$$

$$\int_{-1}^{0}\int_{-y}^{1}(x+y)^2\,dx\,dy = \int_{-1}^{0}\left[\frac{1}{3}(x+y)^3\Big|_{x=-y}^{x=1}\right]dy$$

$$= \int_{-1}^{0}\frac{1}{3}(1+y)^3\,dy = \frac{1}{12}(1+y)^4\Big|_{-1}^{0} = \frac{1}{12} - 0 = \underline{\frac{1}{12}}$$

$$\int_{0}^{1}\int_{y}^{1}(x+y)^2\,dx\,dy = \int_{0}^{1}\left[\frac{1}{3}(x+y)^3\Big|_{x=y}^{x=1}\right]dy$$

$$= \int_{0}^{1}\left[\frac{1}{3}(1+y)^3 - \frac{1}{3}(2y)^3\right]dy = \frac{1}{12}(1+y)^4 - \frac{1}{24}(2y)^4\Big|_{0}^{1}$$

$$= \frac{1}{12}(2)^4 - \frac{1}{24}(2)^4 - \left[\frac{1}{12} - 0\right] = \frac{16}{12} - \frac{16}{24} - \frac{1}{12}$$

$$= \frac{16}{12} - \frac{8}{12} - \frac{1}{12} = \frac{7}{12}$$

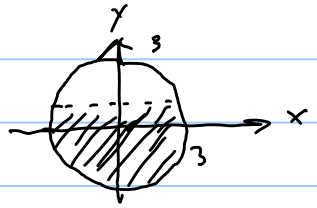
$$\therefore \frac{1}{12} + \frac{7}{12} = \frac{8}{12} = \frac{2}{3}$$

Note: The "change of order" integration would be

$$\int_0^1 \int_{-x}^{x} (x+y)^2 \, dy\, dx \quad \int_0^1 \left[\frac{1}{3}(x+y)^3 \Big|_{y=-x}^{y=x} \right] dx$$

$$= \int_0^1 \frac{1}{3}(2x)^3 \, dx = \int_0^1 \frac{8}{3} x^3 \, dx = \frac{8}{12} x^4 \Big|_0^1 = \frac{8}{12} = \frac{2}{3}$$

(6) (b) $\displaystyle\int_{-3}^{1} \int_{-\sqrt{(9-y^2)}}^{\sqrt{(9-y^2)}} x^2 \, dx\, dy$



$$\int_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} x^2 \, dx = \frac{x^3}{3} \Big|_{-\sqrt{9-y^2}}^{\sqrt{9-y^2}} = 2\frac{(9-y^2)}{3}^{3/2}$$

(using #39 table of integrals from text)

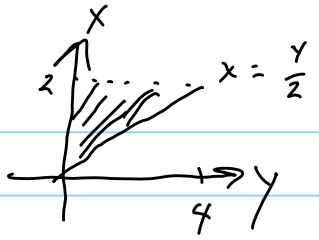
$$\therefore \frac{2}{3} \int_{-3}^{1} (9-y^2)^{3/2} \, dy =$$

$$\frac{2}{3} \left[\frac{y}{8}(5\cdot 9 - 2y^2)\sqrt{9-y^2} + \frac{3(9)}{8} \arcsin\left(\frac{y}{3}\right) \right]_{-3}^{1}$$

$$= \frac{2}{3} \left[\frac{4^3}{8}\sqrt{8} + \frac{27}{8}\arcsin\left(\frac{1}{3}\right) - \left(0 + \frac{27}{8}\left(-\frac{\pi}{2}\right)\right) \right]$$

$$= \frac{4^3}{6}\sqrt{2} + \frac{27}{8}\arcsin\left(\frac{1}{3}\right) + \frac{27}{16}\pi$$

(c) (c) $\int_0^4 \int_{y/2}^2 e^{x^2}\,dx\,dy$



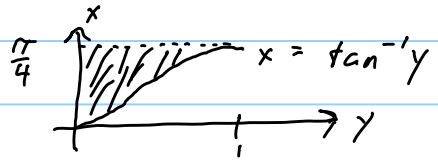
Change order to get an "x" in front of e^{x^2}.

$$\therefore \int_0^4 \int_{y/2}^2 e^{x^2}\,dx\,dy = \int_0^2 \int_0^{2x} e^{x^2}\,dy\,dx$$

$$= \int_0^2 \left[y\,e^{x^2} \Big|_{y=0}^{y=2x} \right] dx = \int_0^2 2x\,e^{x^2}\,dx$$

$$= e^{x^2}\Big|_0^2 = \underline{e^4 - 1}$$

(d) (d) $\int_0^1 \int_{\tan^{-1} y}^{\pi/4} (\sec^5 x)\,dx\,dy$



Change order to get a "tan" in front of $\sec^5 x$.

Over the interval $0 \le y \le 1$, $x = \tan^{-1} y$ is increasing

$\therefore \tan^{-1}(0) = 0$, $\tan^{-1}(1) = \frac{\pi}{4}$ $\Rightarrow$ new outer limits

$$\therefore \int_0^1 \int_{\tan^{-1} y}^{\frac{\pi}{4}} (\sec^5 x)\,dx\,dy = \int_0^{\frac{\pi}{4}} \int_0^{\tan x} (\sec^5 x)\,dy\,dx$$

$$= \int_0^{\frac{\pi}{4}} \tan x \sec^5 x \, dx = \int_0^{\frac{\pi}{4}} (\sec x \tan x) \sec^4(x) \, dx$$

$$= \int_0^{\frac{\pi}{4}} \sec^4(x) \, d(\sec x) = \frac{\sec^5 x}{5} \Big|_0^{\frac{\pi}{4}} = \frac{1}{5\cos^5(x)} \Big|_0^{\frac{\pi}{4}}$$

$$= \frac{1}{5\left(\frac{1}{\sqrt{2}}\right)^5} - \frac{1}{5} = \frac{(\sqrt{2})^5}{5} - \frac{1}{5} = \frac{4\sqrt{2} - 1}{5}$$

5.

5. Change the order of integration and evaluate:

$$\int_0^1 \int_{\sqrt{y}}^1 e^{x^3} \, dx \, dy.$$

$x = \sqrt{y}$ is increasing on $0 \le y \le 1$ (The "outer" limits)

$\therefore$ The new limits become $\int_0^1 \int_0^{x^2} dy \, dx = \int_0^1 \int_0^{x^2} dy \, dx$

$$\therefore \int_0^1 \int_0^{x^2} e^{x^3} \, dy \, dx = \int_0^1 \left[y e^{x^3} \Big|_0^{y=x^2} \right] dx = \int_0^1 x^2 e^{x^3} \, dx$$

$$= \frac{1}{3} e^{x^3} \Big|_0^1 = \frac{1}{3} e - \frac{1}{3}$$

6.

6. Consider the intuitive fact that if a region D in $\mathbb{R}^2$ can be split into a disjoint union of subsets $D = D_1 \cup D_2$, then a double integral over D may also be divided into a sum of two integrals:

$$\iint_D f(x, y)\, dA = \iint_{D_1} f(x, y)\, dA + \iint_{D_2} f(x, y)\, dA.$$

(See Section 5.2 for the analogous statement over a rectangular box.) Are the following attempts to change the order of integration true or false?

(a)

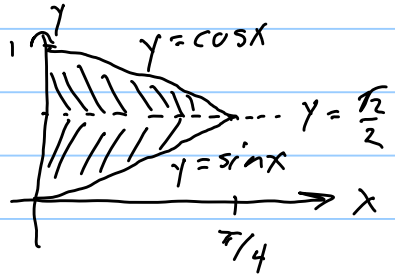
$$\int_0^{\pi/4} \int_{\sin x}^{\cos x} dy\, dx = \int_0^{\sqrt{2}/2} \int_0^{\arcsin y} dx\, dy + \int_{\sqrt{2}/2}^{2} \int_0^{\arccos y} dx\, dy$$

(a)

Over the interval $0 \le x \le \frac{\pi}{4}$, $y = \cos x$ is decreasing, $y = \sin x$ is increasing.

$\cos(0) = 1 \quad \cos(\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$

$\sin(0) = 0 \quad \sin(\pi/4) = \sqrt{2}/2$



∴ To change order, split the integral as:

$$\int_0^{\frac{\pi}{4}} \int_{\sin x}^{\cos x} dy\, dx = \int_0^{\frac{\pi}{4}} \int_{\sin x}^{\frac{\sqrt{2}}{2}} dy\, dx + \int_0^{\frac{\pi}{4}} \int_{\frac{\sqrt{2}}{2}}^{\cos x} dy\, dx$$

$$= \int_{\sin(0)}^{\sin(\frac{\pi}{4})} \int_0^{\arcsin y} dx\, dy + \int_{\cos(\pi/4)}^{\cos(0)} \int_0^{\arccos(y)} dx\, dy$$

$$= \int_0^{\frac{\sqrt{2}}{2}} \int_0^{\arcsin(y)} dx\, dy + \int_{\pi/2}^{1} \int_0^{\arccos(y)} dx\, dy$$

∴ False the second out limit is wrong

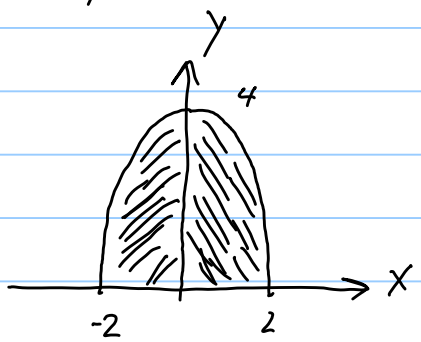
(b) $$\int_{-2}^{2}\int_{0}^{4-x^2} dy\,dx = \int_{0}^{4}\int_{-\sqrt{4-y}}^{\sqrt{4-y}} dx\,dy$$

Over interval $-2 \le x \le 2$, $y = 4-x^2$ is both increasing and decreasing. ∴ Break it up.

$$\int_{-2}^{2}\int_{0}^{4-x^2} dy\,dx = \underbrace{\int_{-2}^{0}\int_{0}^{4-x^2} dy\,dx}_{\text{increasing}} + \underbrace{\int_{0}^{2}\int_{0}^{4-x^2} dy\,dx}_{\text{decreasing}}$$

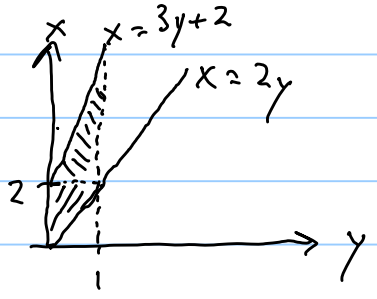
$$= \int_{0}^{4}\int_{-\sqrt{4-y}}^{0} dx\,dy + \int_{0}^{4}\int_{0}^{\sqrt{4-y}} dx\,dy$$

$$= \int_{0}^{4}\int_{-\sqrt{4-y}}^{\sqrt{4-y}} dx\,dy$$



∴ True

(c) $$\int_{0}^{2}\int_{0}^{(1/2)x} dy\,dx + \int_{2}^{5}\int_{(1/3)x-(2/3)}^{1} dy\,dx = \int_{0}^{1}\int_{2y}^{3y+2} dx\,dy$$



Over the interval $0 \le y \le 1$, $x = 2y$ and $x = 3y+2$ are both increasing. Note $2y$ max is at $y = 1$ and $3y+2$ minimum is at $y = 0$.

$x = 2y$ at $y = 1$ is 2
$x = 3y+2$ at $y = 0$ is 2
} ∴ Break up at $x = 2$

For $x = 2y$, $y = \frac{x}{2}$ For $x = 3y+2$, $y = \frac{x-2}{3}$

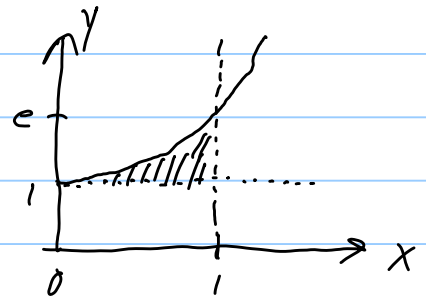
$2(0) = 0$, $2(1) = 2$ $3(0)+2 = 2$, $3(1)+2 = 5$

$$\therefore \int_0^1 \int_{2y}^{3y+2} dx\,dy = \int_0^1 \int_{2y}^{2} dx\,dy + \int_0^1 \int_2^{3y+2} dx\,dy$$

$$= \int_0^2 \int_0^{\frac{x}{2}} dy\,dx + \int_2^5 \int_{\frac{x-2}{3}}^{1} dy\,dx$$

$\therefore$ True

(d) (d) $\int_0^1 \int_1^{e^x} dy\,dx = \int_1^e \int_{\ln y}^1 dx\,dy$



Over the interval $0 \le x \le 1$, $y = e^x$ is increasing

$e^{(0)} = 1$, $e^{(1)} = e$. $y = e^x \Leftrightarrow x = \ln y$

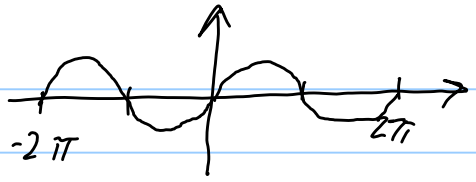
$$\therefore \int_0^1 \int_1^{e^x} dy\,dx = \int_{e^{(0)}}^{e^{(1)}} \int_{\ln y}^1 dx\,dy = \int_1^e \int_{\ln y}^1 dx\,dy$$

$\therefore$ True

7.

7. If $f(x, y) = e^{\sin(x+y)}$ and $D = [-\pi, \pi] \times [-\pi, \pi]$, show that

$$\frac{1}{e} \leq \frac{1}{4\pi^2} \iint_D f(x, y)\, dA \leq e.$$



Over the interval $-2\pi \leq u \leq 2\pi$, $\sin(u)$ takes on
a max at $u = \frac{\pi}{2}$ and $u = -\frac{3}{2}\pi$ $\therefore \sin(u) = 1$
a min at $u = 3/2\,\pi$, $u = -\pi/2$ $\therefore \sin(u) = -1$

$\therefore \sin(x+y)$ over $[-\pi, \pi] \times [-\pi, \pi]$ has
a max of 1 and a min of -1.

$\therefore e^z$ is an increasing function.

$\therefore e^{\sin(x+y)}$ is a max at $e^1 = e$
a min at $e^{-1} = 1/e$.

Area of $\Delta = (2\pi)(2\pi) = 4\pi^2$.

$\therefore \iint_\Delta e^{\sin(x+y)}\, dA$ has a min of $\frac{1}{e}(4\pi^2)$
and a max of $e(4\pi^2)$

$\therefore \frac{1}{e}(4\pi^2) \leq \iint_\Delta f(x,y)\, dA \leq e(4\pi^2)$

$\therefore \frac{1}{e} \leq \frac{1}{4\pi^2} \iint_\Delta f(x,y)\, dA \leq e$

8. **8.** Show that

$$\frac{1}{2}(1-\cos 1) \le \iint_{[0,1]\times[0,1]} \frac{\sin x}{1+(xy)^4}\,dx\,dy \le 1.$$

The min for $1+(xy)^4$ is 1.
The max for $1+(xy)^4$ is 2 ($x=1, y=1$ on $[0,1]\times[0,1]$).

Max for $\sin(x)$ on $[0,1]$ is $\sin(1)$ since
$\sin(x)$ is increasing on $[0,1]$, and $1 \le \frac{\pi}{2}$

$\therefore$ Max for $\frac{\sin(x)}{1+(xy)^4} = \frac{\sin(1)}{1} = \sin(1) \le 1 = \sin(\frac{\pi}{2})$.

Min for $\sin(x)$ on $[0,1]$ is $0 = \sin(0)$.

$\therefore$ Min for $\frac{\sin(x)}{1+(xy)^4} = \frac{\sin(0)}{2} = 0$.

Area of $[0,1]\times[0,1] = 1$.

$\therefore 0(1) \le \iint_D f(x,y)\,dA \le \sin(1) \le 1$

$\therefore 0 \le \iint_{[0,1]\times[0,1]} \frac{\sin(x)}{1+(xy)^4}\,dx\,dy \le 1$

Note also $\frac{\sin(x)}{2} \le \frac{\sin(x)}{1+(xy)^4}$ on $[0,1]\times[0,1]$.

$\therefore \int_0^1\int_0^1 \frac{\sin(x)}{2}\,dx\,dy \le \int_0^1\int_0^1 \frac{\sin(x)}{1+(xy)^4}\,dx\,dy$

But $\int_0^1 \int_0^1 \frac{\sin(x)}{2}\,dx\,dy = \int_0^1 \int_0^1 \frac{\sin(x)}{2}\,dy\,dx$

$= \int_0^1 \left[y\frac{\sin(x)}{2} \Big|_{y=0}^{y=1} \right] dx = \int_0^1 \frac{\sin(x)}{2}\,dx$

$= -\frac{\cos(x)}{2}\Big|_0^1 = -\frac{\cos(1)}{2} + \frac{1}{2} = \frac{1-\cos(1)}{2}$

And $\frac{1-\cos(1)}{2} > 0$.

$\therefore \frac{1-\cos(1)}{2} \leq \iint_{[0,1]\times[0,1]} \frac{\sin x}{1+(xy)^4}\,dx\,dy \leq 1$

---

9. If $D = [-1, 1] \times [-1, 2]$, show that

$$1 \leq \iint_D \frac{dx\,dy}{x^2 + y^2 + 1} \leq 6.$$

For all x, y, $1 \leq x^2+y^2+1$ $\therefore \frac{1}{x^2+y^2+1} \leq 1$

On $[-1,1]\times[-1,2]$, $x^2+y^2+1 \leq (1)^2+(2)^2+1 = 6$

$\therefore \frac{1}{6} \leq \frac{1}{x^2+y^2+1}$

$\therefore \frac{1}{6} \leq \frac{1}{x^2+y^2+1} \leq 6$

$\therefore \frac{1}{6}(D_{Area}) \leq \iint_D \frac{1}{x^2+y^2+1}\,dA \leq 1\,(D_{Area})$

For $[-1,1]\times[-1,2]$, Area $= (2)(3) = 6$

$$\therefore \frac{1}{6}(6) \le \iint_D \frac{1}{x^2+y^2+1}\,dA \le 1(6)$$

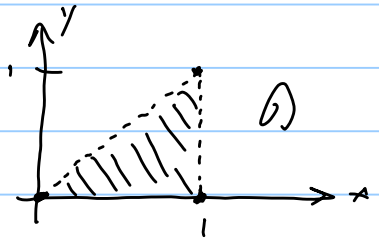
Or,

$$1 \le \iint_D \frac{1}{x^2+y^2+1}\,dx\,dy \le 6$$

10. **10.** Using the mean-value inequality, show that

$$\frac{1}{6} \le \iint_D \frac{dA}{y-x+3} \le \frac{1}{4},$$

where D is the triangle with vertices (0, 0), (1, 1), and (1, 0).



Area of $D = \frac{1}{2}$

Min of $y-x+3 = 2$ $(y=0, x=1)$

Max of $y-x+3 = 3$ $(y=x$, for any $x \in D)$

Note $y-x+3 \ge 0$ on D.

$$\therefore 2 \le y-x+3 \le 3 \Rightarrow \frac{1}{3} \le \frac{1}{y-x+3} \le \frac{1}{2}$$

$$\frac{1}{3}(\text{Area}) = \frac{1}{3}\left(\frac{1}{2}\right) \le \iint_D \frac{dA}{y-x+3} \le \frac{1}{2}(\text{Area}) = \frac{1}{2}\left(\frac{1}{2}\right)$$

$$\therefore \frac{1}{6} \le \iint_D \frac{dA}{y-x+3} \le \frac{1}{4}$$

11. **11.** Compute the volume of an ellipsoid with semiaxes a, b, and c. (HINT: Use symmetry and first find the volume of one half of the ellipsoid.)

A general formula is $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

$\therefore$ Upper half is: $\frac{z}{c} = \sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$, $a, b, c > 0$.

$\therefore$ Upper half volume $= \iint_D f(x,y)\, dA$

where $f(x,y) = c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}$, where D is

$-a \le x \le a$, $-b\sqrt{1 - \frac{x^2}{a^2}} \le y \le b\sqrt{1 - \frac{x^2}{a^2}}$

$$\therefore \frac{1}{2}V = \int_{-a}^{a} \int_{-b\sqrt{1-\frac{x^2}{a^2}}}^{b\sqrt{1-\frac{x^2}{a^2}}} c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}\, dy\, dx$$

Using symmetry again, just find the volume in the positive quadrant.

$$\therefore \frac{1}{8}V = \int_{0}^{a} \int_{0}^{b\sqrt{1-\frac{x^2}{a^2}}} c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}\, dy\, dx$$

(1) First consider $\int_0^{b\sqrt{1-x^2/a^2}} c\sqrt{1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}}\, dy$

Let $u^2 = 1 - \frac{x^2}{a^2}$, valid since $\frac{x^2}{a^2} \le 1$.

Let $v = \frac{y}{b}$. $\therefore b\,dv = dy$

and $y = 0 \Rightarrow v = 0$, $y = bu \Rightarrow v = u$

$$\therefore \int_0^{b\sqrt{1-\frac{x^2}{a^2}}} c\sqrt{1-\frac{x^2}{a^2}-\frac{y^2}{b^2}}\,dy = \int_0^u cb\sqrt{u^2-v^2}\,dv$$

$$= cb\left[\frac{v}{2}\sqrt{u^2-v^2} + \frac{u^2}{2}\arcsin\frac{v}{u}\right]_{v=0}^{v=u}$$

$$= cb\left[0 + \frac{u^2}{2}\arcsin(1) - (0+0)\right]$$

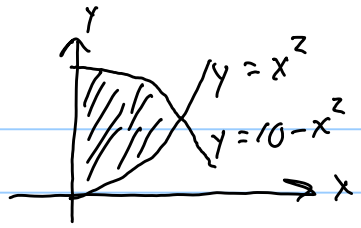
$$= cb\left[\frac{u^2}{2}\left(\frac{\pi}{2}\right)\right] = \frac{\pi}{4}cbu^2 = \frac{\pi bc}{4}\left(1-\frac{x^2}{a^2}\right)$$

(2) $\therefore \frac{\pi bc}{4}\int_0^a \left(1-\frac{x^2}{a^2}\right)dx = \frac{\pi bc}{4}\left[x - \frac{x^3}{3a^2}\right]_0^a$

$$= \frac{\pi bc}{4}\left[a - \frac{a^3}{3a^2}\right] = \frac{\pi bc}{4}\left(\frac{2}{3}a\right) = \frac{\pi abc}{6}$$

$\therefore \frac{1}{8}V = \frac{\pi abc}{6}$, $V = \frac{4}{3}\pi abc$

12. **12.** Compute $\iint_D f(x, y)\, dA$, where $f(x, y) = y^2\sqrt{x}$ and D is the set of (x, y), where $x > 0$, $y > x^2$, and $y < 10 - x^2$.



$x^2 = 10 - x^2 \Rightarrow x^2 = 5, \quad x = \sqrt{5} \quad \therefore 0 \le x \le \sqrt{5}$

$$\therefore \int_0^{\sqrt{5}} \int_{x^2}^{10-x^2} y^2\sqrt{x}\, dy\, dx = \int_0^{\sqrt{5}} \left[\frac{y^3}{3}\sqrt{x}\right]_{y=x^2}^{y=10-x^2} dx$$

$$= \int_0^{\sqrt{5}} \frac{(10-x^2)^3 x^{\frac{1}{2}}}{3} - \frac{x^6 x^{\frac{1}{2}}}{3}\, dx$$

$$= \int_0^{\sqrt{5}} \frac{(10-x^2)^3 x^{\frac{1}{2}}}{3}\, dx - \int_0^{\sqrt{5}} \frac{1}{3} x^{\frac{13}{2}}\, dx$$

(a) $$-\int_0^{\sqrt{5}} \frac{1}{3} x^{\frac{13}{2}}\, dx = -\frac{1}{3}\left(\frac{2}{15}\right) x^{\frac{15}{2}}\Big|_0^{\sqrt{5}} = -\frac{2}{45} 5^{\frac{15}{4}}$$

(b) $$\int_0^{\sqrt{5}} \frac{(10-x^2)^3 x^{\frac{1}{2}}}{3}\, dx = \int_0^{\sqrt{5}} (1000 - 300x^2 + 30x^4 - x^6) x^{\frac{1}{2}}\, dx$$

$$= \int_0^{\sqrt{5}} \frac{1000 x^{\frac{1}{2}} - 300 x^{\frac{5}{2}} + 30 x^{\frac{9}{2}} - x^{\frac{13}{2}}}{3}\, dx$$

$$= \frac{1000}{3}\left(\frac{2}{3}\right) x^{\frac{3}{2}} - 100\left(\frac{2}{7}\right) x^{\frac{7}{2}} + 10\left(\frac{2}{11}\right) x^{\frac{11}{2}} - \frac{1}{3}\left(\frac{2}{15}\right) x^{\frac{15}{2}}\Big|_0^{\sqrt{5}}$$

$$= \frac{2000}{9} s^{\frac{3}{4}} - \frac{200}{7} s^{\frac{7}{4}} + \frac{20}{11} s^{\frac{11}{4}} - \frac{2}{45} s^{\frac{15}{4}}$$

$\therefore$ (a) + (b) =

$$\frac{2000}{9} s^{3/4} - \frac{200}{7} s^{7/4} + \frac{20}{11} s^{\frac{11}{4}} - \frac{4}{45} s^{\frac{15}{4}}$$

13.

13. Find the volume of the region determined by $x^2 + y^2 + z^2 \leq 10$, $z \geq 2$. Use the disk method from one-variable calculus and state how the method is related to Cavalieri's principle.

This is a sphere of radius $\sqrt{10}$. $z \geq 2$ means the volume of the section above the plane $z = 2$.

Using Cavalieri's principle, take a slice, perpendicular to z-axis. The disc slice has a volume of $\pi r^2 \, dz$, since the slice is a flat cylinder.

Here, $r^2 = x^2 + y^2 = 10 - z^2$

$$\therefore V = \int_2^{\sqrt{10}} \pi (10 - z^2) \, dz = \pi 10z - \pi \frac{z^3}{3} \Big|_2^{\sqrt{10}}$$

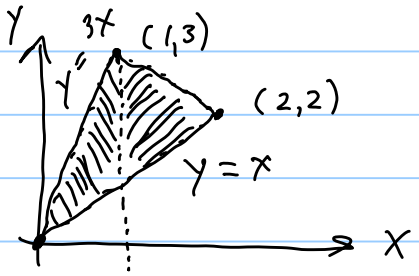
$$= \pi 10\sqrt{10} - \pi \frac{10^{3/2}}{3} - \left(20\pi - \frac{8}{3}\pi\right)$$

$$= \pi\left(\frac{30\sqrt{10} - 10\sqrt{10}}{3}\right) - \pi\left(\frac{60 - 8}{3}\right)$$

$$= \frac{\pi(20\sqrt{10} - 52)}{3}$$

14.

14. Evaluate $\iint_D e^{x-y}\,dx\,dy$, where D is the interior of the triangle with vertices (0, 0), (1, 3), and (2, 2).



Break the triangle into 2 parts, from $0 \le x \le 1$ and $1 \le x \le 2$

The left section is bounded by $y = x$ and $y = 3x$

$$\therefore \int_0^1 \int_x^{3x} dy\,dx$$

The right section is bounded by $y = x$ and

$y = \frac{3-2}{1-2}x + b$, or $y = -x + b$, $(3) = -(1) + b$, $b = 4$

$\therefore y = -x + 4$ $\therefore \int_1^2 \int_x^{-x+4} dy\,dx$

$$\therefore \int_0^1 \int_x^{3x} e^{x-y}\,dy\,dx + \int_1^2 \int_x^{-x+4} e^{x-y}\,dy\,dx$$

(a) $\int_0^1 \int_x^{3x} e^{x-y}\,dy\,dx = \int_0^1 \left[-e^{x-y} \Big|_{y=x}^{y=3x} \right] dx$

$= \int_0^1 \left(-e^{-2x} + e^0 \right) dx = \frac{1}{2} e^{-2x} + x \Big|_0^1$

$= \frac{1}{2} e^{-2} + 1 - \left(\frac{1}{2} + 0 \right) = \frac{1}{2} e^{-2} + \frac{1}{2}$

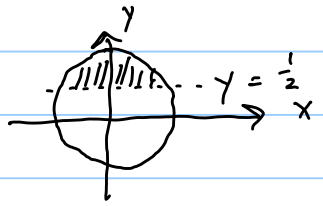
(b) $\int_1^2 \int_x^{-x+4} e^{x-y}\,dy\,dx = \int_1^2 \left[-e^{x-y} \Big|_{y=x}^{y=-x+4} \right] dx$

$= \int_1^2 \left(-e^{2x-4} + 1 \right) dx = -\frac{1}{2} e^{2x-4} + x \Big|_1^2$

$= -\frac{1}{2} + 2 - \left(-\frac{1}{2} e^{-2} + 1 \right) = \frac{1}{2} e^{-2} + \frac{1}{2}$

$\therefore$ (a) + (b) $= e^{-2} + 1$

15. **15.** Evaluate $\iint_D y^3 (x^2 + y^2)^{-3/2}\,dx\,dy$, where D is the region determined by the conditions $\frac{1}{2} \le y \le 1$ and $x^2 + y^2 \le 1$.



The x-simple region is $-\sqrt{1-y^2} \le x \le \sqrt{1-y^2}$

for $\frac{1}{2} \le y \le 1$

$$\therefore \iint_D f(x,y)\,dA = \int_{\frac{1}{2}}^{1} \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} y^3 (x^2+y^2)^{-3/2}\,dx\,dy$$

There looks to be some cancelling of y terms when the limits $x = \pm\sqrt{1-y^2}$ are inserted for x^2+y^2, so proceed without changing order of integration limits.

From a table of integrals: $\int \frac{du}{(a^2+u^2)^{3/2}} = \frac{u}{a^2\sqrt{a^2+u^2}} + C$

$$\therefore \int_{-\sqrt{1-y^2}}^{\sqrt{1-y^2}} \frac{y^3}{(y^2+x^2)^{3/2}}\,dx = \frac{y^3 x}{y^2\sqrt{y^2+x^2}}\Bigg|_{x=-\sqrt{1-y^2}}^{x=\sqrt{1-y^2}}$$

$$= \frac{y^3\sqrt{1-y^2}}{y^2\sqrt{1}} - \left(\frac{-y^3\sqrt{1-y^2}}{y^2\sqrt{1}}\right) = 2y\sqrt{1-y^2}$$

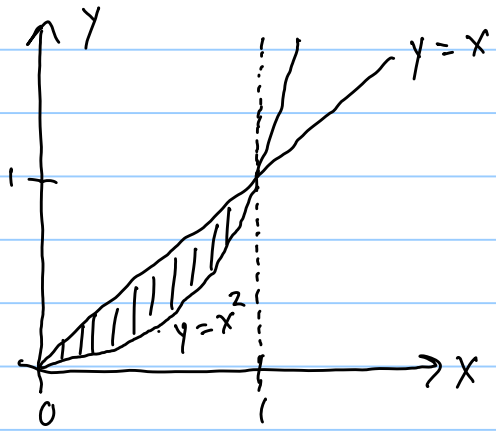
$$\therefore \int_{\frac{1}{2}}^{1} 2y\sqrt{1-y^2}\,dy = -\frac{2}{3}(1-y^2)^{3/2}\Bigg|_{\frac{1}{2}}^{1}$$

$$= -\frac{2}{3}(0) - \left(-\frac{2}{3}\left(1-\left(\frac{1}{2}\right)^2\right)^{3/2}\right)$$

$$= \frac{2}{3}\left(\frac{3}{4}\right)^{3/2} = \frac{2}{3}\left(\frac{\sqrt{3}}{2}\right)^3 = \frac{2}{3}\left(\frac{3\sqrt{3}}{8}\right) = \frac{\sqrt{3}}{4}$$

16.

16. Given that the double integral $\iint_D f(x, y)\, dx\, dy$ of a positive continuous function f equals the iterated integral $\int_0^1 \left[\int_{x^2}^{x} f(x, y)\, dy \right] dx$, sketch the region D and interchange the order of integration.



Change order: $y = x \Rightarrow x = y$

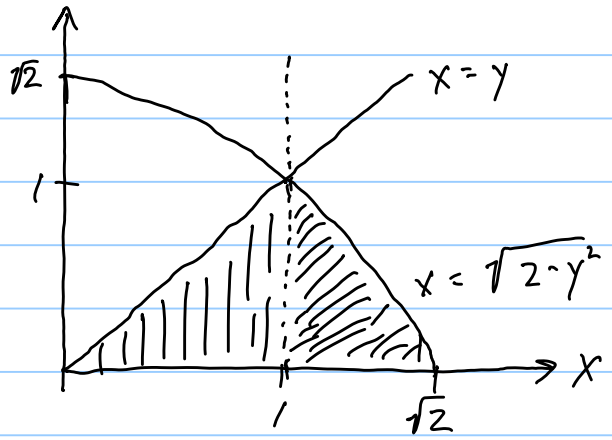
$y = x^2 \Rightarrow x = \pm\sqrt{y}$

use $x = \sqrt{y}$ since $0 \le x \le 1$.

$\therefore \int_0^1 \int_y^{\sqrt{y}} f(x,y)\, dy\, dx$

17.

17. Given that the double integral $\iint_D f(x, y)\, dx\, dy$ of a positive continuous function f equals the iterated integral $\int_0^1 \left[\int_y^{\sqrt{2-y^2}} f(x, y)\, dx \right] dy$, sketch the region D and interchange the order of integration.



Change order: $x = y \Rightarrow y = x$

$x = \sqrt{2-y^2} \Rightarrow$

$y = \pm\sqrt{2-x^2}$

use $y = \sqrt{2-x^2}$

since $0 \le y \le 1$

Over the interval, $0 \le x \le \sqrt{2}$,

there is no simple $y = f(x)$

$\therefore$ Break up into two adjacent y-simple regions,

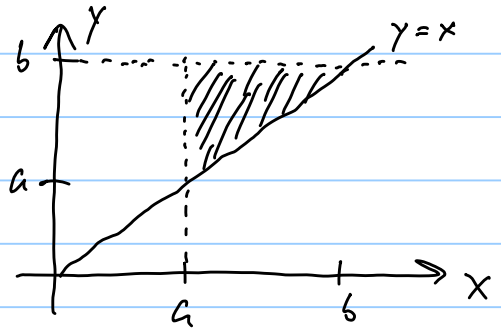
using $y = x$ as one region, $y = \sqrt{2-x^2}$ the other.

$$\therefore \int_0^1 \int_0^x f(x,y)\,dy\,dx + \int_1^{\sqrt{2}} \int_0^{\sqrt{2-x^2}} f(x,y)\,dy\,dx$$

18. **18.** Prove that

$$2\int_a^b \int_x^b f(x)\,f(y)\,dy\,dx = \left(\int_a^b f(x)\,dx\right)^2.$$

[HINT: Notice that $\left(\int_a^b f(x)\,dx\right)^2 = \iint_{[a,b]\times[a,b]} f(x)\,f(y)\,dx\,dy$.]



Using $c\int_a^b f = \int_a^b cf$,

$$\iint_R f(x)f(y)\,dy\,dx = \int_a^b \int_a^b f(x)\,f(y)\,dy\,dx =$$

$$\int_a^b \left[\int_a^b f(x)f(y)\,dy\right] dx = \int_a^b f(x)\left[\int_a^b f(y)\,dy\right] dx$$

But $\int_a^b f(y)\,dy$ is a number (a constant)

$$= \left[\int_a^b f(y)\,dy\right] \int_a^b f(x)\,dx = \left[\int_a^b f\right]^2$$

$$\therefore \left[\int_a^b f(x)\,dx\right]^2 = \int_a^b f(x)\left[\int_a^b f(y)\,dy\right] dx$$

$$= \int_a^b f(x) \left[\int_a^x f(y)\,dy + \int_x^b f(y)\,dy \right] dx$$

as $a \le x \le b$

$$= \int_a^b f(x) \left[\int_a^x f(y)\,dy \right] dx + \int_a^b f(x) \left[\int_x^b f(y)\,dy \right] dx$$

$$= \int_a^b \int_a^x f(x) f(y)\,dy\,dx + \int_a^b \int_x^b f(x) f(y)\,dy\,dx$$

$$= \int_a^b \int_y^b f(x) f(y)\,dy\,dx + \int_a^b \int_x^b f(x) f(y)\,dy\,dx \quad [2]$$

↖ from changing order of integration

But viewed as "dummy variables",

$$\int_a^b \int_y^b f(m) f(n)\,dn\,dm = \int_a^b \int_z^b f(m) f(n)\,dn\,dm$$

$\therefore$ in [2], $\int_a^b \int_y^b f(x) f(y)\,dy\,dx = \int_a^b \int_x^b f(x) f(y)\,dy\,dx$

and so, $\int_a^b \int_y^b f(x) f(y)\,dy\,dx + \int_a^b \int_x^b f(x) f(y)\,dy\,dx = 2 \int_a^b \int_x^b f(x) f(y)\,dy\,dx$

$$\therefore \left[\int_a^b f(x)\,dx\right]^2 = 2\int_a^b \int_x^b f(x)f(y)\,dy\,dx$$

19. **19.** Show that (see Section 2.5, Exercise 29)

$$\frac{d}{dx}\int_a^x \int_c^d f(x, y, z)\,dz\,dy = \int_c^d f(x, y, z)\,dz + \int_a^x \int_c^d f_x(x, y, z)\,dz\,dy.$$

On the left, the "x" variable should be viewed as two separate "dummy variables".

$$\therefore \int_a^x \int_c^d f(x,y,z)\,dz\,dy = \int_a^u \int_c^d f(v,y,z)\,dz\,dy$$

$$\therefore \text{Let } F(u,v) = \int_a^u \int_c^d f(v,y,z)\,dz\,dy$$

By the chain rule, $\frac{\partial F(u,v)}{\partial u} = \frac{\partial F}{\partial u} + \frac{\partial F}{\partial v}\cdot\frac{\partial v}{\partial u}$

When $v(u) = u$, $\frac{\partial v}{\partial u} = 1$.

Letting $u = x$, $v(x) = x$, $\frac{\partial F(x,x)}{\partial x} = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial v}\cdot\frac{\partial v}{\partial x}$

$$= \frac{\partial F}{\partial x} + \frac{\partial F}{\partial v}\cdot(1) = \frac{\partial F}{\partial x} + \frac{\partial F}{\partial v}$$

and $F(x,x) = \int_a^x \int_c^d f(x,y,z)\,dz\,dy$

$$\therefore \frac{d}{dx} F(x,x) = \frac{d}{dx} \int_a^x \int_c^d f(x,y,z)\,dz\,dy$$

$$= \frac{\partial F}{\partial(1\text{st variable})} + \frac{\partial F}{\partial(2\text{nd variable})} \qquad [0]$$

By Fundamental Theorem of Calculus,

$$\frac{d}{dx} \int_a^x g(t)\,dt = g(x)$$

Now view $\int_a^x \left[\int_c^d f(v,y,z)\,dz \right] dy$ as

$$\int_a^x g(y)\,dy, \quad \text{where } g(y) = \int_c^d f(v,y,z)\,dz$$

$$\therefore \frac{d}{dx} \int_a^x g(y)\,dy = g(x) = \int_c^d f(v,x,z)\,dz$$

Letting $v(x) = x$,

$$\frac{d}{dx} \int_a^x g(y)\,dy = \int_c^d f(x,x,z)\,dz$$

$$\therefore \frac{\partial F}{\partial(\text{1st variable})} = \frac{\partial F}{\partial x} = \int_c^d f(x,x,z)\,dz \qquad [1]$$

$$\frac{\partial F}{\partial(\text{2nd variable})} = \frac{\partial F}{\partial v} = \frac{\partial}{\partial v}\int_a^x \int_c^d f(v,y,z)\,dz\,dy$$

$$= \int_a^x \int_c^d \frac{\partial}{\partial v} f(v,y,z)\,dz\,dy$$

$$= \int_a^x \int_c^d f_x(x,y,z)\,dz\,dy \qquad [2]$$

Switching back to $v = x$, and using "differention under the integral".

[0], [1], [2] mean,

$$\frac{d}{dx}\int_a^x \int_c^d f(x,y,z)\,dz\,dy = \int_c^d f(x,x,z)\,dz + \int_a^x \int_c^d f_x(x,y,z)\,dz\,dy$$

5.5 The Triple Integral

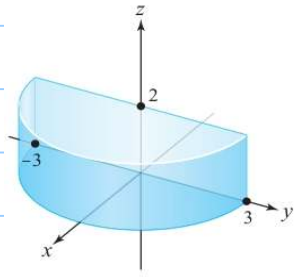
Note Title 11/6/2016

1.

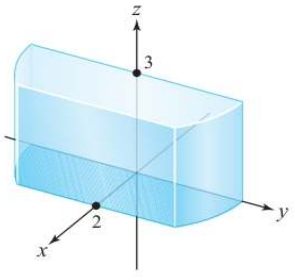
1. In parts (a) through (d) below, each iterated integral is an integral over a region D. Match the integral with the correct region of integration.

(a) $\displaystyle\int_0^2\int_0^3\int_{-\sqrt{9-x}}^{\sqrt{9-x}} dy\,dz\,dx$ (c) $\displaystyle\int_0^1\int_0^x\int_0^y dz\,dy\,dx$

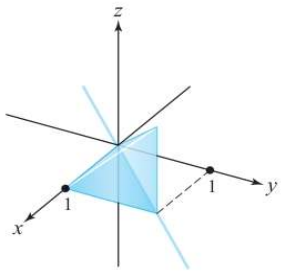
(b) $\displaystyle\int_0^2\int_0^3\int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dy\,dx\,dz$ (d) $\displaystyle\int_0^1\int_0^y\int_0^x dz\,dx\,dy$



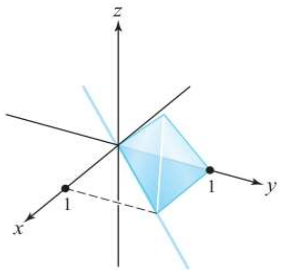
(i)



(ii)



(iii)



(iv)

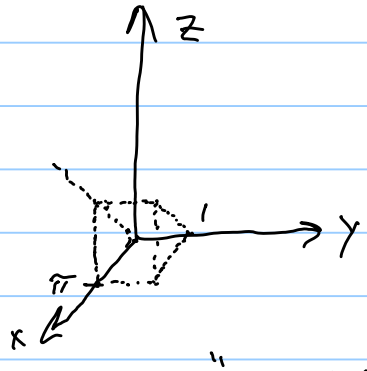
(b) (a) (c) (d)

2.

2. Evaluate the following triple integral:

$$\iiint_W \sin x\,dx\,dy\,dz,$$

where W is the solid given by $0 \le x \le \pi$, $0 \le y \le 1$, and $0 \le z \le x$.



z has "nonconstant" bounds. ∴ Make it "inside".

$$\therefore \int_0^1\int_0^{\pi}\int_0^x \sin x\,dz\,dx\,dy$$

$$\int_0^x \sin x\,dz = z\sin x\Big|_0^x = x\sin x$$

$$\int_0^{\pi} x\sin x\,dx = -x\cos x\Big|_0^{\pi} + \int_0^{\pi}\cos x\,dx$$

$$= \pi + \sin x\Big|_0^{\pi} = \pi$$

$$\int_0^1 \pi\,dy = \pi \qquad \therefore \iiint_W \sin x\,dx\,dy\,dz = \underline{\pi}$$

In Exercises 3 to 6, perform the indicated integration over the given box.

3. **3.** $\iiint_B x^2\,dx\,dy\,dz,\ B = [0,1]\times[0,1]\times[0,1]$

$$\iiint_B x^2\,dx\,dy\,dz = \int_0^1\int_0^1\int_0^1 x^2\,dy\,dz\,dx$$

$$= \int_0^1\int_0^1 x^2\,dz\,dx = \int_0^1 x^2\,dx = \frac{x^3}{3}\Big|_0^1 = \underline{\frac{1}{3}}$$

4. **4.** $\iiint_B e^{-xy}\,y\,dx\,dy\,dz,\ B = [0,1]\times[0,1]\times[0,1]$

$$= \int_0^1\int_0^1\int_0^1 y e^{-xy}\,dz\,dx\,dy = \int_0^1\int_0^1 y e^{-xy}\,dx\,dy$$

$$= \int_0^1 \left[-e^{-xy}\right]_{x=0}^{x=1} dy = \int_0^1 (-e^{-y}+1)\,dy =$$

$$e^{-y} + y \Big|_0^1 = e^{-1} + 1 - (1 + 0) = \frac{1}{e}$$

5. **5.** $\iiint_B (2x+3y+z)\,dx\,dy\,dz,\ B = [0,2]\times[-1,1]\times[0,1]$

$$= \int_{-1}^{1}\int_{0}^{1}\int_{0}^{2} (2x+3y+z)\,dx\,dz\,dy$$

$$= \int_{-1}^{1}\int_{0}^{1} (x^2 + 3xy + xz)\Big|_0^2\,dz\,dy$$

$$= \int_{-1}^{1}\int_{0}^{1} (4 + 6y + 2z)\,dz\,dy = \int_{-1}^{1} (4z + 6yz + z^2)\Big|_0^1\,dy$$

$$= \int_{-1}^{1} (5 + 6y)\,dy = 5y + 3y^2\Big|_{-1}^{1} = (5+3) - (-5+3)$$

$$= \underline{10}$$

6. **6.** $\iiint_B ze^{x+y}\,dx\,dy\,dz,\ B = [0,1]\times[0,1]\times[0,1]$

$$= \int_0^1\int_0^1\int_0^1 ze^{x+y}\,dz\,dx\,dy = \int_0^1\int_0^1 \frac{1}{2}e^{x+y}\,dx\,dy$$

$$= \int_0^1 \left[\frac{1}{2} e^{x+y} \Big|_{x=0}^{x=1} \right] dy = \int_0^1 \frac{1}{2} e^{1+y} - \frac{1}{2} e^{y} \, dy$$

$$= \frac{1}{2} e^{1+y} - \frac{1}{2} e^{y} \Big|_0^1 = \frac{1}{2} e^2 - \frac{1}{2} e - \left(\frac{1}{2} e - \frac{1}{2} \right)$$

$$= \frac{1}{2} e^2 - e + \frac{1}{2}$$

In Exercises 7 to 10, describe the given region as an elementary region.

7.

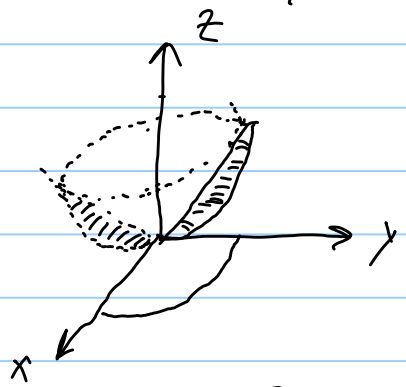
7. The region between the cone $z = \sqrt{x^2 + y^2}$ and the paraboloid $z = x^2 + y^2$

Assuming the region is "underneath" the cone and "above" the paraboloid, they intersect at $x^2+y^2=1$

$$x^2 + y^2 \le z \le \sqrt{x^2 + y^2},$$

$$-\sqrt{1 - x^2} \le y \le \sqrt{1 - x^2},$$

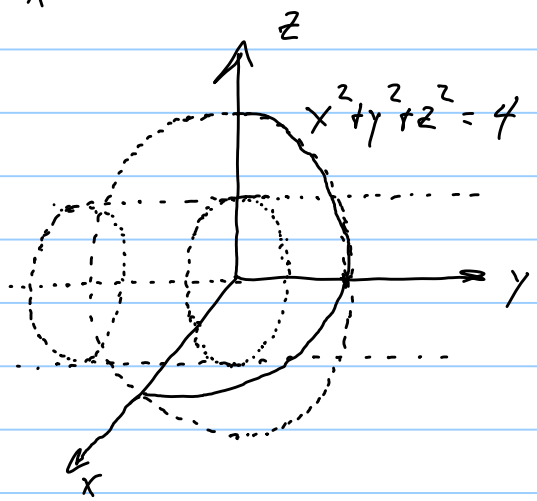
$$-1 \le x \le 1$$



8.

8. The region cut out of the ball $x^2 + y^2 + z^2 \le 4$ by the elliptic cylinder $2x^2 + z^2 = 1$; that is, the region inside the cylinder and the ball

$$\frac{x^2}{\left(\frac{1}{\sqrt{2}}\right)^2} + z^2 = 1$$



The bounds for y are only limited by the sphere.
The sphere has radius $= \sqrt{4} = 2$

$\therefore -\sqrt{4-x^2-z^2} \le y \le \sqrt{4-x^2-z^2}$

The x and z bounds are limited by the cylinder

$\therefore -\sqrt{1-2x^2} \le z \le \sqrt{1-2x^2}$, $-\frac{\sqrt{2}}{2} \le x \le \frac{\sqrt{2}}{2}$

or,

$-\sqrt{(1-z^2)/2} \le x \le \sqrt{(1-z^2)/2}$, $-1 \le z \le 1$

9. **9.** The region inside the sphere $x^2 + y^2 + z^2 = 1$ and above the plane $z = 0$

$0 \le z \le \sqrt{1-x^2-y^2}$, and

$-\sqrt{1-x^2} \le y \le \sqrt{1-x^2}$, $-1 \le x \le 1$

or

$-\sqrt{1-y^2} \le x \le \sqrt{1-y^2}$, $-1 \le y \le 1$

10. **10.** The region bounded by the planes $x = 0$, $y = 0$, $z = 0$, $x + y = 4$, and $x = z - y - 1$

$x=0, y=0, z=0 \Rightarrow$ one of the octants

$x+y=4 \Rightarrow$ intersect x-axis at $(4,0)$, y-axis at $(0,4)$ and so "volume" is in 1st octant

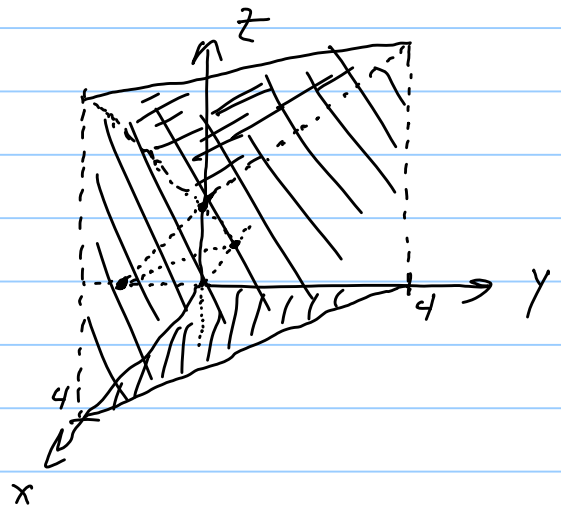
$x = z-y-1 \Leftrightarrow x+y-z=-1$ intersects axes at $(-1,0,0)$, $(0,-1,0)$, $(0,0,1)$ and can be expressed as $z = x+y+1$

$\therefore 0 \le z \le x+y+1$

From $x+y=4$: $0 \le y \le 4-x$

The extremes of x are from $x+y=4$, so $0 \le x \le 4$

$\therefore\ 0 \le z \le x+y+1,\ 0 \le y \le 4-x,\ 0 \le x \le 4$



Find the volume of the region in Exercises 11 to 14.

11. **11.** The region bounded by $z = x^2 + y^2$ and $z = 10 - x^2 - 2y^2$

$z = x^2+y^2$ is an "upward" paraboloid,
$z = 10-x^2-2y^2$ is a "downward paraboloid.

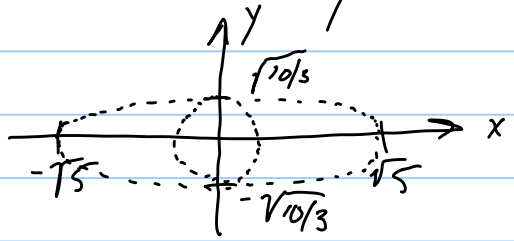
They intersect at $x^2+y^2 = 10-x^2-2y^2$, or

$2x^2+3y^2 = 10$, or $\frac{x^2}{(\sqrt{5})^2} + \frac{y^2}{\left(\sqrt{\frac{10}{3}}\right)^2} = 1$, an

ellipse with borders $\left(\pm\sqrt{5}, 0\right), \left(0, \pm\sqrt{\frac{10}{3}}\right)$

At $\left(\pm\sqrt{5}, 0\right)$, $z = 5$, at $\left(0, \pm\sqrt{\frac{10}{3}}\right)$, $z = \frac{10}{3}$

$\therefore$ Intersection is not in a plane, but the intersection marks the extreme boundary values for x and y.



Notice the $x^2+y^2=z$ circle is limited first by the $y = \pm\sqrt{10/3}$ coordinate before the $x = \pm\sqrt{5}$ coordinate.

$\therefore$ Use $-\sqrt{\frac{10}{3}} \le y \le \sqrt{10/3}$

$$-\sqrt{(10-3y^2)/2} \le x \le \sqrt{(10-3y^2)/2}$$

$$x^2+y^2 \le z \le 10-x^2-2y^2$$

$\therefore$
$$\int_{-\sqrt{\frac{10}{3}}}^{\sqrt{10/3}} \int_{-\sqrt{(10-3y^2)/2}}^{\sqrt{(10-3y^2)/2}} \int_{x^2+y^2}^{10-x^2-2y^2} dz\,dx\,dy$$

$$= \int_{-\sqrt{\frac{10}{3}}}^{\sqrt{\frac{10}{3}}} \int_{-\sqrt{(10-3y^2)/2}}^{\sqrt{(10-3y^2)/2}} (10-2x^2-3y^2)\,dx\,dy$$

$$\int_{-\sqrt{(10-3y^2)/2}}^{\sqrt{(10-3y^2)/2}} (10-2x^2-3y^2)\,dx = 10x - \frac{2}{3}x^3 - 3y^2x \Big|_{x=-\sqrt{(10-3y^2)/2}}^{x=\sqrt{(10-3y^2)/2}}$$

$$= 20\left(\frac{10-3y^2}{2}\right)^{\frac{1}{2}} - \frac{4}{3}\left(\frac{10-3y^2}{2}\right)^{3/2} - 6y^2\left(\frac{10-3y^2}{2}\right)^{\frac{1}{2}}$$

$$= 10\sqrt{2}\sqrt{10-3y^2} - \frac{\sqrt{2}}{3}(10-3y^2)^{3/2} - 3\sqrt{2}\,y^2\sqrt{10-3y^2}$$

Taking each term one at a time, and using $\sqrt{10-3y^2} = \sqrt{3}\sqrt{\frac{10}{3}-y^2}$, using a table of integrals:

(1) $$10\sqrt{6}\int_{-\sqrt{\frac{10}{3}}}^{\sqrt{\frac{10}{3}}} \sqrt{\frac{10}{3}-y^2}\,dy =$$

$$10\sqrt{6}\left[\frac{y}{2}\sqrt{\frac{10}{3}-y^2} + \frac{10}{6}\operatorname{Arcsin}\frac{y}{\sqrt{\frac{10}{3}}}\right]_{y=-\sqrt{\frac{10}{3}}}^{y=\sqrt{\frac{10}{3}}} =$$

$$10\sqrt{6}\left\{0 + \frac{5}{3}\operatorname{Arcsin}(1) - \left(0 - \frac{5}{3}\operatorname{Arcsin}(1)\right)\right\}$$

$$= 10\sqrt{6}\left(\frac{5}{3}\right)(\pi) = \frac{50\sqrt{6}\,\pi}{3} \qquad [1]$$

(2) $$-\frac{\sqrt{2}}{3}\int_{-\sqrt{\frac{10}{3}}}^{\sqrt{\frac{10}{3}}} (10-3y^2)^{3/2}\,dy = -\sqrt{6}\int_{-\sqrt{\frac{10}{3}}}^{\sqrt{\frac{10}{3}}} \left(\frac{10}{3}-y^2\right)^{3/2} dy$$

$$= -\sqrt{6}\left[\frac{y}{8}\left(2y^2 - \frac{50}{3}\right)\sqrt{\frac{10}{3}-y^2} + \frac{3}{8}\left(\frac{100}{9}\right)\operatorname{Arcsin}\left(\frac{y}{\sqrt{\frac{10}{3}}}\right)\right]_{-\sqrt{\frac{10}{3}}}^{\sqrt{\frac{10}{3}}}$$

$$= -\sqrt{6}\left[0 + \frac{25}{2(3)}\operatorname{Arcsin}(1) - \left(0 - \frac{25}{2(3)}\operatorname{Arcsin}(1)\right)\right]$$

$$= -\sqrt{6}\left(\frac{25}{3}\cdot\frac{\pi}{2}\right) = -\frac{25\sqrt{6}\,\pi}{6} \quad [2]$$

(3) $$-3\sqrt{2}\int_{-\sqrt{\frac{10}{3}}}^{\sqrt{\frac{10}{3}}} y^2\sqrt{10-3y^2}\,dy = -3\sqrt{6}\int_{-\sqrt{10/3}}^{\sqrt{10/3}} y^2\sqrt{\frac{10}{3}-y^2}\,dy$$

$$-3\sqrt{6}\left[-\frac{y}{4}\left(\frac{10}{3}-y^2\right)^{3/2} + \frac{y^2}{8}y\sqrt{\frac{10}{3}-y^2} + \left(\frac{10}{3}\right)^2\frac{1}{8}\operatorname{Arcsin}\left(\frac{y}{\sqrt{\frac{10}{3}}}\right)\right]_{-\sqrt{\frac{10}{3}}}^{\sqrt{\frac{10}{3}}}$$

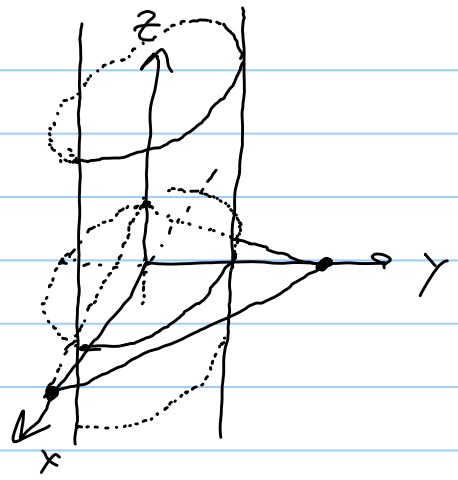
$$= -3\sqrt{6}\left[0 + 0 + \frac{100}{9(8)}\operatorname{Arcsin}(1) - \left(0 + 0 - \frac{100}{9(8)}\operatorname{Arcsin}(1)\right)\right]$$

$$= -3\sqrt{6}\left(\frac{100}{9(8)}\right)(\pi) = -\frac{25\sqrt{6}\,\pi}{6} \quad [3]$$

$$[1] + [2] + [3] = \frac{50\sqrt{6}}{3}\pi + \left(-\frac{25\sqrt{6}}{6}\pi\right) + \left(-\frac{25\sqrt{6}}{6}\right)\pi$$

$$= \pi\left(\frac{100\sqrt{6}}{6} - \frac{50\sqrt{6}}{6}\right) = \underline{\frac{50\sqrt{6}}{6}\pi}$$

12. **12.** The solid bounded by $x^2 + 2y^2 = 2$, $z = 0$, and $x + y + 2z = 2$



$x^2 + 2y^2 = 2 \Leftrightarrow \frac{x^2}{(\sqrt{2})^2} + \frac{y^2}{1} = 1$,

a cylinder parallel to the z-axis, in the shape of an ellipse with margins $(\pm\sqrt{2}, 0)$, $(0, \pm 1)$. $x = \pm\sqrt{2-2y^2}$

The "top" of the volume is a plane $z = 1 - \frac{x}{2} - \frac{y}{2}$, with axis intercepts at $(2,0,0)$, $(0,2,0)$, $(0,0,1)$.

∴ The plane serves as the "top" to the solid, and $z = 0$ serves as the "bottom." Lateral sides are bounded by the ellipse.

∴ $0 \le z \le 1 - \frac{x}{2} - \frac{y}{2}$, $-\sqrt{2-2y^2} \le x \le \sqrt{2-2y^2}$, $-1 \le y \le 1$

∴

$$V = \int_{-1}^{1} \int_{-\sqrt{2-2y^2}}^{\sqrt{2-2y^2}} \int_{0}^{1-\frac{x}{2}-\frac{y}{2}} dz\, dx\, dy$$

$$= \int_{-1}^{1} \int_{-\sqrt{2-2y^2}}^{\sqrt{2-2y^2}} \left(1 - \frac{x}{2} - \frac{y}{2}\right) dx\, dy$$

$$\int_{-1}^{1} \left[\left(x - \frac{x^2}{4} - \frac{xy}{2} \right) \Big|_{x=-\sqrt{2-2y^2}}^{x=\sqrt{2-2y^2}} \right] dx$$

$$= \int_{-1}^{1} \left[\sqrt{2-2y^2} - \frac{(2-2y^2)}{4} - \frac{y\sqrt{2-2y^2}}{2} - \left(-\sqrt{2-2y^2} - \frac{(2-2y^2)}{4} + \frac{y}{2}\sqrt{2-2y^2} \right) \right] dx$$

$$= \int_{-1}^{1} \left(2\sqrt{2-2y^2} - y\sqrt{2-2y^2} \right) dy$$

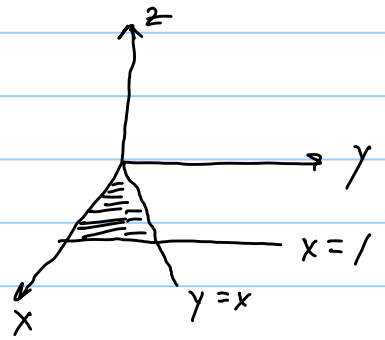
$$= 2\sqrt{2} \int_{-1}^{1} \sqrt{1-y^2}\, dy - \sqrt{2} \int_{-1}^{1} y\sqrt{1-y^2}\, dy$$

$$= 2\sqrt{2} \left[\frac{y}{2}\sqrt{1-y^2} + \frac{1}{2}\operatorname{Arcsin}(y) \right]_{-1}^{1} + \frac{\sqrt{2}}{2}\left(\frac{2}{3}\right)\left(1-y^2\right)^{3/2} \Big|_{-1}^{1}$$

$$= 2\sqrt{2} \left\{ 0 + \frac{1}{2}\left(\frac{\pi}{2}\right) - \left(0 - \frac{1}{2}\left(\frac{\pi}{2}\right)\right) \right\} + 0$$

$$= 2\sqrt{2}\left(\frac{\pi}{2}\right) = \underline{\sqrt{2}\,\pi}$$

13. **13.** The solid bounded by $x = y$, $z = 0$, $y = 0$, $x = 1$, and $x + y + z = 0$



$z = -x - y$ is a plane through $(0,0,0)$, and acts as a "floor" to the solid, $z = 0$ as a "roof", as $z < 0$ for $x, y > 0$.

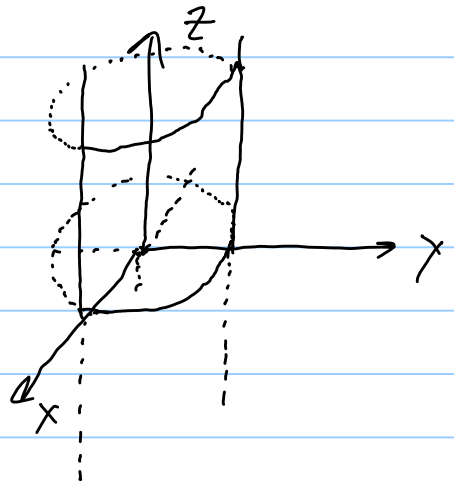
$\therefore\ 0 \le x \le 1,\ 0 \le y \le x,\ -x - y \le z \le 0$

$$\therefore V = \int_0^1 \int_0^x \int_{-x-y}^{0} dz\,dy\,dx$$

$$= \int_0^1 \int_0^x (x+y)\,dy\,dx = \int_0^1 \left[xy + \frac{y^2}{2} \Big|_{y=0}^{y=x} \right] dx$$

$$= \int_0^1 \frac{3}{2} x^2\,dx = \frac{x^3}{2}\Big|_0^1 = \underline{\underline{\frac{1}{2}}}$$

14. **14.** The region common to the intersecting cylinders $x^2 + y^2 \le a^2$ and $x^2 + z^2 \le a^2$



$$-a \le x \le a$$

$$-\sqrt{a^2-x^2} \le y \le \sqrt{a^2-x^2}$$

$$-\sqrt{a^2-x^2} \le z \le \sqrt{a^2-x^2}$$

$$\therefore V = \int_{-a}^{a} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} dz\,dy\,dx$$

$$= \int_{-a}^{a} \int_{-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}} 2\sqrt{a^2-x^2}\,dy\,dx$$

$$= \int_{-a}^{a} \left[2x\sqrt{a^2-x^2} \,\Big|_{y=-\sqrt{a^2-x^2}}^{y=\sqrt{a^2-x^2}} \right] dx$$

$$= \int_{-a}^{a} 4(a^2-x^2)\,dx = 4a^2x - \frac{4}{3}x^3 \Big|_{x=-a}^{x=a}$$

$$= 4a^3 - \frac{4}{3}a^3 - \left(-4a^3 + \frac{4}{3}a^3\right) = 8a^3 - \frac{8}{3}a^3$$

$$= \frac{16}{3}a^3$$

Evaluate the integrals in Exercises 15 to 23.

15.

15. $\int_0^1 \int_1^2 \int_2^3 \cos[\pi(x+y+z)]\,dx\,dy\,dz$

$$\int_2^3 \cos[\pi(x+y+z)]\,dx = \frac{1}{\pi}\sin[\pi(x+y+z)] \Big|_{x=2}^{x=3}$$

$$= \frac{1}{\pi}\left[\sin(3\pi+\pi y+\pi z) - \sin(2\pi+\pi y+\pi z)\right]$$

$$= \frac{1}{\pi}\left[-\sin(\pi y+\pi z) - \sin(\pi y+\pi z)\right]$$

$$= -\frac{2}{\pi}\sin(\pi y+\pi z)$$

$$\int_1^2 -\frac{2}{\pi}\sin(\pi y+\pi z)\,dy = \frac{2}{\pi^2}\cos(\pi y+\pi z) \Big|_{y=1}^{y=2}$$

$$= \frac{2}{\pi^2}\left[\cos(2\pi+\pi z) - \cos(\pi+\pi z)\right]$$

$$= \frac{2}{\pi^2}\left[\cos(\pi z) + \cos(\pi z)\right]$$

$$= \frac{4}{\pi^2}\cos(\pi z)$$

$$\int_0^1 \frac{4}{\pi^2}\cos(\pi z)\,dz = \frac{4}{\pi^3}\sin(\pi z)\Big|_{z=0}^{z=1}$$

$$= \frac{4}{\pi^3}\left[\sin(\pi) - \sin(0)\right] = \underline{0}$$

16. **16.** $\int_0^1 \int_0^x \int_0^y (y + xz)\,dz\,dy\,dx$

$$\int_0^y (y + xz)\,dz = yz + x\frac{z^2}{2}\Big|_{z=0}^{z=y} = y^2 + x\frac{y^2}{2}$$

$$\int_0^x \left(y^2 + x\frac{y^2}{2}\right)dy = \frac{y^3}{3} + \frac{xy^3}{6}\Big|_{y=0}^{y=x} = \frac{x^3}{3} + \frac{x^4}{6}$$

$$\int_0^1 \frac{x^3}{3} + \frac{x^4}{6}\,dx = \frac{x^4}{12} + \frac{x^5}{30}\Big|_0^1 = \frac{1}{12} + \frac{1}{30}$$

$$= \frac{30}{360} + \frac{12}{360} = \frac{42}{360} = \frac{7\cdot 3\cdot 2}{3^2\cdot 2^3\cdot 5} = \frac{7}{3\cdot 2^2\cdot 5} = \underline{\frac{7}{60}}$$

17. **17.** $\iiint_W (x^2+y^2+z^2)\,dx\,dy\,dz$, W is the region bounded by $x+y+z=a$ (where $a>0$), $x=0$, $y=0$, and $z=0$.

The plane $x+y+z=a$ intersects axes at $(a,0,0)$, $(0,a,0)$, $(0,0,a)$. At $z=0$, $x+y=a$, so $y=a-x$.

Top of W is $z=a-x-y$, bottom is $z=0$.

$\therefore$ W: $0\le x\le a$

$0\le y\le a-x$

$0\le z\le a-x-y$

$$\therefore \int_0^a \int_0^{a-x} \int_0^{a-x-y} (x^2+y^2+z^2)\,dz\,dy\,dx$$

$$= \int_0^a \int_0^{a-x} \left(x^2z + y^2z + \frac{z^3}{3}\right)\Big|_{z=0}^{z=a-x-y} dy\,dx$$

$$= \int_0^a \int_0^{a-x} x^2(a-x-y) + y^2(a-x-y) + \frac{(a-x-y)^3}{3}\,dy\,dz$$

$$= \int_0^a \int_0^{a-x} ax^2 - x^3 - x^2y + (a-x)y^2 - y^3 + \frac{(a-x-y)^3}{3}\,dy\,dz$$

$$= \int_0^a \left[ax^2y - x^3y - \frac{x^2y^2}{2} + (a-x)\frac{y^3}{3} - \frac{y^4}{4} - \frac{(a-x-y)^4}{12}\right]\Big|_{y=0}^{y=a-x}$$